CHAPTER 10
Fixed-Axis Rotation

FIGURE 10.1 Brazos wind farm in west Texas. During 2019, wind farms in the United States had an average power output of 34 gigawatts, which is enough to power 28 million homes. (credit: modification of work by U.S. Department of Energy)

CHAPTER OUTLINE

INTRODUCTION In previous chapters, we described motion (kinematics) and how to change motion (dynamics), and we defined important concepts such as energy for objects that can be considered as point masses. Point masses, by definition, have no shape and so can only undergo translational motion. However, we know from everyday life that rotational motion is also very important and that many objects that move have both translation and rotation. The wind turbines in our chapter opening image are a prime example of how rotational motion impacts our daily lives, as the market for clean energy sources continues to grow.

We begin to address rotational motion in this chapter, starting with fixed-axis rotation. Fixed-axis rotation describes the rotation around a fixed axis of a rigid body; that is, an object that does not deform as it moves. We will show how to apply all the ideas we've developed up to this point about translational motion to an object rotating around a fixed axis. In the next chapter, we extend these ideas to more complex rotational motion, including objects that both rotate and translate, and objects that do not have a fixed rotational axis.

10.1 Rotational Variables

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the physical meaning of rotational variables as applied to fixed-axis rotation
- Explain how angular velocity is related to tangential speed
- Calculate the instantaneous angular velocity given the angular position function
- Find the angular velocity and angular acceleration in a rotating system
- Calculate the average angular acceleration when the angular velocity is changing
- Calculate the instantaneous angular acceleration given the angular velocity function

So far in this text, we have mainly studied translational motion, including the variables that describe it: displacement, velocity, and acceleration. Now we expand our description of motion to rotation—specifically, rotational motion about a fixed axis. We will find that rotational motion is described by a set of related variables similar to those we used in translational motion.

Angular Velocity

Uniform circular motion (discussed previously in Motion in Two and Three Dimensions) is motion in a circle at constant speed. Although this is the simplest case of rotational motion, it is very useful for many situations, and we use it here to introduce rotational variables.

In Figure 10.2, we show a particle moving in a circle. The coordinate system is fixed and serves as a frame of reference to define the particle's position. Its position vector from the origin of the circle to the particle sweeps out the angle θ, which increases in the counterclockwise direction as the particle moves along its circular path. The angle θ is called the **angular position** of the particle. As the particle moves in its circular path, it also traces an arc length *s*. The particle may complete more than one revolution around the circle, and so the angle θ may be greater than 2π, and the arc length *s* may be greater than the circumference, $2\pi r$.

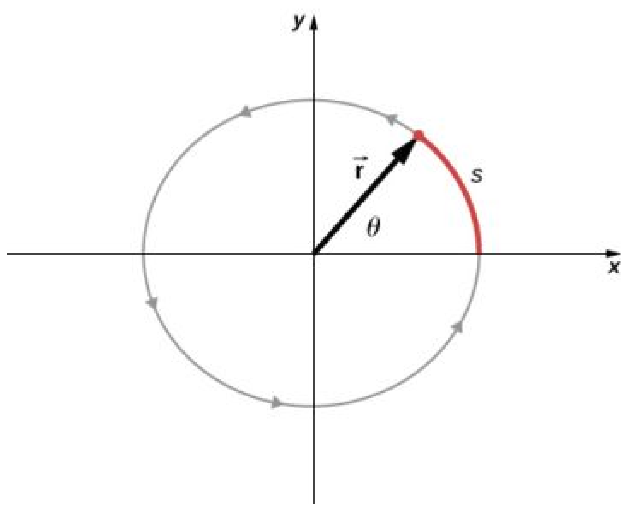


FIGURE 10.2 A particle follows a circular path. As it moves counterclockwise, it sweeps out a positive angle θ with respect to the *x*-axis and traces out an arc length *s*.

The angle is related to the radius of the circle and the arc length by

$$\theta = \frac{s}{r}. \tag{10.1}$$

The angle θ, the angular position of the particle along its path, has units of radians (rad). There are 2π radians in $360°$. Note that the radian measure is a ratio of length measurements, and therefore is a dimensionless quantity. As the particle moves along its circular path, its angular position changes and it undergoes angular displacements $\Delta\theta$.

We can assign vectors to the quantities in Equation 10.1. The angle $\vec{\boldsymbol{\theta}}$ is a vector out of the page in Figure 10.2. The angular position vector $\vec{\mathbf{r}}$ and the arc length $\vec{\mathbf{s}}$ both lie in the plane of the page. These three vectors are related to each other by

$$\vec{\mathbf{s}} = \vec{\boldsymbol{\theta}} \times \vec{\mathbf{r}}. \tag{10.2}$$

That is, the arc length is the cross product of the angle vector and the position vector, as shown in Figure 10.3.

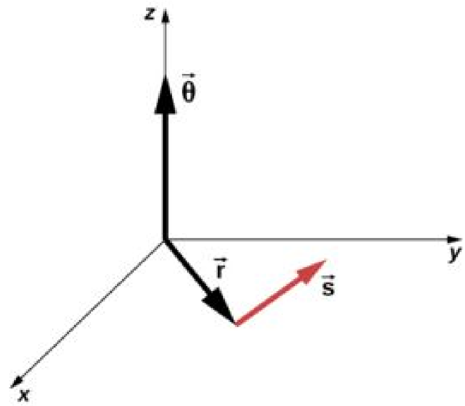


FIGURE 10.3 The angle vector points along the *z*-axis and the position vector and arc length vector both lie in the *xy*-plane. We see that $\vec{\mathbf{s}} = \vec{\boldsymbol{\theta}} \times \vec{\mathbf{r}}$. All three vectors are perpendicular to each other.

The magnitude of the **angular velocity**, denoted by ω, is the time rate of change of the angle θ as the particle moves in its circular path. The **instantaneous angular velocity** is defined as the limit in which $\Delta t \to 0$ in the average angular velocity $\bar{\omega} = \frac{\Delta\theta}{\Delta t}$:

$$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}, \tag{10.3}$$

where θ is the angle of rotation (Figure 10.2). The units of angular velocity are radians per second (rad/s). Angular velocity can also be referred to as the rotation rate in radians per second. In many situations, we are given the rotation rate in revolutions/s or cycles/s. To find the angular velocity, we must multiply revolutions/s by 2π, since there are 2π radians in one complete revolution. Since the direction of a positive angle in a circle is counterclockwise, we take counterclockwise rotations as being positive and clockwise rotations as negative.

We can see how angular velocity is related to the tangential speed of the particle by differentiating Equation 10.1 with respect to time. We rewrite Equation 10.1 as

$$s = r\theta.$$

Taking the derivative with respect to time and noting that the radius *r* is a constant, we have

$$\frac{ds}{dt} = \frac{d}{dt}(r\theta) = \theta\frac{dr}{dt} + r\frac{d\theta}{dt} = r\frac{d\theta}{dt}$$

where $\theta\frac{dr}{dt} = 0$. Here $\frac{ds}{dt}$ is just the tangential speed v_{t} of the particle in Figure 10.2. Thus, by using Equation 10.3, we arrive at

$$v_{\mathrm{t}} = r\omega. \tag{10.4}$$

That is, the tangential speed of the particle is its angular velocity times the radius of the circle. From Equation 10.4, we see that the tangential speed of the particle increases with its distance from the axis of rotation for a constant angular velocity. This effect is shown in Figure 10.4. Two particles are placed at different radii on a rotating disk with a constant angular velocity. As the disk rotates, the tangential speed increases linearly with the radius from the axis of rotation. In Figure 10.4, we see that $v_1 = r_1\omega_1$ and $v_2 = r_2\omega_2$. But the disk has a constant angular velocity, so $\omega_1 = \omega_2$. This means $\frac{v_1}{r_1} = \frac{v_2}{r_2}$ or $v_2 = \left(\frac{r_2}{r_1}\right)v_1$. Thus, since $r_2 > r_1$, $v_2 > v_1$.

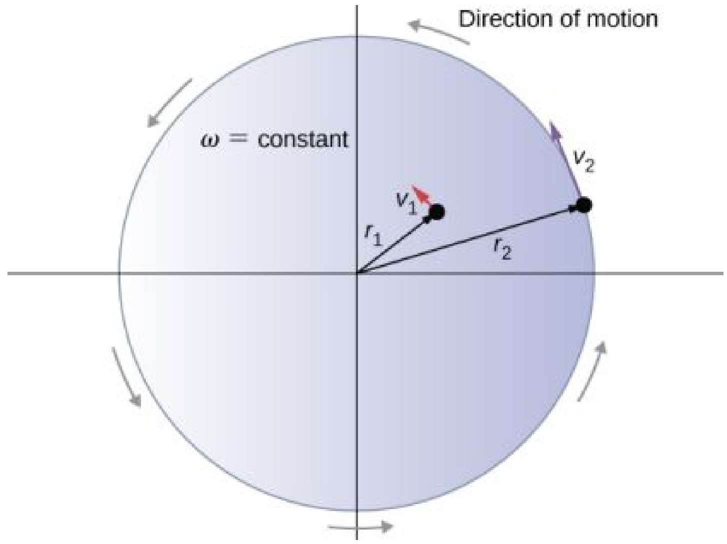


FIGURE 10.4 Two particles on a rotating disk have different tangential speeds, depending on their distance to the axis of rotation.

Up until now, we have discussed the magnitude of the angular velocity $\omega = d\theta/dt$, which is a scalar quantity—the change in angular position with respect to time. The vector $\vec{\boldsymbol{\omega}}$ is the vector associated with the angular velocity and points along the axis of rotation. This is useful because when a rigid body is rotating, we want to know both the axis of rotation and the direction that the body is rotating about the axis, clockwise or counterclockwise. The angular velocity $\vec{\boldsymbol{\omega}}$ gives us this information. The angular velocity $\vec{\boldsymbol{\omega}}$ has a direction determined by what is called the right-hand rule. The right-hand rule is such that if the fingers of your right hand wrap counterclockwise from the *x*-axis (the direction in which θ increases) toward the *y*-axis, your thumb points in the direction of the positive *z*-axis (Figure 10.5). An angular velocity $\vec{\boldsymbol{\omega}}$ that points along the positive *z*-axis therefore corresponds to a counterclockwise rotation, whereas an angular velocity $\vec{\boldsymbol{\omega}}$ that points along the negative *z*-axis corresponds to a clockwise rotation.

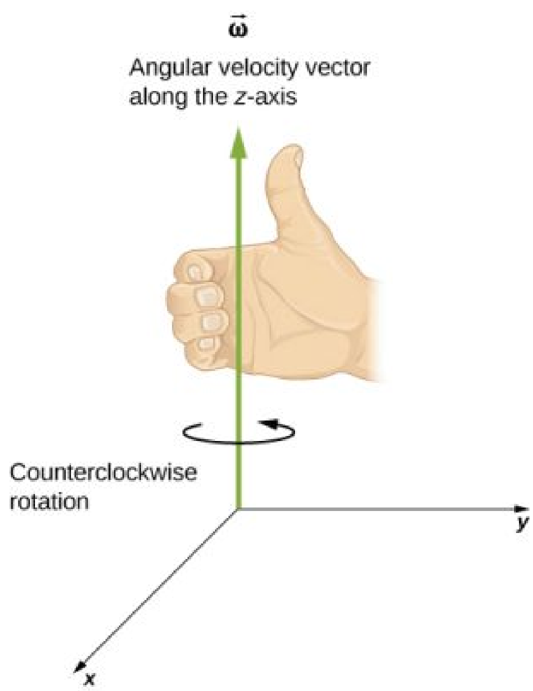


FIGURE 10.5 For counterclockwise rotation in the coordinate system shown, the angular velocity points in the positive *z*-direction by the right-hand-rule.

Similar to Equation 10.2, one can state a cross product relation to the vector of the tangential velocity as stated in Equation 10.4. Therefore, we have

$$\vec{\mathbf{v}} = \vec{\boldsymbol{\omega}} \times \vec{\mathbf{r}}. \tag{10.5}$$

That is, the tangential velocity is the cross product of the angular velocity and the position vector, as shown in Figure 10.6. From part (a) of this figure, we see that with the angular velocity in the positive *z*-direction, the rotation in the *xy*-plane is counterclockwise. In part (b), the angular velocity is in the negative *z*-direction, giving a clockwise rotation in the *xy*-plane.

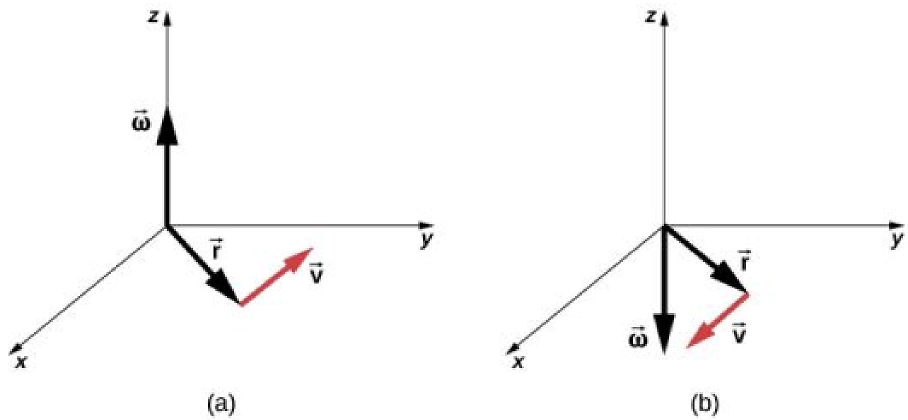


FIGURE 10.6 The vectors shown are the angular velocity, position, and tangential velocity. (a) The angular velocity points in the positive *z*-direction, giving a counterclockwise rotation in the *xy*-plane. (b) The angular velocity points in the negative *z*-direction, giving a clockwise rotation.

EXAMPLE 10.1

Rotation of a Flywheel

A flywheel rotates such that it sweeps out an angle at the rate of $\theta = \omega t = (45.0\ \text{rad/s})t$ radians. The wheel rotates counterclockwise when viewed in the plane of the page. (a) What is the angular velocity of the flywheel? (b) What direction is the angular velocity? (c) How many radians does the flywheel rotate through in 30 s? (d) What is the tangential speed of a point on the flywheel 10 cm from the axis of rotation?

Strategy

The functional form of the angular position of the flywheel is given in the problem as $\theta(t) = \omega t$, so by taking the derivative with respect to time, we can find the angular velocity. We use the right-hand rule to find the angular velocity. To find the angular displacement of the flywheel during 30 s, we seek the angular displacement $\Delta\theta$, where the change in angular position is between 0 and 30 s. To find the tangential speed of a point at a distance from the axis of rotation, we multiply its distance times the angular velocity of the flywheel.

Solution

a. $\omega = \frac{d\theta}{dt} = 45\ \text{rad/s}$. We see that the angular velocity is a constant.
b. By the right-hand rule, we curl the fingers in the direction of rotation, which is counterclockwise in the plane of the page, and the thumb points in the direction of the angular velocity, which is out of the page.
c. $\Delta\theta = \theta(30\ \text{s}) - \theta(0\ \text{s}) = 45.0(30.0\ \text{s}) - 45.0(0\ \text{s}) = 1350.0\ \text{rad}$.
d. $v_\text{t} = r\omega = (0.1\ \text{m})(45.0\ \text{rad/s}) = 4.5\ \text{m/s}$.

Significance

In 30 s, the flywheel has rotated through quite a number of revolutions, about 215 if we divide the angular displacement by 2π. A massive flywheel can be used to store energy in this way, if the losses due to friction are minimal. Recent research has considered superconducting bearings on which the flywheel rests, with zero energy loss due to friction.

Angular Acceleration

We have just discussed angular velocity for uniform circular motion, but not all motion is uniform. Envision an ice skater spinning with his arms outstretched—when he pulls his arms inward, his angular velocity increases. Or think about a computer's hard disk slowing to a halt as the angular velocity decreases. We will explore these situations later, but we can already see a need to define an **angular acceleration** for describing situations where ω changes. The faster the change in ω, the greater the angular acceleration. We define the **instantaneous angular acceleration** α as the derivative of angular velocity with respect to time:

$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta \omega}{\Delta t} = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}, \tag{10.6}$$

where we have taken the limit of the average angular acceleration, $\bar{\alpha} = \frac{\Delta \omega}{\Delta t}$ as $\Delta t \to 0$.

The units of angular acceleration are (rad/s)/s, or rad/s^2.

In the same way as we defined the vector associated with angular velocity $\vec{\boldsymbol{\omega}}$, we can define $\vec{\boldsymbol{\alpha}}$, the vector associated with angular acceleration (Figure 10.7). If the angular velocity is along the positive *z*-axis, as in Figure 10.5, and $\frac{d\omega}{dt}$ is positive, then the angular acceleration $\vec{\boldsymbol{\alpha}}$ is positive and points along the $+z$- axis. Similarly, if the angular velocity $\vec{\boldsymbol{\omega}}$ is along the positive *z*-axis and $\frac{d\omega}{dt}$ is negative, then the angular acceleration is negative and points along the $-z-$ axis.

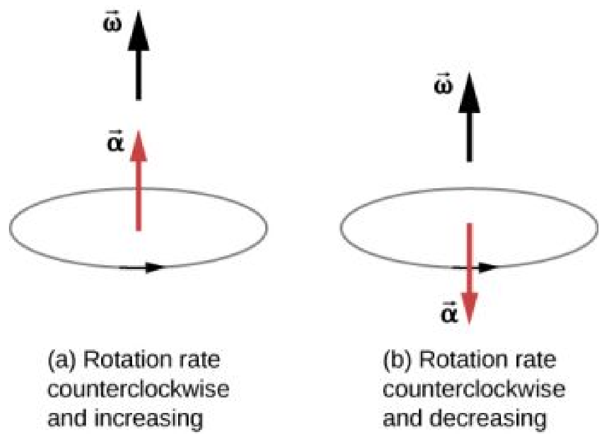


FIGURE 10.7 The rotation is counterclockwise in both (a) and (b) with the angular velocity in the same direction. (a) The angular acceleration is in the same direction as the angular velocity, which increases the rotation rate. (b) The angular acceleration is in the opposite direction to the angular velocity, which decreases the rotation rate.

We can express the tangential acceleration vector as a cross product of the angular acceleration and the position vector. This expression can be found by taking the time derivative of $\vec{\mathbf{v}} = \vec{\boldsymbol{\omega}} \times \vec{\mathbf{r}}$ and is left as an exercise:

$$\vec{\mathbf{a}} = \vec{\boldsymbol{\alpha}} \times \vec{\mathbf{r}}. \tag{10.7}$$

The vector relationships for the angular acceleration and tangential acceleration are shown in Figure 10.8.

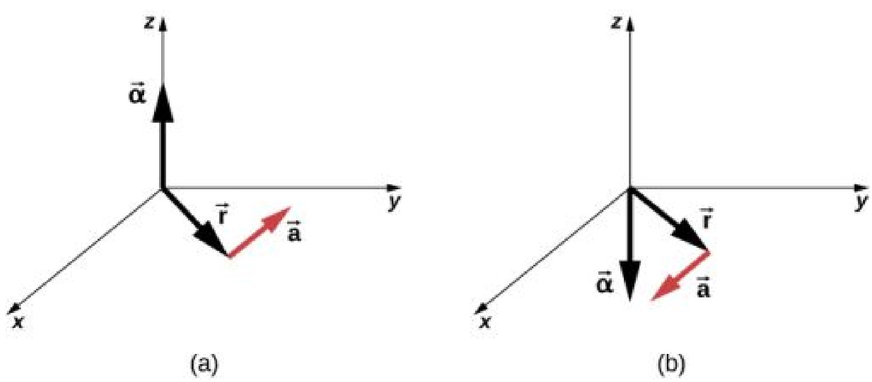


FIGURE 10.8 (a) The angular acceleration is the positive *z*-direction and produces a tangential acceleration in a counterclockwise sense. (b) The angular acceleration is in the negative *z*-direction and produces a tangential acceleration in the clockwise sense.

We can relate the tangential acceleration of a point on a rotating body at a distance from the axis of rotation in the same way that we related the tangential speed to the angular velocity. If we differentiate Equation 10.4 with respect to time, noting that the radius *r* is constant, we obtain

$$a_t = r\alpha. \tag{10.8}$$

Thus, the tangential acceleration a_t is the radius times the angular acceleration. Equation 10.4 and Equation 10.8 are important for the discussion of rolling motion (see Angular Momentum).

Let's apply these ideas to the analysis of a few simple fixed-axis rotation scenarios. Before doing so, we present a problem-solving strategy that can be applied to rotational kinematics: the description of rotational motion.

PROBLEM-SOLVING STRATEGY

Rotational Kinematics

1. Examine the situation to determine that rotational kinematics (rotational motion) is involved.
2. Identify exactly what needs to be determined in the problem (identify the unknowns). A sketch of the situation is useful.
3. Make a complete list of what is given or can be inferred from the problem as stated (identify the knowns).
4. Solve the appropriate equation or equations for the quantity to be determined (the unknown). It can be useful to think in terms of a translational analog, because by now you are familiar with the equations of translational motion.
5. Substitute the known values along with their units into the appropriate equation and obtain numerical solutions complete with units. Be sure to use units of radians for angles.
6. Check your answer to see if it is reasonable: Does your answer make sense?

Now let's apply this problem-solving strategy to a few specific examples.

EXAMPLE 10.2

A Spinning Bicycle Wheel

A bicycle mechanic mounts a bicycle on the repair stand and starts the rear wheel spinning from rest to a final angular velocity of 250 rpm in 5.00 s. (a) Calculate the average angular acceleration in $\mathrm{rad/s^2}$. (b) If she now hits the brakes, causing an angular acceleration of $-87.3\ \mathrm{rad/s^2}$, how long does it take the wheel to stop?

Strategy

The average angular acceleration can be found directly from its definition $\bar{\alpha} = \frac{\Delta\omega}{\Delta t}$ because the final angular velocity and time are given. We see that $\Delta\omega = \omega_{\text{final}} - \omega_{\text{initial}} = 250\ \text{rev/min}$ and Δt is 5.00 s. For part (b), we know the angular acceleration and the initial angular velocity. We can find the stopping time by using the definition of average angular acceleration and solving for Δt, yielding

$$\Delta t = \frac{\Delta\omega}{\alpha}.$$

Solution

a. Entering known information into the definition of angular acceleration, we get

$$\bar{\alpha} = \frac{\Delta\omega}{\Delta t} = \frac{250\ \text{rpm}}{5.00\ \text{s}}.$$

Because $\Delta\omega$ is in revolutions per minute (rpm) and we want the standard units of $\mathrm{rad/s^2}$ for angular acceleration, we need to convert from rpm to rad/s:

$$\Delta\omega = 250\frac{\text{rev}}{\text{min}} \cdot \frac{2\pi\ \text{rad}}{\text{rev}} \cdot \frac{1\ \text{min}}{60\ \text{s}} = 26.2\frac{\text{rad}}{\text{s}}.$$

Entering this quantity into the expression for α, we get

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{26.2\ \text{rad/s}}{5.00\ \text{s}} = 5.24\ \text{rad/s}^2.$$

b. Here the angular velocity decreases from 26.2 rad/s (250 rpm) to zero, so that $\Delta\omega$ is −26.2 rad/s, and α is given to be $-87.3\ \text{rad/s}^2$. Thus,

$$\Delta t = \frac{-26.2\ \text{rad/s}}{-87.3\ \text{rad/s}^2} = 0.300\ \text{s}.$$

Significance

Note that the angular acceleration as the mechanic spins the wheel is small and positive; it takes 5 s to produce an appreciable angular velocity. When she hits the brake, the angular acceleration is large and negative. The angular velocity quickly goes to zero.

CHECK YOUR UNDERSTANDING 10.1

The fan blades on a turbofan jet engine (shown below) accelerate from rest up to a rotation rate of 40.0 rev/s in 20 s. The increase in angular velocity of the fan is constant in time. (The GE90-110B1 turbofan engine mounted on a Boeing 777, as shown, is currently the largest turbofan engine in the world, capable of thrusts of 330–510 kN.)

(a) What is the average angular acceleration?

(b) What is the instantaneous angular acceleration at any time during the first 20 s?

FIGURE 10.9 (credit: “Bubinator”/ Wikimedia Commons)

EXAMPLE 10.3

Wind Turbine

A wind turbine (Figure 10.10) in a wind farm is being shut down for maintenance. It takes 30 s for the turbine to go from its operating angular velocity to a complete stop in which the angular velocity function is $\omega(t) = [(t\text{s}^{-1} - 30.0)^2/100.0]\text{rad/s}$, where t is the time in seconds. If the turbine is rotating counterclockwise looking into the page, (a) what are the directions of the angular velocity and acceleration vectors? (b) What is the average angular acceleration? (c) What is the instantaneous angular acceleration at $t = 0.0, 15.0, 30.0\ \text{s}$?

FIGURE 10.10 A wind turbine that is rotating counterclockwise, as seen head on.

Strategy

a. We are given the rotational sense of the turbine, which is counterclockwise in the plane of the page. Using the right hand rule (Figure 10.5), we can establish the directions of the angular velocity and acceleration vectors.
b. We calculate the initial and final angular velocities to get the average angular acceleration. We establish the sign of the angular acceleration from the results in (a).
c. We are given the functional form of the angular velocity, so we can find the functional form of the angular acceleration function by taking its derivative with respect to time.

Solution

a. Since the turbine is rotating counterclockwise, angular velocity $\vec{\boldsymbol{\omega}}$ points out of the page. But since the angular velocity is decreasing, the angular acceleration $\vec{\boldsymbol{\alpha}}$ points into the page, in the opposite sense to the angular velocity.
b. The initial angular velocity of the turbine, setting $t = 0,$ is $\omega = 9.0\,\text{rad/s}$. The final angular velocity is zero, so the average angular acceleration is
$$\bar{\alpha} = \frac{\Delta\omega}{\Delta t} = \frac{\omega - \omega_0}{t - t_0} = \frac{0 - 9.0\,\text{rad/s}}{30.0 - 0\,\text{s}} = -0.3\,\text{rad/s}^2.$$
c. Taking the derivative of the angular velocity with respect to time gives $\alpha = \frac{d\omega}{dt} = (t - 30.0)\,/50.0\,\text{rad/s}^2$
$$\alpha(0.0\,\text{s}) = -0.6\,\text{rad/s}^2, \alpha(15.0\,\text{s}) = -0.3\,\text{rad/s}^2,\ \text{and}\ \alpha(30.0\,\text{s}) = 0\,\text{rad/s}.$$

Significance

We found from the calculations in (a) and (b) that the angular acceleration α and the average angular acceleration $\bar{\alpha}$ are negative. The turbine has an angular acceleration in the opposite sense to its angular velocity.

We now have a basic vocabulary for discussing fixed-axis rotational kinematics and relationships between rotational variables. We discuss more definitions and connections in the next section.

10.2 Rotation with Constant Angular Acceleration

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Derive the kinematic equations for rotational motion with constant angular acceleration
- Select from the kinematic equations for rotational motion with constant angular acceleration the appropriate equations to solve for unknowns in the analysis of systems undergoing fixed-axis rotation
- Use solutions found with the kinematic equations to verify the graphical analysis of fixed-axis rotation with constant angular acceleration

In the preceding section, we defined the rotational variables of angular displacement, angular velocity, and angular acceleration. In this section, we work with these definitions to derive relationships among these variables and use these relationships to analyze rotational motion for a rigid body about a fixed axis under a constant angular acceleration. This analysis forms the basis for rotational kinematics. If the angular acceleration is constant, the equations of rotational kinematics simplify, similar to the equations of linear kinematics discussed in Motion along a Straight Line and Motion in Two and Three Dimensions. We can then use this simplified set of equations to describe many applications in physics and engineering where the angular acceleration of the system is constant. Rotational kinematics is also a prerequisite to the discussion of rotational dynamics later in this chapter.

Kinematics of Rotational Motion

Using our intuition, we can begin to see how the rotational quantities θ, ω, α, and t are related to one another. For example, we saw in the preceding section that if a flywheel has an angular acceleration in the same direction as its angular velocity vector, its angular velocity increases with time and its angular displacement also increases. On the contrary, if the angular acceleration is opposite to the angular velocity vector, its angular velocity decreases with time. We can describe these physical situations and many others with a consistent set of rotational kinematic equations under a constant angular acceleration. The method to investigate rotational motion in this way is called **kinematics of rotational motion**.

To begin, we note that if the system is rotating under a constant acceleration, then the average angular velocity follows a simple relation because the angular velocity is increasing linearly with time. The average angular velocity is just half the sum of the initial and final values:

$$\bar{\omega} = \frac{\omega_0 + \omega_f}{2}. \tag{10.9}$$

From the definition of the average angular velocity, we can find an equation that relates the angular position, average angular velocity, and time:

$$\bar{\omega} = \frac{\Delta\theta}{\Delta t}.$$

Solving for θ, we have

$$\theta_f = \theta_0 + \bar{\omega} t, \tag{10.10}$$

where we have set $t_0 = 0$. This equation can be very useful if we know the average angular velocity of the system. Then we could find the angular displacement over a given time period. Next, we find an equation relating ω, α, and t. To determine this equation, we start with the definition of angular acceleration:

$$\alpha = \frac{d\omega}{dt}.$$

We rearrange this to get $\alpha dt = d\omega$ and then we integrate both sides of this equation from initial values to final values, that is, from t_0 to t and ω_0 to ω_f. In uniform rotational motion, the angular acceleration is constant so it can be pulled out of the integral, yielding two definite integrals:

$$\alpha \int_{t_0}^{t} dt' = \int_{\omega_0}^{\omega_f} d\omega.$$

Setting $t_0 = 0$, we have

$$\alpha t = \omega_f - \omega_0.$$

We rearrange this to obtain

$$\omega_f = \omega_0 + \alpha t, \tag{10.11}$$

where ω_0 is the initial angular velocity. Equation 10.11 is the rotational counterpart to the linear kinematics equation $v_f = v_0 + at$. With Equation 10.11, we can find the angular velocity of an object at any specified time t given the initial angular velocity and the angular acceleration.

Let's now do a similar treatment starting with the equation $\omega = \frac{d\theta}{dt}$. We rearrange it to obtain $\omega dt = d\theta$ and integrate both sides from initial to final values again, noting that the angular acceleration is constant and does not have a time dependence. However, this time, the angular velocity is not constant (in general), so we substitute in what we derived above:

$$\int_{t_0}^{t_f} (\omega_0 + \alpha t')dt' = \int_{\theta_0}^{\theta_f} d\theta;$$

$$\int_{t_0}^{t} \omega_0 dt + \int_{t_0}^{t} \alpha t' dt' = \int_{\theta_0}^{\theta_f} d\theta = \left[\omega_0 t' + \alpha\left(\frac{(t')^2}{2}\right)\right]_{t_0}^{t} = \omega_0 t + \alpha\left(\frac{t^2}{2}\right) = \theta_f - \theta_0,$$

where we have set $t_0 = 0$. Now we rearrange to obtain

$$\theta_f = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2. \tag{10.12}$$

Equation 10.12 is the rotational counterpart to the linear kinematics equation found in Motion Along a Straight Line for position as a function of time. This equation gives us the angular position of a rotating rigid body at any time t given the initial conditions (initial angular position and initial angular velocity) and the angular acceleration.

We can find an equation that is independent of time by solving for t in Equation 10.11 and substituting into Equation 10.12. Equation 10.12 becomes

$$\begin{aligned}
\theta_f &= \theta_0 + \omega_0\left(\frac{\omega_f - \omega_0}{\alpha}\right) + \frac{1}{2}\alpha\left(\frac{\omega_f - \omega_0}{\alpha}\right)^2 \\
&= \theta_0 + \frac{\omega_0\omega_f}{\alpha} - \frac{\omega_0^2}{\alpha} + \frac{1}{2}\frac{\omega_f^2}{\alpha} - \frac{\omega_0\omega_f}{\alpha} + \frac{1}{2}\frac{\omega_0^2}{\alpha} \\
&= \theta_0 + \frac{1}{2}\frac{\omega_f^2}{\alpha} - \frac{1}{2}\frac{\omega_0^2}{\alpha}, \\
\theta_f - \theta_0 &= \frac{\omega_f^2 - \omega_0^2}{2\alpha}
\end{aligned}$$

or

$$\omega_f^2 = \omega_0^2 + 2\alpha(\Delta\theta). \tag{10.13}$$

Equation 10.10 through Equation 10.13 describe fixed-axis rotation for constant acceleration and are summarized in Table 10.1.

Angular displacement from average angular velocity	$\theta_f = \theta_0 + \bar{\omega} t$
Angular velocity from angular acceleration	$\omega_f = \omega_0 + \alpha t$
Angular displacement from angular velocity and angular acceleration	$\theta_f = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$
Angular velocity from angular displacement and angular acceleration	$\omega_f^2 = \omega_0{}^2 + 2\alpha(\Delta\theta)$

TABLE 10.1 **Kinematic Equations**

Applying the Equations for Rotational Motion

Now we can apply the key kinematic relations for rotational motion to some simple examples to get a feel for how the equations can be applied to everyday situations.

EXAMPLE 10.4

Calculating the Acceleration of a Fishing Reel

A deep-sea fisherman hooks a big fish that swims away from the boat, pulling the fishing line from his fishing reel. The whole system is initially at rest, and the fishing line unwinds from the reel at a radius of 4.50 cm from its axis of rotation. The reel is given an angular acceleration of $110\ \text{rad/s}^2$ for 2.00 s (Figure 10.11).

(a) What is the final angular velocity of the reel after 2 s?

(b) How many revolutions does the reel make?

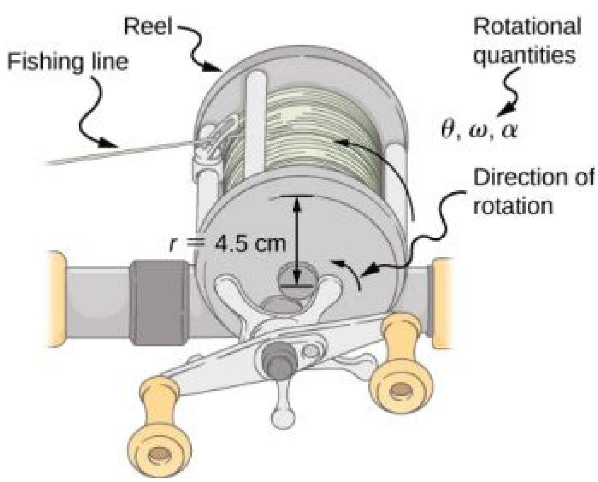


FIGURE 10.11 Fishing line coming off a rotating reel moves linearly.

Strategy

Identify the knowns and compare with the kinematic equations for constant acceleration. Look for the appropriate equation that can be solved for the unknown, using the knowns given in the problem description.

Solution

a. We are given α and t and want to determine ω. The most straightforward equation to use is $\omega_f = \omega_0 + \alpha t$, since all terms are known besides the unknown variable we are looking for. We are given that $\omega_0 = 0$ (it starts from rest), so

$$\omega_f = 0 + (110\ \text{rad/s}^2)(2.00\ \text{s}) = 220\ \text{rad/s}.$$

b. We are asked to find the number of revolutions. Because $1\ \text{rev} = 2\pi\ \text{rad}$, we can find the number of revolutions by finding θ in radians. We are given α and t, and we know ω_0 is zero, so we can obtain θ by using

$$\begin{aligned}\theta_f &= \theta_i + \omega_i t + \tfrac{1}{2}\alpha t^2 \\ &= 0 + 0 + (0.500)\left(110\ \text{rad/s}^2\right)(2.00\ \text{s})^2 = 220\ \text{rad}.\end{aligned}$$

Converting radians to revolutions gives

$$\text{Number of rev} = (220\ \text{rad})\frac{1\ \text{rev}}{2\pi\ \text{rad}} = 35.0\ \text{rev}.$$

Significance

This example illustrates that relationships among rotational quantities are highly analogous to those among linear quantities. The answers to the questions are realistic. After unwinding for two seconds, the reel is found to spin at 220 rad/s, which is 2100 rpm. (No wonder reels sometimes make high-pitched sounds.)

In the preceding example, we considered a fishing reel with a positive angular acceleration. Now let us consider what happens with a negative angular acceleration.

EXAMPLE 10.5

Calculating the Duration When the Fishing Reel Slows Down and Stops

Now the fisherman applies a brake to the spinning reel, achieving an angular acceleration of $-300\ \text{rad/s}^2$. How long does it take the reel to come to a stop?

Strategy

We are asked to find the time *t* for the reel to come to a stop. The initial and final conditions are different from those in the previous problem, which involved the same fishing reel. Now we see that the initial angular velocity is $\omega_0 = 220\ \text{rad/s}$ and the final angular velocity ω is zero. The angular acceleration is given as $\alpha = -300\ \text{rad/s}^2$. Examining the available equations, we see all quantities but *t* are known in $\omega_f = \omega_0 + \alpha t$, making it easiest to use this equation.

Solution

The equation states

$$\omega_f = \omega_0 + \alpha t.$$

We solve the equation algebraically for *t* and then substitute the known values as usual, yielding

$$t = \frac{\omega_f - \omega_0}{\alpha} = \frac{0 - 220.0\ \text{rad/s}}{-300.0\ \text{rad/s}^2} = 0.733\ \text{s}.$$

Significance

Note that care must be taken with the signs that indicate the directions of various quantities. Also, note that the time to stop the reel is fairly small because the acceleration is rather large. Fishing lines sometimes snap because of the accelerations involved, and fishermen often let the fish swim for a while before applying brakes on the reel. A tired fish is slower, requiring a smaller acceleration.

CHECK YOUR UNDERSTANDING 10.2

A centrifuge used in DNA extraction spins at a maximum rate of 7000 rpm, producing a “g-force” on the sample that is 6000 times the force of gravity. If the centrifuge takes 10 seconds to come to rest from the maximum spin rate: (a) What is the angular acceleration of the centrifuge? (b) What is the angular displacement of the centrifuge during this time?

EXAMPLE 10.6

Angular Acceleration of a Propeller

Figure 10.12 shows a graph of the angular velocity of a propeller on an aircraft as a function of time. Its angular velocity starts at 30 rad/s and drops linearly to 0 rad/s over the course of 5 seconds. (a) Find the angular acceleration of the object and verify the result using the kinematic equations. (b) Find the angle through which the propeller rotates during these 5 seconds and verify your result using the kinematic equations.

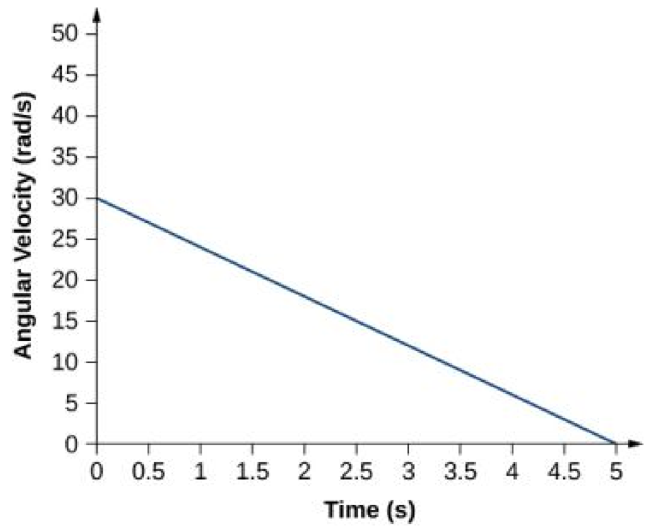


FIGURE 10.12 A graph of the angular velocity of a propeller versus time.

Strategy

a. Since the angular velocity varies linearly with time, we know that the angular acceleration is constant and does not depend on the time variable. The angular acceleration is the slope of the angular velocity vs. time graph, $\alpha = \frac{d\omega}{dt}$. To calculate the slope, we read directly from Figure 10.12, and see that $\omega_0 = 30\,\text{rad/s}$ at $t = 0\,\text{s}$ and $\omega_\text{f} = 0\,\text{rad/s}$ at $t = 5\,\text{s}$. Then, we can verify the result using $\omega = \omega_0 + \alpha t$.
b. We use the equation $\omega = \frac{d\theta}{dt}$; since the time derivative of the angle is the angular velocity, we can find the angular displacement by integrating the angular velocity, which from the figure means taking the area under the angular velocity graph. In other words:

$$\int_{\theta_0}^{\theta_\text{f}} d\theta = \theta_\text{f} - \theta_0 = \int_{t_0}^{t_\text{f}} \omega(t)dt.$$

Then we use the kinematic equations for constant acceleration to verify the result.

Solution

a. Calculating the slope, we get

$$\alpha = \frac{\omega - \omega_0}{t - t_0} = \frac{(0 - 30.0)\,\text{rad/s}}{(5.0 - 0)\,\text{s}} = -6.0\,\text{rad/s}^2.$$

We see that this is exactly Equation 10.11 with a little rearranging of terms.
b. We can find the area under the curve by calculating the area of the right triangle, as shown in Figure 10.13.

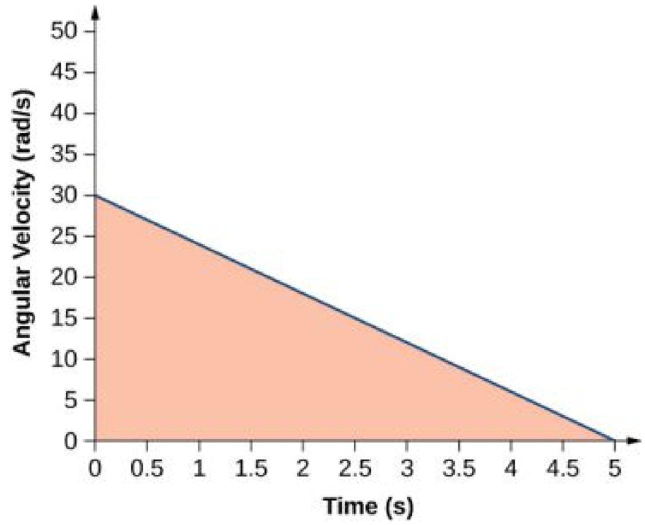


FIGURE 10.13 The area under the curve is the area of the right triangle.

$$\begin{aligned}\Delta\theta &= \text{area (triangle)};\\ \Delta\theta &= \frac{1}{2}(30\text{ rad/s})(5\text{ s}) = 75\text{ rad}.\end{aligned}$$

We verify the solution using Equation 10.12:

$$\theta_\text{f} = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2.$$

Setting $\theta_0 = 0$, we have

$$\theta_f = (30.0\text{ rad/s})(5.0\text{ s}) + \frac{1}{2}(-6.0\text{ rad/s}^2)(5.0\text{ rad/s})^2 = 150.0 - 75.0 = 75.0\text{ rad}.$$

This verifies the solution found from finding the area under the curve.

Significance

We see from part (b) that there are alternative approaches to analyzing fixed-axis rotation with constant acceleration. We started with a graphical approach and verified the solution using the rotational kinematic equations. Since $\alpha = \frac{d\omega}{dt}$, we could do the same graphical analysis on an angular acceleration-vs.-time curve. The area under an α-vs.-t curve gives us the change in angular velocity. Since the angular acceleration is constant in this section, this is a straightforward exercise.

10.3 Relating Angular and Translational Quantities

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Given the linear kinematic equation, write the corresponding rotational kinematic equation
- Calculate the linear distances, velocities, and accelerations of points on a rotating system given the angular velocities and accelerations

In this section, we relate each of the rotational variables to the translational variables defined in Motion Along a Straight Line and Motion in Two and Three Dimensions. This will complete our ability to describe rigid-body rotations.

Angular vs. Linear Variables

In Rotational Variables, we introduced angular variables. If we compare the rotational definitions with the definitions of linear kinematic variables from Motion Along a Straight Line and Motion in Two and Three Dimensions, we find that there is a mapping of the linear variables to the rotational ones. Linear position, velocity, and acceleration have their rotational counterparts, as we can see when we write them side by side:

	Linear	Rotational
Position	x	θ
Velocity	$v = \frac{dx}{dt}$	$\omega = \frac{d\theta}{dt}$
Acceleration	$a = \frac{dv}{dt}$	$\alpha = \frac{d\omega}{dt}$

Let's compare the linear and rotational variables individually. The linear variable of position has physical units of meters, whereas the angular position variable has dimensionless units of radians, as can be seen from the definition of $\theta = \frac{s}{r}$, which is the ratio of two lengths. The linear velocity has units of m/s, and its counterpart, the angular velocity, has units of rad/s. In Rotational Variables, we saw in the case of circular motion that the linear tangential speed of a particle at a radius *r* from the axis of rotation is related to the angular velocity by the relation $v_t = r\omega$. This could also apply to points on a rigid body rotating about a fixed axis. Here, we consider only circular motion. In circular motion, both uniform and nonuniform, there exists a centripetal acceleration (Motion in Two and Three Dimensions). The centripetal acceleration vector points inward from the particle executing circular motion toward the axis of rotation. The derivation of the magnitude of the centripetal acceleration is given in Motion in Two and Three Dimensions. From that derivation, the magnitude of the centripetal acceleration was found to be

$$a_c = \frac{v_t^2}{r}, \tag{10.14}$$

where *r* is the radius of the circle.

Thus, in uniform circular motion when the angular velocity is constant and the angular acceleration is zero, we have a linear acceleration—that is, centripetal acceleration—since the tangential speed in Equation 10.14 is a constant. If nonuniform circular motion is present, the rotating system has an angular acceleration, and we have both a linear centripetal acceleration that is changing (because v_t is changing) as well as a linear tangential acceleration. These relationships are shown in Figure 10.14, where we show the centripetal and tangential accelerations for uniform and nonuniform circular motion.

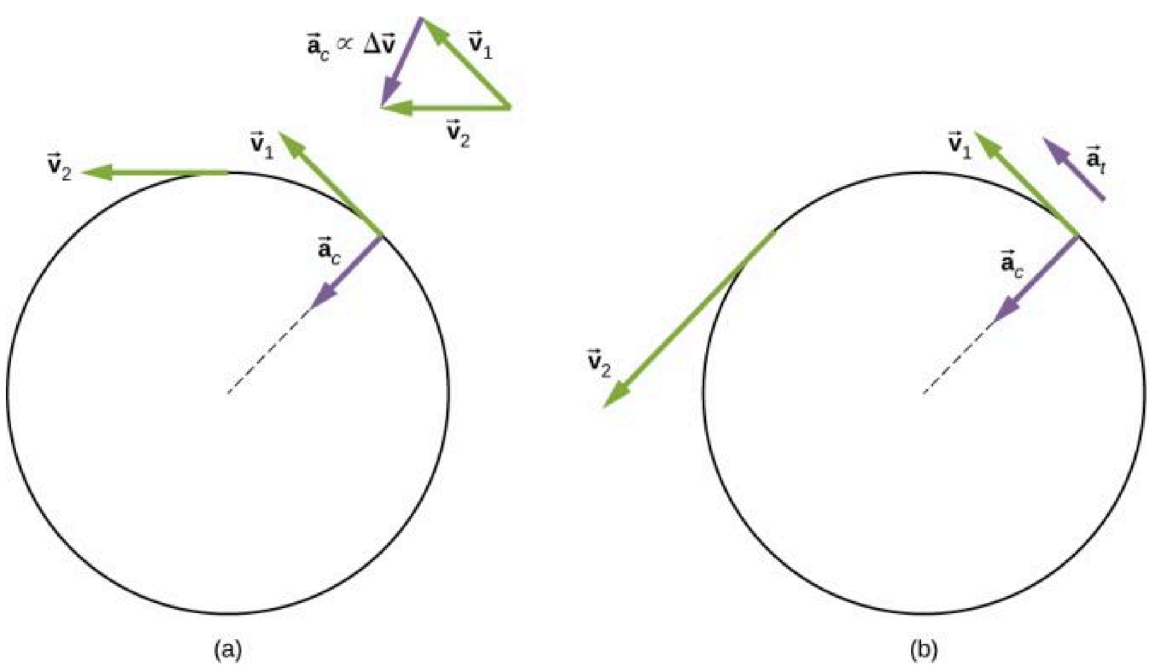


FIGURE 10.14 (a) Uniform circular motion: The centripetal acceleration a_c has its vector inward toward the axis of rotation. There is no tangential acceleration. (b) Nonuniform circular motion: An angular acceleration produces an inward centripetal acceleration that is changing in magnitude, plus a tangential acceleration a_t.

The centripetal acceleration is due to the change in the direction of tangential velocity, whereas the tangential acceleration is due to any change in the magnitude of the tangential velocity. The tangential and centripetal acceleration vectors $\vec{\mathbf{a}}_t$ and $\vec{\mathbf{a}}_c$ are always perpendicular to each other, as seen in Figure 10.14. To complete this description, we can assign a **total linear acceleration** vector to a point on a rotating rigid body or a particle executing circular motion at a radius *r* from a fixed axis. The total linear acceleration vector $\vec{\mathbf{a}}$ is the vector sum of the centripetal and tangential accelerations,

$$\vec{\mathbf{a}} = \vec{\mathbf{a}}_c + \vec{\mathbf{a}}_t. \tag{10.15}$$

The total linear acceleration vector in the case of nonuniform circular motion points at an angle between the centripetal and tangential acceleration vectors, as shown in Figure 10.15. Since $\vec{\mathbf{a}}_c \perp \vec{\mathbf{a}}_t$, the magnitude of the total linear acceleration is

$$|\vec{\mathbf{a}}| = \sqrt{a_c^2 + a_t^2}.$$

Note that if the angular acceleration is zero, the total linear acceleration is equal to the centripetal acceleration.

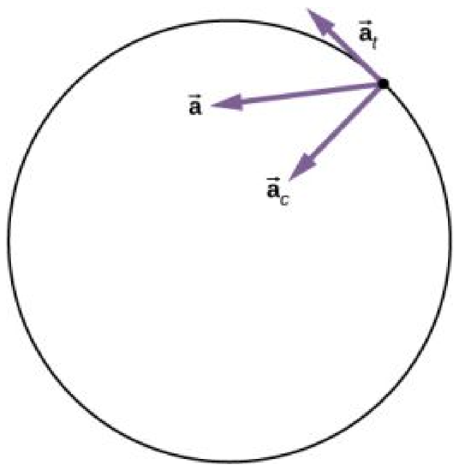


FIGURE 10.15 A particle is executing circular motion and has an angular acceleration. The total linear acceleration of the particle is the vector sum of the centripetal acceleration and tangential acceleration vectors. The total linear acceleration vector is at an angle in between the centripetal and tangential accelerations.

Relationships between Rotational and Translational Motion

We can look at two relationships between rotational and translational motion.

1. Generally speaking, the linear kinematic equations have their rotational counterparts. Table 10.2 lists the four linear kinematic equations and the corresponding rotational counterpart. The two sets of equations look similar to each other, but describe two different physical situations, that is, rotation and translation.

Rotational	Translational
$\theta_f = \theta_0 + \bar{\omega} t$	$x = x_0 + \bar{v} t$
$\omega_f = \omega_0 + \alpha t$	$v_f = v_0 + at$
$\theta_f = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$	$x_f = x_0 + v_0 t + \frac{1}{2}at^2$
$\omega_f^2 = \omega_0^2 + 2\alpha(\Delta\theta)$	$v_f^2 = v_0^2 + 2a(\Delta x)$

TABLE 10.2 **Rotational and Translational Kinematic Equations**

2. The second correspondence has to do with relating linear and rotational variables in the special case of circular motion. This is shown in Table 10.3, where in the third column, we have listed the connecting equation that relates the linear variable to the rotational variable. The rotational variables of angular velocity and

acceleration have subscripts that indicate their definition in circular motion.

Rotational	Translational	Relationship ($r = \text{radius}$)
θ	s	$\theta = \frac{s}{r}$
ω	v_t	$\omega = \frac{v_t}{r}$
α	a_t	$\alpha = \frac{a_t}{r}$
	a_c	$a_c = \frac{v_t^2}{r}$

TABLE 10.3 Rotational and Translational Quantities: Circular Motion

EXAMPLE 10.7

Linear Acceleration of a Centrifuge

A centrifuge has a radius of 20 cm and accelerates from a maximum rotation rate of 10,000 rpm to rest in 30 seconds under a constant angular acceleration. It is rotating counterclockwise. What is the magnitude of the total acceleration of a point at the tip of the centrifuge at $t = 29.0\text{s}$? What is the direction of the total acceleration vector?

Strategy

With the information given, we can calculate the angular acceleration, which then will allow us to find the tangential acceleration. We can find the centripetal acceleration at $t = 0$ by calculating the tangential speed at this time. With the magnitudes of the accelerations, we can calculate the total linear acceleration. From the description of the rotation in the problem, we can sketch the direction of the total acceleration vector.

Solution

The angular acceleration is

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{0 - (1.0 \times 10^4)2\pi/60.0\,\text{s(rad/s)}}{30.0\,\text{s}} = -34.9\,\text{rad/s}^2.$$

Therefore, the tangential acceleration is

$$a_t = r\alpha = 0.2\,\text{m}(-34.9\,\text{rad/s}^2) = -7.0\,\text{m/s}^2.$$

The angular velocity at $t = 29.0\,\text{s}$ is

$$\begin{aligned}\omega &= \omega_0 + \alpha t = 1.0 \times 10^4 \left(\frac{2\pi}{60.0\,\text{s}}\right) + \left(-34.9\,\text{rad/s}^2\right)(29.0\,\text{s}) \\ &= 1047.2\,\text{rad/s} - 1012.71 = 35.1\,\text{rad/s}.\end{aligned}$$

Thus, the tangential speed at $t = 29.0\,\text{s}$ is

$$v_t = r\omega = 0.2\,\text{m}(35.1\,\text{rad/s}) = 7.0\,\text{m/s}.$$

We can now calculate the centripetal acceleration at $t = 29.0\,\text{s}$:

$$a_c = \frac{v^2}{r} = \frac{(7.0\,\text{m/s})^2}{0.2\,\text{m}} = 245.0\,\text{m/s}^2.$$

Since the two acceleration vectors are perpendicular to each other, the magnitude of the total linear acceleration is

$$|\vec{\mathbf{a}}| = \sqrt{a_c^2 + a_t^2} = \sqrt{(245.0)^2 + (-7.0)^2} = 245.1\,\text{m/s}^2.$$

Since the centrifuge has a negative angular acceleration, it is slowing down. The total acceleration vector is as shown in Figure 10.16. The angle with respect to the centripetal acceleration vector is

$$\theta = \tan^{-1}\frac{-7.0}{245.0} = -1.6°.$$

The negative sign means that the total acceleration vector is angled toward the clockwise direction.

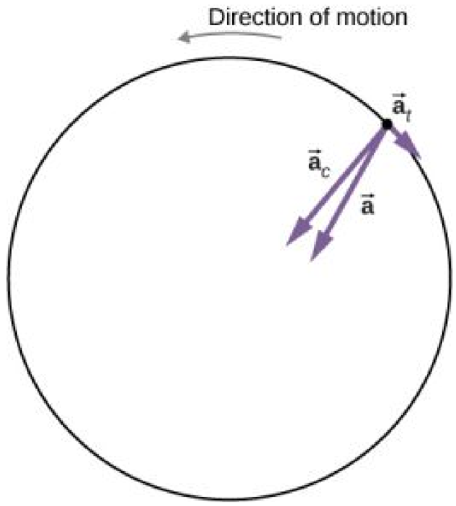


FIGURE 10.16 The centripetal, tangential, and total acceleration vectors. The centrifuge is slowing down, so the tangential acceleration is clockwise, opposite the direction of rotation (counterclockwise).

Significance

From Figure 10.16, we see that the tangential acceleration vector is opposite the direction of rotation. The magnitude of the tangential acceleration is much smaller than the centripetal acceleration, so the total linear acceleration vector will make a very small angle with respect to the centripetal acceleration vector.

CHECK YOUR UNDERSTANDING 10.3

A boy jumps on a merry-go-round with a radius of 5 m that is at rest. It starts accelerating at a constant rate up to an angular velocity of 5 rad/s in 20 seconds. What is the distance travelled by the boy?

INTERACTIVE

Check out this PhET simulation (https://openstax.org/l/28ladybugrevolutionrotation) to change the parameters of a rotating disk (the initial angle, angular velocity, and angular acceleration), and place bugs at different radial distances from the axis. The simulation then lets you explore how circular motion relates to the bugs' *xy*-position, velocity, and acceleration using vectors or graphs.

10.4 Moment of Inertia and Rotational Kinetic Energy

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the differences between rotational and translational kinetic energy
- Define the physical concept of moment of inertia in terms of the mass distribution from the rotational axis
- Explain how the moment of inertia of rigid bodies affects their rotational kinetic energy
- Use conservation of mechanical energy to analyze systems undergoing both rotation and translation
- Calculate the angular velocity of a rotating system when there are energy losses due to nonconservative forces

So far in this chapter, we have been working with rotational kinematics: the description of motion for a rotating rigid body with a fixed axis of rotation. In this section, we define two new quantities that are helpful for analyzing properties of rotating objects: moment of inertia and rotational kinetic energy. With these properties defined, we will have two important tools we need for analyzing rotational dynamics.

Rotational Kinetic Energy

Any moving object has kinetic energy. We know how to calculate this for a body undergoing translational motion, but how about for a rigid body undergoing rotation? This might seem complicated because each point on the rigid body has a different velocity. However, we can make use of angular velocity—which is the same for the entire rigid body—to express the kinetic energy for a rotating object. Figure 10.17 shows an example of a very energetic rotating body: an electric grindstone propelled by a motor. Sparks are flying, and noise and vibration are generated as the grindstone does its work. This system has considerable energy, some of it in the form of heat, light, sound, and vibration. However, most of this energy is in the form of **rotational kinetic energy**.

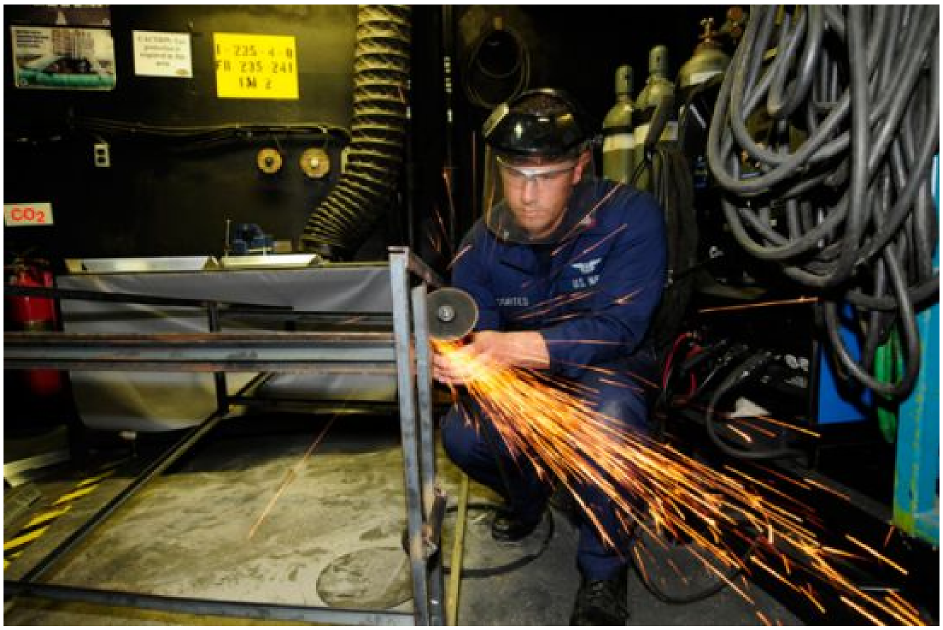


FIGURE 10.17 The rotational kinetic energy of the grindstone is converted to heat, light, sound, and vibration. (credit: Zachary David Bell, US Navy)

Energy in rotational motion is not a new form of energy; rather, it is the energy associated with rotational motion, the same as kinetic energy in translational motion. However, because kinetic energy is given by $K = \frac{1}{2}mv^2$, and velocity is a quantity that is different for every point on a rotating body about an axis, it makes sense to find a way to write kinetic energy in terms of the variable ω, which is the same for all points on a rigid rotating body. For a single particle rotating around a fixed axis, this is straightforward to calculate. We can relate the angular velocity to the magnitude of the translational velocity using the relation $v_t = \omega r$, where r is the distance of the particle from the axis of rotation and v_t is its tangential speed. Substituting into the equation for kinetic energy, we find

$$K = \frac{1}{2}mv_t^2 = \frac{1}{2}m(\omega r)^2 = \frac{1}{2}(mr^2)\omega^2.$$

In the case of a rigid rotating body, we can divide up any body into a large number of smaller masses, each with a mass m_j and distance to the axis of rotation r_j, such that the total mass of the body is equal to the sum of the individual masses: $M = \sum_j m_j$. Each smaller mass has tangential speed v_j, where we have dropped the subscript t for the moment. The total kinetic energy of the rigid rotating body is

$$K = \sum_j \frac{1}{2}m_j v_j^2 = \sum_j \frac{1}{2}m_j(r_j\omega_j)^2$$

and since $\omega_j = \omega$ for all masses,

$$K = \frac{1}{2}\left(\sum_j m_j r_j^2\right)\omega^2. \qquad (10.16)$$

The units of Equation 10.16 are joules (J). The equation in this form is complete, but awkward; we need to find a way to generalize it.

Moment of Inertia

If we compare Equation 10.16 to the way we wrote kinetic energy in Work and Kinetic Energy, $\left(\frac{1}{2}mv^2\right)$, this suggests we have a new rotational variable to add to our list of our relations between rotational and translational variables. The quantity $\sum_j m_j r_j^2$ is the counterpart for mass in the equation for rotational kinetic energy. This is an important new term for rotational motion. This quantity is called the **moment of inertia** I, with units of $\text{kg}\cdot\text{m}^2$:

$$I = \sum_j m_j r_j^2. \qquad (10.17)$$

For now, we leave the expression in summation form, representing the moment of inertia of a system of point particles rotating about a fixed axis. We note that the moment of inertia of a single point particle about a fixed axis is simply mr^2, with *r* being the distance from the point particle to the axis of rotation. In the next section, we explore the integral form of this equation, which can be used to calculate the moment of inertia of some regular-shaped rigid bodies.

The moment of inertia is the quantitative measure of rotational inertia, just as in translational motion, and mass is the quantitative measure of linear inertia—that is, the more massive an object is, the more inertia it has, and the greater is its resistance to change in linear velocity. Similarly, the greater the moment of inertia of a rigid body or system of particles, the greater is its resistance to change in angular velocity about a fixed axis of rotation. It is interesting to see how the moment of inertia varies with *r,* the distance to the axis of rotation of the mass particles in Equation 10.17. Rigid bodies and systems of particles with more mass concentrated at a greater distance from the axis of rotation have greater moments of inertia than bodies and systems of the same mass, but concentrated near the axis of rotation. In this way, we can see that a hollow cylinder has more rotational inertia than a solid cylinder of the same mass when rotating about an axis through the center. Substituting Equation 10.17 into Equation 10.16, the expression for the kinetic energy of a rotating rigid body becomes

$$K = \frac{1}{2}I\omega^2. \qquad (10.18)$$

We see from this equation that the kinetic energy of a rotating rigid body is directly proportional to the moment of inertia and the square of the angular velocity. This is exploited in flywheel energy-storage devices, which are designed to store large amounts of rotational kinetic energy. Many carmakers are now testing flywheel energy storage devices in their automobiles, such as the flywheel, or kinetic energy recovery system, shown in Figure 10.18.

FIGURE 10.18 A KERS (kinetic energy recovery system) flywheel used in cars. (credit: “cmonville”/Flickr)

The rotational and translational quantities for kinetic energy and inertia are summarized in Table 10.4. The relationship column is not included because a constant doesn’t exist by which we could multiply the rotational quantity to get the translational quantity, as can be done for the variables in Table 10.3.

Rotational	Translational
$I = \sum_j m_j r_j^2$	m
$K = \frac{1}{2} I\omega^2$	$K = \frac{1}{2} mv^2$

TABLE 10.4 Rotational and Translational Kinetic Energies and Inertia

EXAMPLE 10.8

Moment of Inertia of a System of Particles

Six small washers are spaced 10 cm apart on a rod of negligible mass and 0.5 m in length. The mass of each washer is 20 g. The rod rotates about an axis located at 25 cm, as shown in Figure 10.19. (a) What is the moment of inertia of the system? (b) If the two washers closest to the axis are removed, what is the moment of inertia of the remaining four washers? (c) If the system with six washers rotates at 5 rev/s, what is its rotational kinetic energy?

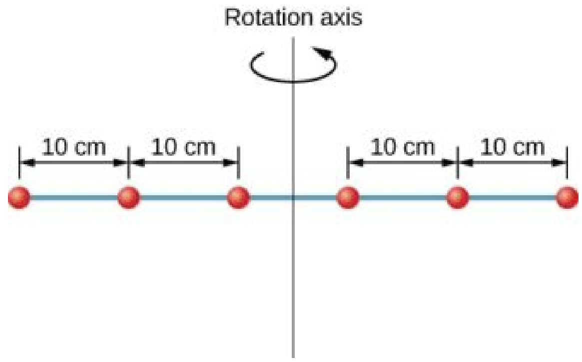


FIGURE 10.19 Six washers are spaced 10 cm apart on a rod of negligible mass and rotating about a vertical axis.

Strategy

a. We use the definition for moment of inertia for a system of particles and perform the summation to evaluate this quantity. The masses are all the same so we can pull that quantity in front of the summation symbol.
b. We do a similar calculation.
c. We insert the result from (a) into the expression for rotational kinetic energy.

Solution

a. $I = \sum_j m_j r_j^2 = (0.02\,\text{kg})(2 \times (0.25\,\text{m})^2 + 2 \times (0.15\,\text{m})^2 + 2 \times (0.05\,\text{m})^2) = 0.0035\,\text{kg} \cdot \text{m}^2.$

b. $I = \sum_j m_j r_j^2 = (0.02\,\text{kg})(2 \times (0.25\,\text{m})^2 + 2 \times (0.15\,\text{m})^2) = 0.0034\,\text{kg} \cdot \text{m}^2.$

c. $K = \frac{1}{2} I\omega^2 = \frac{1}{2}(0.0035\,\text{kg} \cdot \text{m}^2)(5.0 \times 2\pi\,\text{rad/s})^2 = 1.73\,\text{J}.$

Significance

We can see the individual contributions to the moment of inertia. The masses close to the axis of rotation have a very small contribution. When we removed them, it had a very small effect on the moment of inertia.

In the next section, we generalize the summation equation for point particles and develop a method to calculate moments of inertia for rigid bodies. For now, though, Figure 10.20 gives values of moment of inertia for common object shapes around specified axes.

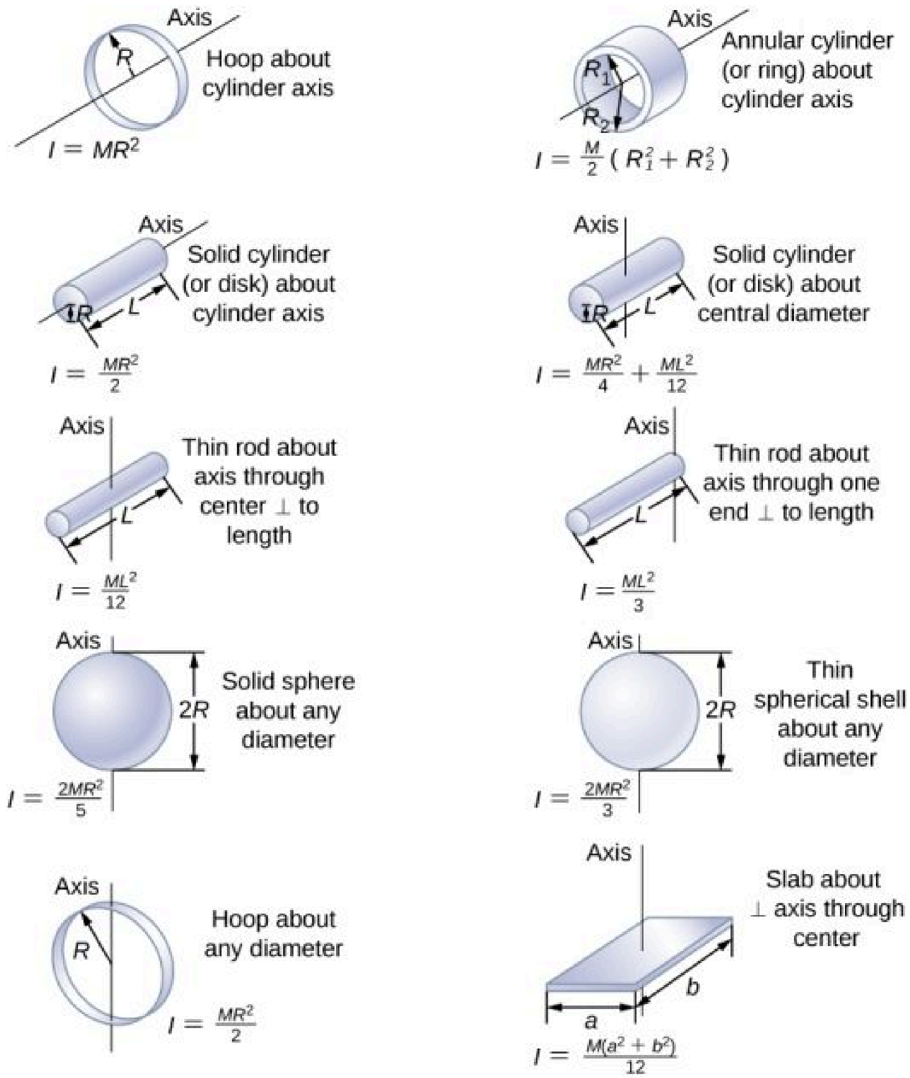


FIGURE 10.20 Moment of inertia for common shapes of objects.

Applying Rotational Kinetic Energy

Now let's apply the ideas of rotational kinetic energy and the moment of inertia table to get a feeling for the energy associated with a few rotating objects. The following examples will also help get you comfortable using these equations. First, let's look at a general problem-solving strategy for rotational energy.

PROBLEM-SOLVING STRATEGY

Rotational Energy

1. Determine that energy or work is involved in the rotation.
2. Determine the system of interest. A sketch usually helps.
3. Analyze the situation to determine the types of work and energy involved.
4. If there are no losses of energy due to friction and other nonconservative forces, mechanical energy is conserved, that is, $K_{\mathrm{i}} + U_{\mathrm{i}} = K_{\mathrm{f}} + U_{\mathrm{f}}$.
5. If nonconservative forces are present, mechanical energy is not conserved, and other forms of energy, such as heat and light, may enter or leave the system. Determine what they are and calculate them as necessary.
6. Eliminate terms wherever possible to simplify the algebra.
7. Evaluate the numerical solution to see if it makes sense in the physical situation presented in the wording of the problem.

EXAMPLE 10.9

Calculating Helicopter Energies

A typical small rescue helicopter has four blades: Each is 4.00 m long and has a mass of 50.0 kg (Figure 10.21). The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length. The helicopter has a total loaded mass of 1000 kg. (a) Calculate the rotational kinetic energy in the blades when they rotate at 300 rpm. (b) Calculate the translational kinetic energy of the helicopter when it flies at 20.0 m/s, and compare it with the rotational energy in the blades.

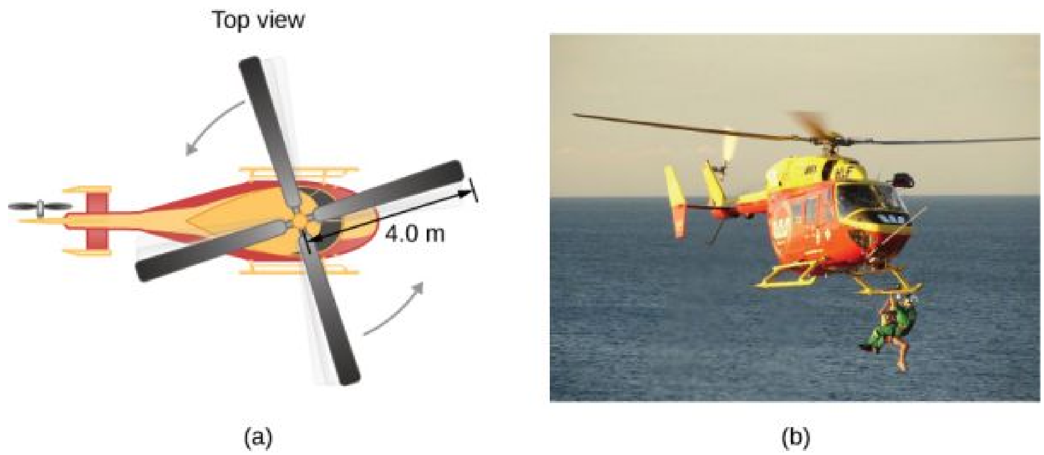


FIGURE 10.21 (a) Sketch of a four-blade helicopter. (b) A water rescue operation featuring a helicopter from the Auckland Westpac Rescue Helicopter Service. (credit b: modification of work by “111 Emergency”/Flickr)

Strategy

Rotational and translational kinetic energies can be calculated from their definitions. The wording of the problem gives all the necessary constants to evaluate the expressions for the rotational and translational kinetic energies.

Solution

a. The rotational kinetic energy is

$$K = \frac{1}{2}I\omega^2.$$

We must convert the angular velocity to radians per second and calculate the moment of inertia before we can find K. The angular velocity ω is

$$\omega = \frac{300\text{ rev}}{1.00\text{ min}}\frac{2\pi\text{ rad}}{1\text{ rev}}\frac{1.00\text{ min}}{60.0\text{ s}} = 31.4\frac{\text{rad}}{\text{s}}.$$

The moment of inertia of one blade is that of a thin rod rotated about its end, listed in Figure 10.20. The total I is four times this moment of inertia because there are four blades. Thus,

$$I = 4\frac{ML^2}{3} = 4 \times \frac{(50.0\text{ kg})(4.00\text{ m})^2}{3} = 1067.0\text{ kg}\cdot\text{m}^2.$$

Entering ω and I into the expression for rotational kinetic energy gives

$$K = 0.5(1067\text{ kg}\cdot\text{m}^2)(31.4\text{ rad/s})^2 = 5.26 \times 10^5\text{ J}.$$

b. Entering the given values into the equation for translational kinetic energy, we obtain

$$K = \frac{1}{2}mv^2 = (0.5)(1000.0\text{ kg})(20.0\text{ m/s})^2 = 2.00 \times 10^5\text{ J}.$$

To compare kinetic energies, we take the ratio of translational kinetic energy to rotational kinetic energy. This ratio is

$$\frac{2.00 \times 10^5\text{ J}}{5.26 \times 10^5\text{ J}} = 0.380.$$

Significance

The ratio of translational energy to rotational kinetic energy is only 0.380. This ratio tells us that most of the kinetic energy of the helicopter is in its spinning blades.

EXAMPLE 10.10

Energy in a Boomerang

A person hurls a boomerang into the air with a velocity of 30.0 m/s at an angle of $40.0°$ with respect to the horizontal (Figure 10.22). It has a mass of 1.0 kg and is rotating at 10.0 rev/s. The moment of inertia of the boomerang is given as $I = \frac{1}{12}mL^2$ where $L = 0.7\text{ m}$. (a) What is the total energy of the boomerang when it leaves the hand? (b) How high does the boomerang go from the elevation of the hand, neglecting air resistance?

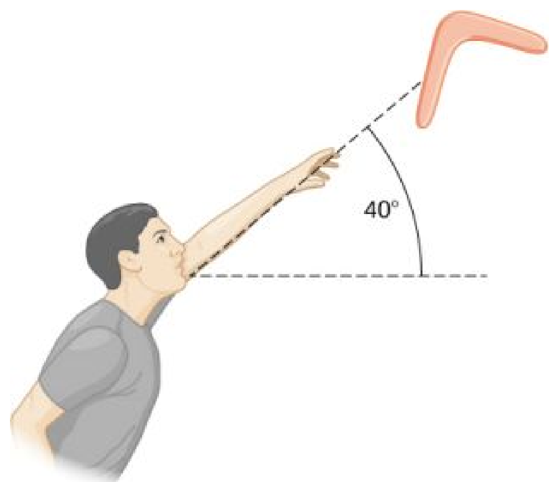


FIGURE 10.22 A boomerang is hurled into the air at an initial angle of $40°$.

Strategy

We use the definitions of rotational and linear kinetic energy to find the total energy of the system. The problem states to neglect air resistance, so we don't have to worry about energy loss. In part (b), we use conservation of mechanical energy to find the maximum height of the boomerang.

Solution

a. Moment of inertia: $I = \frac{1}{12}mL^2 = \frac{1}{12}(1.0\text{ kg})(0.7\text{m})^2 = 0.041\text{ kg} \cdot \text{m}^2$.
Angular velocity: $\omega = (10.0\text{ rev/s})(2\pi) = 62.83\text{ rad/s}$.
The rotational kinetic energy is therefore

$$K_\text{R} = \frac{1}{2}(0.041\text{ kg} \cdot \text{m}^2)(62.83\text{ rad/s})^2 = 80.93\text{ J}.$$

The translational kinetic energy is

$$K_\text{T} = \frac{1}{2}mv^2 = \frac{1}{2}(1.0\text{ kg})(30.0\text{ m/s})^2 = 450.0\text{ J}.$$

Thus, the total energy in the boomerang is

$$K_\text{Total} = K_\text{R} + K_\text{T} = 80.93 + 450.0 = 530.93\text{ J}.$$

b. We use conservation of mechanical energy. Since the boomerang is launched at an angle, we need to write the total energies of the system in terms of its linear kinetic energies using the velocity in the *x*- and *y*-directions. The total energy when the boomerang leaves the hand is

$$E_\text{Before} = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}I\omega^2.$$

The total energy at maximum height is

$$E_{\text{Final}} = \frac{1}{2}mv_x^2 + \frac{1}{2}I\omega^2 + mgh.$$

By conservation of mechanical energy, $E_{\text{Before}} = E_{\text{Final}}$ so we have, after canceling like terms,

$$\frac{1}{2}mv_y^2 = mgh.$$

Since $v_y = 30.0\,\text{m/s}(\sin 40°) = 19.28\,\text{m/s}$, we find

$$h = \frac{(19.28\,\text{m/s})^2}{2(9.8\,\text{m/s}^2)} = 18.97\,\text{m}.$$

Significance

In part (b), the solution demonstrates how energy conservation is an alternative method to solve a problem that normally would be solved using kinematics. In the absence of air resistance, the rotational kinetic energy was not a factor in the solution for the maximum height.

CHECK YOUR UNDERSTANDING 10.4

A nuclear submarine propeller has a moment of inertia of $800.0\,\text{kg}\cdot\text{m}^2$. If the submerged propeller has a rotation rate of 4.0 rev/s when the engine is cut, what is the rotation rate of the propeller after 5.0 s when water resistance has taken 50,000 J out of the system?

10.5 Calculating Moments of Inertia

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Calculate the moment of inertia for uniformly shaped, rigid bodies
- Apply the parallel axis theorem to find the moment of inertia about any axis parallel to one already known
- Calculate the moment of inertia for compound objects

In the preceding section, we defined the moment of inertia but did not show how to calculate it. In this section, we show how to calculate the moment of inertia for several standard types of objects, as well as how to use known moments of inertia to find the moment of inertia for a shifted axis or for a compound object. This section is very useful for seeing how to apply a general equation to complex objects (a skill that is critical for more advanced physics and engineering courses).

Moment of Inertia

We defined the moment of inertia *I* of an object to be $I = \sum_i m_i r_i^2$ for all the point masses that make up the object.

Because *r* is the distance to the axis of rotation from each piece of mass that makes up the object, the moment of inertia for any object depends on the chosen axis. To see this, let's take a simple example of two masses at the end of a massless (negligibly small mass) rod (Figure 10.23) and calculate the moment of inertia about two different axes. In this case, the summation over the masses is simple because the two masses at the end of the barbell can be approximated as point masses, and the sum therefore has only two terms.

In the case with the axis in the center of the barbell, each of the two masses *m* is a distance *R* away from the axis, giving a moment of inertia of

$$I_1 = mR^2 + mR^2 = 2mR^2.$$

In the case with the axis at the end of the barbell—passing through one of the masses—the moment of inertia is

$$I_2 = m(0)^2 + m(2R)^2 = 4mR^2.$$

From this result, we can conclude that it is twice as hard to rotate the barbell about the end than about its center.

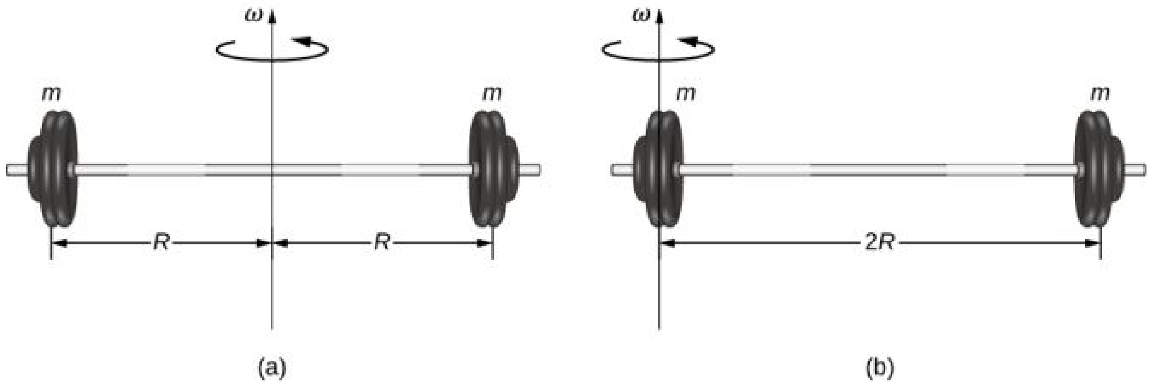


FIGURE 10.23 (a) A barbell with an axis of rotation through its center; (b) a barbell with an axis of rotation through one end.

In this example, we had two point masses and the sum was simple to calculate. However, to deal with objects that are not point-like, we need to think carefully about each of the terms in the equation. The equation asks us to sum over each ‘piece of mass’ a certain distance from the axis of rotation. But what exactly does each ‘piece of mass’ mean? Recall that in our derivation of this equation, each piece of mass had the same magnitude of velocity, which means the whole piece had to have a single distance *r* to the axis of rotation. However, this is not possible unless we take an infinitesimally small piece of mass *dm,* as shown in Figure 10.24.

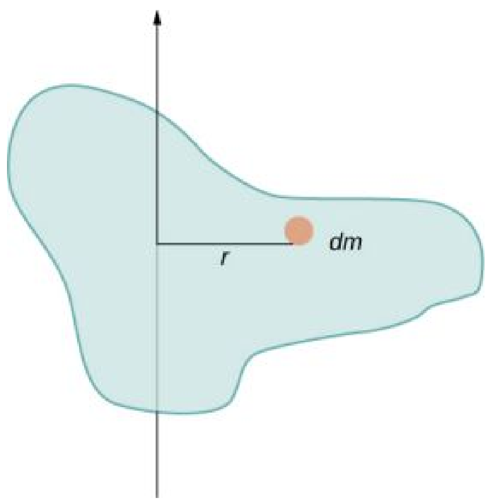


FIGURE 10.24 Using an infinitesimally small piece of mass to calculate the contribution to the total moment of inertia.

The need to use an infinitesimally small piece of mass *dm* suggests that we can write the moment of inertia by evaluating an integral over infinitesimal masses rather than doing a discrete sum over finite masses:

$$I = \sum_i m_i {r_i}^2 \text{ becomes } I = \int r^2 \, dm. \tag{10.19}$$

This, in fact, is the form we need to generalize the equation for complex shapes. It is best to work out specific examples in detail to get a feel for how to calculate the moment of inertia for specific shapes. This is the focus of most of the rest of this section.

A uniform thin rod with an axis through the center

Consider a uniform (density and shape) thin rod of mass *M* and length *L* as shown in Figure 10.25. We want a thin rod so that we can assume the cross-sectional area of the rod is small and the rod can be thought of as a string of masses along a one-dimensional straight line. In this example, the axis of rotation is perpendicular to the rod and passes through the midpoint for simplicity. Our task is to calculate the moment of inertia about this axis. We orient the axes so that the *z*-axis is the axis of rotation and the *x*-axis passes through the length of the rod, as shown in the figure. This is a convenient choice because we can then integrate along the *x*-axis.

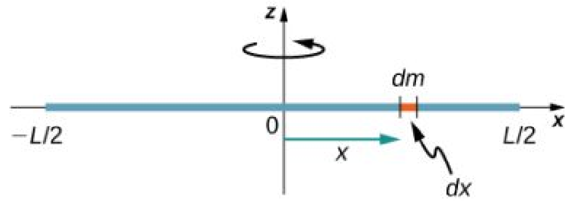


FIGURE 10.25 Calculation of the moment of inertia I for a uniform thin rod about an axis through the center of the rod.

We define *dm* to be a small element of mass making up the rod. The moment of inertia integral is an integral over the mass distribution. However, we know how to integrate over space, not over mass. We therefore need to find a way to relate mass to spatial variables. We do this using the **linear mass density** λ of the object, which is the mass per unit length. Since the mass density of this object is uniform, we can write

$$\lambda = \frac{m}{L} \text{ or } m = \lambda L.$$

If we take the differential of each side of this equation, we find

$$dm = d(\lambda L) = \lambda(dL)$$

since λ is constant. We chose to orient the rod along the *x*-axis for convenience—this is where that choice becomes very helpful. Note that a piece of the rod *dl* lies completely along the *x*-axis and has a length *dx*; in fact, $dL = dx$ in this situation. We can therefore write $dm = \lambda(dx)$, giving us an integration variable that we know how to deal with. The distance of each piece of mass *dm* from the axis is given by the variable *x*, as shown in the figure. Putting this all together, we obtain

$$I = \int r^2 dm = \int x^2 dm = \int x^2 \lambda dx.$$

The last step is to be careful about our limits of integration. The rod extends from $x = -L/2$ to $x = L/2$, since the axis is in the middle of the rod at $x = 0$. This gives us

$$\begin{aligned} I &= \int_{-L/2}^{L/2} x^2 \lambda dx = \lambda \frac{x^3}{3}\Bigg|_{-L/2}^{L/2} = \lambda\left(\tfrac{1}{3}\right)\left[\left(\tfrac{L}{2}\right)^3 - \left(\tfrac{-L}{2}\right)^3\right] \\ &= \lambda\left(\tfrac{1}{3}\right)\tfrac{L^3}{8}(2) = \tfrac{M}{L}\left(\tfrac{1}{3}\right)\tfrac{L^3}{8}(2) = \tfrac{1}{12}ML^2. \end{aligned}$$

Next, we calculate the moment of inertia for the same uniform thin rod but with a different axis choice so we can compare the results. We would expect the moment of inertia to be smaller about an axis through the center of mass than the endpoint axis, just as it was for the barbell example at the start of this section. This happens because more mass is distributed farther from the axis of rotation.

A uniform thin rod with axis at the end

Now consider the same uniform thin rod of mass *M* and length *L*, but this time we move the axis of rotation to the end of the rod. We wish to find the moment of inertia about this new axis (Figure 10.26). The quantity *dm* is again defined to be a small element of mass making up the rod. Just as before, we obtain

$$I = \int r^2 dm = \int x^2 dm = \int x^2 \lambda dx.$$

However, this time we have different limits of integration. The rod extends from $x = 0$ to $x = L$, since the axis is at the end of the rod at $x = 0$. Therefore we find

$$\begin{aligned} I &= \int_{0}^{L} x^2 \lambda dx = \lambda \frac{x^3}{3}\Bigg|_{0}^{L} = \lambda\left(\tfrac{1}{3}\right)[(L)^3 - (0)^3] \\ &= \lambda\left(\tfrac{1}{3}\right)L^3 = \tfrac{M}{L}\left(\tfrac{1}{3}\right)L^3 = \tfrac{1}{3}ML^2. \end{aligned}$$

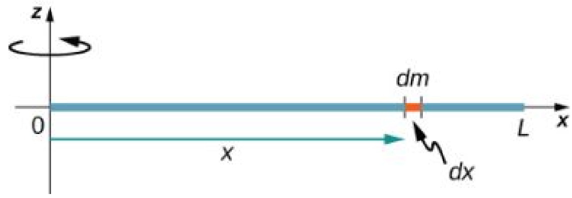


FIGURE 10.26 Calculation of the moment of inertia I for a uniform thin rod about an axis through the end of the rod.

Note the rotational inertia of the rod about its endpoint is larger than the rotational inertia about its center (consistent with the barbell example) by a factor of four.

The Parallel-Axis Theorem

The similarity between the process of finding the moment of inertia of a rod about an axis through its middle and about an axis through its end is striking, and suggests that there might be a simpler method for determining the moment of inertia for a rod about any axis parallel to the axis through the center of mass. Such an axis is called a **parallel axis**. There is a theorem for this, called the **parallel-axis theorem**, which we state here but do not derive in this text.

PARALLEL-AXIS THEOREM

Let *m* be the mass of an object and let *d* be the distance from an axis through the object's center of mass to a new axis. Then we have

$$I_{\text{parallel-axis}} = I_{\text{center of mass}} + md^2. \quad (10.20)$$

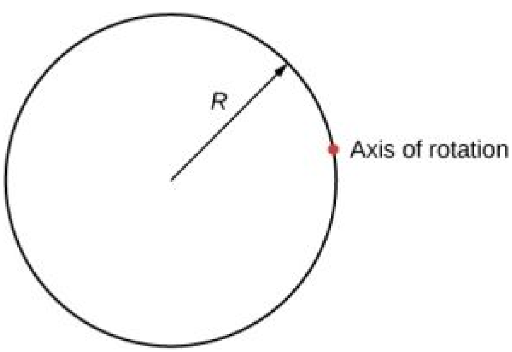
Let's apply this to the rod examples solved above:

$$I_{\text{end}} = I_{\text{center of mass}} + md^2 = \frac{1}{12}mL^2 + m\left(\frac{L}{2}\right)^2 = \left(\frac{1}{12} + \frac{1}{4}\right)mL^2 = \frac{1}{3}mL^2.$$

This result agrees with our more lengthy calculation from above. This is a useful equation that we apply in some of the examples and problems.

CHECK YOUR UNDERSTANDING 10.5

What is the moment of inertia of a cylinder of radius *R* and mass *m* about an axis through a point on the surface, as shown below?



A uniform thin disk about an axis through the center

Integrating to find the moment of inertia of a two-dimensional object is a little bit trickier, but one shape is commonly done at this level of study—a uniform thin disk about an axis through its center (Figure 10.27).

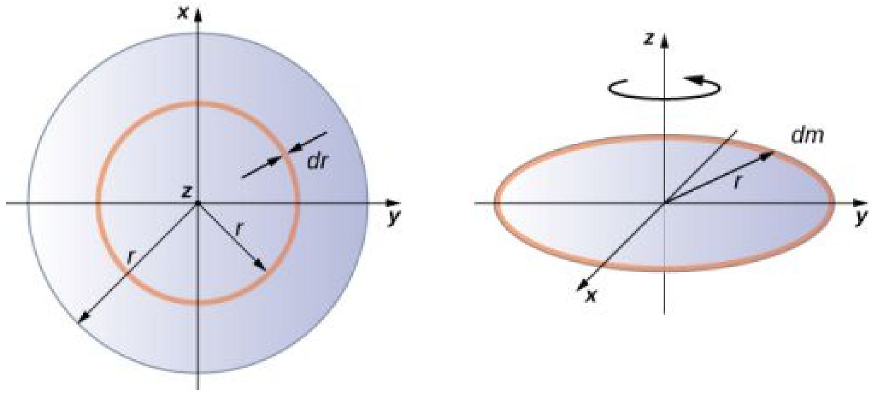


FIGURE 10.27 Calculating the moment of inertia for a thin disk about an axis through its center.

Since the disk is thin, we can take the mass as distributed entirely in the *xy*-plane. We again start with the relationship for the **surface mass density**, which is the mass per unit surface area. Since it is uniform, the surface mass density σ is constant:

$$\sigma = \frac{m}{A} \quad \text{or} \quad \sigma A = m, \text{ so } dm = \sigma(dA).$$

Now we use a simplification for the area. The area can be thought of as made up of a series of thin rings, where each ring is a mass increment *dm* of radius *r* equidistant from the axis, as shown in part (b) of the figure. The infinitesimal area of each ring *dA* is therefore given by the length of each ring ($2\pi r$) times the infinitesimal width of each ring *dr*:

$$A = \pi r^2, dA = d(\pi r^2) = \pi dr^2 = 2\pi r dr.$$

The full area of the disk is then made up from adding all the thin rings with a radius range from 0 to *R*. This radius range then becomes our limits of integration for *dr*, that is, we integrate from $r = 0$ to $r = R$. Putting this all together, we have

$$\begin{aligned} I &= \int_0^R r^2 \sigma(2\pi r)dr = 2\pi\sigma \int_0^R r^3 dr = 2\pi\sigma \frac{r^4}{4}\bigg|_0^R = 2\pi\sigma\left(\frac{R^4}{4} - 0\right) \\ &= 2\pi\frac{m}{A}\left(\frac{R^4}{4}\right) = 2\pi\frac{m}{\pi R^2}\left(\frac{R^4}{4}\right) = \frac{1}{2}mR^2. \end{aligned}$$

Note that this agrees with the value given in Figure 10.20.

Calculating the moment of inertia for compound objects

Now consider a compound object such as that in Figure 10.28, which depicts a thin disk at the end of a thin rod. This cannot be easily integrated to find the moment of inertia because it is not a uniformly shaped object. However, if we go back to the initial definition of moment of inertia as a summation, we can reason that a compound object's moment of inertia can be found from the sum of each part of the object:

$$I_{\text{total}} = \sum_i I_i. \tag{10.21}$$

It is important to note that the moments of inertia of the objects in Equation 10.21 are *about a common axis*. In the case of this object, that would be a rod of length *L* rotating about its end, and a thin disk of radius *R* rotating about an axis shifted off of the center by a distance $L + R$, where *R* is the radius of the disk. Let's define the mass of the rod to be m_{r} and the mass of the disk to be m_{d}.

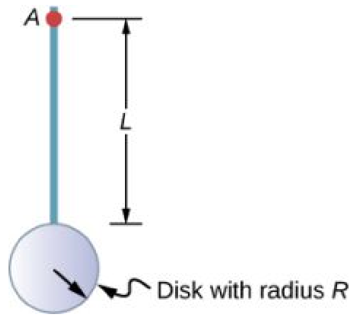


FIGURE 10.28 Compound object consisting of a disk at the end of a rod. The axis of rotation is located at *A*.

The moment of inertia of the rod is simply $\frac{1}{3}m_r L^2$, but we have to use the parallel-axis theorem to find the moment of inertia of the disk about the axis shown. The moment of inertia of the disk about its center is $\frac{1}{2}m_d R^2$ and we apply the parallel-axis theorem $I_{\text{parallel-axis}} = I_{\text{center of mass}} + md^2$ to find

$$I_{\text{parallel-axis}} = \frac{1}{2}m_d R^2 + m_d(L+R)^2.$$

Adding the moment of inertia of the rod plus the moment of inertia of the disk with a shifted axis of rotation, we find the moment of inertia for the compound object to be

$$I_{\text{total}} = \frac{1}{3}m_r L^2 + \frac{1}{2}m_d R^2 + m_d(L+R)^2.$$

Applying moment of inertia calculations to solve problems

Now let's examine some practical applications of moment of inertia calculations.

EXAMPLE 10.11

Person on a Merry-Go-Round

A 25-kg child stands at a distance $r = 1.0\text{ m}$ from the axis of a rotating merry-go-round (Figure 10.29). The merry-go-round can be approximated as a uniform solid disk with a mass of 500 kg and a radius of 2.0 m. Find the moment of inertia of this system.

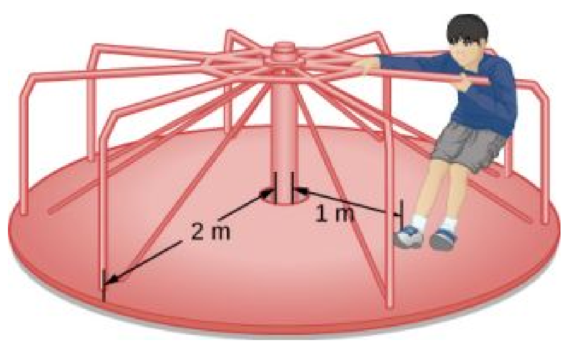


FIGURE 10.29 Calculating the moment of inertia for a child on a merry-go-round.

Strategy

This problem involves the calculation of a moment of inertia. We are given the mass and distance to the axis of rotation of the child as well as the mass and radius of the merry-go-round. Since the mass and size of the child are much smaller than the merry-go-round, we can approximate the child as a point mass. The notation we use is $m_c = 25\text{ kg}, r_c = 1.0\text{ m}, m_m = 500\text{ kg}, r_m = 2.0\text{ m}$.

Our goal is to find $I_{\text{total}} = \sum_i I_i$.

Solution

For the child, $I_c = m_c r^2$, and for the merry-go-round, $I_m = \frac{1}{2} m_m r^2$. Therefore

$$I_{total} = 25(1)^2 + \frac{1}{2}(500)(2)^2 = 25 + 1000 = 1025\ \text{kg} \cdot \text{m}^2.$$

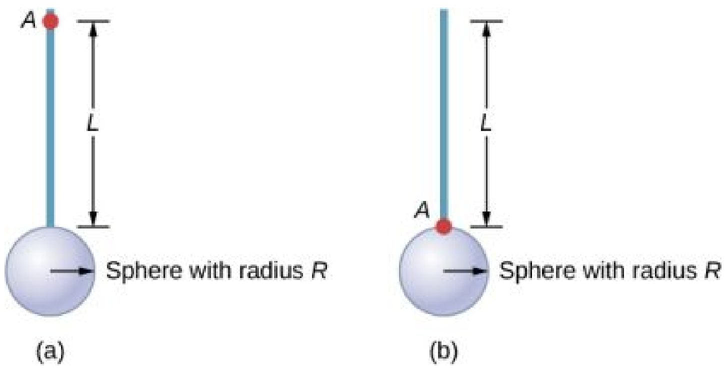
Significance

The value should be close to the moment of inertia of the merry-go-round by itself because it has much more mass distributed away from the axis than the child does.

EXAMPLE 10.12

Rod and Solid Sphere

Find the moment of inertia of the rod and solid sphere combination about the two axes as shown below. The rod has length 0.5 m and mass 2.0 kg. The radius of the sphere is 20.0 cm and has mass 1.0 kg.



Strategy

Since we have a compound object in both cases, we can use the parallel-axis theorem to find the moment of inertia about each axis. In (a), the center of mass of the sphere is located at a distance $L + R$ from the axis of rotation. In (b), the center of mass of the sphere is located a distance R from the axis of rotation. In both cases, the moment of inertia of the rod is about an axis at one end. Refer to Table 10.4 for the moments of inertia for the individual objects.

a. $I_{total} = \sum_i I_i = I_{Rod} + I_{Sphere};$

$$I_{Sphere} = I_{\text{center of mass}} + m_{Sphere}(L+R)^2 = \frac{2}{5} m_{Sphere} R^2 + m_{Sphere}(L+R)^2;$$
$$I_{total} = I_{Rod} + I_{Sphere} = \frac{1}{3} m_{Rod} L^2 + \frac{2}{5} m_{Sphere} R^2 + m_{Sphere}(L+R)^2;$$
$$I_{total} = \frac{1}{3}(2.0\ \text{kg})(0.5\ \text{m})^2 + \frac{2}{5}(1.0\ \text{kg})(0.2\ \text{m})^2 + (1.0\ \text{kg})(0.5\ \text{m} + 0.2\ \text{m})^2;$$
$$I_{total} = (0.167 + 0.016 + 0.490)\ \text{kg} \cdot \text{m}^2 = 0.673\ \text{kg} \cdot \text{m}^2.$$

b. $I_{Sphere} = \frac{2}{5} m_{Sphere} R^2 + m_{Sphere} R^2;$

$$I_{total} = I_{Rod} + I_{Sphere} = \frac{1}{3} m_{Rod} L^2 + \frac{2}{5} m_{Sphere} R^2 + m_{Sphere} R^2;$$
$$I_{total} = \frac{1}{3}(2.0\ \text{kg})(0.5\ \text{m})^2 + \frac{2}{5}(1.0\ \text{kg})(0.2\ \text{m})^2 + (1.0\ \text{kg})(0.2\ \text{m})^2;$$
$$I_{total} = (0.167 + 0.016 + 0.04)\ \text{kg} \cdot \text{m}^2 = 0.223\ \text{kg} \cdot \text{m}^2.$$

Significance

Using the parallel-axis theorem eases the computation of the moment of inertia of compound objects. We see that the moment of inertia is greater in (a) than (b). This is because the axis of rotation is closer to the center of mass of the system in (b). The simple analogy is that of a rod. The moment of inertia about one end is $\frac{1}{3} mL^2$, but the moment of inertia through the center of mass along its length is $\frac{1}{12} mL^2$.

EXAMPLE 10.13

Angular Velocity of a Pendulum

A pendulum in the shape of a rod (Figure 10.30) is released from rest at an angle of $30°$. It has a length 30 cm and mass 300 g. What is its angular velocity at its lowest point?

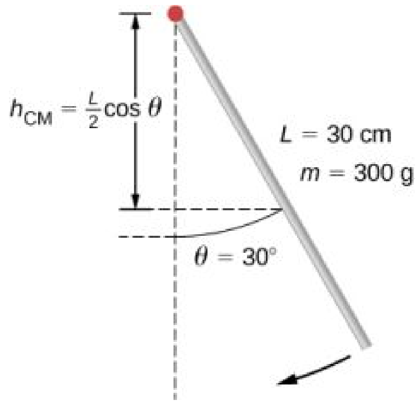


FIGURE 10.30 A pendulum in the form of a rod is released from rest at an angle of $30°$.

Strategy

Use conservation of energy to solve the problem. At the point of release, the pendulum has gravitational potential energy, which is determined from the height of the center of mass above its lowest point in the swing. At the bottom of the swing, all of the gravitational potential energy is converted into rotational kinetic energy.

Solution

The change in potential energy is equal to the change in rotational kinetic energy, $\Delta U + \Delta K = 0$.

At the top of the swing: $U = mgh_{\text{cm}} = mg\frac{L}{2}(\cos\theta)$. At the bottom of the swing, $U = mg\frac{L}{2}$ with respect to the lowest point the rod swings.

At the top of the swing, the rotational kinetic energy is $K = 0$. At the bottom of the swing, $K = \frac{1}{2}I\omega^2$. Therefore:

$$\Delta U + \Delta K = 0 \Rightarrow (mg\frac{L}{2}(1 - \cos\theta) - 0) + (0 - \frac{1}{2}I\omega^2) = 0$$

or

$$\frac{1}{2}I\omega^2 = mg\frac{L}{2}(1 - \cos\theta).$$

Solving for ω, we have

$$\omega = \sqrt{mg\frac{L}{I}(1 - \cos\theta)} = \sqrt{mg\frac{L}{1/3mL^2}(1 - \cos\theta)} = \sqrt{g\frac{3}{L}(1 - \cos\theta)}.$$

Inserting numerical values, we have

$$\omega = \sqrt{9.8\text{ m/s}^2\frac{3}{0.3\text{ m}}(1 - \cos 30)} = 3.6\text{ rad/s}.$$

Significance

Note that the angular velocity of the pendulum does not depend on its mass.

10.6 Torque

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe how the magnitude of a torque depends on the magnitude of the lever arm and the angle the force vector makes with the lever arm
- Determine the sign (positive or negative) of a torque using the right-hand rule
- Calculate individual torques about a common axis and sum them to find the net torque

An important quantity for describing the dynamics of a rotating rigid body is torque. We see the application of torque in many ways in our world. We all have an intuition about torque, as when we use a large wrench to unscrew a stubborn bolt. Torque is at work in unseen ways, as when we press on the accelerator in a car, causing the engine to put additional torque on the drive train. Or every time we move our bodies from a standing position, we apply a torque to our limbs. In this section, we define torque and make an argument for the equation for calculating torque for a rigid body with fixed-axis rotation.

Defining Torque

So far we have defined many variables that are rotational equivalents to their translational counterparts. Let's consider what the counterpart to force must be. Since forces change the translational motion of objects, the rotational counterpart must be related to changing the rotational motion of an object about an axis. We call this rotational counterpart **torque**.

In everyday life, we rotate objects about an axis all the time, so intuitively we already know much about torque. Consider, for example, how we rotate a door to open it. First, we know that a door opens slowly if we push too close to its hinges; it is more efficient to rotate a door open if we push far from the hinges. Second, we know that we should push perpendicular to the plane of the door; if we push parallel to the plane of the door, we are not able to rotate it. Third, the larger the force, the more effective it is in opening the door; the harder you push, the more rapidly the door opens. The first point implies that the farther the force is applied from the axis of rotation, the greater the angular acceleration; the second implies that the effectiveness depends on the angle at which the force is applied; the third implies that the magnitude of the force must also be part of the equation. Note that for rotation in a plane, torque has two possible directions. Torque is either clockwise or counterclockwise relative to the chosen pivot point. Figure 10.31 shows counterclockwise rotations.

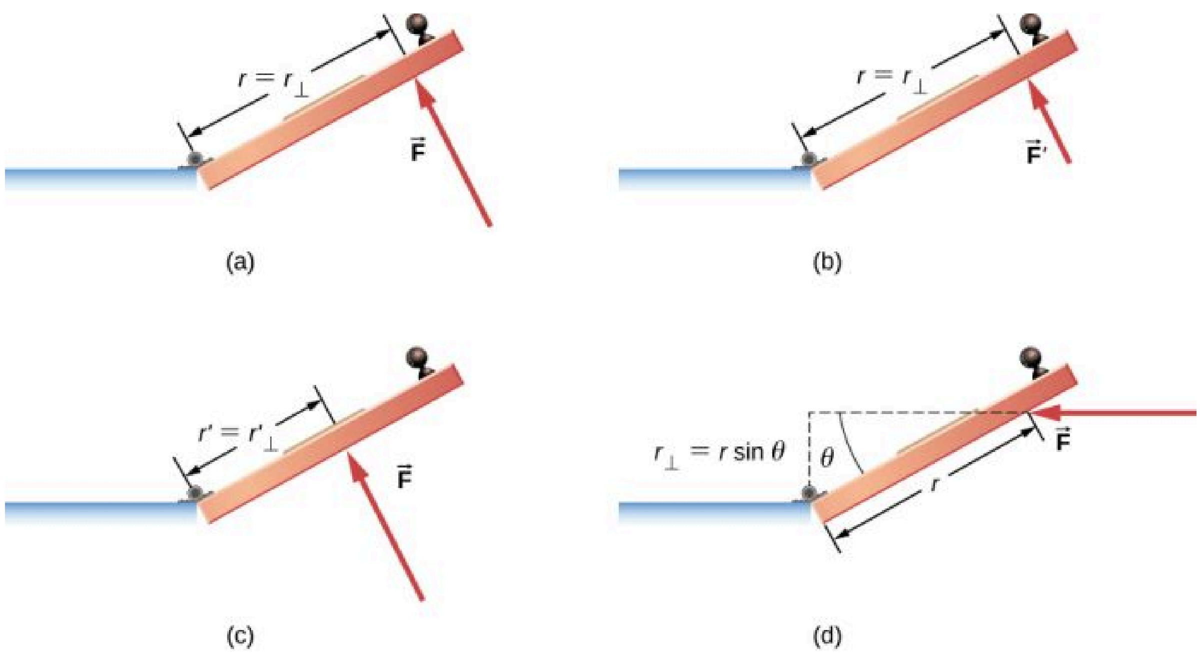


FIGURE 10.31 Torque is the turning or twisting effectiveness of a force, illustrated here for door rotation on its hinges (as viewed from overhead). Torque has both magnitude and direction. (a) A counterclockwise torque is produced by a force $\vec{F}$ acting at a distance r from the hinges (the pivot point). (b) A smaller counterclockwise torque is produced when a smaller force $\vec{F}'$ acts at the same distance r from the

hinges. (c) The same force as in (a) produces a smaller counterclockwise torque when applied at a smaller distance from the hinges. (d) A smaller counterclockwise torque is produced by the same magnitude force as (a) acting at the same distance as (a) but at an angle θ that is less than $90°$.

Now let's consider how to define torques in the general three-dimensional case.

TORQUE

When a force $\vec{\mathbf{F}}$ is applied to a point *P* whose position is $\vec{\mathbf{r}}$ relative to *O* (Figure 10.32), the torque $\vec{\tau}$ around *O* is

$$\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}. \tag{10.22}$$

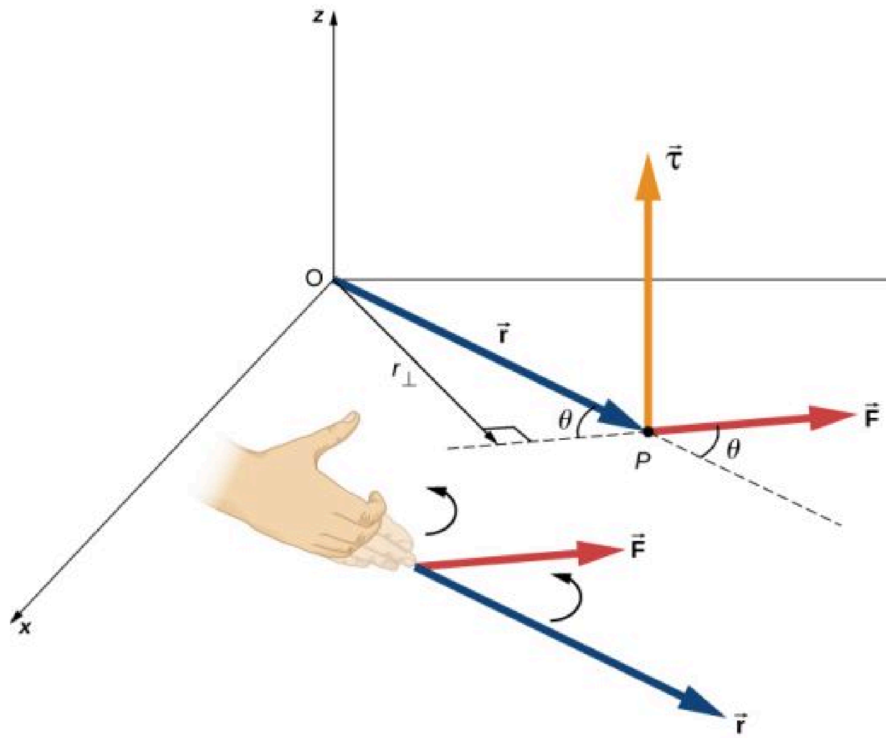


FIGURE 10.32 The torque is perpendicular to the plane defined by $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$ and its direction is determined by the right-hand rule.

From the definition of the cross product, the torque $\vec{\boldsymbol{\tau}}$ is perpendicular to the plane containing $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$ and has magnitude

$$\left|\vec{\boldsymbol{\tau}}\right| = \left|\vec{\mathbf{r}} \times \vec{\mathbf{F}}\right| = rF\sin\theta,$$

where θ is the angle between the vectors $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$. The SI unit of torque is newtons times meters, usually written as $\mathrm{N \cdot m}$. The quantity $r_\perp = r\sin\theta$ is the perpendicular distance from *O* to the line determined by the vector $\vec{\mathbf{F}}$ and is called the **lever arm**. Note that the greater the lever arm, the greater the magnitude of the torque. In terms of the lever arm, the magnitude of the torque is

$$\left|\vec{\boldsymbol{\tau}}\right| = r_\perp F. \tag{10.23}$$

The cross product $\vec{\mathbf{r}} \times \vec{\mathbf{F}}$ also tells us the sign of the torque. In Figure 10.32, the cross product $\vec{\mathbf{r}} \times \vec{\mathbf{F}}$ is along the positive *z*-axis, which by convention is a positive torque. If $\vec{\mathbf{r}} \times \vec{\mathbf{F}}$ is along the negative *z*-axis, this produces a negative torque.

If we consider a disk that is free to rotate about an axis through the center, as shown in Figure 10.33, we can see how the angle between the radius $\vec{\mathbf{r}}$ and the force $\vec{\mathbf{F}}$ affects the magnitude of the torque. If the angle is zero, the torque is zero; if the angle is $90°$, the torque is maximum. The torque in Figure 10.33 is positive because the direction of the torque by the right-hand rule is out of the page along the positive *z*-axis. The disk rotates counterclockwise due to the torque, in the same direction as a positive angular acceleration.

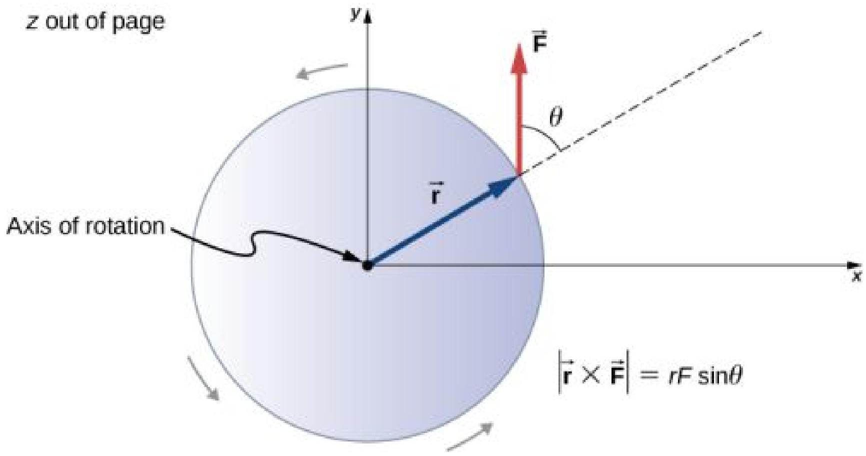


FIGURE 10.33 A disk is free to rotate about its axis through the center. The magnitude of the torque on the disk is $rF\sin\theta$.When $\theta = 0°$, the torque is zero and the disk does not rotate. When $\theta = 90°$, the torque is maximum and the disk rotates with maximum angular acceleration.

Any number of torques can be calculated about a given axis. The individual torques add to produce a net torque about the axis. When the appropriate sign (positive or negative) is assigned to the magnitudes of individual torques about a specified axis, the net torque about the axis is the sum of the individual torques:

$$\vec{\boldsymbol{\tau}}_{\text{net}} = \sum_i \vec{\boldsymbol{\tau}}_i. \tag{10.24}$$

Calculating Net Torque for Rigid Bodies on a Fixed Axis

In the following examples, we calculate the torque both abstractly and as applied to a rigid body.

We first introduce a problem-solving strategy.

PROBLEM-SOLVING STRATEGY

Finding Net Torque

1. Choose a coordinate system with the pivot point or axis of rotation as the origin of the selected coordinate system.
2. Determine the angle between the lever arm $\vec{\mathbf{r}}$ and the force vector.
3. Take the cross product of $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$ to determine if the torque is positive or negative about the pivot point or axis.
4. Evaluate the magnitude of the torque using $r_\perp F$.
5. Assign the appropriate sign, positive or negative, to the magnitude.
6. Sum the torques to find the net torque.

EXAMPLE 10.14

Calculating Torque

Four forces are shown in Figure 10.34 at particular locations and orientations with respect to a given *xy*-coordinate system. Find the torque due to each force about the origin, then use your results to find the net torque about the origin.

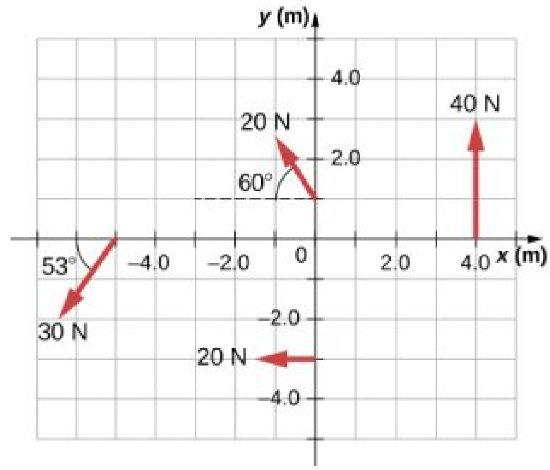


FIGURE 10.34 Four forces producing torques.

Strategy

This problem requires calculating torque. All known quantities––forces with directions and lever arms––are given in the figure. The goal is to find each individual torque and the net torque by summing the individual torques. Be careful to assign the correct sign to each torque by using the cross product of $\vec{\mathbf{r}}$ and the force vector $\vec{\mathbf{F}}$.

Solution

Use $|\vec{\boldsymbol{\tau}}| = r_\perp F = rF\sin\theta$ to find the magnitude and $\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}$ to determine the sign of the torque.

The torque from force 40 N in the first quadrant is given by $(4)(40)\sin 90° = 160\,\mathrm{N \cdot m}$.

The cross product of $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$ is out of the page, positive.

The torque from force 20 N in the third quadrant is given by$-(3)(20)\sin 90° = -60\,\mathrm{N \cdot m}$.

The cross product of $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$ is into the page, so it is negative.

The torque from force 30 N in the third quadrant is given by $(5)(30)\sin 53° = 120\,\mathrm{N \cdot m}$.

The cross product of $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$ is out of the page, positive.

The torque from force 20 N in the second quadrant is given by $(1)(20)\sin 30° = 10\,\mathrm{N \cdot m}$.

The cross product of $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$ is out of the page.

The net torque is therefore $\tau_{\text{net}} = \sum_i |\boldsymbol{\tau}_i| = 160 - 60 + 120 + 10 = 230\,\mathrm{N \cdot m}$.

Significance

Note that each force that acts in the counterclockwise direction has a positive torque, whereas each force that acts in the clockwise direction has a negative torque. The torque is greater when the distance, force, or perpendicular components are greater.

EXAMPLE 10.15

Calculating Torque on a rigid body

Figure 10.35 shows several forces acting at different locations and angles on a flywheel. We have $|\vec{\mathbf{F}}_1| = 20\,\mathrm{N}$, $|\vec{\mathbf{F}}_2| = 30\,\mathrm{N}$, $|\vec{\mathbf{F}}_3| = 30\,\mathrm{N}$, and $r = 0.5\,\mathrm{m}$. Find the net torque on the flywheel about an axis through the center.

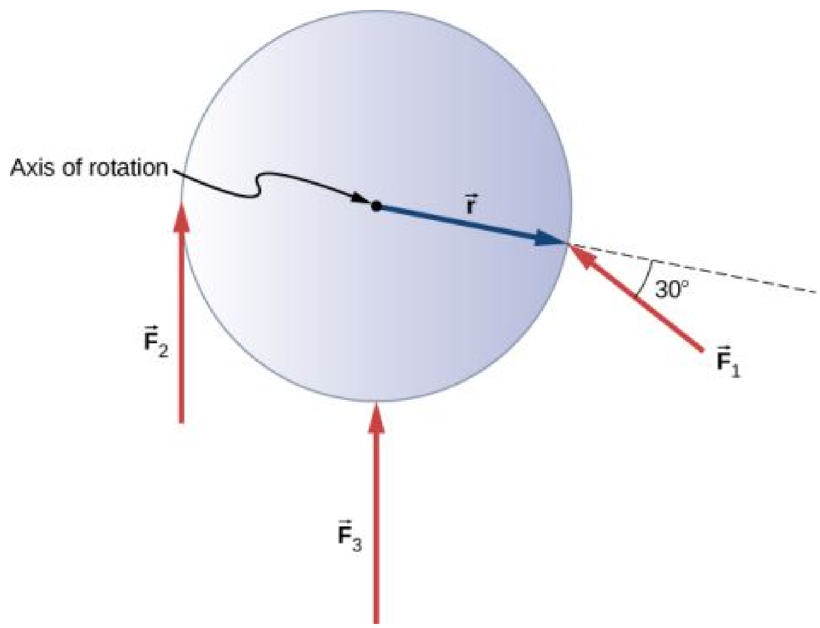


FIGURE 10.35 Three forces acting on a flywheel.

Strategy

We calculate each torque individually, using the cross product, and determine the sign of the torque. Then we sum the torques to find the net torque.

Solution

We start with $\vec{\mathbf{F}}_1$. If we look at Figure 10.35, we see that $\vec{\mathbf{F}}_1$ makes an angle of $90° + 60°$ with the radius vector $\vec{\mathbf{r}}$. Taking the cross product, we see that it is out of the page and so is positive. We also see this from calculating its magnitude:

$$\left|\vec{\boldsymbol{\tau}}_1\right| = rF_1 \sin 150° = 0.5\,\text{m}(20\,\text{N})(0.5) = 5.0\,\text{N} \cdot \text{m}.$$

Next we look at $\vec{\mathbf{F}}_2$. The angle between $\vec{\mathbf{F}}_2$ and $\vec{\mathbf{r}}$ is $90°$ and the cross product is into the page so the torque is negative. Its value is

$$\left|\vec{\boldsymbol{\tau}}_2\right| = -rF_2 \sin 90° = -0.5\,\text{m}(30\,\text{N}) = -15.0\,\text{N} \cdot \text{m}.$$

When we evaluate the torque due to $\vec{\mathbf{F}}_3$, we see that the angle it makes with $\vec{\mathbf{r}}$ is zero so $\vec{\mathbf{r}} \times \vec{\mathbf{F}}_3 = 0$. Therefore, $\vec{\mathbf{F}}_3$ does not produce any torque on the flywheel.

We evaluate the sum of the torques:

$$\tau_{\text{net}} = \sum_i |\boldsymbol{\tau}_i| = 5 - 15 = -10\,\text{N} \cdot \text{m}.$$

Significance

The axis of rotation is at the center of mass of the flywheel. Since the flywheel is on a fixed axis, it is not free to translate. If it were on a frictionless surface and not fixed in place, $\vec{\mathbf{F}}_3$ would cause the flywheel to translate, as well as $\vec{\mathbf{F}}_1$. Its motion would be a combination of translation and rotation.

CHECK YOUR UNDERSTANDING 10.6

A large ocean-going ship runs aground near the coastline, similar to the fate of the *Costa Concordia*, and lies at an angle as shown below. Salvage crews must apply a torque to right the ship in order to float the vessel for transport. A force of 5.0×10^5 N acting at point *A* must be applied to right the ship. What is the torque about the point of contact of the ship with the ground (Figure 10.36)?

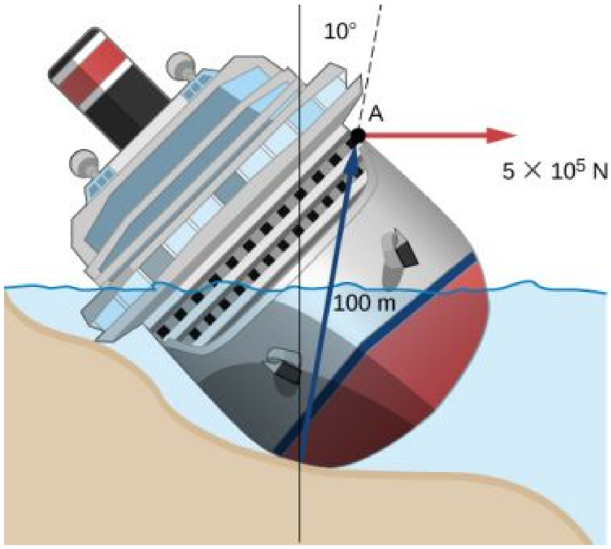


FIGURE 10.36 A ship runs aground and tilts, requiring torque to be applied to return the vessel to an upright position.

10.7 Newton's Second Law for Rotation

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Calculate the torques on rotating systems about a fixed axis to find the angular acceleration
- Explain how changes in the moment of inertia of a rotating system affect angular acceleration with a fixed applied torque

In this section, we put together all the pieces learned so far in this chapter to analyze the dynamics of rotating rigid bodies. We have analyzed motion with kinematics and rotational kinetic energy but have not yet connected these ideas with force and/or torque. In this section, we introduce the rotational equivalent to Newton's second law of motion and apply it to rigid bodies with fixed-axis rotation.

Newton's Second Law for Rotation

We have thus far found many counterparts to the translational terms used throughout this text, most recently, torque, the rotational analog to force. This raises the question: Is there an analogous equation to Newton's second law, $\Sigma\vec{\mathbf{F}} = m\vec{\mathbf{a}}$, which involves torque and rotational motion? To investigate this, we start with Newton's second law for a single particle rotating around an axis and executing circular motion. Let's exert a force $\vec{\mathbf{F}}$ on a point mass m that is at a distance r from a pivot point (Figure 10.37). The particle is constrained to move in a circular path with fixed radius and the force is tangent to the circle. We apply Newton's second law to determine the magnitude of the acceleration $a = F/m$ in the direction of $\vec{\mathbf{F}}$. Recall that the magnitude of the tangential acceleration is proportional to the magnitude of the angular acceleration by $a = r\alpha$. Substituting this expression into Newton's second law, we obtain

$$F = mr\alpha.$$

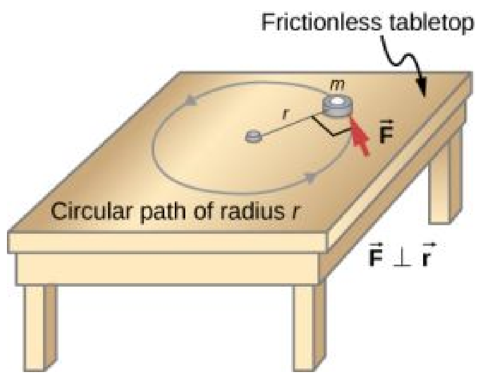


FIGURE 10.37 An object is supported by a horizontal frictionless table and is attached to a pivot point by a cord that supplies centripetal force. A force $\vec{\mathbf{F}}$ is applied to the object perpendicular to the radius *r*, causing it to accelerate about the pivot point. The force is perpendicular to *r*.

Multiply both sides of this equation by *r*,

$$rF = mr^2\alpha.$$

Note that the left side of this equation is the torque about the axis of rotation, where *r* is the lever arm and *F* is the force, perpendicular to *r*. Recall that the moment of inertia for a point particle is $I = mr^2$. The torque applied perpendicularly to the point mass in Figure 10.37 is therefore

$$\tau = I\alpha.$$

The torque on the particle is equal to the moment of inertia about the rotation axis times the angular acceleration. We can generalize this equation to a rigid body rotating about a fixed axis.

NEWTON'S SECOND LAW FOR ROTATION

If more than one torque acts on a rigid body about a fixed axis, then the sum of the torques equals the moment of inertia times the angular acceleration:

$$\tau_{\text{net}} = \sum_i \tau_i = I\alpha. \tag{10.25}$$

The term $I\alpha$ is a scalar quantity and can be positive or negative (counterclockwise or clockwise) depending upon the sign of the net torque. Remember the convention that counterclockwise angular acceleration is positive. Thus, if a rigid body is rotating clockwise and experiences a positive torque (counterclockwise), the angular acceleration is positive.

Equation 10.25 is **Newton's second law for rotation** and tells us how to relate torque, moment of inertia, and rotational kinematics. This is called the equation for **rotational dynamics**. With this equation, we can solve a whole class of problems involving force and rotation. It makes sense that the relationship for how much force it takes to rotate a body would include the moment of inertia, since that is the quantity that tells us how easy or hard it is to change the rotational motion of an object.

Deriving Newton's Second Law for Rotation in Vector Form

As before, when we found the angular acceleration, we may also find the torque vector. The second law $\Sigma\vec{\mathbf{F}} = m\vec{\mathbf{a}}$ tells us the relationship between net force and how to change the translational motion of an object. We have a vector rotational equivalent of this equation, which can be found by using Equation 10.7 and Figure 10.8. Equation 10.7 relates the angular acceleration to the position and tangential acceleration vectors:

$$\vec{\mathbf{a}} = \vec{\boldsymbol{\alpha}} \times \vec{\mathbf{r}}.$$

We form the cross product of this equation with $\vec{\mathbf{r}}$ and use a cross product identity (note that $\vec{\mathbf{r}} \cdot \vec{\boldsymbol{\alpha}} = 0$):

$$\vec{\mathbf{r}} \times \vec{\mathbf{a}} = \vec{\mathbf{r}} \times (\vec{\boldsymbol{\alpha}} \times \vec{\mathbf{r}}) = \vec{\boldsymbol{\alpha}}(\vec{\mathbf{r}} \cdot \vec{\mathbf{r}}) - \vec{\mathbf{r}}(\vec{\mathbf{r}} \cdot \vec{\boldsymbol{\alpha}}) = \vec{\boldsymbol{\alpha}}(\vec{\mathbf{r}} \cdot \vec{\mathbf{r}}) = \vec{\boldsymbol{\alpha}}r^2.$$

We now form the cross product of Newton's second law with the position vector $\vec{\mathbf{r}}$,

$$\Sigma(\vec{\mathbf{r}} \times \vec{\mathbf{F}}) = \vec{\mathbf{r}} \times (m\vec{\mathbf{a}}) = m\vec{\mathbf{r}} \times \vec{\mathbf{a}} = mr^2\vec{\boldsymbol{\alpha}}.$$

Identifying the first term on the left as the sum of the torques, and mr^2 as the moment of inertia, we arrive at Newton's second law of rotation in vector form:

$$\tau_{\text{net}} = \Sigma\vec{\boldsymbol{\tau}} = I\vec{\boldsymbol{\alpha}}. \tag{10.26}$$

This equation is exactly Equation 10.25 but with the torque and angular acceleration as vectors. An important point is that the torque vector is in the same direction as the angular acceleration.

Applying the Rotational Dynamics Equation

Before we apply the rotational dynamics equation to some everyday situations, let's review a general problem-solving strategy for use with this category of problems.

PROBLEM-SOLVING STRATEGY

Rotational Dynamics

1. Examine the situation to determine that torque and mass are involved in the rotation. Draw a careful sketch of the situation.
2. Determine the system of interest.
3. Draw a free-body diagram. That is, draw and label all external forces acting on the system of interest.
4. Identify the pivot point. If the object is in equilibrium, it must be in equilibrium for all possible pivot points––chose the one that simplifies your work the most.
5. Apply $\sum_i \tau_i = I\alpha$, the rotational equivalent of Newton's second law, to solve the problem. Care must be taken to use the correct moment of inertia and to consider the torque about the point of rotation.
6. As always, check the solution to see if it is reasonable.

EXAMPLE 10.16

Calculating the Effect of Mass Distribution on a Merry-Go-Round

Consider the father pushing a playground merry-go-round in Figure 10.38. He exerts a force of 250 N at the edge of the 50.0-kg merry-go-round, which has a 1.50-m radius. Calculate the angular acceleration produced (a) when no one is on the merry-go-round and (b) when an 18.0-kg child sits 1.25 m away from the center. Consider the merry-go-round itself to be a uniform disk with negligible friction.

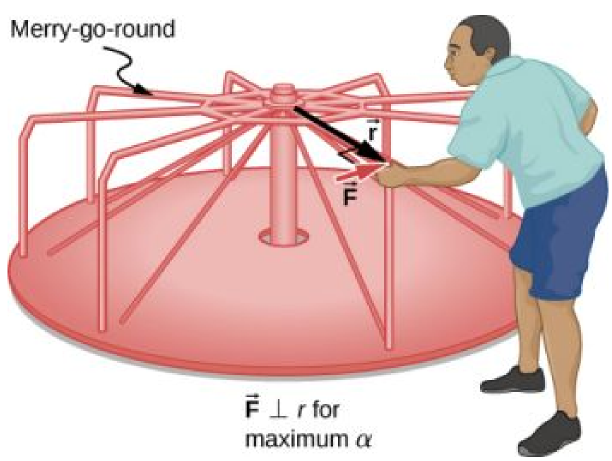


FIGURE 10.38 A father pushes a playground merry-go-round at its edge and perpendicular to its radius to achieve maximum torque.

Strategy

The net torque is given directly by the expression $\sum_i \tau_i = I\alpha$, To solve for α, we must first calculate the net torque τ (which is the same in both cases) and moment of inertia I (which is greater in the second case).

Solution

a. The moment of inertia of a solid disk about this axis is given in Figure 10.20 to be

$$\frac{1}{2}MR^2.$$

We have $M = 50.0\text{ kg}$ and $R = 1.50\text{ m}$, so

$$I = (0.500)(50.0\text{ kg})(1.50\text{ m})^2 = 56.25\text{ kg-m}^2.$$

To find the net torque, we note that the applied force is perpendicular to the radius and friction is negligible, so that

$$\tau = rF\sin\theta = (1.50\text{ m})(250.0\text{ N}) = 375.0\text{ N-m}.$$

Now, after we substitute the known values, we find the angular acceleration to be

$$\alpha = \frac{\tau}{I} = \frac{375.0\text{ N-m}}{56.25\text{ kg-m}^2} = 6.67\frac{\text{rad}}{\text{s}^2}.$$

b. We expect the angular acceleration for the system to be less in this part because the moment of inertia is greater when the child is on the merry-go-round. To find the total moment of inertia I, we first find the child's moment of inertia I_c by approximating the child as a point mass at a distance of 1.25 m from the axis. Then

$$I_c = mR^2 = (18.0\text{ kg})(1.25\text{ m})^2 = 28.13\text{ kg-m}^2.$$

The total moment of inertia is the sum of the moments of inertia of the merry-go-round and the child (about the same axis):

$$I = 28.13\text{ kg-m}^2 + 56.25\text{ kg-m}^2 = 84.38\text{ kg-m}^2.$$

Substituting known values into the equation for α gives

$$\alpha = \frac{\tau}{I} = \frac{375.0\text{ N-m}}{84.38\text{ kg-m}^2} = 4.44\frac{\text{rad}}{\text{s}^2}.$$

Significance

The angular acceleration is less when the child is on the merry-go-round than when the merry-go-round is empty, as expected. The angular accelerations found are quite large, partly due to the fact that friction was considered to be negligible. If, for example, the father kept pushing perpendicularly for 2.00 s, he would give the merry-go-round an angular velocity of 13.3 rad/s when it is empty but only 8.89 rad/s when the child is on it. In terms of revolutions per second, these angular velocities are 2.12 rev/s and 1.41 rev/s, respectively. The father would end up running at about 50 km/h in the first case.

CHECK YOUR UNDERSTANDING 10.7

The fan blades on a jet engine have a moment of inertia 30.0 kg-m^2. In 10 s, they rotate counterclockwise from rest up to a rotation rate of 20 rev/s. (a) What torque must be applied to the blades to achieve this angular acceleration? (b) What is the torque required to bring the fan blades rotating at 20 rev/s to a rest in 20 s?

10.8 Work and Power for Rotational Motion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Use the work-energy theorem to analyze rotation to find the work done on a system when it is rotated about a fixed axis for a finite angular displacement
- Solve for the angular velocity of a rotating rigid body using the work-energy theorem
- Find the power delivered to a rotating rigid body given the applied torque and angular velocity
- Summarize the rotational variables and equations and relate them to their translational counterparts

Thus far in the chapter, we have extensively addressed kinematics and dynamics for rotating rigid bodies around a fixed axis. In this final section, we define work and power within the context of rotation about a fixed axis, which has applications to both physics and engineering. The discussion of work and power makes our treatment of rotational motion almost complete, with the exception of rolling motion and angular momentum, which are discussed in Angular Momentum. We begin this section with a treatment of the work-energy theorem for rotation.

Work for Rotational Motion

Now that we have determined how to calculate kinetic energy for rotating rigid bodies, we can proceed with a discussion of the work done on a rigid body rotating about a fixed axis. Figure 10.39 shows a rigid body that has rotated through an angle $d\theta$ from A to B while under the influence of a force $\vec{\mathbf{F}}$. The external force $\vec{\mathbf{F}}$ is applied to point P, whose position is $\vec{\mathbf{r}}$, and the rigid body is constrained to rotate about a fixed axis that is perpendicular to the page and passes through O. The rotational axis is fixed, so the vector $\vec{\mathbf{r}}$ moves in a circle of radius r, and the vector $d\vec{\mathbf{s}}$ is perpendicular to $\vec{\mathbf{r}}$.

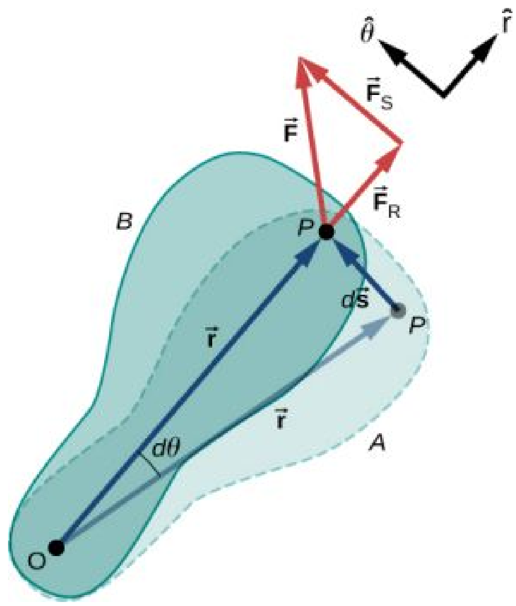


FIGURE 10.39 A rigid body rotates through an angle $d\theta$ from A to B by the action of an external force $\vec{\mathbf{F}}$ applied to point P.

From Equation 10.2, we have

$$\vec{\mathbf{s}} = \vec{\boldsymbol{\theta}} \times \vec{\mathbf{r}}.$$

Thus,

$$d\vec{\mathbf{s}} = d(\vec{\boldsymbol{\theta}} \times \vec{\mathbf{r}}) = d\vec{\boldsymbol{\theta}} \times \vec{\mathbf{r}} + d\vec{\mathbf{r}} \times \vec{\boldsymbol{\theta}} = d\vec{\boldsymbol{\theta}} \times \vec{\mathbf{r}}.$$

Note that $d\vec{\mathbf{r}}$ is zero because $\vec{\mathbf{r}}$ is fixed on the rigid body from the origin O to point P. Using the definition of work, we obtain

$$W = \int \sum \vec{\mathbf{F}} \cdot d\vec{\mathbf{s}} = \int \sum \vec{\mathbf{F}} \cdot (d\vec{\boldsymbol{\theta}} \times \vec{\mathbf{r}}) = \int d\vec{\boldsymbol{\theta}} \cdot (\vec{\mathbf{r}} \times \sum \vec{\mathbf{F}})$$

where we used the identity $\vec{\mathbf{a}} \cdot (\vec{\mathbf{b}} \times \vec{\mathbf{c}}) = \vec{\mathbf{b}} \cdot (\vec{\mathbf{c}} \times \vec{\mathbf{a}})$. Noting that $(\vec{\mathbf{r}} \times \sum \vec{F}) = \sum \vec{\boldsymbol{\tau}}$, we arrive at the expression for the **rotational work** done on a rigid body:

$$W = \int \sum \vec{\boldsymbol{\tau}} \cdot d\vec{\boldsymbol{\theta}}. \qquad \mathbf{10.27}$$

The total work done on a rigid body is the sum of the torques integrated over the angle through which the body rotates. The incremental work is

$$dW = \left(\sum_i \tau_i \right) d\theta \qquad \mathbf{10.28}$$

where we have taken the dot product in Equation 10.27, leaving only torques along the axis of rotation. In a rigid body, all particles rotate through the same angle; thus the work of every external force is equal to the torque times the common incremental angle $d\theta$. The quantity $\left(\sum_i \tau_i \right)$ is the net torque on the body due to external forces.

Similarly, we found the kinetic energy of a rigid body rotating around a fixed axis by summing the kinetic energy of each particle that makes up the rigid body. Since the work-energy theorem $W_i = \Delta K_i$ is valid for each particle, it is valid for the sum of the particles and the entire body.

WORK-ENERGY THEOREM FOR ROTATION

The work-energy theorem for a rigid body rotating around a fixed axis is

$$W_{AB} = K_B - K_A \qquad \mathbf{10.29}$$

where

$$K = \frac{1}{2} I \omega^2$$

and the rotational work done by a net force rotating a body from point *A* to point *B* is

$$W_{AB} = \int_{\theta_A}^{\theta_B} \left(\sum_i \tau_i \right) d\theta. \qquad \mathbf{10.30}$$

We give a strategy for using this equation when analyzing rotational motion.

PROBLEM-SOLVING STRATEGY

Work-Energy Theorem for Rotational Motion

1. Identify the forces on the body and draw a free-body diagram. Calculate the torque for each force.
2. Calculate the work done during the body's rotation by every torque.
3. Apply the work-energy theorem by equating the net work done on the body to the change in rotational kinetic energy.

Let's look at two examples and use the work-energy theorem to analyze rotational motion.

EXAMPLE 10.17

Rotational Work and Energy

A $12.0\,\text{N}\cdot\text{m}$ torque is applied to a flywheel that rotates about a fixed axis and has a moment of inertia of $30.0\,\text{kg}\cdot\text{m}^2$. If the flywheel is initially at rest, what is its angular velocity after it has turned through eight revolutions?

Strategy

We apply the work-energy theorem. We know from the problem description what the torque is and the angular displacement of the flywheel. Then we can solve for the final angular velocity.

Solution

The flywheel turns through eight revolutions, which is 16π radians. The work done by the torque, which is constant and therefore can come outside the integral in Equation 10.30, is

$$W_{AB} = \tau(\theta_B - \theta_A).$$

We apply the work-energy theorem:

$$W_{AB} = \tau(\theta_B - \theta_A) = \frac{1}{2}I\omega_B^2 - \frac{1}{2}I\omega_A^2.$$

With $\tau = 12.0\,\text{N}\cdot\text{m}, \theta_B - \theta_A = 16.0\pi\,\text{rad},\ I = 30.0\,\text{kg}\cdot\text{m}^2,$ and $\omega_A = 0$, we have

$$12.0\,\text{N-m}(16.0\pi\,\text{rad}) = \frac{1}{2}(30.0\,\text{kg}\cdot\text{m}^2)(\omega_B^2) - 0.$$

Therefore,

$$\omega_B = 6.3\,\text{rad/s}.$$

This is the angular velocity of the flywheel after eight revolutions.

Significance

The work-energy theorem provides an efficient way to analyze rotational motion, connecting torque with rotational kinetic energy.

EXAMPLE 10.18

Rotational Work: A Pulley

A string wrapped around the pulley in Figure 10.40 is pulled with a constant downward force $\vec{\mathbf{F}}$ of magnitude 50 N. The radius R and moment of inertia I of the pulley are 0.10 m and $2.5 \times 10^{-3}\text{kg-m}^2$, respectively. If the string does not slip, what is the angular velocity of the pulley after 1.0 m of string has unwound? Assume the pulley starts from rest.

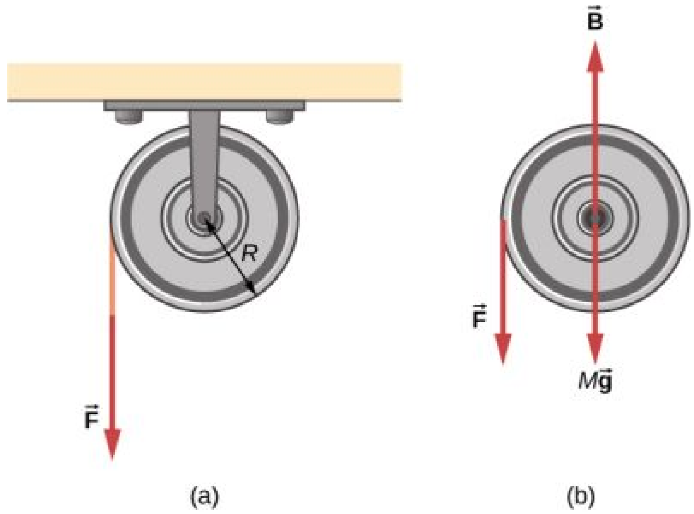


FIGURE 10.40 (a) A string is wrapped around a pulley of radius *R*. (b) The free-body diagram.

Strategy

Looking at the free-body diagram, we see that neither $\vec{\mathbf{B}}$, the force on the bearings of the pulley, nor $M\vec{\mathbf{g}}$, the weight of the pulley, exerts a torque around the rotational axis, and therefore does no work on the pulley. As the pulley rotates through an angle θ, $\vec{\mathbf{F}}$ acts through a distance *d* such that $d = R\theta$.

Solution

Since the torque due to $\vec{\mathbf{F}}$ has magnitude $\tau = RF$, we have

$$W = \tau\theta = (FR)\theta = Fd.$$

If the force on the string acts through a distance of 1.0 m, we have, from the work-energy theorem,

$$\begin{aligned} W_{AB} &= K_B - K_A \\ Fd &= \tfrac{1}{2}I\omega^2 - 0 \\ (50.0\,\text{N})(1.0\,\text{m}) &= \tfrac{1}{2}(2.5 \times 10^{-3}\text{kg-m}^2)\omega^2. \end{aligned}$$

Solving for ω, we obtain

$$\omega = 200.0\,\text{rad/s}.$$

Power for Rotational Motion

Power always comes up in the discussion of applications in engineering and physics. Power for rotational motion is equally as important as power in linear motion and can be derived in a similar way as in linear motion when the force is a constant. The linear power when the force is a constant is $P = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}}$. If the net torque is constant over the angular displacement, Equation 10.25 simplifies and the net torque can be taken out of the integral. In the following discussion, we assume the net torque is constant. We can apply the definition of power derived in Power to rotational motion. From Work and Kinetic Energy, the instantaneous power (or just power) is defined as the rate of doing work,

$$P = \frac{dW}{dt}.$$

If we have a constant net torque, Equation 10.25 becomes $W = \tau\theta$ and the power is

$$P = \frac{dW}{dt} = \frac{d}{dt}(\tau\theta) = \tau\frac{d\theta}{dt}$$

or

$$P = \tau\omega. \tag{10.31}$$

EXAMPLE 10.19

Torque on a Boat Propeller

A boat engine operating at 9.0×10^4 W is running at 300 rev/min. What is the torque on the propeller shaft?

Strategy

We are given the rotation rate in rev/min and the power consumption, so we can easily calculate the torque.

Solution

$$300.0 \text{ rev/min} = 31.4 \text{ rad/s};$$

$$\tau = \frac{P}{\omega} = \frac{9.0 \times 10^4 \text{N} \cdot \text{m/s}}{31.4 \text{ rad/s}} = 2864.8 \text{ N} \cdot \text{m}.$$

Significance

It is important to note the radian is a dimensionless unit because its definition is the ratio of two lengths. It therefore does not appear in the solution.

CHECK YOUR UNDERSTANDING 10.8

A constant torque of $500 \text{ kN} \cdot \text{m}$ is applied to a wind turbine to keep it rotating at 6 rad/s. What is the power required to keep the turbine rotating?

Rotational and Translational Relationships Summarized

The rotational quantities and their linear analog are summarized in three tables. Table 10.5 summarizes the rotational variables for circular motion about a fixed axis with their linear analogs and the connecting equation, except for the centripetal acceleration, which stands by itself. Table 10.6 summarizes the rotational and translational kinematic equations. Table 10.7 summarizes the rotational dynamics equations with their linear analogs.

Rotational	Translational	Relationship
θ	x	$\theta = \frac{s}{r}$
ω	v_t	$\omega = \frac{v_t}{r}$
α	a_t	$\alpha = \frac{a_t}{r}$
	a_c	$a_c = \frac{v_t^2}{r}$

TABLE 10.5 Rotational and Translational Variables: Summary

Rotational	Translational
$\theta_f = \theta_0 + \bar{\omega}t$	$x = x_0 + \bar{v}t$
$\omega_f = \omega_0 + \alpha t$	$v_f = v_0 + at$
$\theta_f = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$	$x_f = x_0 + v_0 t + \frac{1}{2}at^2$
$\omega_f^2 = \omega^2{}_0 + 2\alpha(\Delta\theta)$	$v_f^2 = v^2{}_0 + 2a(\Delta x)$

TABLE 10.6 **Rotational and Translational Kinematic Equations: Summary**

Rotational	Translational
$I = \sum_i m_i r_i^2$	m
$K = \frac{1}{2}I\omega^2$	$K = \frac{1}{2}mv^2$
$\sum_i \tau_i = I\alpha$	$\sum_i \vec{\mathbf{F}}_i = m\vec{\mathbf{a}}$
$W_{AB} = \int_{\theta_A}^{\theta_B} \left(\sum_i \tau_i\right) d\theta$	$W = \int \vec{\mathbf{F}} \cdot d\vec{\mathbf{s}}$
$P = \tau\omega$	$P = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}}$

TABLE 10.7 **Rotational and Translational Equations: Dynamics**

Chapter Review

Key Terms

angular acceleration time rate of change of angular velocity

angular position angle a body has rotated through in a fixed coordinate system

angular velocity time rate of change of angular position

instantaneous angular acceleration derivative of angular velocity with respect to time

instantaneous angular velocity derivative of angular position with respect to time

kinematics of rotational motion describes the relationships among rotation angle, angular velocity, angular acceleration, and time

lever arm perpendicular distance from the line that the force vector lies on to a given axis

linear mass density the mass per unit length λ of a one dimensional object

moment of inertia rotational mass of rigid bodies that relates to how easy or hard it will be to change the angular velocity of the rotating rigid body

Newton's second law for rotation sum of the torques on a rotating system equals its moment of inertia times its angular acceleration

parallel axis axis of rotation that is parallel to an axis about which the moment of inertia of an object is known

parallel-axis theorem if the moment of inertia is known for a given axis, it can be found for any axis parallel to it

rotational dynamics analysis of rotational motion using the net torque and moment of inertia to find the angular acceleration

rotational kinetic energy kinetic energy due to the rotation of an object; this is part of its total kinetic energy

rotational work work done on a rigid body due to the sum of the torques integrated over the angle through with the body rotates

surface mass density mass per unit area σ of a two dimensional object

torque cross product of a force and a lever arm to a given axis

total linear acceleration vector sum of the centripetal acceleration vector and the tangential acceleration vector

Key Equations

Angular position	$\theta = \frac{s}{r}$
Angular velocity	$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$
Tangential speed	$v_t = r\omega$
Angular acceleration	$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$
Tangential acceleration	$a_t = r\alpha$
Average angular velocity	$\bar{\omega} = \frac{\omega_0 + \omega_f}{2}$
Angular displacement	$\theta_f = \theta_0 + \bar{\omega} t$
Angular velocity from constant angular acceleration	$\omega_f = \omega_0 + \alpha t$
Angular velocity from displacement and constant angular acceleration	$\theta_f = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$
Change in angular velocity	$\omega_f^2 = \omega_0^2 + 2\alpha(\Delta\theta)$
Total acceleration	$\vec{\mathbf{a}} = \vec{\mathbf{a}}_c + \vec{\mathbf{a}}_t$
Rotational kinetic energy	$K = \frac{1}{2}\left(\sum_j m_j r_j^2\right)\omega^2$
Moment of inertia	$I = \sum_j m_j r_j^2$
Rotational kinetic energy in terms of the moment of inertia of a rigid body	$K = \frac{1}{2} I \omega^2$
Moment of inertia of a continuous object	$I = \int r^2 \, dm$

Parallel-axis theorem	$I_{\text{parallel-axis}} = I_{\text{center of mass}} + md^2$
Moment of inertia of a compound object	$I_{\text{total}} = \sum_i I_i$
Torque vector	$\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}$
Magnitude of torque	$\lvert\vec{\boldsymbol{\tau}}\rvert = r_\perp F$
Total torque	$\vec{\tau}_{\text{net}} = \sum_i \lvert\vec{\tau}_i\rvert$
Newton's second law for rotation	$\sum_i \tau_i = I\alpha$
Incremental work done by a torque	$dW = \left(\sum_i \tau_i\right) d\theta$
Work-energy theorem	$W_{AB} = K_B - K_A$
Rotational work done by net force	$W_{AB} = \int_{\theta_A}^{\theta_B} \left(\sum_i \tau_i\right) d\theta$
Rotational power	$P = \tau\omega$

Summary

10.1 Rotational Variables

- The angular position θ of a rotating body is the angle the body has rotated through in a fixed coordinate system, which serves as a frame of reference.
- The angular velocity of a rotating body about a fixed axis is defined as $\omega(\text{rad/s})$, the rotational rate of the body in radians per second. The instantaneous angular velocity of a rotating body $\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$ is the derivative with respect to time of the angular position θ, found by taking the limit $\Delta t \to 0$ in the average angular velocity $\bar{\omega} = \frac{\Delta\theta}{\Delta t}$. The angular velocity relates v_{t} to the tangential speed of a point on the rotating body through the relation $v_{\text{t}} = r\omega$, where r is the radius to the point and v_{t} is the tangential speed at the given point.
- The angular velocity $\vec{\boldsymbol{\omega}}$ is found using the right-hand rule. If the fingers curl in the direction of rotation about a fixed axis, the thumb points in the direction of $\vec{\boldsymbol{\omega}}$ (see Figure 10.5).
- If the system's angular velocity is not constant, then the system has an angular acceleration. The average angular acceleration over a given time interval is the change in angular velocity over this time interval, $\bar{\alpha} = \frac{\Delta\omega}{\Delta t}$. The instantaneous angular acceleration is the time derivative of angular velocity, $\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt}$. The angular acceleration $\vec{\boldsymbol{\alpha}}$ is found by locating the angular velocity. If a rotation rate of a rotating body is decreasing, the angular acceleration is in the opposite direction to $\vec{\boldsymbol{\omega}}$. If the rotation rate is increasing, the angular acceleration is in the same direction as $\vec{\boldsymbol{\omega}}$.
- The tangential acceleration of a point at a radius from the axis of rotation is the angular acceleration times the radius to the point.

10.2 Rotation with Constant Angular Acceleration

- The kinematics of rotational motion describes the relationships among rotation angle (angular position), angular velocity, angular acceleration, and time.
- For a constant angular acceleration, the angular velocity varies linearly. Therefore, the average angular velocity is 1/2 the initial plus final angular velocity over a given time period:
$$\bar{\omega} = \frac{\omega_0 + \omega_{\text{f}}}{2}.$$
- We used a graphical analysis to find solutions to fixed-axis rotation with constant angular acceleration. From the relation $\omega = \frac{d\theta}{dt}$, we found that the area under an angular velocity-vs.-time curve gives the angular displacement, $\theta_{\text{f}} - \theta_0 = \Delta\theta = \int_{t_0}^{t} \omega(t)dt$. The results of the graphical analysis were verified using the kinematic equations for constant angular acceleration. Similarly, since $\alpha = \frac{d\omega}{dt}$, the area under an angular acceleration-vs.-time graph

gives the change in angular velocity:

$$\omega_f - \omega_0 = \Delta\omega = \int_{t_0}^{t} \alpha(t)dt.$$

10.3 Relating Angular and Translational Quantities

- The linear kinematic equations have their rotational counterparts such that there is a mapping $x \rightarrow \theta, \quad v \rightarrow \omega, \quad a \rightarrow \alpha$.
- A system undergoing uniform circular motion has a constant angular velocity, but points at a distance *r* from the rotation axis have a linear centripetal acceleration.
- A system undergoing nonuniform circular motion has an angular acceleration and therefore has both a linear centripetal and linear tangential acceleration at a point a distance *r* from the axis of rotation.
- The total linear acceleration is the vector sum of the centripetal acceleration vector and the tangential acceleration vector. Since the centripetal and tangential acceleration vectors are perpendicular to each other for circular motion, the magnitude of the total linear acceleration is $|\vec{\mathbf{a}}| = \sqrt{a_c^2 + a_t^2}$.

10.4 Moment of Inertia and Rotational Kinetic Energy

- The rotational kinetic energy is the kinetic energy of rotation of a rotating rigid body or system of particles, and is given by $K = \frac{1}{2}I\omega^2$, where I is the moment of inertia, or "rotational mass" of the rigid body or system of particles.
- The moment of inertia for a system of point particles rotating about a fixed axis is $I = \sum_j m_j r_j^2$, where m_j is the mass of the point particle and r_j is the distance of the point particle to the rotation axis. Because of the r^2 term, the moment of inertia increases as the square of the distance to the fixed rotational axis. The moment of inertia is the rotational counterpart to the mass in linear motion.
- In systems that are both rotating and translating, conservation of mechanical energy can be used if there are no nonconservative forces at work. The total mechanical energy is then conserved and is the sum of the rotational and translational kinetic energies, and the gravitational potential energy.

10.5 Calculating Moments of Inertia

- Moments of inertia can be found by summing or integrating over every 'piece of mass' that makes up an object, multiplied by the square of the distance of each 'piece of mass' to the axis. In integral form the moment of inertia is $I = \int r^2 dm.$
- Moment of inertia is larger when an object's mass is farther from the axis of rotation.
- It is possible to find the moment of inertia of an object about a new axis of rotation once it is known for a parallel axis. This is called the parallel axis theorem given by $I_{\text{parallel-axis}} = I_{\text{center of mass}} + md^2$, where *d* is the distance from the initial axis to the parallel axis.
- Moment of inertia for a compound object is simply the sum of the moments of inertia for each individual object that makes up the compound object.

10.6 Torque

- The magnitude of a torque about a fixed axis is calculated by finding the lever arm to the point where the force is applied and using the relation $|\vec{\boldsymbol{\tau}}| = r_\perp F$, where $r_\perp$ is the perpendicular distance from the axis to the line upon which the force vector lies.
- The sign of the torque is found using the right hand rule. If the page is the plane containing $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$, then $\vec{\mathbf{r}} \times \vec{\mathbf{F}}$ is out of the page for positive torques and into the page for negative torques.
- The net torque can be found from summing the individual torques about a given axis.

10.7 Newton's Second Law for Rotation

- Newton's second law for rotation, $\sum_i \tau_i = I\alpha$, says that the sum of the torques on a rotating system about a fixed axis equals the product of the moment of inertia and the angular acceleration. This is the rotational analog to Newton's second law of linear motion.
- In the vector form of Newton's second law for rotation, the torque vector $\vec{\boldsymbol{\tau}}$ is in the same direction as the angular acceleration $\vec{\boldsymbol{\alpha}}$. If the angular acceleration of a rotating system is positive, the torque on the system is also positive, and if the angular acceleration is negative, the torque is negative.

10.8 Work and Power for Rotational Motion

- The incremental work *dW* in rotating a rigid body about a fixed axis is the sum of the torques about the axis times the incremental angle $d\theta$.

- The total work done to rotate a rigid body through an angle θ about a fixed axis is the sum of the torques integrated over the angular displacement. If the torque is a constant as a function of θ, then $W_{AB} = \tau(\theta_B - \theta_A)$.
- The work-energy theorem relates the rotational work done to the change in rotational kinetic energy: $W_{AB} = K_B - K_A$ where $K = \frac{1}{2} I\omega^2$.
- The power delivered to a system that is rotating about a fixed axis is the torque times the angular velocity, $P = \tau\omega$.

Conceptual Questions

10.1 Rotational Variables

1. A clock is mounted on the wall. As you look at it, what is the direction of the angular velocity vector of the second hand?

2. What is the value of the angular acceleration of the second hand of the clock on the wall?

3. A baseball bat is swung. Do all points on the bat have the same angular velocity? The same tangential speed?

4. The blades of a blender on a counter are rotating clockwise as you look into it from the top. If the blender is put to a greater speed what direction is the angular acceleration of the blades?

10.2 Rotation with Constant Angular Acceleration

5. If a rigid body has a constant angular acceleration, what is the functional form of the angular velocity in terms of the time variable?

6. If a rigid body has a constant angular acceleration, what is the functional form of the angular position?

7. If the angular acceleration of a rigid body is zero, what is the functional form of the angular velocity?

8. A massless tether with a masses tied to both ends rotates about a fixed axis through the center. Can the total acceleration of the tether/mass combination be zero if the angular velocity is constant?

10.3 Relating Angular and Translational Quantities

9. Explain why centripetal acceleration changes the direction of velocity in circular motion but not its magnitude.

10. In circular motion, a tangential acceleration can change the magnitude of the velocity but not its direction. Explain your answer.

11. Suppose a piece of food is on the edge of a rotating microwave oven plate. Does it experience nonzero tangential acceleration, centripetal acceleration, or both when: (a) the plate starts to spin faster? (b) The plate rotates at constant angular velocity? (c) The plate slows to a halt?

10.4 Moment of Inertia and Rotational Kinetic Energy

12. What if another planet the same size as Earth were put into orbit around the Sun along with Earth. Would the moment of inertia of the system increase, decrease, or stay the same?

13. A solid sphere is rotating about an axis through its center at a constant rotation rate. Another hollow sphere of the same mass and radius is rotating about its axis through the center at the same rotation rate. Which sphere has a greater rotational kinetic energy?

10.5 Calculating Moments of Inertia

14. If a child walks toward the center of a merry-go-round, does the moment of inertia increase or decrease?

15. A discus thrower rotates with a discus in his hand before letting it go. (a) How does his moment of inertia change after releasing the discus? (b) What would be a good approximation to use in calculating the moment of inertia of the discus thrower and discus?

16. Does increasing the number of blades on a propeller increase or decrease its moment of inertia, and why?

17. The moment of inertia of a long rod spun around an axis through one end perpendicular to its length is $mL^2/3$. Why is this moment of inertia greater than it would be if you spun a point mass *m* at the location of the center of mass of the rod (at *L*/2) (that would be $mL^2/4$)?

18. Why is the moment of inertia of a hoop that has a mass *M* and a radius *R* greater than the moment of inertia of a disk that has the same mass and radius?

10.6 Torque

19. What three factors affect the torque created by a force relative to a specific pivot point?

20. Give an example in which a small force exerts a large torque. Give another example in which a large force exerts a small torque.

21. When reducing the mass of a racing bike, the greatest benefit is realized from reducing the mass of the tires and wheel rims. Why does this allow a racer to achieve greater accelerations than would an identical reduction in the mass of the bicycle's frame?

22. Can a single force produce a zero torque?

23. Can a set of forces have a net torque that is zero and a net force that is not zero?

24. Can a set of forces have a net force that is zero and a net torque that is not zero?

25. In the expression $\vec{\mathbf{r}} \times \vec{\mathbf{F}}$ can $|\vec{\mathbf{r}}|$ ever be less than the lever arm? Can it be equal to the lever arm?

10.7 Newton's Second Law for Rotation

26. If you were to stop a spinning wheel with a constant force, where on the wheel would you apply the force to produce the maximum negative acceleration?

27. A rod is pivoted about one end. Two forces $\vec{\mathbf{F}}$and $-\vec{\mathbf{F}}$ are applied to it. Under what circumstances will the rod not rotate?

Problems

10.1 Rotational Variables

28. Calculate the angular velocity of Earth.

29. A track star runs a 400-m race on a 400-m circular track in 45 s. What is his angular velocity assuming a constant speed?

30. A wheel rotates at a constant rate of 2.0×10^3 rev/min . (a) What is its angular velocity in radians per second? (b) Through what angle does it turn in 10 s? Express the solution in radians and degrees.

31. A particle moves 3.0 m along a circle of radius 1.5 m. (a) Through what angle does it rotate? (b) If the particle makes this trip in 1.0 s at a constant speed, what is its angular velocity? (c) What is its acceleration?

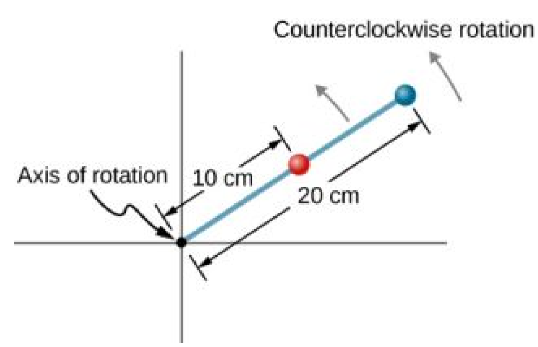
32. A compact disc rotates at 500 rev/min. If the diameter of the disc is 120 mm, (a) what is the tangential speed of a point at the edge of the disc? (b) At a point halfway to the center of the disc?

33. **Unreasonable results.** The propeller of an aircraft is spinning at 10 rev/s when the pilot shuts off the engine. The propeller reduces its angular velocity at a constant $2.0\ \text{rad/s}^2$ for a time period of 40 s. What is the rotation rate of the propeller in 40 s? Is this a reasonable situation?

34. A gyroscope slows from an initial rate of 32.0 rad/s at a rate of $0.700\ \text{rad/s}^2$. How long does it take to come to rest?

35. On takeoff, the propellers on a UAV (unmanned aerial vehicle) increase their angular velocity for 3.0 s from rest at a rate of $\omega = (25.0t)$ rad/s where t is measured in seconds. (a) What is the instantaneous angular velocity of the propellers at $t = 2.0$ s? (b) What is the angular acceleration?

36. The angular position of a rod varies as $20.0t^2$ radians from time $t = 0$. The rod has two beads on it as shown in the following figure, one at 10 cm from the rotation axis and the other at 20 cm from the rotation axis. (a) What is the instantaneous angular velocity of the rod at $t = 5$ s? (b) What is the angular acceleration of the rod? (c) What are the tangential speeds of the beads at $t = 5$ s? (d) What are the tangential accelerations of the beads at $t = 5$ s? (e) What are the centripetal accelerations of the beads at $t = 5$ s?



10.2 Rotation with Constant Angular Acceleration

37. A wheel has a constant angular acceleration of $5.0\ \text{rad/s}^2$. Starting from rest, it turns through

300 rad. (a) What is its final angular velocity? (b) How much time elapses while it turns through the 300 radians?

38. During a 6.0-s time interval, a flywheel with a constant angular acceleration turns through 500 radians that acquire an angular velocity of 100 rad/s. (a) What is the angular velocity at the beginning of the 6.0 s? (b) What is the angular acceleration of the flywheel?

39. The angular velocity of a rotating rigid body increases from 500 to 1500 rev/min in 120 s. (a) What is the angular acceleration of the body? (b) Through what angle does it turn in this 120 s?

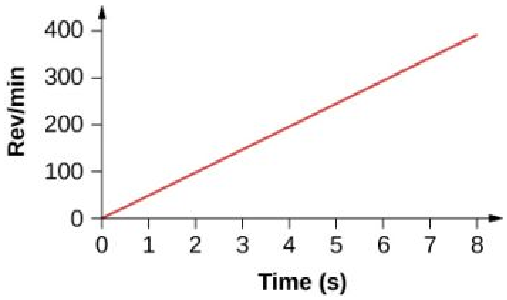
40. A flywheel slows from 600 to 400 rev/min while rotating through 40 revolutions. (a) What is the angular acceleration of the flywheel? (b) How much time elapses during the 40 revolutions?

41. A wheel 1.0 m in radius rotates with an angular acceleration of $4.0\,\text{rad/s}^2$. (a) If the wheel's initial angular velocity is 2.0 rad/s, what is its angular velocity after 10 s? (b) Through what angle does it rotate in the 10-s interval? (c) What are the tangential speed and acceleration of a point on the rim of the wheel at the end of the 10-s interval?

42. A vertical wheel with a diameter of 50 cm starts from rest and rotates with a constant angular acceleration of $5.0\,\text{rad/s}^2$ around a fixed axis through its center counterclockwise. (a) Where is the point that is initially at the bottom of the wheel at $t = 10\,\text{s}$? (b) What is the point's linear acceleration at this instant?

43. A circular disk of radius 10 cm has a constant angular acceleration of $1.0\,\text{rad/s}^2$; at $t = 0$ its angular velocity is 2.0 rad/s. (a) Determine the disk's angular velocity at $t = 5.0\,\text{s}$. (b) What is the angle it has rotated through during this time? (c) What is the tangential acceleration of a point on the disk at $t = 5.0\,\text{s}$?

44. The angular velocity vs. time for a fan on a hovercraft is shown below. (a) What is the angle through which the fan blades rotate in the first 8 seconds? (b) Verify your result using the kinematic equations.



45. A rod of length 20 cm has two beads attached to its ends. The rod with beads starts rotating from rest. If the beads are to have a tangential speed of 20 m/s in 7 s, what is the angular acceleration of the rod to achieve this?

10.3 Relating Angular and Translational Quantities

46. At its peak, a tornado is 60.0 m in diameter and carries 500 km/h winds. What is its angular velocity in revolutions per second?

47. A man stands on a merry-go-round that is rotating at 2.5 rad/s. If the coefficient of static friction between the man's shoes and the merry-go-round is $\mu_S = 0.5$, how far from the axis of rotation can he stand without sliding?

48. An ultracentrifuge accelerates from rest to 100,000 rpm in 2.00 min. (a) What is the average angular acceleration in rad/s^2? (b) What is the tangential acceleration of a point 9.50 cm from the axis of rotation? (c) What is the centripetal acceleration in m/s^2 and multiples of *g* of this point at full rpm? (d) What is the total distance traveled during the acceleration by a point 9.5 cm from the axis of rotation of the ultracentrifuge?

49. A wind turbine is rotating counterclockwise at 0.5 rev/s and slows to a stop in 10 s. Its blades are 20 m in length. (a) What is the angular acceleration of the turbine? (b) What is the centripetal acceleration of the tip of the blades at $t = 0\,\text{s}$? (c) What is the magnitude and direction of the total linear acceleration of the tip of the blade that lies along the positive *x*-axis at $t = 0\,\text{s}$?

50. What is (a) the angular speed and (b) the linear speed of a point on Earth's surface at latitude $30°$ N. Take the radius of the Earth to be 6309 km. (c) At what latitude would your linear speed be 10 m/s?

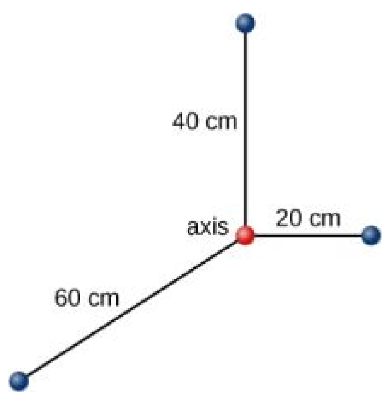
51. A child with mass 40 kg sits on the edge of a merry-go-round at a distance of 3.0 m from its axis of rotation. The merry-go-round accelerates from rest up to 0.4 rev/s in 10 s. If the coefficient of static friction between the child and the surface of the merry-go-round is 0.6, does the child fall off before 5 s?

52. A bicycle wheel with radius 0.3 m rotates from rest to 3 rev/s in 5 s. What is the magnitude and direction of the total acceleration vector at the edge of the wheel at 1.0 s?

53. The angular velocity of a flywheel with radius 1.0 m varies according to $\omega(t) = 2.0t$. Plot $a_c(t)$ and $a_t(t)$ from $t = 0$ to 3.0 s for $r = 1.0$ m. Analyze these results to explain when $a_c \gg a_t$ and when $a_c \ll a_t$ for a point on the flywheel at a radius of 1.0 m.

10.4 Moment of Inertia and Rotational Kinetic Energy

54. A system of point particles is shown in the following figure. Each particle has mass 0.3 kg and they all lie in the same plane. (a) What is the moment of inertia of the system about the given axis? (b) If the system rotates at 5 rev/s, what is its rotational kinetic energy?



55. (a) Calculate the rotational kinetic energy of Earth on its axis. (b) What is the rotational kinetic energy of Earth in its orbit around the Sun?

56. Calculate the rotational kinetic energy of a 12-kg motorcycle wheel if its angular velocity is 120 rad/s and its inner radius is 0.280 m and outer radius 0.330 m.

57. A baseball pitcher throws the ball in a motion where there is rotation of the forearm about the elbow joint as well as other movements. If the linear velocity of the ball relative to the elbow joint is 20.0 m/s at a distance of 0.480 m from the joint and the moment of inertia of the forearm is $0.500\ \text{kg-m}^2$, what is the rotational kinetic energy of the forearm?

58. A diver goes into a somersault during a dive by tucking her limbs. If her rotational kinetic energy is 100 J and her moment of inertia in the tuck is $9.0\ \text{kg}\cdot\text{m}^2$, what is her rotational rate during the somersault?

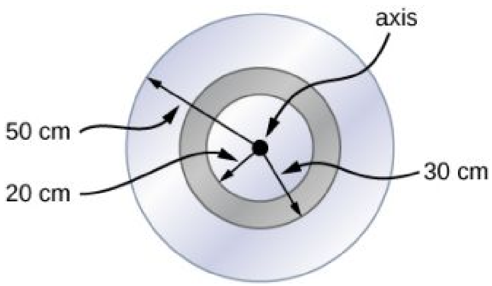
59. An aircraft is coming in for a landing at 300 meters height when the propeller falls off. The aircraft is flying at 40.0 m/s horizontally. The propeller has a rotation rate of 20 rev/s, a moment of inertia of $70.0\ \text{kg-m}^2$, and a mass of 200 kg. Neglect air resistance so the rotation rate does not change as the propeller falls. (a) With what translational velocity does the propeller hit the ground? (b) What is the rotation rate of the propeller at impact?

60. An aircraft is coming in for a landing at 300 meters height when the propeller falls off. When it comes off, the propeller has a rotation rate of 20 rev/s, a moment of inertia of $70.0\ \text{kg-m}^2$, and a mass of 200 kg. If air resistance is present and reduces the propeller's rotational kinetic energy at impact by 30%, what is the propeller's rotation rate at impact?

61. A neutron star of mass 2×10^{30} kg and radius 10 km rotates with a period of 0.02 seconds. What is its rotational kinetic energy?

62. An electric sander consisting of a rotating disk of mass 0.7 kg and radius 10 cm rotates at 15 rev/s. When applied to a rough wooden wall the rotation rate decreases by 20%. (a) What is the final rotational kinetic energy of the rotating disk? (b) How much has its rotational kinetic energy decreased?

63. A system consists of a disk of mass 2.0 kg and radius 50 cm upon which is mounted an annular cylinder of mass 1.0 kg with inner radius 20 cm and outer radius 30 cm (see below). The system rotates about an axis through the center of the disk and annular cylinder at 10 rev/s. (a) What is the moment of inertia of the system? (b) What is its rotational kinetic energy?



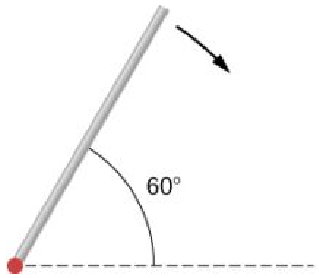
10.5 Calculating Moments of Inertia

64. While punting a football, a kicker rotates their leg about the hip joint. The moment of inertia of the leg is 3.75 kg-m^2 and its rotational kinetic energy is 175 J. (a) What is the angular velocity of the leg? (b) What is the velocity of tip of the punter's shoe if it is 1.05 m from the hip joint?

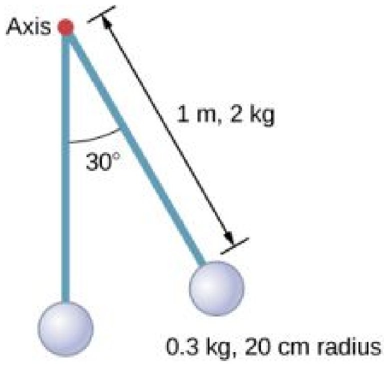
65. Using the parallel axis theorem, what is the moment of inertia of the rod of mass *m* about the axis shown below?

66. Find the moment of inertia of the rod in the previous problem by direct integration.

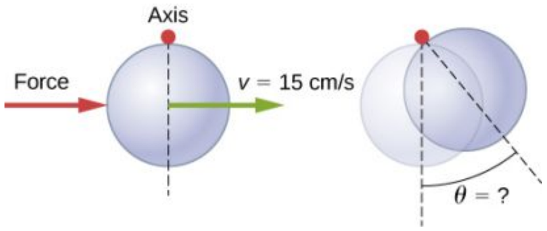
67. A uniform rod of mass 1.0 kg and length 2.0 m is free to rotate about one end (see the following figure). If the rod is released from rest at an angle of $60°$ with respect to the horizontal, what is the speed of the tip of the rod as it passes the horizontal position?



68. A pendulum consists of a rod of mass 2 kg and length 1 m with a solid sphere at one end with mass 0.3 kg and radius 20 cm (see the following figure). If the pendulum is released from rest at an angle of $30°$, what is the angular velocity at the lowest point?



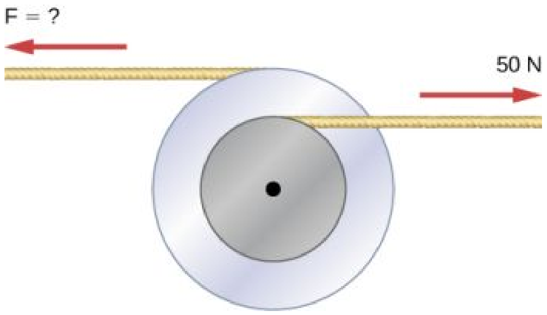
69. A solid sphere of radius 10 cm is allowed to rotate freely about an axis. The sphere is given a sharp blow so that its center of mass starts from the position shown in the following figure with speed 15 cm/s. What is the maximum angle that the diameter makes with the vertical?



70. Calculate the moment of inertia by direct integration of a thin rod of mass *M* and length *L* about an axis through the rod at *L*/3, as shown below. Check your answer with the parallel-axis theorem.

10.6 Torque

71. Two flywheels of negligible mass and different radii are bonded together and rotate about a common axis (see below). The smaller flywheel of radius 30 cm has a cord that has a pulling force of 50 N on it. What pulling force needs to be applied to the cord connecting the larger flywheel of radius 50 cm such that the combination does not rotate?

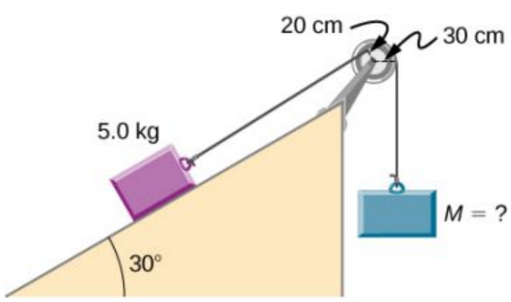


72. The cylinder head bolts on a car are to be tightened with a torque of 62.0 N·m. If a mechanic uses a wrench of length 20 cm, what perpendicular force must he exert on the end of the wrench to tighten a bolt correctly?

73. (a) When opening a door, you push on it perpendicularly with a force of 55.0 N at a distance of 0.850 m from the hinges. What torque are you exerting relative to the hinges? (b) Does it matter if you push at the same height as the hinges? There is only one pair of hinges.

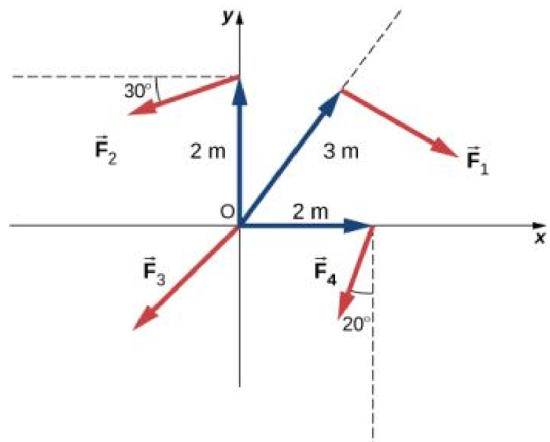
74. When tightening a bolt, you push perpendicularly on a wrench with a force of 165 N at a distance of 0.140 m from the center of the bolt. How much torque are you exerting in newton-meters (relative to the center of the bolt)?

75. What hanging mass must be placed on the cord to keep the pulley from rotating (see the following figure)? The mass on the frictionless plane is 5.0 kg. The mass on the plane is connected to a cord that wraps around the pulley's inner radius of 20 cm. The hanging mass is connected to a cord that wraps around the pulley's outer radius of 30 cm.

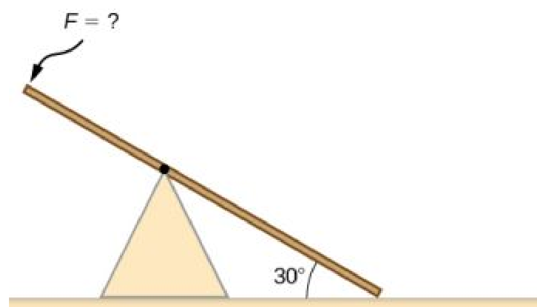


76. A simple pendulum consists of a massless tether 50 cm in length connected to a pivot and a small mass of 1.0 kg attached at the other end. What is the torque about the pivot when the pendulum makes an angle of $40°$ with respect to the vertical?

77. Calculate the torque about the *z*-axis that is out of the page at the origin in the following figure, given that $F_1 = 3\,\text{N},\ \ F_2 = 2\,\text{N},\ \ F_3 = 3\,\text{N},\ \ F_4 = 1.8\,\text{N}.$

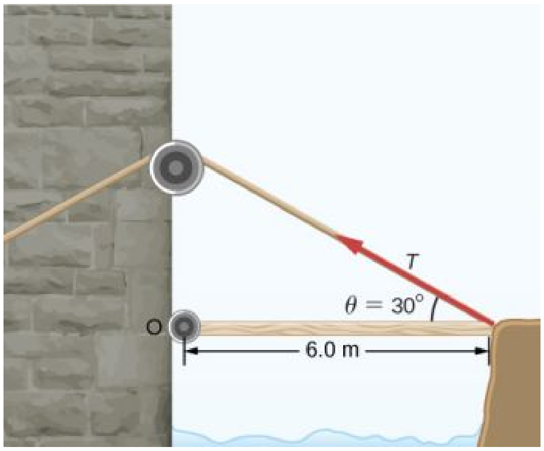


78. A seesaw has length 10.0 m and uniform mass 10.0 kg and is resting at an angle of $30°$ with respect to the ground (see the following figure). The pivot is located at 6.0 m. What magnitude of force needs to be applied perpendicular to the seesaw at the raised end so as to allow the seesaw to barely start to rotate?

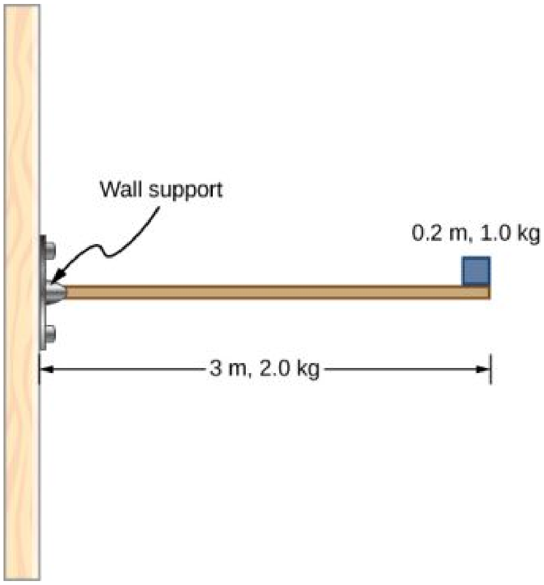


79. A pendulum consists of a rod of mass 1 kg and length 1 m connected to a pivot with a solid sphere attached at the other end with mass 0.5 kg and radius 30 cm. What is the torque about the pivot when the pendulum makes an angle of $30°$ with respect to the vertical?

80. A torque of $5.00 \times 10^3\,\text{N}\cdot\text{m}$ is required to raise a drawbridge (see the following figure). What is the tension necessary to produce this torque? Would it be easier to raise the drawbridge if the angle θ were larger or smaller?



81. A horizontal beam of length 3 m and mass 2.0 kg has a mass of 1.0 kg and width 0.2 m sitting at the end of the beam (see the following figure). What is the torque of the system about the support at the wall?



82. What force must be applied to end of a rod along the *x*-axis of length 2.0 m in order to produce a torque on the rod about the origin of $8.0\hat{\mathbf{k}}\ \mathrm{N \cdot m}$?

83. What is the torque about the origin of the force $(5.0\hat{\mathbf{i}} - 2.0\hat{\mathbf{j}} + 1.0\hat{\mathbf{k}})\ \mathrm{N}$ if it is applied at the point whose position is: $\vec{\mathbf{r}} = \left(-2.0\hat{\mathbf{i}} + 4.0\hat{\mathbf{j}}\right)\ \mathrm{m}$?

10.7 Newton's Second Law for Rotation

84. You have a grindstone (a disk) that is 90.0 kg, has a 0.340-m radius, and is turning at 90.0 rpm, and you press a steel axe against it with a radial force of 20.0 N. (a) Assuming the kinetic coefficient of friction between steel and stone is 0.20, calculate the angular acceleration of the grindstone. (b) How many turns will the stone make before coming to rest?

85. Suppose you exert a force of 180 N tangential to a 0.280-m-radius, 75.0-kg grindstone (a solid disk). (a)What torque is exerted? (b) What is the angular acceleration assuming negligible opposing friction? (c) What is the angular acceleration if there is an opposing frictional force of 20.0 N exerted 1.50 cm from the axis?

86. A flywheel ($I = 50\ \mathrm{kg\text{-}m^2}$) starting from rest acquires an angular velocity of 200.0 rad/s while subject to a constant torque from a motor for 5 s. (a) What is the angular acceleration of the flywheel? (b) What is the magnitude of the torque?

87. A constant torque is applied to a rigid body whose moment of inertia is $4.0\ \mathrm{kg\text{-}m^2}$ around the axis of rotation. If the wheel starts from rest and attains an angular velocity of 20.0 rad/s in 10.0 s, what is the applied torque?

88. A torque of 50.0 N-m is applied to a grinding wheel ($I = 20.0\ \mathrm{kg\text{-}m^2}$) for 20 s. (a) If it starts from rest, what is the angular velocity of the grinding wheel after the torque is removed? (b) Through what angle does the wheel move while the torque is applied?

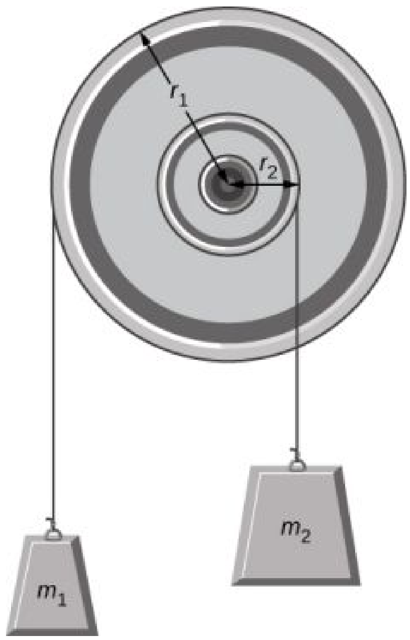
89. A flywheel ($I = 100.0\ \mathrm{kg\text{-}m^2}$) rotating at 500.0 rev/min is brought to rest by friction in 2.0 min. What is the frictional torque on the flywheel?

90. A uniform cylindrical grinding wheel of mass 50.0 kg and diameter 1.0 m is turned on by an electric motor. The friction in the bearings is negligible. (a) What torque must be applied to the wheel to bring it from rest to 120 rev/min in 20 revolutions? (b) A tool whose coefficient of kinetic friction with the wheel is 0.60 is pressed perpendicularly against the wheel with a force of 40.0 N. What torque must be supplied by the motor to keep the wheel rotating at a constant angular velocity?

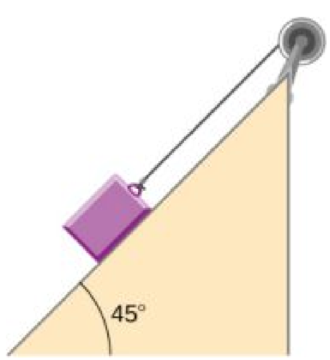
91. Suppose when Earth was created, it was not rotating. However, after the application of a uniform torque for 6 days according to the current length of a day, it was rotating at 1 rev/day. (a) What was the angular acceleration during the 6 days? (b) What torque was applied to Earth during this period? (c) What force tangent to Earth at its equator would produce

this torque?

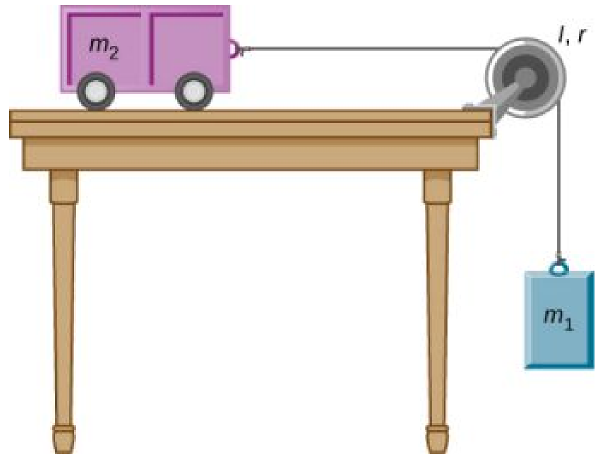
92. A pulley of moment of inertia $2.0\,\text{kg-m}^2$ is mounted on a wall as shown in the following figure. Light strings are wrapped around two circumferences of the pulley and weights are attached. What are (a) the angular acceleration of the pulley and (b) the linear acceleration of the weights? Assume the following data: $r_1 = 50\,\text{cm}, \quad r_2 = 20\,\text{cm}, \quad m_1 = 1.0\,\text{kg}, \quad m_2 = 2.0\,\text{kg}.$



93. A block of mass 3 kg slides down an inclined plane at an angle of $45°$ with a massless tether attached to a pulley with mass 1 kg and radius 0.5 m at the top of the incline (see the following figure). The pulley can be approximated as a disk. The coefficient of kinetic friction on the plane is 0.4. What is the acceleration of the block?

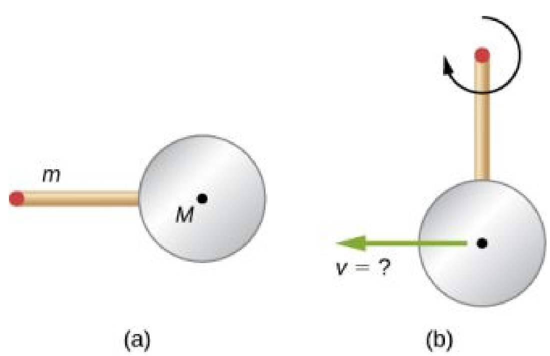


94. The cart shown below moves across the table top as the block falls. What is the acceleration of the cart? Neglect friction and assume the following data:$m_1 = 2.0\,\text{kg}, m_2 = 4.0\,\text{kg}, I = 0.4\,\text{kg-m}^2, r = 20\,\text{cm}$



95. A uniform rod of mass and length is held vertically by two strings of negligible mass, as shown below. (a) Immediately after the string is cut, what is the linear acceleration of the free end of the stick? (b) Of the middle of the stick?

96. A thin stick of mass 0.2 kg and length $L = 0.5\,\text{m}$ is attached to the rim of a metal disk of mass $M = 2.0\,\text{kg}$ and radius $R = 0.3\,\text{m}$. The stick is free to rotate around a horizontal axis through its other end (see the following figure). (a) If the combination is released with the stick horizontal, what is the speed of the center of the disk when the stick is vertical? (b) What is the acceleration of the center of the disk at the instant the stick is released? (c) At the instant the stick passes through the vertical?



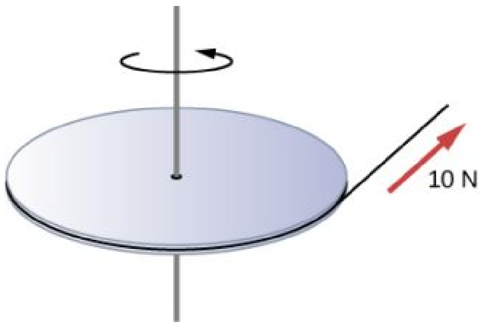
10.8 Work and Power for Rotational Motion

97. A wind turbine rotates at 20 rev/min. If its power output is 2.0 MW, what is the torque produced on the turbine from the wind?

98. A clay cylinder of radius 20 cm on a potter's wheel spins at a constant rate of 10 rev/s. The potter applies a force of 10 N to the clay with his hands where the coefficient of friction is 0.1 between his hands and the clay. What is the power that the potter has to deliver to the wheel to keep it rotating at this constant rate?

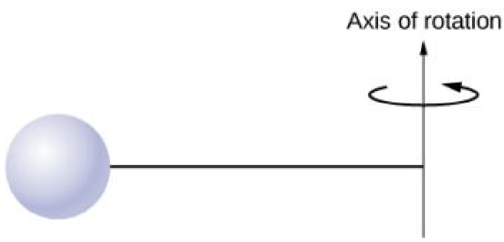
99. A uniform cylindrical grindstone has a mass of 10 kg and a radius of 12 cm. (a) What is the rotational kinetic energy of the grindstone when it is rotating at $1.5 \times 10^3 \text{rev/min}$? (b) After the grindstone's motor is turned off, a knife blade is pressed against the outer edge of the grindstone with a perpendicular force of 5.0 N. The coefficient of kinetic friction between the grindstone and the blade is 0.80. Use the work energy theorem to determine how many turns the grindstone makes before it stops.

100. A uniform disk of mass 500 kg and radius 0.25 m is mounted on frictionless bearings so it can rotate freely around a vertical axis through its center (see the following figure). A cord is wrapped around the rim of the disk and pulled with a force of 10 N. (a) How much work has the force done at the instant the disk has completed three revolutions, starting from rest? (b) Determine the torque due to the force, then calculate the work done by this torque at the instant the disk has completed three revolutions? (c) What is the angular velocity at that instant? (d) What is the power output of the force at that instant?

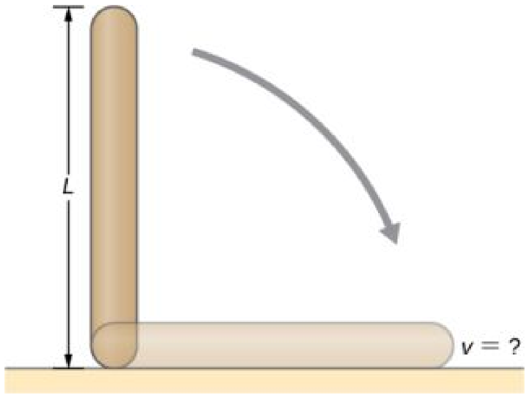


101. A propeller is accelerated from rest to an angular velocity of 1000 rev/min over a period of 6.0 seconds by a constant torque of $2.0 \times 10^3 \text{N} \cdot \text{m}$. (a) What is the moment of inertia of the propeller? (b) What power is being provided to the propeller 3.0 s after it starts rotating?

102. A sphere of mass 1.0 kg and radius 0.5 m is attached to the end of a massless rod of length 3.0 m. The rod rotates about an axis that is at the opposite end of the sphere (see below). The system rotates horizontally about the axis at a constant 400 rev/min. After rotating at this angular speed in a vacuum, air resistance is introduced and provides a force 0.15 N on the sphere opposite to the direction of motion. What is the power provided by air resistance to the system 100.0 s after air resistance is introduced?

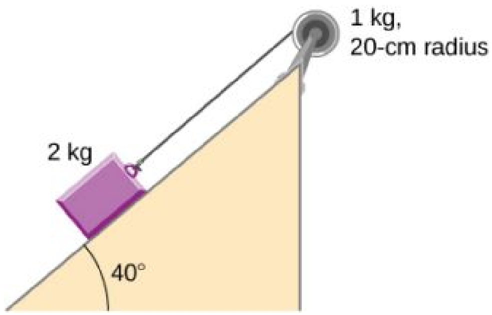


103. A uniform rod of length *L* and mass *M* is held vertically with one end resting on the floor as shown below. When the rod is released, it rotates around its lower end until it hits the floor. Assuming the lower end of the rod does not slip, what is the linear velocity of the upper end when it hits the floor?

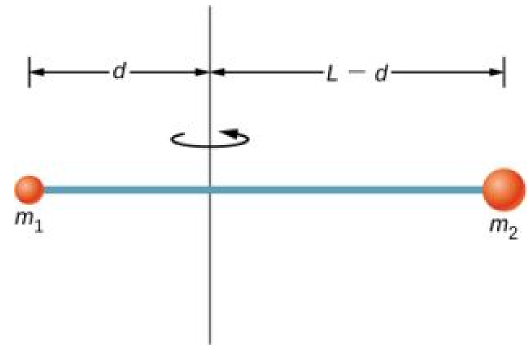


104. An athlete in a gym applies a constant force of 50 N to the pedals of a bicycle at a rate of the pedals moving 60 rev/min. The length of the pedal arms is 30 cm. What is the power delivered to the bicycle by the athlete?

105. A 2-kg block on a frictionless inclined plane at $40°$ has a cord attached to a pulley of mass 1 kg and radius 20 cm (see the following figure). The block slides a distance of 0.50 m. (a) What is the acceleration of the block down the plane? (b) What is the work done by the cord on the pulley?



106. Small bodies of mass m_1 and m_2 are attached to opposite ends of a thin rigid rod of length *L* and mass *M*. The rod is mounted so that it is free to rotate in a horizontal plane around a vertical axis (see below). What distance *d* from m_1 should the rotational axis be so that a minimum amount of work is required to set the rod rotating at an angular velocity ω?



Additional Problems

107. A cyclist is riding such that the wheels of the bicycle have a rotation rate of 3.0 rev/s. If the cyclist brakes such that the rotation rate of the wheels decrease at a rate of $0.3\ \text{rev/s}^2$, how long does it take for the cyclist to come to a complete stop?

108. Calculate the angular velocity of the orbital motion of Earth around the Sun.

109. A phonograph turntable rotating at 33 1/3 rev/min slows down and stops in 1.0 min. (a) What is the turntable's angular acceleration, in radians/s^2, assuming it is constant? (b) How many revolutions does the turntable make while stopping?

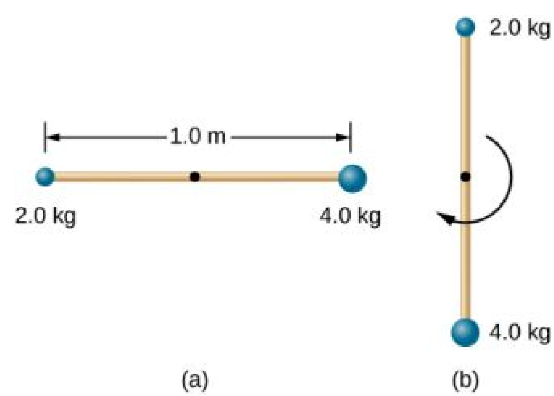
110. With the aid of a string, a gyroscope is accelerated from rest to 32 rad/s in 0.40 s under a constant angular acceleration. (a) What is its angular acceleration in rad/s^2? (b) How many revolutions does it go through in the process?

111. Suppose a piece of dust has fallen on a CD. If the spin rate of the CD is 500 rpm, and the piece of dust is 4.3 cm from the center, what is the total distance traveled by the dust in 3 minutes? (Ignore accelerations due to getting the CD rotating.)

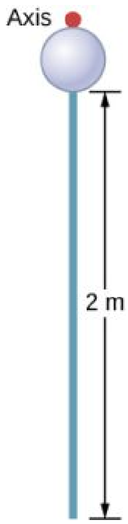
112. A system of point particles is rotating about a fixed axis at 4 rev/s. The particles are fixed with respect to each other. The masses and distances to the axis of the point particles are $m_1 = 0.1\ \text{kg}, r_1 = 0.2\ \text{m}$, $m_2 = 0.05\ \text{kg}, r_2 = 0.4\ \text{m}$, $m_3 = 0.5\ \text{kg}, r_3 = 0.01\ \text{m}$. (a) What is the moment of inertia of the system? (b) What is the rotational kinetic energy of the system?

113. Calculate the moment of inertia of a skater given the following information. (a) The 60.0-kg skater is approximated as a cylinder that has a 0.110-m radius. (b) The skater with arms extended is approximated by a cylinder that is 52.5 kg, has a 0.110-m radius, and has two 0.900-m-long arms which are 3.75 kg each and extend straight out from the cylinder like rods rotated about their ends.

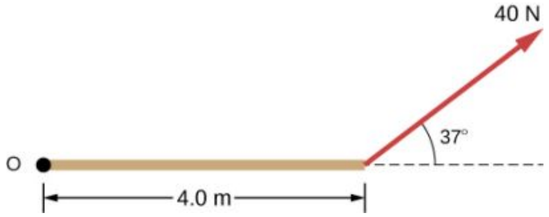
114. A stick of length 1.0 m and mass 6.0 kg is free to rotate about a horizontal axis through the center. Small bodies of masses 4.0 and 2.0 kg are attached to its two ends (see the following figure). The stick is released from the horizontal position. What is the angular velocity of the stick when it swings through the vertical?



115. A pendulum consists of a rod of length 2 m and mass 3 kg with a solid sphere of mass 1 kg and radius 0.3 m attached at one end. The axis of rotation is as shown below. What is the angular velocity of the pendulum at its lowest point if it is released from rest at an angle of $30°$?



116. Calculate the torque of the 40-N force around the axis through *O* and perpendicular to the plane of the page as shown below.



117. Two children push on opposite sides of a door during play. Both push horizontally and perpendicular to the door. One child pushes with a force of 17.5 N at a distance of 0.600 m from the hinges, and the second child pushes at a distance of 0.450 m. What force must the second child exert to keep the door from moving? Assume friction is negligible.

118. The force of $20\hat{\mathbf{j}}\,\text{N}$ is applied at $\vec{\mathbf{r}} = (4.0\hat{\mathbf{i}} - 2.0\hat{\mathbf{j}})\,\text{m}$. What is the torque of this force about the origin?

119. An automobile engine can produce 200 N· m of torque. Calculate the angular acceleration produced if 95.0% of this torque is applied to the drive shaft, axle, and rear wheels of a car, given the following information. The car is suspended so that the wheels can turn freely. Each wheel acts like a 15.0-kg disk that has a 0.180-m radius. The walls of each tire act like a 2.00-kg annular ring that has inside radius of 0.180 m and outside radius of 0.320 m. The tread of each tire acts like a 10.0-kg hoop of radius 0.330 m. The 14.0-kg axle acts like a rod that has a 2.00-cm radius. The 30.0-kg drive shaft acts like a rod that has a 3.20-cm radius.

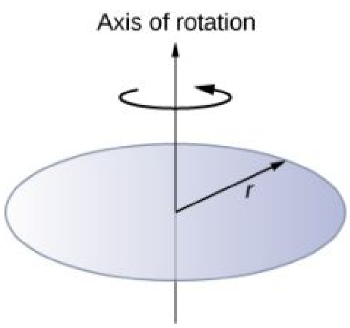
120. A grindstone with a mass of 50 kg and radius 0.8 m maintains a constant rotation rate of 4.0 rev/s by a motor while a knife is pressed against the edge with a force of 5.0 N. The coefficient of kinetic friction between the grindstone and the blade is 0.8. What is the power provided by the motor to keep the grindstone at the constant rotation rate?

Challenge Problems

121. The angular acceleration of a rotating rigid body is given by $\alpha = (2.0 - 3.0t)\ \text{rad/s}^2$. If the body starts rotating from rest at $t = 0$, (a) what is the angular velocity? (b) Angular position? (c) What angle does it rotate through in 10 s? (d) Where does the vector perpendicular to the axis of rotation indicating $0°$ at $t = 0$ lie at $t = 10\ \text{s}$?

122. Earth's day has increased by 0.002 s in the last century. If this increase in Earth's period is constant, how long will it take for Earth to come to rest?

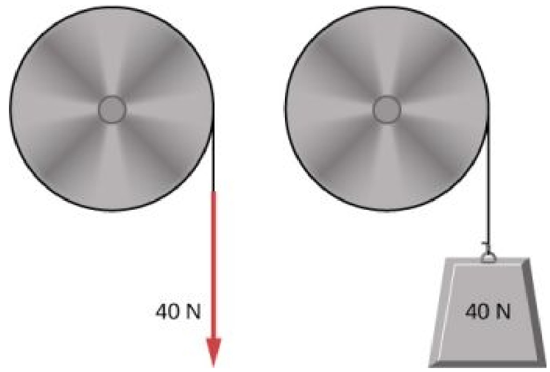
123. A disk of mass *m*, radius *R*, and area *A* has a surface mass density $\sigma = \frac{mr}{AR}$ (see the following figure). What is the moment of inertia of the disk about an axis through the center?



124. Zorch, an archenemy of Rotation Man, decides to slow Earth's rotation to once per 28.0 h by exerting an opposing force at and parallel to the equator. Rotation Man is not immediately concerned, because he knows Zorch can only exert a force of $4.00 \times 10^7\,\text{N}$ (a little greater than a Saturn V rocket's thrust). How long must

Zorch push with this force to accomplish his goal? (This period gives Rotation Man time to devote to other villains.)

125. A cord is wrapped around the rim of a solid cylinder of radius 0.25 m, and a constant force of 40 N is exerted on the cord shown, as shown in the following figure. The cylinder is mounted on frictionless bearings, and its moment of inertia is $6.0\,\text{kg} \cdot \text{m}^2$. (a) Use the work energy theorem to calculate the angular velocity of the cylinder after 5.0 m of cord have been removed. (b) If the 40-N force is replaced by a 40-N weight, what is the angular velocity of the cylinder after 5.0 m of cord have unwound?



CHAPTER 11
Angular Momentum

FIGURE 11.1 A helicopter has its main lift blades rotating to keep the aircraft airborne. Due to conservation of angular momentum, the body of the helicopter would want to rotate in the opposite sense to the blades, if it were not for the small rotor on the tail of the aircraft, which provides thrust to stabilize it.

CHAPTER OUTLINE

11.1 Rolling Motion

11.2 Angular Momentum

11.3 Conservation of Angular Momentum

11.4 Precession of a Gyroscope

INTRODUCTION Angular momentum is the rotational counterpart of linear momentum. Any massive object that rotates about an axis carries angular momentum, including rotating flywheels, planets, stars, hurricanes, tornadoes, whirlpools, and so on. The helicopter shown in the chapter-opening picture can be used to illustrate the concept of angular momentum. The lift blades spin about a vertical axis through the main body and carry angular momentum. The body of the helicopter tends to rotate in the opposite sense in order to conserve angular momentum. The small rotors at the tail of the aircraft provide a counter thrust against the body to prevent this from happening, and the helicopter stabilizes itself. The concept of conservation of angular momentum is discussed later in this chapter. In the main part of this chapter, we explore the intricacies of angular momentum of rigid bodies such as a top, and also of point particles and systems of particles. But to be complete, we start with a discussion of rolling motion, which builds upon the concepts of the previous chapter.

11.1 Rolling Motion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the physics of rolling motion without slipping
- Explain how linear variables are related to angular variables for the case of rolling motion without slipping
- Find the linear and angular accelerations in rolling motion with and without slipping
- Calculate the static friction force associated with rolling motion without slipping
- Use energy conservation to analyze rolling motion

Rolling motion is that common combination of rotational and translational motion that we see everywhere, every day. Think about the different situations of wheels moving on a car along a highway, or wheels on a plane landing on a runway, or wheels on a robotic explorer on another planet. Understanding the forces and torques involved in **rolling motion** is a crucial factor in many different types of situations.

For analyzing rolling motion in this chapter, refer to Figure 10.20 in Fixed-Axis Rotation to find moments of inertia of some common geometrical objects. You may also find it useful in other calculations involving rotation.

Rolling Motion without Slipping

People have observed rolling motion without slipping ever since the invention of the wheel. For example, we can look at the interaction of a car's tires and the surface of the road. If the driver depresses the accelerator to the floor, such that the tires spin without the car moving forward, there must be kinetic friction between the wheels and the surface of the road. If the driver depresses the accelerator slowly, causing the car to move forward, then the tires roll without slipping. It is surprising to most people that, in fact, the bottom of the wheel is at rest with respect to the ground, indicating there must be static friction between the tires and the road surface. In Figure 11.2, the bicycle is in motion with the rider staying upright. The tires have contact with the road surface, and, even though they are rolling, the bottoms of the tires deform slightly, do not slip, and are at rest with respect to the road surface for a measurable amount of time. There must be static friction between the tire and the road surface for this to be so.

FIGURE 11.2 (a) The bicycle moves forward, and its tires do not slip. The bottom of the slightly deformed tire is at rest with respect to the road surface for a measurable amount of time. (b) This image shows that the top of a rolling wheel appears blurred by its motion, but the bottom of the wheel is instantaneously at rest. (credit a: modification of work by Nelson Lourenço; credit b: modification of work by Colin Rose)

To analyze rolling without slipping, we first derive the linear variables of velocity and acceleration of the center of mass of the wheel in terms of the angular variables that describe the wheel's motion. The situation is shown in Figure 11.3.

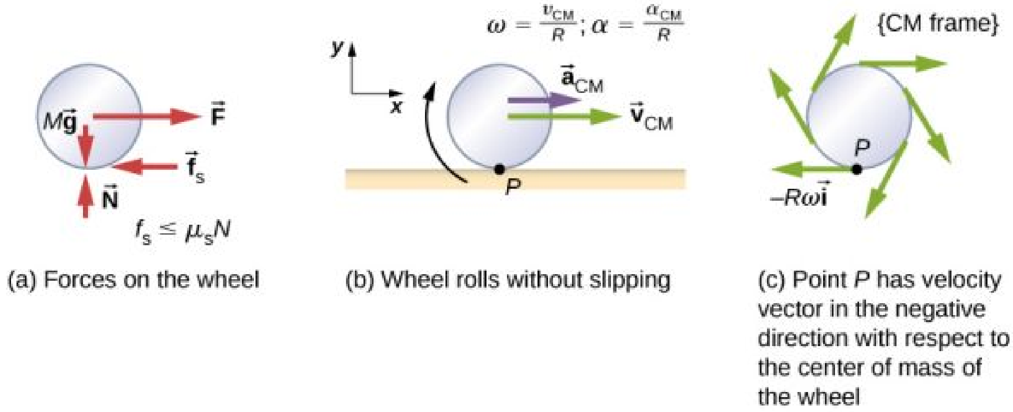


FIGURE 11.3 (a) A wheel is pulled across a horizontal surface by a force $\vec{\mathbf{F}}$. The force of static friction $\vec{\mathbf{f}}_s, \left|\vec{\mathbf{f}}_s\right| \le \mu_s N$ is large enough to keep it from slipping. (b) The linear velocity and acceleration vectors of the center of mass and the relevant expressions for ω and α. Point *P* is at rest relative to the surface. (c) Relative to the center of mass (CM) frame, point *P* has linear velocity $-R\omega\hat{\mathbf{i}}$.

From Figure 11.3(a), we see the force vectors involved in preventing the wheel from slipping. In (b), point *P* that touches the surface is at rest relative to the surface. Relative to the center of mass, point *P* has velocity $-R\omega\hat{\mathbf{i}}$, where *R* is the radius of the wheel and ω is the wheel's angular velocity about its axis. Since the wheel is rolling, the velocity of *P* with respect to the surface is its velocity with respect to the center of mass plus the velocity of the center of mass with respect to the surface:

$$\vec{\mathbf{v}}_P = -R\omega\hat{\mathbf{i}} + v_{CM}\hat{\mathbf{i}}.$$

Since the velocity of *P* relative to the surface is zero, $v_P = 0$, this says that

$$v_{CM} = R\omega. \quad (11.1)$$

Thus, the velocity of the wheel's center of mass is its radius times the angular velocity about its axis. We show the correspondence of the linear variable on the left side of the equation with the angular variable on the right side of the equation. This is done below for the linear acceleration.

If we differentiate Equation 11.1 on the left side of the equation, we obtain an expression for the linear acceleration of the center of mass. On the right side of the equation, *R* is a constant and since $\alpha = \frac{d\omega}{dt}$, we have

$$a_{CM} = R\alpha. \quad (11.2)$$

Furthermore, we can find the distance the wheel travels in terms of angular variables by referring to Figure 11.4. As the wheel rolls from point *A* to point *B*, its outer surface maps onto the ground by exactly the distance travelled, which is d_{CM}. We see from Figure 11.4 that the length of the outer surface that maps onto the ground is the arc length $R\theta$. Equating the two distances, we obtain

$$d_{CM} = R\theta. \quad (11.3)$$

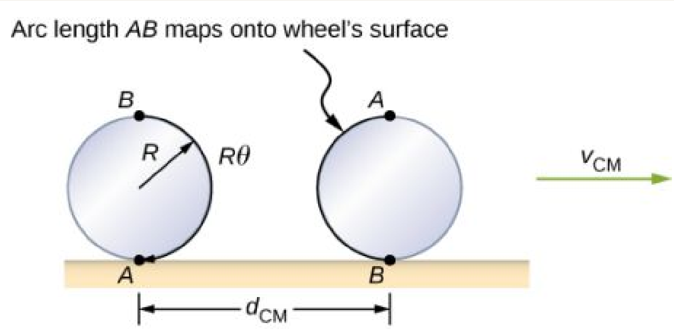


FIGURE 11.4 As the wheel rolls on the surface, the arc length $R\theta$ from *A* to *B* maps onto the surface, corresponding to the distance d_{CM}

that the center of mass has moved.

EXAMPLE 11.1

Rolling Down an Inclined Plane

A solid cylinder rolls down an inclined plane without slipping, starting from rest. It has mass *m* and radius *r*. (a) What is its acceleration? (b) What condition must the coefficient of static friction μ_S satisfy so the cylinder does not slip?

Strategy

Draw a sketch and free-body diagram, and choose a coordinate system. We put *x* in the direction down the plane and *y* upward perpendicular to the plane. Identify the forces involved. These are the normal force, the force of gravity, and the force due to friction. Write down Newton's laws in the *x*- and *y*-directions, and Newton's law for rotation, and then solve for the acceleration and force due to friction.

Solution

a. The free-body diagram and sketch are shown in Figure 11.5, including the normal force, components of the weight, and the static friction force. There is barely enough friction to keep the cylinder rolling without slipping. Since there is no slipping, the magnitude of the friction force is less than or equal to $\mu_S N$. Writing down Newton's laws in the *x*- and *y*-directions, we have

$$\sum F_x = ma_x; \quad \sum F_y = ma_y.$$

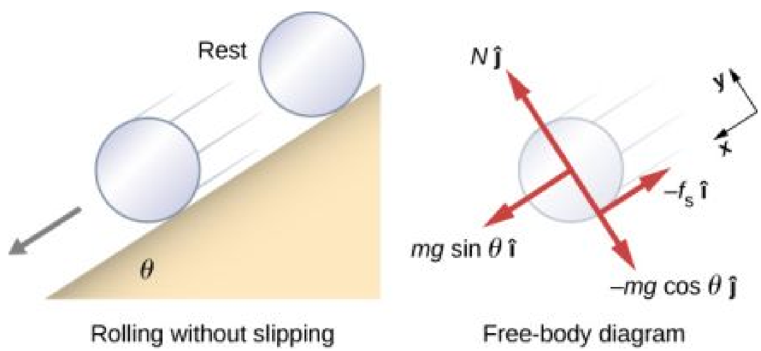


FIGURE 11.5 A solid cylinder rolls down an inclined plane without slipping from rest. The coordinate system has *x* in the direction down the inclined plane and *y* perpendicular to the plane. The free-body diagram is shown with the normal force, the static friction force, and the components of the weight $m\vec{g}$. Friction makes the cylinder roll down the plane rather than slip.

Substituting in from the free-body diagram,

$$\begin{aligned} mg\sin\theta - f_S &= m(a_{CM})_x, \\ N - mg\cos\theta &= 0 \end{aligned}$$

we can then solve for the linear acceleration of the center of mass from these equations:

$$a_{CM} = g\sin\theta - \frac{f_S}{m}$$

However, it is useful to express the linear acceleration in terms of the moment of inertia. For this, we write down Newton's second law for rotation,

$$\sum \tau_{CM} = I_{CM}\alpha.$$

The torques are calculated about the axis through the center of mass of the cylinder. The only nonzero torque is provided by the friction force. We have

$$f_S r = I_{CM}\alpha.$$

Finally, the linear acceleration is related to the angular acceleration by

$$(a_{\text{CM}})_x = r\alpha.$$

These equations can be used to solve for $a_{\text{CM}}, \alpha,$ and f_{S} in terms of the moment of inertia, where we have dropped the *x*-subscript. We rewrite a_{CM} in terms of the vertical component of gravity and the friction force, and make the following substitutions.

$$f_{\text{S}} = \frac{I_{\text{CM}}\alpha}{r} = \frac{I_{\text{CM}}a_{\text{CM}}}{r^2}$$

From this we obtain

$$\begin{aligned} a_{\text{CM}} &= g\sin\theta - \frac{I_{\text{CM}}a_{\text{CM}}}{mr^2}, \\ &= \frac{mg\sin\theta}{m+(I_{\text{CM}}/r^2)}. \end{aligned}$$

Note that this result is independent of the coefficient of static friction, μ_{S}.

Since we have a solid cylinder, from Figure 10.20, we have $I_{\text{CM}} = mr^2/2$ and

$$a_{\text{CM}} = \frac{mg\sin\theta}{m+(mr^2/2r^2)} = \frac{2}{3}g\sin\theta.$$

Therefore, we have

$$\alpha = \frac{a_{\text{CM}}}{r} = \frac{2}{3r}g\sin\theta.$$

b. Because slipping does not occur, $f_{\text{S}} \le \mu_{\text{S}}N$. Solving for the friction force,

$$f_{\text{S}} = I_{\text{CM}}\frac{\alpha}{r} = I_{\text{CM}}\frac{(a_{\text{CM}})}{r^2} = \frac{I_{\text{CM}}}{r^2}\left(\frac{mg\sin\theta}{m+(I_{\text{CM}}/r^2)}\right) = \frac{mgI_{\text{CM}}\sin\theta}{mr^2+I_{\text{CM}}}.$$

Substituting this expression into the condition for no slipping, and noting that $N = mg\cos\theta$, we have

$$\frac{mgI_{\text{CM}}\sin\theta}{mr^2+I_{\text{CM}}} \le \mu_{\text{S}}mg\cos\theta$$

or

$$\mu_{\text{S}} \ge \frac{\tan\theta}{1+(mr^2/I_{\text{CM}})}.$$

For the solid cylinder, this becomes

$$\mu_{\text{S}} \ge \frac{\tan\theta}{1+(2mr^2/mr^2)} = \frac{1}{3}\tan\theta.$$

Significance

a. The linear acceleration is linearly proportional to $\sin\theta$. Thus, the greater the angle of the incline, the greater the linear acceleration, as would be expected. The angular acceleration, however, is linearly proportional to $\sin\theta$ and inversely proportional to the radius of the cylinder. Thus, the larger the radius, the smaller the angular acceleration.
b. For no slipping to occur, the coefficient of static friction must be greater than or equal to $(1/3)\tan\theta$. Thus, the greater the angle of incline, the greater the coefficient of static friction must be to prevent the cylinder from slipping.

CHECK YOUR UNDERSTANDING 11.1

A hollow cylinder is on an incline at an angle of $60°$. The coefficient of static friction on the surface is $\mu_s = 0.6$. (a) Does the cylinder roll without slipping? (b) Will a solid cylinder roll without slipping?

It is worthwhile to repeat the equation derived in this example for the acceleration of an object rolling without slipping:

$$a_{\text{CM}} = \frac{mg \sin \theta}{m + (I_{\text{CM}}/r^2)}. \quad (11.4)$$

This is a very useful equation for solving problems involving rolling without slipping. Note that the acceleration is less than that of an object sliding down a frictionless plane with no rotation. The acceleration will also be different for two rotating objects with different rotational inertias.

Rolling Motion with Slipping

In the case of rolling motion with slipping, we must use the coefficient of kinetic friction, which gives rise to the kinetic friction force since static friction is not present. The situation is shown in Figure 11.6. In the case of slipping, $v_{\text{CM}} - R\omega \neq 0$, because point *P* on the wheel is not at rest on the surface, and $v_P \neq 0$. Thus, $\omega \neq \frac{v_{\text{CM}}}{R}, \alpha \neq \frac{a_{\text{CM}}}{R}$.

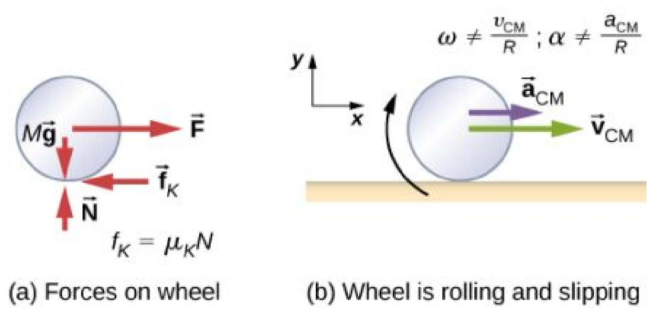


FIGURE 11.6 (a) Kinetic friction arises between the wheel and the surface because the wheel is slipping. (b) The simple relationships between the linear and angular variables are no longer valid.

EXAMPLE 11.2

Rolling Down an Inclined Plane with Slipping

A solid cylinder rolls down an inclined plane from rest and undergoes slipping (Figure 11.7). It has mass *m* and radius *r*. (a) What is its linear acceleration? (b) What is its angular acceleration about an axis through the center of mass?

Strategy

Draw a sketch and free-body diagram showing the forces involved. The free-body diagram is similar to the no-slipping case except for the friction force, which is kinetic instead of static. Use Newton's second law to solve for the acceleration in the *x*-direction. Use Newton's second law of rotation to solve for the angular acceleration.

Solution

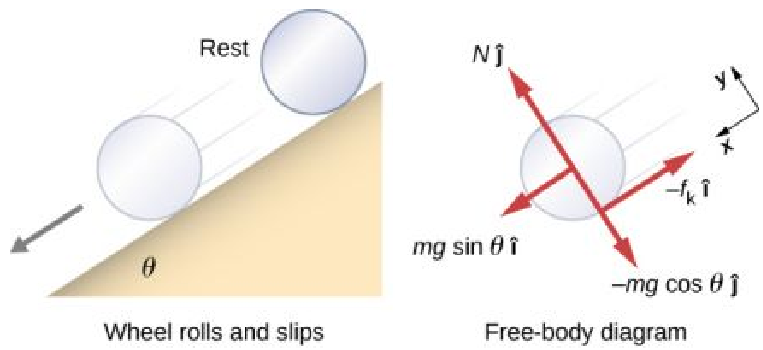


FIGURE 11.7 A solid cylinder rolls down an inclined plane from rest and undergoes slipping. The coordinate system has *x* in the direction down the inclined plane and *y* upward perpendicular to the plane. The free-body diagram shows the normal force, kinetic friction force, and the components of the weight $m\vec{g}$.

The sum of the forces in the *y*-direction is zero, so the friction force is now $f_k = \mu_k N = \mu_k mg\cos\theta$.

Newton's second law in the *x*-direction becomes

$$\sum F_x = ma_x,$$

$$mg\sin\theta - \mu_k mg\cos\theta = m(a_{CM})_x,$$

or

$$(a_{CM})_x = g(\sin\theta - \mu_k \cos\theta).$$

The friction force provides the only torque about the axis through the center of mass, so Newton's second law of rotation becomes

$$\sum \tau_{CM} = I_{CM}\alpha,$$

$$f_k r = I_{CM}\alpha = \frac{1}{2}mr^2\alpha.$$

Solving for α, we have

$$\alpha = \frac{2f_k}{mr} = \frac{2\mu_k g\cos\theta}{r}.$$

Significance

We write the linear and angular accelerations in terms of the coefficient of kinetic friction. The linear acceleration is the same as that found for an object sliding down an inclined plane with kinetic friction. The angular acceleration about the axis of rotation is linearly proportional to the normal force, which depends on the cosine of the angle of inclination. As $\theta \to 90°$, this force goes to zero, and, thus, the angular acceleration goes to zero.

Conservation of Mechanical Energy in Rolling Motion

In the preceding chapter, we introduced rotational kinetic energy. Any rolling object carries rotational kinetic energy, as well as translational kinetic energy and potential energy if the system requires. Including the gravitational potential energy, the total mechanical energy of an object rolling is

$$E_T = \frac{1}{2}mv_{CM}^2 + \frac{1}{2}I_{CM}\omega^2 + mgh.$$

In the absence of any nonconservative forces that would take energy out of the system in the form of heat, the total energy of a rolling object without slipping is conserved and is constant throughout the motion. Examples where energy is not conserved are a rolling object that is slipping, production of heat as a result of kinetic friction, and a rolling object encountering air resistance.

You may ask why a rolling object that is not slipping conserves energy, since the static friction force is nonconservative. The answer can be found by referring back to Figure 11.3. Point *P* in contact with the surface is at rest with respect to the surface. Therefore, its infinitesimal displacement $d\vec{\mathbf{r}}$ with respect to the surface is zero, and the incremental work done by the static friction force is zero. We can apply energy conservation to our study of rolling motion to bring out some interesting results.

EXAMPLE 11.3

Curiosity Rover

The *Curiosity* rover, shown in Figure 11.8, was deployed on Mars on August 6, 2012. The wheels of the rover have a radius of 25 cm. Suppose astronauts arrive on Mars in the year 2050 and find the now-inoperative *Curiosity* on the side of a basin. While they are dismantling the rover, an astronaut accidentally loses a grip on one of the wheels, which rolls without slipping down into the bottom of the basin 25 meters below. If the wheel has a mass of 5 kg, what is its velocity at the bottom of the basin?

FIGURE 11.8 The NASA Mars Science Laboratory rover *Curiosity* during testing on June 3, 2011. The location is inside the Spacecraft Assembly Facility at NASA's Jet Propulsion Laboratory in Pasadena, California. (credit: NASA/JPL-Caltech)

Strategy

We use mechanical energy conservation to analyze the problem. At the top of the hill, the wheel is at rest and has only potential energy. At the bottom of the basin, the wheel has rotational and translational kinetic energy, which must be equal to the initial potential energy by energy conservation. Since the wheel is rolling without slipping, we use the relation $v_{\text{CM}} = r\omega$ to relate the translational variables to the rotational variables in the energy conservation equation. We then solve for the velocity. From Figure 11.8, we see that a hollow cylinder is a good approximation for the wheel, so we can use this moment of inertia to simplify the calculation.

Solution

Energy at the top of the basin equals energy at the bottom:

$$mgh = \frac{1}{2}mv_{\text{CM}}^2 + \frac{1}{2}I_{\text{CM}}\omega^2.$$

The known quantities are $I_{\text{CM}} = mr^2$, $r = 0.25\text{ m}$, and $h = 25.0\text{ m}$.

We rewrite the energy conservation equation eliminating ω by using $\omega = \frac{v_{\text{CM}}}{r}$. We have

$$mgh = \frac{1}{2}mv_{\text{CM}}^2 + \frac{1}{2}mr^2\frac{v_{\text{CM}}^2}{r^2}$$

or

$$gh = \frac{1}{2}v_{\text{CM}}^2 + \frac{1}{2}v_{\text{CM}}^2 \Rightarrow v_{\text{CM}} = \sqrt{gh}.$$

On Mars, the acceleration of gravity is $3.71\ \text{m/s}^2$, which gives the magnitude of the velocity at the bottom of the basin as

$$v_{\text{CM}} = \sqrt{(3.71\text{ m/s}^2)25.0\text{ m}} = 9.63\text{ m/s}.$$

Significance

This is a fairly accurate result considering that Mars has very little atmosphere, and the loss of energy due to air resistance would be minimal. The result also assumes that the terrain is smooth, such that the wheel wouldn't encounter rocks and bumps along the way.

Also, in this example, the kinetic energy, or energy of motion, is equally shared between linear and rotational motion. If we look at the moments of inertia in Figure 10.20, we see that the hollow cylinder has the largest moment of inertia for a given radius and mass. If the wheels of the rover were solid and approximated by solid cylinders, for example, there would be more kinetic energy in linear motion than in rotational motion. This would give the wheel a larger linear velocity than the hollow cylinder approximation. Thus, the solid cylinder would reach the bottom of the basin faster than the hollow cylinder.

11.2 Angular Momentum

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the vector nature of angular momentum
- Find the total angular momentum and torque about a designated origin of a system of particles
- Calculate the angular momentum of a rigid body rotating about a fixed axis
- Calculate the torque on a rigid body rotating about a fixed axis
- Use conservation of angular momentum in the analysis of objects that change their rotation rate

Why does Earth keep on spinning? What started it spinning to begin with? Why doesn't Earth's gravitational attraction not bring the Moon crashing in toward Earth? And how does an ice skater manage to spin faster and faster simply by pulling her arms in? Why does she not have to exert a torque to spin faster?

Questions like these have answers based in angular momentum, the rotational analog to linear momentum. In this chapter, we first define and then explore angular momentum from a variety of viewpoints. First, however, we investigate the angular momentum of a single particle. This allows us to develop angular momentum for a system of particles and for a rigid body that is cylindrically symmetric.

Angular Momentum of a Single Particle

Figure 11.9 shows a particle at a position $\vec{\mathbf{r}}$ with linear momentum $\vec{\mathbf{p}} = m\vec{\mathbf{v}}$ with respect to the origin. Even if the particle is not rotating about the origin, we can still define an angular momentum in terms of the position vector and the linear momentum.

ANGULAR MOMENTUM OF A PARTICLE

The **angular momentum** $\vec{\mathbf{l}}$ of a particle is defined as the cross-product of $\vec{\mathbf{r}}$ and $\vec{\mathbf{p}}$, and is perpendicular to the plane containing $\vec{\mathbf{r}}$ and $\vec{\mathbf{p}}$:

$$\vec{\mathbf{l}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}. \tag{11.5}$$

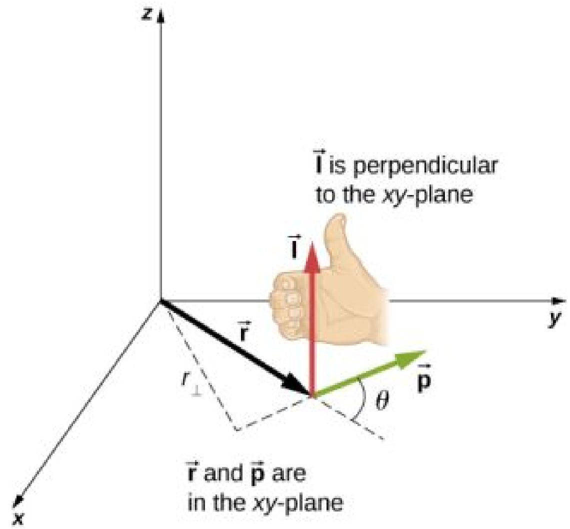


FIGURE 11.9 In three-dimensional space, the position vector $\vec{\mathbf{r}}$ locates a particle in the *xy*-plane with linear momentum $\vec{\mathbf{p}}$. The angular momentum with respect to the origin is $\vec{\mathbf{l}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$, which is in the *z*-direction. The direction of $\vec{\mathbf{l}}$ is given by the right-hand rule, as shown.

The intent of choosing the direction of the angular momentum to be perpendicular to the plane containing $\vec{\mathbf{r}}$ and $\vec{\mathbf{p}}$ is similar to choosing the direction of torque to be perpendicular to the plane of $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$, as discussed in Fixed-Axis Rotation. The magnitude of the angular momentum is found from the definition of the cross-product,

$$l = rp\sin\theta,$$

where θ is the angle between $\vec{\mathbf{r}}$ and $\vec{\mathbf{p}}$. The units of angular momentum are $\mathrm{kg \cdot m^2/s}$.

As with the definition of torque, we can define a lever arm $r_\perp$ that is the perpendicular distance from the momentum vector $\vec{\mathbf{p}}$ to the origin, $r_\perp = r\sin\theta$. With this definition, the magnitude of the angular momentum becomes

$$l = r_\perp p = r_\perp mv.$$

We see that if the direction of $\vec{\mathbf{p}}$ is such that it passes through the origin, then $\theta = 0$, and the angular momentum is zero because the lever arm is zero. In this respect, the magnitude of the angular momentum depends on the choice of origin.

If we take the time derivative of the angular momentum, we arrive at an expression for the torque on the particle:

$$\frac{d\vec{\mathbf{l}}}{dt} = \frac{d\vec{\mathbf{r}}}{dt} \times \vec{\mathbf{p}} + \vec{\mathbf{r}} \times \frac{d\vec{\mathbf{p}}}{dt} = \vec{\mathbf{v}} \times m\vec{\mathbf{v}} + \vec{\mathbf{r}} \times \frac{d\vec{\mathbf{p}}}{dt} = \vec{\mathbf{r}} \times \frac{d\vec{\mathbf{p}}}{dt}.$$

Here we have used the definition of $\vec{\mathbf{p}}$ and the fact that a vector crossed into itself is zero. From Newton's second law, $\frac{d\vec{\mathbf{p}}}{dt} = \sum \vec{\mathbf{F}}$, the net force acting on the particle, and the definition of the net torque, we can write

$$\frac{d\vec{\mathbf{l}}}{dt} = \sum \vec{\boldsymbol{\tau}}. \tag{11.6}$$

Note the similarity with the linear result of Newton's second law, $\frac{d\vec{\mathbf{p}}}{dt} = \sum \vec{\mathbf{F}}$. The following problem-solving strategy can serve as a guideline for calculating the angular momentum of a particle.

PROBLEM-SOLVING STRATEGY

Angular Momentum of a Particle

1. Choose a coordinate system about which the angular momentum is to be calculated.
2. Write down the radius vector to the point particle in unit vector notation.
3. Write the linear momentum vector of the particle in unit vector notation.
4. Take the cross product $\vec{\mathbf{l}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$ and use the right-hand rule to establish the direction of the angular momentum vector.
5. See if there is a time dependence in the expression of the angular momentum vector. If there is, then a torque exists about the origin, and use $\frac{d\vec{\mathbf{l}}}{dt} = \sum \vec{\boldsymbol{\tau}}$ to calculate the torque. If there is no time dependence in the expression for the angular momentum, then the net torque is zero.

EXAMPLE 11.4

Angular Momentum and Torque on a Meteor

A meteor enters Earth's atmosphere (Figure 11.10) and is observed by someone on the ground before it burns up in the atmosphere. The vector $\vec{\mathbf{r}} = 25\text{ km}\hat{\mathbf{i}} + 25\text{ km}\hat{\mathbf{j}}$ gives the position of the meteor with respect to the observer. At the instant the observer sees the meteor, it has linear momentum $\vec{\mathbf{p}} = 15.0\text{ kg}(-2.0\text{km/s}\hat{\mathbf{j}})$, and it is accelerating at a constant $2.0\text{ m/s}^2(-\hat{\mathbf{j}})$ along its path, which for our purposes can be taken as a straight line. (a) What is the angular momentum of the meteor about the origin, which is at the location of the observer? (b) What is the torque on the meteor about the origin?

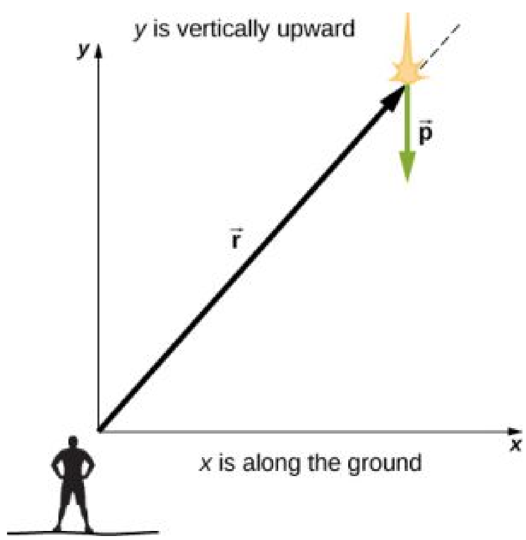


FIGURE 11.10 An observer on the ground sees a meteor at position $\vec{\mathbf{r}}$ with linear momentum $\vec{\mathbf{p}}$.

Strategy

We resolve the acceleration into *x*- and *y*-components and use the kinematic equations to express the velocity as a function of acceleration and time. We insert these expressions into the linear momentum and then calculate the angular momentum using the cross-product. Since the position and momentum vectors are in the *xy*-plane, we expect the angular momentum vector to be along the *z*-axis. To find the torque, we take the time derivative of the angular momentum.

Solution

The meteor is entering Earth's atmosphere at an angle of $90.0°$ below the horizontal, so the components of the acceleration in the *x*- and *y*-directions are

$$a_x = 0, \quad a_y = -2.0\,\text{m/s}^2.$$

We write the velocities using the kinematic equations.

$$v_x = 0, \quad v_y = -2.0 \times 10^3\ \text{m/s} - (2.0\,\text{m/s}^2)t.$$

a. The angular momentum is

$$\begin{aligned}\vec{\mathbf{l}} &= \vec{\mathbf{r}} \times \vec{\mathbf{p}} = (25.0\,\text{km}\hat{\mathbf{i}} + 25.0\,\text{km}\hat{\mathbf{j}}) \times 15.0\,\text{kg}(0\hat{\mathbf{i}} + v_y\hat{\mathbf{j}}) \\ &= 15.0\,\text{kg}[25.0\,\text{km}(v_y)\hat{\mathbf{k}}] \\ &= 15.0\,\text{kg}[2.50 \times 10^4\ \text{m}(-2.0 \times 10^3\ \text{m/s} - (2.0\,\text{m/s}^2)t)\hat{\mathbf{k}}].\end{aligned}$$

At $t = 0$, the angular momentum of the meteor about the origin is

$$\vec{\mathbf{l}}_0 = 15.0\,\text{kg}[2.50 \times 10^4\ \text{m}(-2.0 \times 10^3\ \text{m/s})\hat{\mathbf{k}}] = 7.50 \times 10^8\ \text{kg} \cdot \text{m}^2/\text{s}(-\hat{\mathbf{k}}).$$

This is the instant that the observer sees the meteor.

b. To find the torque, we take the time derivative of the angular momentum. Taking the time derivative of $\vec{\mathbf{l}}$ as a function of time, which is the second equation immediately above, we have

$$\frac{d\vec{\mathbf{l}}}{dt} = -15.0\,\text{kg}(2.50 \times 10^4\ \text{m})(2.0\,\text{m/s}^2)\hat{\mathbf{k}}.$$

Then, since $\frac{d\vec{\mathbf{l}}}{dt} = \sum \vec{\boldsymbol{\tau}}$, we have

$$\sum \vec{\boldsymbol{\tau}} = -7.5 \times 10^5\text{N} \cdot \text{m}\hat{\mathbf{k}}.$$

The units of torque are given as newton-meters, not to be confused with joules. As a check, we note that the lever arm is the *x*-component of the vector $\vec{\mathbf{r}}$ in Figure 11.10 since it is perpendicular to the force acting on the meteor, which is along its path. By Newton's second law, this force is

$$\vec{\mathbf{F}} = ma(-\hat{\mathbf{j}}) = 15.0\,\text{kg}(2.0\,\text{m/s}^2)(-\hat{\mathbf{j}}) = 30.0\,\text{kg} \cdot \text{m/s}^2(-\hat{\mathbf{j}}).$$

The lever arm is

$$\vec{\mathbf{r}}_\perp = 2.5 \times 10^4\ \text{m}\,\hat{\mathbf{i}}.$$

Thus, the torque is

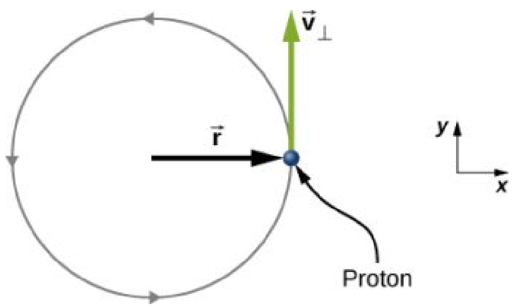
$$\begin{aligned}\sum \vec{\boldsymbol{\tau}} = \vec{\mathbf{r}}_\perp \times \vec{\mathbf{F}} &= (2.5 \times 10^4\ \text{m}\,\hat{\mathbf{i}}) \times (-30.0\,\text{kg} \cdot \text{m/s}^2\hat{\mathbf{j}}), \\ &= 7.5 \times 10^5\ \text{N} \cdot \text{m}(-\hat{\mathbf{k}}).\end{aligned}$$

Significance

Since the meteor is accelerating downward toward Earth, its radius and velocity vector are changing. Therefore, since $\vec{\mathbf{l}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$, the angular momentum is changing as a function of time. The torque on the meteor about the origin, however, is constant, because the lever arm $\vec{\mathbf{r}}_\perp$ and the force on the meteor are constants. This example is important in that it illustrates that the angular momentum depends on the choice of origin about which it is calculated. The methods used in this example are also important in developing angular momentum for a system of particles and for a rigid body.

CHECK YOUR UNDERSTANDING 11.2

A proton spiraling around a magnetic field executes circular motion in the plane of the paper, as shown below. The circular path has a radius of 0.4 m and the proton has velocity 4.0×10^6 m/s. What is the angular momentum of the proton about the origin?



Angular Momentum of a System of Particles

The angular momentum of a system of particles is important in many scientific disciplines, one being astronomy. Consider a spiral galaxy, a rotating island of stars like our own Milky Way. The individual stars can be treated as point particles, each of which has its own angular momentum. The vector sum of the individual angular momenta give the total angular momentum of the galaxy. In this section, we develop the tools with which we can calculate the total angular momentum of a system of particles.

In the preceding section, we introduced the angular momentum of a single particle about a designated origin. The expression for this angular momentum is $\vec{\mathbf{l}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$, where the vector $\vec{\mathbf{r}}$ is from the origin to the particle, and $\vec{\mathbf{p}}$ is the particle's linear momentum. If we have a system of *N* particles, each with position vector from the origin given by $\vec{\mathbf{r}}_i$ and each having momentum $\vec{\mathbf{p}}_i$, then the total angular momentum of the system of particles about the origin is the vector sum of the individual angular momenta about the origin. That is,

$$\vec{\mathbf{L}} = \vec{\mathbf{l}}_1 + \vec{\mathbf{l}}_2 + \cdots + \vec{\mathbf{l}}_N. \quad (11.7)$$

Similarly, if particle *i* is subject to a net torque $\vec{\tau}_i$ about the origin, then we can find the net torque about the origin due to the system of particles by differentiating Equation 11.7:

$$\frac{d\vec{\mathbf{L}}}{dt} = \sum_i \frac{d\vec{\mathbf{l}}_i}{dt} = \sum_i \vec{\boldsymbol{\tau}}_i.$$

The sum of the individual torques produces a net external torque on the system, which we designate $\sum \vec{\boldsymbol{\tau}}$. Thus,

$$\frac{d\vec{\mathbf{L}}}{dt} = \sum \vec{\boldsymbol{\tau}}. \quad (11.8)$$

Equation 11.8 states that *the rate of change of the total angular momentum of a system is equal to the net external torque acting on the system when both quantities are measured with respect to a given origin.* Equation 11.8 can be applied to any system that has net angular momentum, including rigid bodies, as discussed in the next section.

EXAMPLE 11.5

Angular Momentum of Three Particles

Referring to Figure 11.11(a), determine the total angular momentum due to the three particles about the origin. (b) What is the rate of change of the angular momentum?

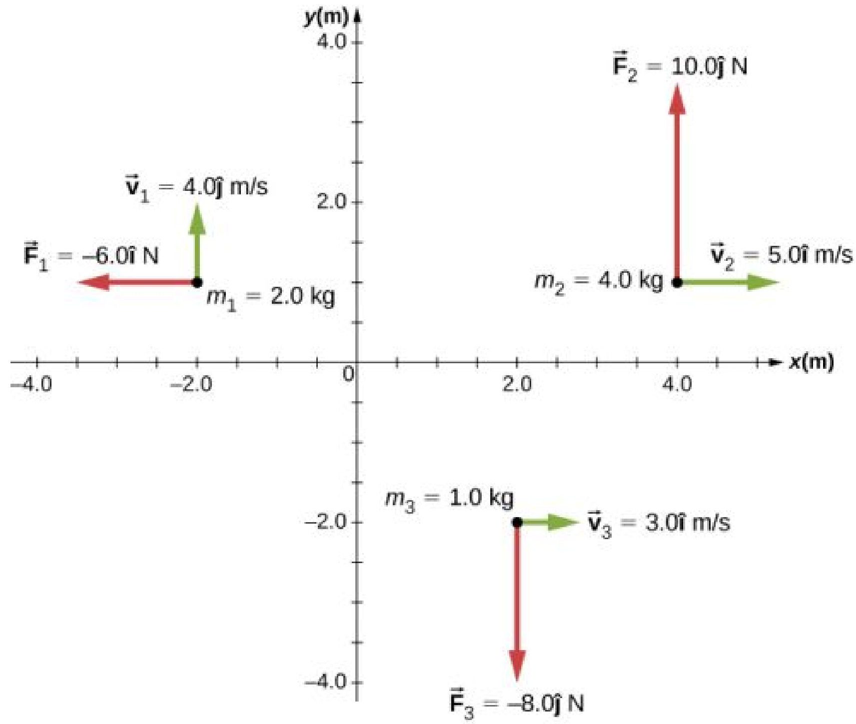


FIGURE 11.11 Three particles in the *xy*-plane with different position and momentum vectors.

Strategy

Write down the position and momentum vectors for the three particles. Calculate the individual angular momenta and add them as vectors to find the total angular momentum. Then do the same for the torques.

Solution

a. Particle 1: $\vec{\mathbf{r}}_1 = -2.0\,\mathrm{m}\hat{\mathbf{i}} + 1.0\,\mathrm{m}\hat{\mathbf{j}},\ \ \vec{\mathbf{p}}_1 = 2.0\,\mathrm{kg}(4.0\,\mathrm{m/s}\hat{\mathbf{j}}) = 8.0\,\mathrm{kg}\cdot\mathrm{m/s}\hat{\mathbf{j}},$

$$\vec{\mathbf{l}}_1 = \vec{\mathbf{r}}_1 \times \vec{\mathbf{p}}_1 = -16.0\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}\hat{\mathbf{k}}.$$

Particle 2: $\vec{\mathbf{r}}_2 = 4.0\,\mathrm{m}\hat{\mathbf{i}} + 1.0\,\mathrm{m}\hat{\mathbf{j}},\ \ \vec{\mathbf{p}}_2 = 4.0\,\mathrm{kg}(5.0\,\mathrm{m/s}\hat{\mathbf{i}}) = 20.0\,\mathrm{kg}\cdot\mathrm{m/s}\hat{\mathbf{i}},$

$$\vec{\mathbf{l}}_2 = \vec{\mathbf{r}}_2 \times \vec{\mathbf{p}}_2 = -20.0\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}\hat{\mathbf{k}}.$$

Particle 3: $\vec{\mathbf{r}}_3 = 2.0\,\mathrm{m}\hat{\mathbf{i}} - 2.0\,\mathrm{m}\hat{\mathbf{j}},\ \ \vec{\mathbf{p}}_3 = 1.0\,\mathrm{kg}(3.0\,\mathrm{m/s}\hat{\mathbf{i}}) = 3.0\,\mathrm{kg}\cdot\mathrm{m/s}\hat{\mathbf{i}},$

$$\vec{\mathbf{l}}_3 = \vec{\mathbf{r}}_3 \times \vec{\mathbf{p}}_3 = 6.0\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}\hat{\mathbf{k}}.$$

We add the individual angular momenta to find the total about the origin:

$$\vec{\mathbf{l}}_T = \vec{\mathbf{l}}_1 + \vec{\mathbf{l}}_2 + \vec{\mathbf{l}}_3 = -30\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}\hat{\mathbf{k}}.$$

b. The individual forces and lever arms are

$$\vec{\mathbf{r}}_{1\perp} = 1.0\,\mathrm{m}\hat{\mathbf{j}},\ \ \vec{\mathbf{F}}_1 = -6.0\,\mathrm{N}\hat{\mathbf{i}},\ \ \vec{\boldsymbol{\tau}}_1 = 6.0\mathbf{N\cdot m}\hat{\mathbf{k}}$$
$$\vec{\mathbf{r}}_{2\perp} = 4.0\,\mathrm{m}\hat{\mathbf{i}},\ \ \vec{\mathbf{F}}_2 = 10.0\,\mathrm{N}\hat{\mathbf{j}},\ \ \vec{\tau}_2 = 40.0\,\mathrm{N}\cdot\mathrm{m}\hat{\mathbf{k}}$$
$$\vec{\mathbf{r}}_{3\perp} = 2.0\,\mathrm{m}\hat{\mathbf{i}},\ \ \vec{\mathbf{F}}_3 = -8.0\,\mathrm{N}\hat{\mathbf{j}},\ \ \vec{\tau}_3 = -16.0\,\mathrm{N}\cdot\mathrm{m}\hat{\mathbf{k}}.$$

Therefore:

$$\sum_i \vec{\boldsymbol{\tau}}_i = \vec{\boldsymbol{\tau}}_1 + \vec{\boldsymbol{\tau}}_2 + \vec{\boldsymbol{\tau}}_3 = 30\,\mathrm{N}\cdot\mathrm{m}\hat{\mathbf{k}}.$$

Significance

This example illustrates the superposition principle for angular momentum and torque of a system of particles. Care must be taken when evaluating the radius vectors $\vec{\mathbf{r}}_i$ of the particles to calculate the angular momenta, and the lever arms, $\vec{\mathbf{r}}_{i\perp}$ to calculate the torques, as they are completely different quantities.

Angular Momentum of a Rigid Body

We have investigated the angular momentum of a single particle, which we generalized to a system of particles. Now we can use the principles discussed in the previous section to develop the concept of the angular momentum of a rigid body. Celestial objects such as planets have angular momentum due to their spin and orbits around stars. In engineering, anything that rotates about an axis carries angular momentum, such as flywheels, propellers, and rotating parts in engines. Knowledge of the angular momenta of these objects is crucial to the design of the system in which they are a part.

To develop the angular momentum of a rigid body, we model a rigid body as being made up of small mass segments, Δm_i. In Figure 11.12, a rigid body is constrained to rotate about the *z*-axis with angular velocity ω. All mass segments that make up the rigid body undergo circular motion about the *z*-axis with the same angular velocity. Part (a) of the figure shows mass segment Δm_i with position vector $\vec{\mathbf{r}}_i$ from the origin and radius R_i to the *z*-axis. The magnitude of its tangential velocity is $v_i = R_i\omega$. Because the vectors $\vec{\mathbf{v}}_i$ and $\vec{\mathbf{r}}_i$ are perpendicular to each other, the magnitude of the angular momentum of this mass segment is

$$l_i = r_i(\Delta m v_i)\sin 90°.$$

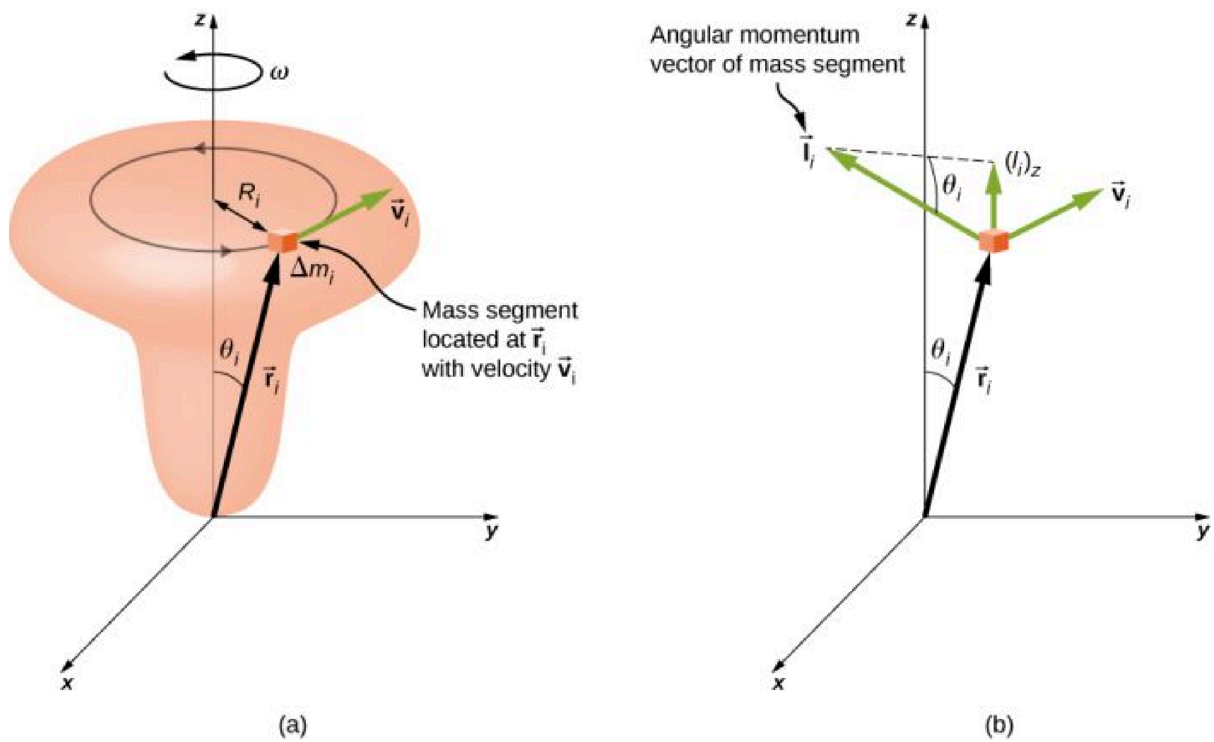


FIGURE 11.12 (a) A rigid body is constrained to rotate around the *z*-axis. The rigid body is symmetrical about the *z*-axis. A mass segment Δm_i is located at position $\vec{\mathbf{r}}_i$, which makes angle θ_i with respect to the *z*-axis. The circular motion of an infinitesimal mass segment is shown. (b) $\vec{\mathbf{l}}_i$ is the angular momentum of the mass segment and has a component along the *z*-axis $(\vec{\mathbf{l}}_i)_z$.

Using the right-hand rule, the angular momentum vector points in the direction shown in part (b). The sum of the angular momenta of all the mass segments contains components both along and perpendicular to the axis of rotation. Every mass segment has a perpendicular component of the angular momentum that will be cancelled by the perpendicular component of an identical mass segment on the opposite side of the rigid body, because it is cylindrically symmetric. Thus, the component along the axis of rotation is the only component that gives a nonzero value when summed over all the mass segments. From part (b), the component of $\vec{\mathbf{l}}_i$ along the axis of rotation is

$$\begin{aligned}(l_i)_z &= l_i \sin\theta_i = (r_i \Delta m_i v_i)\sin\theta_i, \\ &= (r_i \sin\theta_i)(\Delta m_i v_i) = R_i \Delta m_i v_i.\end{aligned}$$

The net angular momentum of the rigid body along the axis of rotation is

$$L = \sum_i (\vec{\mathbf{l}}_i)_z = \sum_i R_i \Delta m_i v_i = \sum_i R_i \Delta m_i (R_i \omega) = \omega \sum_i \Delta m_i (R_i)^2.$$

The summation $\sum_i \Delta m_i (R_i)^2$ is simply the moment of inertia I of the rigid body about the axis of rotation. For a thin hoop rotating about an axis perpendicular to the plane of the hoop, all of the R_i's are equal to R so the summation reduces to $R^2 \sum_i \Delta m_i = mR^2$, which is the moment of inertia for a thin hoop found in Figure 10.20. Thus, the magnitude of the angular momentum along the axis of rotation of a rigid body rotating with angular velocity ω about the axis is

$$L = I\omega. \tag{11.9}$$

This equation is analogous to the magnitude of the linear momentum $p = mv$. The direction of the angular momentum vector is directed along the axis of rotation given by the right-hand rule.

EXAMPLE 11.6

Angular Momentum of a Robot Arm

A robot arm on a Mars rover like *Curiosity* shown in Figure 11.8 is 1.0 m long and has forceps at the free end to pick up rocks. The mass of the arm is 2.0 kg and the mass of the forceps is 1.0 kg. See Figure 11.13. The robot arm and forceps move from rest to $\omega = 0.1\pi \text{ rad/s}$ in 0.1 s. It rotates down and picks up a Mars rock that has mass 1.5 kg. The axis of rotation is the point where the robot arm connects to the rover. (a) What is the angular momentum of the robot arm by itself about the axis of rotation after 0.1 s when the arm has stopped accelerating? (b) What is the angular momentum of the robot arm when it has the Mars rock in its forceps and is rotating upwards? (c) When the arm does not have a rock in the forceps, what is the torque about the point where the arm connects to the rover when it is accelerating from rest to its final angular velocity?

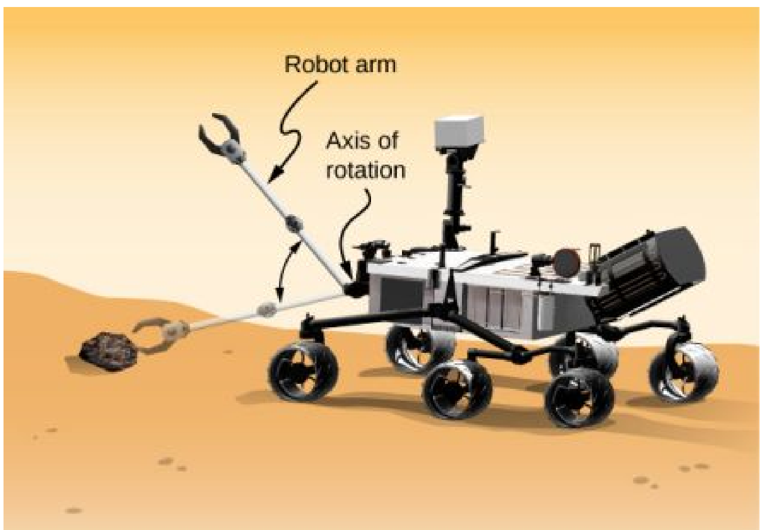


FIGURE 11.13 A robot arm on a Mars rover swings down and picks up a Mars rock. (credit: modification of work by NASA/JPL-Caltech)

Strategy

We use Equation 11.9 to find angular momentum in the various configurations. When the arm is rotating downward, the right-hand rule gives the angular momentum vector directed out of the page, which we will call the positive *z*-direction. When the arm is rotating upward, the right-hand rule gives the direction of the angular momentum vector into the page or in the negative *z*-direction. The moment of inertia is the sum of the individual moments of inertia. The arm can be approximated with a solid rod, and the forceps and Mars rock can be approximated as point masses located at a distance of 1 m from the origin. For part (c), we use Newton's second law of motion for rotation to find the torque on the robot arm.

Solution

a. Writing down the individual moments of inertia, we have
Robot arm: $I_R = \frac{1}{3} m_R r^2 = \frac{1}{3}(2.00\,\text{kg})(1.00\,\text{m})^2 = \frac{2}{3}\,\text{kg} \cdot \text{m}^2$.
Forceps: $I_F = m_F r^2 = (1.0\,\text{kg})(1.0\,\text{m})^2 = 1.0\,\text{kg} \cdot \text{m}^2$.
Mars rock: $I_{MR} = m_{MR} r^2 = (1.5\,\text{kg})(1.0\,\text{m})^2 = 1.5\,\text{kg} \cdot \text{m}^2$.
Therefore, without the Mars rock, the total moment of inertia is
$$I_{\text{Total}} = I_R + I_F = 1.67\,\text{kg} \cdot \text{m}^2$$
and the magnitude of the angular momentum is
$$L = I\omega = 1.67\,\text{kg} \cdot \text{m}^2(0.1\pi\,\text{rad/s}) = 0.17\pi\,\text{kg} \cdot \text{m}^2/\text{s}.$$
The angular momentum vector is directed out of the page in the $\hat{\mathbf{k}}$ direction since the robot arm is rotating counterclockwise.

b. We must include the Mars rock in the calculation of the moment of inertia, so we have
$$I_{\text{Total}} = I_R + I_F + I_{MR} = 3.17\,\text{kg} \cdot \text{m}^2$$
and
$$L = I\omega = 3.17\,\text{kg} \cdot \text{m}^2(0.1\pi\,\text{rad/s}) = 0.32\pi\,\text{kg} \cdot \text{m}^2/\text{s}.$$
Now the angular momentum vector is directed into the page in the $-\hat{\mathbf{k}}$ direction, by the right-hand rule, since the robot arm is now rotating clockwise.

c. We find the torque when the arm does not have the rock by taking the derivative of the angular momentum using Equation 11.8 $\frac{d\vec{\mathbf{L}}}{dt} = \sum \vec{\boldsymbol{\tau}}$. But since $L = I\omega$, and understanding that the direction of the angular momentum and torque vectors are along the axis of rotation, we can suppress the vector notation and find
$$\frac{dL}{dt} = \frac{d(I\omega)}{dt} = I\frac{d\omega}{dt} = I\alpha = \sum \tau,$$
which is Newton's second law for rotation. Since $\alpha = \frac{0.1\pi\,\text{rad/s}}{0.1\,\text{s}} = \pi\,\text{rad/s}^2$, we can calculate the net torque:
$$\sum \tau = I\alpha = 1.67\,\text{kg} \cdot \text{m}^2(\pi\,\text{rad/s}^2) = 1.67\pi\,\text{N} \cdot \text{m}.$$

Significance

The angular momentum in (a) is less than that of (b) due to the fact that the moment of inertia in (b) is greater than (a), while the angular velocity is the same.

CHECK YOUR UNDERSTANDING 11.3

Which has greater angular momentum: a solid sphere of mass *m* rotating at a constant angular frequency ω_0 about the *z*-axis, or a solid cylinder of same mass and rotation rate about the *z*-axis?

INTERACTIVE

Visit the University of Colorado's Interactive Simulation of Angular Momentum (https://openstax.org/l/21angmomintsim) to learn more about angular momentum.

11.3 Conservation of Angular Momentum

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Apply conservation of angular momentum to determine the angular velocity of a rotating system in which the moment of inertia is changing
- Explain how the rotational kinetic energy changes when a system undergoes changes in both moment of inertia and angular velocity

So far, we have looked at the angular momentum of systems consisting of point particles and rigid bodies. We have also analyzed the torques involved, using the expression that relates the external net torque to the change in angular momentum, Equation 11.8. Examples of systems that obey this equation include a freely spinning bicycle tire that slows over time due to torque arising from friction, or the slowing of Earth's rotation over millions of years due to frictional forces exerted on tidal deformations.

However, suppose there is no net external torque on the system, $\sum \vec{\boldsymbol{\tau}} = 0$. In this case, Equation 11.8 becomes the **law of conservation of angular momentum**.

LAW OF CONSERVATION OF ANGULAR MOMENTUM

The angular momentum of a system of particles around a point in a fixed inertial reference frame is conserved if there is no net external torque around that point:

$$\frac{d\vec{\mathbf{L}}}{dt} = 0 \tag{11.10}$$

or

$$\vec{\mathbf{L}} = \vec{\mathbf{l}}_1 + \vec{\mathbf{l}}_2 + \cdots + \vec{\mathbf{l}}_N = \text{constant.} \tag{11.11}$$

Note that the *total* angular momentum $\vec{\mathbf{L}}$ is conserved. Any of the individual angular momenta can change as long as their sum remains constant. This law is analogous to linear momentum being conserved when the external force on a system is zero.

As an example of conservation of angular momentum, Figure 11.14 shows an ice skater executing a spin. The net torque on her is very close to zero because there is relatively little friction between her skates and the ice. Also, the friction is exerted very close to the pivot point. Both $\left|\vec{\mathbf{F}}\right|$ and $\left|\vec{\mathbf{r}}\right|$ are small, so $\left|\vec{\boldsymbol{\tau}}\right|$ is negligible. Consequently, she can spin for quite some time. She can also increase her rate of spin by pulling her arms and legs in. Why does pulling her arms and legs in increase her rate of spin? The answer is that her angular momentum is constant, so that

$$L' = L$$

or

$$I'\omega' = I\omega,$$

where the primed quantities refer to conditions after she has pulled in her arms and reduced her moment of inertia. Because I' is smaller, the angular velocity ω' must increase to keep the angular momentum constant.

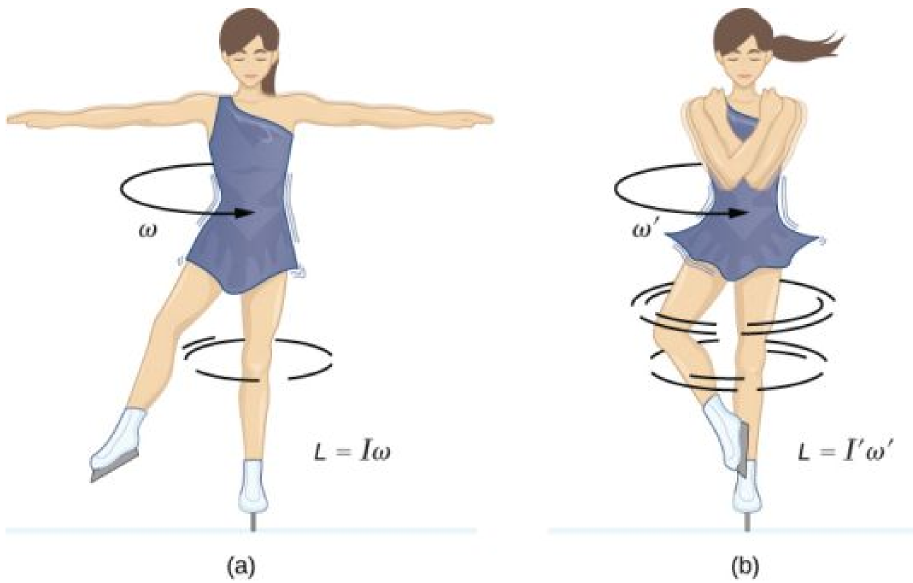


FIGURE 11.14 (a) An ice skater is spinning on the tip of her skate with her arms extended. Her angular momentum is conserved because the net torque on her is negligibly small. (b) Her rate of spin increases greatly when she pulls in her arms, decreasing her moment of inertia. The work she does to pull in her arms results in an increase in rotational kinetic energy.

It is interesting to see how the rotational kinetic energy of the skater changes when she pulls her arms in. Her initial rotational energy is

$$K_{\text{Rot}} = \frac{1}{2} I \omega^2,$$

whereas her final rotational energy is

$$K'_{\text{Rot}} = \frac{1}{2} I' (\omega')^2.$$

Since $I'\omega' = I\omega$, we can substitute for ω' and find

$$K'_{\text{Rot}} = \frac{1}{2} I' (\omega')^2 = \frac{1}{2} I' \left(\frac{I}{I'}\omega\right)^2 = \frac{1}{2} I \omega^2 \left(\frac{I}{I'}\right) = K_{\text{Rot}} \left(\frac{I}{I'}\right) \quad .$$

Because her moment of inertia has decreased, $I' < I$, her final rotational kinetic energy has increased. The source of this additional rotational kinetic energy is the work required to pull her arms inward. Note that the skater's arms do not move in a perfect circle—they spiral inward. This work causes an increase in the rotational kinetic energy, while her angular momentum remains constant. Since she is in a frictionless environment, no energy escapes the system. Thus, if she were to extend her arms to their original positions, she would rotate at her original angular velocity and her kinetic energy would return to its original value.

The solar system is another example of how conservation of angular momentum works in our universe. Our solar system was born from a huge cloud of gas and dust that initially had rotational energy. Gravitational forces caused the cloud to contract, and the rotation rate increased as a result of conservation of angular momentum (Figure 11.15).

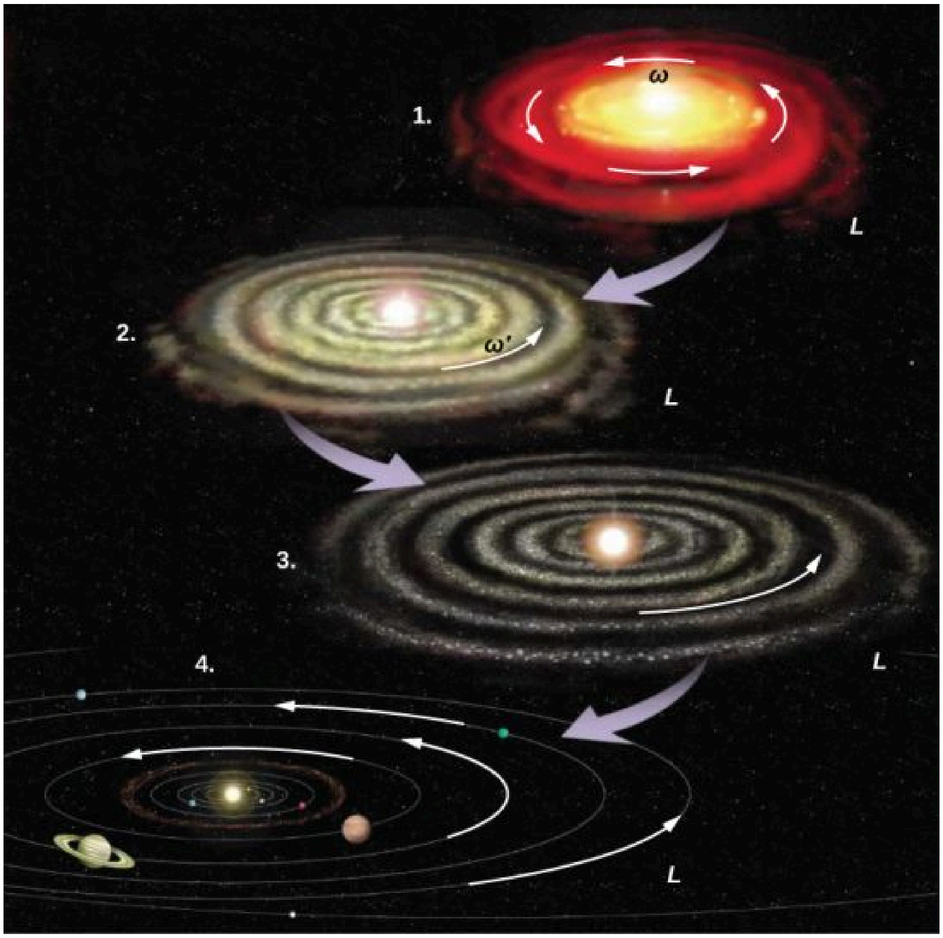


FIGURE 11.15 The solar system coalesced from a cloud of gas and dust that was originally rotating. The orbital motions and spins of the planets are in the same direction as the original spin and conserve the angular momentum of the parent cloud. (credit: modification of work by NASA)

We continue our discussion with an example that has applications to engineering.

EXAMPLE 11.7

Coupled Flywheels

A flywheel rotates without friction at an angular velocity $\omega_0 = 600 \text{ rev/min}$ on a frictionless, vertical shaft of negligible rotational inertia. A second flywheel, which is at rest and has a moment of inertia three times that of the rotating flywheel, is dropped onto it (Figure 11.16). Because friction exists between the surfaces, the flywheels very quickly reach the same rotational velocity, after which they spin together. (a) Use the law of conservation of angular momentum to determine the angular velocity ω of the combination. (b) What fraction of the initial kinetic energy is lost in the coupling of the flywheels?

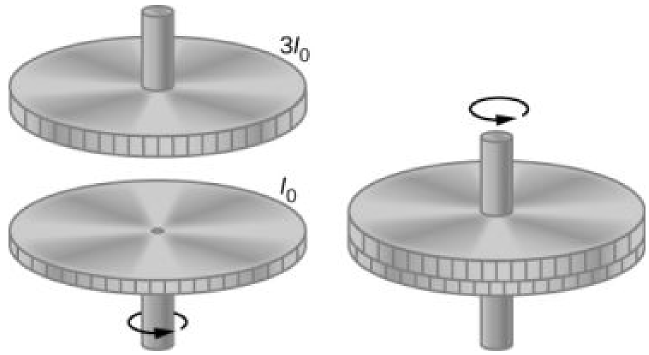


FIGURE 11.16 Two flywheels are coupled and rotate together.

Strategy

Part (a) is straightforward to solve for the angular velocity of the coupled system. We use the result of (a) to compare the initial and final kinetic energies of the system in part (b).

Solution

a. No external torques act on the system. The force due to friction produces an internal torque, which does not affect the angular momentum of the system. Therefore conservation of angular momentum gives

$$I_0\omega_0 = (I_0 + 3I_0)\omega,$$
$$\omega = \tfrac{1}{4}\omega_0 = 150\ \text{rev/min} = 15.7\ \text{rad/s}.$$

b. Before contact, only one flywheel is rotating. The rotational kinetic energy of this flywheel is the initial rotational kinetic energy of the system, $\frac{1}{2}I_0\omega_0^2$. The final kinetic energy is $\frac{1}{2}(4I_0)\omega^2 = \frac{1}{2}(4I_0)\left(\frac{\omega_0}{4}\right)^2 = \frac{1}{8}I_0\omega_0^2$.

Therefore, the ratio of the final kinetic energy to the initial kinetic energy is

$$\frac{\frac{1}{8}I_0\omega_0^2}{\frac{1}{2}I_0\omega_0^2} = \frac{1}{4}.$$

Thus, 3/4 of the initial kinetic energy is lost to the coupling of the two flywheels.

Significance

Since the rotational inertia of the system increased, the angular velocity decreased, as expected from the law of conservation of angular momentum. In this example, we see that the final kinetic energy of the system has decreased, as energy is lost to the coupling of the flywheels. Compare this to the example of the skater in Figure 11.14 doing work to bring her arms inward and adding rotational kinetic energy.

CHECK YOUR UNDERSTANDING 11.4

A merry-go-round at a playground is rotating at 4.0 rev/min. Three children jump on and increase the moment of inertia of the merry-go-round/children rotating system by 25%. What is the new rotation rate?

EXAMPLE 11.8

Dismount from a High Bar

An 80.0-kg gymnast dismounts from a high bar. He starts the dismount at full extension, then tucks to complete a number of revolutions before landing. His moment of inertia when fully extended can be approximated as a rod of length 1.8 m and when in the tuck a rod of half that length. If his rotation rate at full extension is 1.0 rev/s and he enters the tuck when his center of mass is at 3.0 m height moving horizontally to the floor, how many revolutions can he execute if he comes out of the tuck at 1.8 m height? See Figure 11.17.

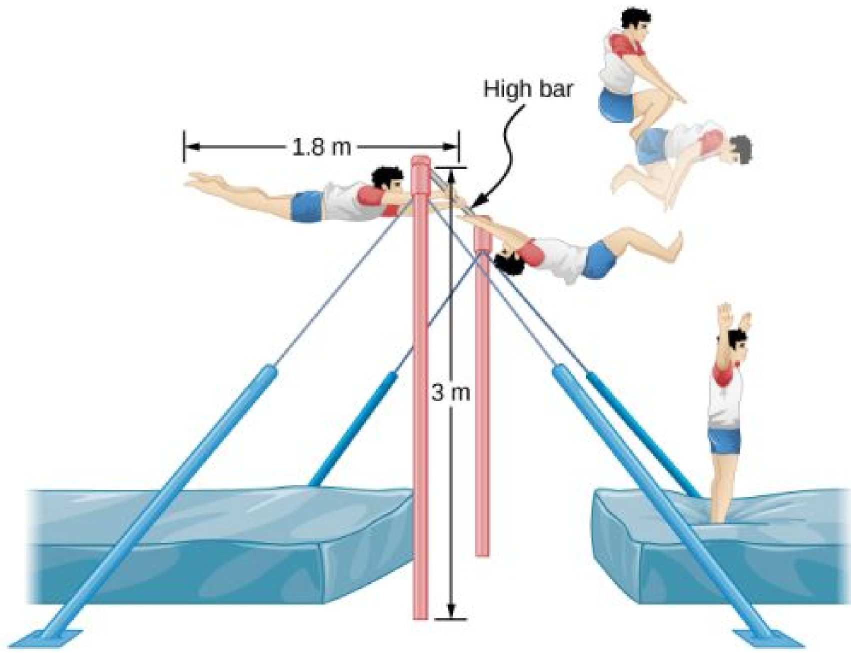


FIGURE 11.17 A gymnast dismounts from a high bar and executes a number of revolutions in the tucked position before landing upright.

Strategy

Using conservation of angular momentum, we can find his rotation rate when in the tuck. Using the equations of kinematics, we can find the time interval from a height of 3.0 m to 1.8 m. Since he is moving horizontally with respect to the ground, the equations of free fall simplify. This will allow the number of revolutions that can be executed to be calculated. Since we are using a ratio, we can keep the units as rev/s and don't need to convert to radians/s.

Solution

The moment of inertia at full extension is $I_0 = \frac{1}{12}mL^2 = \frac{1}{12}80.0\,\text{kg}(1.8\,\text{m})^2 = 21.6\,\text{kg}\cdot\text{m}^2$.

The moment of inertia in the tuck is $I_\text{f} = \frac{1}{12}mL_\text{f}^2 = \frac{1}{12}80.0\,\text{kg}(0.9\,\text{m})^2 = 5.4\,\text{kg}\cdot\text{m}^2$.

Conservation of angular momentum: $I_\text{f}\omega_\text{f} = I_0\omega_0 \Rightarrow \omega_\text{f} = \frac{I_0\omega_0}{I_\text{f}} = \frac{21.6\,\text{kg·m}^2(1.0\,\text{rev/s})}{5.4\,\text{kg·m}^2} = 4.0\,\text{rev/s}$.

Time interval in the tuck: $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2(3.0-1.8)\text{m}}{9.8\,\text{m/s}}} = 0.5\,\text{s}$.

In 0.5 s, he will be able to execute two revolutions at 4.0 rev/s.

Significance

Note that the number of revolutions he can complete will depend on how long he is in the air. In the problem, he is exiting the high bar horizontally to the ground. He could also exit at an angle with respect to the ground, giving him more or less time in the air depending on the angle, positive or negative, with respect to the ground. Gymnasts must take this into account when they are executing their dismounts.

EXAMPLE 11.9

Conservation of Angular Momentum of a Collision

A bullet of mass $m = 2.0\,\text{g}$ is moving horizontally with a speed of $500.0\,\text{m/s}$. The bullet strikes and becomes embedded in the edge of a solid disk of mass $M = 3.2\,\text{kg}$ and radius $R = 0.5\,\text{m}$. The cylinder is free to rotate around its axis and is initially at rest (Figure 11.18). What is the angular velocity of the disk immediately after the bullet is embedded?

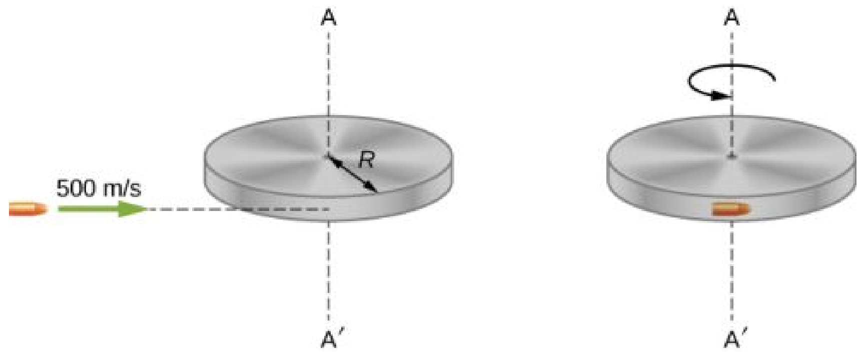


FIGURE 11.18 A bullet is fired horizontally and becomes embedded in the edge of a disk that is free to rotate about its vertical axis.

Strategy

For the system of the bullet and the cylinder, no external torque acts along the vertical axis through the center of the disk. Thus, the angular momentum along this axis is conserved. The initial angular momentum of the bullet is mvR, which is taken about the rotational axis of the disk the moment before the collision. The initial angular momentum of the cylinder is zero. Thus, the net angular momentum of the system is mvR. Since angular momentum is conserved, the initial angular momentum of the system is equal to the angular momentum of the bullet embedded in the disk immediately after impact.

Solution

The initial angular momentum of the system is

$$L_i = mvR.$$

The moment of inertia of the system with the bullet embedded in the disk is

$$I = mR^2 + \frac{1}{2}MR^2 = \left(m + \frac{M}{2}\right)R^2.$$

The final angular momentum of the system is

$$L_f = I\omega_f.$$

Thus, by conservation of angular momentum, $L_i = L_f$ and

$$mvR = \left(m + \frac{M}{2}\right)R^2\omega_f.$$

Solving for ω_f,

$$\omega_f = \frac{mvR}{(m + M/2)R^2} = \frac{(2.0 \times 10^{-3}\ \text{kg})(500.0\ \text{m/s})}{(2.0 \times 10^{-3}\ \text{kg} + 1.6\ \text{kg})(0.50\ \text{m})} = 1.2\ \text{rad/s}.$$

Significance

The system is composed of both a point particle and a rigid body. Care must be taken when formulating the angular momentum before and after the collision. Just before impact the angular momentum of the bullet is taken about the rotational axis of the disk.

11.4 Precession of a Gyroscope

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the physical processes underlying the phenomenon of precession
- Calculate the precessional angular velocity of a gyroscope

Figure 11.19 shows a gyroscope, defined as a spinning disk in which the axis of rotation is free to assume any orientation. When spinning, the orientation of the spin axis is unaffected by the orientation of the body that encloses it. The body or vehicle enclosing the gyroscope can be moved from place to place and the orientation of the spin axis will remain the same. This makes gyroscopes very useful in navigation, especially where magnetic compasses can't be used, such as in piloted and unpiloted spacecrafts, intercontinental ballistic missiles, unmanned aerial vehicles, and satellites like the Hubble Space Telescope.

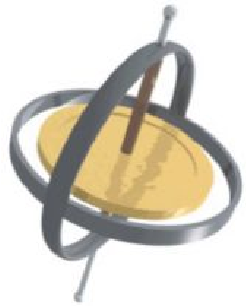


FIGURE 11.19 A gyroscope consists of a spinning disk about an axis that is free to assume any orientation.

We illustrate the **precession** of a gyroscope with an example of a top in the next two figures. If the top is placed on a flat surface near the surface of Earth at an angle to the vertical and is not spinning, it will fall over, due to the force of gravity producing a torque acting on its center of mass. This is shown in Figure 11.20(a). However, if the top is spinning on its axis, rather than topple over due to this torque, it precesses about the vertical, shown in part (b) of the figure. This is due to the torque on the center of mass, which provides the change in angular momentum.

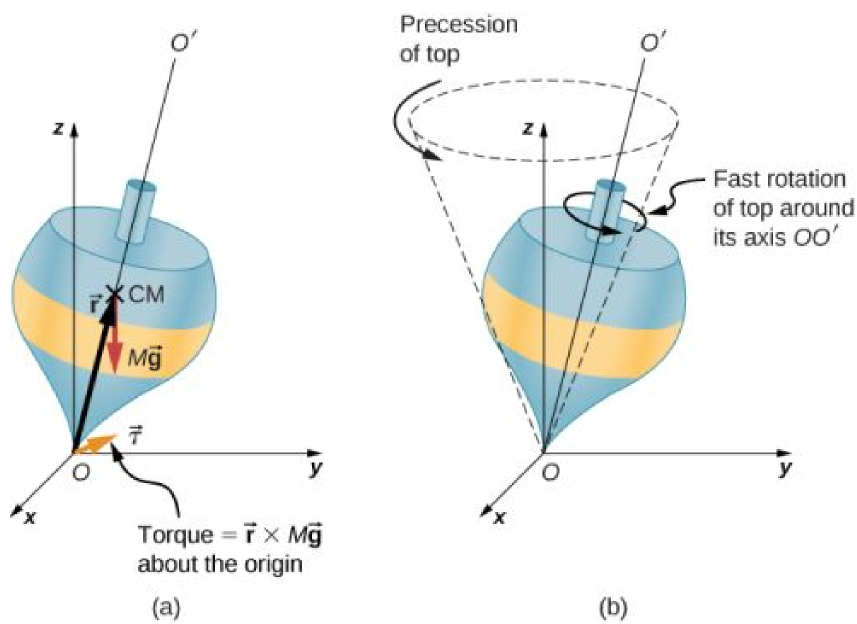


FIGURE 11.20 (a) If the top is not spinning, there is a torque $\vec{\mathbf{r}} \times M\vec{\mathbf{g}}$ about the origin, and the top falls over. (b) If the top is spinning about its axis OO', it doesn't fall over but precesses about the *z*-axis.

Figure 11.21 shows the forces acting on a spinning top. The torque produced is perpendicular to the angular momentum vector. This changes the direction of the angular momentum vector $\vec{\mathbf{L}}$ according to $d\vec{\mathbf{L}} = \vec{\boldsymbol{\tau}}dt$, but not its magnitude. The top *precesses* around a vertical axis, since the torque is always horizontal and perpendicular to $\vec{\mathbf{L}}$. If the top is *not* spinning, it acquires angular momentum in the direction of the torque, and it rotates around a horizontal axis, falling over just as we would expect.

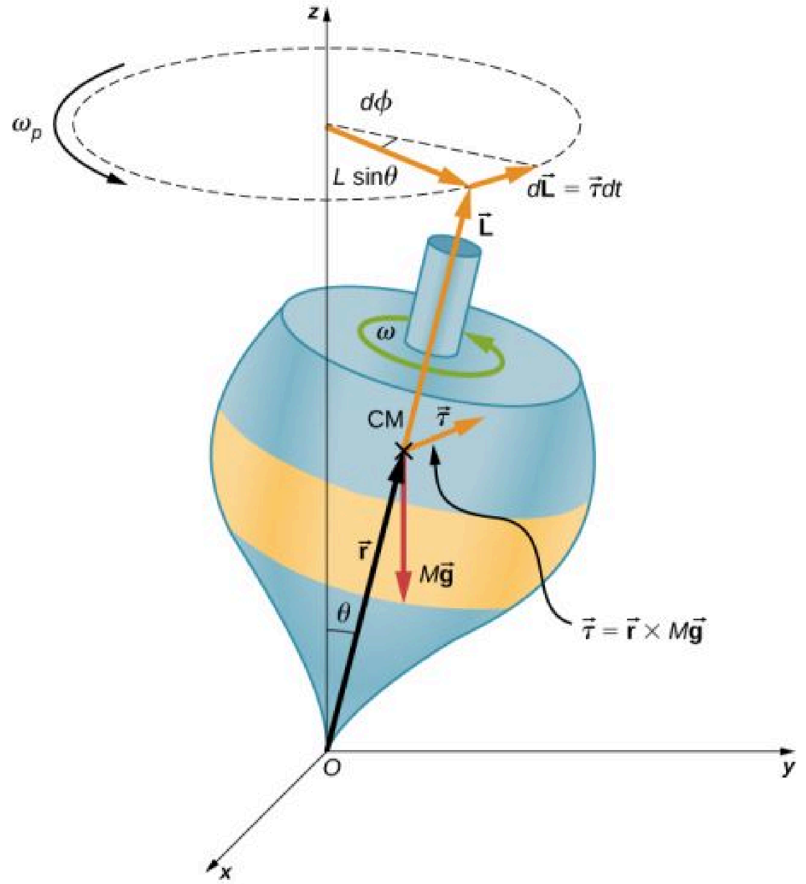


FIGURE 11.21 The force of gravity acting on the center of mass produces a torque $\vec{\boldsymbol{\tau}}$ in the direction perpendicular to $\vec{\mathbf{L}}$. The magnitude of $\vec{\mathbf{L}}$ doesn't change but its direction does, and the top precesses about the *z*-axis.

We can experience this phenomenon first hand by holding a spinning bicycle wheel and trying to rotate it about an axis perpendicular to the spin axis. As shown in Figure 11.22, the person applies forces perpendicular to the spin axis in an attempt to rotate the wheel, but instead, the wheel axis starts to change direction to her left due to the applied torque.

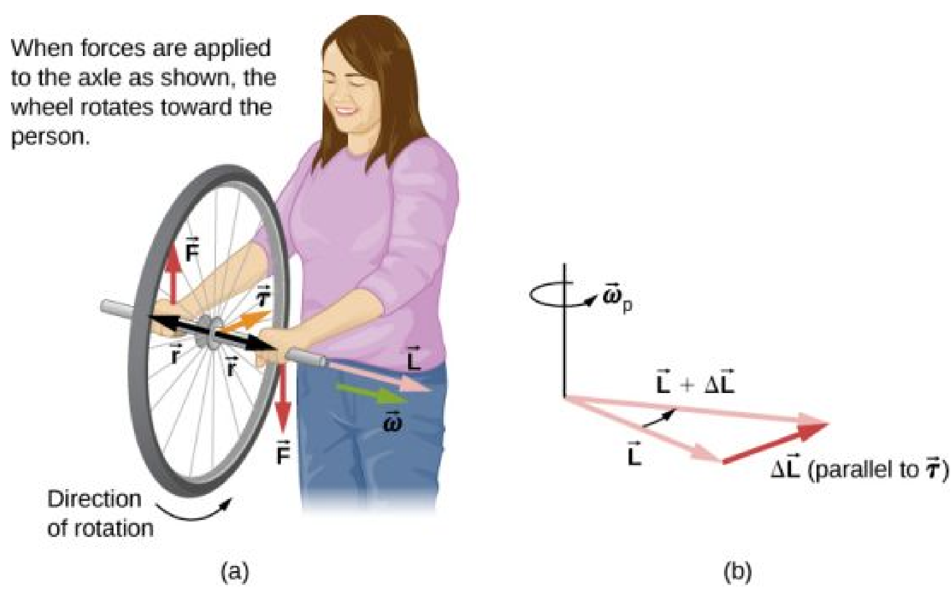


FIGURE 11.22 (a) A person holding the spinning bike wheel lifts it with her right hand and pushes down with her left hand in an attempt to rotate the wheel. This action creates a torque directly toward her. This torque causes a change in angular momentum $\Delta\vec{\mathbf{L}}$ in exactly the same direction. (b) A vector diagram depicting how $\Delta\vec{\mathbf{L}}$ and $\vec{\mathbf{L}}$ add, producing a new angular momentum pointing more toward the person.

The wheel moves toward the person, perpendicular to the forces she exerts on it.

We all know how easy it is for a bicycle to tip over when sitting on it at rest. But when riding the bicycle at a good pace, tipping it over involves changing the angular momentum vector of the spinning wheels.

INTERACTIVE

View the video on gyroscope precession (https://openstax.org/l/21gyrovideo) for a complete demonstration of precession of the bicycle wheel.

Also, when a spinning disk is put in a box such as a Blu-Ray player, try to move it. It is easy to translate the box in a given direction but difficult to rotate it about an axis perpendicular to the axis of the spinning disk, since we are putting a torque on the box that will cause the angular momentum vector of the spinning disk to precess.

We can calculate the precession rate of the top in Figure 11.21. From Figure 11.21, we see that the magnitude of the torque is

$$\tau = rMg\sin\theta.$$

Thus,

$$dL = rMg\sin\theta\, dt.$$

The angle the top precesses through in time *dt* is

$$d\phi = \frac{dL}{L\sin\theta} = \frac{rMg\sin\theta}{L\sin\theta}dt = \frac{rMg}{L}dt.$$

The precession angular velocity is $\omega_P = \frac{d\phi}{dt}$ and from this equation we see that

$$\omega_P = \frac{rMg}{L}. \text{ or, since } L = I\omega,$$

$$\omega_P = \frac{rMg}{I\omega}. \quad (11.12)$$

In this derivation, we assumed that $\omega_P \ll \omega$, that is, that the precession angular velocity is much less than the angular velocity of the gyroscope disk. The precession angular velocity adds a small component to the angular momentum along the *z*-axis. This is seen in a slight bob up and down as the gyroscope precesses, referred to as nutation.

Earth itself acts like a gigantic gyroscope. Its angular momentum is along its axis and currently points at Polaris, the North Star. But Earth is slowly precessing (once in about 26,000 years) due to the torque of the Sun and the Moon on its nonspherical shape.

EXAMPLE 11.10

Period of Precession

A gyroscope spins with its tip on the ground and is spinning with negligible frictional resistance. The disk of the gyroscope has mass 0.3 kg and is spinning at 20 rev/s. Its center of mass is 5.0 cm from the pivot and the radius of the disk is 5.0 cm. What is the precessional period of the gyroscope?

Strategy

We use Equation 11.12 to find the precessional angular velocity of the gyroscope. This allows us to find the period of precession.

Solution

The moment of inertia of the disk is

$$I = \frac{1}{2}mr^2 = \frac{1}{2}(0.30\,\text{kg})(0.05\,\text{m})^2 = 3.75 \times 10^{-4}\,\text{kg} \cdot \text{m}^2.$$

The angular velocity of the disk is

$$20.0\,\text{rev/s} = 20.0(2\pi)\,\text{rad/s} = 125.66\,\text{rad/s}.$$

We can now substitute in Equation 11.12. The precessional angular velocity is

$$\omega_P = \frac{rMg}{I\omega} = \frac{(0.05\,\text{m})(0.3\,\text{kg})(9.8\,\text{m/s}^2)}{(3.75 \times 10^{-4}\,\text{kg} \cdot \text{m}^2)(125.66\,\text{rad/s})} = 3.12\,\text{rad/s}.$$

The precessional period of the gyroscope is

$$T_P = \frac{2\pi}{3.12\,\text{rad/s}} = 2.0\,\text{s}.$$

Significance

The precessional angular frequency of the gyroscope, 3.12 rad/s, or about 0.5 rev/s, is much less than the angular velocity 20 rev/s of the gyroscope disk. Therefore, we don't expect a large component of the angular momentum to arise due to precession, and Equation 11.12 is a good approximation of the precessional angular velocity.

CHECK YOUR UNDERSTANDING 11.5

A top has a precession frequency of 5.0 rad/s on Earth. What is its precession frequency on the Moon?

Chapter Review

Key Terms

angular momentum rotational analog of linear momentum, found by taking the product of moment of inertia and angular velocity

law of conservation of angular momentum angular momentum is conserved, that is, the initial angular momentum is equal to the final angular momentum when no external torque is applied to the system

precession circular motion of the pole of the axis of a spinning object around another axis due to a torque

rolling motion combination of rotational and translational motion with or without slipping

Key Equations

Velocity of center of mass of rolling object	$v_{\text{CM}} = R\omega$
Acceleration of center of mass of rolling object	$a_{\text{CM}} = R\alpha$
Displacement of center of mass of rolling object	$d_{\text{CM}} = R\theta$
Acceleration of an object rolling without slipping	$a_{\text{CM}} = \frac{mg\sin\theta}{m+(I_{\text{CM}}/r^2)}$
Angular momentum	$\vec{\mathbf{l}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$
Derivative of angular momentum equals torque	$\frac{d\vec{\mathbf{l}}}{dt} = \sum \vec{\boldsymbol{\tau}}$
Angular momentum of a system of particles	$\vec{\mathbf{L}} = \vec{\mathbf{l}}_1 + \vec{\mathbf{l}}_2 + \cdots + \vec{\mathbf{l}}_N$
For a system of particles, derivative of angular momentum equals torque	$\frac{d\vec{\mathbf{L}}}{dt} = \sum \vec{\boldsymbol{\tau}}$
Angular momentum of a rotating rigid body	$L = I\omega$
Conservation of angular momentum	$\frac{d\vec{\mathbf{L}}}{dt} = 0$
Conservation of angular momentum	$\vec{\mathbf{L}} = \vec{\mathbf{l}}_1 + \vec{\mathbf{l}}_2 + \cdots + \vec{\mathbf{l}}_N = \text{constant}$
Precessional angular velocity	$\omega_P = \frac{rMg}{I\omega}$

Summary

11.1 Rolling Motion

- In rolling motion without slipping, a static friction force is present between the rolling object and the surface. The relations $v_{\text{CM}} = R\omega, a_{\text{CM}} = R\alpha,$ and $d_{\text{CM}} = R\theta$ all apply, such that the linear velocity, acceleration, and distance of the center of mass are the angular variables multiplied by the radius of the object.
- In rolling motion with slipping, a kinetic friction force arises between the rolling object and the surface. In this case, $v_{\text{CM}} \neq R\omega, a_{\text{CM}} \neq R\alpha,$ and $d_{\text{CM}} \neq R\theta.$
- Energy conservation can be used to analyze rolling motion. Energy is conserved in rolling motion without slipping. Energy is not conserved in rolling motion with slipping due to the heat generated by kinetic friction.

11.2 Angular Momentum

- The angular momentum $\vec{\mathbf{l}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$ of a single particle about a designated origin is the vector product of the position vector in the given coordinate system and the particle's linear momentum.
- The angular momentum $\vec{\mathbf{l}} = \sum_i \vec{\mathbf{l}}_i$ of a system of particles about a designated origin is the vector sum of the individual momenta of the particles that make up the system.
- The net torque on a system about a given origin is the time derivative of the angular momentum about that origin: $\frac{d\vec{\mathbf{L}}}{dt} = \sum \vec{\boldsymbol{\tau}}.$
- A rigid rotating body has angular momentum $L = I\omega$ directed along the axis of rotation. The time derivative of the angular momentum $\frac{dL}{dt} = \sum \tau$ gives the net torque on a rigid body and is directed along the axis of rotation.

11.3 Conservation of Angular Momentum

- In the absence of external torques, a system's total angular momentum is conserved. This is the rotational counterpart to linear momentum being conserved when the external force on a system is zero.

- For a rigid body that changes its angular momentum in the absence of a net external torque, conservation of angular momentum gives $I_f \omega_f = I_i \omega_i$. This equation says that the angular velocity is inversely proportional to the moment of inertia. Thus, if the moment of inertia decreases, the angular velocity must increase to conserve angular momentum.
- Systems containing both point particles and rigid bodies can be analyzed using conservation of angular momentum. The angular momentum of all bodies in the system must be taken about a common axis.

11.4 Precession of a Gyroscope

- When a gyroscope is set on a pivot near the surface of Earth, it precesses around a vertical axis, since the torque is always horizontal and perpendicular to $\vec{\mathbf{L}}$. If the gyroscope is not spinning, it acquires angular momentum in the direction of the torque, and it rotates about a horizontal axis, falling over just as we would expect.
- The precessional angular velocity is given by $\omega_P = \frac{rMg}{I\omega}$, where *r* is the distance from the pivot to the center of mass of the gyroscope, I is the moment of inertia of the gyroscope's spinning disk, *M* is its mass, and ω is the angular frequency of the gyroscope disk.

Conceptual Questions

11.1 Rolling Motion

1. Can a round object released from rest at the top of a frictionless incline undergo rolling motion?

2. A cylindrical can of radius *R* is rolling across a horizontal surface without slipping. (a) After one complete revolution of the can, what is the distance that its center of mass has moved? (b) Would this distance be greater or smaller if slipping occurred?

3. A wheel is released from the top on an incline. Is the wheel most likely to slip if the incline is steep or gently sloped?

4. Which rolls down an inclined plane faster, a hollow cylinder or a solid sphere? Both have the same mass and radius.

5. A hollow sphere and a hollow cylinder of the same radius and mass roll up an incline without slipping and have the same initial center of mass velocity. Which object reaches a greater height before stopping?

11.2 Angular Momentum

6. Can you assign an angular momentum to a particle without first defining a reference point?

7. For a particle traveling in a straight line, are there any points about which the angular momentum is zero? Assume the line intersects the origin.

8. Under what conditions does a rigid body have angular momentum but not linear momentum?

9. If a particle is moving with respect to a chosen origin it has linear momentum. What conditions must exist for this particle's angular momentum to be zero about the chosen origin?

10. If you know the velocity of a particle, can you say anything about the particle's angular momentum?

11.3 Conservation of Angular Momentum

11. What is the purpose of the small propeller at the back of a helicopter that rotates in the plane perpendicular to the large propeller?

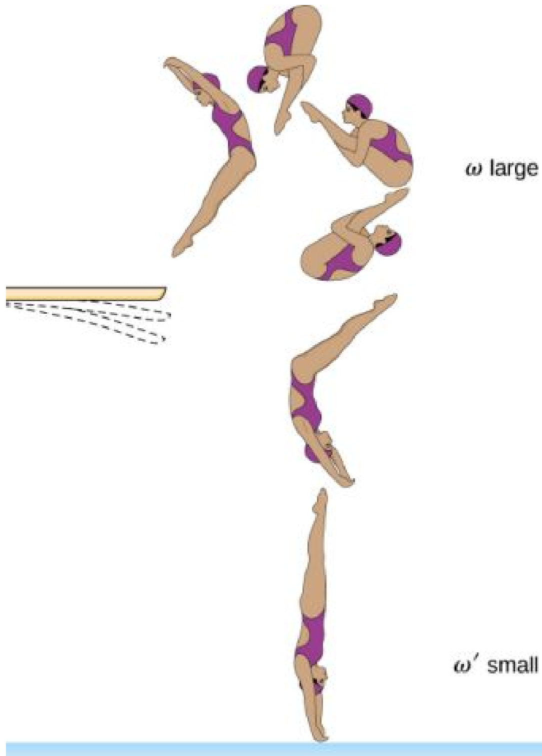
12. Suppose a child walks from the outer edge of a rotating merry-go-round to the inside. Does the angular velocity of the merry-go-round increase, decrease, or remain the same? Explain your answer. Assume the merry-go-round is spinning without friction.

13. As the rope of a tethered ball winds around a pole, what happens to the angular velocity of the ball?

14. Suppose the polar ice sheets broke free and floated toward Earth's equator without melting. What would happen to Earth's angular velocity?

15. Explain why stars spin faster when they collapse.

16. Competitive divers pull their limbs in and curl up their bodies when they do flips. Just before entering the water, they fully extend their limbs to enter straight down (see below). Explain the effect of both actions on their angular velocities. Also explain the effect on their angular momentum.



11.4 Precession of a Gyroscope

17. Gyroscopes used in guidance systems to indicate directions in space must have an angular momentum that does not change in direction. When placed in the vehicle, they are put in a compartment that is separated from the main fuselage, such that changes in the orientation of the fuselage does not affect the orientation of the gyroscope. If the space vehicle is subjected to large forces and accelerations how can the direction of the gyroscopes angular momentum be constant at all times?

18. Earth precesses about its vertical axis with a period of 26,000 years. Discuss whether Equation 11.12 can be used to calculate the precessional angular velocity of Earth.

Problems

11.1 Rolling Motion

19. What is the angular velocity of a 75.0-cm-diameter tire on an automobile traveling at 90.0 km/h?

20. A boy rides his bicycle 2.00 km. The wheels have radius 30.0 cm. What is the total angle the tires rotate through during his trip?

21. If the boy on the bicycle in the preceding problem accelerates from rest to a speed of 10.0 m/s in 10.0 s, what is the angular acceleration of the tires?

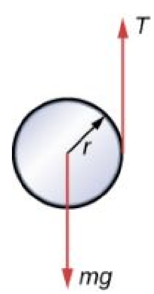
22. Formula One race cars have 66-cm-diameter tires. If a Formula One averages a speed of 300 km/h during a race, what is the angular displacement in revolutions of the wheels if the race car maintains this speed for 1.5 hours?

23. A marble rolls down an incline at $30°$ from rest. (a) What is its acceleration? (b) How far does it go in 3.0 s?

24. Repeat the preceding problem replacing the marble with a solid cylinder. Explain the new result.

25. A rigid body with a cylindrical cross-section is released from the top of a $30°$ incline. It rolls 10.0 m to the bottom in 2.60 s. Find the moment of inertia of the body in terms of its mass *m* and radius *r.*

26. A yo-yo can be thought of a solid cylinder of mass *m* and radius *r* that has a light string wrapped around its circumference (see below). One end of the string is held fixed in space. If the cylinder falls as the string unwinds without slipping, what is the acceleration of the cylinder?



27. A solid cylinder of radius 10.0 cm rolls down an incline with slipping. The angle of the incline is $30°$. The coefficient of kinetic friction on the surface is 0.400. What is the angular acceleration of the solid cylinder? What is the linear acceleration?

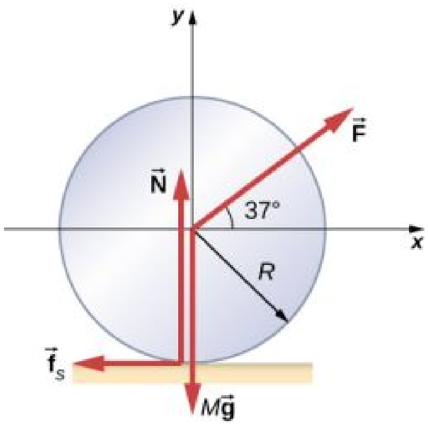
28. A bowling ball rolls up a ramp 0.5 m high without slipping to storage. It has an initial velocity of its center of mass of 3.0 m/s. (a) What is its velocity at the top of the ramp? (b) If the ramp is 1 m high does it make it to the top?

29. A 40.0-kg solid cylinder is rolling across a horizontal surface at a speed of 6.0 m/s. How much work is required to stop it?

30. A 40.0-kg solid sphere is rolling across a horizontal surface with a speed of 6.0 m/s. How much work is required to stop it? Compare results with the preceding problem.

31. A solid cylinder rolls up an incline at an angle of $20°$. If it starts at the bottom with a speed of 10 m/s, how far up the incline does it travel?

32. A solid cylindrical wheel of mass *M* and radius *R* is pulled by a force $\vec{\mathbf{F}}$ applied to the center of the wheel at $37°$ to the horizontal (see the following figure). If the wheel is to roll without slipping, what is the maximum value of $\left|\vec{\mathbf{F}}\right|$? The coefficients of static and kinetic friction are $\mu_S = 0.40$ and $\mu_k = 0.30$.



33. A hollow cylinder that is rolling without slipping is given an initial velocity and rolls up an incline to a vertical height of 1.0 m. If a hollow sphere of the same mass and radius is given the same initial velocity, how high vertically does it roll up the incline?

11.2 Angular Momentum

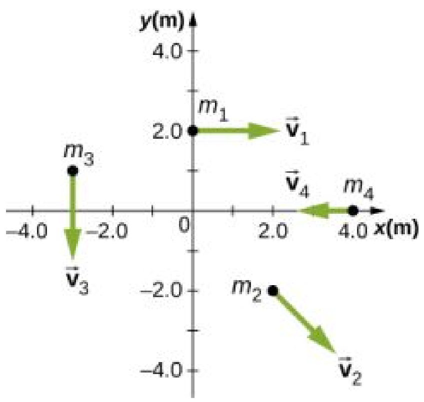
34. A 0.2-kg particle is travelling along the line $y = 2.0$ m with a velocity 5.0 m/s. What is the angular momentum of the particle about the origin?

35. A bird flies overhead from where you stand at an altitude of 300.0 m and at a speed horizontal to the ground of 20.0 m/s. The bird has a mass of 2.0 kg. The radius vector to the bird makes an angle θ with respect to the ground. The radius vector to the bird and its momentum vector lie in the *xy*-plane. What is the bird's angular momentum about the point where you are standing?

36. A Formula One race car with mass 750.0 kg is speeding through a course in Monaco and enters a circular turn at 220.0 km/h in the counterclockwise direction about the origin of the circle. At another part of the course, the car enters a second circular turn at 180 km/h also in the counterclockwise direction. If the radius of curvature of the first turn is 130.0 m and that of the second is 100.0 m, compare the angular momenta of the race car in each turn taken about the origin of the circular turn.

37. A particle of mass 5.0 kg has position vector $\vec{\mathbf{r}} = (2.0\hat{\mathbf{i}} - 3.0\hat{\mathbf{j}})\text{m}$ at a particular instant of time when its velocity is $\vec{\mathbf{v}} = (3.0\hat{\mathbf{i}})\text{m/s}$ with respect to the origin. (a) What is the angular momentum of the particle? (b) If a force $\vec{\mathbf{F}} = 5.0\hat{\mathbf{j}}\ \text{N}$ acts on the particle at this instant, what is the torque about the origin?

38. Use the right-hand rule to determine the directions of the angular momenta about the origin of the particles as shown below. The *z*-axis is out of the page.



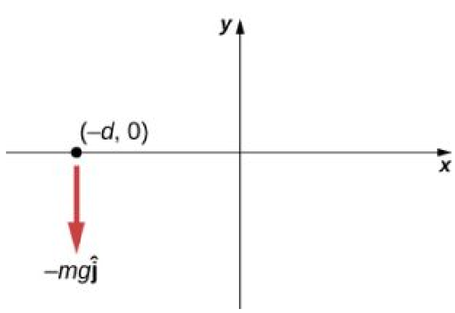
39. Suppose the particles in the preceding problem have masses $m_1 = 0.10\,\text{kg},\ m_2 = 0.20\,\text{kg},\ m_3 = 0.30\,\text{kg},$ $m_4 = 0.40\,\text{kg}$. The velocities of the particles are $v_1 = 2.0\hat{\mathbf{i}}\text{m/s}$, $v_2 = (3.0\hat{\mathbf{i}} - 3.0\hat{\mathbf{j}})\text{m/s}$, $v_3 = -1.5\hat{\mathbf{j}}\text{m/s}$, $v_4 = -4.0\hat{\mathbf{i}}\text{m/s}$. (a) Calculate the angular momentum of each particle about the origin. (b) What is the total angular momentum of the four-particle system about the origin?

40. Two particles of equal mass travel with the same speed in opposite directions along parallel lines separated by a distance *d*. Show that the angular momentum of this two-particle system is the same no matter what point is used as the reference for calculating the angular momentum.

41. An airplane of mass 4.0×10^4 kg flies horizontally at an altitude of 10 km with a constant speed of 250 m/s relative to Earth. (a) What is the magnitude of the airplane's angular momentum relative to a ground observer directly below the plane? (b) Does the angular momentum change as the airplane flies along a constant altitude?

42. At a particular instant, a 1.0-kg particle's position is $\vec{\mathbf{r}} = (2.0\hat{\mathbf{i}} - 4.0\hat{\mathbf{j}} + 6.0\hat{\mathbf{k}})\text{m}$, its velocity is $\vec{\mathbf{v}} = (-1.0\hat{\mathbf{i}} + 4.0\hat{\mathbf{j}} + 1.0\hat{\mathbf{k}})\text{m/s}$, and the force on it is $\vec{\mathbf{F}} = (10.0\hat{\mathbf{i}} + 15.0\hat{\mathbf{j}})\text{N}$. (a) What is the angular momentum of the particle about the origin? (b) What is the torque on the particle about the origin? (c) What is the time rate of change of the particle's angular momentum at this instant?

43. A particle of mass *m* is dropped at the point $(-d, 0)$ and falls vertically in Earth's gravitational field $-g\hat{\mathbf{j}}$. (a) What is the expression for the angular momentum of the particle around the *z*-axis, which points directly out of the page as shown below? (b) Calculate the torque on the particle around the *z*-axis. (c) Is the torque equal to the time rate of change of the angular momentum?



44. (a) Calculate the angular momentum of Earth in its orbit around the Sun. (b) Compare this angular momentum with the angular momentum of Earth about its axis.

45. A boulder of mass 20 kg and radius 20 cm rolls down a hill 15 m high from rest. What is its angular momentum when it is half way down the hill? (b) At the bottom?

46. A satellite is spinning at 6.0 rev/s. The satellite consists of a main body in the shape of a sphere of radius 2.0 m and mass 10,000 kg, and two antennas projecting out from the center of mass of the main body that can be approximated with rods of length 3.0 m each and mass 10 kg. The antenna's lie in the plane of rotation. What is the angular momentum of the satellite?

47. A propeller consists of two blades each 3.0 m in length and mass 120 kg each. The propeller can be approximated by a single rod rotating about its center of mass. The propeller starts from rest and rotates up to 1200 rpm in 30 seconds at a constant rate. (a) What is the angular momentum of the propeller at $t = 10\text{ s}; t = 20\text{ s}$? (b) What is the torque on the propeller?

48. A pulsar is a rapidly rotating neutron star. The Crab nebula pulsar in the constellation Taurus has a period of 33.5×10^{-3} s, radius 10.0 km, and mass 2.8×10^{30} kg. The pulsar's rotational period will increase over time due to the release of electromagnetic radiation, which doesn't change its radius but reduces its rotational energy. (a) What is the angular momentum of the pulsar? (b) Suppose the angular velocity decreases at a rate of 10^{-14} rad/s^2. What is the torque on the pulsar?

49. The blades of a wind turbine are 30 m in length and rotate at a maximum rotation rate of 20 rev/min. (a) If the blades are 6000 kg each and the rotor assembly has three blades, calculate the angular momentum of the turbine at this rotation rate. (b) What is the torque require to rotate the blades up to the maximum rotation rate in 5 minutes?

50. A roller coaster has mass 3000.0 kg and needs to make it safely through a vertical circular loop of radius 50.0 m. What is the minimum angular momentum of the coaster at the bottom of the loop to make it safely through? Neglect friction on the track. Take the coaster to be a point particle.

51. A mountain biker takes a jump in a race and goes airborne. The mountain bike is travelling at 10.0 m/s before it goes airborne. If the mass of the front wheel on the bike is 750 g and has radius 35 cm, what is the angular momentum of the spinning wheel in the air the moment the bike leaves the ground?

11.3 Conservation of Angular Momentum

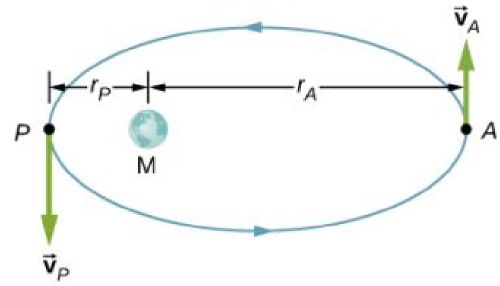
52. A disk of mass 2.0 kg and radius 60 cm with a small mass of 0.05 kg attached at the edge is rotating at 2.0 rev/s. The small mass, while attached to the disk, slides gradually to the center of the disk. What is the disk's final rotation rate?

53. The Sun's mass is $2.0 \times 10^{30} \text{kg}$, its radius is 7.0×10^{5} km, and it has a rotational period of approximately 28 days. If the Sun should collapse into a white dwarf of radius 3.5×10^{3} km, what would its period be if no mass were ejected and a sphere of uniform density can model the Sun both before and after?

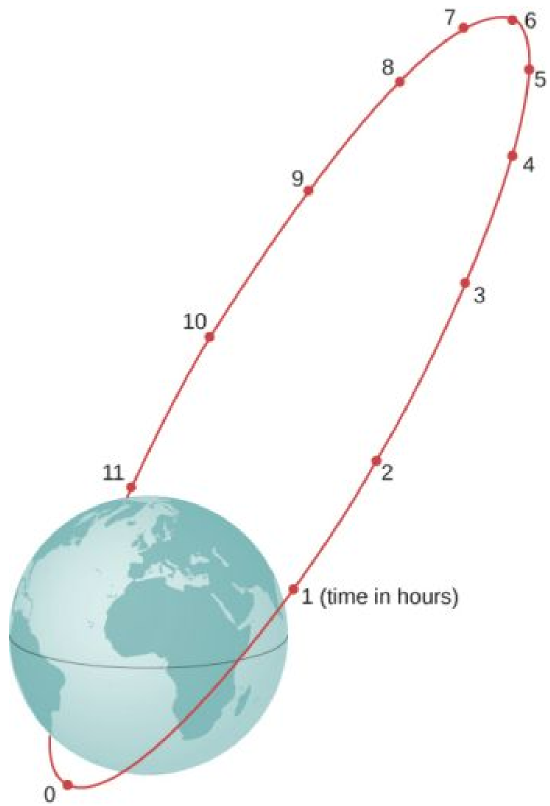
54. A cylinder with rotational inertia $I_1 = 2.0 \,\text{kg} \cdot \text{m}^2$ rotates clockwise about a vertical axis through its center with angular speed $\omega_1 = 5.0 \,\text{rad/s}$. A second cylinder with rotational inertia $I_2 = 1.0 \,\text{kg} \cdot \text{m}^2$ rotates counterclockwise about the same axis with angular speed $\omega_2 = 8.0 \,\text{rad/s}$. If the cylinders couple so they have the same rotational axis what is the angular speed of the combination? What percentage of the original kinetic energy is lost to friction?

55. A diver off the high board imparts an initial rotation with their body fully extended before going into a tuck and executing three back somersaults before hitting the water. If their moment of inertia before the tuck is $16.9 \,\text{kg} \cdot \text{m}^2$ and after the tuck during the somersaults is $4.2 \,\text{kg} \cdot \text{m}^2$, what rotation rate must the diver impart to their body directly off the board and before the tuck if they take 1.4 s to execute the somersaults before hitting the water?

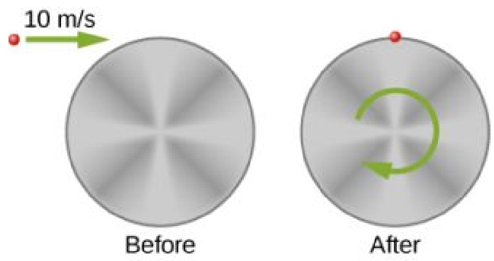
56. An Earth satellite has its apogee at 2500 km above the surface of Earth and perigee at 500 km above the surface of Earth. At apogee its speed is 6260 m/s. What is its speed at perigee? Earth's radius is 6370 km (see below).



57. A Molniya orbit is a highly eccentric orbit of a communication satellite so as to provide continuous communications coverage for Scandinavian countries and adjacent Russia. The orbit is positioned so that these countries have the satellite in view for extended periods in time (see below). If a satellite in such an orbit has an apogee at 40,000.0 km as measured from the center of Earth and a velocity of 1.68 km/s, what would be its velocity at perigee measured at 200.0 km altitude?



58. Shown below is a small particle of mass 20 g that is moving at a speed of 10.0 m/s when it collides and sticks to the edge of a uniform solid cylinder. The cylinder is free to rotate about its axis through its center and is perpendicular to the page. The cylinder has a mass of 0.5 kg and a radius of 10 cm, and is initially at rest. (a) What is the angular velocity of the system after the collision? (b) How much kinetic energy is lost in the collision?



59. A bug of mass 0.020 kg is at rest on the edge of a solid cylindrical disk $(M = 0.10\,\text{kg}, R = 0.10\,\text{m})$ rotating in a horizontal plane around the vertical axis through its center. The disk is rotating at 10.0 rad/s. The bug crawls to the center of the disk. (a) What is the new angular velocity of the disk? (b) What is the change in the kinetic energy of the system? (c) If the bug crawls back to the outer edge of the disk, what is the angular velocity of the disk then? (d) What is the new kinetic energy of the system? (e) What is the cause of the increase and decrease of kinetic energy?

60. A uniform rod of mass 200 g and length 100 cm is free to rotate in a horizontal plane around a fixed vertical axis through its center, perpendicular to its length. Two small beads, each of mass 20 g, are mounted in grooves along the rod. Initially, the two beads are held by catches on opposite sides of the rod's center, 10 cm from the axis of rotation. With the beads in this position, the rod is rotating with an angular velocity of 10.0 rad/s. When the catches are released, the beads slide outward along the rod. (a) What is the rod's angular velocity when the beads reach the ends of the rod? (b) What is the rod's angular velocity if the beads fly off the rod?

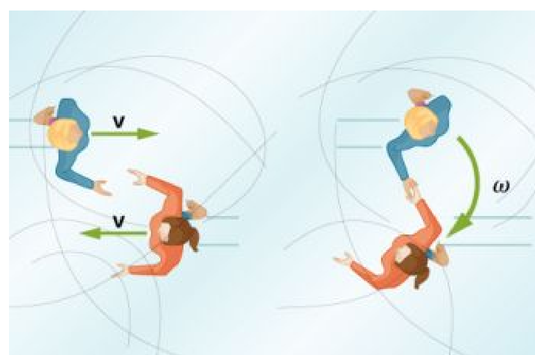
61. A merry-go-round has a radius of 2.0 m and a moment of inertia $300\,\text{kg}\cdot\text{m}^2$. A boy of mass 50 kg runs tangent to the rim at a speed of 4.0 m/s and jumps on. If the merry-go-round is initially at rest, what is the angular velocity after the boy jumps on?

62. A playground merry-go-round has a mass of 120 kg and a radius of 1.80 m and it is rotating with an angular velocity of 0.500 rev/s. What is its angular velocity after a 22.0-kg child gets onto it by grabbing its outer edge? The child is initially at rest.

63. Three children are riding on the edge of a merry-go-round that is 100 kg, has a 1.60-m radius, and is spinning at 20.0 rpm. The children have masses of 22.0, 28.0, and 33.0 kg. If the child who has a mass of 28.0 kg moves to the center of the merry-go-round, what is the new angular velocity in rpm?

64. (a) Calculate the angular momentum of an ice skater spinning at 6.00 rev/s given his moment of inertia is $0.400\,\text{kg}\cdot\text{m}^2$. (b) He reduces his rate of spin (his angular velocity) by extending his arms and increasing his moment of inertia. Find the value of his moment of inertia if his angular velocity decreases to 1.25 rev/s. (c) Suppose instead he keeps his arms in and allows friction of the ice to slow him to 3.00 rev/s. What average torque was exerted if this takes 15.0 s?

65. Twin skaters approach one another as shown below and lock hands. (a) Calculate their final angular velocity, given each had an initial speed of 2.50 m/s relative to the ice. Each has a mass of 70.0 kg, and each has a center of mass located 0.800 m from their locked hands. You may approximate their moments of inertia to be that of point masses at this radius. (b) Compare the initial kinetic energy and final kinetic energy.



(a) (b)

66. A baseball catcher extends his arm straight up to catch a fast ball with a speed of 40 m/s. The baseball is 0.145 kg and the catcher's arm length is 0.5 m and mass 4.0 kg. (a) What is the angular velocity of the arm immediately after catching the ball as measured from the arm socket? (b) What is the torque applied if the catcher stops the rotation of his arm 0.3 s after catching the ball?

67. In 2015, in Warsaw, Poland, Olivia Oliver of Nova Scotia broke the world record for being the fastest spinner on ice skates. She achieved a record 342 rev/min, beating the existing

Guinness World Record by 34 rotations. If an ice skater extends her arms at that rotation rate, what would be her new rotation rate? Assume she can be approximated by a 45-kg rod that is 1.7 m tall with a radius of 15 cm in the record spin. With her arms stretched take the approximation of a rod of length 130 cm with 10% of her body mass aligned perpendicular to the spin axis. Neglect frictional forces.

68. A satellite in a geosynchronous circular orbit is 42,164.0 km from the center of Earth. A small asteroid collides with the satellite sending it into an elliptical orbit of apogee 45,000.0 km. What is the speed of the satellite at apogee? Assume its angular momentum is conserved.

69. A gymnast does cartwheels along the floor and then launches herself into the air and executes several flips in a tuck while she is airborne. If her moment of inertia when executing the cartwheels is $13.5\ \text{kg} \cdot \text{m}^2$ and her spin rate is 0.5 rev/s, how many revolutions does she do in the air if her moment of inertia in the tuck is $3.4\ \text{kg} \cdot \text{m}^2$ and she has 2.0 s to do the flips in the air?

70. The centrifuge at NASA Ames Research Center has a radius of 8.8 m and can produce forces on its payload of 20 *g*s or 20 times the force of gravity on Earth. (a) What is the angular momentum of a 20-kg payload that experiences 10 *g*s in the centrifuge? (b) If the driver motor was turned off in (a) and the payload lost 10 kg, what would be its new spin rate, taking into account there are no frictional forces present?

71. A ride at a carnival has four spokes to which pods are attached that can hold two people. The spokes are each 15 m long and are attached to a central axis. Each spoke has mass 200.0 kg, and the pods each have mass 100.0 kg. If the ride spins at 0.2 rev/s with each pod containing two 50.0-kg children, what is the new spin rate if all the children jump off the ride?

72. An ice skater is preparing for a jump with turns and has his arms extended. His moment of inertia is $1.8\ \text{kg} \cdot \text{m}^2$ while his arms are extended, and he is spinning at 0.5 rev/s. If he launches himself into the air at 9.0 m/s at an angle of $45°$ with respect to the ice, how many revolutions can he execute while airborne if his moment of inertia in the air is $0.5\ \text{kg} \cdot \text{m}^2$?

73. A space station consists of a giant rotating hollow cylinder of mass $10^6\ \text{kg}$ including people on the station and a radius of 100.00 m. It is rotating in space at 3.30 rev/min in order to produce artificial gravity. If 100 people of an average mass of 65.00 kg spacewalk to an awaiting spaceship, what is the new rotation rate when all the people are off the station?

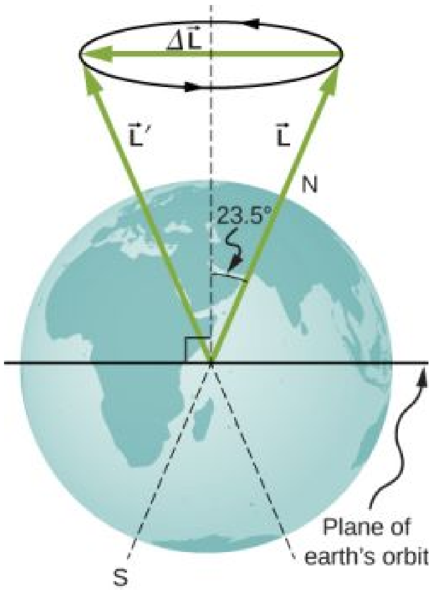
74. Neptune has a mass of $1.0 \times 10^{26}\ \text{kg}$ and is $4.5 \times 10^9 \text{km}$ from the Sun with an orbital period of 165 years. Planetesimals in the outer primordial solar system 4.5 billion years ago coalesced into Neptune over hundreds of millions of years. If the primordial disk that evolved into our present day solar system had a radius of 10^{11} km and if the matter that made up these planetesimals that later became Neptune was spread out evenly on the edges of it, what was the orbital period of the outer edges of the primordial disk?

11.4 Precession of a Gyroscope

75. A gyroscope has a 0.5-kg disk that spins at 40 rev/s. The center of mass of the disk is 15 cm from a pivot with a radius of the disk of 10 cm. What is the precession angular velocity?

76. The precession angular velocity of a gyroscope is 1.0 rad/s. If the mass of the rotating disk is 0.4 kg and its radius is 30 cm, and the distance from the center of mass to the pivot is 40 cm, what is the rotation rate in rev/s of the disk?

77. The axis of Earth makes a $23.5°$ angle with a direction perpendicular to the plane of Earth's orbit. As shown below, this axis precesses, making one complete rotation in 25,780 y.
(a) Calculate the change in angular momentum in half this time.
(b) What is the average torque producing this change in angular momentum?
(c) If this torque were created by a pair of forces acting at the most effective point on the equator, what would the magnitude of each force be?



Additional Problems

78. A marble is rolling across the floor at a speed of 7.0 m/s when it starts up a plane inclined at $30°$ to the horizontal. (a) How far along the plane does the marble travel before coming to a rest? (b) How much time elapses while the marble moves up the plane?

79. Repeat the preceding problem replacing the marble with a hollow sphere. Explain the new results.

80. The mass of a hoop of radius 1.0 m is 6.0 kg. It rolls across a horizontal surface with a speed of 10.0 m/s. (a) How much work is required to stop the hoop? (b) If the hoop starts up a surface at $30°$ to the horizontal with a speed of 10.0 m/s, how far along the incline will it travel before stopping and rolling back down?

81. Repeat the preceding problem for a hollow sphere of the same radius and mass and initial speed. Explain the differences in the results.

82. A particle has mass 0.5 kg and is traveling along the line $x = 5.0 \text{ m}$ at 2.0 m/s in the positive *y*-direction. What is the particle's angular momentum about the origin?

83. A 4.0-kg particle moves in a circle of radius 2.0 m. The angular momentum of the particle varies in time according to $l = 5.0t^2$. (a) What is the torque on the particle about the center of the circle at $t = 3.4 \text{ s}$? (b) What is the angular velocity of the particle at $t = 3.4 \text{ s}$?

84. A proton is accelerated in a cyclotron to $5.0 \times 10^6 \text{ m/s}$ in 0.01 s. The proton follows a circular path. If the radius of the cyclotron is 0.5 km, (a) What is the angular momentum of the proton about the center at its maximum speed? (b) What is the torque on the proton about the center as it accelerates to maximum speed?

85. (a) What is the angular momentum of the Moon in its orbit around Earth? (b) How does this angular momentum compare with the angular momentum of the Moon on its axis? Remember that the Moon keeps one side toward Earth at all times.

86. A DVD is rotating at 500 rpm. What is the angular momentum of the DVD if has a radius of 6.0 cm and mass 20.0 g?

87. A potter's disk spins from rest up to 10 rev/s in 15 s. The disk has a mass 3.0 kg and radius 30.0 cm. What is the angular momentum of the disk at $t = 5 \text{ s}, t = 10 \text{ s}$?

88. Suppose you start an antique car by exerting a force of 300 N on its crank for 0.250 s. What is the angular momentum given to the engine if the handle of the crank is 0.300 m from the pivot and the force is exerted to create maximum torque the entire time?

89. A solid cylinder of mass 2.0 kg and radius 20 cm is rotating counterclockwise around a vertical axis through its center at 600 rev/min. A second solid cylinder of the same mass and radius is rotating clockwise around the same vertical axis at 900 rev/min. If the cylinders couple so that they rotate about the same vertical axis, what is the angular velocity of the combination?

90. A boy stands at the center of a platform that is rotating without friction at 1.0 rev/s. The boy holds weights as far from his body as possible. At this position the total moment of inertia of the boy, platform, and weights is $5.0\,\text{kg}\cdot\text{m}^2$. The boy draws the weights in close to his body, thereby decreasing the total moment of inertia to $1.5\,\text{kg}\cdot\text{m}^2$. (a) What is the final angular velocity of the platform? (b) By how much does the rotational kinetic energy increase?

91. Eight children, each of mass 40 kg, climb on a small merry-go-round. They position themselves evenly on the outer edge and join hands. The merry-go-round has a radius of 4.0 m and a moment of inertia $1000.0\,\text{kg}\cdot\text{m}^2$. After the merry-go-round is given an angular velocity of 6.0 rev/min, the children walk inward and stop when they are 0.75 m from the axis of rotation. What is the new angular velocity of the merry-go-round? Assume there is negligible frictional torque on the structure.

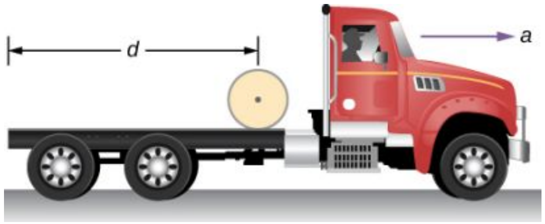
92. A thin meter stick of mass 150 g rotates around an axis perpendicular to the stick's long axis at an angular velocity of 240 rev/min. What is the angular momentum of the stick if the rotation axis (a) passes through the center of the stick? (b) Passes through one end of the stick?

93. A satellite in the shape of a sphere of mass 20,000 kg and radius 5.0 m is spinning about an axis through its center of mass. It has a rotation rate of 8.0 rev/s. Two antennas deploy in the plane of rotation extending from the center of mass of the satellite. Each antenna can be approximated as a rod has mass 200.0 kg and length 7.0 m. What is the new rotation rate of the satellite?

94. A top has moment of inertia $3.2\times10^{-4}\,\text{kg}\cdot\text{m}^2$ and radius 4.0 cm from the center of mass to the pivot point. If it spins at 20.0 rev/s and is precessing, how many revolutions does it precess in 10.0 s?

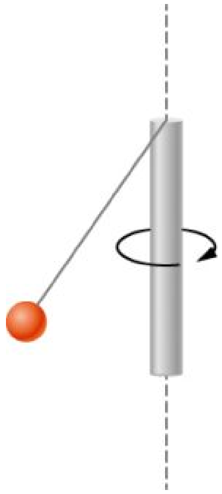
Challenge Problems

95. The truck shown below is initially at rest with solid cylindrical roll of paper sitting on its bed. If the truck moves forward with a uniform acceleration *a*, what distance *s* does it move before the paper rolls off its back end? (*Hint*: If the roll accelerates forward with a', then is accelerates backward relative to the truck with an acceleration $a-a'$. Also, $R\alpha = a - a'$.)



96. A bowling ball of radius 8.5 cm is tossed onto a bowling lane with speed 9.0 m/s. The direction of the toss is to the left, as viewed by the observer, so the bowling ball starts to rotate counterclockwise when in contact with the floor. The coefficient of kinetic friction on the lane is 0.3. (a) What is the time required for the ball to come to the point where it is not slipping? What is the distance d to the point where the ball is rolling without slipping?

97. A small ball of mass 0.50 kg is attached by a massless string to a vertical rod that is spinning as shown below. When the rod has an angular velocity of 6.0 rad/s, the string makes an angle of $30°$ with respect to the vertical. (a) If the angular velocity is increased to 10.0 rad/s, what is the new angle of the string? (b) Calculate the initial and final angular momenta of the ball. (c) Can the rod spin fast enough so that the ball is horizontal?



98. A bug flying horizontally at 1.0 m/s collides and sticks to the end of a uniform stick hanging vertically. After the impact, the stick swings out to a maximum angle of 5.0° from the vertical before rotating back. If the mass of the stick is 10 times that of the bug, calculate the length of the stick.

CHAPTER 12
Static Equilibrium and Elasticity

FIGURE 12.1 Two stilt walkers in standing position. All forces acting on each stilt walker balance out; neither changes its translational motion. In addition, all torques acting on each person balance out, and thus neither of them changes its rotational motion. The result is static equilibrium. (credit: modification of work by Stuart Redler)

CHAPTER OUTLINE

INTRODUCTION In earlier chapters, you learned about forces and Newton's laws for translational motion. You then studied torques and the rotational motion of a body about a fixed axis of rotation. You also learned that static equilibrium means no motion at all and that dynamic equilibrium means motion without acceleration.

In this chapter, we combine the conditions for static translational equilibrium and static rotational equilibrium to describe situations typical for any kind of construction. What type of cable will support a suspension bridge? What type of foundation will support an office building? Will this prosthetic arm function correctly? These are examples of questions that contemporary engineers must be able to answer.

The elastic properties of materials are especially important in engineering applications, including bioengineering. For example, materials that can stretch or compress and then return to their original form or position make good shock absorbers. In this chapter, you will learn about some applications that combine equilibrium with elasticity to construct real structures that last.

12.1 Conditions for Static Equilibrium

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify the physical conditions of static equilibrium.
- Draw a free-body diagram for a rigid body acted on by forces.
- Explain how the conditions for equilibrium allow us to solve statics problems.

We say that a rigid body is in **equilibrium** when both its linear and angular acceleration are zero relative to an inertial frame of reference. This means that a body in equilibrium can be moving, but if so, its linear and angular velocities must be constant. We say that a rigid body is in **static equilibrium** when it is at rest *in our selected frame of reference*. Notice that the distinction between the state of rest and a state of uniform motion is artificial—that is, an object may be at rest in our selected frame of reference, yet to an observer moving at constant velocity relative to our frame, the same object appears to be in uniform motion with constant velocity. Because the motion is *relative*, what is in static equilibrium to us is in dynamic equilibrium to the moving observer, and vice versa. Since the laws of physics are identical for all inertial reference frames, in an inertial frame of reference, there is no distinction between static equilibrium and equilibrium.

According to Newton's second law of motion, the linear acceleration of a rigid body is caused by a net force acting on it, or

$$\sum_k \vec{\mathbf{F}}_k = m\vec{\mathbf{a}}_{\mathrm{CM}}. \tag{12.1}$$

Here, the sum is of all external forces acting on the body, where m is its mass and $\vec{\mathbf{a}}_{\mathrm{CM}}$ is the linear acceleration of its center of mass (a concept we discussed in Linear Momentum and Collisions on linear momentum and collisions). In equilibrium, the linear acceleration is zero. If we set the acceleration to zero in Equation 12.1, we obtain the following equation:

FIRST EQUILIBRIUM CONDITION

The first equilibrium condition for the static equilibrium of a rigid body expresses *translational* equilibrium:

$$\sum_k \vec{\mathbf{F}}_k = \vec{\mathbf{0}}. \tag{12.2}$$

The first equilibrium condition, Equation 12.2, is the equilibrium condition for forces, which we encountered when studying applications of Newton's laws.

This vector equation is equivalent to the following three scalar equations for the components of the net force:

$$\sum_k F_{kx} = 0, \quad \sum_k F_{ky} = 0, \quad \sum_k F_{kz} = 0. \tag{12.3}$$

Analogously to Equation 12.1, we can state that the rotational acceleration $\vec{\boldsymbol{\alpha}}$ of a rigid body about a fixed axis of rotation is caused by the net torque acting on the body, or

$$\sum_k \vec{\boldsymbol{\tau}}_k = I\vec{\boldsymbol{\alpha}}. \tag{12.4}$$

Here I is the rotational inertia of the body in rotation about this axis and the summation is over *all* torques $\vec{\boldsymbol{\tau}}_k$ of external forces in Equation 12.2. In equilibrium, the rotational acceleration is zero. By setting to zero the right-hand side of Equation 12.4, we obtain the second equilibrium condition:

SECOND EQUILIBRIUM CONDITION

The second equilibrium condition for the static equilibrium of a rigid body expresses *rotational* equilibrium:

$$\sum_k \vec{\boldsymbol{\tau}}_k = \vec{\mathbf{0}}. \quad \text{12.5}$$

The second equilibrium condition, Equation 12.5, is the equilibrium condition for torques that we encountered when we studied rotational dynamics. It is worth noting that this equation for equilibrium is generally valid for rotational equilibrium about any axis of rotation (fixed or otherwise). Again, this vector equation is equivalent to three scalar equations for the vector components of the net torque:

$$\sum_k \tau_{kx} = 0, \quad \sum_k \tau_{ky} = 0, \quad \sum_k \tau_{kz} = 0. \quad \text{12.6}$$

The second equilibrium condition means that in equilibrium, there is no net external torque to cause rotation about any axis.

The first and second equilibrium conditions are stated in a particular reference frame. The first condition involves only forces and is therefore independent of the origin of the reference frame. However, the second condition involves torque, which is defined as a cross product, $\vec{\boldsymbol{\tau}}_k = \vec{\mathbf{r}}_k \times \vec{\mathbf{F}}_k$, where the position vector $\vec{\mathbf{r}}_k$ with respect to the axis of rotation of the point where the force is applied enters the equation. Therefore, torque depends on the location of the axis in the reference frame. However, when rotational and translational equilibrium conditions hold simultaneously in one frame of reference, then they also hold in any other inertial frame of reference, so that the net torque about any axis of rotation is still zero. The explanation for this is fairly straightforward.

Suppose vector $\vec{\mathbf{R}}$ is the position of the origin of a new inertial frame of reference S' in the old inertial frame of reference *S*. From our study of relative motion, we know that in the new frame of reference S', the position vector $\vec{\mathbf{r}}'_k$ of the point where the force $\vec{\mathbf{F}}_k$ is applied is related to $\vec{\mathbf{r}}_k$ via the equation

$$\vec{\mathbf{r}}'_k = \vec{\mathbf{r}}_k - \vec{\mathbf{R}}.$$

Now, we can sum all torques $\vec{\boldsymbol{\tau}}'_k = \vec{\mathbf{r}}'_k \times \vec{\mathbf{F}}_k$ of all external forces in a new reference frame, S' :

$$\sum_k \vec{\boldsymbol{\tau}}'_k = \sum_k \vec{\mathbf{r}}'_k \times \vec{\mathbf{F}}_k = \sum_k (\vec{\mathbf{r}}_k - \vec{\mathbf{R}}) \times \vec{\mathbf{F}}_k = \sum_k \vec{\mathbf{r}}_k \times \vec{\mathbf{F}}_k - \sum_k \vec{\mathbf{R}} \times \vec{\mathbf{F}}_k = \sum_k \vec{\boldsymbol{\tau}}_k - \vec{\mathbf{R}} \times \sum_k \vec{\mathbf{F}}_k = \vec{\mathbf{0}}.$$

In the final step in this chain of reasoning, we used the fact that in equilibrium in the old frame of reference, *S*, the first term vanishes because of Equation 12.5 and the second term vanishes because of Equation 12.2. Hence, we see that the net torque in any inertial frame of reference S' is zero, provided that both conditions for equilibrium hold in an inertial frame of reference *S*.

The practical implication of this is that when applying equilibrium conditions for a rigid body, we are free to choose any point as the origin of the reference frame. Our choice of reference frame is dictated by the physical specifics of the problem we are solving. In one frame of reference, the mathematical form of the equilibrium conditions may be quite complicated, whereas in another frame, the same conditions may have a simpler mathematical form that is easy to solve. The origin of a selected frame of reference is called the pivot point.

In the most general case, equilibrium conditions are expressed by the six scalar equations (Equation 12.3 and Equation 12.6). For planar equilibrium problems with rotation about a fixed axis, which we consider in this chapter, we can reduce the number of equations to three. The standard procedure is to adopt a frame of reference where the *z*-axis is the axis of rotation. With this choice of axis, the net torque has only a *z*-component, all forces that have non-zero torques lie in the *xy*-plane, and therefore contributions to the net torque come from only the *x*- and *y*-components of external forces. Thus, for planar problems with the axis of rotation perpendicular to the *xy*-plane, we have the following three equilibrium conditions for forces and torques:

$$F_{1x} + F_{2x} + \cdots + F_{Nx} = 0 \quad \text{12.7}$$

$$F_{1y} + F_{2y} + \cdots + F_{Ny} = 0 \quad \text{12.8}$$

$$\tau_1 + \tau_2 + \cdots + \tau_N = 0 \quad \text{12.9}$$

where the summation is over all *N* external forces acting on the body and over their torques. In Equation 12.9, we simplified the notation by dropping the subscript *z*, but we understand here that the summation is over all contributions along the *z*-axis, which is the axis of rotation. In Equation 12.9, the *z*-component of torque $\vec{\tau}_k$ from the force $\vec{\mathbf{F}}_k$ is

$$\tau_k = r_k F_k \sin\theta \tag{12.10}$$

where r_k is the length of the lever arm of the force and F_k is the magnitude of the force (as you saw in Fixed-Axis Rotation). The angle θ is the angle between vectors $\vec{\mathbf{r}}_k$ and $\vec{\mathbf{F}}_k$, measuring *from vector* $\vec{\mathbf{r}}_k$ *to vector* $\vec{\mathbf{F}}_k$ in the *counterclockwise* direction (Figure 12.2). When using Equation 12.10, we often compute the magnitude of torque and assign its sense as either positive $(+)$ or negative $(-)$, depending on the direction of rotation caused by this torque alone. In Equation 12.9, net torque is the sum of terms, with each term computed from Equation 12.10, and each term must have the correct *sense*. Similarly, in Equation 12.7, we assign the $+$ sign to force components in the $+$ *x*-direction and the $-$ sign to components in the $-$ *x*-direction. The same rule must be consistently followed in Equation 12.8, when computing force components along the *y*-axis.

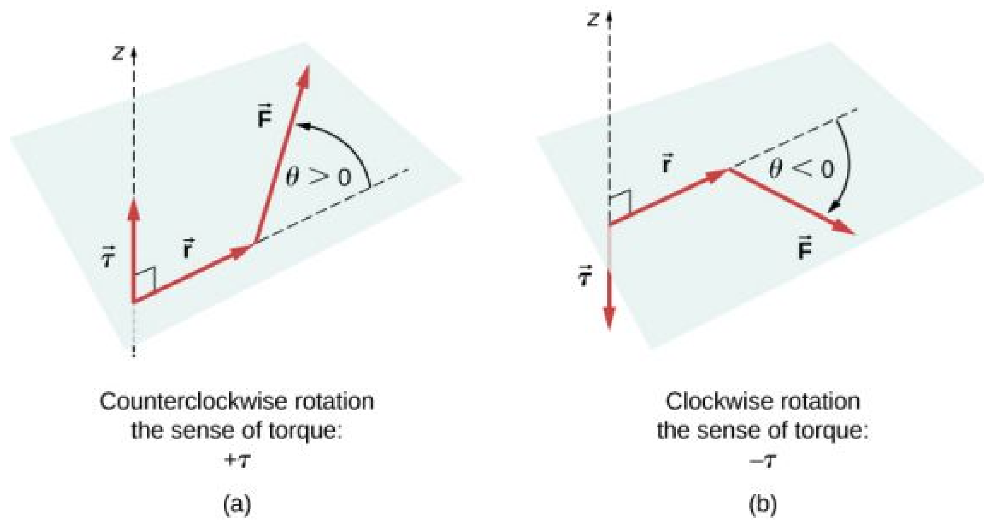


FIGURE 12.2 Torque of a force: (a) When the torque of a force causes counterclockwise rotation about the axis of rotation, we say that its *sense* is positive, which means the torque vector is parallel to the axis of rotation. (b) When torque of a force causes clockwise rotation about the axis, we say that its sense is negative, which means the torque vector is antiparallel to the axis of rotation.

INTERACTIVE

View this demonstration (https://openstax.org/l/21rigsquare) to see two forces act on a rigid square in two dimensions. At all times, the static equilibrium conditions given by Equation 12.7 through Equation 12.9 are satisfied. You can vary magnitudes of the forces and their lever arms and observe the effect these changes have on the square.

In many equilibrium situations, one of the forces acting on the body is its weight. In free-body diagrams, the weight vector is attached to the **center of gravity** of the body. For all practical purposes, the center of gravity is identical to the center of mass, as you learned in Linear Momentum and Collisions on linear momentum and collisions. Only in situations where a body has a large spatial extension so that the gravitational field is nonuniform throughout its volume, are the center of gravity and the center of mass located at different points. In practical situations, however, even objects as large as buildings or cruise ships are located in a uniform gravitational field on Earth's surface, where the acceleration due to gravity has a constant magnitude of $g = 9.8\ \text{m/s}^2$. In these situations, the center of gravity is identical to the center of mass. Therefore, throughout this chapter, we use the center of mass (CM) as the point where the weight vector is attached. Recall that the CM has a special physical meaning: When an external force is applied to a body at exactly its CM, the body as a whole undergoes translational motion and such a force does not cause rotation.

When the CM is located off the axis of rotation, a net **gravitational torque** occurs on an object. Gravitational torque is the torque caused by weight. This gravitational torque may rotate the object if there is no support present to balance it. The magnitude of the gravitational torque depends on how far away from the pivot the CM is located. For example, in the case of a tipping truck (Figure 12.3), the pivot is located on the line where the tires make contact with the road's surface. If the CM is located high above the road's surface, the gravitational torque may be large enough to turn the truck over. Passenger cars with a low-lying CM, close to the pavement, are more resistant to tipping over than are trucks.

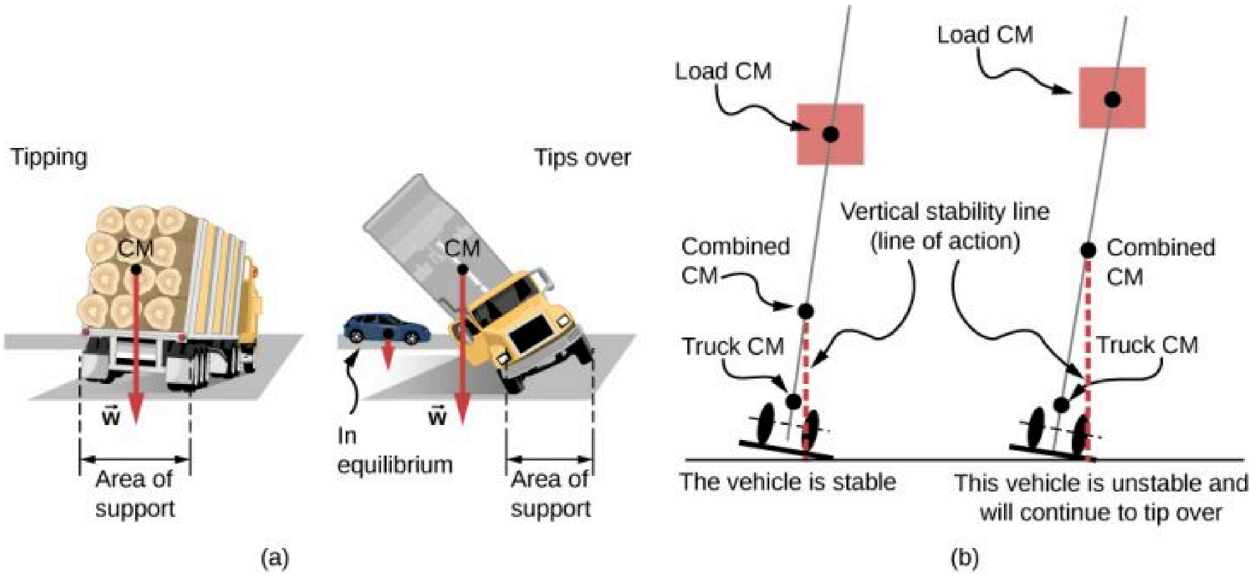


FIGURE 12.3 The distribution of mass affects the position of the center of mass (CM), where the weight vector $\vec{w}$ is attached. If the center of gravity is within the area of support, the truck returns to its initial position after tipping [see the left panel in (b)]. But if the center of gravity lies outside the area of support, the truck turns over [see the right panel in (b)]. Both vehicles in (b) are out of equilibrium. Notice that the car in (a) is in equilibrium: The low location of its center of gravity makes it hard to tip over.

INTERACTIVE

If you tilt a box so that one edge remains in contact with the table beneath it, then one edge of the base of support becomes a pivot. As long as the center of gravity of the box remains over the base of support, gravitational torque rotates the box back toward its original position of stable equilibrium. When the center of gravity moves outside of the base of support, gravitational torque rotates the box in the opposite direction, and the box rolls over. View this demonstration (https://openstax.org/l/21unstable) to experiment with stable and unstable positions of a box.

EXAMPLE 12.1

Center of Gravity of a Car

A passenger car with a 2.5-m wheelbase has 52% of its weight on the front wheels on level ground, as illustrated in Figure 12.4. Where is the CM of this car located with respect to the rear axle?

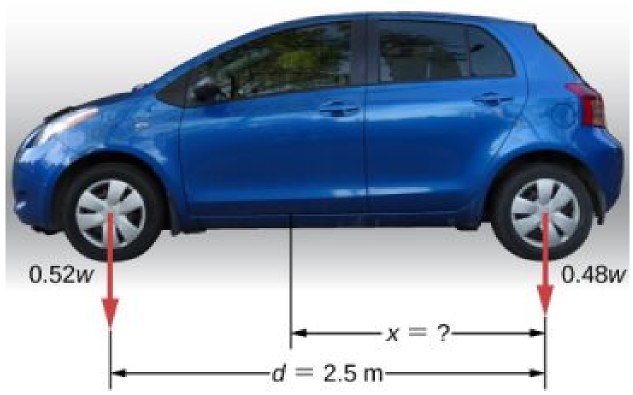


FIGURE 12.4 The weight distribution between the axles of a car. Where is the center of gravity located? (credit "car": modification of work by Jane Whitney)

Strategy

We do not know the weight *w* of the car. All we know is that when the car rests on a level surface, 0.52*w* pushes down on the surface at contact points of the front wheels and 0.48*w* pushes down on the surface at contact points of the rear wheels. Also, the contact points are separated from each other by the distance $d = 2.5\text{ m}$. At these contact points, the car experiences normal reaction forces with magnitudes $F_{\text{F}} = 0.52w$ and $F_{\text{R}} = 0.48w$ on the front and rear axles, respectively. We also know that the car is an example of a rigid body in equilibrium whose entire weight *w* acts at its CM. The CM is located somewhere between the points where the normal reaction forces act, somewhere at a distance *x* from the point where F_R acts. Our task is to find *x*. Thus, we identify three forces acting on the body (the car), and we can draw a free-body diagram for the extended rigid body, as shown in Figure 12.5.

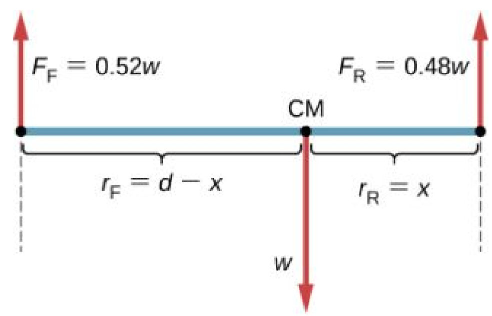


FIGURE 12.5 The free-body diagram for the car clearly indicates force vectors acting on the car and distances to the center of mass (CM). When CM is selected as the pivot point, these distances are lever arms of normal reaction forces. Notice that vector magnitudes and lever arms do not need to be drawn to scale, but all quantities of relevance must be clearly labeled.

We are almost ready to write down equilibrium conditions Equation 12.7 through Equation 12.9 for the car, but first we must decide on the reference frame. Suppose we choose the *x*-axis along the length of the car, the *y*-axis vertical, and the *z*-axis perpendicular to this *xy*-plane. With this choice we only need to write Equation 12.7 and Equation 12.9 because all the *y*-components are identically zero. Now we need to decide on the location of the pivot point. We can choose any point as the location of the axis of rotation (*z*-axis). Suppose we place the axis of rotation at CM, as indicated in the free-body diagram for the car. At this point, we are ready to write the equilibrium conditions for the car.

Solution

Each equilibrium condition contains only three terms because there are $N = 3$ forces acting on the car. The first equilibrium condition, Equation 12.7, reads

$$+F_{\text{F}} - w + F_{\text{R}} = 0. \tag{12.11}$$

This condition is trivially satisfied because when we substitute the data, Equation 12.11 becomes $+0.52w - w + 0.48w = 0$. The second equilibrium condition, Equation 12.9, reads

$$\tau_{\text{F}} + \tau_w + \tau_{\text{R}} = 0 \tag{12.12}$$

where τ_F is the torque of force F_F, τ_w is the gravitational torque of force *w*, and τ_R is the torque of force F_R. When the pivot is located at CM, the gravitational torque is identically zero because the lever arm of the weight with respect to an axis that passes through CM is zero. The lines of action of both normal reaction forces are perpendicular to their lever arms, so in Equation 12.10, we have $|\sin\theta| = 1$ for both forces. From the free-body diagram, we read that torque τ_F causes clockwise rotation about the pivot at CM, so its sense is negative; and torque τ_R causes counterclockwise rotation about the pivot at CM, so its sense is positive. With this information, we write the second equilibrium condition as

$$-r_F F_F + r_R F_R = 0. \tag{12.13}$$

With the help of the free-body diagram, we identify the force magnitudes $F_R = 0.48w$ and $F_F = 0.52w$, and their corresponding lever arms $r_R = x$ and $r_F = d - x$. We can now write the second equilibrium condition, Equation 12.13, explicitly in terms of the unknown distance *x*:

$$-0.52(d - x)w + 0.48xw = 0. \tag{12.14}$$

Here the weight *w* cancels and we can solve the equation for the unknown position *x* of the CM. The answer is $x = 0.52d = 0.52(2.5\text{ m}) = 1.3\text{ m}$.

Solution

Choosing the pivot at the position of the front axle does not change the result. The free-body diagram for this pivot location is presented in Figure 12.6. For this choice of pivot point, the second equilibrium condition is

$$-r_w w + r_R F_R = 0. \tag{12.15}$$

When we substitute the quantities indicated in the diagram, we obtain

$$-(d - x)w + 0.48dw = 0. \tag{12.16}$$

The answer obtained by solving Equation 12.13 is, again, $x = 0.52d = 1.3\text{ m}$.

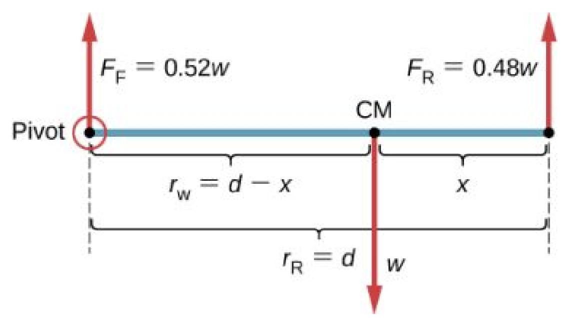


FIGURE 12.6 The equivalent free-body diagram for the car; the pivot is clearly indicated.

Significance

This example shows that when solving static equilibrium problems, we are free to choose the pivot location. For different choices of the pivot point we have different sets of equilibrium conditions to solve. However, all choices lead to the same solution to the problem.

CHECK YOUR UNDERSTANDING 12.1

Solve Example 12.1 by choosing the pivot at the location of the rear axle.

CHECK YOUR UNDERSTANDING 12.2

Explain which one of the following situations satisfies both equilibrium conditions: (a) a tennis ball that does not spin as it travels in the air; (b) a pelican that is gliding in the air at a constant velocity at one altitude; or (c) a crankshaft in the engine of a parked car.

A special case of static equilibrium occurs when all external forces on an object act at or along the axis of rotation or when the spatial extension of the object can be disregarded. In such a case, the object can be effectively treated like a point mass. In this special case, we need not worry about the second equilibrium condition, Equation 12.9, because all torques are identically zero and the first equilibrium condition (for forces) is the only condition to be satisfied. The free-body diagram and problem-solving strategy for this special case were outlined in Newton's Laws of Motion and Applications of Newton's Laws. You will see a typical equilibrium situation involving only the first equilibrium condition in the next example.

INTERACTIVE

View this demonstration (https://openstax.org/l/21pulleyknot) to see three weights that are connected by strings over pulleys and tied together in a knot. You can experiment with the weights to see how they affect the equilibrium position of the knot and, at the same time, see the vector-diagram representation of the first equilibrium condition at work.

EXAMPLE 12.2

A Breaking Tension

A small pan of mass 42.0 g is supported by two strings, as shown in Figure 12.7. The maximum tension that the string can support is 2.80 N. Mass is added gradually to the pan until one of the strings snaps. Which string is it? How much mass must be added for this to occur?

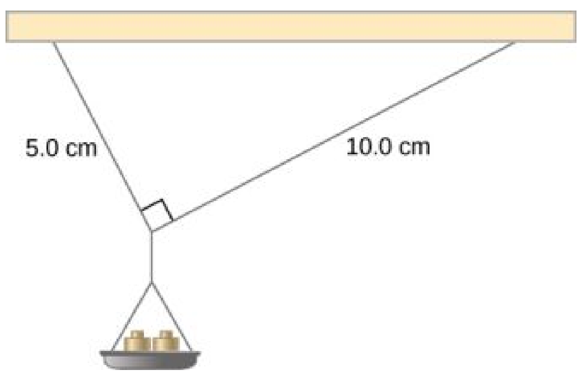


FIGURE 12.7 Mass is added gradually to the pan until one of the strings snaps.

Strategy

This mechanical system consisting of strings, masses, and the pan is in static equilibrium. Specifically, the knot that ties the strings to the pan is in static equilibrium. The knot can be treated as a point; therefore, we need only the first equilibrium condition. The three forces pulling at the knot are the tension $\vec{\mathbf{T}}_1$ in the 5.0-cm string, the tension $\vec{\mathbf{T}}_2$ in the 10.0-cm string, and the weight $\vec{\mathbf{w}}$ of the pan holding the masses. We adopt a rectangular coordinate system with the *y*-axis pointing opposite to the direction of gravity and draw the free-body diagram for the knot (see Figure 12.8). To find the tension components, we must identify the direction angles α_1 and α_2 that the strings make with the horizontal direction that is the *x*-axis. As you can see in Figure 12.7, the strings make two sides of a right triangle. We can use the Pythagorean theorem to solve this triangle, shown in Figure 12.8, and find the sine and cosine of the angles α_1 and α_2. Then we can resolve the tensions into their rectangular components, substitute in the first condition for equilibrium (Equation 12.7 and Equation 12.8), and solve for the tensions in the strings. The string with a greater tension will break first.

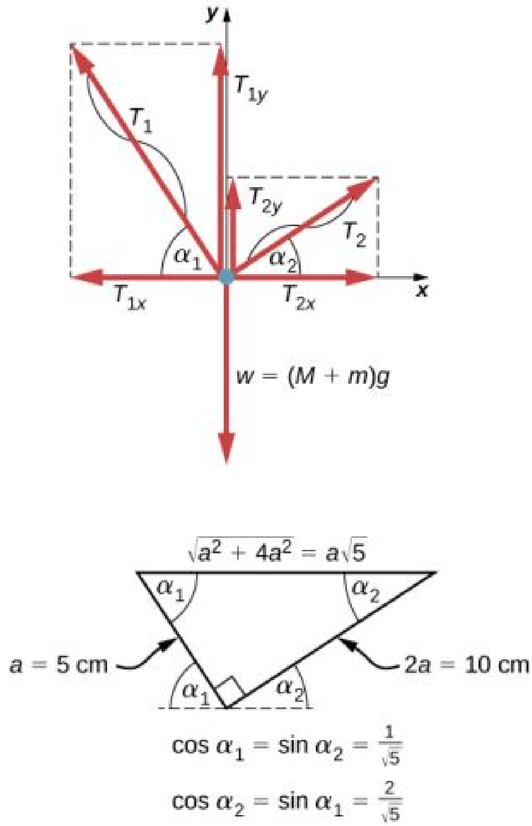


FIGURE 12.8 Free-body diagram for the knot in Example 12.2.

Solution

The weight w pulling on the knot is due to the mass M of the pan and mass m added to the pan, or $w = (M + m)g$. With the help of the free-body diagram in Figure 12.8, we can set up the equilibrium conditions for the knot:

$$\begin{aligned} \text{in the } x\text{-direction,} \qquad -T_{1x} + T_{2x} &= 0 \\ \text{in the } y\text{-direction,} \qquad +T_{1y} + T_{2y} - w &= 0. \end{aligned}$$

From the free-body diagram, the magnitudes of components in these equations are

$$\begin{aligned} T_{1x} &= T_1 \cos\alpha_1 = T_1/\sqrt{5}, \qquad T_{1y} = T_1 \sin\alpha_1 = 2T_1/\sqrt{5} \\ T_{2x} &= T_2 \cos\alpha_2 = 2T_2/\sqrt{5}, \qquad T_{2y} = T_2 \sin\alpha_2 = T_2/\sqrt{5}. \end{aligned}$$

We substitute these components into the equilibrium conditions and simplify. We then obtain two equilibrium equations for the tensions:

$$\begin{aligned} \text{in } x\text{-direction,} \qquad T_1 &= 2T_2 \\ \text{in } y\text{-direction,} \qquad \frac{2T_1}{\sqrt{5}} + \frac{T_2}{\sqrt{5}} &= (M + m)g. \end{aligned}$$

The equilibrium equation for the x-direction tells us that the tension T_1 in the 5.0-cm string is twice the tension T_2 in the 10.0-cm string. Therefore, the shorter string will snap. When we use the first equation to eliminate T_2 from the second equation, we obtain the relation between the mass m on the pan and the tension T_1 in the shorter string:

$$2.5T_1/\sqrt{5} = (M + m)g.$$

The string breaks when the tension reaches the critical value of $T_1 = 2.80\text{ N}$. The preceding equation can be solved for the critical mass m that breaks the string:

$$m = \frac{2.5}{\sqrt{5}} \frac{T_1}{g} - M = \frac{2.5}{\sqrt{5}} \frac{2.80\,\text{N}}{9.8\,\text{m/s}^2} - 0.042\,\text{kg} = 0.277\,\text{kg} = 277.0\,\text{g}.$$

Significance

Suppose that the mechanical system considered in this example is attached to a ceiling inside an elevator going up. As long as the elevator moves up at a constant speed, the result stays the same because the weight w does not change. If the elevator moves up with acceleration, the critical mass is smaller because the weight of $M + m$ becomes larger by an apparent weight due to the acceleration of the elevator. Still, in all cases the shorter string breaks first.

12.2 Examples of Static Equilibrium

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify and analyze static equilibrium situations
- Set up a free-body diagram for an extended object in static equilibrium
- Set up and solve static equilibrium conditions for objects in equilibrium in various physical situations

All examples in this chapter are planar problems. Accordingly, we use equilibrium conditions in the component form of Equation 12.7 to Equation 12.9. We introduced a problem-solving strategy in Example 12.1 to illustrate the physical meaning of the equilibrium conditions. Now we generalize this strategy in a list of steps to follow when solving static equilibrium problems for extended rigid bodies. We proceed in five practical steps.

PROBLEM-SOLVING STRATEGY

Static Equilibrium

1. Identify the object to be analyzed. For some systems in equilibrium, it may be necessary to consider more than one object. Identify all forces acting on the object. Identify the questions you need to answer. Identify the information given in the problem. In realistic problems, some key information may be implicit in the situation rather than provided explicitly.
2. Set up a free-body diagram for the object. (a) Choose the *xy*-reference frame for the problem. Draw a free-body diagram for the object, including only the forces that act on it. When suitable, represent the forces in terms of their components in the chosen reference frame. As you do this for each force, cross out the original force so that you do not erroneously include the same force twice in equations. Label all forces—you will need this for correct computations of net forces in the *x*- and *y*-directions. For an unknown force, the direction must be assigned arbitrarily; think of it as a 'working direction' or 'suspected direction.' The correct direction is determined by the sign that you obtain in the final solution. A plus sign $(+)$ means that the working direction is the actual direction. A minus sign $(-)$ means that the actual direction is opposite to the assumed working direction. (b) Choose the location of the rotation axis; in other words, choose the pivot point with respect to which you will compute torques of acting forces. On the free-body diagram, indicate the location of the pivot and the lever arms of acting forces—you will need this for correct computations of torques. In the selection of the pivot, keep in mind that the pivot can be placed anywhere you wish, but the guiding principle is that the best choice will simplify as much as possible the calculation of the net torque along the rotation axis.
3. Set up the equations of equilibrium for the object. (a) Use the free-body diagram to write a correct equilibrium condition Equation 12.7 for force components in the *x*-direction. (b) Use the free-body diagram to write a correct equilibrium condition Equation 12.11 for force components in the *y*-direction. (c) Use the free-body diagram to write a correct equilibrium condition Equation 12.9 for torques along the axis of rotation. Use Equation 12.10 to evaluate torque magnitudes and senses.
4. Simplify and solve the system of equations for equilibrium to obtain unknown quantities. At this point, your work involves algebra only. Keep in mind that the number of equations must be the same as the number of unknowns. If the number of unknowns is larger than the number of equations, the problem cannot be solved.
5. Evaluate the expressions for the unknown quantities that you obtained in your solution. Your final answers

should have correct numerical values and correct physical units. If they do not, then use the previous steps to track back a mistake to its origin and correct it. Also, you may independently check for your numerical answers by shifting the pivot to a different location and solving the problem again, which is what we did in Example 12.1.

Note that setting up a free-body diagram for a rigid-body equilibrium problem is the most important component in the solution process. Without the correct setup and a correct diagram, you will not be able to write down correct conditions for equilibrium. Also note that a free-body diagram for an extended rigid body that may undergo rotational motion is different from a free-body diagram for a body that experiences only translational motion (as you saw in the chapters on Newton's laws of motion). In translational dynamics, a body is represented as its CM, where all forces on the body are attached and no torques appear. This does not hold true in rotational dynamics, where an extended rigid body cannot be represented by one point alone. The reason for this is that in analyzing rotation, we must identify torques acting on the body, and torque depends both on the acting force and on its lever arm. Here, the free-body diagram for an extended rigid body helps us identify external torques.

EXAMPLE 12.3

The Torque Balance

Three masses are attached to a uniform meter stick, as shown in Figure 12.9. The mass of the meter stick is 150.0 g and the masses to the left of the fulcrum are $m_1 = 50.0\text{ g}$ and $m_2 = 75.0\text{ g}$. Find the mass m_3 that balances the system when it is attached at the right end of the stick, and the normal reaction force at the fulcrum when the system is balanced.

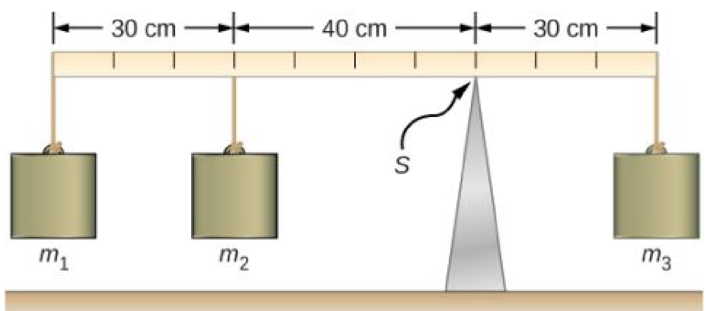


FIGURE 12.9 In a torque balance, a horizontal beam is supported at a fulcrum (indicated by *S*) and masses are attached to both sides of the fulcrum. The system is in static equilibrium when the beam does not rotate. It is balanced when the beam remains level.

Strategy

For the arrangement shown in the figure, we identify the following five forces acting on the meter stick:

$w_1 = m_1 g$ is the weight of mass m_1; $w_2 = m_2 g$ is the weight of mass m_2;

$w = mg$ is the weight of the entire meter stick; $w_3 = m_3 g$ is the weight of unknown mass m_3;

F_S is the normal reaction force at the support point *S*.

We choose a frame of reference where the direction of the *y*-axis is the direction of gravity, the direction of the *x*-axis is along the meter stick, and the axis of rotation (the *z*-axis) is perpendicular to the *x*-axis and passes through the support point *S*. In other words, we choose the pivot at the point where the meter stick touches the support. This is a natural choice for the pivot because this point does not move as the stick rotates. Now we are ready to set up the free-body diagram for the meter stick. We indicate the pivot and attach five vectors representing the five forces along the line representing the meter stick, locating the forces with respect to the pivot Figure 12.10. At this stage, we can identify the lever arms of the five forces given the information provided in the problem. For the three hanging masses, the problem is explicit about their locations along the stick, but the information about the location of the weight *w* is given implicitly. The key word here is "uniform." We know from our previous studies that the CM of a uniform stick is located at its midpoint, so this is where we attach the weight *w*, at the 50-cm mark.

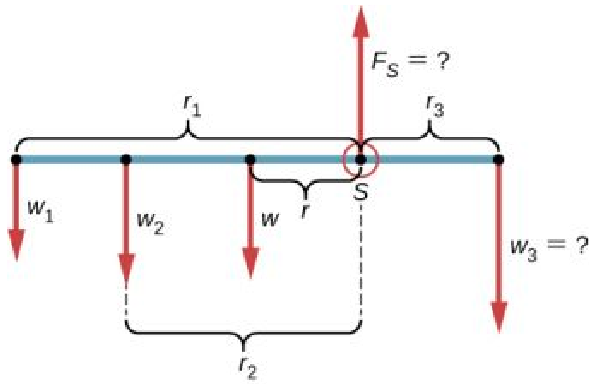


FIGURE 12.10 Free-body diagram for the meter stick. The pivot is chosen at the support point *S*.

Solution

With Figure 12.9 and Figure 12.10 for reference, we begin by finding the lever arms of the five forces acting on the stick:

$$\begin{aligned} r_1 &= 30.0\,\text{cm} + 40.0\,\text{cm} = 70.0\,\text{cm} \\ r_2 &= 40.0\,\text{cm} \\ r &= 50.0\,\text{cm} - 30.0\,\text{cm} = 20.0\,\text{cm} \\ r_S &= 0.0\,\text{cm (because } F_S \text{ is attached at the pivot)} \\ r_3 &= 30.0\,\text{cm.} \end{aligned}$$

Now we can find the five torques with respect to the chosen pivot:

$$\begin{aligned} \tau_1 &= +r_1 w_1 \sin 90° = +r_1 m_1 g && \text{(counterclockwise rotation, positive sense)} \\ \tau_2 &= +r_2 w_2 \sin 90° = +r_2 m_2 g && \text{(counterclockwise rotation, positive sense)} \\ \tau &= +rw \sin 90° = +rmg && \text{(gravitational torque)} \\ \tau_S &= r_S F_S \sin \theta_S = 0 && \text{(because } r_S = 0\,\text{cm)} \\ \tau_3 &= -r_3 w_3 \sin 90° = -r_3 m_3 g && \text{(clockwise rotation, negative sense)} \end{aligned}$$

The second equilibrium condition (equation for the torques) for the meter stick is

$$\tau_1 + \tau_2 + \tau + \tau_S + \tau_3 = 0.$$

When substituting torque values into this equation, we can omit the torques giving zero contributions. In this way the second equilibrium condition is

$$+r_1 m_1 g + r_2 m_2 g + rmg - r_3 m_3 g = 0. \tag{12.17}$$

Selecting the $+y$-direction to be parallel to $\vec{F}_S$, the first equilibrium condition for the stick is

$$-w_1 - w_2 - w + F_S - w_3 = 0.$$

Substituting the forces, the first equilibrium condition becomes

$$-m_1 g - m_2 g - mg + F_S - m_3 g = 0. \tag{12.18}$$

We solve these equations simultaneously for the unknown values m_3 and F_S. In Equation 12.17, we cancel the g factor and rearrange the terms to obtain

$$r_3 m_3 = r_1 m_1 + r_2 m_2 + rm.$$

To obtain m_3 we divide both sides by r_3, so we have

$$\begin{aligned} m_3 &= \frac{r_1}{r_3} m_1 + \frac{r_2}{r_3} m_2 + \frac{r}{r_3} m \\ &= \frac{70}{30}(50.0\text{ g}) + \frac{40}{30}(75.0\text{ g}) + \frac{20}{30}(150.0\text{ g}) = 316.0\tfrac{2}{3}\text{ g} \simeq 317\text{ g}. \end{aligned} \tag{12.19}$$

To find the normal reaction force, we rearrange the terms in Equation 12.18, converting grams to kilograms:

$$\begin{aligned} F_S &= (m_1 + m_2 + m + m_3)g \\ &= (50.0 + 75.0 + 150.0 + 316.7) \times 10^{-3}\text{kg} \times 9.8\,\frac{\text{m}}{\text{s}^2} = 5.8\text{ N}. \end{aligned} \tag{12.20}$$

Significance

Notice that Equation 12.17 is independent of the value of *g*. The torque balance may therefore be used to measure mass, since variations in *g*-values on Earth's surface do not affect these measurements. This is not the case for a spring balance because it measures the force.

CHECK YOUR UNDERSTANDING 12.3

Repeat Example 12.3 using the left end of the meter stick to calculate the torques; that is, by placing the pivot at the left end of the meter stick.

In the next example, we show how to use the first equilibrium condition (equation for forces) in the vector form given by Equation 12.7 and Equation 12.8. We present this solution to illustrate the importance of a suitable choice of reference frame. Although all inertial reference frames are equivalent and numerical solutions obtained in one frame are the same as in any other, an unsuitable choice of reference frame can make the solution quite lengthy and convoluted, whereas a wise choice of reference frame makes the solution straightforward. We show this in the equivalent solution to the same problem. This particular example illustrates an application of static equilibrium to biomechanics.

EXAMPLE 12.4

Forces in the Forearm

A weightlifter is holding a 50.0-lb weight (equivalent to 222.4 N) with his forearm, as shown in Figure 12.11. His forearm is positioned at $\beta = 60^\circ$ with respect to his upper arm. The forearm is supported by a contraction of the biceps muscle, which causes a torque around the elbow. Assuming that the tension in the biceps acts along the vertical direction given by gravity, what tension must the muscle exert to hold the forearm at the position shown? What is the force on the elbow joint? Assume that the forearm's weight is negligible. Give your final answers in SI units.

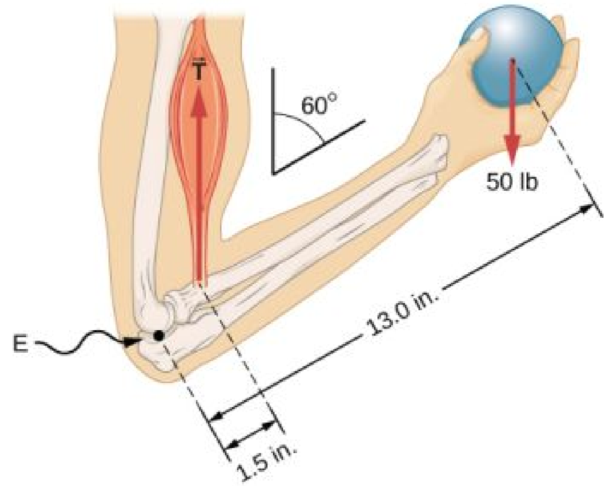


FIGURE 12.11 The forearm is rotated around the elbow (*E*) by a contraction of the biceps muscle, which causes tension $\vec{\mathbf{T}}_{\mathrm{M}}$.

Strategy

We identify three forces acting on the forearm: the unknown force $\vec{\mathbf{F}}$ at the elbow; the unknown tension $\vec{\mathbf{T}}_{\mathrm{M}}$ in the muscle; and the weight $\vec{\mathbf{w}}$ with magnitude $w = 50\ \mathrm{lb}$. We adopt the frame of reference with the *x*-axis along the forearm and the pivot at the elbow. The vertical direction is the direction of the weight, which is the same as the direction of the upper arm. The *x*-axis makes an angle $\beta = 60°$ with the vertical. The *y*-axis is perpendicular to the *x*-axis. Now we set up the free-body diagram for the forearm. First, we draw the axes, the pivot, and the three vectors representing the three identified forces. Then we locate the angle β and represent each force by its *x*- and *y*-components, remembering to cross out the original force vector to avoid double counting. Finally, we label the forces and their lever arms. The free-body diagram for the forearm is shown in Figure 12.12. At this point, we are ready to set up equilibrium conditions for the forearm. Each force has *x*- and *y*-components; therefore, we have two equations for the first equilibrium condition, one equation for each component of the net force acting on the forearm.

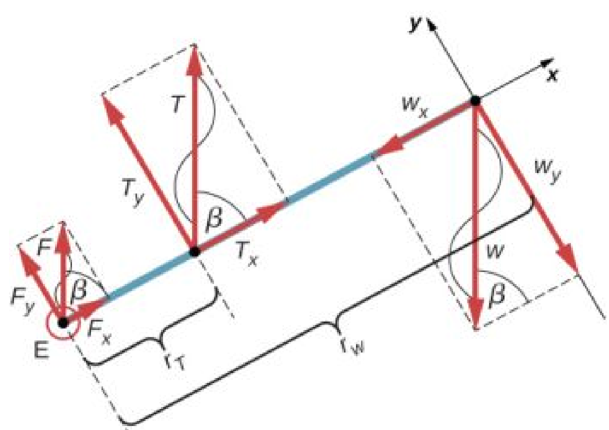


FIGURE 12.12 Free-body diagram for the forearm: The pivot is located at point *E* (elbow).

Notice that in our frame of reference, contributions to the second equilibrium condition (for torques) come only from the *y*-components of the forces because the *x*-components of the forces are all parallel to their lever arms, so that for any of them we have $\sin\theta = 0$ in Equation 12.10. For the *y*-components we have $\theta = \pm 90°$ in Equation 12.10. Also notice that the torque of the force at the elbow is zero because this force is attached at the pivot. So the contribution to the net torque comes only from the torques of T_y and of w_y.

Solution

We see from the free-body diagram that the *x*-component of the net force satisfies the equation

$$+F_x + T_x - w_x = 0 \tag{12.21}$$

and the y-component of the net force satisfies

$$+F_y + T_y - w_y = 0. \tag{12.22}$$

Equation 12.21 and Equation 12.22 are two equations of the first equilibrium condition (for forces). Next, we read from the free-body diagram that the net torque along the axis of rotation is

$$+r_T T_y - r_w w_y = 0. \tag{12.23}$$

Equation 12.23 is the second equilibrium condition (for torques) for the forearm. The free-body diagram shows that the lever arms are $r_T = 1.5\text{ in.}$ and $r_w = 13.0\text{ in.}$ At this point, we do not need to convert inches into SI units, because as long as these units are consistent in Equation 12.23, they cancel out. Using the free-body diagram again, we find the magnitudes of the component forces:

$$\begin{aligned}
F_x &= F\cos\beta = F\cos 60° = F/2 \\
T_x &= T\cos\beta = T\cos 60° = T/2 \\
w_x &= w\cos\beta = w\cos 60° = w/2 \\
F_y &= F\sin\beta = F\sin 60° = F\sqrt{3}/2 \\
T_y &= T\sin\beta = T\sin 60° = T\sqrt{3}/2 \\
w_y &= w\sin\beta = w\sin 60° = w\sqrt{3}/2.
\end{aligned}$$

We substitute these magnitudes into Equation 12.21, Equation 12.22, and Equation 12.23 to obtain, respectively,

$$\begin{aligned}
F/2 + T/2 - w/2 &= 0 \\
F\sqrt{3}/2 + T\sqrt{3}/2 - w\sqrt{3}/2 &= 0 \\
r_T T\sqrt{3}/2 - r_w w\sqrt{3}/2 &= 0.
\end{aligned}$$

When we simplify these equations, we see that we are left with only two independent equations for the two unknown force magnitudes, *F* and *T*, because Equation 12.21 for the *x*-component is equivalent to Equation 12.22 for the *y*-component. In this way, we obtain the first equilibrium condition for forces

$$F + T - w = 0 \tag{12.24}$$

and the second equilibrium condition for torques

$$r_T T - r_w w = 0. \tag{12.25}$$

The magnitude of tension in the muscle is obtained by solving Equation 12.25:

$$T = \frac{r_w}{r_T} w = \frac{13.0}{1.5}(50\text{ lb}) = 433\frac{1}{3}\text{lb} \simeq 433.3\text{ lb}.$$

The force at the elbow is obtained by solving Equation 12.24:

$$F = w - T = 50.0\text{ lb} - 433.3\text{ lb} = -383.3\text{ lb}.$$

The negative sign in the equation tells us that the actual force at the elbow is antiparallel to the working direction adopted for drawing the free-body diagram. In the final answer, we convert the forces into SI units of force. The answer is

$$\begin{aligned}
F &= 383.3\text{ lb} = 383.3(4.448\text{ N}) = 1705\text{ N downward} \\
T &= 433.3\text{ lb} = 433.3(4.448\text{ N}) = 1927\text{ N upward.}
\end{aligned}$$

Significance

Two important issues here are worth noting. The first concerns conversion into SI units, which can be done at the very end of the solution as long as we keep consistency in units. The second important issue concerns the hinge joints such as the elbow. In the initial analysis of a problem, hinge joints should always be assumed to exert a force in an *arbitrary direction*, and then you must solve for all components of a hinge force independently. In this example, the elbow force happens to be vertical because the problem assumes the tension by the biceps to be vertical as well. Such a simplification, however, is not a general rule.

Solution

Suppose we adopt a reference frame with the direction of the *y*-axis along the 50-lb weight and the pivot placed at the elbow. In this frame, all three forces have only *y*-components, so we have only one equation for the first equilibrium condition (for forces). We draw the free-body diagram for the forearm as shown in Figure 12.13, indicating the pivot, the acting forces and their lever arms with respect to the pivot, and the angles θ_T and θ_w that the forces $\vec{\mathbf{T}}_{\mathrm{M}}$ and $\vec{\mathbf{w}}$ (respectively) make with their lever arms. In the definition of torque given by Equation 12.10, the angle θ_T is the direction angle of the vector $\vec{T}_{\mathrm{M}}$, counted *counterclockwise* from the radial direction of the lever arm that always points away from the pivot. By the same convention, the angle θ_w is measured *counterclockwise* from the radial direction of the lever arm to the vector $\vec{\mathbf{w}}$. Done this way, the non-zero torques are most easily computed by directly substituting into Equation 12.10 as follows:

$$\tau_T = r_T T\sin\theta_T = r_T T\sin\beta = r_T T\sin 60° = +r_T T\sqrt{3}/2$$
$$\tau_w = r_w w\sin\theta_w = r_w w\sin(\beta + 180°) = -r_w w\sin\beta = -r_w w\sqrt{3}/2.$$

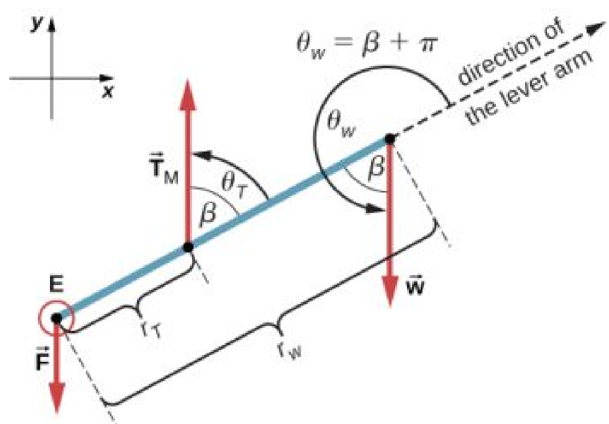


FIGURE 12.13 Free-body diagram for the forearm for the equivalent solution. The pivot is located at point E (elbow).

The second equilibrium condition, $\tau_T + \tau_w = 0$, can be now written as

$$r_T T\sqrt{3}/2 - r_w w\sqrt{3}/2 = 0. \tag{12.26}$$

From the free-body diagram, the first equilibrium condition (for forces) is

$$-F + T - w = 0. \tag{12.27}$$

Equation 12.26 is identical to Equation 12.25 and gives the result $T = 433.3\text{ lb}$. Equation 12.27 gives

$$F = T - w = 433.3\text{ lb} - 50.0\text{ lb} = 383.3\text{ lb}.$$

We see that these answers are identical to our previous answers, but the second choice for the frame of reference leads to an equivalent solution that is simpler and quicker because it does not require that the forces be resolved into their rectangular components.

CHECK YOUR UNDERSTANDING 12.4

Repeat Example 12.4 assuming that the forearm is an object of uniform density that weighs 8.896 N.

EXAMPLE 12.5

A Ladder Resting Against a Wall

A uniform ladder is $L = 5.0\text{ m}$ long and weighs 400.0 N. The ladder rests against a slippery vertical wall, as shown in Figure 12.14. The minimum inclination angle between the ladder and the rough floor below which the ladder slips is $\beta = 53°$. Find the reaction forces from the floor and from the wall on the ladder and the coefficient of static friction μ_s at the interface of the ladder with the floor that prevents the ladder from slipping.

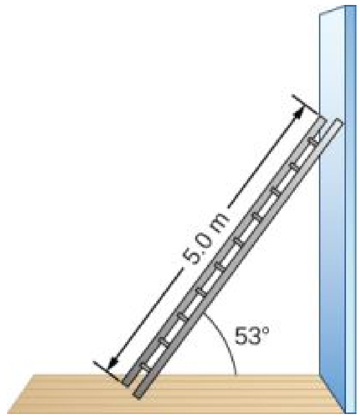


FIGURE 12.14 A 5.0-m-long ladder rests against a frictionless wall.

Strategy

We can identify four forces acting on the ladder. The first force is the normal reaction force *N* from the floor in the upward vertical direction. The second force is the static friction force $f = \mu_s N$ directed horizontally along the floor toward the wall—this force prevents the ladder from slipping. These two forces act on the ladder at its contact point with the floor. The third force is the weight *w* of the ladder, attached at its CM located midway between its ends. The fourth force is the normal reaction force *F* from the wall in the horizontal direction away from the wall, attached at the contact point with the wall. There are no other forces because the wall is slippery, which means there is no friction between the wall and the ladder. Based on this analysis, we adopt the frame of reference with the *y*-axis in the vertical direction (parallel to the wall) and the *x*-axis in the horizontal direction (parallel to the floor). In this frame, each force has either a horizontal component or a vertical component but not both, which simplifies the solution. We select the pivot at the contact point with the floor. In the free-body diagram for the ladder, we indicate the pivot, all four forces and their lever arms, and the angles between lever arms and the forces, as shown in Figure 12.15. With our choice of the pivot location, there is no torque either from the normal reaction force *N* or from the static friction *f* because they both act at the pivot.

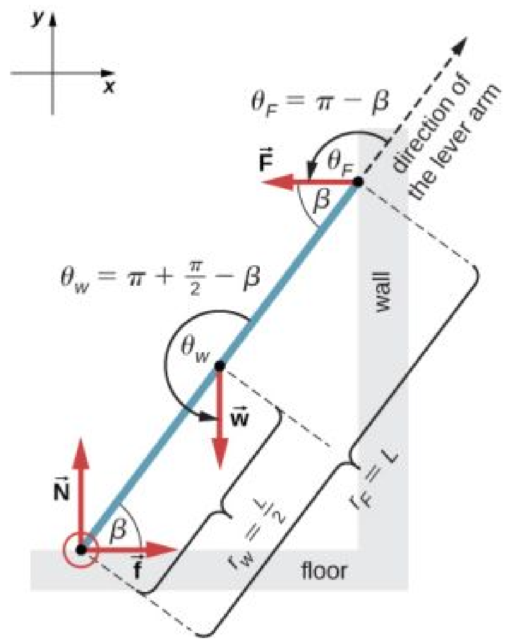


FIGURE 12.15 Free-body diagram for a ladder resting against a frictionless wall.

Solution

From the free-body diagram, the net force in the *x*-direction is

$$+f - F = 0 \tag{12.28}$$

the net force in the *y*-direction is

$$+N - w = 0 \tag{12.29}$$

and the net torque along the rotation axis at the pivot point is

$$\tau_w + \tau_F = 0. \tag{12.30}$$

where τ_w is the torque of the weight *w* and τ_F is the torque of the reaction *F*. From the free-body diagram, we identify that the lever arm of the reaction at the wall is $r_F = L = 5.0\ \text{m}$ and the lever arm of the weight is $r_w = L/2 = 2.5\ \text{m}$. With the help of the free-body diagram, we identify the angles to be used in Equation 12.10 for torques: $\theta_F = 180° - \beta$ for the torque from the reaction force with the wall, and $\theta_w = 180° + (90° - \beta)$ for the torque due to the weight. Now we are ready to use Equation 12.10 to compute torques:

$$\tau_w = r_w w \sin\theta_w = r_w w \sin(180° + 90° - \beta) = -\frac{L}{2} w \sin(90° - \beta) = -\frac{L}{2} w \cos\beta$$
$$\tau_F = r_F F \sin\theta_F = r_F F \sin(180° - \beta) = LF \sin\beta.$$

We substitute the torques into Equation 12.30 and solve for F :

$$\begin{aligned} -\frac{L}{2} w \cos\beta + LF \sin\beta &= 0 \\ F = \frac{w}{2} \cot\beta = \frac{400.0\ \text{N}}{2} \cot 53° &= 150.7\ \text{N} \end{aligned} \tag{12.31}$$

We obtain the normal reaction force with the floor by solving Equation 12.29: $N = w = 400.0\ \text{N}$. The magnitude of friction is obtained by solving Equation 12.28: $f = F = 150.7\ \text{N}$. Since the ladder will slip if the angle is any smaller, the static friction force is at its maximum angle and the coefficient of static friction is $\mu_s = f/N = 150.7/400.0 = 0.377$.

The net force on the ladder at the contact point with the floor is the vector sum of the normal reaction from the floor and the static friction forces:

$$\vec{\mathbf{F}}_{\text{floor}} = \vec{\mathbf{f}} + \vec{\mathbf{N}} = (150.7\text{ N})(-\hat{\mathbf{i}}) + (400.0\text{ N})(+\hat{\mathbf{j}}) = (-150.7\hat{\mathbf{i}} + 400.0\hat{\mathbf{j}})\text{ N}.$$

Its magnitude is

$$F_{\text{floor}} = \sqrt{f^2 + N^2} = \sqrt{150.7^2 + 400.0^2}\text{ N} = 427.4\text{ N}$$

and its direction is

$$\varphi = \tan^{-1}(N/f) = \tan^{-1}(400.0/150.7) = 69.3° \text{ above the floor.}$$

We should emphasize here two general observations of practical use. First, notice that when we choose a pivot point, there is no expectation that the system will actually pivot around the chosen point. The ladder in this example is not rotating at all but firmly stands on the floor; nonetheless, its contact point with the floor is a good choice for the pivot. Second, notice when we use Equation 12.10 for the computation of individual torques, we do not need to resolve the forces into their normal and parallel components with respect to the direction of the lever arm, and we do not need to consider a sense of the torque. As long as the angle in Equation 12.10 is correctly identified—with the help of a free-body diagram—as the angle measured counterclockwise from the direction of the lever arm to the direction of the force vector, Equation 12.10 gives both the magnitude and the sense of the torque. This is because torque is the vector product of the lever-arm vector crossed with the force vector, and Equation 12.10 expresses the rectangular component of this vector product along the axis of rotation.

Significance

This result is independent of the length of the ladder because L is cancelled in the second equilibrium condition, Equation 12.31. No matter how long or short the ladder is, as long as its weight is 400 N and the angle with the floor is $53°$, our results hold. But the ladder will slip if the net torque becomes negative in Equation 12.31. This happens for some angles when the coefficient of static friction is not great enough to prevent the ladder from slipping.

CHECK YOUR UNDERSTANDING 12.5

For the situation described in Example 12.5, determine the values of the coefficient μ_s of static friction for which the ladder starts slipping, given that β is the angle that the ladder makes with the floor.

EXAMPLE 12.6

Forces on Door Hinges

A swinging door that weighs $w = 400.0\text{ N}$ is supported by hinges *A* and *B* so that the door can swing about a vertical axis passing through the hinges Figure 12.16. The door has a width of $b = 1.00\text{ m}$, and the door slab has a uniform mass density. The hinges are placed symmetrically at the door's edge in such a way that the door's weight is evenly distributed between them. The hinges are separated by distance $a = 2.00\text{ m}$. Find the forces on the hinges when the door rests half-open.

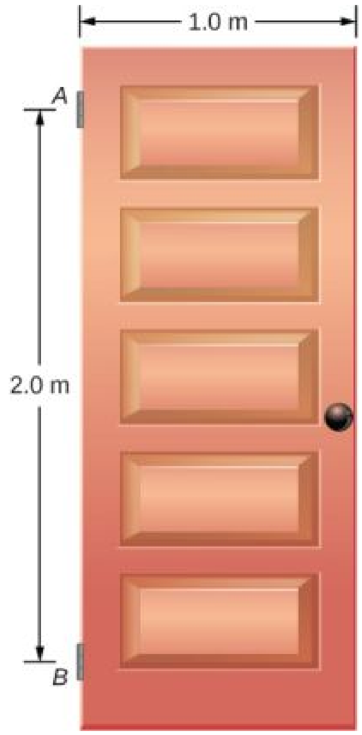


FIGURE 12.16 A 400-N swinging vertical door is supported by two hinges attached at points *A* and *B*.

Strategy

The forces that the door exerts on its hinges can be found by simply reversing the directions of the forces that the hinges exert on the door. Hence, our task is to find the forces from the hinges on the door. Three forces act on the door slab: an unknown force $\vec{\mathbf{A}}$ from hinge A, an unknown force $\vec{\mathbf{B}}$ from hinge B, and the known weight $\vec{\mathbf{w}}$ attached at the center of mass of the door slab. The CM is located at the geometrical center of the door because the slab has a uniform mass density. We adopt a rectangular frame of reference with the *y*-axis along the direction of gravity and the *x*-axis in the plane of the slab, as shown in panel (a) of Figure 12.17, and resolve all forces into their rectangular components. In this way, we have four unknown component forces: two components of force $\vec{\mathbf{A}}$ $\left(A_x \text{ and } A_y\right)$, and two components of force $\vec{\mathbf{B}}$ $\left(B_x \text{ and } B_y\right)$. In the free-body diagram, we represent the two forces at the hinges by their vector components, whose assumed orientations are arbitrary. Because there are four unknowns $\left(A_x, B_x, A_y, \text{ and } B_y\right)$, we must set up four independent equations. One equation is the equilibrium condition for forces in the *x*-direction. The second equation is the equilibrium condition for forces in the *y*-direction. The third equation is the equilibrium condition for torques in rotation about a hinge. Because the weight is evenly distributed between the hinges, we have the fourth equation, $A_y = B_y$. To set up the equilibrium conditions, we draw a free-body diagram and choose the pivot point at the upper hinge, as shown in panel (b) of Figure 12.17. Finally, we solve the equations for the unknown force components and find the forces.

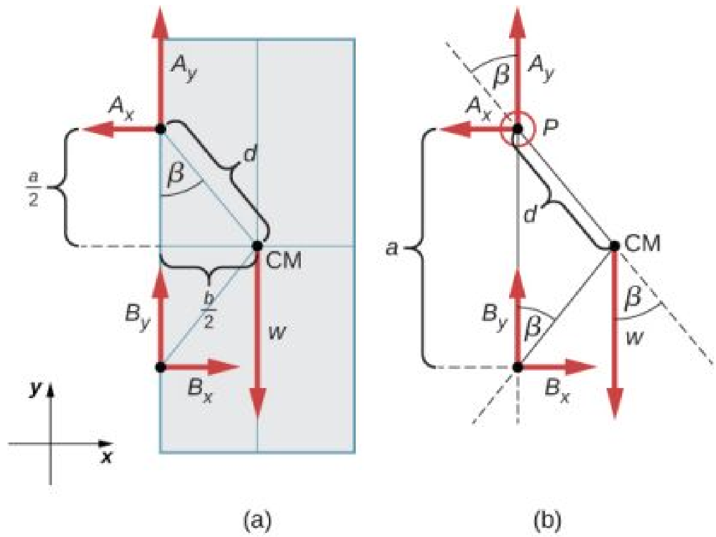


FIGURE 12.17 (a) Geometry and (b) free-body diagram for the door.

Solution

From the free-body diagram for the door we have the first equilibrium condition for forces:

$$\begin{aligned}
&\text{in } x\text{-direction:} \quad -A_x + B_x = 0 \quad \Rightarrow \quad A_x = B_x \\
&\text{in } y\text{-direction:} \quad +A_y + B_y - w = 0 \quad \Rightarrow \quad A_y = B_y = \tfrac{w}{2} = \tfrac{400.0\ \text{N}}{2} = 200.0\ \text{N}.
\end{aligned}$$

We select the pivot at point P (upper hinge, per the free-body diagram) and write the second equilibrium condition for torques in rotation about point P:

$$\text{pivot at } P\text{: } \tau_w + \tau_{Bx} + \tau_{By} = 0. \tag{12.32}$$

We use the free-body diagram to find all the terms in this equation:

$$\begin{aligned}
\tau_w &= dw\sin(-\beta) = -dw\sin\beta = -dw\tfrac{b/2}{d} = -w\tfrac{b}{2} \\
\tau_{Bx} &= aB_x \sin 90^\circ = +aB_x \\
\tau_{By} &= aB_y \sin 180^\circ = 0.
\end{aligned}$$

In evaluating $\sin\beta$, we use the geometry of the triangle shown in part (a) of the figure. Now we substitute these torques into Equation 12.32 and compute B_x :

$$\text{pivot at } P\text{: } -w\frac{b}{2} + aB_x = 0 \quad \Rightarrow \quad B_x = w\frac{b}{2a} = (400.0\ \text{N})\frac{1}{2 \cdot 2} = 100.0\ \text{N}.$$

Therefore the magnitudes of the horizontal component forces are $A_x = B_x = 100.0\ \text{N}$. The forces on the door are

$$\begin{aligned}
&\text{at the upper hinge: } \vec{\mathbf{F}}_{A\text{ on door}} = -100.0\ \text{N}\hat{\mathbf{i}} + 200.0\ \text{N}\hat{\mathbf{j}} \\
&\text{at the lower hinge:} \vec{\mathbf{F}}_{B\text{ on door}} = +100.0\ \text{N}\hat{\mathbf{i}} + 200.0\ \text{N}\hat{\mathbf{j}}.
\end{aligned}$$

The forces on the hinges are found from Newton's third law as

$$\begin{aligned}
&\text{on the upper hinge: } \vec{\mathbf{F}}_{\text{door on } A} = 100.0\ \text{N}\hat{\mathbf{i}} - 200.0\ \text{N}\hat{\mathbf{j}} \\
&\text{on the lower hinge: } \vec{\mathbf{F}}_{\text{door on } B} = -100.0\ \text{N}\hat{\mathbf{i}} - 200.0\ \text{N}\hat{\mathbf{j}}.
\end{aligned}$$

Significance

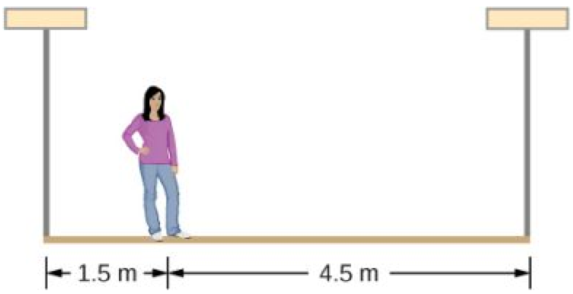
Note that if the problem were formulated without the assumption of the weight being equally distributed between the two hinges, we wouldn't be able to solve it because the number of the unknowns would be greater than the number of equations expressing equilibrium conditions.

CHECK YOUR UNDERSTANDING 12.6

Solve the problem in Example 12.6 by taking the pivot position at the center of mass.

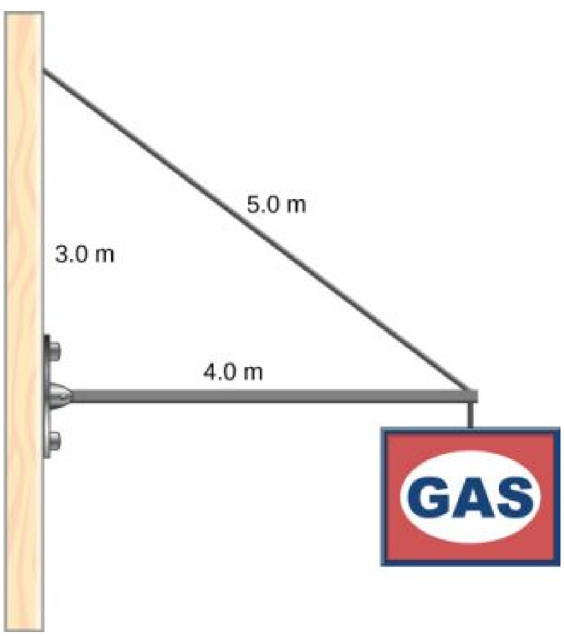
CHECK YOUR UNDERSTANDING 12.7

A 50-kg person stands 1.5 m away from one end of a uniform 6.0-m-long scaffold of mass 70.0 kg. Find the tensions in the two vertical ropes supporting the scaffold.



CHECK YOUR UNDERSTANDING 12.8

A 400.0-N sign hangs from the end of a uniform strut. The strut is 4.0 m long and weighs 600.0 N. The strut is supported by a hinge at the wall and by a cable whose other end is tied to the wall at a point 3.0 m above the left end of the strut. Find the tension in the supporting cable and the force of the hinge on the strut.



12.3 Stress, Strain, and Elastic Modulus

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the concepts of stress and strain in describing elastic deformations of materials
- Describe the types of elastic deformation of objects and materials

A model of a rigid body is an idealized example of an object that does not deform under the actions of external forces. It is very useful when analyzing mechanical systems—and many physical objects are indeed rigid to a great extent. The extent to which an object can be *perceived* as rigid depends on the physical properties of the material from which it is made. For example, a ping-pong ball made of plastic is brittle, and a tennis ball made of rubber is elastic when acted upon by squashing forces. However, under other circumstances, both a ping-pong ball and a tennis ball may bounce well as rigid bodies. Similarly, someone who designs prosthetic limbs may be able to approximate the mechanics of human limbs by modeling them as rigid bodies; however, the actual combination of bones and tissues is an elastic medium.

For the remainder of this chapter, we move from consideration of forces that affect the motion of an object to those that affect an object's shape. A change in shape due to the application of a force is known as a deformation. Even very small forces are known to cause some deformation. Deformation is experienced by objects or physical media under the action of external forces—for example, this may be squashing, squeezing, ripping, twisting, shearing, or pulling the objects apart. In the language of physics, two terms describe the forces on objects undergoing deformation: *stress* and *strain*.

Stress is a quantity that describes the magnitude of forces that cause deformation. Stress is generally defined as *force per unit area*. When forces pull on an object and cause its elongation, like the stretching of an elastic band, we call such stress a **tensile stress**. When forces cause a compression of an object, we call it a **compressive stress**. When an object is being squeezed from all sides, like a submarine in the depths of an ocean, we call this kind of stress a **bulk stress** (or **volume stress**). In other situations, the acting forces may be neither tensile nor compressive, and still produce a noticeable deformation. For example, suppose you hold a book tightly between the palms of your hands, then with one hand you press-and-pull on the front cover away from you, while with the other hand you press-and-pull on the back cover toward you. In such a case, when deforming forces act tangentially to the object's surface, we call them 'shear' forces and the stress they cause is called **shear stress**.

The SI unit of stress is the pascal (Pa). When one newton of force presses on a unit surface area of one meter squared, the resulting stress is one pascal:

$$\text{one pascal} = 1.0\,\text{Pa} = \frac{1.0\,\text{N}}{1.0\,\text{m}^2}.$$

In the Imperial system of units, the unit of stress is 'psi,' which stands for 'pound per square inch' $\left(\text{lb/in}^2\right)$. Another unit that is often used for bulk stress is the atm (atmosphere). Conversion factors are

$$1\,\text{psi} = 6895\,\text{Pa} \;\;\text{and}\;\; 1\,\text{Pa} = 1.450 \times 10^{-4}\,\text{psi}$$
$$1\,\text{atm} = 1.013 \times 10^{5}\,\text{Pa} = 14.7\,\text{psi}.$$

An object or medium under stress becomes deformed. The quantity that describes this deformation is called **strain**. Strain is given as a fractional change in either length (under tensile stress) or volume (under bulk stress) or geometry (under shear stress). Therefore, strain is a dimensionless number. Strain under a tensile stress is called **tensile strain**, strain under bulk stress is called **bulk strain** (or **volume strain**), and that caused by shear stress is called **shear strain**.

The greater the stress, the greater the strain; however, the relation between strain and stress does not need to be linear. Only when stress is sufficiently low is the deformation it causes in direct proportion to the stress value. The proportionality constant in this relation is called the **elastic modulus**. In the linear limit of low stress values, the general relation between stress and strain is

$$\text{stress} = (\text{elastic modulus}) \times \text{strain}. \tag{12.33}$$

As we can see from dimensional analysis of this relation, the elastic modulus has the same physical unit as stress because strain is dimensionless.

We can also see from Equation 12.33 that when an object is characterized by a large value of elastic modulus, the effect of stress is small. On the other hand, a small elastic modulus means that stress produces large strain and noticeable deformation. For example, a stress on a rubber band produces larger strain (deformation) than the same stress on a steel band of the same dimensions because the elastic modulus for rubber is two orders of magnitude smaller than the elastic modulus for steel.

The elastic modulus for tensile stress is called **Young's modulus**; that for the bulk stress is called the **bulk modulus**; and that for shear stress is called the **shear modulus**. Note that the relation between stress and strain is an *observed* relation, measured in the laboratory. Elastic moduli for various materials are measured under various physical conditions, such as varying temperature, and collected in engineering data tables for reference (Table 12.1). These tables are valuable references for industry and for anyone involved in engineering or construction. In the next section, we discuss strain-stress relations beyond the linear limit represented by Equation 12.33, in the full range of stress values up to a fracture point. In the remainder of this section, we study the linear limit expressed by Equation 12.33.

Material	Young's modulus $\times 10^{10} \mathrm{Pa}$	Bulk modulus $\times 10^{10} \mathrm{Pa}$	Shear modulus $\times 10^{10} \mathrm{Pa}$
Aluminum	7.0	7.5	2.5
Bone (tension)	1.6	0.8	8.0
Bone (compression)	0.9		
Brass	9.0	6.0	3.5
Brick	1.5		
Concrete	2.0		
Copper	11.0	14.0	4.4
Crown glass	6.0	5.0	2.5
Granite	4.5	4.5	2.0
Hair (human)	1.0		
Hardwood	1.5		1.0
Iron	21.0	16.0	7.7
Lead	1.6	4.1	0.6
Marble	6.0	7.0	2.0
Nickel	21.0	17.0	7.8
Polystyrene	3.0		

TABLE 12.1 **Approximate Elastic Moduli for Selected Materials**

Material	Young's modulus $\times 10^{10}\,\mathrm{Pa}$	Bulk modulus $\times 10^{10}\,\mathrm{Pa}$	Shear modulus $\times 10^{10}\,\mathrm{Pa}$
Silk	6.0		
Spider thread	3.0		
Steel	20.0	16.0	7.5
Acetone		0.07	
Ethanol		0.09	
Glycerin		0.45	
Mercury		2.5	
Water		0.22	

TABLE 12.1 **Approximate Elastic Moduli for Selected Materials**

Tensile or Compressive Stress, Strain, and Young's Modulus

Tension or compression occurs when two antiparallel forces of equal magnitude act on an object along only one of its dimensions, in such a way that the object does not move. One way to envision such a situation is illustrated in Figure 12.18. A rod segment is either stretched or squeezed by a pair of forces acting along its length and perpendicular to its cross-section. The net effect of such forces is that the rod changes its length from the original length L_0 that it had before the forces appeared, to a new length *L* that it has under the action of the forces. This change in length $\Delta L = L - L_0$ may be either elongation (when *L* is larger than the original length L_0) or contraction (when *L* is smaller than the original length L_0). Tensile stress and strain occur when the forces are stretching an object, causing its elongation, and the length change ΔL is positive. Compressive stress and strain occur when the forces are contracting an object, causing its shortening, and the length change ΔL is negative.

In either of these situations, we define stress as the ratio of the deforming force $F_\perp$ to the cross-sectional area *A* of the object being deformed. The symbol $F_\perp$ that we reserve for the deforming force means that this force acts perpendicularly to the cross-section of the object. Forces that act parallel to the cross-section do not change the length of an object. The definition of the tensile stress is

$$\text{tensile stress} = \frac{F_\perp}{A}. \tag{12.34}$$

Tensile strain is the measure of the deformation of an object under tensile stress and is defined as the fractional change of the object's length when the object experiences tensile stress

$$\text{tensile strain} = \frac{\Delta L}{L_0}. \tag{12.35}$$

Compressive stress and strain are defined by the same formulas, Equation 12.34 and Equation 12.35, respectively. The only difference from the tensile situation is that for compressive stress and strain, we take absolute values of the right-hand sides in Equation 12.34 and Equation 12.35.

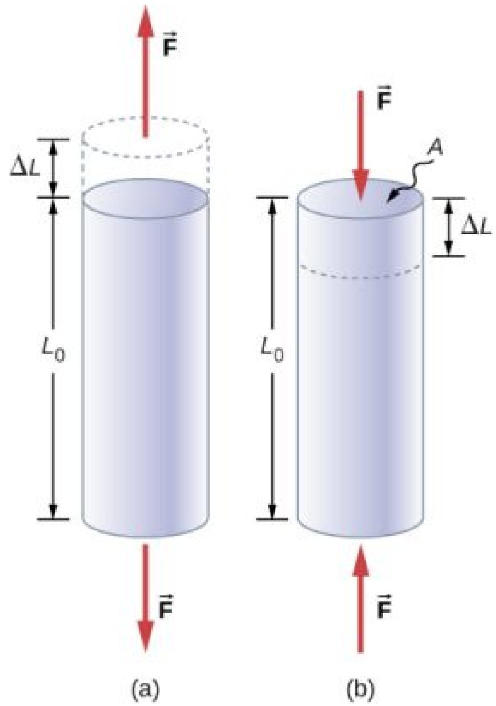


FIGURE 12.18 When an object is in either tension or compression, the net force on it is zero, but the object deforms by changing its original length L_0. (a) Tension: The rod is elongated by ΔL. (b) Compression: The rod is contracted by ΔL. In both cases, the deforming force acts along the length of the rod and perpendicular to its cross-section. In the linear range of low stress, the cross-sectional area of the rod does not change.

Young's modulus Y is the elastic modulus when deformation is caused by either tensile or compressive stress, and is defined by Equation 12.33. Dividing this equation by tensile strain, we obtain the expression for Young's modulus:

$$Y = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{F_{\perp}/A}{\Delta L/L_0} = \frac{F_{\perp}}{A}\frac{L_0}{\Delta L}. \tag{12.36}$$

EXAMPLE 12.7

Compressive Stress in a Pillar

A sculpture weighing 10,000 N rests on a horizontal surface at the top of a 6.0-m-tall vertical pillar Figure 12.19. The pillar's cross-sectional area is $0.20\ \mathrm{m}^2$ and it is made of granite with a mass density of $2700\ \mathrm{kg/m}^3$. Find the compressive stress at the cross-section located 3.0 m below the top of the pillar and the value of the compressive strain of the top 3.0-m segment of the pillar.

FIGURE 12.19 Nelson’s Column in Trafalgar Square, London, England. (credit: modification of work by Cristian Bortes)

Strategy

First we find the weight of the 3.0-m-long top section of the pillar. The normal force that acts on the cross-section located 3.0 m down from the top is the sum of the pillar’s weight and the sculpture’s weight. Once we have the normal force, we use Equation 12.34 to find the stress. To find the compressive strain, we find the value of Young’s modulus for granite in Table 12.1 and invert Equation 12.36.

Solution

The volume of the pillar segment with height $h = 3.0\ \text{m}$ and cross-sectional area $A = 0.20\ \text{m}^2$ is

$$V = Ah = (0.20\ \text{m}^2)(3.0\ \text{m}) = 0.60\ \text{m}^3.$$

With the density of granite $\rho = 2.7 \times 10^3\ \text{kg/m}^3$, the mass of the pillar segment is

$$m = \rho V = (2.7 \times 10^3\ \text{kg/m}^3)(0.60\ \text{m}^3) = 1.60 \times 10^3\,\text{kg}.$$

The weight of the pillar segment is

$$w_p = mg = (1.60 \times 10^3\,\text{kg})(9.80\ \text{m/s}^2) = 1.568 \times 10^4\,\text{N}.$$

The weight of the sculpture is $w_s = 1.0 \times 10^4\,\text{N}$, so the normal force on the cross-sectional surface located 3.0 m below the sculpture is

$$F_\perp = w_p + w_s = (1.568 + 1.0) \times 10^4\,\text{N} = 2.568 \times 10^4\,\text{N}.$$

Therefore, the stress is

$$\text{stress} = \frac{F_\perp}{A} = \frac{2.568 \times 10^4\,\text{N}}{0.20\ \text{m}^2} = 1.284 \times 10^5\,\text{Pa} = 128.4\ \text{kPa}.$$

Young’s modulus for granite is $Y = 4.5 \times 10^{10}\,\text{Pa} = 4.5 \times 10^7\,\text{kPa}$. Therefore, the compressive strain at this position is

$$\text{strain} = \frac{\text{stress}}{Y} = \frac{128.4\ \text{kPa}}{4.5 \times 10^7\,\text{kPa}} = 2.85 \times 10^{-6}.$$

Significance

Notice that the normal force acting on the cross-sectional area of the pillar is not constant along its length, but varies from its smallest value at the top to its largest value at the bottom of the pillar. Thus, if the pillar has a uniform cross-sectional area along its length, the stress is largest at its base.

CHECK YOUR UNDERSTANDING 12.9

Find the compressive stress and strain at the base of Nelson's column.

EXAMPLE 12.8

Stretching a Rod

A 2.0-m-long steel rod has a cross-sectional area of $0.30\ \text{cm}^2$. The rod is a part of a vertical support that holds a heavy 550-kg platform that hangs attached to the rod's lower end. Ignoring the weight of the rod, what is the tensile stress in the rod and the elongation of the rod under the stress?

Strategy

First we compute the tensile stress in the rod under the weight of the platform in accordance with Equation 12.34. Then we invert Equation 12.36 to find the rod's elongation, using $L_0 = 2.0\ \text{m}$. From Table 12.1, Young's modulus for steel is $Y = 2.0 \times 10^{11}\ \text{Pa}$.

Solution

Substituting numerical values into the equations gives us

$$\begin{aligned}\frac{F_\perp}{A} &= \frac{(550\ \text{kg})(9.8\ \text{m/s}^2)}{3.0 \times 10^{-5}\ \text{m}^2} = 1.8 \times 10^8\ \text{Pa} \\ \Delta L &= \frac{F_\perp}{A}\frac{L_0}{Y} = (1.8 \times 10^8\ \text{Pa})\frac{2.0\ \text{m}}{2.0 \times 10^{11}\ \text{Pa}} = 1.8 \times 10^{-3}\ \text{m} = 1.8\ \text{mm}.\end{aligned}$$

Significance

Similarly as in the example with the column, the tensile stress in this example is not uniform along the length of the rod. Unlike in the previous example, however, if the weight of the rod is taken into consideration, the stress in the rod is largest at the top and smallest at the bottom of the rod where the equipment is attached.

CHECK YOUR UNDERSTANDING 12.10

A 2.0-m-long wire stretches 1.0 mm when subjected to a load. What is the tensile strain in the wire?

Objects can often experience both compressive stress and tensile stress simultaneously Figure 12.20. One example is a long shelf loaded with heavy books that sags between the end supports under the weight of the books. The top surface of the shelf is in compressive stress and the bottom surface of the shelf is in tensile stress. Similarly, long and heavy beams sag under their own weight. In modern building construction, such bending strains can be almost eliminated with the use of I-beams Figure 12.21.

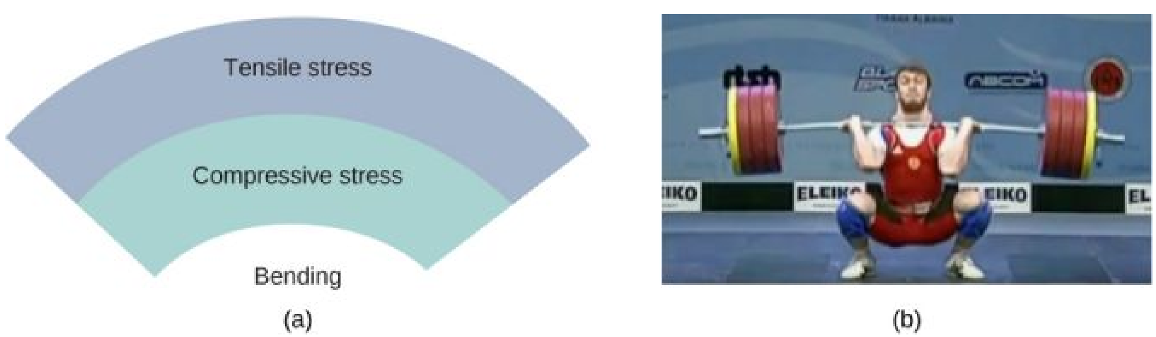


FIGURE 12.20 (a) An object bending downward experiences tensile stress (stretching) in the upper section and compressive stress (compressing) in the lower section. (b) Elite weightlifters often bend iron bars temporarily during lifting, as in the 2012 Olympics

competition. (credit b: modification of work by Oleksandr Kocherzhenko)

FIGURE 12.21 Steel I-beams are used in construction to reduce bending strains. (credit: modification of work by “US Army Corps of Engineers Europe District”/Flickr)

INTERACTIVE

A heavy box rests on a table supported by three columns. View this demonstration (https://openstax.org/l/21movebox) to move the box to see how the compression (or tension) in the columns is affected when the box changes its position.

Bulk Stress, Strain, and Modulus

When you dive into water, you feel a force pressing on every part of your body from all directions. What you are experiencing then is bulk stress, or in other words, **pressure**. Bulk stress always tends to decrease the volume enclosed by the surface of a submerged object. The forces of this “squeezing” are always perpendicular to the submerged surface Figure 12.22. The effect of these forces is to decrease the volume of the submerged object by an amount ΔV compared with the volume V_0 of the object in the absence of bulk stress. This kind of deformation is called bulk strain and is described by a change in volume relative to the original volume:

$$\text{bulk strain} = \frac{\Delta V}{V_0}. \tag{12.37}$$

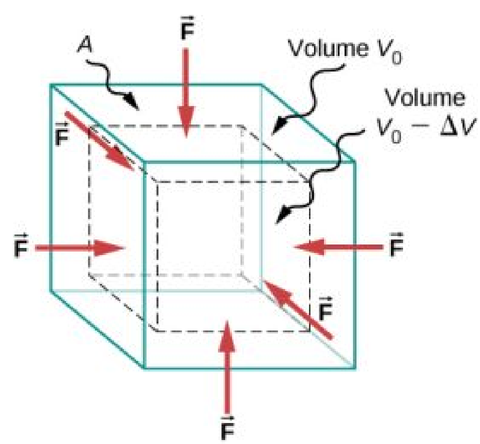


FIGURE 12.22 An object under increasing bulk stress always undergoes a decrease in its volume. Equal forces perpendicular to the surface act from all directions. The effect of these forces is to decrease the volume by the amount ΔV compared to the original volume, V_0.

The bulk strain results from the bulk stress, which is a force $F_\perp$ normal to a surface that presses on the unit surface area *A* of a submerged object. This kind of physical quantity, or pressure *p*, is defined as

$$\text{pressure} = p \equiv \frac{F_\perp}{A}. \tag{12.38}$$

We will study pressure in fluids in greater detail in Fluid Mechanics. An important characteristic of pressure is that it is a scalar quantity and does not have any particular direction; that is, pressure acts equally in all possible directions. When you submerge your hand in water, you sense the same amount of pressure acting on the top surface of your hand as on the bottom surface, or on the side surface, or on the surface of the skin between your fingers. What you are perceiving in this case is an increase in pressure Δp over what you are used to feeling when your hand is not submerged in water. What you feel when your hand is not submerged in the water is the **normal pressure** p_0 of one atmosphere, which serves as a reference point. The bulk stress is this increase in pressure, or Δp, over the normal level, p_0.

When the bulk stress increases, the bulk strain increases in response, in accordance with Equation 12.33. The proportionality constant in this relation is called the bulk modulus, *B*, or

$$B = \frac{\text{bulk stress}}{\text{bulk strain}} = -\frac{\Delta p}{\Delta V / V_0} = -\Delta p \frac{V_0}{\Delta V}. \quad (12.39)$$

The minus sign that appears in Equation 12.39 is for consistency, to ensure that *B* is a positive quantity. Note that the minus sign $(-)$ is necessary because an increase Δp in pressure (a positive quantity) always causes a decrease ΔV in volume, and decrease in volume is a negative quantity. The reciprocal of the bulk modulus is called **compressibility** k, or

$$k = \frac{1}{B} = -\frac{\Delta V / V_0}{\Delta p}. \quad (12.40)$$

The term 'compressibility' is used in relation to fluids (gases and liquids). Compressibility describes the change in the volume of a fluid per unit increase in pressure. Fluids characterized by a large compressibility are relatively easy to compress. For example, the compressibility of water is $4.64 \times 10^{-5}/\text{atm}$ and the compressibility of acetone is $1.45 \times 10^{-4}/\text{atm}$. This means that under a 1.0-atm increase in pressure, the relative decrease in volume is approximately three times as large for acetone as it is for water.

EXAMPLE 12.9

Hydraulic Press

In a hydraulic press Figure 12.23, a 250-liter volume of oil is subjected to a 2300-psi pressure increase. If the compressibility of oil is $2.0 \times 10^{-5}/\text{atm}$, find the bulk strain and the absolute decrease in the volume of oil when the press is operating.

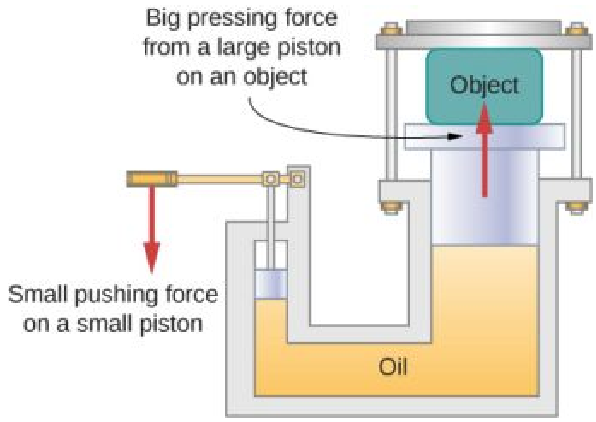


FIGURE 12.23 In a hydraulic press, when a small piston is displaced downward, the pressure in the oil is transmitted throughout the oil to the large piston, causing the large piston to move upward. A small force applied to a small piston causes a large pressing force, which the large piston exerts on an object that is either lifted or squeezed. The device acts as a mechanical lever.

Strategy

We must invert Equation 12.40 to find the bulk strain. First, we convert the pressure increase from psi to atm, $\Delta p = 2300 \text{ psi} = 2300/14.7 \text{ atm} \approx 160 \text{ atm}$, and identify $V_0 = 250 \text{ L}$.

Solution

Substituting values into the equation, we have

$$\text{bulk strain} = \frac{\Delta V}{V_0} = \frac{\Delta p}{B} = k\Delta p = (2.0 \times 10^{-5}/\text{atm})(160\ \text{atm}) = 0.0032$$
$$\text{answer:}\ \ \Delta V = 0.0032\ V_0 = 0.0032(250\ \text{L}) = 0.78\ \text{L}.$$

Significance

Notice that since the compressibility of water is 2.32 times larger than that of oil, if the working substance in the hydraulic press of this problem were changed to water, the bulk strain as well as the volume change would be 2.32 times larger.

CHECK YOUR UNDERSTANDING 12.11

If the normal force acting on each face of a cubical 1.0-m^3 piece of steel is changed by $1.0 \times 10^7\,\text{N}$, find the resulting change in the volume of the piece of steel.

Shear Stress, Strain, and Modulus

The concepts of shear stress and strain concern only solid objects or materials. Buildings and tectonic plates are examples of objects that may be subjected to shear stresses. In general, these concepts do not apply to fluids.

Shear deformation occurs when two antiparallel forces of equal magnitude are applied tangentially to opposite surfaces of a solid object, causing no deformation in the transverse direction to the line of force, as in the typical example of shear stress illustrated in Figure 12.24. Shear deformation is characterized by a gradual shift Δx of layers in the direction tangent to the acting forces. This gradation in Δx occurs in the transverse direction along some distance L_0. Shear strain is defined by the ratio of the largest displacement Δx to the transverse distance L_0

$$\text{shear strain} = \frac{\Delta x}{L_0}. \tag{12.41}$$

Shear strain is caused by shear stress. Shear stress is due to forces that act *parallel* to the surface. We use the symbol $F_{\parallel}$ for such forces. The magnitude $F_{\parallel}$ per surface area *A* where shearing force is applied is the measure of shear stress

$$\text{shear stress} = \frac{F_{\parallel}}{A}. \tag{12.42}$$

The shear modulus is the proportionality constant in Equation 12.33 and is defined by the ratio of stress to strain. Shear modulus is commonly denoted by *S*:

$$S = \frac{\text{shear stress}}{\text{shear strain}} = \frac{F_{\parallel}/A}{\Delta x/L_0} = \frac{F_{\parallel}}{A}\,\frac{L_0}{\Delta x}. \tag{12.43}$$

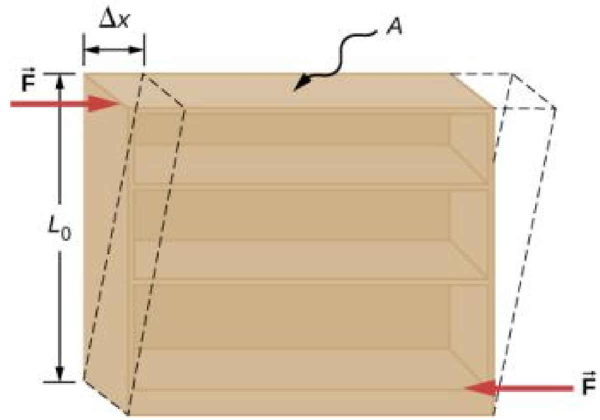


FIGURE 12.24 An object under shear stress: Two antiparallel forces of equal magnitude are applied tangentially to opposite parallel surfaces of the object. The dashed-line contour depicts the resulting deformation. There is no change in the direction transverse to the acting forces and the transverse length L_0 is unaffected. Shear deformation is characterized by a gradual shift Δx of layers in the direction tangent to the forces.

EXAMPLE 12.10

An Old Bookshelf

A cleaning person tries to move a heavy, old bookcase on a carpeted floor by pushing tangentially on the surface of the very top shelf. However, the only noticeable effect of this effort is similar to that seen in Figure 12.24, and it disappears when the person stops pushing. The bookcase is 180.0 cm tall and 90.0 cm wide with four 30.0-cm-deep shelves, all partially loaded with books. The total weight of the bookcase and books is 600.0 N. If the person gives the top shelf a 50.0-N push that displaces the top shelf horizontally by 15.0 cm relative to the motionless bottom shelf, find the shear modulus of the bookcase.

Strategy

The only pieces of relevant information are the physical dimensions of the bookcase, the value of the tangential force, and the displacement this force causes. We identify $F_{\parallel} = 50.0\,\text{N},\ \Delta x = 15.0\,\text{cm},\ L_0 = 180.0\,\text{cm}$, and $A = (30.0\,\text{cm})(90.0\,\text{cm}) = 2700.0\,\text{cm}^2$, and we use Equation 12.43 to compute the shear modulus.

Solution

Substituting numbers into the equations, we obtain for the shear modulus

$$S = \frac{F_{\parallel}}{A}\frac{L_0}{\Delta x} = \frac{50.0\,\text{N}}{2700.0\,\text{cm}^2}\frac{180.0\,\text{cm.}}{15.0\,\text{cm.}} = \frac{2}{9}\frac{\text{N}}{\text{cm}^2} = \frac{2}{9} \times 10^4\frac{\text{N}}{\text{m}^2} = \frac{20}{9} \times 10^3\,\text{Pa} = 2.222\,\text{kPa.}$$

We can also find shear stress and strain, respectively:

$$\frac{F_{\parallel}}{A} = \frac{50.0\,\text{N}}{2700.0\,\text{cm}^2} = \frac{5}{27}\,\text{kPa} = 185.2\,\text{Pa}$$

$$\frac{\Delta x}{L_0} = \frac{15.0\,\text{cm}}{180.0\,\text{cm}} = \frac{1}{12} = 0.083.$$

Significance

If the person in this example gave the shelf a healthy push, it might happen that the induced shear would collapse it to a pile of rubbish. Much the same shear mechanism is responsible for failures of earth-filled dams and levees; and, in general, for landslides.

CHECK YOUR UNDERSTANDING 12.12

Explain why the concepts of Young’s modulus and shear modulus do not apply to fluids.

12.4 Elasticity and Plasticity

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the limit where a deformation of material is elastic
- Describe the range where materials show plastic behavior
- Analyze elasticity and plasticity on a stress-strain diagram

We referred to the proportionality constant between stress and strain as the elastic modulus. But why do we call it that? What does it mean for an object to be elastic and how do we describe its behavior?

Elasticity is the tendency of solid objects and materials to return to their original shape after the external forces (load) causing a deformation are removed. An object is **elastic** when it comes back to its original size and shape when the load is no longer present. Physical reasons for elastic behavior vary among materials and depend on the microscopic structure of the material. For example, the elasticity of polymers and rubbers is caused by stretching polymer chains under an applied force. In contrast, the elasticity of metals is caused by resizing and reshaping the crystalline cells of the lattices (which are the material structures of metals) under the action of externally applied forces.

The two parameters that determine the elasticity of a material are its *elastic modulus* and its *elastic limit*. A high elastic modulus is typical for materials that are hard to deform; in other words, materials that require a high load to achieve a significant strain. An example is a steel band. A low elastic modulus is typical for materials that are easily deformed under a load; for example, a rubber band. If the stress under a load becomes too high, then when the load is removed, the material no longer comes back to its original shape and size, but relaxes to a different shape and size: The material becomes permanently deformed. The **elastic limit** is the stress value beyond which the material no longer behaves elastically but becomes permanently deformed.

Our perception of an elastic material depends on both its elastic limit and its elastic modulus. For example, all rubbers are characterized by a low elastic modulus and a high elastic limit; hence, it is easy to stretch them and the stretch is noticeably large. Among materials with identical elastic limits, the most elastic is the one with the lowest elastic modulus.

When the load increases from zero, the resulting stress is in direct proportion to strain in the way given by Equation 12.33, but only when stress does not exceed some limiting value. For stress values within this linear limit, we can describe elastic behavior in analogy with Hooke's law for a spring. According to Hooke's law, the stretch value of a spring under an applied force is directly proportional to the magnitude of the force. Conversely, the response force from the spring to an applied stretch is directly proportional to the stretch. In the same way, the deformation of a material under a load is directly proportional to the load, and, conversely, the resulting stress is directly proportional to strain. The linearity limit (or the **proportionality limit**) is the largest stress value beyond which stress is no longer proportional to strain. Beyond the linearity limit, the relation between stress and strain is no longer linear. When stress becomes larger than the linearity limit but still within the elasticity limit, behavior is still elastic, but the relation between stress and strain becomes nonlinear.

For stresses beyond the elastic limit, a material exhibits **plastic behavior**. This means the material deforms irreversibly and does not return to its original shape and size, even when the load is removed. When stress is gradually increased beyond the elastic limit, the material undergoes plastic deformation. Rubber-like materials show an increase in stress with the increasing strain, which means they become more difficult to stretch and, eventually, they reach a fracture point where they break. Ductile materials such as metals show a gradual decrease in stress with the increasing strain, which means they become easier to deform as stress-strain values approach the breaking point. Microscopic mechanisms responsible for plasticity of materials are different for different materials.

We can graph the relationship between stress and strain on a **stress-strain diagram**. Each material has its own characteristic strain-stress curve. A typical stress-strain diagram for a ductile metal under a load is shown in Figure 12.25. In this figure, strain is a fractional elongation (not drawn to scale). When the load is gradually increased, the linear behavior (red line) that starts at the no-load point (the origin) ends at the linearity limit at point *H*. For further load increases beyond point *H*, the stress-strain relation is nonlinear but still elastic. In the figure, this nonlinear region is seen between points *H* and *E*. Ever larger loads take the stress to the elasticity limit *E*, where elastic behavior ends and plastic deformation begins. Beyond the elasticity limit, when the load is removed, for example at *P*, the material relaxes to a new shape and size along the green line. This is to say that the material becomes permanently deformed and does not come back to its initial shape and size when stress becomes zero.

The material undergoes plastic deformation for loads large enough to cause stress to go beyond the elasticity limit at *E*. The material continues to be plastically deformed until the stress reaches the fracture point (breaking point). Beyond the fracture point, we no longer have one sample of material, so the diagram ends at the fracture point. For the completeness of this qualitative description, it should be said that the linear, elastic, and plasticity limits denote a range of values rather than one sharp point.

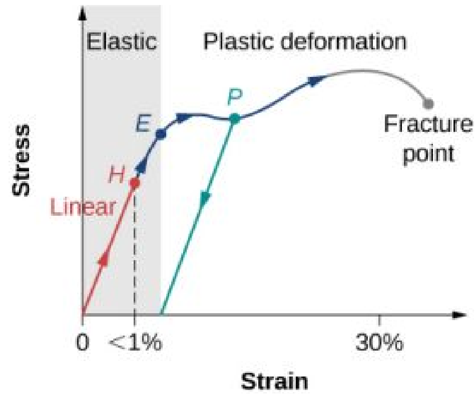


FIGURE 12.25 Typical stress-strain plot for a metal under a load: The graph ends at the fracture point. The arrows show the direction of changes under an ever-increasing load. Points *H* and *E* are the linearity and elasticity limits, respectively. Between points *H* and *E*, the behavior is nonlinear. The green line originating at *P* illustrates the metal's response when the load is removed. The permanent deformation has a strain value at the point where the green line intercepts the horizontal axis.

The value of stress at the fracture point is called breaking stress (or **ultimate stress**). Materials with similar elastic properties, such as two metals, may have very different breaking stresses. For example, ultimate stress for aluminum is $2.2 \times 10^8 \mathrm{Pa}$ and for steel it may be as high as $20.0 \times 10^8 \mathrm{Pa}$, depending on the kind of steel. We can make a quick estimate, based on Equation 12.34, that for rods with a 1-in^2 cross-sectional area, the breaking load for an aluminum rod is $3.2 \times 10^4 \text{ lb}$, and the breaking load for a steel rod is about nine times larger.

Chapter Review

Key Terms

breaking stress (ultimate stress) value of stress at the fracture point

bulk modulus elastic modulus for the bulk stress

bulk strain (or **volume strain**) strain under the bulk stress, given as fractional change in volume

bulk stress (or **volume stress**) stress caused by compressive forces, in all directions

center of gravity point where the weight vector is attached

compressibility reciprocal of the bulk modulus

compressive strain strain that occurs when forces are contracting an object, causing its shortening

compressive stress stress caused by compressive forces, only in one direction

elastic object that comes back to its original size and shape when the load is no longer present

elastic limit stress value beyond which material no longer behaves elastically and becomes permanently deformed

elastic modulus proportionality constant in linear relation between stress and strain, in SI pascals

equilibrium body is in equilibrium when its linear and angular accelerations are both zero relative to an inertial frame of reference

first equilibrium condition expresses translational equilibrium; all external forces acting on the body balance out and their vector sum is zero

gravitational torque torque on the body caused by its weight; it occurs when the center of gravity of the body is not located on the axis of rotation

linearity limit (proportionality limit) largest stress value beyond which stress is no longer proportional to strain

normal pressure pressure of one atmosphere, serves as a reference level for pressure

pascal (Pa) SI unit of stress, SI unit of pressure

plastic behavior material deforms irreversibly, does not go back to its original shape and size when load is removed and stress vanishes

pressure force pressing in normal direction on a surface per the surface area, the bulk stress in fluids

second equilibrium condition expresses rotational equilibrium; all torques due to external forces acting on the body balance out and their vector sum is zero

shear modulus elastic modulus for shear stress

shear strain strain caused by shear stress

shear stress stress caused by shearing forces

static equilibrium body is in static equilibrium when it is at rest in our selected inertial frame of reference

strain dimensionless quantity that gives the amount of deformation of an object or medium under stress

stress quantity that contains information about the magnitude of force causing deformation, defined as force per unit area

stress-strain diagram graph showing the relationship between stress and strain, characteristic of a material

tensile strain strain under tensile stress, given as fractional change in length, which occurs when forces are stretching an object, causing its elongation

tensile stress stress caused by tensile forces, only in one direction, which occurs when forces are stretching an object, causing its elongation

Young's modulus elastic modulus for tensile or compressive stress

Key Equations

First Equilibrium Condition	$\sum_k \vec{\mathbf{F}}_k = \vec{\mathbf{0}}$
Second Equilibrium Condition	$\sum_k \vec{\tau}_k = \vec{\mathbf{0}}$
Linear relation between stress and strain	$\text{stress} = (\text{elastic modulus}) \times \text{strain}$
Young's modulus	$Y = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{F_\perp}{A}\frac{L_0}{\Delta L}$
Bulk modulus	$B = \frac{\text{bulk stress}}{\text{bulk strain}} = -\Delta p \frac{V_0}{\Delta V}$
Shear modulus	$S = \frac{\text{shear stress}}{\text{shear strain}} = \frac{F_\parallel}{A}\frac{L_0}{\Delta x}$

Summary

12.1 Conditions for Static Equilibrium

- A body is in equilibrium when it remains either in uniform motion (both translational and rotational) or at rest. When a body in a selected inertial frame of reference neither rotates nor moves in translational motion, we say the body is in static equilibrium in this frame of reference.
- Conditions for equilibrium require that the sum of all external forces acting on the body is zero (first condition of equilibrium), and the sum of all external torques from external forces is zero (second condition of equilibrium). These two conditions must be simultaneously satisfied in equilibrium. If one of them is not satisfied, the body is not in equilibrium.
- The free-body diagram for a body is a useful tool that allows us to count correctly all contributions from all external forces and torques acting on the body. Free-body diagrams for the equilibrium of an extended rigid body must indicate a pivot point and lever arms of acting forces with respect to the pivot.

12.2 Examples of Static Equilibrium

- A variety of engineering problems can be solved by applying equilibrium conditions for rigid bodies.
- In applications, identify all forces that act on a rigid body and note their lever arms in rotation about a chosen rotation axis. Construct a free-body diagram for the body. Net external forces and torques can be clearly identified from a correctly constructed free-body diagram. In this way, you can set up the first equilibrium condition for forces and the second equilibrium condition for torques.
- In setting up equilibrium conditions, we are free to adopt any inertial frame of reference and any position of the pivot point. All choices lead to one answer. However, some choices can make the process of finding the solution unduly complicated. We reach the same answer no matter what choices we make. The only way to master this skill is to practice.

12.3 Stress, Strain, and Elastic Modulus

- External forces on an object (or medium) cause its deformation, which is a change in its size and shape. The strength of the forces that cause deformation is expressed by stress, which in SI units is measured in the unit of pressure (pascal). The extent of deformation under stress is expressed by strain, which is dimensionless.
- For a small stress, the relation between stress and strain is linear. The elastic modulus is the proportionality constant in this linear relation.
- Tensile (or compressive) strain is the response of an object or medium to tensile (or compressive) stress. Here, the elastic modulus is called Young's modulus. Tensile (or compressive) stress causes elongation (or shortening) of the object or medium and is due to an external forces acting along only one direction perpendicular to the cross-section.
- Bulk strain is the response of an object or medium to bulk stress. Here, the elastic modulus is called the bulk modulus. Bulk stress causes a change in the volume of the object or medium and is caused by forces acting on the body from all directions, perpendicular to its surface. Compressibility of an object or medium is the reciprocal of its bulk modulus.
- Shear strain is the deformation of an object or medium under shear stress. The shear modulus is the elastic modulus in this case. Shear stress is caused by forces acting along the object's two parallel surfaces.

12.4 Elasticity and Plasticity

- An object or material is elastic if it comes back to its original shape and size when the stress vanishes. In elastic deformations with stress values lower than the proportionality limit, stress is proportional to strain. When stress goes beyond the proportionality limit, the deformation is still elastic but nonlinear up to the elasticity limit.
- An object or material has plastic behavior when stress is larger than the elastic limit. In the plastic region, the object or material does not come back to its original size or shape when stress vanishes but acquires a permanent deformation. Plastic behavior ends at the breaking point.

Conceptual Questions

12.1 Conditions for Static Equilibrium

1. What can you say about the velocity of a moving body that is in dynamic equilibrium?
2. Under what conditions can a rotating body be in equilibrium? Give an example.
3. What three factors affect the torque created by a force relative to a specific pivot point?
4. Mechanics sometimes put a length of pipe over the handle of a wrench when trying to remove a very tight bolt. How does this help?

For the next four problems, evaluate the statement as either true or false and explain your answer.

5. If there is only one external force (or torque) acting on an object, it cannot be in equilibrium.
6. If an object is in equilibrium there must be an even number of forces acting on it.
7. If an odd number of forces act on an object, the object cannot be in equilibrium.
8. A body moving in a circle with a constant speed is in rotational equilibrium.
9. What purpose is served by a long and flexible pole carried by wire-walkers?

12.2 Examples of Static Equilibrium

10. Is it possible to rest a ladder against a rough wall when the floor is frictionless?
11. Show how a spring scale and a simple fulcrum can be used to weigh an object whose weight is larger than the maximum reading on the scale.
12. A painter climbs a ladder. Is the ladder more likely to slip when the painter is near the bottom or near the top?

12.3 Stress, Strain, and Elastic Modulus

Note: Unless stated otherwise, the weights of the wires, rods, and other elements are assumed to be negligible. Elastic moduli of selected materials are given in Table 12.1.

13. Why can a squirrel jump from a tree branch to the ground and run away undamaged, while a human could break a bone in such a fall?
14. When a glass bottle full of vinegar warms up, both the vinegar and the glass expand, but the vinegar expands significantly more with temperature than does the glass. The bottle will break if it is filled up to its very tight cap. Explain why and how a pocket of air above the vinegar prevents the bottle from breaking.
15. A thin wire strung between two nails in the wall is used to support a large picture. Is the wire likely to snap if it is strung tightly or if it is strung so that it sags considerably?
16. Review the relationship between stress and strain. Can you find any similarities between the two quantities?
17. What type of stress are you applying when you press on the ends of a wooden rod? When you pull on its ends?
18. Can compressive stress be applied to a rubber band?
19. Can Young's modulus have a negative value? What about the bulk modulus?
20. If a hypothetical material has a negative bulk modulus, what happens when you squeeze a piece of it?
21. Discuss how you might measure the bulk modulus of a liquid.

12.4 Elasticity and Plasticity

Note: Unless stated otherwise, the weights of the wires, rods, and other elements are assumed to be negligible. Elastic moduli of selected materials are given in Table 12.1.

22. What is meant when a fishing line is designated as "a 10-lb test?"
23. Steel rods are commonly placed in concrete before it sets. What is the purpose of these rods?

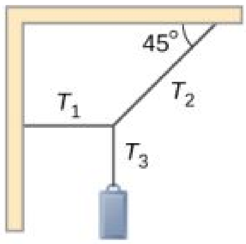
Problems

12.1 Conditions for Static Equilibrium

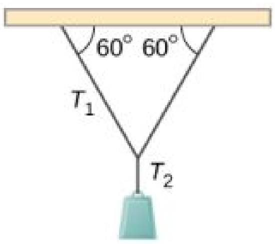
24. When tightening a bolt, you push perpendicularly on a wrench with a force of 165 N at a distance of 0.140 m from the center of the bolt. How much torque are you exerting relative to the center of the bolt?

25. When opening a door, you push on it perpendicularly with a force of 55.0 N at a distance of 0.850 m from the hinges. What torque are you exerting relative to the hinges?

26. Find the magnitude of the tension in each supporting cable shown below. In each case, the weight of the suspended body is 100.0 N and the masses of the cables are negligible.



(a)



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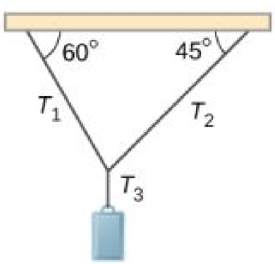
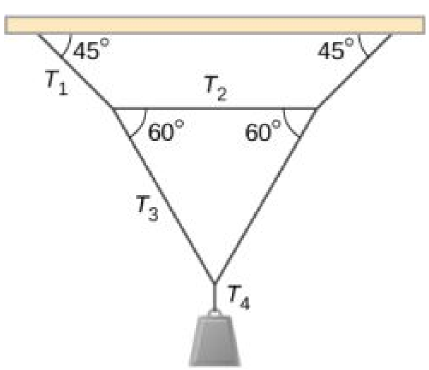
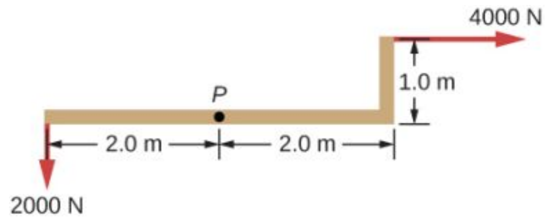


(c)

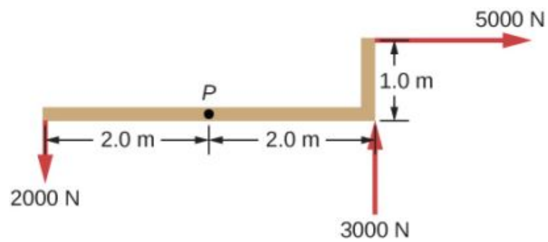


(d)

27. What force must be applied at point *P* to keep the structure shown in equilibrium? The weight of the structure is negligible.



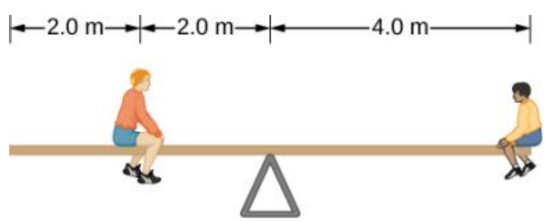
28. Is it possible to apply a force at *P* to keep in equilibrium the structure shown? The weight of the structure is negligible.



29. Two children push on opposite sides of a door during play. Both push horizontally and perpendicular to the door. One child pushes with a force of 17.5 N at a distance of 0.600 m from the hinges, and the second child pushes at a distance of 0.450 m. What force must the second child exert to keep the door from moving? Assume friction is negligible.

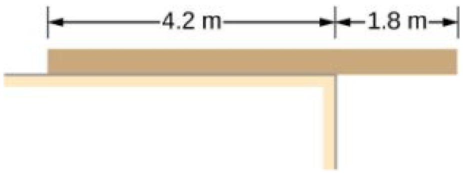
30. A small 1000-kg SUV has a wheel base of 3.0 m. If 60% if its weight rests on the front wheels, how far behind the front wheels is the wagon's center of mass?

31. The uniform seesaw is balanced at its center of mass, as seen below. The smaller boy on the right has a mass of 40.0 kg. What is the mass of his friend?

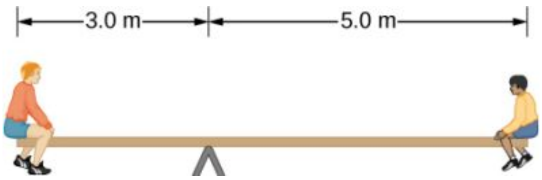


12.2 Examples of Static Equilibrium

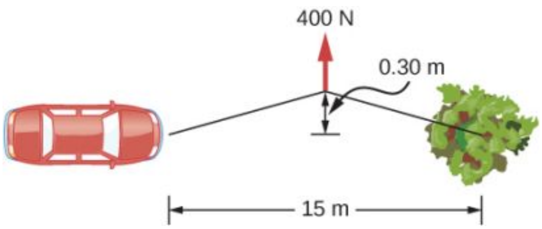
32. A uniform plank rests on a level surface as shown below. The plank has a mass of 30 kg and is 6.0 m long. How much mass can be placed at its right end before it tips? (*Hint:* When the board is about to tip over, it makes contact with the surface only along the edge that becomes a momentary axis of rotation.)



33. The uniform seesaw shown below is balanced on a fulcrum located 3.0 m from the left end. The smaller boy on the right has a mass of 40 kg and the bigger boy on the left has a mass 80 kg. What is the mass of the board?

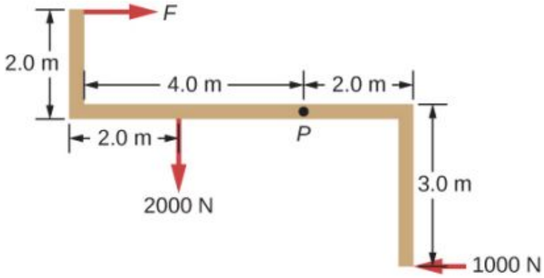


34. In order to get his car out of the mud, a man ties one end of a rope to the front bumper and the other end to a tree 15 m away, as shown below. He then pulls on the center of the rope with a force of 400 N, which causes its center to be displaced 0.30 m, as shown. What is the force of the rope on the car?



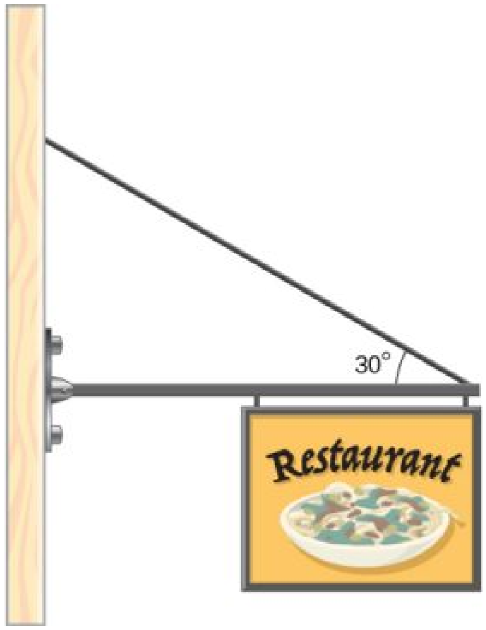
35. A uniform 40.0-kg scaffold of length 6.0 m is supported by two light cables, as shown below. An 80.0-kg painter stands 1.0 m from the left end of the scaffold, and his painting equipment is 1.5 m from the right end. If the tension in the left cable is twice that in the right cable, find the tensions in the cables and the mass of the equipment.

36. When the structure shown below is supported at point *P*, it is in equilibrium. Find the magnitude of force *F* and the force applied at *P*. The weight of the structure is negligible.



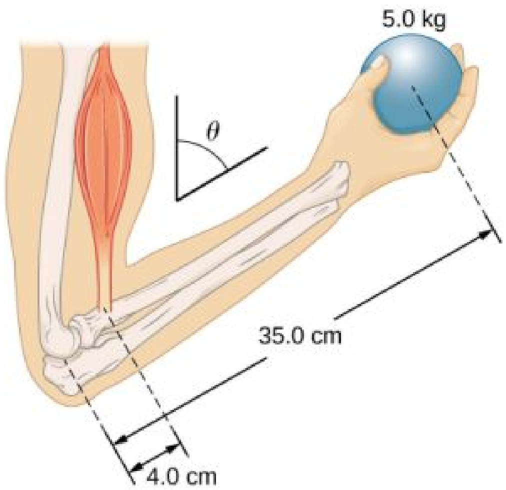
37. To get up on the roof, a person (mass 70.0 kg) places a 6.00-m aluminum ladder (mass 10.0 kg) against the house on a concrete pad with the base of the ladder 2.00 m from the house. The ladder rests against a plastic rain gutter, which we can assume to be frictionless. The center of mass of the ladder is 2.00 m from the bottom. The person is standing 3.00 m from the bottom. Find the normal reaction and friction forces on the ladder at its base.

38. A uniform horizontal strut weighs 400.0 N. One end of the strut is attached to a hinged support at the wall, and the other end of the strut is attached to a sign that weighs 200.0 N. The strut is also supported by a cable attached between the end of the strut and the wall. Assuming that the entire weight of the sign is attached at the very end of the strut, find the tension in the cable and the force at the hinge of the strut.

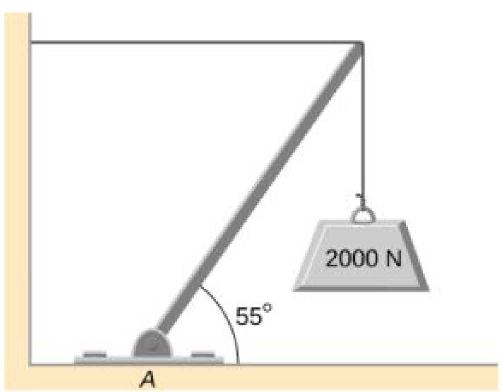


39. The forearm shown below is positioned at an angle θ with respect to the upper arm, and a 5.0-kg mass is held in the hand. The total mass of the forearm and hand is 3.0 kg, and their

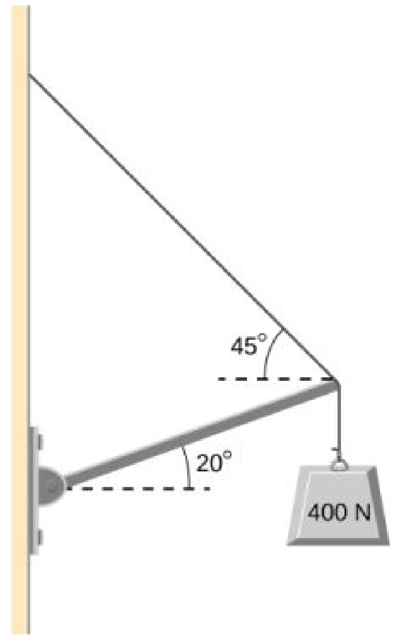
center of mass is 15.0 cm from the elbow. (a) What is the magnitude of the force that the biceps muscle exerts on the forearm for $\theta = 60°$? (b) What is the magnitude of the force on the elbow joint for the same angle? (c) How do these forces depend on the angle θ?



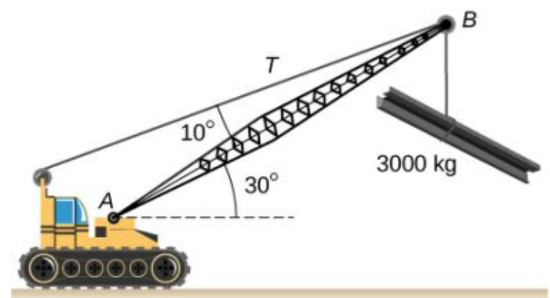
40. The uniform boom shown below weighs 3000 N. It is supported by the horizontal guy wire and by the hinged support at point *A*. What are the forces on the boom due to the wire and due to the support at *A*? Does the force at *A* act along the boom?



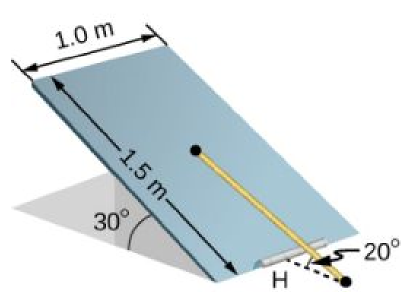
41. The uniform boom shown below weighs 700 N, and the object hanging from its right end weighs 400 N. The boom is supported by a light cable and by a hinge at the wall. Calculate the tension in the cable and the force on the hinge on the boom. Does the force on the hinge act along the boom?



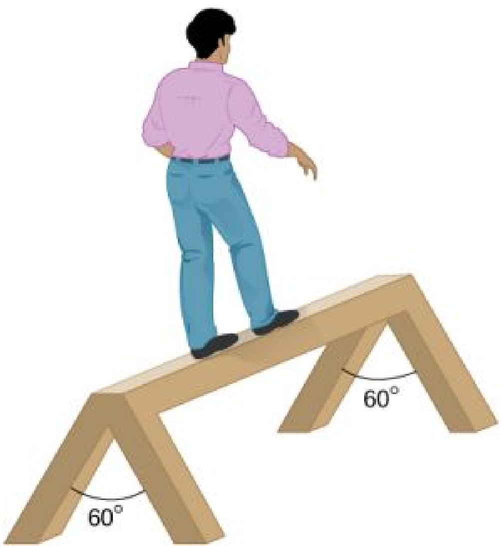
42. A 12.0-m boom, *AB*, of a crane lifting a 3000-kg load is shown below. The center of mass of the boom is at its geometric center, and the mass of the boom is 1000 kg. For the position shown, calculate tension *T* in the cable and the force at the axle *A*.



43. A uniform trapdoor shown below is 1.0 m by 1.5 m and weighs 300 N. It is supported by a single hinge (H), and by a light rope tied between the middle of the door and the floor. The door is held at the position shown, where its slab makes a $30°$ angle with the horizontal floor and the rope makes a $20°$ angle with the floor. Find the tension in the rope and the force at the hinge.



44. A 90-kg man walks on a sawhorse, as shown below. The sawhorse is 2.0 m long and 1.0 m high, and its mass is 25.0 kg. Calculate the normal reaction force on each leg at the contact point with the floor when the man is 0.5 m from the far end of the sawhorse. (*Hint:* At each end, find the total reaction force first. This reaction force is the vector sum of two reaction forces, each acting along one leg. The normal reaction force at the contact point with the floor is the normal (with respect to the floor) component of this force.)



12.3 Stress, Strain, and Elastic Modulus

45. The "lead" in pencils is a graphite composition with a Young's modulus of approximately $1.0 \times 10^9 \mathrm{N/m^2}$. Calculate the change in length of the lead in an automatic pencil if you tap it straight into the pencil with a force of 4.0 N. The lead is 0.50 mm in diameter and 60 mm long.

46. TV broadcast antennas are the tallest artificial structures on Earth. In 1987, a 72.0-kg physicist placed himself and 400 kg of equipment at the top of a 610-m-high antenna to perform gravity experiments. By how much was the antenna compressed, if we consider it to be equivalent to a steel cylinder 0.150 m in radius?

47. By how much does a 65.0-kg mountain climber stretch her 0.800-cm diameter nylon rope when she hangs 35.0 m below a rock outcropping? (For nylon, $Y = 1.35 \times 10^9 \mathrm{Pa}$.)

48. When water freezes, its volume increases by 9.05%. What force per unit area is water capable of exerting on a container when it freezes?

49. A farmer making grape juice fills a glass bottle to the brim and caps it tightly. The juice expands more than the glass when it warms up, in such a way that the volume increases by 0.2%. Calculate the force exerted by the juice per square centimeter if its bulk modulus is $1.8 \times 10^9 \mathrm{N/m^2}$, assuming the bottle does not break.

50. A disk between vertebrae in the spine is subjected to a shearing force of 600.0 N. Find its shear deformation, using the shear modulus of $1.0 \times 10^9\ \mathrm{N/m^2}$. The disk is equivalent to a solid cylinder 0.700 cm high and 4.00 cm in diameter.

51. A vertebra is subjected to a shearing force of 500.0 N. Find the shear deformation, taking the vertebra to be a cylinder 3.00 cm high and 4.00 cm in diameter.

52. Calculate the force a piano tuner applies to stretch a steel piano wire by 8.00 mm, if the wire is originally 1.35 m long and its diameter is 0.850 mm.

53. A 20.0-m-tall hollow aluminum flagpole is equivalent in strength to a solid cylinder 4.00 cm in diameter. A strong wind bends the pole as much as a horizontal 900.0-N force on the top would do. How far to the side does the top of the pole flex?

54. A copper wire of diameter 1.0 cm stretches 1.0% when it is used to lift a load upward with an acceleration of $2.0\ \mathrm{m/s^2}$. What is the weight of the load?

55. As an oil well is drilled, each new section of drill pipe supports its own weight and the weight of the pipe and the drill bit beneath it. Calculate the stretch in a new 6.00-m-long steel pipe that supports a 100-kg drill bit and a 3.00-km length of pipe with a linear mass density of 20.0 kg/m. Treat the pipe as a solid cylinder with a 5.00-cm diameter.

56. A large uniform cylindrical steel rod of density $\rho = 7.8\ \mathrm{g/cm^3}$ is 2.0 m long and has a diameter of 5.0 cm. The rod is fastened to a concrete floor with its long axis vertical. What is the normal stress in the rod at the cross-section located at (a) 1.0 m from its lower end? (b) 1.5 m from the lower end?

57. A 90-kg mountain climber hangs from a nylon rope and stretches it by 25.0 cm. If the rope was originally 30.0 m long and its diameter is 1.0 cm, what is Young's modulus for the nylon?

58. A suspender rod of a suspension bridge is 25.0 m long. If the rod is made of steel, what must its diameter be so that it does not stretch more than 1.0 cm when a 2.5×10^4-kg truck passes by it? Assume that the rod supports all of the weight of the truck.

59. A copper wire is 1.0 m long and its diameter is 1.0 mm. If the wire hangs vertically, how much weight must be added to its free end in order to stretch it 3.0 mm?

60. A 100-N weight is attached to a free end of a metallic wire that hangs from the ceiling. When a second 100-N weight is added to the wire, it stretches 3.0 mm. The diameter and the length of the wire are 1.0 mm and 2.0 m, respectively. What is Young's modulus of the metal used to manufacture the wire?

61. The bulk modulus of a material is $1.0 \times 10^{11}\ \text{N/m}^2$. What fractional change in volume does a piece of this material undergo when it is subjected to a bulk stress increase of $10^7\ \text{N/m}^2$? Assume that the force is applied uniformly over the surface.

62. Normal forces of magnitude $1.0 \times 10^6\text{N}$ are applied uniformly to a spherical surface enclosing a volume of a liquid. This causes the radius of the surface to decrease from 50.000 cm to 49.995 cm. What is the bulk modulus of the liquid?

63. During a walk on a rope, a tightrope walker creates a tension of $3.94 \times 10^3\ N$ in a wire that is stretched between two supporting poles that are 15.0 m apart. The wire has a diameter of 0.50 cm when it is not stretched. When the walker is on the wire in the middle between the poles the wire makes an angle of $5.0°$ below the horizontal. How much does this tension stretch the steel wire when the walker is this position?

64. When using a pencil eraser, you exert a vertical force of 6.00 N at a distance of 2.00 cm from the hardwood-eraser joint. The pencil is 6.00 mm in diameter and is held at an angle of $20.0°$ to the horizontal. (a) By how much does the wood flex perpendicular to its length? (b) How much is it compressed lengthwise?

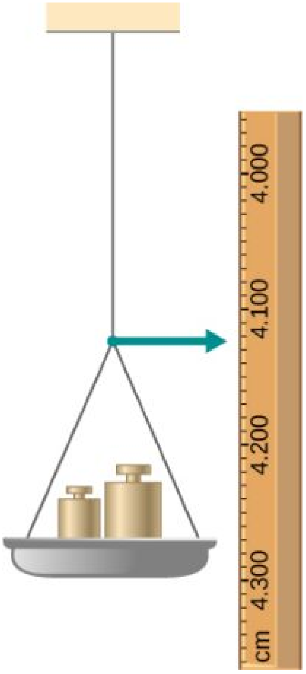
65. Normal forces are applied uniformly over the surface of a spherical volume of water whose radius is 20.0 cm. If the pressure on the surface is increased by 200 MPa, by how much does the radius of the sphere decrease?

12.4 Elasticity and Plasticity

66. A uniform rope of cross-sectional area $0.50\ \text{cm}^2$ breaks when the tensile stress in it reaches $6.00 \times 10^6\text{N/m}^2$. (a) What is the maximum load that can be lifted slowly at a constant speed by the rope? (b) What is the maximum load that can be lifted by the rope with an acceleration of $4.00\ \text{m/s}^2$?

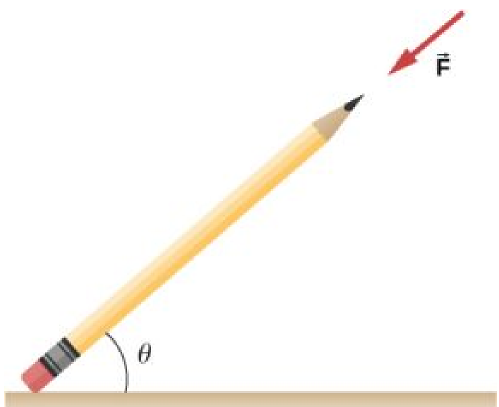
67. One end of a vertical metallic wire of length 2.0 m and diameter 1.0 mm is attached to a ceiling, and the other end is attached to a 5.0-N weight pan, as shown below. The position of the pointer before the pan is 4.000 cm. Different weights are then added to the pan area, and the position of the pointer is recorded in the table shown. Plot stress versus strain for this wire, then use the resulting curve to determine Young's modulus and the proportionality limit of the metal. What metal is this most likely to be?

Added load (including pan) (N)	Scale reading (cm)
0	4.000
15	4.036
25	4.073
35	4.109
45	4.146
55	4.181
65	4.221
75	4.266
85	4.316



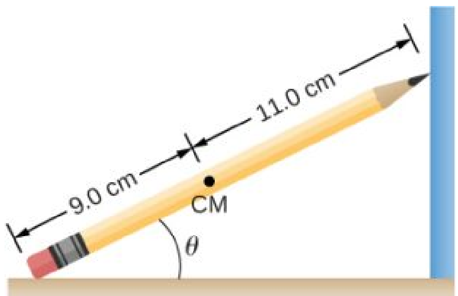
Additional Problems

69. The coefficient of static friction between the rubber eraser of the pencil and the tabletop is $\mu_s = 0.80$. If the force $\vec{\mathbf{F}}$ is applied along the axis of the pencil, as shown below, what is the minimum angle at which the pencil can stand without slipping? Ignore the weight of the pencil.

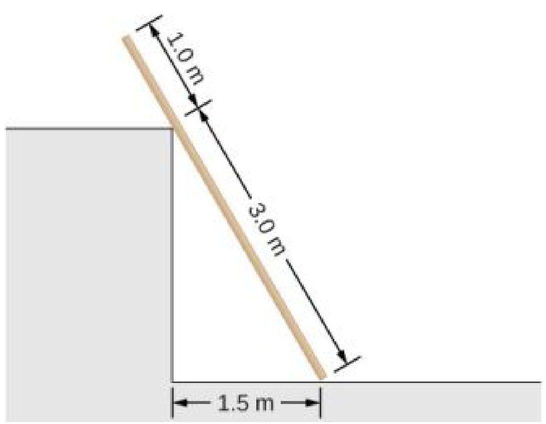


70. A pencil rests against a corner, as shown below. The sharpened end of the pencil touches a smooth vertical surface and the eraser end touches a rough horizontal floor. The coefficient of static friction between the eraser and the floor is $\mu_s = 0.80$. The center of mass of the pencil is located 9.0 cm from the tip of the eraser and 11.0 cm from the tip of the pencil lead. Find the minimum angle θ for which the pencil does not slip.

68. An aluminum $\left(\rho = 2.7 \text{ g/cm}^3\right)$ wire is suspended from the ceiling and hangs vertically. How long must the wire be before the stress at its upper end reaches the proportionality limit, which is $8.0 \times 10^7 \text{N/m}^2$?



71. A uniform 4.0-m plank weighing 200.0 N rests against the corner of a wall, as shown below. There is no friction at the point where the plank meets the corner. (a) Find the forces that the corner and the floor exert on the plank. (b) What is the minimum coefficient of static friction between the floor and the plank to prevent the plank from slipping?



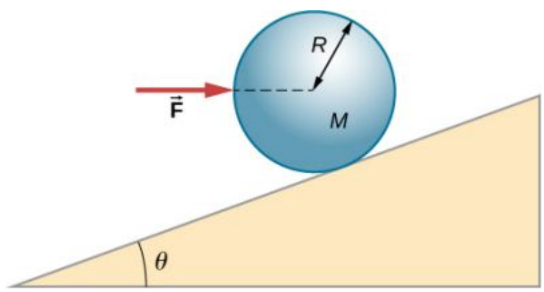
72. A 40-kg boy jumps from a height of 3.0 m, lands on one foot and comes to rest in 0.10 s after he hits the ground. Assume that he comes to rest with a constant acceleration opposite to the motion. If the total cross-sectional area of the bones in his legs just above his ankles is 3.0 cm^2, what is the compression stress in these bones? Leg bones can be fractured when they are subjected to stress greater than 1.7×10^8 Pa. Is the boy in danger of breaking his leg?

73. Two thin rods, one made of steel and the other of aluminum, are joined end to end. Each rod is 2.0 m long and has cross-sectional area 9.1 mm^2. If a 10,000-N tensile force is applied at each end of the combination, find: (a) stress in each rod; (b) strain in each rod; and, (c) elongation of each rod.

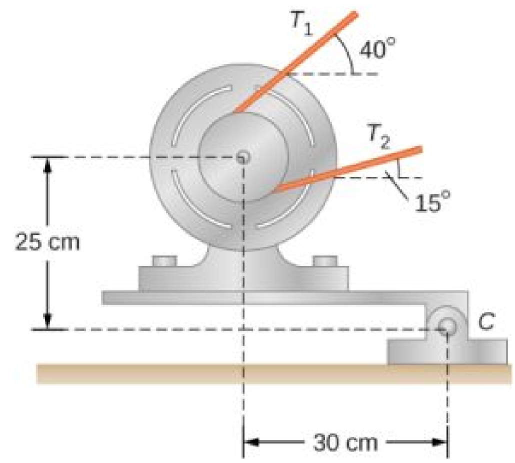
74. Two rods, one made of copper and the other of steel, have the same dimensions. If the copper rod stretches by 0.15 mm under some stress, how much does the steel rod stretch under the same stress?

Challenge Problems

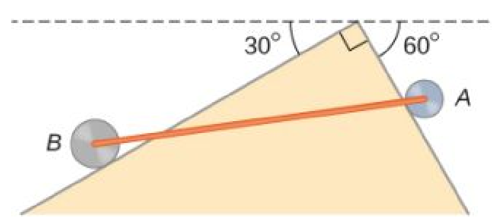
75. A horizontal force $\vec{\mathbf{F}}$ is applied to a uniform sphere in direction exact toward the center of the sphere, as shown below. Find the magnitude of this force so that the sphere remains in static equilibrium. What is the frictional force of the incline on the sphere?



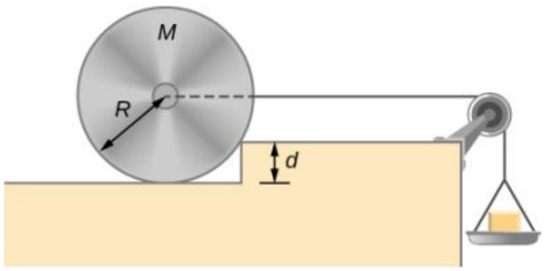
76. When a motor is set on a pivoted mount seen below, its weight can be used to maintain tension in the drive belt. When the motor is not running the tensions T_1 and T_2 are equal. The total mass of the platform and the motor is 100.0 kg, and the diameter of the drive belt pulley is 16.0 cm. when the motor is off, find: (a) the tension in the belt, and (b) the force at the hinged platform support at point *C*. Assume that the center of mass of the motor plus platform is at the center of the motor.



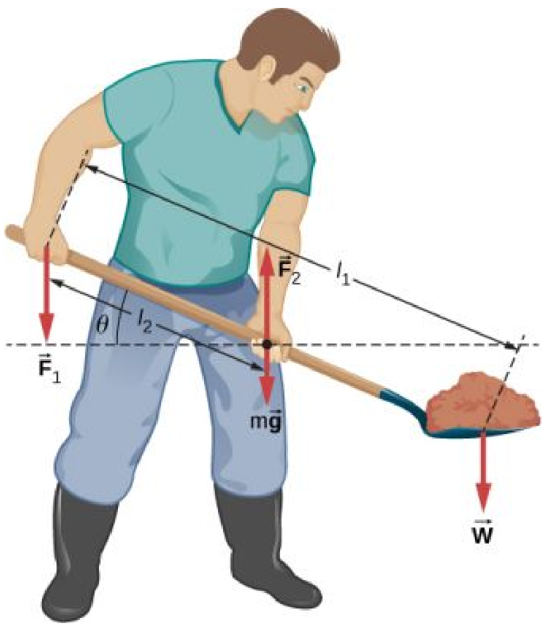
77. Two wheels *A* and *B* with weights *w* and 2*w*, respectively, are connected by a uniform rod with weight *w*/2, as shown below. The wheels are free to roll on the sloped surfaces. Determine the angle that the rod forms with the horizontal when the system is in equilibrium. *Hint:* There are five forces acting on the rod, which is two weights of the wheels, two normal reaction forces at points where the wheels make contacts with the wedge, and the weight of the rod.



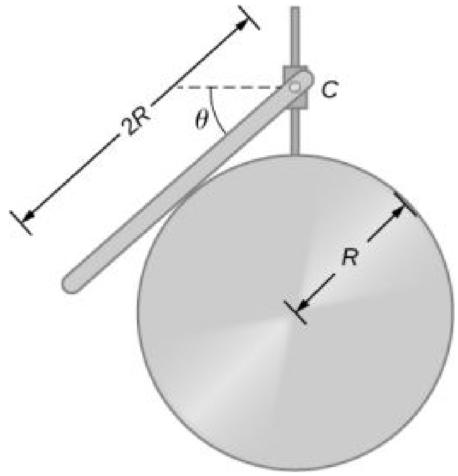
78. Weights are gradually added to a pan until a wheel of mass *M* and radius *R* is pulled over an obstacle of height *d*, as shown below. What is the minimum mass of the weights plus the pan needed to accomplish this?



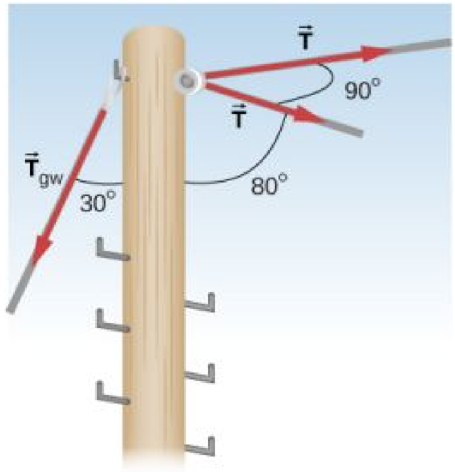
79. In order to lift a shovelful of dirt, a gardener pushes downward on the end of the shovel and pulls upward at distance l_2 from the end, as shown below. The weight of the shovel is $m\vec{g}$ and acts at the point of application of $\vec{F}_2$. Calculate the magnitudes of the forces $\vec{F}_1$ and $\vec{F}_2$ as functions of l_1, l_2, *mg*, and the weight *W* of the load. Why do your answers not depend on the angle θ that the shovel makes with the horizontal?



80. A uniform rod of length *2R* and mass *M* is attached to a small collar *C* and rests on a cylindrical surface of radius *R*, as shown below. If the collar can slide without friction along the vertical guide, find the angle θ for which the rod is in static equilibrium.



81. The pole shown below is at a $90.0°$ bend in a power line and is therefore subjected to more shear force than poles in straight parts of the line. The tension in each line is $4.00 \times 10^4 \text{N}$, at the angles shown. The pole is 15.0 m tall, has an 18.0 cm diameter, and can be considered to have half the strength of hardwood. (a) Calculate the compression of the pole. (b) Find how much it bends and in what direction. (c) Find the tension in a guy wire used to keep the pole straight if it is attached to the top of the pole at an angle of $30.0°$ with the vertical. The guy wire is in the opposite direction of the bend.



CHAPTER 13
Gravitation

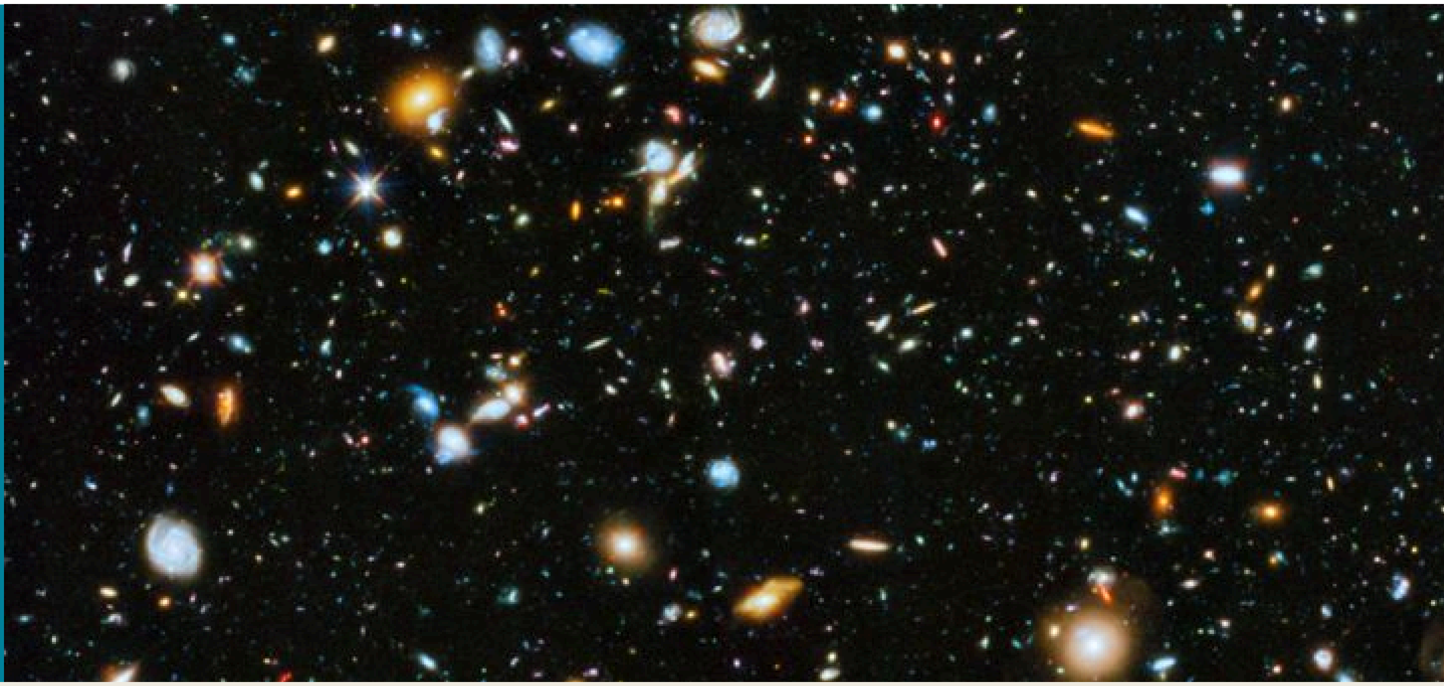


FIGURE 13.1 Our visible Universe contains billions of galaxies, whose very existence is due to the force of gravity. Gravity is ultimately responsible for the energy output of all stars—initiating thermonuclear reactions in stars, allowing the Sun to heat Earth, and making galaxies visible from unfathomable distances. Most of the dots you see in this image are not stars, but galaxies. (credit: modification of work by NASA/ESA)

CHAPTER OUTLINE

INTRODUCTION In this chapter, we study the nature of the gravitational force for objects as small as ourselves and for systems as massive as entire galaxies. We show how the gravitational force affects objects on Earth and the motion of the Universe itself. Gravity is the first force to be postulated as an action-at-a-distance force, that is, objects exert a gravitational force on one another without physical contact and that force falls to zero only at an infinite distance. Earth exerts a gravitational force on you, but so do our Sun, the Milky Way galaxy, and the billions of galaxies, like those shown above, which are so distant that we cannot see them with the naked eye.

13.1 Newton's Law of Universal Gravitation

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- List the significant milestones in the history of gravitation
- Calculate the gravitational force between two point masses
- Estimate the gravitational force between collections of mass

We first review the history of the study of gravitation, with emphasis on those phenomena that for thousands of years have inspired philosophers and scientists to search for an explanation. Then we examine the simplest form of Newton's law of universal gravitation and how to apply it.

The History of Gravitation

The earliest philosophers wondered why objects naturally tend to fall toward the ground. Aristotle (384–322 BCE) believed that it was the nature of rocks to seek Earth and the nature of fire to seek the Heavens. Brahmagupta (598~665 CE) postulated that Earth was a sphere and that objects possessed a natural affinity for it, falling toward the center from wherever they were located.

The motions of the Sun, our Moon, and the planets have been studied for thousands of years as well. These motions were described with amazing accuracy by Ptolemy (90–168 CE), whose method of epicycles described the paths of the planets as circles within circles. However, there is little evidence that anyone connected the motion of astronomical bodies with the motion of objects falling to Earth—until the seventeenth century.

The idea of heliocentrism, in which the Sun is at the center of the solar system, dates at least as far back as Aristarchus of Samos in the third century BCE, but attracted little attention, and Ptolemy's geocentric (Earth-centered) model held sway in Europe until the sixteenth century. Nicolaus Copernicus (1473–1543) is generally credited as being the first to produce a mathematical model of a heliocentric system and to challenge Ptolemy's geocentric system. This idea was supported by the incredibly precise naked-eye measurements of planetary motions by Tycho Brahe and their analysis by Johannes Kepler and Galileo Galilei. Kepler showed that the motion of each planet is an ellipse (the first of his three laws, discussed in Kepler's Laws of Planetary Motion), and Robert Hooke (the same Hooke who formulated Hooke's law for springs) intuitively suggested that these motions are due to the planets being attracted to the Sun. However, it was Isaac Newton who connected the acceleration of objects near Earth's surface with the centripetal acceleration of the Moon in its orbit about Earth.

Other prominent scientists and mathematicians of the time, particularly those outside of England, were reluctant to accept Newton's principles. It took the work of another prominent philosopher, writer, and scientist, Émilie du Châtelet, to establish the Newtonian gravitation as the accurate and overarching law. Du Châtelet, who had earlier laid the foundation for the understanding of conservation of energy as well as the principle that light had no mass, translated and augmented Newton's key work. She also utilized calculus to explain gravity, which helped lead to its acceptance.

Finally, in Einstein's Theory of Gravity, we look at the theory of general relativity proposed by Albert Einstein in 1916. His theory comes from a vastly different perspective, in which gravity is a manifestation of mass warping space and time. The consequences of his theory gave rise to many remarkable predictions, essentially all of which have been confirmed over the many decades following the publication of the theory (including the 2015 measurement of gravitational waves from the merger of two black holes).

Newton's Law of Universal Gravitation

Newton noted that objects at Earth's surface (hence at a distance of R_{E} from the center of Earth) have an acceleration of *g*, but the Moon, at a distance of about $60\ R_{\text{E}}$, has a centripetal acceleration about $(60)^2$ times smaller than *g*. He could explain this by postulating that a force exists between any two objects, whose magnitude is given by the product of the two masses divided by the square of the distance between them. We now know that this inverse square law is ubiquitous in nature, a function of geometry for point sources. The strength of any source at a distance *r* is spread over the surface of a sphere centered about the mass. The surface area of that sphere is proportional to r^2. In later chapters, we see this same form in the electromagnetic force.

NEWTON'S LAW OF GRAVITATION

Newton's law of gravitation can be expressed as

$$\vec{\mathbf{F}}_{12} = G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}}_{12} \qquad 13.1$$

where $\vec{\mathbf{F}}_{12}$ is the force on object 1 exerted by object 2 and $\hat{\mathbf{r}}_{12}$ is a unit vector that points from object 1 toward object 2.

As shown in Figure 13.2, the $\vec{\mathbf{F}}_{12}$ vector points from object 1 toward object 2, and hence represents an attractive force between the objects. The equal but opposite force $\vec{\mathbf{F}}_{21}$ is the force on object 2 exerted by object 1.

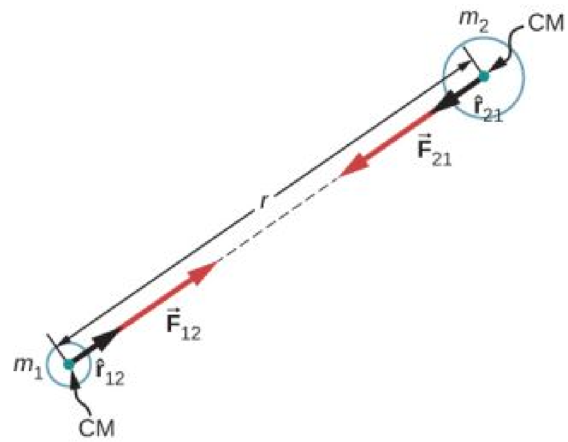


FIGURE 13.2 Gravitational force acts along a line joining the centers of mass of two objects.

These equal but opposite forces reflect Newton's third law, which we discussed earlier. Note that strictly speaking, Equation 13.1 applies to point masses—all the mass is located at one point. But it applies equally to any spherically symmetric objects, where *r* is the distance between the centers of mass of those objects. In many cases, it works reasonably well for nonsymmetrical objects, if their separation is large compared to their size, and we take *r* to be the distance between the center of mass of each body.

The Cavendish Experiment

A century after Newton published his law of universal gravitation, Henry Cavendish determined the proportionality constant *G* by performing a painstaking experiment. He constructed a device similar to that shown in Figure 13.3, in which small masses are suspended from a wire. Once in equilibrium, two fixed, larger masses are placed symmetrically near the smaller ones. The gravitational attraction creates a torsion (twisting) in the supporting wire that can be measured.

The constant *G* is called the **universal gravitational constant** and Cavendish determined it to be $G = 6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2$. The word 'universal' indicates that scientists think that this constant applies to masses of any composition and that it is the same throughout the Universe. The value of *G* is an incredibly small number, showing that the force of gravity is very weak. The attraction between masses as small as our bodies, or even objects the size of skyscrapers, is incredibly small. For example, two 1.0-kg masses located 1.0 meter apart exert a force of $6.7 \times 10^{-11}\ \text{N}$ on each other. This is the weight of a typical grain of pollen.

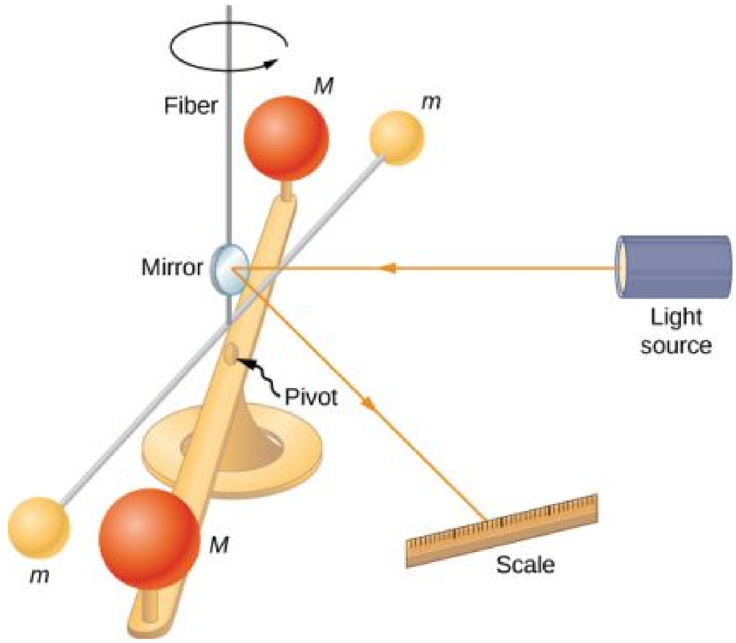


FIGURE 13.3 Cavendish used an apparatus similar to this to measure the gravitational attraction between two spheres (*m*) suspended from a wire and two stationary spheres (*M*). This is a common experiment performed in undergraduate laboratories, but it is quite challenging. Passing trucks outside the laboratory can create vibrations that overwhelm the gravitational forces.

Although gravity is the weakest of the four fundamental forces of nature, its attractive nature is what holds us to Earth, causes the planets to orbit the Sun and the Sun to orbit our galaxy, and binds galaxies into clusters, ranging from a few to millions. Gravity is the force that forms the Universe.

PROBLEM-SOLVING STRATEGY

Newton's Law of Gravitation

To determine the motion caused by the gravitational force, follow these steps:

1. Identify the two masses, one or both, for which you wish to find the gravitational force.
2. Draw a free-body diagram, sketching the force acting on each mass and indicating the distance between their centers of mass.
3. Apply Newton's second law of motion to each mass to determine how it will move.

EXAMPLE 13.1

A Collision in Orbit

Consider two nearly spherical *Soyuz* payload vehicles, in orbit about Earth, each with mass 9000 kg and diameter 4.0 m. They are initially at rest relative to each other, 10.0 m from center to center. (As we will see in Kepler's Laws of Planetary Motion, both orbit Earth at the same speed and interact nearly the same as if they were isolated in deep space.) Determine the gravitational force between them and their initial acceleration. Estimate how long it takes for them to drift together, and how fast they are moving upon impact.

Strategy

We use Newton's law of gravitation to determine the force between them and then use Newton's second law to find the acceleration of each. For the *estimate*, we find the final acceleration as well and assume the acceleration is constant at its average value, and we use the constant-acceleration equations from Motion along a Straight Line to find the time and speed of the collision.

Solution

The magnitude of the force is

$$\left|\vec{\mathbf{F}}_{12}\right| = F_{12} = G\frac{m_1 m_2}{r^2} = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}\frac{(9000\ \mathrm{kg})(9000\ \mathrm{kg})}{(10\ \mathrm{m})^2} = 5.4 \times 10^{-5}\mathrm{N}.$$

The initial acceleration of each payload is

$$a = \frac{F}{m} = \frac{5.4 \times 10^{-5}\ \mathrm{N}}{9000\ \mathrm{kg}} = 6.0 \times 10^{-9}\mathrm{m/s^2}.$$

The vehicles are 4.0 m in diameter, so the vehicles move from 10.0 m to 4.0 m apart, or a distance of 3.0 m each. A similar calculation to that above, for when the vehicles are 4.0 m apart, yields an acceleration of $3.8 \times 10^{-8}\ \mathrm{m/s^2}$, and the average of these two values is $2.2 \times 10^{-8}\ \mathrm{m/s^2}$. If we assume a constant acceleration of this value and they start from rest, then the vehicles collide with speed given by

$$v^2 = v_0^2 + 2a(x - x_0),\ \text{where } v_0 = 0,$$

so

$$v = \sqrt{2(2.2 \times 10^{-9}\ \mathrm{m/s^2})(3.0\ \mathrm{m})} = 3.6 \times 10^{-4}\ \mathrm{m/s}.$$

We use $v = v_0 + at$ to find $t = v/a = 1.7 \times 10^4$ s or about 4.6 hours.

Significance

These calculations—including the initial force—are only estimates, as the vehicles are probably not spherically symmetrical. But you can see that the force is incredibly small. Astronauts must tether themselves when doing work outside even the massive International Space Station (ISS), as in Figure 13.4, because the gravitational attraction cannot save them from even the smallest push away from the station.

FIGURE 13.4 This photo shows Ed White tethered to the Space Shuttle during a spacewalk. (credit: NASA)

CHECK YOUR UNDERSTANDING 13.1

What happens to force and acceleration as the vehicles fall together? What will our estimate of the velocity at a collision higher or lower than the speed actually be? And finally, what would happen if the masses were not identical? Would the force on each be the same or different? How about their accelerations?

The effect of gravity between two objects with masses on the order of these space vehicles is indeed small. Yet, the effect of gravity on you from Earth is significant enough that a fall into Earth of only a few feet can be dangerous. We examine the force of gravity near Earth's surface in the next section.

EXAMPLE 13.2

Attraction between Galaxies

Find the acceleration of our galaxy, the Milky Way, due to the nearest comparably sized galaxy, the Andromeda galaxy (Figure 13.5). The approximate mass of each galaxy is 800 billion solar masses (a solar mass is the mass of our Sun), and they are separated by 2.5 million light-years. (Note that the mass of Andromeda is not so well known but is believed to be slightly larger than our galaxy.) Each galaxy has a diameter of roughly 100,000 light-years $(1 \text{ light-year} = 9.5 \times 10^{15} \text{ m})$.

FIGURE 13.5 Galaxies interact gravitationally over immense distances. The Andromeda galaxy is the nearest spiral galaxy to the Milky Way, and they will eventually collide. (credit: Boris Štromar)

Strategy

As in the preceding example, we use Newton's law of gravitation to determine the force between them and then use Newton's second law to find the acceleration of the Milky Way. We can consider the galaxies to be point masses, since their sizes are about 25 times smaller than their separation. The mass of the Sun (see Appendix D) is 2.0×10^{30} kg and a light-year is the distance light travels in one year, 9.5×10^{15} m.

Solution

The magnitude of the force is

$$F_{12} = G\frac{m_1 m_2}{r^2} = (6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)\frac{[(800 \times 10^9)(2.0 \times 10^{30} \text{ kg})]^2}{[(2.5 \times 10^6)(9.5 \times 10^{15} \text{ m})]^2} = 3.0 \times 10^{29} \text{ N}.$$

The acceleration of the Milky Way is

$$a = \frac{F}{m} = \frac{3.0 \times 10^{29} \text{ N}}{(800 \times 10^9)(2.0 \times 10^{30} \text{ kg})} = 1.9 \times 10^{-13} \text{m/s}^2.$$

Significance

Does this value of acceleration seem astoundingly small? If they start from rest, then they would accelerate directly toward each other, "colliding" at their center of mass. Let's estimate the time for this to happen. The initial acceleration is ~$10^{-13}\ \text{m/s}^2$, so using $v = at$, we see that it would take ~$10^{13}\ \text{s}$ for each galaxy to reach a speed of 1.0 m/s, and they would be only ~$0.5\ \times\ 10^{13}\ \text{m}$ closer. That is nine orders of magnitude smaller than the initial distance between them. In reality, such motions are rarely simple. These two galaxies, along with about 50 other smaller galaxies, are all gravitationally bound into our local cluster. Our local cluster is gravitationally bound to other clusters in what is called a supercluster. All of this is part of the great cosmic dance that results from gravitation, as shown in Figure 13.6.

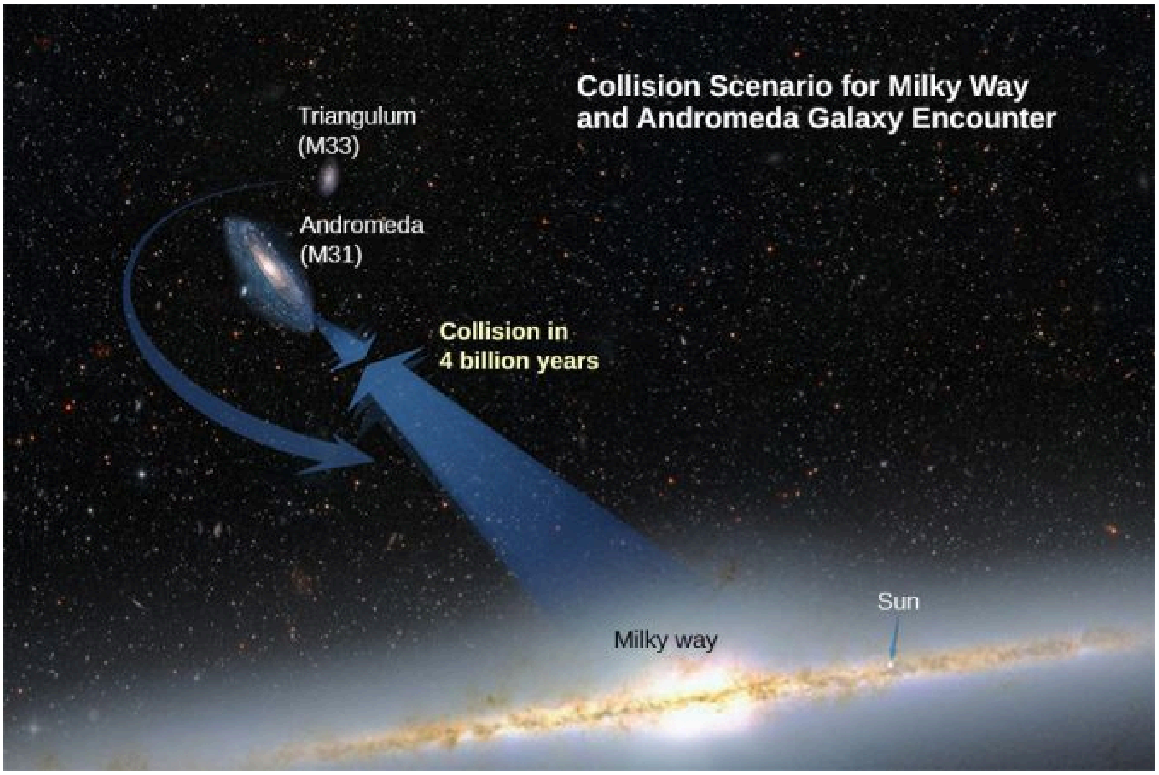


FIGURE 13.6 Based on the results of this example, plus what astronomers have observed elsewhere in the Universe, our galaxy will collide with the Andromeda Galaxy in about 4 billion years. (credit: modification of work by NASA; ESA; A. Feild and R. van der Marel, STScI)

13.2 Gravitation Near Earth's Surface

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the connection between the constants G and g
- Determine the mass of an astronomical body from free-fall acceleration at its surface
- Describe how the value of g varies due to location and Earth's rotation

In this section, we observe how Newton's law of gravitation applies at the surface of a planet and how it connects with what we learned earlier about free fall. We also examine the gravitational effects within spherical bodies.

Weight

Recall that the acceleration of a free-falling object near Earth's surface is approximately $g = 9.80\ \text{m/s}^2$. The force causing this acceleration is called the weight of the object, and from Newton's second law, it has the value *mg*. This weight is present regardless of whether the object is in free fall. We now know that this force is the gravitational force between the object and Earth. If we substitute *mg* for the magnitude of $\vec{\mathbf{F}}_{12}$ in Newton's law of universal gravitation, *m* for m_1, and M_E for m_2, we obtain the scalar equation

$$mg = G\frac{mM_E}{r^2}$$

where r is the distance between the centers of mass of the object and Earth. The mass m of the object cancels, leaving

$$g = G\frac{M_E}{r^2}. \tag{13.2}$$

The average radius of Earth is about 6370 km. Hence, for objects within a few kilometers of Earth's surface, we can take $r = R_E$ (Figure 13.7). This explains why all masses free fall with the same acceleration. We have ignored the fact that Earth also accelerates toward the falling object, but that is acceptable as long as the mass of Earth is much larger than that of the object.

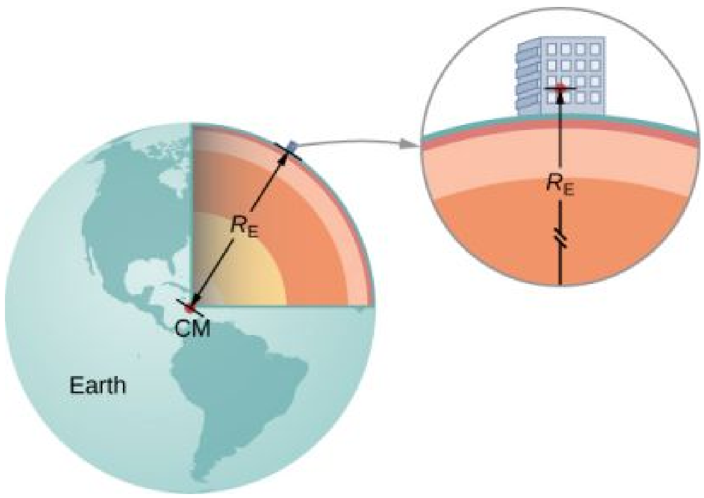


FIGURE 13.7 We can take the distance between the centers of mass of Earth and an object on its surface to be the radius of Earth, provided that its size is much less than the radius of Earth.

EXAMPLE 13.3

Masses of Earth and Moon

Have you ever wondered how we know the mass of Earth? We certainly can't place it on a scale. The values of g and the radius of Earth were measured with reasonable accuracy centuries ago.

a. Use the standard values of g, R_E, and Equation 13.2 to find the mass of Earth.
b. Estimate the value of g on the Moon. Use the fact that the Moon has a radius of about 1700 km (a value of this accuracy was determined many centuries ago) and assume it has the same average density as Earth, $5500\ \text{kg/m}^3$.

Strategy

With the known values of g and R_E, we can use Equation 13.2 to find M_E. For the Moon, we use the assumption of equal average density to determine the mass from a ratio of the volumes of Earth and the Moon.

Solution

a. Rearranging Equation 13.2, we have

$$M_E = \frac{gR_E^2}{G} = \frac{9.80\ \text{m/s}^2(6.37 \times 10^6\ \text{m})^2}{6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2} = 5.95 \times 10^{24}\ \text{kg}.$$

b. The volume of a sphere is proportional to the radius cubed, so a simple ratio gives us

$$\frac{M_{\mathrm{M}}}{M_{\mathrm{E}}} = \frac{R_{\mathrm{M}}^3}{R_{\mathrm{E}}^3} \rightarrow M_{\mathrm{M}} = \left(\frac{(1.7 \times 10^6 \text{ m})^3}{(6.37 \times 10^6 \text{ m})^3} \right) (5.95 \times 10^{24} \text{ kg}) = 1.1 \times 10^{23} \text{ kg}.$$

We now use Equation 13.2.

$$g_{\mathrm{M}} = G \frac{M_{\mathrm{M}}}{r_{\mathrm{M}}^2} = (6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2) \frac{(1.1 \times 10^{23} \text{ kg})}{(1.7 \times 10^6 \text{ m})^2} = 2.5 \text{m/s}^2$$

Significance

As soon as Cavendish determined the value of *G* in 1798, the mass of Earth could be calculated. (In fact, that was the ultimate purpose of Cavendish's experiment in the first place.) The value we calculated for *g* of the Moon is incorrect. The average density of the Moon is actually only 3340 kg/m^3 and $g = 1.6 \text{ m/s}^2$ at the surface. Newton attempted to measure the mass of the Moon by comparing the effect of the Sun on Earth's ocean tides compared to that of the Moon. His value was a factor of two too small. The most accurate values for *g* and the mass of the Moon come from tracking the motion of spacecraft that have orbited the Moon. But the mass of the Moon can actually be determined accurately without going to the Moon. Earth and the Moon orbit about a common center of mass, and careful astronomical measurements can determine that location. The ratio of the Moon's mass to Earth's is the ratio of [the distance from the common center of mass to the Moon's center] to [the distance from the common center of mass to Earth's center].

Later in this chapter, we will see that the mass of other astronomical bodies also can be determined by the period of small satellites orbiting them. But until Cavendish determined the value of *G*, the masses of all these bodies were unknown.

EXAMPLE 13.4

Gravity above Earth's Surface

What is the value of *g* 400 km above Earth's surface, where the International Space Station is in orbit?

Strategy

Using the value of M_{E} and noting the radius is $r = R_{\mathrm{E}} + 400 \text{ km}$, we use Equation 13.2 to find *g*.

From Equation 13.2 we have

$$g = G \frac{M_{\mathrm{E}}}{r^2} = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2 \frac{5.96 \times 10^{24} \text{ kg}}{(6.37 \times 10^6 + 400 \times 10^3 \text{ m})^2} = 8.67 \text{ m/s}^2.$$

Significance

We often see video of astronauts in space stations, apparently weightless. But clearly, the force of gravity is acting on them. Comparing the value of *g* we just calculated to that on Earth (9.80 m/s^2), we see that the astronauts in the International Space Station still have 88% of their weight. They only appear to be weightless because they are in free fall. We will come back to this in Satellite Orbits and Energy.

CHECK YOUR UNDERSTANDING 13.2

How does your weight at the top of a tall building compare with that on the first floor? Do you think engineers need to take into account the change in the value of *g* when designing structural support for a very tall building?

The Gravitational Field

Equation 13.2 is a scalar equation, giving the magnitude of the gravitational acceleration as a function of the distance from the center of the mass that causes the acceleration. But we could have retained the vector form for the force of gravity in Equation 13.1, and written the acceleration in vector form as

$$\vec{\mathbf{g}} = G\frac{M}{r^2}\hat{\mathbf{r}}.$$

We identify the vector field represented by $\vec{\mathbf{g}}$ as the **gravitational field** caused by mass M. We can picture the field as shown Figure 13.8. The lines are directed radially inward and are symmetrically distributed about the mass.

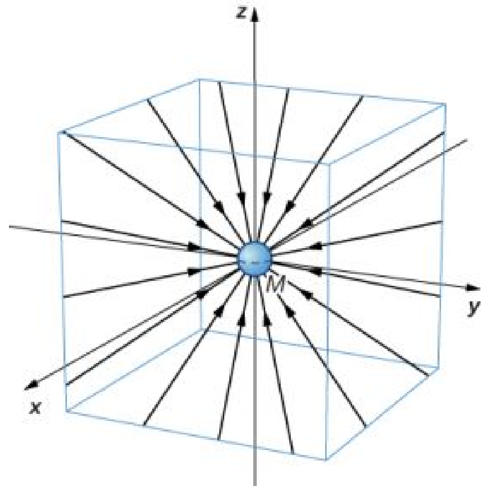


FIGURE 13.8 Field line representation of the gravitational field created by Earth.

The direction of $\vec{\mathbf{g}}$ is parallel to the field lines at any point. The strength of $\vec{\mathbf{g}}$ at any point is inversely proportional to the line spacing. Another way to state this is that the magnitude of the field in any region is proportional to the number of lines that pass through a unit surface area, effectively a density of lines. Since the lines are equally spaced in all directions, the number of lines per unit surface area at a distance *r* from the mass is the total number of lines divided by the surface area of a sphere of radius *r*, which is proportional to r^2. Hence, this picture perfectly represents the inverse square law, in addition to indicating the direction of the field. In the field picture, we say that a mass *m* interacts with the gravitational field of mass *M*. We will use the concept of fields to great advantage in the later chapters on electromagnetism.

Apparent Weight: Accounting for Earth's Rotation

As we saw in Applications of Newton's Laws, objects moving at constant speed in a circle have a centripetal acceleration directed toward the center of the circle, which means that there must be a net force directed toward the center of that circle. Since all objects on the surface of Earth move through a circle every 24 hours, there must be a net centripetal force on each object directed toward the center of that circle.

Let's first consider an object of mass *m* located at the equator, suspended from a scale (Figure 13.9). The scale exerts an upward force $\vec{\mathbf{F}}_{\text{SE}}$ away from Earth's center. This is the reading on the scale, and hence it is the **apparent weight** of the object. The weight (*mg*) points toward Earth's center. If Earth were not rotating, the acceleration would be zero and, consequently, the net force would be zero, resulting in $F_{\text{s}} = mg$. This would be the true reading of the weight.

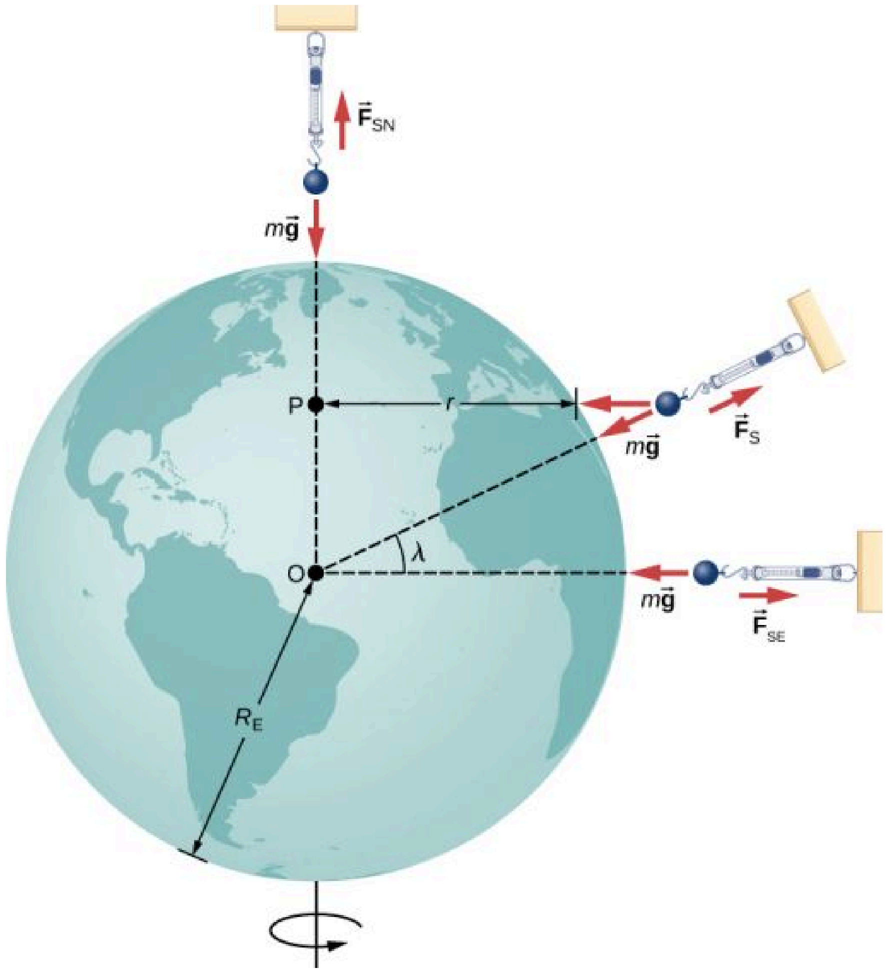


FIGURE 13.9 For a person standing at the equator, the centripetal acceleration (a_c) is in the same direction as the force of gravity. At latitude λ, the angle the between a_c and the force of gravity is λ and the magnitude of a_c decreases with $\cos\lambda$.

With rotation, the sum of these forces must provide the centripetal acceleration, a_c. Using Newton's second law, we have

$$\sum F = F_{SE} - mg = ma_c \quad \text{where} \quad a_c = -\frac{v^2}{r}. \tag{13.3}$$

Note that a_c points in the same direction as the weight; hence, it is negative. The tangential speed v is the speed at the equator and r is R_E. We can calculate the speed simply by noting that objects on the equator travel the circumference of Earth in 24 hours. Instead, let's use the alternative expression for a_c from Motion in Two and Three Dimensions. Recall that the tangential speed is related to the angular speed (ω) by $v = r\omega$. Hence, we have $a_c = -r\omega^2$. By rearranging Equation 13.3 and substituting $r = R_E$, the apparent weight at the equator is

$$F_{SE} = m\left(g - R_E\omega^2\right).$$

The angular speed of Earth everywhere is

$$\omega = \frac{2\pi \text{ rad}}{24 \text{ hr} \times 3600 \text{ s/hr}} = 7.27 \times 10^{-5} \text{ rad/s}.$$

Substituting for the values or R_E and ω, we have $R_E\omega^2 = 0.0337 \text{ m/s}^2$. This is only 0.34% of the value of gravity, so it is clearly a small correction.

EXAMPLE 13.5

Zero Apparent Weight

How fast would Earth need to spin for those at the equator to have zero apparent weight? How long would the length of the day be?

Strategy

Using Equation 13.3, we can set the apparent weight (F_{SE}) to zero and determine the centripetal acceleration required. From that, we can find the speed at the equator. The length of day is the time required for one complete rotation.

Solution

From Equation 13.2, we have $\sum F = F_{\text{SE}} - mg = ma_{\text{c}}$, so setting $F_{\text{SE}} = 0$, we get $g = a_{\text{c}}$. Using the expression for a_{c}, substituting for Earth's radius and the standard value of gravity, we get

$$a_{\text{c}} = \frac{v^2}{r} = g$$
$$v = \sqrt{gr} = \sqrt{(9.80\ \text{m/s}^2)(6.37 \times 10^6\ \text{m})} = 7.91 \times 10^3\ \text{m/s}.$$

The period T is the time for one complete rotation. Therefore, the tangential speed is the circumference divided by T, so we have

$$v = \frac{2\pi r}{T}$$
$$T = \frac{2\pi r}{v} = \frac{2\pi(6.37 \times 10^6\ \text{m})}{7.91 \times 10^3\ \text{m/s}} = 5.06 \times 10^3\ \text{s}.$$

This is about 84 minutes.

Significance

We will see later in this chapter that this speed and length of day would also be the orbital speed and period of a satellite in orbit at Earth's surface. While such an orbit would not be possible near Earth's surface due to air resistance, it certainly is possible only a few hundred miles above Earth.

Results Away from the Equator

At the pole, $a_{\text{c}} \to 0$ and $F_{\text{SN}} = F_{\text{SS}} = mg$, just as is the case without rotation. At any other latitude λ, the situation is more complicated. The centripetal acceleration is directed toward point P in the figure, and the radius becomes $r = R_{\text{E}} \cos\lambda$. The *vector* sum of the weight and $\vec{\mathbf{F}}_{\text{s}}$ must point toward point P, hence $\vec{\mathbf{F}}_{\text{s}}$ no longer points away from the center of Earth. (The difference is small and exaggerated in the figure.) A plumb bob will always point along this deviated direction. All buildings are built aligned along this deviated direction, not along a radius through the center of Earth. For the tallest buildings, this represents a deviation of a few feet at the top.

It is also worth noting that Earth is not a perfect sphere. The interior is partially liquid, and this enhances Earth bulging at the equator due to its rotation. The radius of Earth is about 30 km greater at the equator compared to the poles. It is left as an exercise to compare the strength of gravity at the poles to that at the equator using Equation 13.2. The difference is comparable to the difference due to rotation and is in the same direction. Apparently, you really can lose "weight" by moving to the tropics.

Gravity Away from the Surface

Earlier we stated without proof that the law of gravitation applies to spherically symmetrical objects, where the mass of each body acts as if it were at the center of the body. Since Equation 13.2 is derived from Equation 13.1, it is also valid for symmetrical mass distributions, but both equations are valid only for values of $r \geq R_{\text{E}}$. As we saw in Example 13.4, at 400 km above Earth's surface, where the International Space Station orbits, the value of *g* is 8.67 m/s^2. (We will see later that this is also the centripetal acceleration of the ISS.)

For $r < R_{\text{E}}$, Equation 13.1 and Equation 13.2 are not valid. However, we can determine *g* for these cases using a principle that comes from Gauss's law, which is a powerful mathematical tool that we study in more detail later in the course. A consequence of Gauss's law, applied to gravitation, is that only the mass *within r* contributes to the gravitational force. Also, that mass, just as before, can be considered to be located at the center. The gravitational effect of the mass *outside r* has zero net effect.

Two very interesting special cases occur. For a spherical planet with constant density, the mass within *r* is the density times the volume within *r*. This mass can be considered located at the center. Replacing M_{E} with only the mass within *r*, $M = \rho \times (\text{volume of a sphere})$, and R_{E} with *r*, Equation 13.2 becomes

$$g = G\frac{M_{\text{E}}}{R_{\text{E}}^2} = G\frac{\rho\left(4/3\pi r^3\right)}{r^2} = \frac{4}{3}G\rho\pi r.$$

The value of *g*, and hence your weight, decreases linearly as you descend down a hole to the center of the spherical planet. At the center, you are weightless, as the mass of the planet pulls equally in all directions. Actually, Earth's density is not constant, nor is Earth solid throughout. Figure 13.10 shows the profile of *g* if Earth had constant density and the more likely profile based upon estimates of density derived from seismic data.

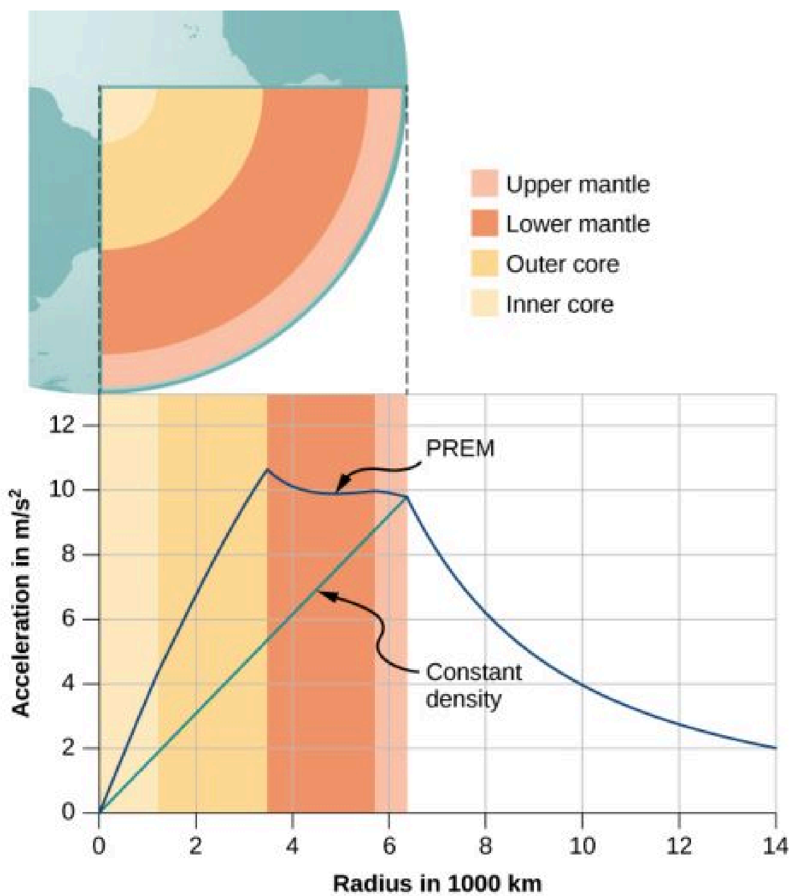


FIGURE 13.10 For $r < R_{\text{E}}$, the value of *g* for the case of constant density is the straight green line. The blue line from the PREM (Preliminary Reference Earth Model) is probably closer to the actual profile for *g*.

The second interesting case concerns living on a spherical shell planet. This scenario has been proposed in many science fiction stories. Ignoring significant engineering issues, the shell could be constructed with a desired radius and total mass, such that *g* at the surface is the same as Earth's. Can you guess what happens once you descend in an elevator to the inside of the shell, where there is no mass between you and the center? What benefits would this provide for traveling great distances from one point on the sphere to another? And finally, what effect would there be if the planet was spinning?

13.3 Gravitational Potential Energy and Total Energy

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Determine changes in gravitational potential energy over great distances
- Apply conservation of energy to determine escape velocity
- Determine whether astronomical bodies are gravitationally bound

We studied gravitational potential energy in Potential Energy and Conservation of Energy, where the value of *g* remained constant. We now develop an expression that works over distances such that *g* is not constant. This is necessary to correctly calculate the energy needed to place satellites in orbit or to send them on missions in space.

Gravitational Potential Energy beyond Earth

We defined work and potential energy in Work and Kinetic Energy and Potential Energy and Conservation of Energy. The usefulness of those definitions is the ease with which we can solve many problems using conservation of energy. Potential energy is particularly useful for forces that change with position, as the gravitational force does over large distances. In Potential Energy and Conservation of Energy, we showed that the change in gravitational potential energy near Earth's surface is $\Delta U = mg(y_2 - y_1)$. This works very well if *g* does not change significantly between y_1 and y_2. We return to the definition of work and potential energy to derive an expression that is correct over larger distances.

Recall that work (*W*) is the integral of the dot product between force and distance. Essentially, it is the product of the component of a force along a displacement times that displacement. We define ΔU as the *negative* of the work done by the force we associate with the potential energy. For clarity, we derive an expression for moving a mass *m* from distance r_1 from the center of Earth to distance r_2. However, the result can easily be generalized to any two objects changing their separation from one value to another.

Consider Figure 13.11, in which we take *m* from a distance r_1 from Earth's center to a distance that is r_2 from the center. Gravity is a conservative force (its magnitude and direction are functions of location only), so we can take any path we wish, and the result for the calculation of work is the same. We take the path shown, as it greatly simplifies the integration. We first move *radially* outward from distance r_1 to distance r_2, and then move along the arc of a circle until we reach the final position. During the radial portion, $\vec{\mathbf{F}}$ is opposite to the direction we travel along $d\vec{\mathbf{r}}$, so $E = K_1 + U_1 = K_2 + U_2$. Along the arc, $\vec{\mathbf{F}}$ is perpendicular to $d\vec{\mathbf{r}}$, so $\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = 0$. No work is done as we move along the arc. Using the expression for the gravitational force and noting the values for $\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ along the two segments of our path, we have

$$\Delta U = -\int_{r_1}^{r_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = GM_\text{E}m \int_{r_1}^{r_2} \frac{dr}{r^2} = GM_\text{E}m\left(\frac{1}{r_1} - \frac{1}{r_2}\right).$$

Since $\Delta U = U_2 - U_1$, we can adopt a simple expression for U:

$$U = -\frac{GM_\text{E}m}{r}. \tag{13.4}$$

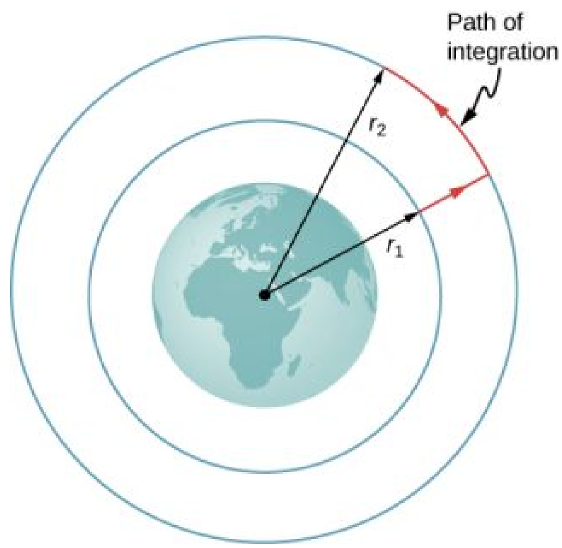


FIGURE 13.11 The work integral, which determines the change in potential energy, can be evaluated along the path shown in red.

Note two important items with this definition. First, $U \to 0$ as $r \to \infty$. The potential energy is zero when the two masses are infinitely far apart. Only the difference in *U* is important, so the choice of $U = 0$ for $r = \infty$ is merely one of convenience. (Recall that in earlier gravity problems, you were free to take $U = 0$ at the top or bottom of a building, or anywhere.) Second, note that *U* becomes increasingly more negative as the masses get closer. That is consistent with what you learned about potential energy in Potential Energy and Conservation of Energy. As the two masses are separated, positive work must be done against the force of gravity, and hence, *U* increases (becomes less negative). All masses naturally fall together under the influence of gravity, falling from a higher to a lower potential energy.

EXAMPLE 13.6

Lifting a Payload

How much energy is required to lift the 9000-kg *Soyuz* vehicle from Earth's surface to the height of the ISS, 400 km above the surface?

Strategy

Use Equation 13.2 to find the change in potential energy of the payload. That amount of work or energy must be supplied to lift the payload.

Solution

Paying attention to the fact that we start at Earth's surface and end at 400 km above the surface, the change in *U* is

$$\Delta U = U_{\text{orbit}} - U_{\text{Earth}} = -\frac{GM_{\text{E}}m}{R_{\text{E}} + 400\,\text{km}} - \left(-\frac{GM_{\text{E}}m}{R_{\text{E}}}\right).$$

We insert the values

$$m = 9000\,\text{kg}, \quad M_{\text{E}} = 5.96 \times 10^{24}\,\text{kg}, \quad R_{\text{E}} = 6.37 \times 10^{6}\,\text{m}$$

and convert 400 km into 4.00×10^{5} m. We find $\Delta U = 3.32 \times 10^{10}$ J. It is positive, indicating an increase in potential energy, as we would expect.

Significance

For perspective, consider that the average US household energy use in 2013 was 909 kWh per month. That is energy of

$$909\,\text{kWh} \times 1000\,\text{W/kW} \times 3600\,\text{s/h} = 3.27 \times 10^{9}\,\text{J per month.}$$

So our result is an energy expenditure equivalent to 10 months. But this is just the energy needed to raise the payload 400 km. If we want the *Soyuz* to be in orbit so it can rendezvous with the ISS and not just fall back to Earth, it needs a lot of kinetic energy. As we see in the next section, that kinetic energy is about five times that of ΔU. In addition, far more energy is expended lifting the propulsion system itself. Space travel is not cheap.

CHECK YOUR UNDERSTANDING 13.3

Why not use the simpler expression $\Delta U = mg(y_2 - y_1)$? How significant would the error be? (Recall the previous result, in Example 13.4, that the value *g* at 400 km above the Earth is 8.67 m/s^2.)

Conservation of Energy

In Potential Energy and Conservation of Energy, we described how to apply conservation of energy for systems with conservative forces. We were able to solve many problems, particularly those involving gravity, more simply using conservation of energy. Those principles and problem-solving strategies apply equally well here. The only change is to place the new expression for potential energy into the conservation of energy equation, $E = K_1 + U_1 = K_2 + U_2$.

$$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2} \tag{13.5}$$

Note that we use *M*, rather than M_E, as a reminder that we are not restricted to problems involving Earth. However, we still assume that $m << M$. (For problems in which this is not true, we need to include the kinetic energy of both masses and use conservation of momentum to relate the velocities to each other. But the principle remains the same.)

Escape velocity

Escape velocity is often defined to be the *minimum* initial velocity of an object that is required to escape the surface of a planet (or any large body like a moon) and never return. As usual, we assume no energy lost to an atmosphere, should there be any.

Consider the case where an object is launched from the surface of a planet with an initial velocity directed away from the planet. With the *minimum* velocity needed to escape, the object would *just* come to rest infinitely far away, that is, the object gives up the last of its kinetic energy just as it reaches infinity, where the force of gravity becomes zero. Since $U \to 0 \text{ as } r \to \infty$, this means the total energy is zero. Thus, we find the escape velocity from the surface of an astronomical body of mass *M* and radius *R* by setting the total energy equal to zero. At the surface of the body, the object is located at $r_1 = R$ and it has escape velocity $v_1 = v_{esc}$. It reaches $r_2 = \infty$ with velocity $v_2 = 0$. Substituting into Equation 13.5, we have

$$\frac{1}{2}mv_{esc}^2 - \frac{GMm}{R} = \frac{1}{2}m0^2 - \frac{GMm}{\infty} = 0.$$

Solving for the escape velocity,

$$v_{esc} = \sqrt{\frac{2GM}{R}}. \tag{13.6}$$

Notice that *m* has canceled out of the equation. The escape velocity is the same for all objects, regardless of mass. Also, we are not restricted to the surface of the planet; *R* can be any starting point beyond the surface of the planet.

EXAMPLE 13.7

Escape from Earth

What is the escape speed from the surface of Earth? Assume there is no energy loss from air resistance. Compare this to the escape speed from the Sun, starting from Earth's orbit.

Strategy

We use Equation 13.6, clearly defining the values of *R* and *M*. To escape Earth, we need the mass and radius of Earth. For escaping the Sun, we need the mass of the Sun, and the orbital distance between Earth and the Sun.

Solution

Substituting the values for Earth's mass and radius directly into Equation 13.6, we obtain

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2)(5.96 \times 10^{24}\ \text{kg})}{6.37 \times 10^{6}\ \text{m}}} = 1.12 \times 10^{4}\ \text{m/s}.$$

That is about 11 km/s or 25,000 mph. To escape the Sun, starting from Earth's orbit, we use $R = R_{\text{ES}} = 1.50 \times 10^{11}\ \text{m}$ and $M_{\text{Sun}} = 1.99 \times 10^{30}\ \text{kg}$. The result is $v_{\text{esc}} = 4.21 \times 10^{4}\ \text{m/s}$ or about 42 km/s.

Significance

The speed needed to escape the Sun (leave the solar system) is nearly four times the escape speed from Earth's surface. But there is help in both cases. Earth is rotating, at a speed of nearly 460 m/s at the equator, and we can use that velocity to help escape, or to achieve orbit. For this reason, many commercial space companies maintain launch facilities near the equator. To escape the Sun, there is even more help. Earth revolves about the Sun at a speed of approximately 30 km/s. By launching in the direction that Earth is moving, we need only an additional 12 km/s. The use of gravitational assist from other planets, essentially a gravity slingshot technique, allows space probes to reach even greater speeds. In this slingshot technique, the vehicle approaches the planet and is accelerated by the planet's gravitational attraction. It has its greatest speed at the closest point of approach, although it accelerates opposite to the motion in equal measure as it moves away. But relative to the planet, the vehicle's speed far before the approach, and long after, are the same. If the directions are chosen correctly, that can result in a significant increase (or decrease if needed) in the vehicle's speed relative to the rest of the solar system.

INTERACTIVE

Visit this website (https://openstax.org/l/21escapevelocit) to learn more about escape velocity.

CHECK YOUR UNDERSTANDING 13.4

If we send a probe out of the solar system starting from Earth's surface, do we only have to escape the Sun?

Energy and gravitationally bound objects

As stated previously, escape velocity can be defined as the initial velocity of an object that can escape the surface of a moon or planet. More generally, it is the speed at *any* position such that the *total* energy is zero. If the total energy is zero or greater, the object escapes. If the total energy is negative, the object cannot escape. Let's see why that is the case.

As noted earlier, we see that $U \to 0$ as $r \to \infty$. If the total energy is zero, then as *m* reaches a value of *r* that approaches infinity, *U* becomes zero and so must the kinetic energy. Hence, *m* comes to rest infinitely far away from *M*. It has "just escaped" *M*. If the total energy is positive, then kinetic energy remains at $r = \infty$ and certainly *m* does not return. When the total energy is zero or greater, then we say that *m* is not gravitationally bound to *M*.

On the other hand, if the total energy is negative, then the kinetic energy must reach zero at some finite value of *r*, where U is negative and equal to the total energy. The object can never exceed this finite distance from M, since to do so would require the kinetic energy to become negative, which is not possible. We say m is **gravitationally bound** to M.

We have simplified this discussion by assuming that the object was headed directly away from the planet. What is remarkable is that the result applies for any velocity. Energy is a scalar quantity and hence Equation 13.5 is a scalar equation—the direction of the velocity plays no role in conservation of energy. It is possible to have a gravitationally bound system where the masses do not “fall together,” but maintain an orbital motion about each other.

We have one important final observation. Earlier we stated that if the total energy is zero or greater, the object escapes. Strictly speaking, Equation 13.5 and Equation 13.6 apply for point objects. They apply to finite-sized, spherically symmetric objects as well, provided that the value for r in Equation 13.5 is always greater than the sum of the radii of the two objects. If r becomes less than this sum, then the objects collide. (Even for greater values of r, but near the sum of the radii, gravitational tidal forces could create significant effects if both objects are planet sized. We examine tidal effects in Tidal Forces.) Neither positive nor negative total energy precludes finite-sized masses from colliding. For real objects, direction is important.

EXAMPLE 13.8

How Far Can an Object Escape?

Let’s consider the preceding example again, where we calculated the escape speed from Earth and the Sun, starting from Earth’s orbit. We noted that Earth already has an orbital speed of 30 km/s. As we see in the next section, that is the tangential speed needed to stay in circular orbit. If an object had this speed at the distance of Earth’s orbit, but was headed directly away from the Sun, how far would it travel before coming to rest? Ignore the gravitational effects of any other bodies.

Strategy

The object has initial kinetic and potential energies that we can calculate. When its speed reaches zero, it is at its maximum distance from the Sun. We use Equation 13.5, conservation of energy, to find the distance at which kinetic energy is zero.

Solution

The initial position of the object is Earth’s radius of orbit and the intial speed is given as 30 km/s. The final velocity is zero, so we can solve for the distance at that point from the conservation of energy equation. Using $R_{\text{ES}} = 1.50 \times 10^{11}\ \text{m}$ and $M_{\text{Sun}} = 1.99 \times 10^{30}\ \text{kg}$, we have

$$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$$

$$\frac{1}{2}\cancel{m}(3.0 \times 10^3\,\text{m/s})^2 - \frac{(6.67 \times 10^{-11}\ \text{N·m/kg}^2)(1.99 \times 10^{30}\ \text{kg})\cancel{m}}{1.50 \times 10^{11}\ \text{m}}$$

$$= \frac{1}{2}\cancel{m}0^2 - \frac{(6.67 \times 10^{-11}\ \text{N·m/kg}^2)(1.99 \times 10^{30}\ \text{kg})\cancel{m}}{r_2}$$

where the mass m cancels. Solving for r_2 we get $r_2 = 3.0 \times 10^{11}\ \text{m}$. Note that this is twice the initial distance from the Sun and takes us past Mars’s orbit, but not quite to the asteroid belt.

Significance

The object in this case reached a distance *exactly* twice the initial orbital distance. We will see the reason for this in the next section when we calculate the speed for circular orbits.

CHECK YOUR UNDERSTANDING 13.5

Assume you are in a spacecraft in orbit about the Sun at Earth's orbit, but far away from Earth (so that it can be ignored). How could you redirect your tangential velocity to the radial direction such that you could then pass by Mars's orbit? What would be required to change just the direction of the velocity?

13.4 Satellite Orbits and Energy

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the mechanism for circular orbits
- Find the orbital periods and speeds of satellites
- Determine whether objects are gravitationally bound

The Moon orbits Earth. In turn, Earth and the other planets orbit the Sun. The space directly above our atmosphere is filled with artificial satellites in orbit. We examine the simplest of these orbits, the circular orbit, to understand the relationship between the speed and period of planets and satellites in relation to their positions and the bodies that they orbit.

Circular Orbits

As noted at the beginning of this chapter, Nicolaus Copernicus first suggested that Earth and all other planets orbit the Sun in circles. He further noted that orbital periods increased with distance from the Sun. Later analysis by Kepler showed that these orbits are actually ellipses, but the orbits of most planets in the solar system are nearly circular. Earth's orbital distance from the Sun varies a mere 2%. The exception is the eccentric orbit of Mercury, whose orbital distance varies nearly 40%.

Determining the **orbital speed** and **orbital period** of a satellite is much easier for circular orbits, so we make that assumption in the derivation that follows. As we described in the previous section, an object with negative total energy is gravitationally bound and therefore is in orbit. Our computation for the special case of circular orbits will confirm this. We focus on objects orbiting Earth, but our results can be generalized for other cases.

Consider a satellite of mass m in a circular orbit about Earth at distance r from the center of Earth (Figure 13.12). It has centripetal acceleration directed toward the center of Earth. Earth's gravity is the only force acting, so Newton's second law gives

$$\frac{GmM_{\text{E}}}{r^2} = ma_{\text{c}} = \frac{mv_{\text{orbit}}^2}{r}.$$

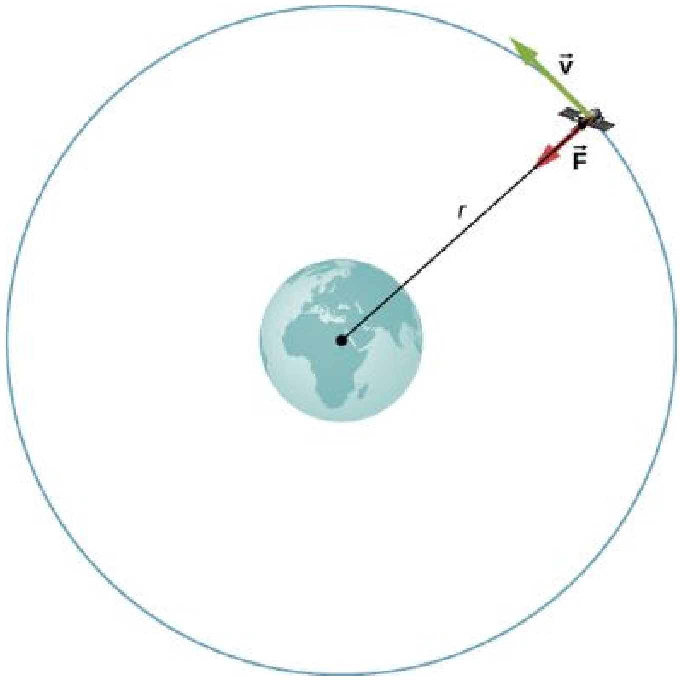


FIGURE 13.12 A satellite of mass *m* orbiting at radius *r* from the center of Earth. The gravitational force supplies the centripetal acceleration.

We solve for the speed of the orbit, noting that *m* cancels, to get the orbital speed

$$v_{\text{orbit}} = \sqrt{\frac{GM_{\text{E}}}{r}}. \tag{13.7}$$

Consistent with what we saw in Equation 13.2 and Equation 13.6, *m* does not appear in Equation 13.7. The value of *g*, the escape velocity, and orbital velocity depend only upon the distance from the center of the planet, and *not* upon the mass of the object being acted upon. Notice the similarity in the equations for v_{orbit} and v_{esc}. The escape velocity is exactly $\sqrt{2}$ times greater, about 40%, than the orbital velocity. This comparison was noted in Example 13.7, and it is true for a satellite at any radius.

To find the period of a circular orbit, we note that the satellite travels the circumference of the orbit $2\pi r$ in one period *T*. Using the definition of speed, we have $v_{\text{orbit}} = 2\pi r/T$. We substitute this into Equation 13.7 and rearrange to get

$$T = 2\pi\sqrt{\frac{r^3}{GM_{\text{E}}}}. \tag{13.8}$$

We see in the next section that this represents Kepler's third law for the case of circular orbits. It also confirms Copernicus's observation that the period of a planet increases with increasing distance from the Sun. We need only replace M_{E} with M_{Sun} in Equation 13.8.

We conclude this section by returning to our earlier discussion about astronauts in orbit appearing to be weightless, as if they were free-falling towards Earth. In fact, they are in free fall. Consider the trajectories shown in Figure 13.13. (This figure is based on a drawing by Newton in his *Principia* and also appeared earlier in Motion in Two and Three Dimensions.) All the trajectories shown that hit the surface of Earth have less than orbital velocity. The astronauts would accelerate toward Earth along the noncircular paths shown and feel weightless. (Astronauts actually train for life in orbit by riding in airplanes that free fall for 30 seconds at a time.) But with the correct orbital velocity, Earth's surface curves away from them at exactly the same rate as they fall toward Earth. Of course, staying the same distance from the surface is the point of a circular orbit.

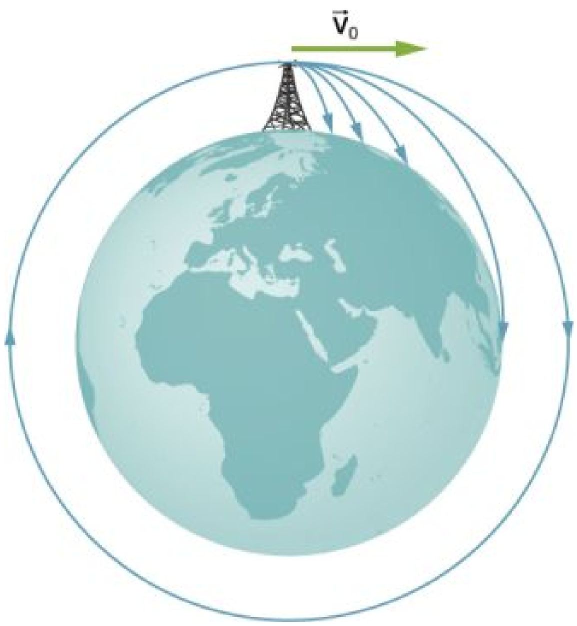


FIGURE 13.13 A circular orbit is the result of choosing a tangential velocity such that Earth's surface curves away at the same rate as the object falls toward Earth.

We can summarize our discussion of orbiting satellites in the following Problem-Solving Strategy.

PROBLEM-SOLVING STRATEGY

Orbits and Conservation of Energy

1. Determine whether the equations for speed, energy, or period are valid for the problem at hand. If not, start with the first principles we used to derive those equations.
2. To start from first principles, draw a free-body diagram and apply Newton's law of gravitation and Newton's second law.
3. Along with the definitions for speed and energy, apply Newton's second law of motion to the bodies of interest.

EXAMPLE 13.9

The International Space Station

Determine the orbital speed and period for the International Space Station (ISS).

Strategy

Since the ISS orbits $4.00 \times 10^2 \text{km}$ above Earth's surface, the radius at which it orbits is $R_E + 4.00 \times 10^2 \text{km}$. We use Equation 13.7 and Equation 13.8 to find the orbital speed and period, respectively.

Solution

Using Equation 13.7, the orbital velocity is

$$v_{\text{orbit}} = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2 (5.96 \times 10^{24} \text{ kg})}{(6.36 \times 10^6 + 4.00 \times 10^5 \text{ m})}} = 7.67 \times 10^3 \text{ m/s}$$

which is about 17,000 mph. Using Equation 13.8, the period is

$$T = 2\pi\sqrt{\frac{r^3}{GM_E}} = 2\pi\sqrt{\frac{(6.37 \times 10^6 + 4.00 \times 10^5 \text{ m})^3}{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.96 \times 10^{24} \text{ kg})}} = 5.55 \times 10^3 \text{ s}$$

which is just over 90 minutes.

Significance

The ISS is considered to be in low Earth orbit (LEO). Nearly all satellites are in LEO, including most weather satellites. GPS satellites, at about 20,000 km, are considered medium Earth orbit. The higher the orbit, the more energy is required to put it there and the more energy is needed to reach it for repairs. Of particular interest are the satellites in geosynchronous orbit. All fixed satellite dishes on the ground pointing toward the sky, such as TV reception dishes, are pointed toward geosynchronous satellites. These satellites are placed at the exact distance, and just above the equator, such that their period of orbit is 1 day. They remain in a fixed position relative to Earth's surface.

CHECK YOUR UNDERSTANDING 13.6

By what factor must the radius change to reduce the orbital velocity of a satellite by one-half? By what factor would this change the period?

EXAMPLE 13.10

Determining the Mass of Earth

Determine the mass of Earth from the orbit of the Moon.

Strategy

We use Equation 13.8, solve for M_E, and substitute for the period and radius of the orbit. The radius and period of the Moon's orbit was measured with reasonable accuracy thousands of years ago. From the astronomical data in Appendix D, the period of the Moon is 27.3 days $= 2.36 \times 10^6$ s, and the *average* distance between the centers of Earth and the Moon is 384,000 km.

Solution

Solving for M_E,

$$\begin{aligned} T &= 2\pi\sqrt{\frac{r^3}{GM_E}} \\ M_E &= \frac{4\pi^2 r^3}{GT^2} = \frac{4\pi^2(3.84\times10^8\text{ m})^3}{(6.67\times10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2)(2.36\times10^6\text{ s})^2} = 6.01\times10^{24}\text{ kg}. \end{aligned}$$

Significance

Compare this to the value of 5.96×10^{24} kg that we obtained in Example 13.5, using the value of *g* at the surface of Earth. Although these values are very close (~0.8%), both calculations use average values. The value of *g* varies from the equator to the poles by approximately 0.5%. But the Moon has an elliptical orbit in which the value of *r* varies just over 10%. (The apparent size of the full Moon actually varies by about this amount, but it is difficult to notice through casual observation as the time from one extreme to the other is many months.)

CHECK YOUR UNDERSTANDING 13.7

There is another consideration to this last calculation of M_E. We derived Equation 13.8 assuming that the satellite orbits around the center of the astronomical body at the same radius used in the expression for the gravitational force between them. What assumption is made to justify this? Earth is about 81 times more massive than the Moon. Does the Moon orbit about the exact center of Earth?

EXAMPLE 13.11

Galactic Speed and Period

Let's revisit Example 13.2. Assume that the Milky Way and Andromeda galaxies are in a circular orbit about each other. What would be the velocity of each and how long would their orbital period be? Assume the mass of each is 800 billion solar masses and their centers are separated by 2.5 million light years.

Strategy

We cannot use Equation 13.7 and Equation 13.8 directly because they were derived assuming that the object of mass *m* orbited about the center of a much larger planet of mass *M*. We determined the gravitational force in Example 13.2 using Newton's law of universal gravitation. We can use Newton's second law, applied to the centripetal acceleration of either galaxy, to determine their tangential speed. From that result we can determine the period of the orbit.

Solution

In Example 13.2, we found the force between the galaxies to be

$$F_{12} = G\frac{m_1 m_2}{r^2} = (6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2})\frac{[(800 \times 10^9)(2.0 \times 10^{30}\ \mathrm{kg})]^2}{[(2.5 \times 10^6)(9.5 \times 10^{15}\ \mathrm{m})]^2} = 3.0 \times 10^{29}\ \mathrm{N}$$

and that the acceleration of each galaxy is

$$a = \frac{F}{m} = \frac{3.0 \times 10^{29}\ \mathrm{N}}{(800 \times 10^9)(2.0 \times 10^{30}\ \mathrm{kg})} = 1.9 \times 10^{-13}\,\mathrm{m/s^2}.$$

Since the galaxies are in a circular orbit, they have centripetal acceleration. If we ignore the effect of other galaxies, then, as we learned in Linear Momentum and Collisions and Fixed-Axis Rotation, the centers of mass of the two galaxies remain fixed. Hence, the galaxies must orbit about this common center of mass. For equal masses, the center of mass is exactly half way between them. So the radius of the orbit, r_{orbit}, is not the same as the distance between the galaxies, but one-half that value, or 1.25 million light-years. These two different values are shown in Figure 13.14.

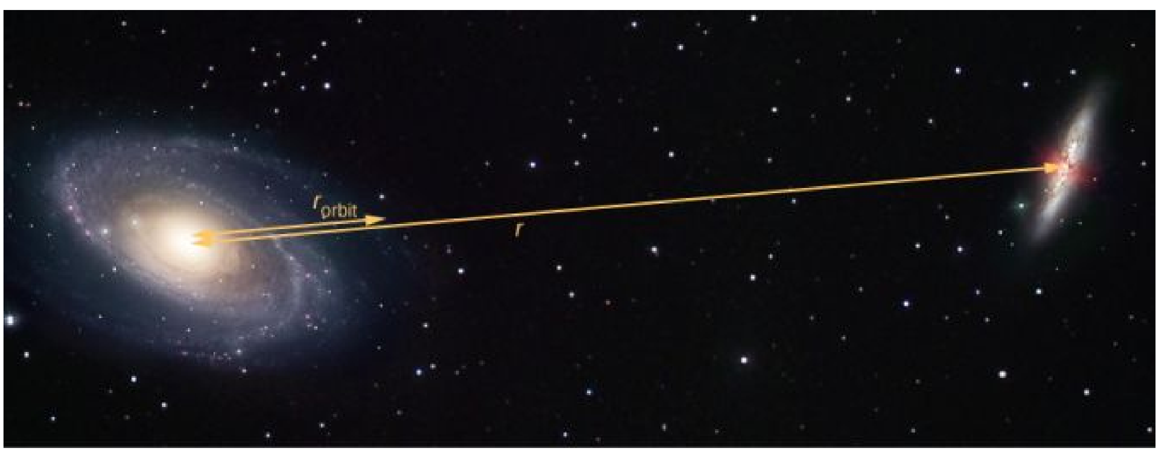


FIGURE 13.14 The distance between two galaxies, which determines the gravitational force between them, is *r*, and is different from r_{orbit}, which is the radius of orbit for each. For equal masses, $r_{\mathrm{orbit}} = r/2$. (credit: modification of work by Marc Van Norden)

Using the expression for centripetal acceleration, we have

$$\begin{aligned} a_{\mathrm{c}} &= \frac{v_{\mathrm{orbit}}^2}{r_{\mathrm{orbit}}} \\ 1.9 \times 10^{-13}\,\mathrm{m/s^2} &= \frac{v_{\mathrm{orbit}}^2}{(1.25 \times 10^6)(9.5 \times 10^{15}\ \mathrm{m})}. \end{aligned}$$

Solving for the orbit velocity, we have $v_{\text{orbit}} = 47\ \text{km/s}$. Finally, we can determine the period of the orbit directly from $T = 2\pi r / v_{\text{orbit}}$, to find that the period is $T = 1.6 \times 10^{18}\ \text{s}$, about 50 billion years.

Significance

The orbital speed of 47 km/s might seem high at first. But this speed is comparable to the escape speed from the Sun, which we calculated in an earlier example. To give even more perspective, this period is nearly four times longer than the time that the Universe has been in existence.

In fact, the present relative motion of these two galaxies is such that they are expected to collide in about 4 billion years. Although the density of stars in each galaxy makes a direct collision of any two stars unlikely, such a collision will have a dramatic effect on the shape of the galaxies. Examples of such collisions are well known in astronomy.

CHECK YOUR UNDERSTANDING 13.8

Galaxies are not single objects. How does the gravitational force of one galaxy exerted on the "closer" stars of the other galaxy compare to those farther away? What effect would this have on the shape of the galaxies themselves?

INTERACTIVE

See the Sloan Digital Sky Survey page (https://openstax.org/l/21sloandigskysu) for more information on colliding galaxies.

Engage the Phet simulation below to move the Sun, Earth, Moon, and space station to see the effects on their gravitational forces and orbital paths. Visualize the sizes and distances between different heavenly bodies, and turn off gravity to see what would happen without it.

Access multimedia content (https://openstax.org/books/university-physics-volume-1/pages/13-4-satellite-orbits-and-energy)

Energy in Circular Orbits

In Gravitational Potential Energy and Total Energy, we argued that objects are gravitationally bound if their total energy is negative. The argument was based on the simple case where the velocity was directly away or toward the planet. We now examine the total energy for a circular orbit and show that indeed, the total energy is negative. As we did earlier, we start with Newton's second law applied to a circular orbit,

$$\begin{aligned} \frac{GmM_{\text{E}}}{r^2} &= ma_c = \frac{mv^2}{r} \\ \frac{GmM_{\text{E}}}{r} &= mv^2. \end{aligned}$$

In the last step, we multiplied by r on each side. The right side is just twice the kinetic energy, so we have

$$K = \frac{1}{2}mv^2 = \frac{GmM_{\text{E}}}{2r}.$$

The total energy is the sum of the kinetic and potential energies, so our final result is

$$E = K + U = \frac{GmM_{\text{E}}}{2r} - \frac{GmM_{\text{E}}}{r} = -\frac{GmM_{\text{E}}}{2r}. \qquad \textbf{13.9}$$

We can see that the total energy is negative, with the same magnitude as the kinetic energy. For circular orbits, the magnitude of the kinetic energy is exactly one-half the magnitude of the potential energy. Remarkably, this result applies to any two masses in circular orbits about their common center of mass, at a distance r from each other. The proof of this is left as an exercise. We will see in the next section that a very similar expression applies in the case of elliptical orbits.

EXAMPLE 13.12

Energy Required to Orbit

In Example 13.8, we calculated the energy required to simply lift the 9000-kg *Soyuz* vehicle from Earth's surface to the height of the ISS, 400 km above the surface. In other words, we found its *change* in potential energy. We now ask, what total energy change in the *Soyuz* vehicle is required to take it from Earth's surface and put it in orbit with the ISS for a rendezvous (Figure 13.15)? How much of that total energy is kinetic energy?

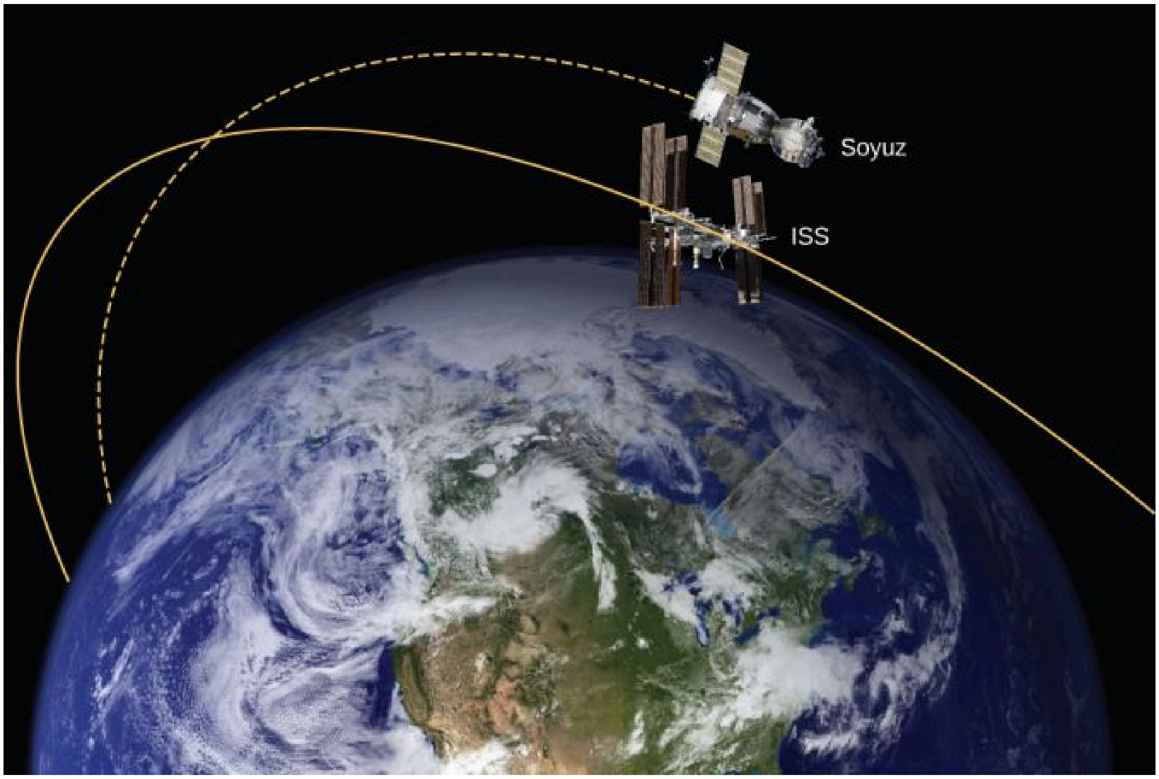


FIGURE 13.15 The *Soyuz* in a rendezvous with the ISS. Note that this diagram is not to scale; the *Soyuz* is very small compared to the ISS and its orbit is much closer to Earth. (credit: modification of works by NASA)

Strategy

The energy required is the difference in the *Soyuz*'s total energy in orbit and that at Earth's surface. We can use Equation 13.9 to find the total energy of the *Soyuz* at the ISS orbit. But the total energy at the surface is simply the potential energy, since it starts from rest. [Note that we *do not* use Equation 13.9 at the surface, since we are not in orbit at the surface.] The kinetic energy can then be found from the difference in the total energy change and the change in potential energy found in Example 13.8. Alternatively, we can use Equation 13.7 to find v_{orbit} and calculate the kinetic energy directly from that. The total energy required is then the kinetic energy plus the change in potential energy found in Example 13.8.

Solution

From Equation 13.9, the total energy of the *Soyuz* in the same orbit as the ISS is

$$\begin{aligned} E_{\text{orbit}} &= K_{\text{orbit}} + U_{\text{orbit}} = -\frac{GmM_{\text{E}}}{2r} \\ &= \frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(9000\ \text{kg})(5.96\times10^{24}\ \text{kg})}{2(6.36\times10^{6}+4.00\times10^{5}\ \text{m})} = -2.65\times10^{11}\ \text{J}. \end{aligned}$$

The total energy at Earth's surface is

$$\begin{aligned} E_{\text{surface}} &= K_{\text{surface}} + U_{\text{surface}} = 0 - \frac{GmM_{\text{E}}}{r} \\ &= -\frac{(6.67 \times 10^{-11}\ \text{N·m}^2/\text{kg}^2)(9000\ \text{kg})(5.96 \times 10^{24}\ \text{kg})}{(6.36 \times 10^6\ \text{m})} \\ &= -5.63 \times 10^{11}\ \text{J}. \end{aligned}$$

The change in energy is $\Delta E = E_{\text{orbit}} - E_{\text{surface}} = 2.98 \times 10^{11}$ J. To get the kinetic energy, we subtract the change in potential energy from Example 13.6, $\Delta U = 3.32 \times 10^{10}$ J. That gives us $K_{\text{orbit}} = 2.98 \times 10^{11} - 3.32 \times 10^{10} = 2.65 \times 10^{11}$ J. As stated earlier, the kinetic energy of a circular orbit is always one-half the magnitude of the potential energy, and the same as the magnitude of the total energy. Our result confirms this.

The second approach is to use Equation 13.7 to find the orbital speed of the *Soyuz*, which we did for the ISS in Example 13.9.

$$v_{\text{orbit}} = \sqrt{\frac{GM_{\text{E}}}{r}} = \sqrt{\frac{(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(5.96 \times 10^{24}\ \text{kg})}{(6.36 \times 10^6 + 4.00 \times 10^5\ \text{m})}} = 7.67 \times 10^3\ \text{m/s}.$$

So the kinetic energy of the *Soyuz* in orbit is

$$K_{\text{orbit}} = \frac{1}{2}mv_{\text{orbit}}^2 = \frac{1}{2}(9000\ \text{kg})(7.67 \times 10^3\ \text{m/s})^2 = 2.65 \times 10^{11}\ \text{J},$$

the same as in the previous method. The total energy is just

$$\Delta E = K_{\text{orbit}} + \Delta U = 2.65 \times 10^{11} + 3.32 \times 10^{10} = 2.98 \times 10^{11}\ \text{J}.$$

Significance

The kinetic energy of the *Soyuz* is nearly eight times the change in its potential energy, or 90% of the total energy needed for the rendezvous with the ISS. And it is important to remember that this energy represents only the energy that must be given to the *Soyuz*. With our present rocket technology, the mass of the propulsion system (the rocket fuel, its container and combustion system) far exceeds that of the payload, and a tremendous amount of kinetic energy must be given to that mass. So the actual cost in energy is many times that of the change in energy of the payload itself.

13.5 Kepler's Laws of Planetary Motion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the conic sections and how they relate to orbital motion
- Describe how orbital velocity is related to conservation of angular momentum
- Determine the period of an elliptical orbit from its major axis

Using the precise data collected by Tycho Brahe, Johannes Kepler carefully analyzed the positions in the sky of all the known planets and the Moon, plotting their positions at regular intervals of time. From this analysis, he formulated three laws, which we address in this section.

Kepler's First Law

The prevailing view during the time of Kepler was that all planetary orbits were circular. The data for Mars presented the greatest challenge to this view and that eventually encouraged Kepler to give up the popular idea. **Kepler's first law** states that every planet moves along an ellipse, with the Sun located at a focus of the ellipse. An ellipse is defined as the set of all points such that the sum of the distance from each point to two foci is a constant. Figure 13.16 shows an ellipse and describes a simple way to create it.

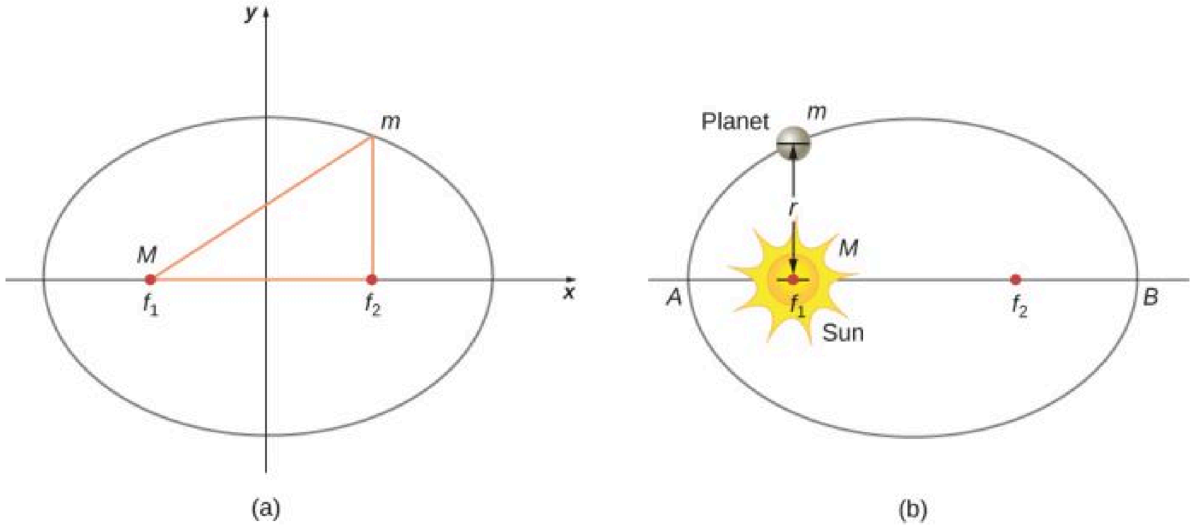


FIGURE 13.16 (a) An ellipse is a curve in which the sum of the distances from a point on the curve to two foci (f_1 and f_2) is a constant. From this definition, you can see that an ellipse can be created in the following way. Place a pin at each focus, then place a loop of string around a pencil and the pins. Keeping the string taught, move the pencil around in a complete circuit. If the two foci occupy the same place, the result is a circle—a special case of an ellipse. (b) For an elliptical orbit, if $m \ll M$, then *m* follows an elliptical path with *M* at one focus. More exactly, both *m* and *M* move in their own ellipse about the common center of mass.

For elliptical orbits, the point of closest approach of a planet to the Sun is called the **perihelion**. It is labeled point *A* in Figure 13.16. The farthest point is the **aphelion** and is labeled point *B* in the figure. For the Moon's orbit about Earth, those points are called the perigee and apogee, respectively.

An ellipse has several mathematical forms, but all are a specific case of the more general equation for conic sections. There are four different conic sections, all given by the equation

$$\frac{\alpha}{r} = 1 + e\cos\theta. \tag{13.10}$$

The variables *r* and θ are shown in Figure 13.17 in the case of an ellipse. The constants α and *e* are determined by the total energy and angular momentum of the satellite at a given point. The constant *e* is called the eccentricity. The values of α and *e* determine which of the four conic sections represents the path of the satellite.

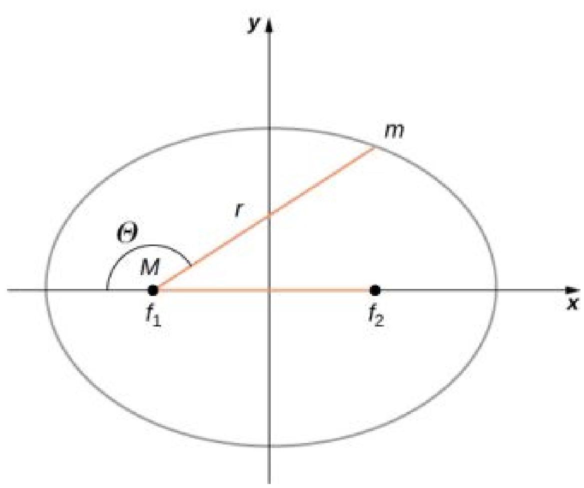


FIGURE 13.17 As before, the distance between the planet and the Sun is *r*, and the angle measured from the *x*-axis, which is along the major axis of the ellipse, is θ.

One of the real triumphs of Newton's law of universal gravitation, with the force proportional to the inverse of the distance squared, is that when it is combined with his second law, the solution for the path of any satellite is a conic section. Every path taken by *m* is one of the four conic sections: a circle or an ellipse for bound or closed orbits, or a parabola or hyperbola for unbounded or open orbits. These conic sections are shown in Figure 13.18.

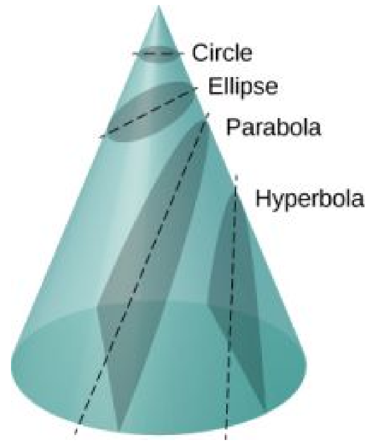


FIGURE 13.18 All motion caused by an inverse square force is one of the four conic sections and is determined by the energy and direction of the moving body.

If the total energy is negative, then $0 \leq e < 1$, and Equation 13.10 represents a bound or closed orbit of either an ellipse or a circle, where $e = 0$. [You can see from Equation 13.10 that for $e = 0$, $r = \alpha$, and hence the radius is constant.] For ellipses, the eccentricity is related to how oblong the ellipse appears. A circle has zero eccentricity, whereas a very long, drawn-out ellipse has an eccentricity near one.

If the total energy is exactly zero, then $e = 1$ and the path is a parabola. Recall that a satellite with zero total energy has exactly the escape velocity. (The parabola is formed only by slicing the cone parallel to the tangent line along the surface.) Finally, if the total energy is positive, then $e > 1$ and the path is a hyperbola. These last two paths represent unbounded orbits, where *m* passes by *M* once and only once. This situation has been observed for several comets that approach the Sun and then travel away, never to return.

We have confined ourselves to the case in which the smaller mass (planet) orbits a much larger, and hence stationary, mass (Sun), but Equation 13.10 also applies to any two gravitationally interacting masses. Each mass traces out the exact same-shaped conic section as the other. That shape is determined by the total energy and angular momentum of the system, with the center of mass of the system located at the focus. The ratio of the dimensions of the two paths is the inverse of the ratio of their masses.

INTERACTIVE

You can see an animation of two interacting objects at the *My Solar System* page at Phet (https://openstax.org/l/21mysolarsys). Choose the Sun and Planet preset option. You can also view the more complicated multiple body problems as well. You may find the actual path of the Moon quite surprising, yet is obeying Newton's simple laws of motion.

Orbital Transfers

People have imagined traveling to the other planets of our solar system since they were discovered. But how can we best do this? The most efficient method was discovered in 1925 by Walter Hohmann, inspired by a popular science fiction novel of that time. The method is now called a Hohmann transfer. For the case of traveling between two circular orbits, the transfer is along a "transfer" ellipse that perfectly intercepts those orbits at the aphelion and perihelion of the ellipse. Figure 13.19 shows the case for a trip from Earth's orbit to that of Mars. As before, the Sun is at the focus of the ellipse.

For any ellipse, the semi-major axis is defined as one-half the sum of the perihelion and the aphelion. In Figure 13.17, the semi-major axis is the distance from the origin to either side of the ellipse along the *x*-axis, or just one-half the longest axis (called the major axis). Hence, to travel from one circular orbit of radius r_1 to another circular orbit of radius r_2, the aphelion of the transfer ellipse will be equal to the value of the larger orbit, while the perihelion will be the smaller orbit. The semi-major axis, denoted *a*, is therefore given by $a = \frac{1}{2}(r_1 + r_2)$.

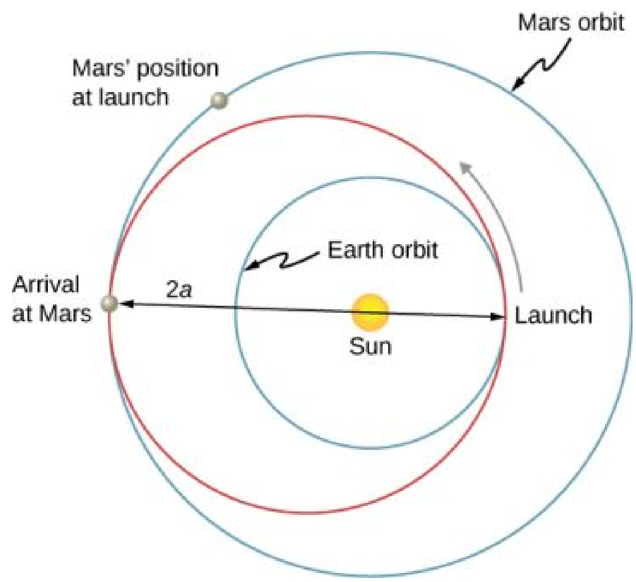


FIGURE 13.19 The transfer ellipse has its perihelion at Earth's orbit and aphelion at Mars' orbit.

Let's take the case of traveling from Earth to Mars. For the moment, we ignore the planets and assume we are alone in Earth's orbit and wish to move to Mars' orbit. From Equation 13.9, the expression for total energy, we can see that the total energy for a spacecraft in the larger orbit (Mars) is greater (less negative) than that for the smaller orbit (Earth). To move onto the transfer ellipse from Earth's orbit, we will need to increase our kinetic energy, that is, we need a velocity boost. The most efficient method is a very quick acceleration along the circular orbital path, which is also along the path of the ellipse at that point. (In fact, the acceleration should be instantaneous, such that the circular and elliptical orbits are congruent during the acceleration. In practice, the finite acceleration is short enough that the difference is not a significant consideration.) Once you have arrived at Mars orbit, you will need another velocity boost to move into that orbit, or you will stay on the elliptical orbit and simply fall back to perihelion where you started. For the return trip, you simply reverse the process with a retro-boost at each transfer point.

To make the move onto the transfer ellipse and then off again, we need to know each circular orbit velocity and the transfer orbit velocities at perihelion and aphelion. The velocity boost required is simply the difference between the circular orbit velocity and the elliptical orbit velocity at each point. We can find the circular orbital velocities from Equation 13.7. To determine the velocities for the ellipse, we state without proof (as it is beyond the scope of this course) that total energy for an elliptical orbit is

$$E = -\frac{GmM_S}{2a}$$

where M_S is the mass of the Sun and a is the semi-major axis. Remarkably, this is the same as Equation 13.9 for circular orbits, but with the value of the semi-major axis replacing the orbital radius. Since we know the potential energy from Equation 13.4, we can find the kinetic energy and hence the velocity needed for each point on the ellipse. We leave it as a challenge problem to find those transfer velocities for an Earth-to-Mars trip.

We end this discussion by pointing out a few important details. First, we have not accounted for the gravitational potential energy due to Earth and Mars, or the mechanics of landing on Mars. In practice, that must be part of the calculations. Second, timing is everything. You do not want to arrive at the orbit of Mars to find out it isn't there. We must leave Earth at precisely the correct time such that Mars will be at the aphelion of our transfer ellipse just as we arrive. That opportunity comes about every 2 years. And returning requires correct timing as well. The total trip would take just under 3 years! There are other options that provide for a faster transit, including a gravity assist flyby of Venus. But these other options come with an additional cost in energy and danger to the astronauts.

INTERACTIVE

Visit this site (https://openstax.org/l/21plantripmars) for more details about planning a trip to Mars.

Kepler's Second Law

Kepler's second law states that a planet sweeps out equal areas in equal times, that is, the area divided by time, called the areal velocity, is constant. Consider Figure 13.20. The time it takes a planet to move from position *A* to *B*, sweeping out area A_1, is exactly the time taken to move from position *C* to *D*, sweeping area A_2, and to move from *E* to *F*, sweeping out area A_3. These areas are the same: $A_1 = A_2 = A_3$.

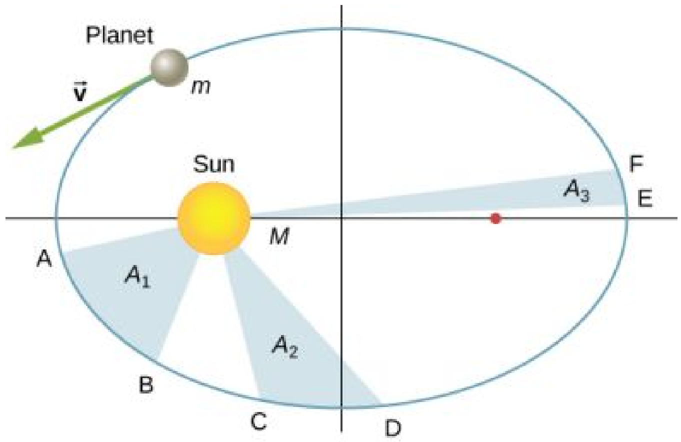


FIGURE 13.20 The shaded regions shown have equal areas and represent the same time interval.

Comparing the areas in the figure and the distance traveled along the ellipse in each case, we can see that in order for the areas to be equal, the planet must speed up as it gets closer to the Sun and slow down as it moves away. This behavior is completely consistent with our conservation equation, Equation 13.5. But we will show that Kepler's second law is actually a consequence of the conservation of angular momentum, which holds for any system with only radial forces.

Recall the definition of angular momentum from Angular Momentum, $\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$. For the case of orbiting motion, $\vec{\mathbf{L}}$ is the angular momentum of the planet about the Sun, $\vec{\mathbf{r}}$ is the position vector of the planet measured from the Sun, and $\vec{\mathbf{p}} = m\vec{\mathbf{v}}$ is the instantaneous linear momentum at any point in the orbit. Since the planet moves along the ellipse, $\vec{\mathbf{p}}$ is always tangent to the ellipse.

We can resolve the linear momentum into two components: a radial component $\vec{\mathbf{p}}_{\text{rad}}$ along the line to the Sun, and a component $\vec{\mathbf{p}}_{\text{perp}}$ perpendicular to $\vec{\mathbf{r}}$. The cross product for angular momentum can then be written as

$$\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}} = \vec{\mathbf{r}} \times \left(\vec{\mathbf{p}}_{\text{rad}} + \vec{\mathbf{p}}_{\text{perp}}\right) = \vec{\mathbf{r}} \times \vec{\mathbf{p}}_{\text{rad}} + \vec{\mathbf{r}} \times \vec{\mathbf{p}}_{\text{perp}}.$$

The first term on the right is zero because $\vec{\mathbf{r}}$ is parallel to $\vec{\mathbf{p}}_{\text{rad}}$, and in the second term $\vec{\mathbf{r}}$ is perpendicular to $\vec{\mathbf{p}}_{\text{perp}}$, so the magnitude of the cross product reduces to $L = rp_{\text{perp}} = rmv_{\text{perp}}$. Note that the angular momentum does *not* depend upon p_{rad}. Since the gravitational force is only in the radial direction, it can change only p_{rad} and not p_{perp}; hence, the angular momentum must remain constant.

Now consider Figure 13.21. A small triangular area ΔA is swept out in time Δt. The velocity is along the path and it makes an angle θ with the radial direction. Hence, the perpendicular velocity is given by $v_{\text{perp}} = v\sin\theta$. The planet moves a distance $\Delta s = v\Delta t\sin\theta$ projected along the direction perpendicular to *r*. Since the area of a triangle is one-half the base (*r*) times the height (Δs), for a small displacement, the area is given by $\Delta A = \frac{1}{2}r\Delta s$. Substituting for Δs, multiplying by *m* in the numerator and denominator, and rearranging, we obtain

$$\Delta A = \frac{1}{2}r\Delta s = \frac{1}{2}r(v\Delta t\sin\theta) = \frac{1}{2m}r(mv\sin\theta\Delta t) = \frac{1}{2m}r(mv_{\text{perp}}\Delta t) = \frac{L}{2m}\Delta t.$$

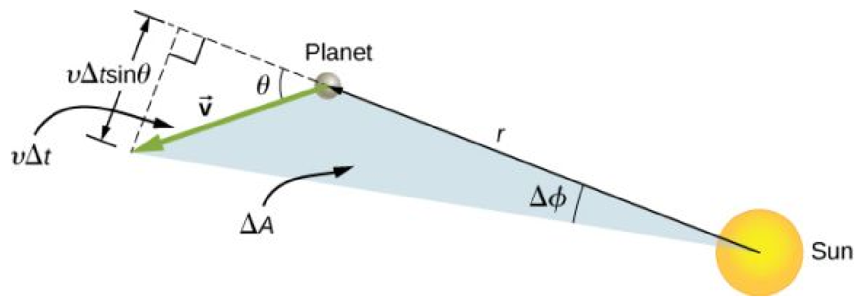


FIGURE 13.21 The element of area ΔA swept out in time Δt as the planet moves through angle $\Delta\phi$. The angle between the radial direction and $\vec{v}$ is θ.

The areal velocity is simply the rate of change of area with time, so we have

$$\text{areal velocity} = \frac{\Delta A}{\Delta t} = \frac{L}{2m}.$$

Since the angular momentum is constant, the areal velocity must also be constant. This is exactly Kepler's second law. As with Kepler's first law, Newton showed it was a natural consequence of his law of gravitation.

INTERACTIVE

You can view an animated version (https://openstax.org/l/21animationgrav) of Figure 13.20, and many other interesting animations as well, at the School of Physics (University of New South Wales) site.

Kepler's Third Law

Kepler's third law states that the square of the period is proportional to the cube of the semi-major axis of the orbit. In Satellite Orbits and Energy, we derived Kepler's third law for the special case of a circular orbit. Equation 13.8 gives us the period of a circular orbit of radius *r* about Earth:

$$T = 2\pi\sqrt{\frac{r^3}{GM_E}}.$$

For an ellipse, recall that the semi-major axis is one-half the sum of the perihelion and the aphelion. For a circular orbit, the semi-major axis (*a*) is the same as the radius for the orbit. In fact, Equation 13.8 gives us Kepler's third law if we simply replace *r* with *a* and square both sides.

$$T^2 = \frac{4\pi^2}{GM}a^3 \tag{13.11}$$

We have changed the mass of Earth to the more general *M*, since this equation applies to satellites orbiting any large mass.

EXAMPLE 13.13

Orbit of Halley's Comet

Determine the semi-major axis of the orbit of Halley's comet, given that it arrives at perihelion every 75.3 years. If the perihelion is 0.586 AU, what is the aphelion?

Strategy

We are given the period, so we can rearrange Equation 13.11, solving for the semi-major axis. Since we know the value for the perihelion, we can use the definition of the semi-major axis, given earlier in this section, to find the aphelion. We note that 1 Astronomical Unit (AU) is the average radius of Earth's orbit and is defined to be $1\text{ AU} = 1.50 \times 10^{11}\text{ m}.$

Solution

Rearranging Equation 13.11 and inserting the values of the period of Halley's comet and the mass of the Sun, we have

$$\begin{aligned} a &= \left(\frac{GM}{4\pi^2}T^2\right)^{1/3} \\ &= \left(\frac{(6.67 \times 10^{-11}\ \text{N·m}^2/\text{kg}^2)(2.00 \times 10^{30}\ \text{kg})}{4\pi^2}(75.3\ \text{yr} \times 365\ \text{days/yr} \times 24\ \text{hr/day} \times 3600\ \text{s/hr})^2\right)^{1/3}. \end{aligned}$$

This yields a value of 2.67×10^{12} m or 17.8 AU for the semi-major axis.

The semi-major axis is one-half the sum of the aphelion and perihelion, so we have

$$\begin{aligned} a &= \frac{1}{2}(\text{aphelion} + \text{perihelion}) \\ \text{aphelion} &= 2a - \text{perihelion}. \end{aligned}$$

Substituting for the values, we found for the semi-major axis and the value given for the perihelion, we find the value of the aphelion to be 35.0 AU.

Significance

Edmond Halley, a contemporary of Newton, first suspected that three comets, reported in 1531, 1607, and 1682, were actually the same comet. Before Tycho Brahe made measurements of comets, it was believed that they were one-time events, perhaps disturbances in the atmosphere, and that they were not affected by the Sun. Halley used Newton's new mechanics to predict his namesake comet's return in 1758.

CHECK YOUR UNDERSTANDING 13.9

The nearly circular orbit of Saturn has an average radius of about 9.5 AU and has a period of 30 years, whereas Uranus averages about 19 AU and has a period of 84 years. Is this consistent with our results for Halley's comet?

13.6 Tidal Forces

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the origins of Earth's ocean tides
- Describe how neap and leap tides differ
- Describe how tidal forces affect binary systems

The origin of Earth's ocean tides has been a subject of continuous investigation for over 2000 years. But the work of Newton is considered to be the beginning of the true understanding of the phenomenon. Ocean tides are the result of gravitational tidal forces. These same tidal forces are present in any astronomical body. They are responsible for the internal heat that creates the volcanic activity on Io, one of Jupiter's moons, and the breakup of stars that get too close to black holes.

Lunar Tides

If you live on an ocean shore almost anywhere in the world, you can observe the rising and falling of the sea level about twice per day. This is caused by a combination of Earth's rotation about its axis and the gravitational attraction of both the Moon and the Sun.

Let's consider the effect of the Moon first. In Figure 13.22, we are looking "down" onto Earth's North Pole. One side of Earth is closer to the Moon than the other side, by a distance equal to Earth's diameter. Hence, the gravitational force is greater on the near side than on the far side. The magnitude at the center of Earth is between these values.

This is why a tidal bulge appears on both sides of Earth.

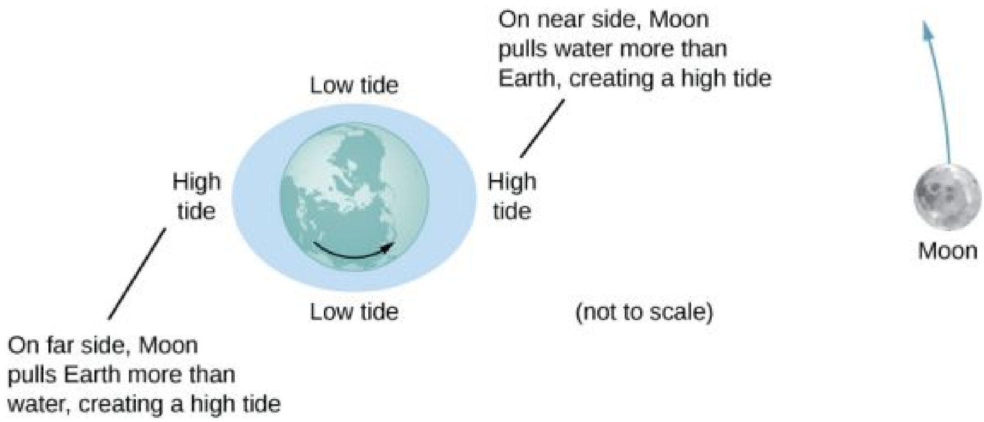


FIGURE 13.22 The tidal force stretches Earth along the line between Earth and the Moon. It is the *difference* between the gravitational force from the far side to the near side, along with the orbit of Earth and the Moon around each other, that creates the tidal bulge on both sides of the planet. Tidal variations of the oceans are on the order of few meters; hence, this diagram is greatly exaggerated.

The net force on Earth causes it to orbit about the Earth-Moon center of mass, located about 1600 km below Earth's surface along the line between Earth and the Moon. The **tidal force** can be viewed as the *difference* between the force at the center of Earth and that at any other location. In Figure 13.23, this difference is shown at sea level, where we observe the ocean tides. (Note that the change in sea level caused by these tidal forces is measured from the baseline sea level. We saw earlier that Earth bulges many kilometers at the equator due to its rotation. This defines the baseline sea level and here we consider only the much smaller tidal bulge measured from that baseline sea level.)

FIGURE 13.23 The tidal force is the *difference* between the gravitational force at the center and that elsewhere. In this figure, the tidal forces are shown at the ocean surface. These forces would diminish to zero as you approach Earth's center.

Why does the rise and fall of the tides occur twice per day? Look again at Figure 13.22. If Earth were not rotating and the Moon was fixed, then the bulges would remain in the same location on Earth. Relative to the Moon, the bulges stay fixed—along the line connecting Earth and the Moon. But Earth rotates (in the direction shown by the blue arrow) approximately every 24 hours. In 6 hours, the near and far locations of Earth move to where the low tides are occurring, and 6 hours later, those locations are back to the high-tide position. Since the Moon also orbits Earth approximately every 28 days, and in the same direction as Earth rotates, the time between high (and low) tides is actually about 12.5 hours. The actual timing of the tides is complicated by numerous factors, the most important of which is another astronomical body—the Sun.

The Effect of the Sun on Tides

In addition to the Moon's tidal forces on Earth's oceans, the Sun exerts a tidal force as well. The gravitational attraction of the Sun on any object on Earth is nearly 200 times that of the Moon. However, as we show later in an example, the *tidal* effect of the Sun is less than that of the Moon, but a significant effect nevertheless. Depending upon the positions of the Moon and Sun relative to Earth, the net tidal effect can be amplified or attenuated.

Figure 13.22 illustrates the relative positions of the Sun and the Moon that create the largest tides, called **spring tides** (or leap tides). During spring tides, Earth, the Moon, and the Sun are aligned and the tidal effects add. (Recall that the tidal forces cause bulges on both sides.) Figure 13.22(c) shows the relative positions for the smallest tides, called **neap tides**. The extremes of both high and low tides are affected. Spring tides occur during the new or full moon, and neap tides occur at half-moon.

INTERACTIVE

You can see an animation (https://openstax.org/l/21tidesinmot01) of the tides in motion.

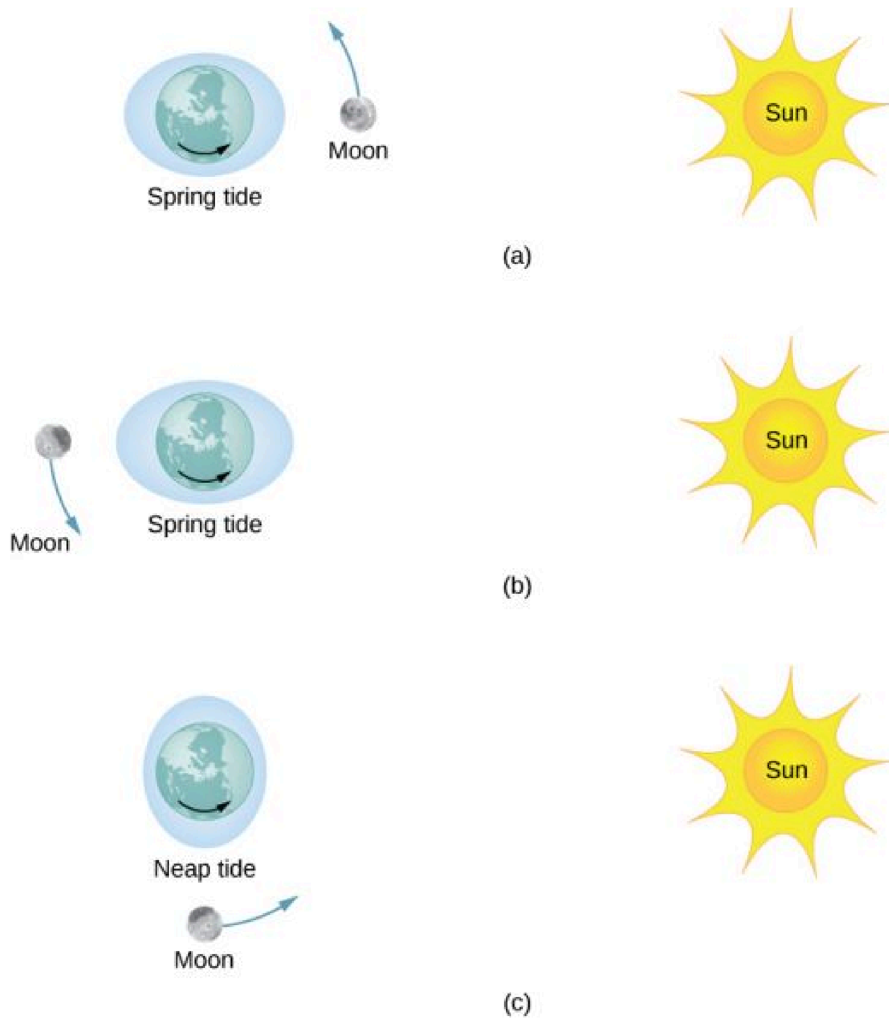


FIGURE 13.24 (a and b) The spring tides occur when the Sun and the Moon are aligned, whereas (c) the neap tides occur when the Sun and Moon make a right triangle with Earth. (Figure is not drawn to scale.)

The Magnitude of the Tides

With accurate data for the positions of the Moon and the Sun, the time of maximum and minimum tides at most locations on our planet can be predicted accurately.

INTERACTIVE

Visit this site (https://openstax.org/l/21tidepredic) to generate tide predictions for up to 2 years in the past or future, at more than 3000 locations around the United States.

The magnitude of the tides, however, is far more complicated. The relative angles of Earth and the Moon determine spring and neap tides, but the magnitudes of these tides are affected by the distances from Earth as well. Tidal forces are greater when the distances are smaller. Both the Moon's orbit about Earth and Earth's orbit about the Sun are elliptical, so a spring tide is exceptionally large if it occurs when the Moon is at perigee and Earth is at perihelion. Conversely, it is relatively small if it occurs when the Moon is at apogee and Earth is at aphelion.

The greatest causes of tide variation are the topography of the local shoreline and the bathymetry (the profile of the depth) of the ocean floor. The range of tides due to these effects is astounding. Although ocean tides are much smaller than a meter in many places around the globe, the tides at the Bay of Fundy (Figure 13.25), on the east coast of Canada, can be as much as 16.3 meters.

FIGURE 13.25 Boats in the Bay of Fundy at high and low tides. The twice-daily change in sea level creates a real challenge to the safe mooring of boats. (credit: modification of works by Dylan Kereluk)

EXAMPLE 13.14

Comparing Tidal Forces

Compare the Moon's gravitational force on a 1.0-kg mass located on the near side and another on the far side of Earth. Repeat for the Sun and then compare the results to confirm that the Moon's tidal forces are about twice that of the Sun.

Strategy

We use Newton's law of gravitation given by Equation 13.1. We need the masses of the Moon and the Sun and their distances from Earth, as well as the radius of Earth. We use the astronomical data from Appendix D.

Solution

Substituting the mass of the Moon and mean distance from Earth to the Moon, we have

$$F_{12} = G\frac{m_1 m_2}{r^2} = (6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)\frac{(1.0\ \text{kg})(7.35 \times 10^{22}\ \text{kg})}{(3.84 \times 10^8 \pm 6.37 \times 10^6\ \text{m})^2}.$$

In the denominator, we use the minus sign for the near side and the plus sign for the far side. The results are

$$F_{\text{near}} = 3.44 \times 10^{-5}\ \text{N} \quad \text{and} \quad F_{\text{far}} = 3.22 \times 10^{-5}\ \text{N}.$$

The Moon's gravitational force is nearly 7% higher at the near side of Earth than at the far side, but both forces are much less than that of Earth itself on the 1.0-kg mass. Nevertheless, this small difference creates the tides. We now repeat the problem, but substitute the mass of the Sun and the mean distance between the Earth and Sun. The results are

$$F_{\text{near}} = 5.89975 \times 10^{-3}\ \text{N} \quad \text{and} \quad F_{\text{far}} = 5.89874 \times 10^{-3}\ \text{N}.$$

We have to keep six significant digits since we wish to compare the difference between them to the difference for the Moon. (Although we can't justify the absolute value to this accuracy, since all values in the calculation are the same except the distances, the accuracy in the difference is still valid to three digits.) The difference between the near and far forces on a 1.0-kg mass due to the Moon is

$$F_{\text{near}} = 3.44 \times 10^{-5}\ \text{N} - 3.22 \times 10^{-5}\ \text{N} = 0.22 \times 10^{-5}\ \text{N},$$

whereas the difference for the Sun is

$$F_{\text{near}} - F_{\text{far}} = 5.89975 \times 10^{-3}\ \text{N} - 5.89874 \times 10^{-3}\ \text{N} = 0.101 \times 10^{-5}\ \text{N}.$$

Note that a more proper approach is to write the difference in the two forces with the difference between the near and far distances explicitly expressed. With just a bit of algebra we can show that

$$F_{\text{tidal}} = \frac{GMm}{r_1^2} - \frac{GMm}{r_2^2} = GMm\left(\frac{(r_2 - r_1)(r_2 + r_1)}{r_1^2 r_2^2}\right),$$

where r_1 and r_2 are the same to three significant digits, but their difference $(r_2 - r_1)$, equal to the diameter of Earth, is also known to three significant digits. The results of the calculation are the same. This approach would be necessary if the number of significant digits needed exceeds that available on your calculator or computer.

Significance

Note that the forces exerted by the Sun are nearly 200 times greater than the forces exerted by the Moon. But the *difference* in those forces for the Sun is half that for the Moon. This is the nature of tidal forces. The Moon has a greater tidal effect because the fractional change in distance from the near side to the far side is so much greater for the Moon than it is for the Sun.

CHECK YOUR UNDERSTANDING 13.10

Earth exerts a tidal force on the Moon. Is it greater than, the same as, or less than that of the Moon on Earth? Be careful in your response, as tidal forces arise from the *difference* in gravitational forces between one side and the other. Look at the calculations we performed for the tidal force on Earth and consider the values that would change significantly for the Moon. The diameter of the Moon is one-fourth that of Earth. Tidal forces on the Moon are not easy to detect, since there is no liquid on the surface.

Other Tidal Effects

Tidal forces exist between any two bodies. The effect stretches the bodies along the line between their centers. Although the tidal effect on Earth's seas is observable on a daily basis, long-term consequences cannot be observed so easily. One consequence is the dissipation of rotational energy due to friction during flexure of the bodies themselves. Earth's rotation rate is slowing down as the tidal forces transfer rotational energy into heat. The other effect, related to this dissipation and conservation of angular momentum, is called "locking" or tidal synchronization. It has already happened to most moons in our solar system, including Earth's Moon. The Moon keeps one face toward Earth—its rotation rate has locked into the orbital rate about Earth. The same process is happening to Earth, and eventually it will keep one face toward the Moon. If that does happen, we would no longer see tides, as the tidal bulge would remain in the same place on Earth, and half the planet would never see the Moon. However, this locking will take many billions of years, perhaps not before our Sun expires.

One of the more dramatic example of tidal effects is found on Io, one of Jupiter's moons. In 1979, the *Voyager* spacecraft sent back dramatic images of volcanic activity on Io. It is the only other astronomical body in our solar system on which we have found such activity. Figure 13.26 shows a more recent picture of Io taken by the *New Horizons* spacecraft on its way to Pluto, while using a gravity assist from Jupiter.

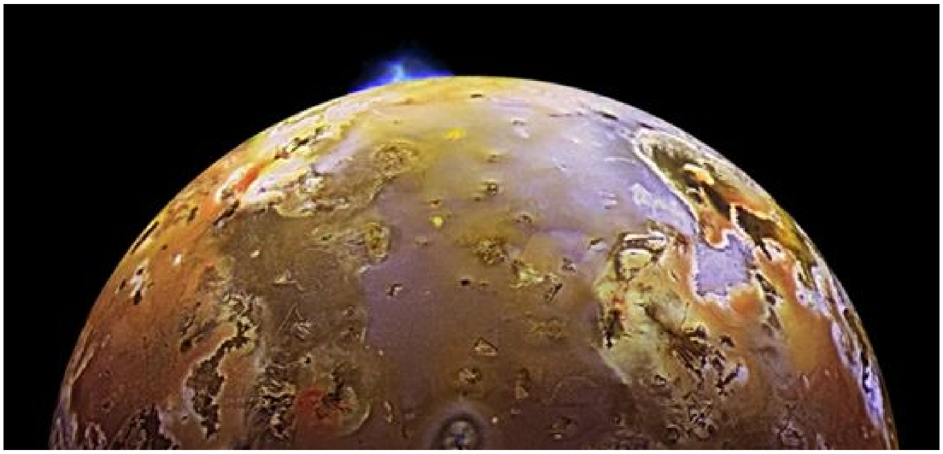


FIGURE 13.26 Dramatic evidence of tidal forces can be seen on Io. The eruption seen in blue is due to the internal heat created by the tidal forces exerted on Io by Jupiter. (credit: modification of work by NASA/JPL/University of Arizona)

For some stars, the effect of tidal forces can be catastrophic. The tidal forces in very close binary systems can be strong enough to rip matter from one star to the other, once the tidal forces exceed the cohesive self-gravitational forces that hold the stars together. This effect can be seen in normal stars that orbit nearby compact stars, such as neutron stars or black holes. Figure 13.27 shows an artist's rendition of this process. As matter falls into the compact star, it forms an accretion disc that becomes super-heated and radiates in the X-ray spectrum.

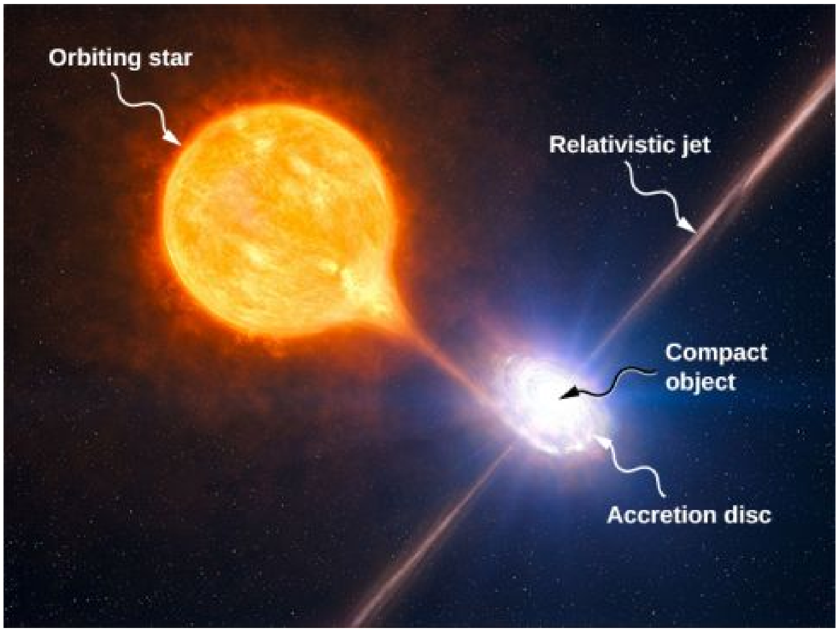


FIGURE 13.27 Tidal forces from a compact object can tear matter away from an orbiting star. In addition to the accretion disc orbiting the compact object, material is often ejected along relativistic jets as shown. (credit: modification of work by ESO/L. Calçada/M. Kornmesser)

The energy output of these binary systems can exceed the typical output of thousands of stars. Another example might be a quasar. Quasars are very distant and immensely bright objects, often exceeding the energy output of entire galaxies. It is the general consensus among astronomers that they are, in fact, massive black holes producing radiant energy as matter that has been tidally ripped from nearby stars falls into them.

13.7 Einstein's Theory of Gravity

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe how the theory of general relativity approaches gravitation
- Explain the principle of equivalence
- Calculate the Schwarzschild radius of an object
- Summarize the evidence for black holes

Newton's law of universal gravitation accurately predicts much of what we see within our solar system. Indeed, only Newton's laws have been needed to accurately send every space vehicle on its journey. The paths of Earth-crossing asteroids, and most other celestial objects, can be accurately determined solely with Newton's laws. Nevertheless, many phenomena have shown a discrepancy from what Newton's laws predict, including the orbit of Mercury and the effect that gravity has on light. In this section, we examine a different way of envisioning gravitation.

A Revolution in Perspective

In 1905, Albert Einstein published his theory of special relativity. This theory is discussed in great detail in Relativity (http://openstax.org/books/university-physics-volume-3/pages/5-introduction), so we say only a few words here. In this theory, no motion can exceed the speed of light—it is the speed limit of the Universe. This simple fact has been verified in countless experiments. However, it has incredible consequences—space and time are no longer absolute. Two people moving relative to one another do not agree on the length of objects or the passage of time. Almost all of the mechanics you learned in previous chapters, while remarkably accurate even for speeds of many thousands of miles per second, begin to fail when approaching the speed of light.

This speed limit on the Universe was also a challenge to the inherent assumption in Newton's law of gravitation that gravity is an **action-at-a-distance force**. That is, without physical contact, any change in the position of one mass is instantly communicated to all other masses. This assumption does not come from any first principle, as Newton's theory simply does not address the question. (The same was believed of electromagnetic forces, as well. It is fair to say that most scientists were not completely comfortable with the action-at-a-distance concept.)

A second assumption also appears in Newton's law of gravitation Equation 13.1. The masses are assumed to be exactly the same as those used in Newton's second law, $\vec{\mathbf{F}} = m\vec{\mathbf{a}}$. We made that assumption in many of our derivations in this chapter. Again, there is no underlying principle that this must be, but experimental results are consistent with this assumption. In Einstein's subsequent **theory of general relativity** (1916), both of these issues were addressed. His theory was a theory of **space-time** geometry and how mass (and acceleration) distort and interact with that space-time. It was not a theory of gravitational forces. The mathematics of the general theory is beyond the scope of this text, but we can look at some underlying principles and their consequences.

The Principle of Equivalence

Einstein came to his general theory in part by wondering why someone who was free falling did not feel their weight. Indeed, it is common to speak of astronauts orbiting Earth as being weightless, despite the fact that Earth's gravity is still quite strong there. In Einstein's general theory, there is no difference between free fall and being weightless. This is called the **principle of equivalence**. The equally surprising corollary to this is that there is no difference between a uniform gravitational field and a uniform acceleration in the absence of gravity. Let's focus on this last statement. Although a perfectly uniform gravitational field is not feasible, we can approximate it very well.

Within a reasonably sized laboratory on Earth, the gravitational field $\vec{\mathbf{g}}$ is essentially uniform. The corollary states that any physical experiments performed there have the identical results as those done in a laboratory accelerating at $\vec{\mathbf{a}} = \vec{\mathbf{g}}$ in deep space, well away from all other masses. Figure 13.28 illustrates the concept.

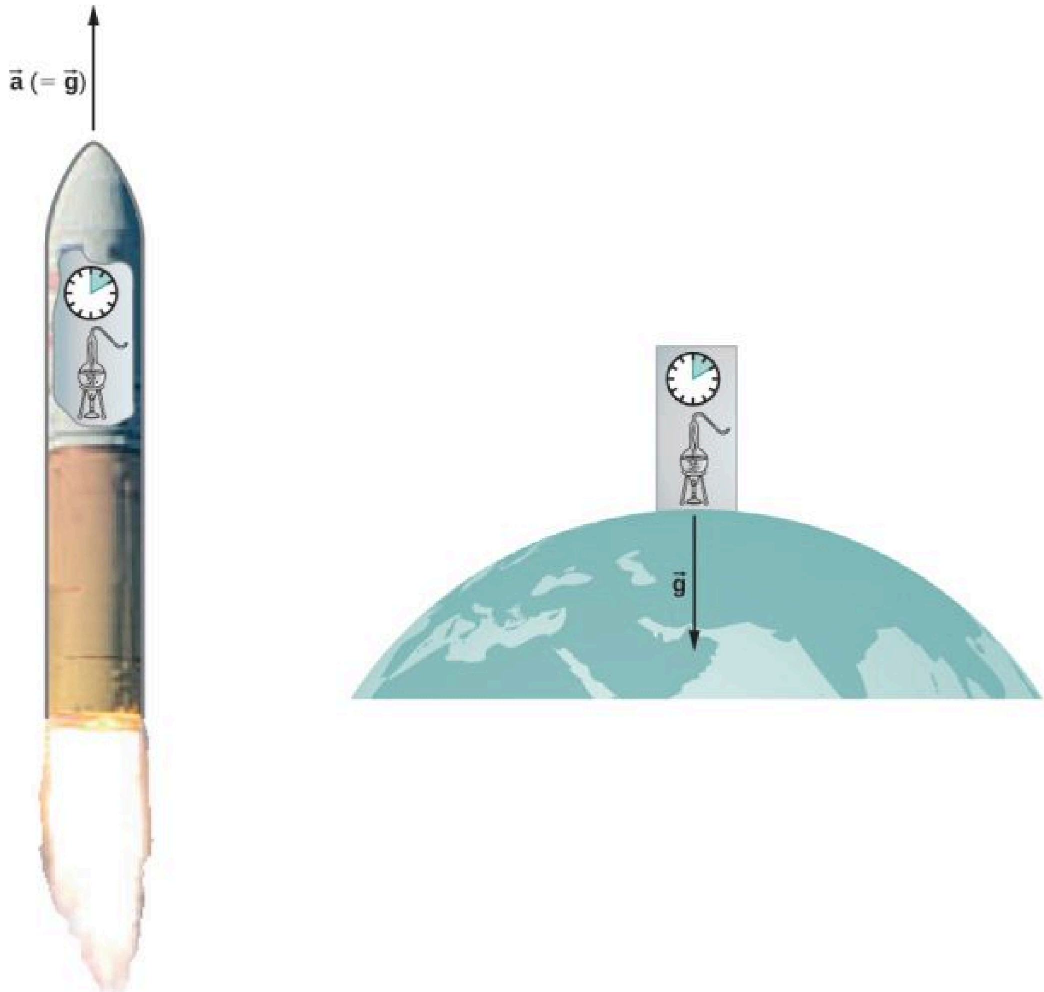


FIGURE 13.28 According to the principle of equivalence, the results of all experiments performed in a laboratory in a uniform gravitational field are identical to the results of the same experiments performed in a uniformly accelerating laboratory.

How can these two apparently fundamentally different situations be the same? The answer is that gravitation is not a force between two objects but is the result of each object responding to the effect that the other has on the space-time surrounding it. A uniform gravitational field and a uniform acceleration have exactly the same effect on space-time.

A Geometric Theory of Gravity

Euclidian geometry assumes a “flat” space in which, among the most commonly known attributes, a straight line is the shortest distance between two points, the sum of the angles of all triangles must be 180 degrees, and parallel lines never intersect. **Non-Euclidean geometry** was not seriously investigated until the nineteenth century, so it is not surprising that Euclidean space is inherently assumed in all of Newton’s laws.

The general theory of relativity challenges this long-held assumption. Only empty space is flat. The presence of mass—or energy, since relativity does not distinguish between the two—distorts or curves space and time, or space-time, around it. The motion of any other mass is simply a response to this curved space-time. Figure 13.29 is a two-dimensional representation of a smaller mass orbiting in response to the distorted space created by the presence of a larger mass. In a more precise but confusing picture, we would also see space distorted by the orbiting mass, and both masses would be in motion in response to the total distortion of space. Note that the figure is a representation to help visualize the concept. These are distortions in our three-dimensional space and time. We do not see them as we would a dimple on a ball. We see the distortion only by careful measurements of the motion of objects and light as they move through space.

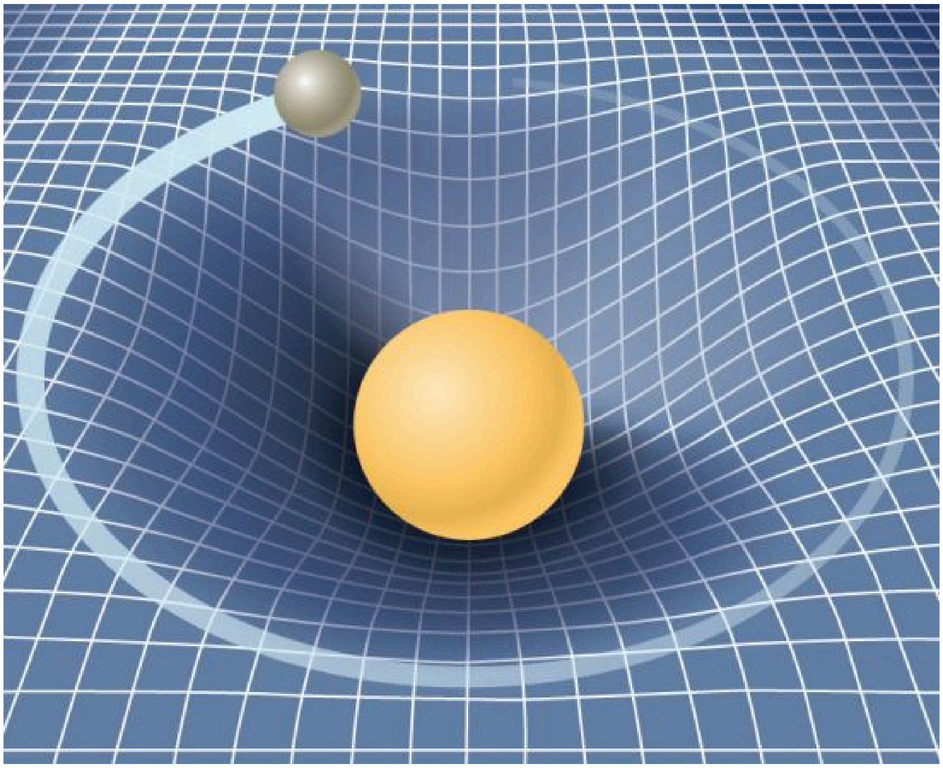


FIGURE 13.29 A smaller mass orbiting in the distorted space-time of a larger mass. In fact, all mass or energy distorts space-time.

For weak gravitational fields, the results of general relativity do not differ significantly from Newton's law of gravitation. But for intense gravitational fields, the results diverge, and general relativity has been shown to predict the correct results. Even in our Sun's relatively weak gravitational field at the distance of Mercury's orbit, we can observe the effect. Starting in the mid-1800s, Mercury's elliptical orbit has been carefully measured. However, although it is elliptical, its motion is complicated by the fact that the perihelion position of the ellipse slowly advances. Most of the advance is due to the gravitational pull of other planets, but a small portion of that advancement could not be accounted for by Newton's law. At one time, there was even a search for a "companion" planet that would explain the discrepancy. But general relativity correctly predicts the measurements. Since then, many measurements, such as the deflection of light of distant objects by the Sun, have verified that general relativity correctly predicts the observations.

We close this discussion with one final comment. We have often referred to distortions of space-time or distortions in both space and time. In both special and general relativity, the dimension of time has equal footing with each spatial dimension (differing in its place in both theories only by an ultimately unimportant scaling factor). Near a very large mass, not only is the nearby space "stretched out," but time is dilated or "slowed." We discuss these effects more in the next section.

Black Holes

Einstein's theory of gravitation is expressed in one deceptively simple-looking tensor equation (tensors are a generalization of scalars and vectors), which expresses how a mass determines the curvature of space-time around it. The solutions to that equation yield one of the most fascinating predictions: the **black hole**. The prediction is that if an object is sufficiently dense, it will collapse in upon itself and be surrounded by an **event horizon** from which nothing can escape. The name "black hole," which was coined by astronomer John Wheeler in 1969, refers to the fact that light cannot escape such an object. Karl Schwarzschild was the first person to note this phenomenon in 1916, but at that time, it was considered mostly to be a mathematical curiosity.

Surprisingly, the idea of a massive body from which light cannot escape dates back to the late 1700s. Independently, John Michell and Pierre-Simon Laplace used Newton's law of gravitation to show that light leaving the surface of a star with enough mass could not escape. Their work was based on the fact that the speed of light had been measured by Ole Rømer in 1676. He noted discrepancies in the data for the orbital period of the moon Io about Jupiter. Rømer realized that the difference arose from the relative positions of Earth and Jupiter at different times and that he could find the speed of light from that difference. Michell and Laplace both realized that since light had a finite speed, there could be a star massive enough that the escape speed from its surface could exceed that speed. Hence, light always would fall back to the star. Oddly, observers far enough away from the very largest stars would not be able to see them, yet they could see a smaller star from the same distance.

Recall that in Gravitational Potential Energy and Total Energy, we found that the escape speed, given by Equation 13.6, is independent of the mass of the object escaping. Even though the nature of light was not fully understood at the time, the mass of light, if it had any, was not relevant. Hence, Equation 13.6 should be valid for light. Substituting c, the speed of light, for the escape velocity, we have

$$v_{\text{esc}} = c = \sqrt{\frac{2GM}{R}}.$$

Thus, we only need values for R and M such that the escape velocity exceeds c, and then light will not be able to escape. Michell posited that if a star had the density of our Sun and a radius that extended just beyond the orbit of Mars, then light would not be able to escape from its surface. He also conjectured that we would still be able to detect such a star from the gravitational effect it would have on objects around it. This was an insightful conclusion, as this is precisely how we infer the existence of such objects today. While we have yet to, and may never, visit a black hole, the circumstantial evidence for them has become so compelling that few astronomers doubt their existence.

Before we examine some of that evidence, we turn our attention back to Schwarzschild's solution to the tensor equation from general relativity. In that solution arises a critical radius, now called the **Schwarzschild radius** (R_{S}). For any mass M, if that mass were compressed to the extent that its radius becomes less than the Schwarzschild radius, then the mass will collapse to a singularity, and anything that passes inside that radius cannot escape. Once inside R_{S}, the arrow of time takes all things to the singularity. (In a broad mathematical sense, a singularity is where the value of a function goes to infinity. In this case, it is a point in space of zero volume with a finite mass. Hence, the mass density and gravitational energy become infinite.) The Schwarzschild radius is given by

$$R_{\text{S}} = \frac{2GM}{c^2}. \tag{13.12}$$

If you look at our escape velocity equation with $v_{\text{esc}} = c$, you will notice that it gives precisely this result. But that is merely a fortuitous accident caused by several incorrect assumptions. One of these assumptions is the use of the *incorrect* classical expression for the kinetic energy for light. Just how dense does an object have to be in order to turn into a black hole?

EXAMPLE 13.15

Calculating the Schwarzschild Radius

Calculate the Schwarzschild radius for both the Sun and Earth. Compare the density of the nucleus of an atom to the density required to compress Earth's mass uniformly to its Schwarzschild radius. The density of a nucleus is about $2.3 \times 10^{17}\ \text{kg/m}^3$.

Strategy

We use Equation 13.12 for this calculation. We need only the masses of Earth and the Sun, which we obtain from the astronomical data given in Appendix D.

Solution

Substituting the mass of the Sun, we have

$$R_{\text{S}} = \frac{2GM}{c^2} = \frac{2(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(1.99 \times 10^{30}\ \text{kg})}{(3.0 \times 10^8\ \text{m/s})^2} = 2.95 \times 10^3\ \text{m}.$$

This is a diameter of only about 6 km. If we use the mass of Earth, we get $R_S = 8.85 \times 10^{-3}$ m. This is a diameter of less than 2 cm! If we pack Earth's mass into a sphere with the radius $R_S = 8.85 \times 10^{-3}$ m, we get a density of

$$\rho = \frac{\text{mass}}{\text{volume}} = \frac{5.97 \times 10^{24}\ \text{kg}}{\left(4/3\,\pi\right)\left(8.85 \times 10^{-3}\ \text{m}\right)^3} = 2.06 \times 10^{30}\ \text{kg/m}^3.$$

Significance

A **neutron star** is the most compact object known—outside of a black hole itself. The neutron star is composed of neutrons, with the density of an atomic nucleus, and, like many black holes, is believed to be the remnant of a supernova—a star that explodes at the end of its lifetime. To create a black hole from Earth, we would have to compress it to a density thirteen orders of magnitude greater than that of a neutron star. This process would require unimaginable force. There is no known mechanism that could cause an Earth-sized object to become a black hole. For the Sun, you should be able to show that it would have to be compressed to a density only about 80 times that of a nucleus. (Note: Once the mass is compressed within its Schwarzschild radius, general relativity dictates that it will collapse to a singularity. These calculations merely show the density we must achieve to initiate that collapse.)

CHECK YOUR UNDERSTANDING 13.11

Consider the density required to make Earth a black hole compared to that required for the Sun. What conclusion can you draw from this comparison about what would be required to create a black hole? Would you expect the Universe to have many black holes with small mass?

The event horizon

The Schwarzschild radius is also called the event horizon of a black hole. We noted that both space and time are stretched near massive objects, such as black holes. Figure 13.30 illustrates that effect on space. The distortion caused by our Sun is actually quite small, and the diagram is exaggerated for clarity. Consider the neutron star, described in Example 13.15. Although the distortion of space-time at the surface of a neutron star is very high, the radius is still larger than its Schwarzschild radius. Objects could still escape from its surface.

However, if a neutron star gains additional mass, it would eventually collapse, shrinking beyond the Schwarzschild radius. Once that happens, the entire mass would be pulled, inevitably, to a singularity. In the diagram, space is stretched to infinity. Time is also stretched to infinity. As objects fall toward the event horizon, we see them approaching ever more slowly, but never reaching the event horizon. As outside observers, we never see objects pass through the event horizon—effectively, time is stretched to a stop.

INTERACTIVE

Visit this site (https://openstax.org/l/21spacetelescop) to view an animated example of these spatial distortions.

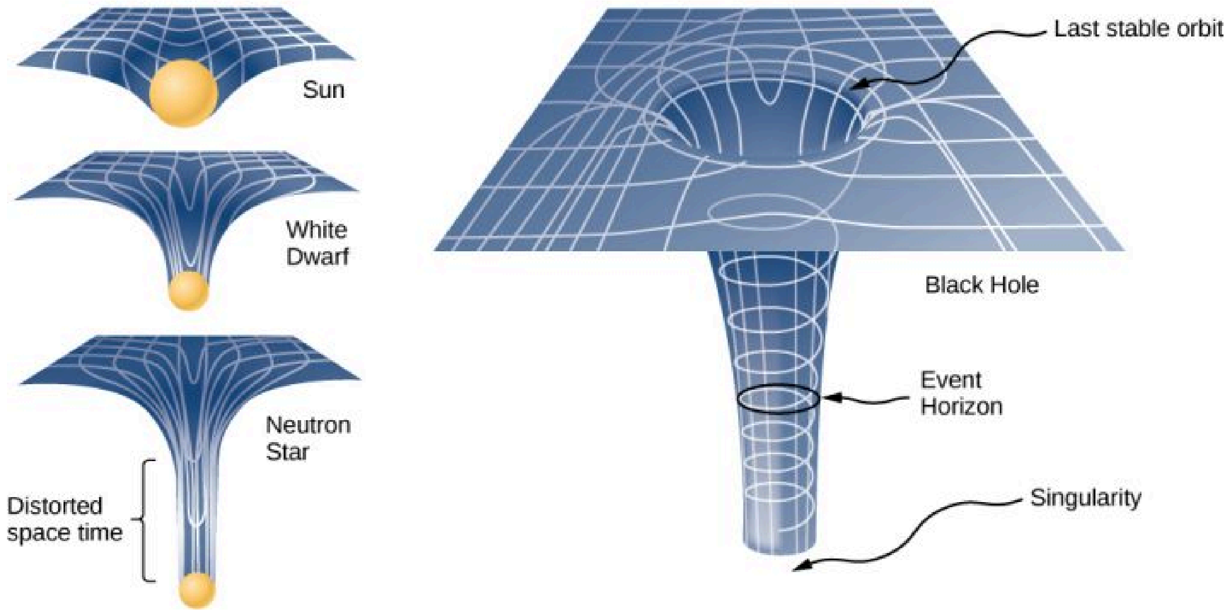


FIGURE 13.30 The space distortion becomes more noticeable around increasingly larger masses. Once the mass density reaches a critical level, a black hole forms and the fabric of space-time is torn. The curvature of space is greatest at the surface of each of the first three objects shown and is finite. The curvature then decreases (not shown) to zero as you move to the center of the object. But the black hole is different. The curvature becomes infinite: The surface has collapsed to a singularity, and the cone extends to infinity. (Note: These diagrams are not to any scale.) (credit: modification of work by NASA)

Observational evidence of neutron stars and black holes

The first validated neutron star was discovered by Jocelyn Bell Burnell in 1967. Burnell observed a regular radio pulse from an distant object, a phenomenon not previously observed. After further analysis and the discovery of a second pulsar (as the objects would later be called), the researchers and scientific community recognized that the pulses were caused by rapidly rotating neutron stars. At about the same time, a number of other astronomers, including Iosof Shklovsky, observed radio sources that would be proven to be neutron stars. These discoveries renewed interest in black holes. Evidence for black holes is based upon several types of observations, such as radiation analysis of X-ray binaries, gravitational lensing of the light from distant galaxies, and the motion of visible objects around invisible partners. We will focus on these later observations as they relate to what we have learned in this chapter. Although light cannot escape from a black hole for us to see, we can nevertheless see the gravitational effect of the black hole on surrounding masses.

The closest, and perhaps most dramatic, evidence for a black hole is at the center of our Milky Way galaxy. The UCLA Galactic Group, using data obtained by the W. M. Keck telescopes, has determined the orbits of several stars near the center of our galaxy. Some of that data is shown in Figure 13.31. The orbits of two stars are highlighted. From measurements of the periods and sizes of their orbits, it is estimated that they are orbiting a mass of approximately 4 million solar masses. Note that the mass must reside in the region created by the intersection of the ellipses of the stars. The region in which that mass must reside would fit inside the orbit of Mercury—yet nothing is seen there in the visible spectrum. In 2020, UCLA astrophysicist Andrea Ghez and Plank Institute's (Germany) Reinhardt Genzel shared the Nobel prize for the discovery of the supermassive object. They shared the prize with Richard Penrose, whose mathematical models proved that black holes can actually form and align with Einstein's theory. Since their work, the world has actually seen a black hole, thanks to the Event Horizon Telescope. The global network of observatories combined massive amounts of data in order to produce remarkable images of two black holes: first at the center of galaxy Messier 87, and second at the center of the Milky Way, Saggitarious A.

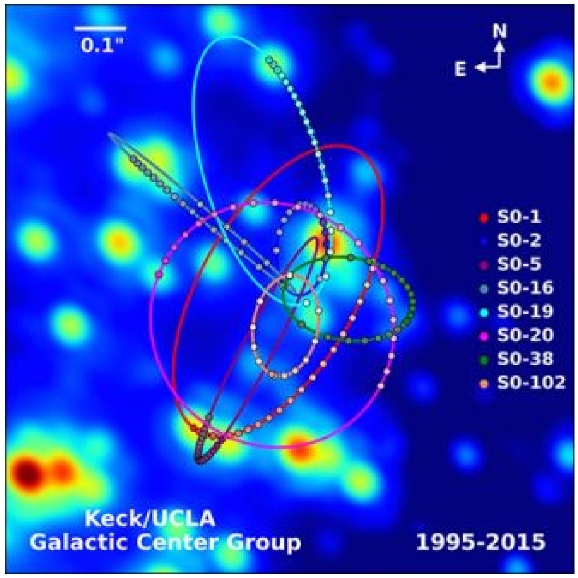


FIGURE 13.31 Paths of stars orbiting about a mass at the center of our Milky Way galaxy. From their motion, it is estimated that a black hole of about 4 million solar masses resides at the center. (credit: modification of work by UCLA Galactic Center Group – W.M. Keck Observatory Laser Team)

The physics of stellar creation and evolution is well established. The ultimate source of energy that makes stars shine is the self-gravitational energy that triggers fusion. The general behavior is that the more massive a star, the brighter it shines and the shorter it lives. The logical inference is that a mass that is 4 million times the mass of our Sun, confined to a very small region, and that cannot be seen, has no viable interpretation other than a black hole. Extragalactic observations strongly suggest that black holes are common at the center of galaxies.

INTERACTIVE

Visit the UCLA Galactic Center Group main page (https://openstax.org/l/21galacenter01) for information on X-ray binaries and gravitational lensing. Visit this page (https://openstax.org/l/21galacenter02) to view the background and the image of the black hole at the center of the Milky Way.

Dark matter

Stars orbiting near the very heart of our galaxy provide strong evidence for a black hole there, but the orbits of stars far from the center suggest another intriguing phenomenon that is observed indirectly as well. Recall from Gravitation Near Earth's Surface that we can consider the mass for spherical objects to be located at a point at the center for calculating their gravitational effects on other masses. Similarly, we can treat the total mass that lies within the orbit of any star in our galaxy as being located at the center of the Milky Way disc. We can estimate that mass from counting the visible stars and include in our estimate the mass of the black hole at the center as well.

But when we do that, we find the orbital speed of the stars is far too fast to be caused by that amount of matter. Astronomer Vera Rubin discovered this phenomenon in the 1970s while researching the movement of spiral galaxies. She observed that their outermost reaches were rotating as quickly as their centers, which was not the predicted outcome. Figure 13.32 shows the orbital velocities of stars as a function of their distance from the center of the Milky Way. The blue line represents the velocities we would expect from our estimates of the mass, whereas the green curve is what we get from direct measurements. Rubin calculated that the rotational velocity of galaxies should have been enough to cause them to fly apart, unless there was a significant discrepancy between their observable matter and their actual matter. This became known as the galaxy rotation problem. Rubin proposed that massive amounts of unseen matter must be present in and around the galaxies for the rotation speeds to occur. Furthermore, the velocity profile does not follow what we expect from the observed distribution of visible stars. Not only is the estimate of the total mass inconsistent with the data, but the expected distribution is inconsistent as well. Rubin's 1978 discovery provided much more concrete and widespread evidence of dark matter, a term Fritz Zwicky coined in 1933 when he observed a similar anomaly while examining a galaxy cluster.

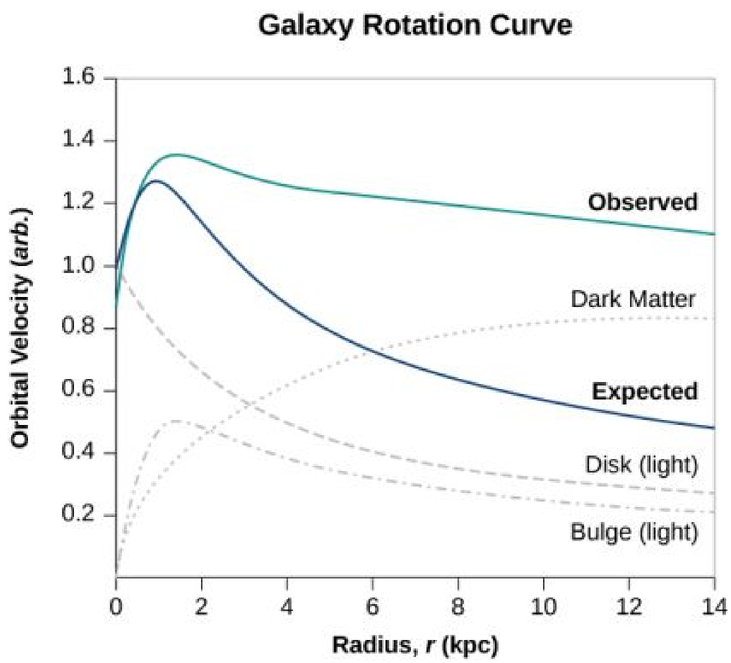


FIGURE 13.32 The blue curve shows the expected orbital velocity of stars in the Milky Way based upon the visible stars we can see. The green curve shows that the actually velocities are higher, suggesting additional matter that cannot be seen. (credit: modification of work by Matthew Newby)

There are two prevailing ideas of what this matter could be—WIMPs and MACHOs. WIMPs stands for weakly interacting massive particles. These particles (neutrinos are one example) interact very weakly with ordinary matter and, hence, are very difficult to detect directly. MACHOs stands for massive compact halo objects, which are composed of ordinary baryonic matter, such as neutrons and protons. There are unresolved issues with both of these ideas, and far more research will be needed to solve the mystery.

Chapter Review

Key Terms

action-at-a-distance force type of force exerted without physical contact

aphelion farthest point from the Sun of an orbiting body; the corresponding term for the Moon's farthest point from Earth is the apogee

apparent weight reading of the weight of an object on a scale that does not account for acceleration

black hole mass that becomes so dense, that it collapses in on itself, creating a singularity at the center surround by an event horizon

escape velocity initial velocity an object needs to escape the gravitational pull of another; it is more accurately defined as the velocity of an object with zero total mechanical energy

event horizon location of the Schwarzschild radius and is the location near a black hole from within which no object, even light, can escape

gravitational field vector field that surrounds the mass creating the field; the field is represented by field lines, in which the direction of the field is tangent to the lines, and the magnitude (or field strength) is inversely proportional to the spacing of the lines; other masses respond to this field

gravitationally bound two object are gravitationally bound if their orbits are closed; gravitationally bound systems have a negative total mechanical energy

Kepler's first law law stating that every planet moves along an ellipse, with the Sun located at a focus of the ellipse

Kepler's second law law stating that a planet sweeps out equal areas in equal times, meaning it has a constant areal velocity

Kepler's third law law stating that the square of the period is proportional to the cube of the semi-major axis of the orbit

neap tide low tide created when the Moon and the Sun form a right triangle with Earth

neutron star most compact object known—outside of a black hole itself

Newton's law of gravitation every mass attracts every other mass with a force proportional to the product of their masses, inversely proportional to the square of the distance between them, and with direction along the line connecting the center of mass of each

non-Euclidean geometry geometry of curved space, describing the relationships among angles and lines on the surface of a sphere, hyperboloid, etc.

orbital period time required for a satellite to complete one orbit

orbital speed speed of a satellite in a circular orbit; it can be also be used for the instantaneous speed for noncircular orbits in which the speed is not constant

perihelion point of closest approach to the Sun of an orbiting body; the corresponding term for the Moon's closest approach to Earth is the perigee

principle of equivalence part of the general theory of relativity, it states that there no difference between free fall and being weightless, or a uniform gravitational field and uniform acceleration

Schwarzschild radius critical radius (R_S) such that if a mass were compressed to the extent that its radius becomes less than the Schwarzschild radius, then the mass will collapse to a singularity, and anything that passes inside that radius cannot escape

space-time concept of space-time is that time is essentially another coordinate that is treated the same way as any individual spatial coordinate; in the equations that represent both special and general relativity, time appears in the same context as do the spatial coordinates

spring tide high tide created when the Moon, the Sun, and Earth are along one line

theory of general relativity Einstein's theory for gravitation and accelerated reference frames; in this theory, gravitation is the result of mass and energy distorting the space-time around it; it is also often referred to as Einstein's theory of gravity

tidal force *difference* between the gravitational force at the center of a body and that at any other location on the body; the tidal force stretches the body

universal gravitational constant constant representing the strength of the gravitational force, that is believed to be the same throughout the universe

Key Equations

Newton's law of gravitation	$\vec{\mathbf{F}}_{12} = G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}}_{12}$
Acceleration due to gravity at the surface of Earth	$g = G\frac{M_E}{r^2}$

Gravitational potential energy beyond Earth	$U = -\frac{GM_E m}{r}$
Conservation of energy	$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$
Escape velocity	$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$
Orbital speed	$v_{\text{orbit}} = \sqrt{\frac{\text{GM}_E}{r}}$
Orbital period	$T = 2\pi\sqrt{\frac{r^3}{\text{GM}_E}}$
Energy in circular orbit	$E = K + U = -\frac{Gm\text{M}_E}{2r}$
Conic sections	$\frac{\alpha}{r} = 1 + e\cos\theta$
Kepler’s third law	$T^2 = \frac{4\pi^2}{GM}a^3$
Schwarzschild radius	$R_S = \frac{2GM}{c^2}$

Summary

13.1 Newton's Law of Universal Gravitation

- All masses attract one another with a gravitational force proportional to their masses and inversely proportional to the square of the distance between them.
- Spherically symmetrical masses can be treated as if all their mass were located at the center.
- Nonsymmetrical objects can be treated as if their mass were concentrated at their center of mass, provided their distance from other masses is large compared to their size.

13.2 Gravitation Near Earth's Surface

- The weight of an object is the gravitational attraction between Earth and the object.
- The gravitational field is represented as lines that indicate the direction of the gravitational force; the line spacing indicates the strength of the field.
- Apparent weight differs from actual weight due to the acceleration of the object.

13.3 Gravitational Potential Energy and Total Energy

- The acceleration due to gravity changes as we move away from Earth, and the expression for gravitational potential energy must reflect this change.
- The total energy of a system is the sum of kinetic and gravitational potential energy, and this total energy is conserved in orbital motion.
- Objects must have a minimum velocity, the escape velocity, to leave a planet and not return.
- Objects with total energy less than zero are bound; those with zero or greater are unbounded.

13.4 Satellite Orbits and Energy

- Orbital velocities are determined by the mass of the body being orbited and the distance from the center of that body, and not by the mass of a much smaller orbiting object.
- The period of the orbit is likewise independent of the orbiting object’s mass.
- Bodies of comparable masses orbit about their common center of mass and their velocities and periods should be determined from Newton’s second law and law of gravitation.

13.5 Kepler's Laws of Planetary Motion

- All orbital motion follows the path of a conic section. Bound or closed orbits are either a circle or an ellipse; unbounded or open orbits are either a parabola or a hyperbola.
- The areal velocity of any orbit is constant, a reflection of the conservation of angular momentum.
- The square of the period of an elliptical orbit is proportional to the cube of the semi-major axis of that orbit.

13.6 Tidal Forces

- Earth’s tides are caused by the difference in gravitational forces from the Moon and the Sun on the different sides of Earth.
- Spring or neap (high) tides occur when Earth, the Moon, and the Sun are aligned, and neap or (low) tides occur when they form a right triangle.
- Tidal forces can create internal heating, changes in orbital motion, and even destruction of orbiting bodies.

13.7 Einstein's Theory of Gravity

- According to the theory of general relativity, gravity is the result of distortions in space-time created by mass and energy.
- The principle of equivalence states that that both mass and acceleration distort space-time and are indistinguishable in comparable circumstances.
- Black holes, the result of gravitational collapse, are singularities with an event horizon that is proportional to their mass.
- Evidence for the existence of black holes is still circumstantial, but the amount of that evidence is overwhelming.

Conceptual Questions

13.1 Newton's Law of Universal Gravitation

1. Action at a distance, such as is the case for gravity, was once thought to be illogical and therefore untrue. What is the ultimate determinant of the truth in science, and why was this action at a distance ultimately accepted?

2. In the law of universal gravitation, Newton assumed that the force was proportional to the product of the two masses ($\sim m_1 m_2$). While all scientific conjectures must be experimentally verified, can you provide arguments as to why this must be? (You may wish to consider simple examples in which any other form would lead to contradictory results.)

13.2 Gravitation Near Earth's Surface

3. Must engineers take Earth's rotation into account when constructing very tall buildings at any location other than the equator or very near the poles?

13.3 Gravitational Potential Energy and Total Energy

4. It was stated that a satellite with negative total energy is in a bound orbit, whereas one with zero or positive total energy is in an unbounded orbit. Why is this true? What choice for gravitational potential energy was made such that this is true?

5. It was shown that the energy required to lift a satellite into a *low* Earth orbit (the change in potential energy) is only a small fraction of the kinetic energy needed to keep it in orbit. Is this true for larger orbits? Is there a trend to the ratio of kinetic energy to change in potential energy as the size of the orbit increases?

13.4 Satellite Orbits and Energy

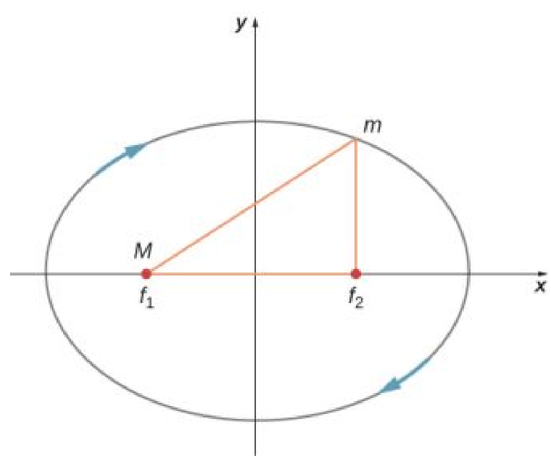
6. One student argues that a satellite in orbit is in free fall because the satellite keeps falling toward Earth. Another says a satellite in orbit is not in free fall because the acceleration due to gravity is not 9.80 m/s^2. With whom do you agree with and why?

7. Many satellites are placed in geosynchronous orbits. What is special about these orbits? For a global communication network, how many of these satellites would be needed?

13.5 Kepler's Laws of Planetary Motion

8. Are Kepler's laws purely descriptive, or do they contain causal information?

9. In the diagram below for a satellite in an elliptical orbit about a much larger mass, indicate where its speed is the greatest and where it is the least. What conservation law dictates this behavior? Indicate the directions of the force, acceleration, and velocity at these points. Draw vectors for these same three quantities at the two points where the *y*-axis intersects (along the semi-minor axis) and from this determine whether the speed is increasing decreasing, or at a max/min.



13.6 Tidal Forces

10. As an object falls into a black hole, tidal forces increase. Will these tidal forces always tear the object apart as it approaches the Schwarzschild radius? How does the mass of the black hole and size of the object affect your answer?

13.7 Einstein's Theory of Gravity

11. The principle of equivalence states that all experiments done in a lab in a uniform gravitational field cannot be distinguished from those done in a lab that is not in a gravitational field but is uniformly accelerating. For the latter case, consider what happens to a laser beam at some height shot perfectly horizontally to the floor, across the accelerating lab. (View this from a nonaccelerating frame outside the lab.) Relative to the height of the laser, where will the laser beam hit the far wall? What does this say about the effect of a gravitational field on light? Does the fact that light has no mass make any difference to the argument?

12. As a person approaches the Schwarzschild radius of a black hole, outside observers see all the processes of that person (their clocks, their heart rate, etc.) slowing down, and coming to a halt as they reach the Schwarzschild radius. (The person falling into the black hole sees their own processes unaffected.) But the speed of light is the same everywhere for all observers. What does this say about space as you approach the black hole?

Problems

13.1 Newton's Law of Universal Gravitation

13. Evaluate the magnitude of gravitational force between two 5-kg spherical steel balls separated by a center-to-center distance of 15 cm.

14. Estimate the gravitational force between two sumo wrestlers, with masses 220 kg and 240 kg, when they are embraced and their centers are 1.2 m apart.

15. Astrology makes much of the position of the planets at the moment of one's birth. The only known force a planet exerts on Earth is gravitational. (a) Calculate the gravitational force exerted on a 4.20-kg baby by a 100-kg father 0.200 m away at birth (he is assisting, so he is close to the child). (b) Calculate the force on the baby due to Jupiter if it is at its closest distance to Earth, some 6.29×10^{11} m away. How does the force of Jupiter on the baby compare to the force of the father on the baby? Other objects in the room and the hospital building also exert similar gravitational forces. (Of course, there could be an unknown force acting, but scientists first need to be convinced that there is even an effect, much less that an unknown force causes it.)

16. A mountain 10.0 km from a person exerts a gravitational force on the person equal to 2.00% of their weight. (a) Calculate the mass of the mountain. (b) Compare the mountain's mass with that of Earth. (c) What is unreasonable about these results? (d) Which premises are unreasonable or inconsistent? (Note that accurate gravitational measurements can easily detect the effect of nearby mountains and variations in local geology.)

17. The International Space Station has a mass of approximately 370,000 kg. (a) What is the force on a 150-kg suited astronaut if she is 20 m from the center of mass of the station? (b) How accurate do you think your answer would be?

FIGURE 13.33 (credit: ©ESA–David Ducros)

18. Asteroid Toutatis passed near Earth in 2006 at four times the distance to our Moon. This was the closest approach we will have until 2060. If it has mass of 5.0×10^{13} kg, what force did it exert on Earth at its closest approach?

19. (a) What was the acceleration of Earth caused by asteroid Toutatis (see previous problem) at its closest approach? (b) What was the acceleration of Toutatis at this point?

13.2 Gravitation Near Earth's Surface

20. (a) Calculate Earth's mass given the acceleration due to gravity at the North Pole is measured to be $9.832\ \mathrm{m/s^2}$ and the radius of the Earth at the pole is 6356 km. (b) Compare this with the NASA's Earth Fact Sheet value of

5.9726×10^{24} kg.

21. (a) What is the acceleration due to gravity on the surface of the Moon? (b) On the surface of Mars? The mass of Mars is 6.418×10^{23} kg and its radius is 3.38×10^{6} m.

22. (a) Calculate the acceleration due to gravity on the surface of the Sun. (b) By what factor would your weight increase if you could stand on the Sun? (Never mind that you cannot.)

23. The mass of a particle is 15 kg. (a) What is its weight on Earth? (b) What is its weight on the Moon? (c) What is its mass on the Moon? (d) What is its weight in outer space far from any celestial body? (e) What is its mass at this point?

24. On a planet whose radius is 1.2×10^{7} m, the acceleration due to gravity is 18 m/s^2. What is the mass of the planet?

25. The mean diameter of the planet Saturn is 1.2×10^{8} m, and its mean mass density is 0.69 g/cm^3. Find the acceleration due to gravity at Saturn's surface.

26. The mean diameter of the planet Mercury is 4.88×10^{6} m, and the acceleration due to gravity at its surface is 3.78 m/s^2. Estimate the mass of this planet.

27. The acceleration due to gravity on the surface of a planet is three times as large as it is on the surface of Earth. The mass density of the planet is known to be twice that of Earth. What is the radius of this planet in terms of Earth's radius?

28. A body on the surface of a planet with the same radius as Earth's weighs 10 times more than it does on Earth. What is the mass of this planet in terms of Earth's mass?

13.3 Gravitational Potential Energy and Total Energy

29. Find the escape speed of a projectile from the surface of Mars.

30. Find the escape speed of a projectile from the surface of Jupiter.

31. What is the escape speed of a satellite located at the Moon's orbit about Earth? Assume the Moon is not nearby.

32. (a) Evaluate the gravitational potential energy between two 5.00-kg spherical steel balls separated by a center-to-center distance of 15.0 cm. (b) Assuming that they are both initially at rest relative to each other in deep space, use conservation of energy to find how fast will they be traveling upon impact. Each sphere has a radius of 5.10 cm.

33. An average-sized asteroid located 5.0×10^{7} km from Earth with mass 2.0×10^{13} kg is detected headed directly toward Earth with speed of 2.0 km/s. What will its speed be just before it hits our atmosphere? (You may ignore the size of the asteroid.)

34. (a) What will be the kinetic energy of the asteroid in the previous problem just before it hits Earth? b) Compare this energy to the output of the largest fission bomb, 2100 TJ. What impact would this have on Earth?

35. (a) What is the change in energy of a 1000-kg payload taken from rest at the surface of Earth and placed at rest on the surface of the Moon? (b) What would be the answer if the payload were taken from the Moon's surface to Earth? Is this a reasonable calculation of the energy needed to move a payload back and forth?

13.4 Satellite Orbits and Energy

36. If a planet with 1.5 times the mass of Earth was traveling in Earth's orbit, what would its period be?

37. Two planets in circular orbits around a star have speeds of *v* and 2*v*. (a) What is the ratio of the orbital radii of the planets? (b) What is the ratio of their periods?

38. Using the average distance of Earth from the Sun, and the orbital period of Earth, (a) find the centripetal acceleration of Earth in its motion about the Sun. (b) Compare this value to that of the centripetal acceleration at the equator due to Earth's rotation.

39. (a) What is the orbital radius of an Earth satellite having a period of 1.00 h? (b) What is unreasonable about this result?

40. Calculate the mass of the Sun based on data for Earth's orbit and compare the value obtained with the Sun's actual mass.

41. Find the mass of Jupiter based on the fact that Io, its innermost moon, has an average orbital radius of 421,700 km and a period of 1.77 days.

42. Astronomical observations of light from the Milky Way galaxy's stars indicate that it has a mass of about 8.0×10^{11} solar masses. A star

orbiting on the galaxy's periphery is about 6.0×10^4 light-years from its center. (a) What should the orbital period of that star be? (b) If its period is 6.0×10^7 years instead, what is the mass of the galaxy? Such calculations are used to imply the existence of mass that does not emit any light. Scientists do not know what it is comprised of, so this mass is known simply as "dark matter" and remains an enduring astronomical mystery.

43. (a) In order to keep a small satellite from drifting into a nearby asteroid, it is placed in orbit with a period of 3.02 hours and radius of 2.0 km. What is the mass of the asteroid? (b) Does this mass seem reasonable for the size of the orbit?

44. The Moon and Earth rotate about their common center of mass, which is located about 4700 km from the center of Earth. (This is 1690 km below the surface.) (a) Calculate the acceleration due to the Moon's gravity at that point. (b) Calculate the centripetal acceleration of the center of Earth as it rotates about that point once each lunar month (about 27.3 d) and compare it with the acceleration found in part (a). Comment on whether or not they are equal and why they should or should not be.

45. The Sun orbits the Milky Way galaxy once each 2.60×10^8 years, with a roughly circular orbit averaging a radius of 3.00×10^4 light-years. (A light-year is the distance traveled by light in 1 year.) Calculate the centripetal acceleration of the Sun in its galactic orbit. Does your result support the contention that a nearly inertial frame of reference can be located at the Sun? (b) Calculate the average speed of the Sun in its galactic orbit. Does the answer surprise you?

46. A geosynchronous Earth satellite is one that has an orbital period of precisely 1 day. Such orbits are useful for communication and weather observation because the satellite remains above the same point on Earth (provided it orbits in the equatorial plane in the same direction as Earth's rotation). Calculate the radius of such an orbit based on the data for Earth in Appendix D.

13.5 Kepler's Laws of Planetary Motion

47. Calculate the mass of the Sun based on data for average Earth's orbit and compare the value obtained with the Sun's commonly listed value of 1.989×10^{30} kg.

48. Io orbits Jupiter with an average radius of 421,700 km and a period of 1.769 days. Based upon these data, what is the mass of Jupiter?

49. The "mean" orbital radius listed for astronomical objects orbiting the Sun is typically not an integrated average but is calculated such that it gives the correct period when applied to the equation for circular orbits. Given that, what is the mean orbital radius in terms of aphelion and perihelion?

50. The perihelion of Halley's comet is 0.586 AU and the aphelion is 17.8 AU. Given that its speed at perihelion is 55 km/s, what is the speed at aphelion ($1\ \text{AU} = 1.496 \times 10^{11}$ m)? (*Hint:* You may use either conservation of energy or angular momentum, but the latter is much easier.)

51. The perihelion of the comet Lagerkvist is 2.61 AU and it has a period of 7.36 years. Show that the aphelion for this comet is 4.95 AU.

52. What is the ratio of the speed at perihelion to that at aphelion for the comet Lagerkvist in the previous problem?

53. Eros has an elliptical orbit about the Sun, with a perihelion distance of 1.13 AU and aphelion distance of 1.78 AU. What is the period of its orbit?

13.6 Tidal Forces

54. (a) What is the difference between the forces on a 1.0-kg mass on the near side of Io and far side due to Jupiter? Io has a mean radius of 1821 km and a mean orbital radius about Jupiter of 421,700 km. (b) Compare this difference to that calculated for the difference for Earth due to the Moon calculated in Example 13.14. Tidal forces are the cause of Io's volcanic activity.

55. If the Sun were to collapse into a black hole, the point of no return for an investigator would be approximately 3 km from the center singularity. Would the investigator be able to survive visiting even 300 km from the center? Answer this by finding the difference in the gravitational attraction the black holes exerts on a 1.0-kg mass at the head and at the feet of the investigator.

56. Consider Figure 13.23 in Tidal Forces. This diagram represents the tidal forces for spring tides. Sketch a similar diagram for neap tides. (*Hint:* For simplicity, imagine that the Sun and the Moon contribute equally. Your diagram would be the vector sum of two force fields (as in Figure 13.23), reduced by a factor of two, and superimposed at right angles.)

13.7 Einstein's Theory of Gravity

57. What is the Schwarzschild radius for the black hole at the center of our galaxy if it has the mass of 4 million solar masses?

58. What would be the Schwarzschild radius, in light years, if our Milky Way galaxy of 100 billion stars collapsed into a black hole? Compare this to our distance from the center, about 13,000 light years.

Additional Problems

59. A neutron star is a cold, collapsed star with nuclear density. A particular neutron star has a mass twice that of our Sun with a radius of 12.0 km. (a) What would be the weight of a 100-kg astronaut on standing on its surface? (b) What does this tell us about landing on a neutron star?

60. (a) How far from the center of Earth would the net gravitational force of Earth and the Moon on an object be zero? (b) Setting the *magnitudes* of the forces equal should result in two answers from the quadratic. Do you understand why there are two positions, but only one where the net force is zero?

61. How far from the center of the Sun would the net gravitational force of Earth and the Sun on a spaceship be zero?

62. Calculate the values of *g* at Earth's surface for the following changes in Earth's properties: (a) its mass is doubled and its radius is halved; (b) its mass density is doubled and its radius is unchanged; (c) its mass density is halved and its mass is unchanged.

63. Suppose you can communicate with the inhabitants of a planet in another solar system. They tell you that on their planet, whose diameter and mass are 5.0×10^3 km and 3.6×10^{23} kg, respectively, the record for the high jump is 2.0 m. Given that this record is close to 2.4 m on Earth, what would you conclude about your extraterrestrial friends' jumping ability?

64. (a) Suppose that your measured weight at the equator is one-half your measured weight at the pole on a planet whose mass and diameter are equal to those of Earth. What is the rotational period of the planet? (b) Would you need to take the shape of this planet into account?

65. A body of mass 100 kg is weighed at the North Pole and at the equator with a spring scale. What is the scale reading at these two points? Assume that $g = 9.83\ \text{m/s}^2$ at the pole.

66. Find the speed needed to escape from the solar system starting from the surface of Earth. Assume there are no other bodies involved and do not account for the fact that Earth is moving in its orbit. [*Hint:* Equation 13.6 does not apply. Use Equation 13.5 and include the potential energy of both Earth and the Sun.

67. Consider the previous problem and include the fact that Earth has an orbital speed about the Sun of 29.8 km/s. (a) What speed relative to Earth would be needed and in what direction should you leave Earth? (b) What will be the shape of the trajectory?

68. A comet is observed 1.50 AU from the Sun with a speed of 24.3 km/s. Is this comet in a bound or unbound orbit?

69. An asteroid has speed 15.5 km/s when it is located 2.00 AU from the sun. At its closest approach, it is 0.400 AU from the Sun. What is its speed at that point?

70. Space debris left from old satellites and their launchers is becoming a hazard to other satellites. (a) Calculate the speed of a satellite in an orbit 900 km above Earth's surface. (b) Suppose a loose rivet is in an orbit of the same radius that intersects the satellite's orbit at an angle of $90°$. What is the velocity of the rivet relative to the satellite just before striking it? (c) If its mass is 0.500 g, and it comes to rest inside the satellite, how much energy in joules is generated by the collision? (Assume the satellite's velocity does not change appreciably, because its mass is much greater than the rivet's.)

71. A satellite of mass 1000 kg is in circular orbit about Earth. The radius of the orbit of the satellite is equal to two times the radius of Earth. (a) How far away is the satellite? (b) Find the kinetic, potential, and total energies of the satellite.

72. After Ceres was promoted to a dwarf planet, we now recognize the largest known asteroid to be Vesta, with a mass of 2.67×10^{20} kg and a diameter ranging from 458 km to 578 km. Assuming that Vesta is spherical with radius 520 km, find the approximate escape velocity from its surface.

73. (a) Given the asteroid Vesta which has a diameter of 520 km and mass of 2.67×10^{20} kg, what would be the orbital period for a space probe in a circular orbit of 10.0 km from its surface? (b) Why is this calculation marginally useful at best?

74. What is the orbital velocity of our solar system about the center of the Milky Way? Assume that the mass within a sphere of radius equal to our distance away from the center is about a 100 billion solar masses. Our distance from the center is 27,000 light years.

75. (a) Using the information in the previous problem, what velocity do you need to escape the Milky Way galaxy from our present position? (b) Would you need to accelerate a spaceship to this speed relative to Earth?

76. Circular orbits in Equation 13.10 for conic sections must have eccentricity zero. From this, and using Newton's second law applied to centripetal acceleration, show that the value of α in Equation 13.10 is given by $\alpha = \frac{L^2}{GMm^2}$ where *L* is the angular momentum of the orbiting body. The value of α is constant and given by this expression regardless of the type of orbit.

77. Show that for eccentricity equal to one in Equation 13.10 for conic sections, the path is a parabola. Do this by substituting Cartesian coordinates, *x* and *y*, for the polar coordinates, *r* and θ, and showing that it has the general form for a parabola, $x = ay^2 + by + c$.

78. Using the technique shown in Satellite Orbits and Energy, show that two masses m_1 and m_2 in circular orbits about their common center of mass, will have total energy $E = K + E = K_1 + K_2 - \frac{Gm_1m_2}{r} = -\frac{Gm_1m_2}{2r}$. We have shown the kinetic energy of both masses explicitly. (*Hint:* The masses orbit at radii r_1 and r_2, respectively, where $r = r_1 + r_2$. Be sure not to confuse the radius needed for centripetal acceleration with that for the gravitational force.)

79. Given the perihelion distance, *p*, and aphelion distance, *q*, for an elliptical orbit, show that the velocity at perihelion, v_p, is given by $v_p = \sqrt{\frac{2GM_{\text{Sun}}}{(q+p)} \frac{q}{p}}$. (*Hint:* Use conservation of angular momentum to relate v_p and v_q, and then substitute into the conservation of energy equation.)

80. Comet P/1999 R1 has a perihelion of 0.0570 AU and aphelion of 4.99 AU. Using the results of the previous problem, find its speed at aphelion. (*Hint:* The expression is for the perihelion. Use symmetry to rewrite the expression for aphelion.)

Challenge Problems

81. A tunnel is dug through the center of a perfectly spherical and airless planet of radius *R*. Using the expression for *g* derived in Gravitation Near Earth's Surface for a uniform density, show that a particle of mass *m* dropped in the tunnel will execute simple harmonic motion. Deduce the period of oscillation of *m* and show that it has the same period as an orbit at the surface.

82. Following the technique used in Gravitation Near Earth's Surface, find the value of *g* as a function of the radius *r* from the center of a spherical shell planet of constant density ρ with inner and outer radii R_{in} and R_{out}. Find *g* for both $R_{\text{in}} < r < R_{\text{out}}$ and for $r < R_{\text{in}}$. Assuming the inside of the shell is kept airless, describe travel inside the spherical shell planet.

83. Show that the areal velocity for a circular orbit of radius *r* about a mass *M* is $\frac{\Delta A}{\Delta t} = \frac{1}{2}\sqrt{GMr}$. Does your expression give the correct value for Earth's areal velocity about the Sun?

84. Show that the period of orbit for two masses, m_1 and m_2, in circular orbits of radii r_1 and r_2, respectively, about their common center-of-mass, is given by $T = 2\pi\sqrt{\frac{r^3}{G(m_1+m_2)}}$ where $r = r_1 + r_2$. (*Hint:* The masses orbit at radii r_1 and r_2, respectively where $r = r_1 + r_2$. Use the expression for the center-of-mass to relate the two radii and note that the two masses must have equal but opposite momenta. Start with the relationship of the period to the circumference and speed of orbit for one of the masses. Use the result of the previous problem using momenta in the expressions for the kinetic energy.)

85. Show that for small changes in height *h*, such that $h << R_E$, Equation 13.4 reduces to the expression $\Delta U = mgh$.

86. Using Figure 13.9, carefully sketch a free body diagram for the case of a simple pendulum hanging at latitude lambda, labeling all forces acting on the point mass, *m*. Set up the equations of motion for equilibrium, setting one coordinate in the direction of the centripetal acceleration (toward *P* in the diagram), the other perpendicular to that. Show that the deflection angle ε, defined as the angle between the pendulum string and the radial direction toward the center of Earth, is given by the expression below. What is the deflection angle at latitude 45 degrees? Assume that Earth is a perfect sphere. $\tan(\lambda + \varepsilon) = \frac{g}{(g - \omega^2 R_E)} \tan\lambda$, where ω is the angular velocity of Earth.

87. (a) Show that tidal force on a small object of mass *m*, defined as the *difference* in the gravitational force that would be exerted on *m* at a distance at the near and the far side of the object, due to the gravitation at a distance *R* from *M*, is given by $F_{\text{tidal}} = \frac{2GMm}{R^3} \Delta r$ where Δr is the distance between the near and far side and $\Delta r << R$. (b) Assume you are falling feet first into the black hole at the center of our galaxy. It has mass of 4 million solar masses. What would be the difference between the force at your head and your feet at the Schwarzschild radius (event horizon)? Assume your feet and head each have mass 5.0 kg and are 2.0 m apart. Would you survive passing through the event horizon?

88. Find the Hohmann transfer velocities, $\Delta v_{\text{EllipseEarth}}$ and $\Delta v_{\text{EllipseMars}}$, needed for a trip to Mars. Use Equation 13.7 to find the circular orbital velocities for Earth and Mars. Using Equation 13.4 and the total energy of the ellipse (with semi-major axis *a*), given by $E = -\frac{GmM_S}{2a}$, find the velocities at Earth (perihelion) and at Mars (aphelion) required to be on the transfer ellipse. The difference, Δv, at each point is the velocity boost or transfer velocity needed.

CHAPTER 14
Fluid Mechanics

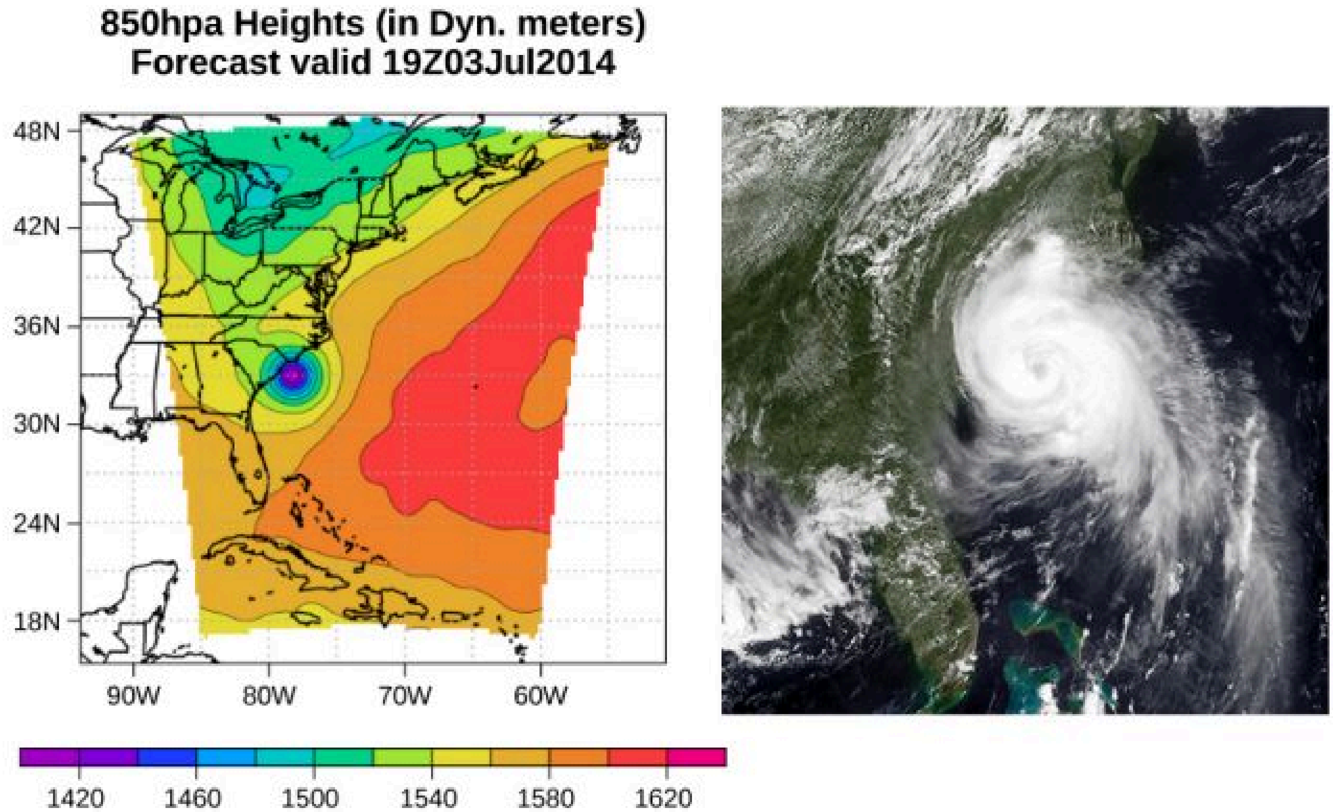


FIGURE 14.1 This pressure map (left) and satellite photo (right) were used to model the path and impact of Hurricane Arthur as it traveled up the East Coast of the United States in July 2014. Computer models use force and energy equations to predict developing weather patterns. Scientists numerically integrate these time-dependent equations, along with the energy budgets of long- and short-wave solar energy, to model changes in the atmosphere. The pressure map on the left was created using the Weather Research and Forecasting Model designed at the National Center for Atmospheric Research. The colors represent the height of the 850-mbar pressure surface. (credit left: modification of work by The National Center for Atmospheric Research; credit right: modification of work by NRL Monterey Marine Meteorology Division, The National Oceanic and Atmospheric Administration)

CHAPTER OUTLINE

INTRODUCTION Picture yourself walking along a beach on the eastern shore of the United States. The air smells of sea salt and the sun warms your body. Suddenly, an alert appears on your cell phone. A tropical depression has formed into a hurricane. Atmospheric pressure has fallen to nearly 15% below average. As a result, forecasters expect torrential rainfall, winds in excess of 100 mph, and millions of dollars in damage. As you prepare to evacuate, you wonder: How can such a small drop in pressure lead to such a severe change in the weather?

Pressure is a physical phenomenon that is responsible for much more than just the weather. Changes in pressure cause ears to “pop” during takeoff in an airplane. Changes in pressure can also cause scuba divers to suffer a sometimes fatal disorder known as the “bends,” which occurs when nitrogen dissolved in the water of the body at extreme depths returns to a gaseous state in the body as the diver surfaces. Pressure lies at the heart of the phenomena called buoyancy, which causes hot air balloons to rise and ships to float. Before we can fully understand the role that pressure plays in these phenomena, we need to discuss the states of matter and the concept of density.

14.1 Fluids, Density, and Pressure

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- State the different phases of matter
- Describe the characteristics of the phases of matter at the molecular or atomic level
- Distinguish between compressible and incompressible materials
- Define density and its related SI units
- Compare and contrast the densities of various substances
- Define pressure and its related SI units
- Explain the relationship between pressure and force
- Calculate force given pressure and area

Matter most commonly exists as a solid, liquid, or gas; these states are known as the three common phases of matter. We will look at each of these phases in detail in this section.

Characteristics of Solids

Solids are rigid and have specific shapes and definite volumes. The atoms or molecules in a solid are in close proximity to each other, and there is a significant force between these molecules. Solids will take a form determined by the nature of these forces between the molecules. Although true solids are not incompressible, it nevertheless requires a large force to change the shape of a solid. In some cases, the force between molecules can cause the molecules to organize into a lattice as shown in Figure 14.2. The structure of this three-dimensional lattice is represented as molecules connected by rigid bonds (modeled as stiff springs), which allow limited freedom for movement. Even a large force produces only small displacements in the atoms or molecules of the lattice, and the solid maintains its shape. Solids also resist shearing forces. (Shearing forces are forces applied tangentially to a surface, as described in Static Equilibrium and Elasticity.)

Characteristics of Fluids

Liquids and gases are considered to be **fluids** because they yield to shearing forces, whereas solids resist them. Like solids, the molecules in a liquid are bonded to neighboring molecules, but possess many fewer of these bonds. The molecules in a liquid are not locked in place and can move with respect to each other. The distance between molecules is similar to the distances in a solid, and so liquids have definite volumes, but the shape of a liquid changes, depending on the shape of its container. Gases are not bonded to neighboring atoms and can have large separations between molecules. Gases have neither specific shapes nor definite volumes, since their molecules move to fill the container in which they are held (Figure 14.2).

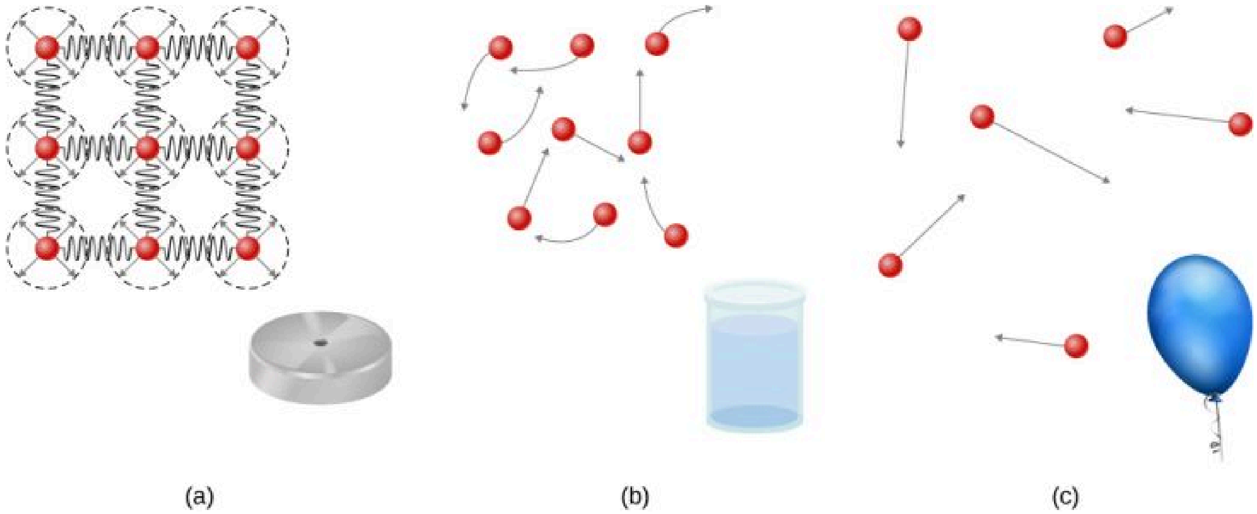


FIGURE 14.2 (a) Atoms in a solid are always in close contact with neighboring atoms, held in place by forces represented here by springs. (b) Atoms in a liquid are also in close contact but can slide over one another. Forces between the atoms strongly resist attempts to compress the atoms. (c) Atoms in a gas move about freely and are separated by large distances. A gas must be held in a closed container to prevent it from expanding freely and escaping.

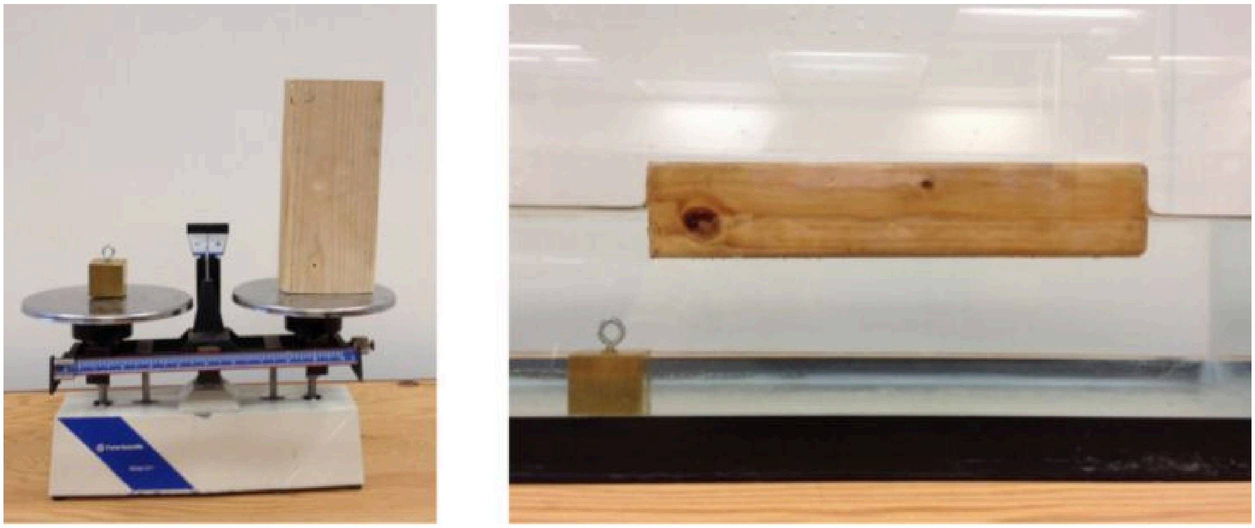
Liquids deform easily when stressed and do not spring back to their original shape once a force is removed. This occurs because the atoms or molecules in a liquid are free to slide about and change neighbors. That is, liquids flow (so they are a type of fluid), with the molecules held together by mutual attraction. When a liquid is placed in a container with no lid, it remains in the container. Because the atoms are closely packed, liquids, like solids, resist compression; an extremely large force is necessary to change the volume of a liquid.

In contrast, atoms in gases are separated by large distances, and the forces between atoms in a gas are therefore very weak, except when the atoms collide with one another. This makes gases relatively easy to compress and allows them to flow (which makes them fluids). When placed in an open container, gases, unlike liquids, will escape.

In this chapter, we generally refer to both gases and liquids simply as fluids, making a distinction between them only when they behave differently. There exists one other phase of matter, plasma, which exists at very high temperatures. At high temperatures, molecules may disassociate into atoms, and atoms disassociate into electrons (with negative charges) and protons (with positive charges), forming a plasma. Plasma will not be discussed in depth in this chapter because plasma has very different properties from the three other common phases of matter, discussed in this chapter, due to the strong electrical forces between the charges.

Density

Suppose a block of brass and a block of wood have exactly the same mass. If both blocks are dropped in a tank of water, why does the wood float and the brass sink (Figure 14.3)? This occurs because the brass has a greater density than water, whereas the wood has a lower density than water.



(a) (b)

FIGURE 14.3 (a) A block of brass and a block of wood both have the same weight and mass, but the block of wood has a much greater volume. (b) When placed in a fish tank filled with water, the cube of brass sinks and the block of wood floats. (The block of wood is the same in both pictures; it was turned on its side to fit on the scale.) (credit: modification of works by Joseph J. Trout, Stockton University)

Density is an important characteristic of substances. It is crucial, for example, in determining whether an object sinks or floats in a fluid.

DENSITY

The average density of a substance or object is defined as its mass per unit volume,

$$\rho = \frac{m}{V} \tag{14.1}$$

where the Greek letter ρ (rho) is the symbol for density, m is the mass, and V is the volume.

The SI unit of density is kg/m^3. Table 14.1 lists some representative values. The cgs unit of density is the gram per cubic centimeter, g/cm^3, where

$$1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3.$$

The metric system was originally devised so that water would have a density of 1 g/cm^3, equivalent to 10^3 kg/m^3. Thus, the basic mass unit, the kilogram, was first devised to be the mass of 1000 mL of water, which has a volume of 1000 cm^3.

Solids ($0.0°\text{C}$)		Liquids ($0.0°\text{C}$)		Gases ($0.0°\text{C}$, **101.3 kPa**)	
Substance	$\rho(\text{kg/m}^3)$	**Substance**	$\rho(\text{kg/m}^3)$	**Substance**	$\rho(\text{kg/m}^3)$
Aluminum	2.70×10^3	Benzene	8.79×10^2	Air	1.29×10^0
Bone	1.90×10^3	Blood	1.05×10^3	Carbon dioxide	1.98×10^0
Brass	8.44×10^3	Ethyl alcohol	8.06×10^2	Carbon monoxide	1.25×10^0
Concrete	2.40×10^3	Gasoline	6.80×10^2	Helium	1.80×10^{-1}

TABLE 14.1 **Densities of Some Common Substances**

Solids ($0.0°C$)		Liquids ($0.0°C$)		Gases ($0.0°C$, 101.3 kPa)	
Copper	8.92×10^3	Glycerin	1.26×10^3	Hydrogen	9.00×10^{-2}
Cork	2.40×10^2	Mercury	1.36×10^4	Methane	7.20×10^{-1}
Earth's crust	3.30×10^3	Olive oil	9.20×10^2	Nitrogen	1.25×10^0
Glass	2.60×10^3			Nitrous oxide	1.98×10^0
Gold	1.93×10^4			Oxygen	1.43×10^0
Granite	2.70×10^3				
Iron	7.86×10^3				
Lead	1.13×10^4				
Oak	7.10×10^2				
Pine	3.73×10^2				
Platinum	2.14×10^4				
Polystyrene	1.00×10^2				
Tungsten	1.93×10^4				
Uranium	1.87×10^4				

TABLE 14.1 **Densities of Some Common Substances**

As you can see by examining Table 14.1, the density of an object may help identify its composition. The density of gold, for example, is about 2.5 times the density of iron, which is about 2.5 times the density of aluminum. Density also reveals something about the phase of the matter and its substructure. Notice that the densities of liquids and solids are roughly comparable, consistent with the fact that their atoms are in close contact. The densities of gases are much less than those of liquids and solids, because the atoms in gases are separated by large amounts of empty space. The gases are displayed for a standard temperature of $0.0°C$ and a standard pressure of 101.3 kPa, and there is a strong dependence of the densities on temperature and pressure. The densities of the solids and liquids displayed are given for the standard temperature of $0.0°C$ and the densities of solids and liquids depend on the temperature. The density of solids and liquids normally increase with decreasing temperature.

Table 14.2 shows the density of water in various phases and temperature. The density of water increases with decreasing temperature, reaching a maximum at $4.0°C$, and then decreases as the temperature falls below $4.0°C$. This behavior of the density of water explains why ice forms at the top of a body of water.

Substance	$\rho (kg/m^3)$
Ice ($0°C$)	9.17×10^2
Water ($0°C$)	9.998×10^2

TABLE 14.2 **Densities of Water**

Substance	$\rho(\text{kg/m}^3)$
Water (4°C)	1.000×10^3
Water (20°C)	9.982×10^2
Water (100°C)	9.584×10^2
Steam (100°C, 101.3 kPa)	1.670×10^2
Sea water (0°C)	1.030×10^3

TABLE 14.2 **Densities of Water**

The density of a substance is not necessarily constant throughout the volume of a substance. If the density is constant throughout a substance, the substance is said to be a homogeneous substance. A solid iron bar is an example of a homogeneous substance. The density is constant throughout, and the density of any sample of the substance is the same as its average density. If the density of a substance were not constant, the substance is said to be a heterogeneous substance. A chunk of Swiss cheese is an example of a heterogeneous material containing both the solid cheese and gas-filled voids. The density at a specific location within a heterogeneous material is called *local density*, and is given as a function of location, $\rho = \rho(x, y, z)$ (Figure 14.4).

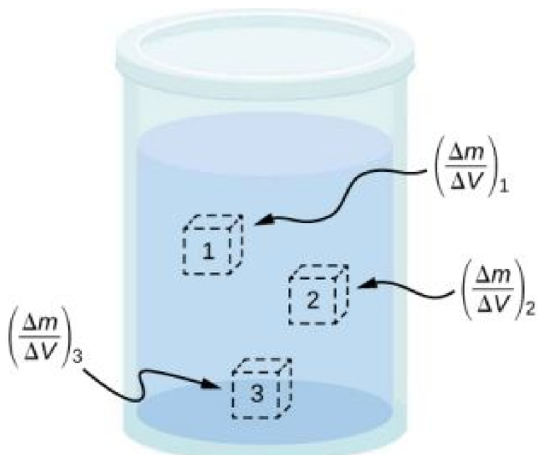


FIGURE 14.4 Density may vary throughout a heterogeneous mixture. Local density at a point is obtained from dividing mass by volume in a small volume around a given point.

Local density can be obtained by a limiting process, based on the average density in a small volume around the point in question, taking the limit where the size of the volume approaches zero,

$$\rho = \lim_{\Delta V \to 0} \frac{\Delta m}{\Delta V} \tag{14.2}$$

where ρ is the density, m is the mass, and V is the volume.

Since gases are free to expand and contract, the densities of the gases vary considerably with temperature, whereas the densities of liquids vary little with temperature. Therefore, the densities of liquids are often treated as constant, with the density equal to the average density.

Density is a dimensional property; therefore, when comparing the densities of two substances, the units must be taken into consideration. For this reason, a more convenient, dimensionless quantity called the **specific gravity** is often used to compare densities. Specific gravity is defined as the ratio of the density of the material to the density of water at $4.0\ °\text{C}$ and one atmosphere of pressure, which is $1000\ \text{kg/m}^3$:

$$\text{Specific gravity} = \frac{\text{Density of material}}{\text{Density of water}}.$$

The comparison uses water because the density of water is 1 g/cm^3, which was originally used to define the kilogram. Specific gravity, being dimensionless, provides a ready comparison among materials without having to worry about the unit of density. For instance, the density of aluminum is 2.7 in g/cm^3 (2700 in kg/m^3), but its specific gravity is 2.7, regardless of the unit of density. Specific gravity is a particularly useful quantity with regard to buoyancy, which we will discuss later in this chapter.

Pressure

You have no doubt heard the word ‘pressure’ used in relation to blood (high or low blood pressure) and in relation to weather (high- and low-pressure weather systems). These are only two of many examples of pressure in fluids. (Recall that we introduced the idea of pressure in Static Equilibrium and Elasticity, in the context of bulk stress and strain.)

PRESSURE

Pressure (p) is defined as the normal force F per unit area A over which the force is applied, or

$$p = \frac{F}{A}. \tag{14.3}$$

To define the pressure at a specific point, the pressure is defined as the force dF exerted by a fluid over an infinitesimal element of area dA containing the point, resulting in $p = \frac{dF}{dA}$.

A given force can have a significantly different effect, depending on the area over which the force is exerted. For instance, a force applied to an area of 1 mm^2 has a pressure that is 100 times as great as the same force applied to an area of 1 cm^2. That is why a sharp needle is able to poke through skin when a small force is exerted, but applying the same force with a finger does not puncture the skin (Figure 14.5).

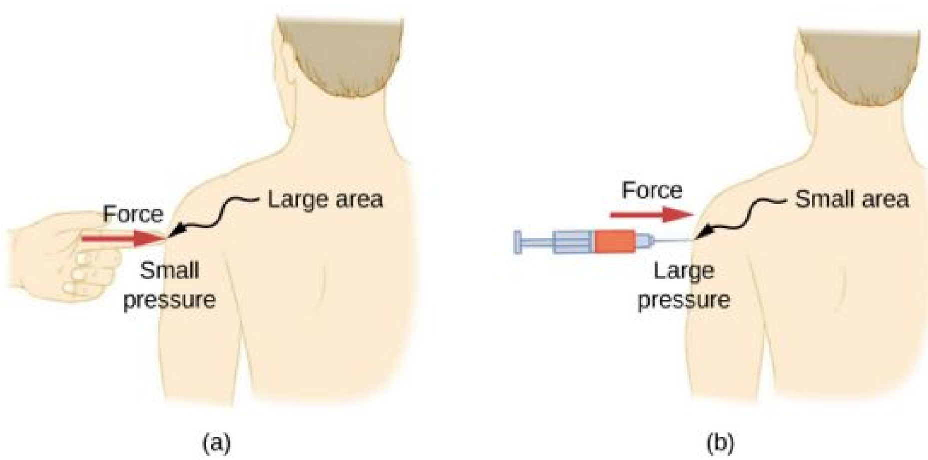


FIGURE 14.5 (a) A person being poked with a finger might be irritated, but the force has little lasting effect. (b) In contrast, the same force applied to an area the size of the sharp end of a needle is enough to break the skin.

Note that although force is a vector, pressure is a scalar. Pressure is a scalar quantity because it is defined to be proportional to the magnitude of the force acting perpendicular to the surface area. The SI unit for pressure is the *pascal* (Pa), named after the French mathematician and physicist Blaise Pascal (1623–1662), where

$$1 \text{ Pa} = 1 \text{ N/m}^2.$$

Several other units are used for pressure, which we discuss later in the chapter.

Variation of pressure with depth in a fluid of constant density

Pressure is defined for all states of matter, but it is particularly important when discussing fluids. An important characteristic of fluids is that there is no significant resistance to the component of a force applied parallel to the surface of a fluid. The molecules of the fluid simply flow to accommodate the horizontal force. A force applied perpendicular to the surface compresses or expands the fluid. If you try to compress a fluid, you find that a reaction force develops at each point inside the fluid in the outward direction, balancing the force applied on the molecules at the boundary.

Consider a fluid of constant density as shown in Figure 14.6. The pressure at the bottom of the container is due to the pressure of the atmosphere (p_0) plus the pressure due to the weight of the fluid. The pressure due to the fluid is equal to the weight of the fluid divided by the area. The weight of the fluid is equal to its mass times the acceleration due to gravity.

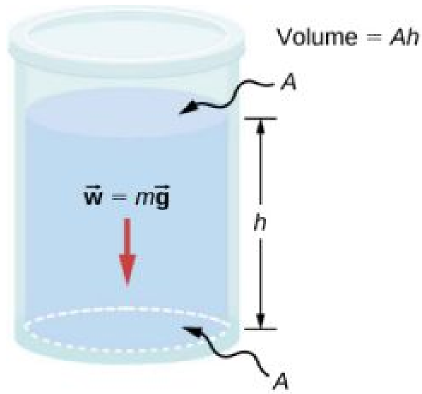


FIGURE 14.6 The bottom of this container supports the entire weight of the fluid in it. The vertical sides cannot exert an upward force on the fluid (since it cannot withstand a shearing force), so the bottom must support it all.

Since the density is constant, the weight can be calculated using the density:

$$w = mg = \rho V g = \rho A h g.$$

The pressure at the bottom of the container is therefore equal to atmospheric pressure added to the weight of the fluid divided by the area:

$$p = p_0 + \frac{\rho A h g}{A} = p_0 + \rho h g.$$

This equation is only good for pressure at a depth for a fluid of constant density.

PRESSURE AT A DEPTH FOR A FLUID OF CONSTANT DENSITY

The pressure at a depth in a fluid of constant density is equal to the pressure of the atmosphere plus the pressure due to the weight of the fluid, or

$$p = p_0 + \rho h g, \tag{14.4}$$

Where p is the pressure at a particular depth, p_0 is the pressure of the atmosphere, ρ is the density of the fluid, g is the acceleration due to gravity, and h is the depth.

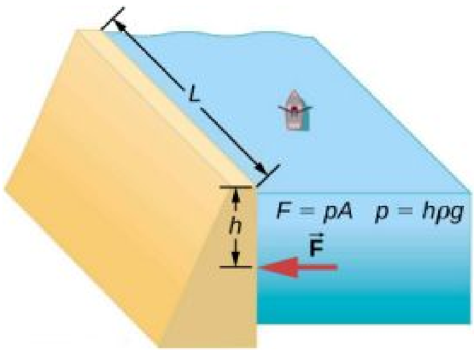
FIGURE 14.7 The Three Gorges Dam, erected on the Yangtze River in central China in 2008, created a massive reservoir that displaced

more than one million people. (credit: modification of work by “Le Grand Portage”/Flickr)

EXAMPLE 14.1

What Force Must a Dam Withstand?

Consider the pressure and force acting on the dam retaining a reservoir of water (Figure 14.7). Suppose the dam is 500-m wide and the water is 80.0-m deep at the dam, as illustrated below. (a) What is the average pressure on the dam due to the water? (b) Calculate the force exerted against the dam.



The average pressure *p* due to the weight of the water is the pressure at the average depth *h* of 40.0 m, since pressure increases linearly with depth. The force exerted on the dam by the water is the average pressure times the area of contact, $F = pA$.

solution

a. The average pressure due to the weight of a fluid is

$$p = h\rho g. \tag{14.5}$$

Entering the density of water from Table 14.2 and taking *h* to be the average depth of 40.0 m, we obtain

$$\begin{aligned} p &= (40.0\,\text{m})\left(10^3\,\frac{\text{kg}}{\text{m}^3}\right)\left(9.80\frac{\text{m}}{\text{s}^2}\right) \\ &= 3.92 \times 10^5\,\frac{\text{N}}{\text{m}^2} = 392\,\text{kPa}. \end{aligned}$$

b. We have already found the value for *p*. The area of the dam is

$$A = 80.0\,\text{m} \times 500\,\text{m} = 4.00 \times 10^4\,\text{m}^2,$$

so that

$$\begin{aligned} F &= (3.92 \times 10^5\,\text{N/m}^2)(4.00 \times 10^4\,\text{m}^2) \\ &= 1.57 \times 10^{10}\,\text{N}. \end{aligned}$$

Significance

Although this force seems large, it is small compared with the $1.96 \times 10^{13}\text{N}$ weight of the water in the reservoir. In fact, it is only 0.0800% of the weight.

CHECK YOUR UNDERSTANDING 14.1

If the reservoir in Example 14.1 covered twice the area, but was kept to the same depth, would the dam need to be redesigned?

Pressure in a static fluid in a uniform gravitational field

A *static fluid* is a fluid that is not in motion. At any point within a static fluid, the pressure on all sides must be equal—otherwise, the fluid at that point would react to a net force and accelerate.

The pressure at any point in a static fluid depends only on the depth at that point. As discussed, pressure in a fluid near Earth varies with depth due to the weight of fluid above a particular level. In the above examples, we assumed density to be constant and the average density of the fluid to be a good representation of the density. This is a reasonable approximation for liquids like water, where large forces are required to compress the liquid or change the volume. In a swimming pool, for example, the density is approximately constant, and the water at the bottom is compressed very little by the weight of the water on top. Traveling up in the atmosphere is quite a different situation, however. The density of the air begins to change significantly just a short distance above Earth's surface.

To derive a formula for the variation of pressure with depth in a tank containing a fluid of density ρ on the surface of Earth, we must start with the assumption that the density of the fluid is not constant. Fluid located at deeper levels is subjected to more force than fluid nearer to the surface due to the weight of the fluid above it. Therefore, the pressure calculated at a given depth is different than the pressure calculated using a constant density.

Imagine a thin element of fluid at a depth *h*, as shown in Figure 14.8. Let the element have a cross-sectional area *A* and height Δy. The forces acting upon the element are due to the pressures *p*(*y*) above and $p\,(y + \Delta y)$ below it. The weight of the element itself is also shown in the free-body diagram.

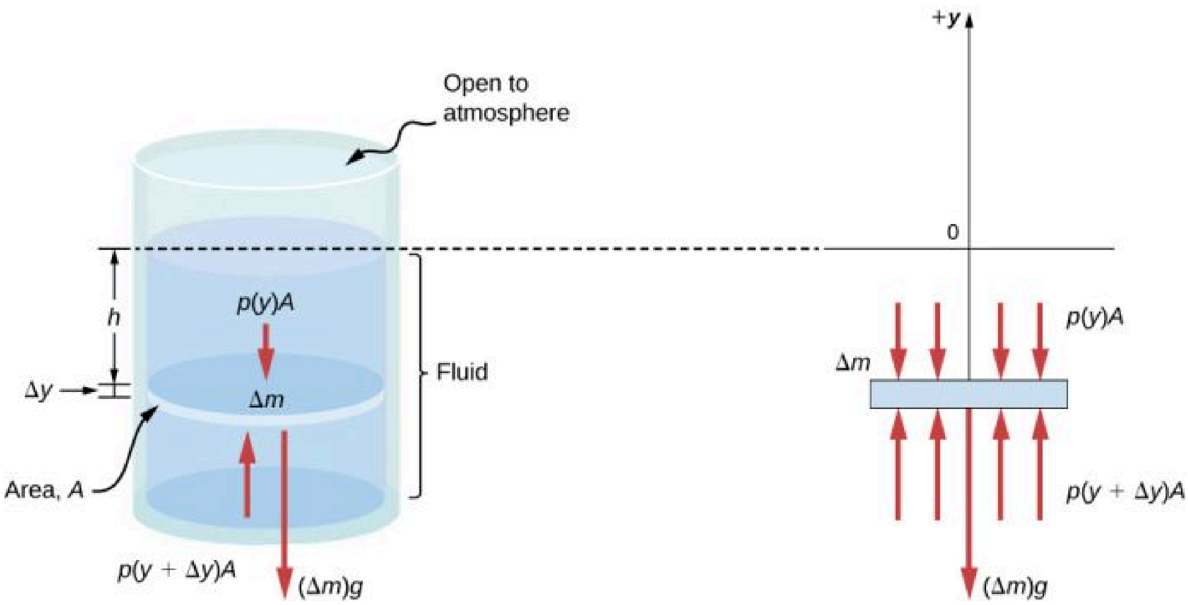


FIGURE 14.8 Forces on a mass element inside a fluid. The weight of the element itself is shown in the free-body diagram.

Since the element of fluid between *y* and $y + \Delta y$ is not accelerating, the forces are balanced. Using a Cartesian *y*-axis oriented up, we find the following equation for the *y*-component:

$$p\,(y + \Delta y)\,A - p\,(y)\,A - g\Delta m = 0(\Delta y < 0). \tag{14.6}$$

Note that if the element had a non-zero *y*-component of acceleration, the right-hand side would not be zero but would instead be the mass times the *y*-acceleration. The mass of the element can be written in terms of the density of the fluid and the volume of the elements:

$$\Delta m = |\rho A \Delta y| = -\rho A \Delta y \quad (\Delta y < 0)\,.$$

Putting this expression for Δm into Equation 14.6 and then dividing both sides by $A\Delta y$, we find

$$\frac{p(y + \Delta y) - p(y)}{\Delta y} = -\rho g. \tag{14.7}$$

Taking the limit of the infinitesimally thin element $\Delta y \to 0$, we obtain the following differential equation, which gives the variation of pressure in a fluid:

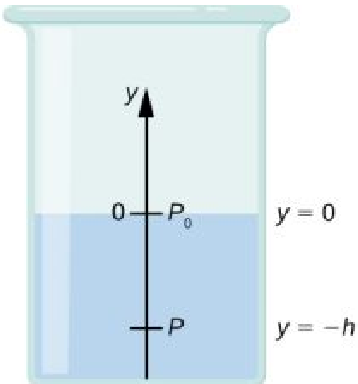
$$\frac{dp}{dy} = -\rho g. \tag{14.8}$$

This equation tells us that the rate of change of pressure in a fluid is proportional to the density of the fluid. The solution of this equation depends upon whether the density ρ is constant or changes with depth; that is, the function $\rho(y)$.

If the range of the depth being analyzed is not too great, we can assume the density to be constant. But if the range of depth is large enough for the density to vary appreciably, such as in the case of the atmosphere, there is significant change in density with depth. In that case, we cannot use the approximation of a constant density.

Pressure in a fluid with a constant density

Let's use Equation 14.9 to work out a formula for the pressure at a depth h from the surface in a tank of a liquid such as water, where the density of the liquid can be taken to be constant.



We need to integrate Equation 14.9 from $y = 0$, where the pressure is atmospheric pressure (p_0) , to $y = -h$, the y-coordinate of the depth:

$$\begin{aligned} \int_{p_0}^{p} dp &= -\int_{0}^{-h} \rho g dy \\ p - p_0 &= \rho g h \\ p &= p_0 + \rho g h. \end{aligned} \qquad 14.9$$

Hence, pressure at a depth of fluid on the surface of Earth is equal to the atmospheric pressure plus ρgh if the density of the fluid is constant over the height, as we found previously.

Note that the pressure in a fluid depends only on the depth from the surface and not on the shape of the container. Thus, in a container where a fluid can freely move in various parts, the liquid stays at the same level in every part, regardless of the shape, as shown in Figure 14.9.

FIGURE 14.9 If a fluid can flow freely between parts of a container, it rises to the same height in each part. In the container pictured, the pressure at the bottom of each column is the same; if it were not the same, the fluid would flow until the pressures became equal.

Variation of atmospheric pressure with height

The change in atmospheric pressure with height is of particular interest. Assuming the temperature of air to be constant, and that the ideal gas law of thermodynamics describes the atmosphere to a good approximation, we can find the variation of atmospheric pressure with height, when the temperature is constant. (We discuss the ideal gas law in a later chapter, but we assume you have some familiarity with it from high school and chemistry.) Let $p(y)$ be the atmospheric pressure at height y. The density ρ at y, the temperature T in the Kelvin scale (K), and the mass m

of a molecule of air are related to the absolute pressure by the ideal gas law, in the form

$$p = \rho \frac{k_B T}{m} \text{ (atmosphere)}, \quad (14.10)$$

where k_B is Boltzmann's constant, which has a value of 1.38×10^{-23} J/K.

You may have encountered the ideal gas law in the form $pV = nRT$, where *n* is the number of moles and *R* is the gas constant. Here, the same law has been written in a different form, using the density ρ instead of volume *V*. Therefore, if pressure *p* changes with height, so does the density ρ. Using density from the ideal gas law, the rate of variation of pressure with height is given as

$$\frac{dp}{dy} = -p\left(\frac{mg}{k_B T}\right),$$

where constant quantities have been collected inside the parentheses. Replacing these constants with a single symbol α, the equation looks much simpler:

$$\begin{aligned} \frac{dp}{dy} &= -\alpha p \\ \frac{dp}{p} &= -\alpha dy \\ \int_{p_0}^{p(y)} \frac{dp}{p} &= \int_0^y -\alpha dy \\ [\ln(p)]_{p_0}^{p(y)} &= [-\alpha y]_0^y \\ \ln(p) - \ln(p_0) &= -\alpha y \\ \ln\left(\frac{p}{p_0}\right) &= -\alpha y \end{aligned}$$

This gives the solution

$$p(y) = p_0 \exp(-\alpha y).$$

Thus, atmospheric pressure drops exponentially with height, since the *y*-axis is pointed up from the ground and *y* has positive values in the atmosphere above sea level. The pressure drops by a factor of $\frac{1}{e}$ when the height is $\frac{1}{\alpha}$, which gives us a physical interpretation for α: The constant $\frac{1}{\alpha}$ is a length scale that characterizes how pressure varies with height and is often referred to as the pressure scale height.

We can obtain an approximate value of α by using the mass of a nitrogen molecule as a proxy for an air molecule. At temperature $27\ °\text{C}$, or 300 K, we find

$$\alpha = -\frac{mg}{k_B T} = \frac{4.8 \times 10^{-26} \text{ kg} \times 9.81 \text{ m/s}^2}{1.38 \times 10^{-23} \text{ J/K} \times 300 \text{ K}} = \frac{1}{8800 \text{ m}}.$$

Therefore, for every 8800 meters, the air pressure drops by a factor 1/*e*, or approximately one-third of its value. This gives us only a rough estimate of the actual situation, since we have assumed both a constant temperature and a constant *g* over such great distances from Earth, neither of which is correct in reality.

Direction of pressure in a fluid

Fluid pressure has no direction, being a scalar quantity, whereas the forces due to pressure have well-defined directions: They are always exerted perpendicular to any surface. The reason is that fluids cannot withstand or exert shearing forces. Thus, in a static fluid enclosed in a tank, the force exerted on the walls of the tank is exerted perpendicular to the inside surface. Likewise, pressure is exerted perpendicular to the surfaces of any object within the fluid. Figure 14.10 illustrates the pressure exerted by air on the walls of a tire and by water on the body of a swimmer.

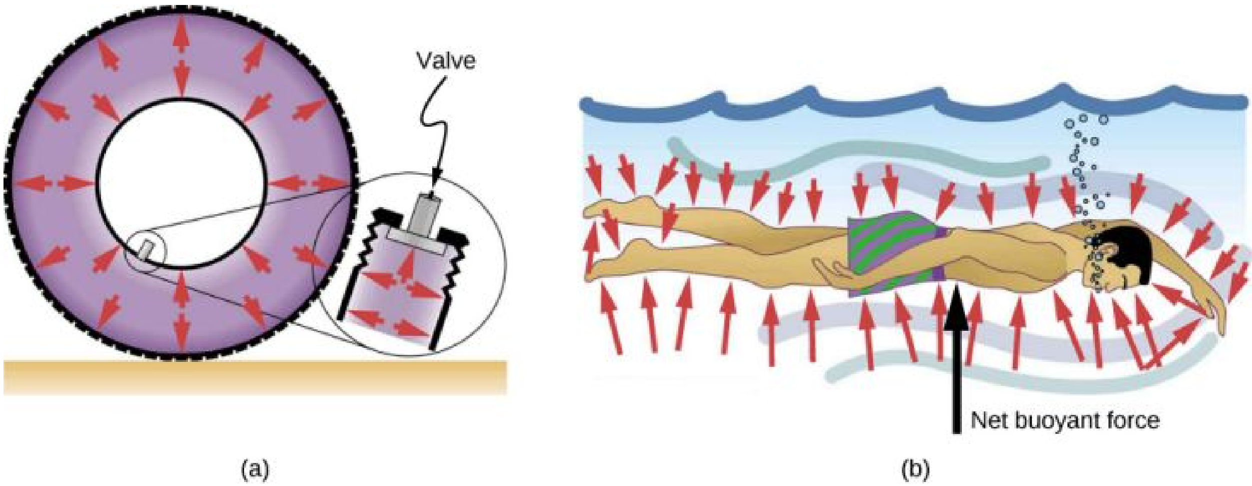


FIGURE 14.10 (a) Pressure inside this tire exerts forces perpendicular to all surfaces it contacts. The arrows represent directions and magnitudes of the forces exerted at various points. (b) Pressure is exerted perpendicular to all sides of this swimmer, since the water would flow into the space he occupies if he were not there. The arrows represent the directions and magnitudes of the forces exerted at various points on the swimmer. Note that the forces are larger underneath, due to greater depth, giving a net upward or buoyant force. The net vertical force on the swimmer is equal to the sum of the buoyant force and the weight of the swimmer.

14.2 Measuring Pressure

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define gauge pressure and absolute pressure
- Explain various methods for measuring pressure
- Understand the working of open-tube barometers
- Describe in detail how manometers and barometers operate

In the preceding section, we derived a formula for calculating the variation in pressure for a fluid in hydrostatic equilibrium. As it turns out, this is a very useful calculation. Measurements of pressure are important in daily life as well as in science and engineering applications. In this section, we discuss different ways that pressure can be reported and measured.

Gauge Pressure vs. Absolute Pressure

Suppose the pressure gauge on a full scuba tank reads 3000 psi, which is approximately 207 atmospheres. When the valve is opened, air begins to escape because the pressure inside the tank is greater than the atmospheric pressure outside the tank. Air continues to escape from the tank until the pressure inside the tank equals the pressure of the atmosphere outside the tank. At this point, the pressure gauge on the tank reads zero, even though the pressure inside the tank is actually 1 atmosphere—the same as the air pressure outside the tank.

Most pressure gauges, like the one on the scuba tank, are calibrated to read zero at atmospheric pressure. Pressure readings from such gauges are called **gauge pressure**, which is the pressure relative to the atmospheric pressure. When the pressure inside the tank is greater than atmospheric pressure, the gauge reports a positive value.

Some gauges are designed to measure negative pressure. For example, many physics experiments must take place in a vacuum chamber, a rigid chamber from which some of the air is pumped out. The pressure inside the vacuum chamber is less than atmospheric pressure, so the pressure gauge on the chamber reads a negative value.

Unlike gauge pressure, **absolute pressure** accounts for atmospheric pressure, which in effect adds to the pressure in any fluid not enclosed in a rigid container.

ABSOLUTE PRESSURE

The absolute pressure, or total pressure, is the sum of gauge pressure and atmospheric pressure:

$$p_{abs} = p_g + p_{atm} \tag{14.11}$$

where p_{abs} is absolute pressure, p_g is gauge pressure, and p_{atm} is atmospheric pressure.

For example, if a tire gauge reads 34 psi, then the absolute pressure is 34 psi plus 14.7 psi (p_{atm} in psi), or 48.7 psi (equivalent to 336 kPa).

In most cases, the absolute pressure in fluids cannot be negative. Fluids push rather than pull, so the smallest absolute pressure in a fluid is zero (a negative absolute pressure is a pull). Thus, the smallest possible gauge pressure is $p_g = -p_{atm}$ (which makes p_{abs} zero). There is no theoretical limit to how large a gauge pressure can be.

Measuring Pressure

A host of devices are used for measuring pressure, ranging from tire gauges to blood pressure monitors. Many other types of pressure gauges are commonly used to test the pressure of fluids, such as mechanical pressure gauges. We will explore some of these in this section.

Any property that changes with pressure in a known way can be used to construct a pressure gauge. Some of the most common types include strain gauges, which use the change in the shape of a material with pressure; capacitance pressure gauges, which use the change in electric capacitance due to shape change with pressure; piezoelectric pressure gauges, which generate a voltage difference across a piezoelectric material under a pressure difference between the two sides; and ion gauges, which measure pressure by ionizing molecules in highly evacuated chambers. Different pressure gauges are useful in different pressure ranges and under different physical situations. Some examples are shown in Figure 14.11.

(a) (b) (c)

FIGURE 14.11 (a) Gauges are used to measure and monitor pressure in gas cylinders. Compressed gases are used in many industrial as well as medical applications. (b) Tire pressure gauges come in many different models, but all are meant for the same purpose: to measure the internal pressure of the tire. This enables the driver to keep the tires inflated at optimal pressure for load weight and driving conditions. (c) An ionization gauge is a high-sensitivity device used to monitor the pressure of gases in an enclosed system. Neutral gas molecules are ionized by the release of electrons, and the current is translated into a pressure reading. Ionization gauges are commonly used in industrial applications that rely on vacuum systems.

Manometers

One of the most important classes of pressure gauges applies the property that pressure due to the weight of a fluid of constant density is given by $p = h\rho g$. The U-shaped tube shown in Figure 14.12 is an example of a *manometer*; in part (a), both sides of the tube are open to the atmosphere, allowing atmospheric pressure to push down on each side equally so that its effects cancel.

A manometer with only one side open to the atmosphere is an ideal device for measuring gauge pressures. The gauge pressure is $p_g = h\rho g$ and is found by measuring *h*. For example, suppose one side of the U-tube is connected to some source of pressure p_{abs}, such as the balloon in part (b) of the figure or the vacuum-packed peanut jar shown in part (c). Pressure is transmitted undiminished to the manometer, and the fluid levels are no longer equal. In part (b), p_{abs} is greater than atmospheric pressure, whereas in part (c), p_{abs} is less than atmospheric pressure. In both cases, p_{abs} differs from atmospheric pressure by an amount $h\rho g$, where ρ is the density of the fluid in the manometer. In part (b), p_{abs} can support a column of fluid of height *h*, so it must exert a pressure $h\rho g$ greater than atmospheric pressure (the gauge pressure p_g is positive). In part (c), atmospheric pressure can support a column of fluid of height *h*, so p_{abs} is less than atmospheric pressure by an amount $h\rho g$ (the gauge pressure p_g is negative).

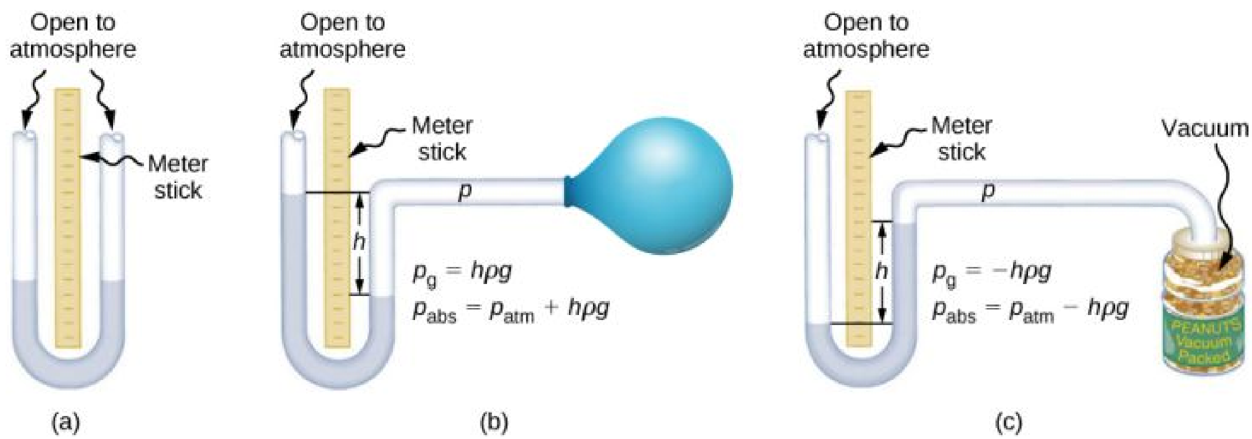


FIGURE 14.12 An open-tube manometer has one side open to the atmosphere. (a) Fluid depth must be the same on both sides, or the pressure each side exerts at the bottom will be unequal and liquid will flow from the deeper side. (b) A positive gauge pressure $p_g = h\rho g$ transmitted to one side of the manometer can support a column of fluid of height *h*. (c) Similarly, atmospheric pressure is greater than a negative gauge pressure p_g by an amount $h\rho g$. The jar's rigidity prevents atmospheric pressure from being transmitted to the peanuts.

Barometers

Manometers typically use a U-shaped tube of a fluid (often mercury) to measure pressure. A *barometer* (see Figure 14.13) is a device that typically uses a single column of mercury to measure atmospheric pressure. The barometer, invented by the Italian mathematician and physicist Evangelista Torricelli (1608–1647) in 1643, is constructed from a glass tube closed at one end and filled with mercury. The tube is then inverted and placed in a pool of mercury. This device measures atmospheric pressure, rather than gauge pressure, because there is a nearly pure vacuum above the mercury in the tube. The height of the mercury is such that $h\rho g = p_{atm}$. When atmospheric pressure varies, the mercury rises or falls.

Weather forecasters closely monitor changes in atmospheric pressure (often reported as barometric pressure), as rising mercury typically signals improving weather and falling mercury indicates deteriorating weather. The barometer can also be used as an altimeter, since average atmospheric pressure varies with altitude. Mercury barometers and manometers are so common that units of mm Hg are often quoted for atmospheric pressure and blood pressures.

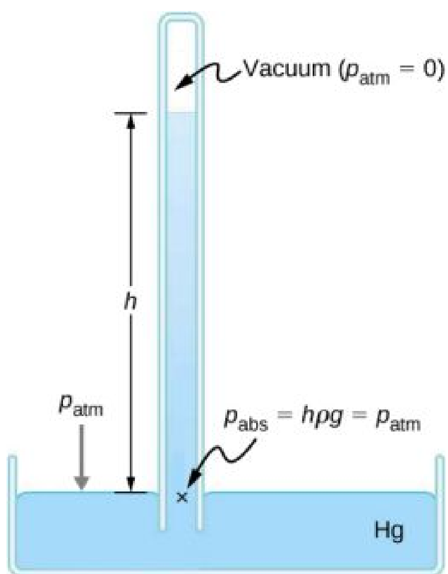


FIGURE 14.13 A mercury barometer measures atmospheric pressure. The pressure due to the mercury's weight, $h\rho g$, equals atmospheric pressure. The atmosphere is able to force mercury in the tube to a height *h* because the pressure above the mercury is zero.

EXAMPLE 14.2

Fluid Heights in an Open U-Tube

A U-tube with both ends open is filled with a liquid of density ρ_1 to a height *h* on both sides (Figure 14.14). A liquid of density $\rho_2 < \rho_1$ is poured into one side and Liquid 2 settles on top of Liquid 1. The heights on the two sides are different. The height to the top of Liquid 2 from the interface is h_2 and the height to the top of Liquid 1 from the level of the interface is h_1. Derive a formula for the height difference.

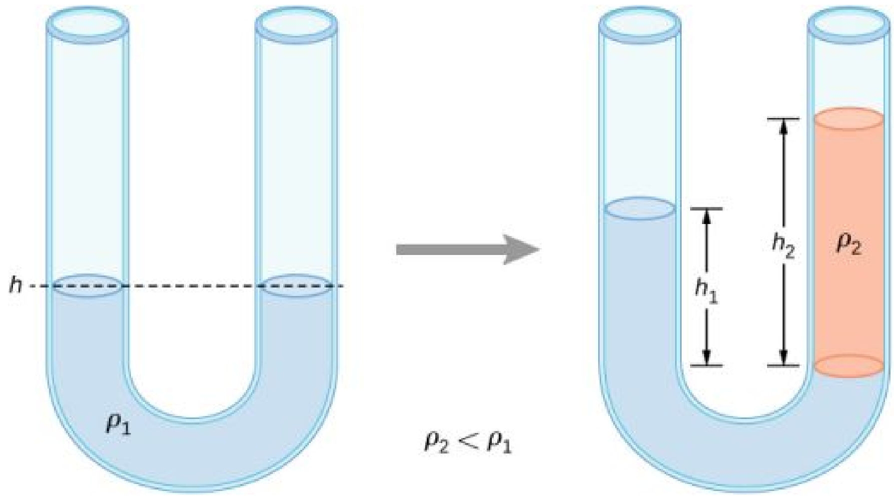


FIGURE 14.14 Two liquids of different densities are shown in a U-tube.

Strategy

The pressure at points at the same height on the two sides of a U-tube must be the same as long as the two points are in the same liquid. Therefore, we consider two points at the same level in the two arms of the tube: One point is the interface on the side of the Liquid 2 and the other is a point in the arm with Liquid 1 that is at the same level as the interface in the other arm. The pressure at each point is due to atmospheric pressure plus the weight of the liquid above it.

$$\text{Pressure on the side with Liquid 1} = p_0 + \rho_1 g h_1$$
$$\text{Pressure on the side with Liquid 2} = p_0 + \rho_2 g h_2$$

Solution

Since the two points are in Liquid 1 and are at the same height, the pressure at the two points must be the same. Therefore, we have

$$p_0 + \rho_1 g h_1 = p_0 + \rho_2 g h_2.$$

Hence,

$$\rho_1 h_1 = \rho_2 h_2.$$

This means that the difference in heights on the two sides of the U-tube is

$$h_2 - h_1 = \left(1 - \frac{\rho_2}{\rho_1}\right) h_2.$$

The result makes sense if we set $\rho_2 = \rho_1$, which gives $h_2 = h_1$. If the two sides have the same density, they have the same height.

CHECK YOUR UNDERSTANDING 14.2

Mercury is a hazardous substance. Why do you suppose mercury is typically used in barometers instead of a safer fluid such as water?

Units of pressure

As stated earlier, the SI unit for pressure is the pascal (Pa), where

$$1\ \text{Pa} = 1\ \text{N/m}^2.$$

In addition to the pascal, many other units for pressure are in common use (Table 14.3). In meteorology, atmospheric pressure is often described in the unit of millibars (mbar), where

$$1000\ \text{mbar} = 1\ \times\ 10^5\ \text{Pa}.$$

The millibar is a convenient unit for meteorologists because the average atmospheric pressure at sea level on Earth is $1.013\ \times\ 10^5\ \text{Pa} = 1013\ \text{mbar} = 1\ \text{atm}$. Using the equations derived when considering pressure at a depth in a fluid, pressure can also be measured as millimeters or inches of mercury. The pressure at the bottom of a 760-mm column of mercury at $0\ °\text{C}$ in a container where the top part is evacuated is equal to the atmospheric pressure. Thus, 760 mm Hg is also used in place of 1 atmosphere of pressure. In vacuum physics labs, scientists often use another unit called the torr, named after Torricelli, who, as we have just seen, invented the mercury manometer for measuring pressure. One torr is equal to a pressure of 1 mm Hg.

Unit	Definition
SI unit: the Pascal	$1\ \text{Pa} = 1\ \text{N/m}^2$
English unit: pounds per square inch (lb/in.^2 or psi)	$1\ \text{psi} = 6.895\ \times\ 10^3\ \text{Pa}$
Other units of pressure	$1\ \text{atm} = 760\ \text{mmHg} = 1.013\ \times\ 10^5\ \text{Pa} = 14.7\ \text{psi} = 29.9\ \text{inches of Hg} = 1013\ \text{mbar}$
	$1\ \text{bar} = 10^5\ \text{Pa}$
	$1\ \text{torr} = 1\ \text{mm Hg} = 133.3\ \text{Pa}$

TABLE 14.3 **Summary of the Units of Pressure**

14.3 Pascal's Principle and Hydraulics

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- State Pascal's principle
- Describe applications of Pascal's principle
- Derive relationships between forces in a hydraulic system

In 1653, the French philosopher and scientist Blaise Pascal published his *Treatise on the Equilibrium of Liquids*, in which he discussed principles of static fluids. A static fluid is a fluid that is not in motion. When a fluid is not flowing, we say that the fluid is in static equilibrium. If the fluid is water, we say it is in **hydrostatic equilibrium**. For a fluid in static equilibrium, the net force on any part of the fluid must be zero; otherwise the fluid will start to flow.

Pascal's observations—since proven experimentally—provide the foundation for hydraulics, one of the most important developments in modern mechanical technology. Pascal observed that a change in pressure applied to an enclosed fluid is transmitted undiminished throughout the fluid and to the walls of its container. Because of this, we often know more about pressure than other physical quantities in fluids. Moreover, Pascal's principle implies that the total pressure in a fluid is the sum of the pressures from different sources. A good example is the fluid at a depth depends on the depth of the fluid and the pressure of the atmosphere.

Pascal's Principle

Pascal's principle (also known as Pascal's law) states that when a change in pressure is applied to an enclosed fluid, it is transmitted undiminished to all portions of the fluid and to the walls of its container. In an enclosed fluid, since atoms of the fluid are free to move about, they transmit pressure to all parts of the fluid *and* to the walls of the container. Any change in pressure is transmitted undiminished.

Note that this principle does not say that the pressure is the same at all points of a fluid—which is not true, since the pressure in a fluid near Earth varies with height. Rather, this principle applies to the *change* in pressure. Suppose you place some water in a cylindrical container of height *H* and cross-sectional area *A* that has a movable piston of mass *m* (Figure 14.15). Adding weight *Mg* at the top of the piston increases the pressure at the top by *Mg*/*A*, since the additional weight also acts over area *A* of the lid:

$$\Delta p_{\text{top}} = \frac{Mg}{A}.$$

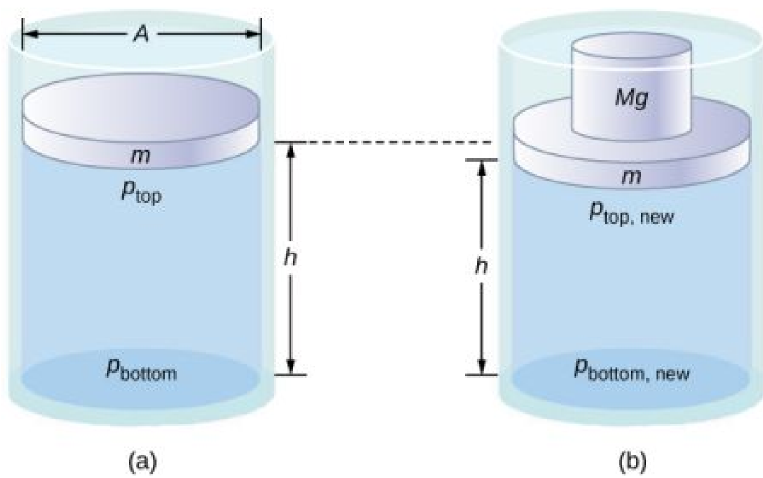


FIGURE 14.15 Pressure in a fluid changes when the fluid is compressed. (a) The pressure at the top layer of the fluid is different from pressure at the bottom layer. (b) The increase in pressure by adding weight to the piston is the same everywhere, for example, $p_{\text{top new}} - p_{\text{top}} = p_{\text{bottom new}} - p_{\text{bottom}}$.

According to Pascal's principle, the pressure at all points in the water changes by the same amount, *Mg*/*A*. Thus, the pressure at the bottom also increases by *Mg*/*A*. The pressure at the bottom of the container is equal to the sum of the atmospheric pressure, the pressure due to the fluid, and the pressure supplied by the mass. The change in pressure at the bottom of the container due to the mass is

$$\Delta p_{\text{bottom}} = \frac{Mg}{A}.$$

Since the pressure changes are the same everywhere in the fluid, we no longer need subscripts to designate the pressure change for top or bottom:

$$\Delta p = \Delta p_{\text{top}} = \Delta p_{\text{bottom}} = \Delta p_{\text{everywhere}}.$$

INTERACTIVE

Pascal's Barrel is a great demonstration of Pascal's principle. Watch a simulation (https://openstax.org/l/21pascalbarrel) of Pascal's 1646 experiment, in which he demonstrated the effects of changing pressure in a fluid.

Applications of Pascal's Principle and Hydraulic Systems

Hydraulic systems are used to operate automotive brakes, hydraulic jacks, and numerous other mechanical systems (Figure 14.16).

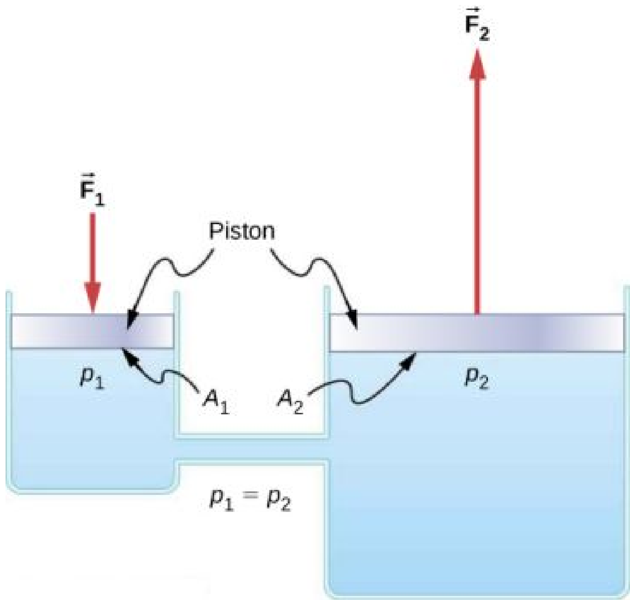


FIGURE 14.16 A typical hydraulic system with two fluid-filled cylinders, capped with pistons and connected by a tube called a hydraulic line. A downward force $\vec{\mathbf{F}}_1$ on the left piston creates a change in pressure that is transmitted undiminished to all parts of the enclosed fluid. This results in an upward force $\vec{\mathbf{F}}_2$ on the right piston that is larger than $\vec{\mathbf{F}}_1$ because the right piston has a larger surface area.

We can derive a relationship between the forces in this simple hydraulic system by applying Pascal's principle. Note first that the two pistons in the system are at the same height, so there is no difference in pressure due to a difference in depth. The pressure due to F_1 acting on area A_1 is simply

$$p_1 = \frac{F_1}{A_1}, \text{as defined by } p = \frac{F}{A}.$$

According to Pascal's principle, this pressure is transmitted undiminished throughout the fluid and to all walls of the container. Thus, a pressure p_2 is felt at the other piston that is equal to p_1. That is, $p_1 = p_2$. However, since$p_2 = F_2/A_2$, we see that

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}. \tag{14.12}$$

This equation relates the ratios of force to area in any hydraulic system, provided that the pistons are at the same vertical height and that friction in the system is negligible.

Hydraulic systems can increase or decrease the force applied to them. To make the force larger, the pressure is applied to a larger area. For example, if a 100-N force is applied to the left cylinder in Figure 14.16 and the right cylinder has an area five times greater, then the output force is 500 N. Hydraulic systems are analogous to simple levers, but they have the advantage that pressure can be sent through tortuously curved lines to several places at once.

The **hydraulic jack** is such a hydraulic system. A hydraulic jack is used to lift heavy loads, such as the ones used by auto mechanics to raise an automobile. It consists of an incompressible fluid in a U-tube fitted with a movable piston on each side. One side of the U-tube is narrower than the other. A small force applied over a small area can balance a much larger force on the other side over a larger area (Figure 14.17).

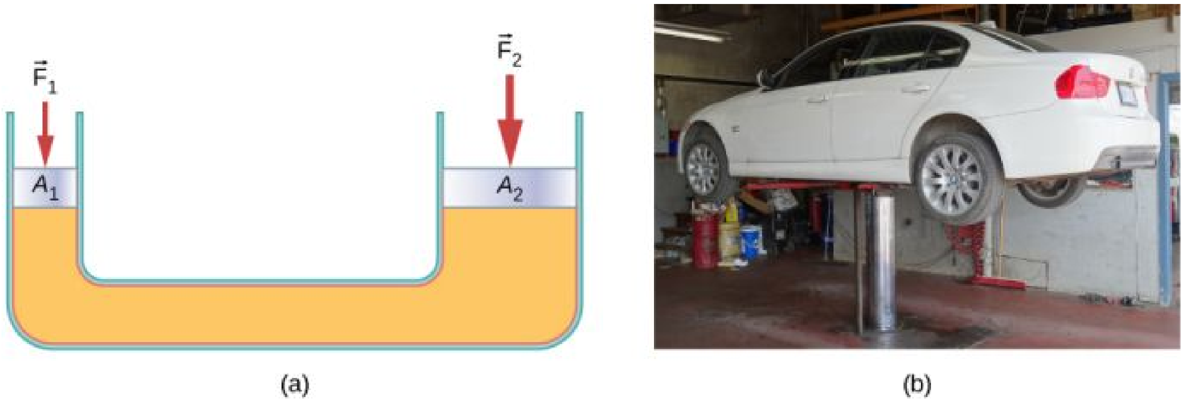


FIGURE 14.17 (a) A hydraulic jack operates by applying forces (F_1, F_2) to an incompressible fluid in a U-tube, using a movable piston (A_1, A_2) on each side of the tube. (b) Hydraulic jacks are commonly used by car mechanics to lift vehicles so that repairs and maintenance can be performed. (credit b: modification of work by Jane Whitney)

From Pascal's principle, it can be shown that the force needed to lift the car is less than the weight of the car:

$$F_1 = \frac{A_1}{A_2} F_2,$$

where F_1 is the force applied to lift the car, A_1 is the cross-sectional area of the smaller piston, A_2 is the cross sectional area of the larger piston, and F_2 is the weight of the car.

EXAMPLE 14.3

Calculating Force on Wheel Cylinders: Pascal Puts on the Brakes

Consider the automobile hydraulic system shown in Figure 14.18. Suppose a force of 100 N is applied to the brake pedal, which acts on the pedal cylinder (acting as a "master" cylinder) through a lever. A force of 500 N is exerted on the pedal cylinder. Pressure created in the pedal cylinder is transmitted to the four wheel cylinders. The pedal cylinder has a diameter of 0.500 cm and each wheel cylinder has a diameter of 2.50 cm. Calculate the magnitude of the force F_2 created at each of the wheel cylinders.

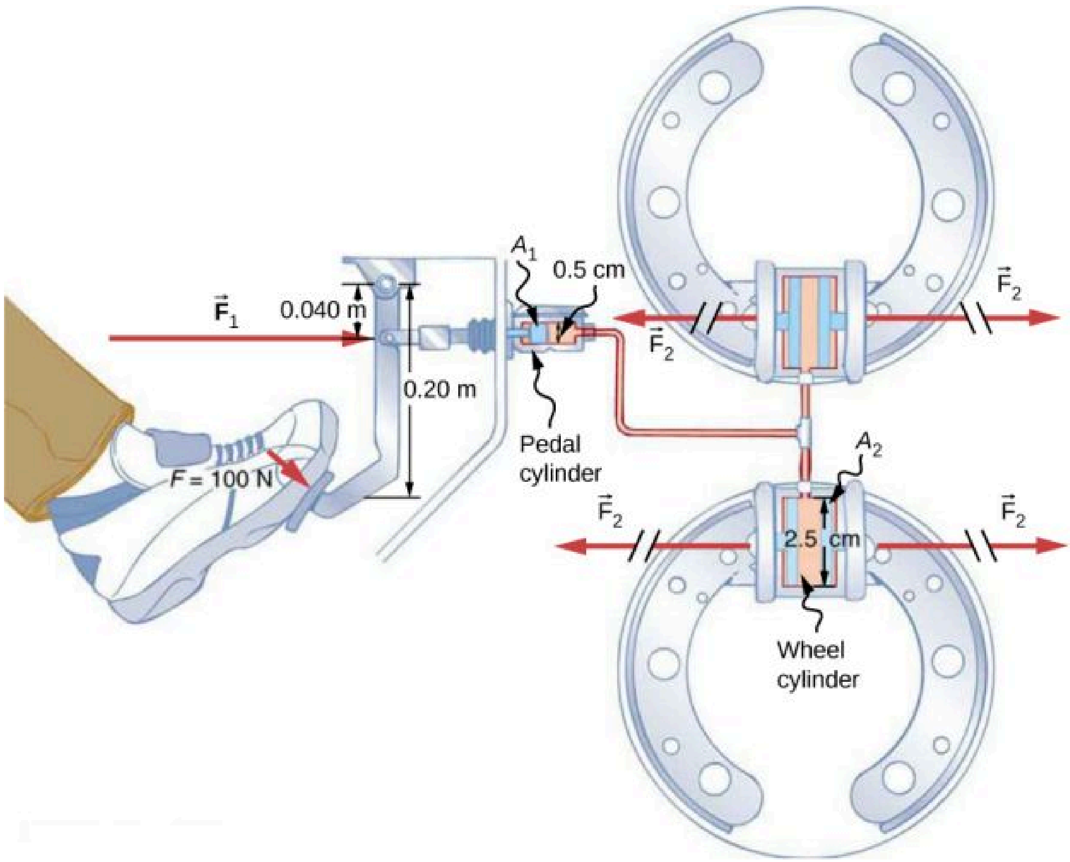


FIGURE 14.18 Hydraulic brakes use Pascal's principle. The driver pushes the brake pedal, exerting a force that is increased by the simple lever and again by the hydraulic system. Each of the identical wheel cylinders receives the same pressure and, therefore, creates the same force output F_2. The circular cross-sectional areas of the pedal and wheel cylinders are represented by A_1 and A_2, respectively.

Strategy

We are given the force F_1 applied to the pedal cylinder. The cross-sectional areas A_1 and A_2 can be calculated from their given diameters. Then we can use the following relationship to find the force F_2:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}.$$

Manipulate this algebraically to get F_2 on one side and substitute known values.

Solution

Pascal's principle applied to hydraulic systems is given by $\frac{F_1}{A_1} = \frac{F_2}{A_2}$:

$$\begin{aligned} F_2 &= \frac{A_2}{A_1} F_1 = \frac{\pi r_2^2}{\pi r_1^2} F_1 \\ &= \frac{(1.25 \text{ cm})^2}{(0.250 \text{ cm})^2} \times 500 \text{ N} = 1.25 \times 10^4 \text{ N}. \end{aligned}$$

Significance

This value is the force exerted by each of the four wheel cylinders. Note that we can add as many wheel cylinders as we wish. If each has a 2.50-cm diameter, each will exert 1.25×10^4 N. A simple hydraulic system, as an example of a simple machine, can increase force but cannot do more work than is done on it. Work is force times distance moved, and the wheel cylinder moves through a smaller distance than the pedal cylinder. Furthermore, the more wheels added, the smaller the distance each one moves. Many hydraulic systems—such as power brakes and those in bulldozers—have a motorized pump that actually does most of the work in the system.

CHECK YOUR UNDERSTANDING 14.3

Would a hydraulic press still operate properly if a gas is used instead of a liquid?

14.4 Archimedes' Principle and Buoyancy

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define buoyant force
- State Archimedes' principle
- Describe the relationship between density and Archimedes' principle

When placed in a fluid, some objects float due to a buoyant force. Where does this buoyant force come from? Why is it that some things float and others do not? Do objects that sink get any support at all from the fluid? Is your body buoyed by the atmosphere, or are only helium balloons affected (Figure 14.19)?

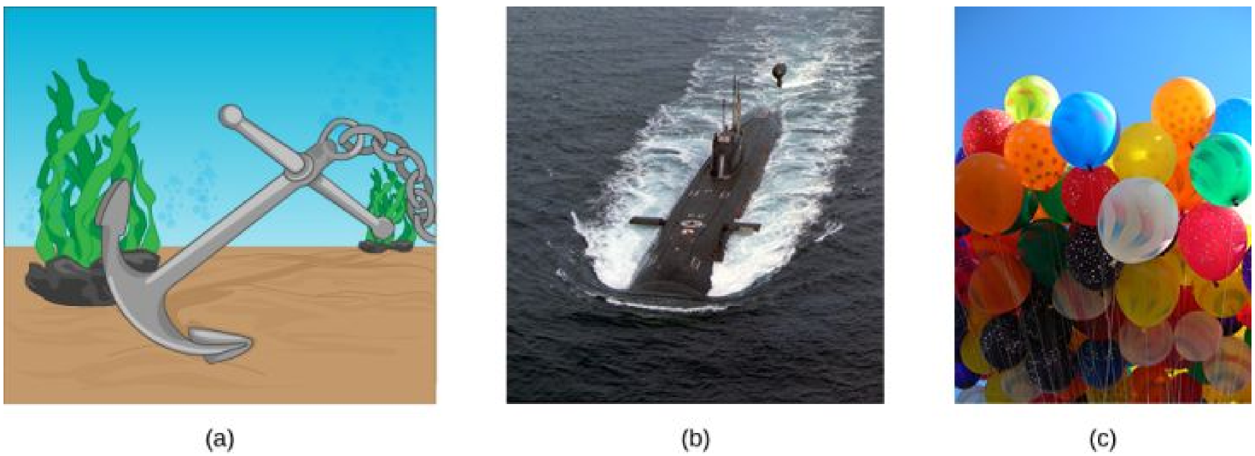


FIGURE 14.19 (a) Even objects that sink, like this anchor, are partly supported by water when submerged. (b) Submarines have adjustable density (ballast tanks) so that they may float or sink as desired. (c) Helium-filled balloons tug upward on their strings, demonstrating air's buoyant effect. (credit b: modification of work by Allied Navy; credit c: modification of work by "Crystl"/Flickr)

Answers to all these questions, and many others, are based on the fact that pressure increases with depth in a fluid. This means that the upward force on the bottom of an object in a fluid is greater than the downward force on top of the object. There is an upward force, or **buoyant force**, on any object in any fluid (Figure 14.20). If the buoyant force is greater than the object's weight, the net force on the object is upward. If we release such an object, it will rise in the fluid. If the buoyant force is less than the object's weight, the object sinks when released. If the buoyant force equals the object's weight, the object can remain suspended at its present depth, either immersed in the fluid (as for a neutrally buoyant diver) or partly above the surface (as with a floating object.) If the buoyant force equals the object's weight, the object can remain suspended at its present depth. The buoyant force is always present, whether the object floats, sinks, or is suspended in a fluid.

BUOYANT FORCE

The buoyant force is the upward force on any object in any fluid.

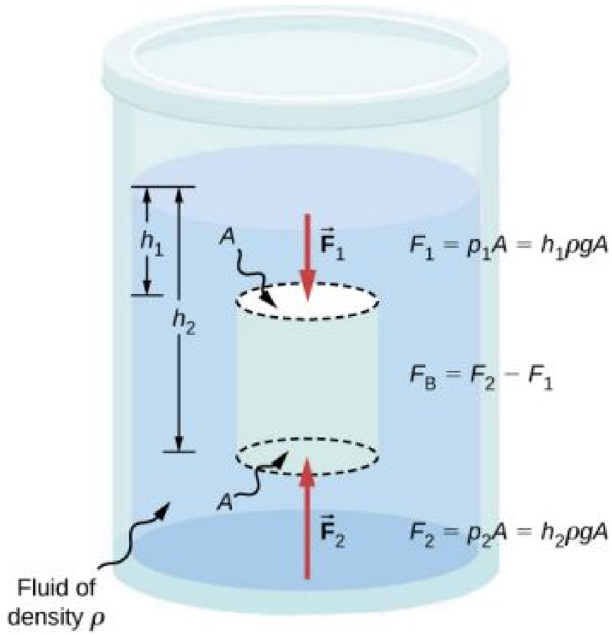


FIGURE 14.20 Pressure due to the weight of a fluid increases with depth because $p = hpg$. This change in pressure and associated upward force on the bottom of the cylinder are greater than the downward force on the top of the cylinder. The differences in the force results in the buoyant force F_B. (Horizontal forces cancel.)

Archimedes' Principle

Just how large a force is buoyant force? To answer this question, think about what happens when a submerged object is removed from a fluid, as in Figure 14.21. If the object were not in the fluid, the space the object occupied would be filled by fluid having a weight w_{fl}. This weight is supported by the surrounding fluid, so the buoyant force must equal w_{fl}, the weight of the fluid displaced by the object.

ARCHIMEDES' PRINCIPLE

The buoyant force on an object equals the weight of the fluid it displaces. In equation form, **Archimedes' principle** is

$$F_B = w_{fl},$$

where F_B is the buoyant force and w_{fl} is the weight of the fluid displaced by the object.

This principle is named after the Greek mathematician and inventor Archimedes (ca. 287–212 BCE), who stated this principle long before concepts of force were well established.

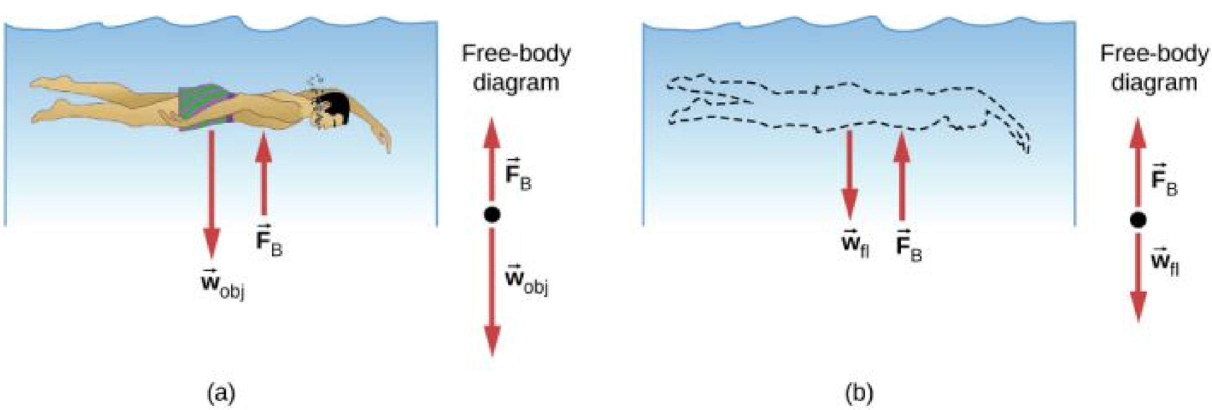


FIGURE 14.21 (a) An object submerged in a fluid experiences a buoyant force F_B. If F_B is greater than the weight of the object, the object rises. If F_B is less than the weight of the object, the object sinks. (b) If the object is removed, it is replaced by fluid having weight w_{fl}. Since this weight is supported by surrounding fluid, the buoyant force must equal the weight of the fluid displaced.

Archimedes' principle refers to the force of buoyancy that results when a body is submerged in a fluid, whether partially or wholly. The force that provides the pressure of a fluid acts on a body perpendicular to the surface of the body. In other words, the force due to the pressure at the bottom is pointed up, while at the top, the force due to the pressure is pointed down; the forces due to the pressures at the sides are pointing into the body.

Since the bottom of the body is at a greater depth than the top of the body, the pressure at the lower part of the body is higher than the pressure at the upper part, as shown in Figure 14.20. Therefore a net upward force acts on the body. This upward force is the force of buoyancy, or simply *buoyancy*.

INTERACTIVE

The exclamation "Eureka" (meaning "I found it") has often been credited to Archimedes as he made the discovery that would lead to Archimedes' principle. Some say it all started in a bathtub. To hear this story, watch this video (https://openstax.org/l/21archNASA) or explore Scientific American (https://openstax.org/l/21archsciamer) to learn more.

Density and Archimedes' Principle

If you drop a lump of clay in water, it will sink. But if you mold the same lump of clay into the shape of a boat, it will float. Because of its shape, the clay boat displaces more water than the lump and experiences a greater buoyant force, even though its mass is the same. The same is true of steel ships.

The average density of an object is what ultimately determines whether it floats. If an object's average density is less than that of the surrounding fluid, it will float. The reason is that the fluid, having a higher density, contains more mass and hence more weight in the same volume. The buoyant force, which equals the weight of the fluid displaced, is thus greater than the weight of the object. Likewise, an object denser than the fluid will sink.

The extent to which a floating object is submerged depends on how the object's density compares to the density of the fluid. In Figure 14.22, for example, the unloaded ship has a lower density and less of it is submerged compared with the same ship when loaded. We can derive a quantitative expression for the fraction submerged by considering density. The fraction submerged is the ratio of the volume submerged to the volume of the object, or

$$\text{fraction submerged} = \frac{V_{\text{sub}}}{V_{\text{obj}}} = \frac{V_{\text{fl}}}{V_{\text{obj}}}.$$

The volume submerged equals the volume of fluid displaced, which we call V_{fl}. Now we can obtain the relationship between the densities by substituting $\rho = \frac{m}{V}$ into the expression. This gives

$$\frac{V_{\text{fl}}}{V_{\text{obj}}} = \frac{m_{\text{fl}}/\rho_{\text{fl}}}{m_{\text{obj}}/\rho_{\text{obj}}},$$

where ρ_{obj} is the average density of the object and ρ_{fl} is the density of the fluid. Since the object floats, its mass and that of the displaced fluid are equal, so they cancel from the equation, leaving

$$\text{fraction submerged} = \frac{\rho_{\text{obj}}}{\rho_{\text{fl}}}.$$

We can use this relationship to measure densities.

FIGURE 14.22 An unloaded ship (a) floats higher in the water than a loaded ship (b).

EXAMPLE 14.4

Calculating Average Density

Suppose a 60.0-kg woman floats in fresh water with 97.0% of her volume submerged when her lungs are full of air. What is her average density?

Strategy

We can find the woman's density by solving the equation

$$\text{fraction submerged} = \frac{\rho_{\text{obj}}}{\rho_{\text{fl}}}$$

for the density of the object. This yields

$$\rho_{\text{obj}} = \rho_{\text{person}} = (\text{fraction submerged}) \cdot \rho_{\text{fl}}.$$

We know both the fraction submerged and the density of water, so we can calculate the woman's density.

Solution

Entering the known values into the expression for her density, we obtain

$$\rho_{\text{person}} = 0.970 \cdot \left(10^3 \, \frac{\text{kg}}{\text{m}^3}\right) = 970 \frac{\text{kg}}{\text{m}^3}.$$

Significance

The woman's density is less than the fluid density. We expect this because she floats.

Numerous lower-density objects or substances float in higher-density fluids: oil on water, a hot-air balloon in the atmosphere, a bit of cork in wine, an iceberg in salt water, and hot wax in a "lava lamp," to name a few. A less obvious example is mountain ranges floating on the higher-density crust and mantle beneath them. Even seemingly solid Earth has fluid characteristics.

Measuring Density

One of the most common techniques for determining density is shown in Figure 14.23.

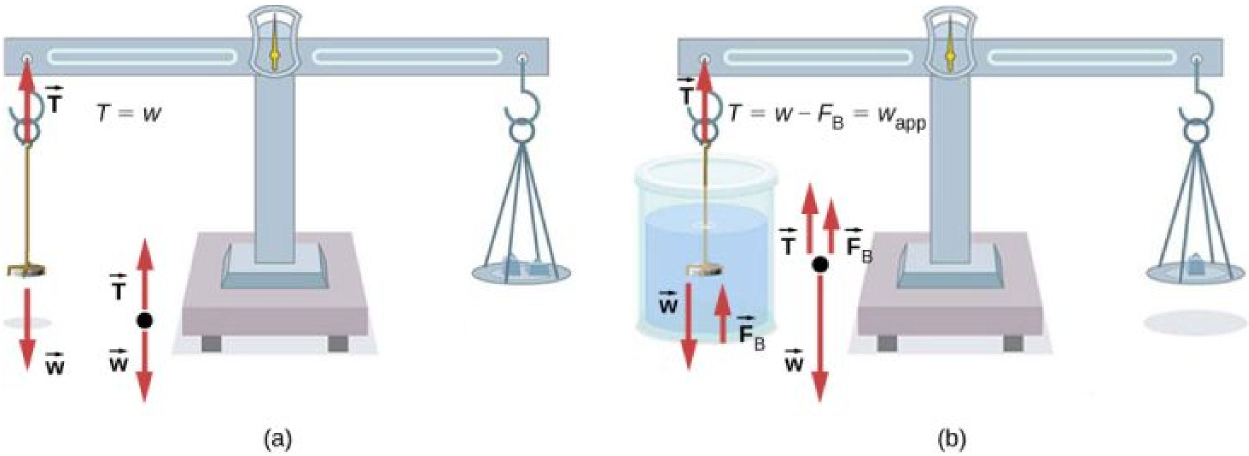


FIGURE 14.23 (a) A coin is weighed in air. (b) The apparent weight of the coin is determined while it is completely submerged in a fluid of known density. These two measurements are used to calculate the density of the coin.

An object, here a coin, is weighed in air and then weighed again while submerged in a liquid. The density of the coin, an indication of its authenticity, can be calculated if the fluid density is known. We can use this same technique to determine the density of the fluid if the density of the coin is known.

All of these calculations are based on Archimedes' principle, which states that the buoyant force on the object equals the weight of the fluid displaced. This, in turn, means that the object appears to weigh less when submerged; we call this measurement the object's apparent weight. The object suffers an apparent weight loss equal to the weight of the fluid displaced. Alternatively, on balances that measure mass, the object suffers an apparent mass loss equal to the mass of fluid displaced. That is, apparent weight loss equals weight of fluid displaced, or apparent mass loss equals mass of fluid displaced.

14.5 Fluid Dynamics

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the characteristics of flow
- Calculate flow rate
- Describe the relationship between flow rate and velocity
- Explain the consequences of the equation of continuity to the conservation of mass

The first part of this chapter dealt with fluid statics, the study of fluids at rest. The rest of this chapter deals with fluid dynamics, the study of fluids in motion. Even the most basic forms of fluid motion can be quite complex. For this reason, we limit our investigation to **ideal fluids** in many of the examples. An ideal fluid is a fluid with negligible **viscosity**. Viscosity is a measure of the internal friction in a fluid; we examine it in more detail in Viscosity and Turbulence. In a few examples, we examine an incompressible fluid—one for which an extremely large force is required to change the volume—since the density in an incompressible fluid is constant throughout.

Characteristics of Flow

Velocity vectors are often used to illustrate fluid motion in applications like meteorology. For example, wind—the fluid motion of air in the atmosphere—can be represented by vectors indicating the speed and direction of the wind at any given point on a map. Figure 14.24 shows velocity vectors describing the winds during Hurricane Arthur in 2014.

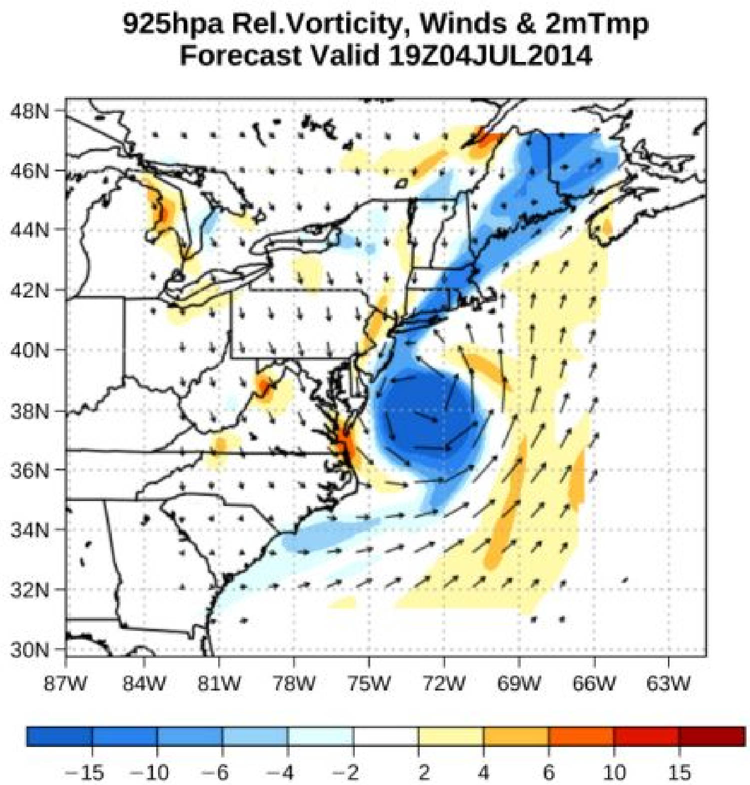


FIGURE 14.24 The velocity vectors show the flow of wind in Hurricane Arthur. Notice the circulation of the wind around the eye of the hurricane. Wind speeds are highest near the eye. The colors represent the relative vorticity, a measure of turning or spinning of the air. (credit: modification of work by Joseph Trout, Stockton University)

Another method for representing fluid motion is a *streamline*. A streamline represents the path of a small volume of fluid as it flows. The velocity is always tangential to the streamline. The diagrams in Figure 14.25 use streamlines to illustrate two examples of fluids moving through a pipe. The first fluid exhibits a **laminar flow** (sometimes described as a steady flow), represented by smooth, parallel streamlines. Note that in the example shown in part (a), the velocity of the fluid is greatest in the center and decreases near the walls of the pipe due to the viscosity of the fluid and friction between the pipe walls and the fluid. This is a special case of laminar flow, where the friction between the pipe and the fluid is high, known as no slip boundary conditions. The second diagram represents **turbulent flow**, in which streamlines are irregular and change over time. In turbulent flow, the paths of the fluid flow are irregular as different parts of the fluid mix together or form small circular regions that resemble whirlpools. This can occur when the speed of the fluid reaches a certain critical speed.

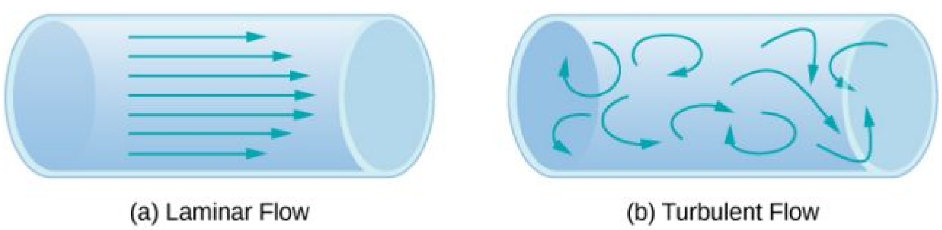


FIGURE 14.25 (a) Laminar flow can be thought of as layers of fluid moving in parallel, regular paths. (b) In turbulent flow, regions of fluid move in irregular, colliding paths, resulting in mixing and swirling.

Flow Rate and its Relation to Velocity

The volume of fluid passing by a given location through an area during a period of time is called **flow rate** Q, or more precisely, volume flow rate. In symbols, this is written as

$$Q = \frac{dV}{dt} \tag{14.13}$$

where V is the volume and t is the elapsed time. In Figure 14.26, the volume of the cylinder is Ax, so the flow rate is

$$Q = \frac{dV}{dt} = \frac{d}{dt}(Ax) = A\frac{dx}{dt} = Av.$$

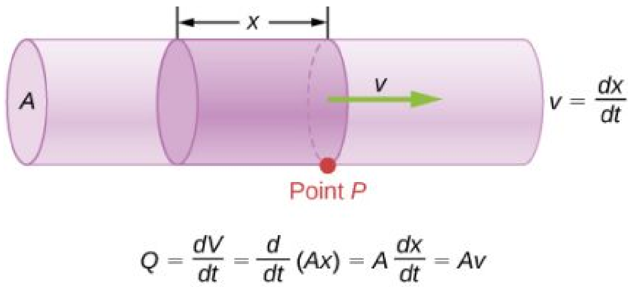


FIGURE 14.26 Flow rate is the volume of fluid flowing past a point through the area A per unit time. Here, the shaded cylinder of fluid flows past point P in a uniform pipe in time t.

The SI unit for flow rate is m^3/s, but several other units for Q are in common use, such as liters per minute (L/min). Note that a liter (L) is 1/1000 of a cubic meter or 1000 cubic centimeters $(10^{-3}\ m^3 \text{or}\ 10^3\ cm^3)$.

Flow rate and velocity are related, but quite different, physical quantities. To make the distinction clear, consider the flow rate of a river. The greater the velocity of the water, the greater the flow rate of the river. But flow rate also depends on the size and shape of the river. A rapid mountain stream carries far less water than the Amazon River in Brazil, for example. Figure 14.26 illustrates the volume flow rate. The volume flow rate is $Q = \frac{dV}{dt} = Av$, where A is the cross-sectional area of the pipe and v is the magnitude of the velocity.

The precise relationship between flow rate Q and average speed v is

$$Q = Av,$$

where A is the cross-sectional area and v is the average speed. The relationship tells us that flow rate is directly proportional to both the average speed of the fluid and the cross-sectional area of a river, pipe, or other conduit. The larger the conduit, the greater its cross-sectional area. Figure 14.26 illustrates how this relationship is obtained. The shaded cylinder has a volume $V = Ad$, which flows past the point P in a time t. Dividing both sides of this relationship by t gives

$$\frac{V}{t} = \frac{Ad}{t}.$$

We note that $Q = V/t$ and the average speed is $v = d/t$. Thus the equation becomes $Q = Av$.

Figure 14.27 shows an incompressible fluid flowing along a pipe of decreasing radius. Because the fluid is incompressible, the same amount of fluid must flow past any point in the tube in a given time to ensure continuity of flow. The flow is continuous because they are no sources or sinks that add or remove mass, so the mass flowing into the pipe must be equal the mass flowing out of the pipe. In this case, because the cross-sectional area of the pipe decreases, the velocity must necessarily increase. This logic can be extended to say that the flow rate must be the same at all points along the pipe. In particular, for arbitrary points 1 and 2,

$$\begin{aligned} Q_1 &= Q_2, \\ A_1 v_1 &= A_2 v_2. \end{aligned} \tag{14.14}$$

This is called the *equation of continuity* and is valid for any incompressible fluid (with constant density). The consequences of the equation of continuity can be observed when water flows from a hose into a narrow spray nozzle: It emerges with a large speed—that is the purpose of the nozzle. Conversely, when a river empties into one end of a reservoir, the water slows considerably, perhaps picking up speed again when it leaves the other end of the reservoir. In other words, speed increases when cross-sectional area decreases, and speed decreases when cross-sectional area increases.

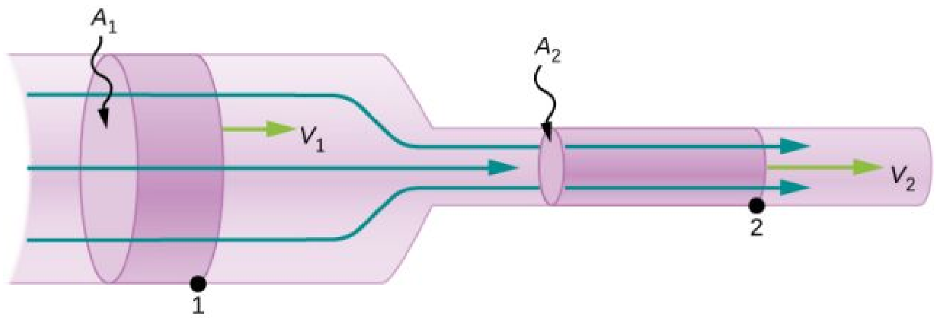


FIGURE 14.27 When a tube narrows, the same volume occupies a greater length. For the same volume to pass points 1 and 2 in a given time, the speed must be greater at point 2. The process is exactly reversible. If the fluid flows in the opposite direction, its speed decreases when the tube widens. (Note that the relative volumes of the two cylinders and the corresponding velocity vector arrows are not drawn to scale.)

Since liquids are essentially incompressible, the equation of continuity is valid for all liquids. However, gases are compressible, so the equation must be applied with caution to gases if they are subjected to compression or expansion.

EXAMPLE 14.5

Calculating Fluid Speed through a Nozzle

A nozzle with a diameter of 0.500 cm is attached to a garden hose with a radius of 0.900 cm. The flow rate through hose and nozzle is 0.500 L/s. Calculate the speed of the water (a) in the hose and (b) in the nozzle.

Strategy

We can use the relationship between flow rate and speed to find both speeds. We use the subscript 1 for the hose and 2 for the nozzle.

Solution

a. We solve the flow rate equation for speed and use $\pi {r_1}^2$ for the cross-sectional area of the hose, obtaining

$$v = \frac{Q}{A} = \frac{Q}{\pi r_1^2}.$$

Substituting values and using appropriate unit conversions yields

$$v = \frac{(0.500\ \text{L/s})(10^{-3}\ \text{m}^3/\text{L})}{3.14(9.00 \times 10^{-3}\ \text{m})^2} = 1.96\ \text{m/s}.$$

b. We could repeat this calculation to find the speed in the nozzle v_2, but we use the equation of continuity to give a somewhat different insight. The equation states

$$A_1 v_1 = A_2 v_2.$$

Solving for v_2 and substituting πr^2 for the cross-sectional area yields

$$v_2 = \frac{A_1}{A_2} v_1 = \frac{\pi r_1^2}{\pi r_2^2} v_1 = \frac{r_1^2}{r_2^2} v_1.$$

Substituting known values,

$$v_2 = \frac{(0.900\ \text{cm})^2}{(0.250\ \text{cm})^2}\, 1.96\ \text{m/s} = 25.5\ \text{m/s}.$$

Significance

A speed of 1.96 m/s is about right for water emerging from a hose with no nozzle. The nozzle produces a considerably faster stream merely by constricting the flow to a narrower tube.

The solution to the last part of the example shows that speed is inversely proportional to the square of the radius of the tube, making for large effects when radius varies. We can blow out a candle at quite a distance, for example, by pursing our lips, whereas blowing on a candle with our mouth wide open is quite ineffective.

Mass Conservation

The rate of flow of a fluid can also be described by the *mass flow rate* or mass rate of flow. This is the rate at which a mass of the fluid moves past a point. Refer once again to Figure 14.26, but this time consider the mass in the shaded volume. The mass can be determined from the density and the volume:

$$m = \rho V = \rho A x.$$

The mass flow rate is then

$$\frac{dm}{dt} = \frac{d}{dt}(\rho A x) = \rho A \frac{dx}{dt} = \rho A v,$$

where ρ is the density, A is the cross-sectional area, and v is the magnitude of the velocity. The mass flow rate is an important quantity in fluid dynamics and can be used to solve many problems. Consider Figure 14.27. The pipe in the figure starts at the inlet with a cross sectional area of A_1 and constricts to an outlet with a smaller cross sectional area of A_2. The mass of fluid entering the pipe has to be equal to the mass of fluid leaving the pipe. For this reason the velocity at the outlet (v_2) is greater than the velocity of the inlet (v_1). Using the fact that the mass of fluid entering the pipe must be equal to the mass of fluid exiting the pipe, we can find a relationship between the velocity and the cross-sectional area by taking the rate of change of the mass in and the mass out:

$$\begin{aligned} \left(\frac{dm}{dt}\right)_1 &= \left(\frac{dm}{dt}\right)_2 \\ \rho_1 A_1 v_1 &= \rho_2 A_2 v_2. \end{aligned} \tag{14.15}$$

Equation 14.15 is also known as the continuity equation in general form. If the density of the fluid remains constant through the constriction—that is, the fluid is incompressible—then the density cancels from the continuity equation,

$$A_1 v_1 = A_2 v_2.$$

The equation reduces to show that the volume flow rate into the pipe equals the volume flow rate out of the pipe.

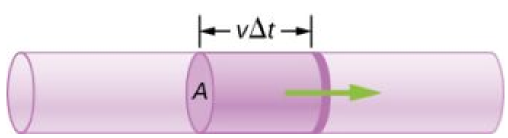


FIGURE 14.28 Geometry for deriving the equation of continuity. The amount of liquid entering the cross-sectional (shaded) area must equal the amount of liquid leaving the cross-sectional area if the liquid is incompressible.

14.6 Bernoulli's Equation

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the terms in Bernoulli's equation
- Explain how Bernoulli's equation is related to the conservation of energy
- Describe how to derive Bernoulli's principle from Bernoulli's equation
- Perform calculations using Bernoulli's principle
- Describe some applications of Bernoulli's principle

As we showed in Figure 14.27, when a fluid flows into a narrower channel, its speed increases. That means its kinetic energy also increases. The increased kinetic energy comes from the net work done on the fluid to push it into the channel. Also, if the fluid changes vertical position, work is done on the fluid by the gravitational force.

A pressure difference occurs when the channel narrows. This pressure difference results in a net force on the fluid because the pressure times the area equals the force, and this net force does work. Recall the work-energy theorem,

$$W_{\text{net}} = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2.$$

The net work done increases the fluid's kinetic energy. As a result, the pressure drops in a rapidly moving fluid whether or not the fluid is confined to a tube.

There are many common examples of pressure dropping in rapidly moving fluids. For instance, shower curtains have a disagreeable habit of bulging into the shower stall when the shower is on. The reason is that the high-velocity stream of water and air creates a region of lower pressure inside the shower, whereas the pressure on the other side remains at the standard atmospheric pressure. This pressure difference results in a net force, pushing the curtain inward. Similarly, when a car passes a truck on the highway, the two vehicles seem to pull toward each other. The reason is the same: The high velocity of the air between the car and the truck creates a region of lower pressure between the vehicles, and they are pushed together by greater pressure on the outside (Figure 14.29). This effect was observed as far back as the mid-1800s, when it was found that trains passing in opposite directions tipped precariously toward one another.

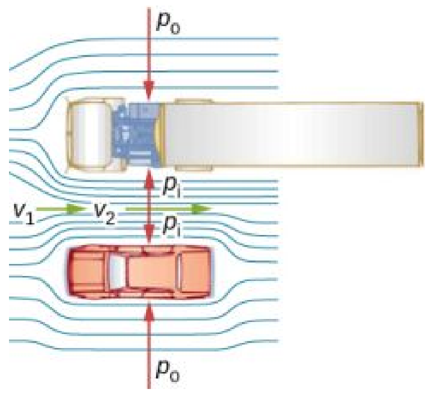


FIGURE 14.29 An overhead view of a car passing a truck on a highway. Air passing between the vehicles flows in a narrower channel and must increase its speed (v_2 is greater than v_1), causing the pressure between them to drop (p_i is less than p_o). Greater pressure on the outside pushes the car and truck together.

Energy Conservation and Bernoulli's Equation

The application of the principle of conservation of energy to frictionless laminar flow leads to a very useful relation between pressure and flow speed in a fluid. This relation is called **Bernoulli's equation**, named after Daniel Bernoulli (1700–1782), who published his studies on fluid motion in his book *Hydrodynamica* (1738).

Consider an incompressible fluid flowing through a pipe that has a varying diameter and height, as shown in Figure 14.30. Subscripts 1 and 2 in the figure denote two locations along the pipe and illustrate the relationships between the areas of the cross sections *A*, the speed of flow *v*, the height from ground *y*, and the pressure *p* at each point. We assume here that the density at the two points is the same—therefore, density is denoted by ρ without any subscripts—and since the fluid is incompressible, the shaded volumes must be equal.

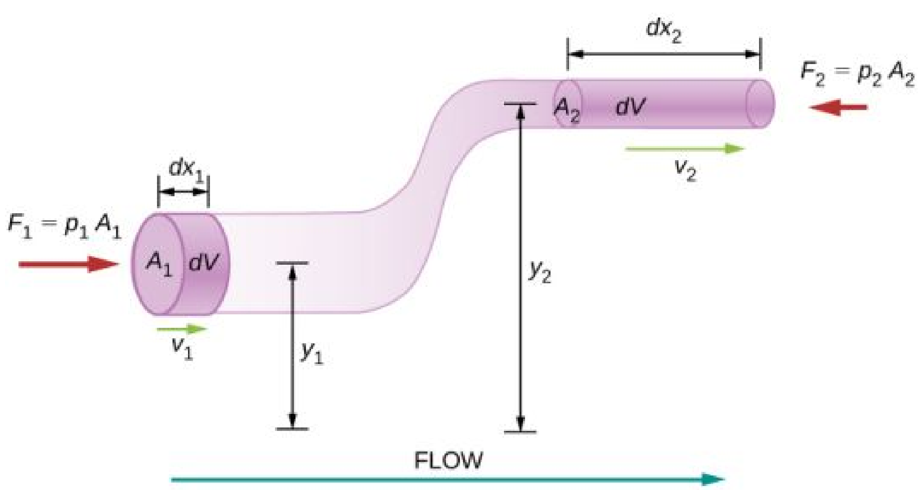


FIGURE 14.30 The geometry used for the derivation of Bernoulli's equation.

We also assume that there are no viscous forces in the fluid, so the energy of any part of the fluid will be conserved.

To derive Bernoulli's equation, we first calculate the work that was done on the fluid:

$$dW = F_1 dx_1 - F_2 dx_2$$
$$dW = p_1 A_1 dx_1 - p_2 A_2 dx_2 = p_1 dV - p_2 dV = (p_1 - p_2) dV.$$

The work done was due to the conservative force of gravity and the change in the kinetic energy of the fluid. The change in the kinetic energy of the fluid is equal to

$$dK = \frac{1}{2}m_2 v_2^2 - \frac{1}{2}m_1 v_1^2 = \frac{1}{2}\rho dV \left(v_2^2 - v_1^2\right).$$

The change in potential energy is

$$dU = mgy_2 - mgy_1 = \rho dVg(y_2 - y_1).$$

The energy equation then becomes

$$\begin{aligned} dW &= dK + dU \\ (p_1 - p_2) dV &= \tfrac{1}{2}\rho dV \left(v_2^2 - v_1^2\right) + \rho dVg(y_2 - y_1) \\ (p_1 - p_2) &= \tfrac{1}{2}\rho \left(v_2^2 - v_1^2\right) + \rho g(y_2 - y_1). \end{aligned}$$

Rearranging the equation gives Bernoulli's equation:

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2.$$

This relation states that the mechanical energy of any part of the fluid changes as a result of the work done by the fluid external to that part, due to varying pressure along the way. Since the two points were chosen arbitrarily, we can write Bernoulli's equation more generally as a conservation principle along the flow.

BERNOULLI'S EQUATION

For an incompressible, frictionless fluid, the combination of pressure and the sum of kinetic and potential energy densities is constant not only over time, but also along a streamline:

$$p + \frac{1}{2}\rho v^2 + \rho g y = \text{constant} \qquad (14.16)$$

A special note must be made here of the fact that in a dynamic situation, the pressures at the same height in different parts of the fluid may be different if they have different speeds of flow.

Analyzing Bernoulli's Equation

According to Bernoulli's equation, if we follow a small volume of fluid along its path, various quantities in the sum may change, but the total remains constant. Bernoulli's equation is, in fact, just a convenient statement of conservation of energy for an incompressible fluid in the absence of friction.

The general form of Bernoulli's equation has three terms in it, and it is broadly applicable. To understand it better, let us consider some specific situations that simplify and illustrate its use and meaning.

Bernoulli's equation for static fluids

First consider the very simple situation where the fluid is static—that is, $v_1 = v_2 = 0$. Bernoulli's equation in that case is

$$p_1 + \rho g h_1 = p_2 + \rho g h_2.$$

We can further simplify the equation by setting $h_2 = 0$. (Any height can be chosen for a reference height of zero, as is often done for other situations involving gravitational force, making all other heights relative.) In this case, we get

$$p_2 = p_1 + \rho g h_1.$$

This equation tells us that, in static fluids, pressure increases with depth. As we go from point 1 to point 2 in the fluid, the depth increases by h_1, and consequently, p_2 is greater than p_1 by an amount $\rho g h_1$. In the very simplest case, p_1 is zero at the top of the fluid, and we get the familiar relationship $p = \rho g h$. (Recall that $p = \rho g h$ and $\Delta U_g = -mgh$.) Thus, Bernoulli's equation confirms the fact that the pressure change due to the weight of a fluid is $\rho g h$. Although we introduce Bernoulli's equation for fluid motion, it includes much of what we studied for static fluids earlier.

Bernoulli's principle

Suppose a fluid is moving but its depth is constant—that is, $h_1 = h_2$. Under this condition, Bernoulli's equation becomes

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2.$$

Situations in which fluid flows at a constant depth are so common that this equation is often also called **Bernoulli's principle**, which is simply Bernoulli's equation for fluids at constant depth. (Note again that this applies to a small volume of fluid as we follow it along its path.) Bernoulli's principle reinforces the fact that pressure drops as speed increases in a moving fluid: If v_2 is greater than v_1 in the equation, then p_2 must be less than p_1 for the equality to hold.

EXAMPLE 14.6

Calculating Pressure

In Example 14.5, we found that the speed of water in a hose increased from 1.96 m/s to 25.5 m/s going from the hose to the nozzle. Calculate the pressure in the hose, given that the absolute pressure in the nozzle is $1.01 \times 10^5 \text{ N/m}^2$ (atmospheric, as it must be) and assuming level, frictionless flow.

Strategy

Level flow means constant depth, so Bernoulli's principle applies. We use the subscript 1 for values in the hose and 2 for those in the nozzle. We are thus asked to find p_1.

Solution

Solving Bernoulli's principle for p_1 yields

$$p_1 = p_2 + \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho(v_2^2 - v_1^2).$$

Substituting known values,

$$\begin{aligned} p_1 &= 1.01 \times 10^5 \text{ N/m}^2 + \tfrac{1}{2}(10^3 \text{ kg/m}^3)[(25.5 \text{ m/s})^2 - (1.96 \text{ m/s})^2] \\ &= 4.24 \times 10^5 \text{ N/m}^2. \end{aligned}$$

Significance

This absolute pressure in the hose is greater than in the nozzle, as expected, since *v* is greater in the nozzle. The pressure p_2 in the nozzle must be atmospheric, because the water emerges into the atmosphere without other changes in conditions.

Applications of Bernoulli's Principle

Many devices and situations occur in which fluid flows at a constant height and thus can be analyzed with Bernoulli's principle.

Entrainment

People have long put the Bernoulli principle to work by using reduced pressure in high-velocity fluids to move things about. With a higher pressure on the outside, the high-velocity fluid forces other fluids into the stream. This process is called *entrainment*. Entrainment devices have been in use since ancient times as pumps to raise water to small heights, as is necessary for draining swamps, fields, or other low-lying areas. Some other devices that use the concept of entrainment are shown in Figure 14.31.

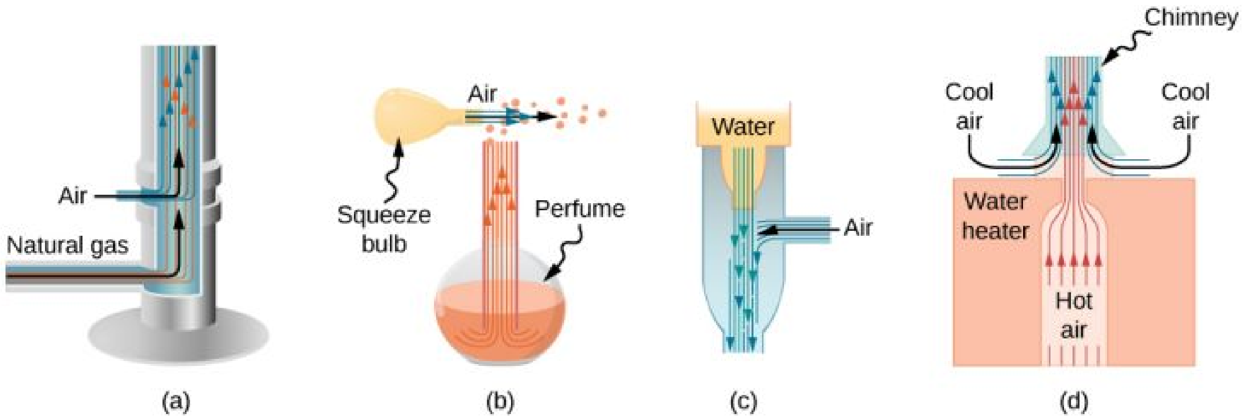


FIGURE 14.31 Entrainment devices use increased fluid speed to create low pressures, which then entrain one fluid into another. (a) A Bunsen burner uses an adjustable gas nozzle, entraining air for proper combustion. (b) An atomizer uses a squeeze bulb to create a jet of air that entrains drops of perfume. Paint sprayers and carburetors use very similar techniques to move their respective liquids. (c) A common aspirator uses a high-speed stream of water to create a region of lower pressure. Aspirators may be used as suction pumps in dental and surgical situations or for draining a flooded basement or producing a reduced pressure in a vessel. (d) The chimney of a water heater is designed to entrain air into the pipe leading through the ceiling.

Velocity measurement

Figure 14.32 shows two devices that apply Bernoulli's principle to measure fluid velocity. The manometer in part (a) is connected to two tubes that are small enough not to appreciably disturb the flow. The tube facing the oncoming fluid creates a dead spot having zero velocity ($v_1 = 0$) in front of it, while fluid passing the other tube has velocity v_2. This means that Bernoulli's principle as stated in

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

becomes

$$p_1 = p_2 + \frac{1}{2}\rho v_2^2.$$

Thus pressure p_2 over the second opening is reduced by $\frac{1}{2}\rho v_2^2$, so the fluid in the manometer rises by h on the side connected to the second opening, where

$$h \propto \frac{1}{2}\rho v_2^2.$$

(Recall that the symbol $\propto$ means "proportional to.") Solving for v_2, we see that

$$v_2 \propto \sqrt{h}.$$

Part (b) shows a version of this device that is in common use for measuring various fluid velocities; such devices are frequently used as air-speed indicators in aircraft.

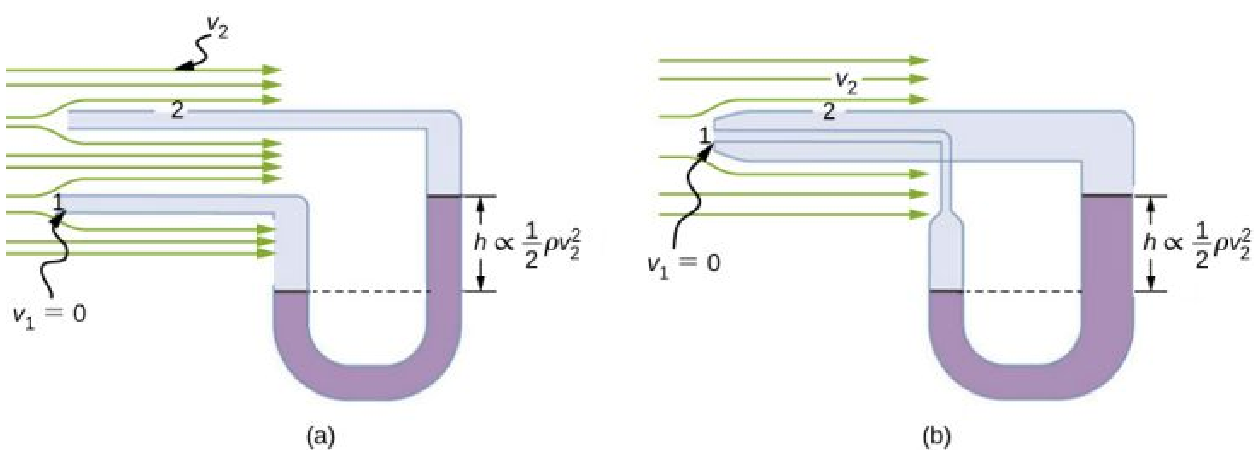


FIGURE 14.32 Measurement of fluid speed based on Bernoulli's principle. (a) A manometer is connected to two tubes that are close

together and small enough not to disturb the flow. Tube 1 is open at the end facing the flow. A dead spot having zero speed is created there. Tube 2 has an opening on the side, so the fluid has a speed *v* across the opening; thus, pressure there drops. The difference in pressure at the manometer is $\frac{1}{2}\rho v_2^2$, so *h* is proportional to $\frac{1}{2}\rho v_2^2$. (b) This type of velocity measuring device is a Prandtl tube, also known as a pitot tube.

A fire hose

All preceding applications of Bernoulli's equation involved simplifying conditions, such as constant height or constant pressure. The next example is a more general application of Bernoulli's equation in which pressure, velocity, and height all change.

EXAMPLE 14.7

Calculating Pressure: A Fire Hose Nozzle

Fire hoses used in major structural fires have an inside diameter of 6.40 cm (Figure 14.33). Suppose such a hose carries a flow of 40.0 L/s, starting at a gauge pressure of $1.62 \times 10^6 \text{N/m}^2$. The hose rises up 10.0 m along a ladder to a nozzle having an inside diameter of 3.00 cm. What is the pressure in the nozzle?

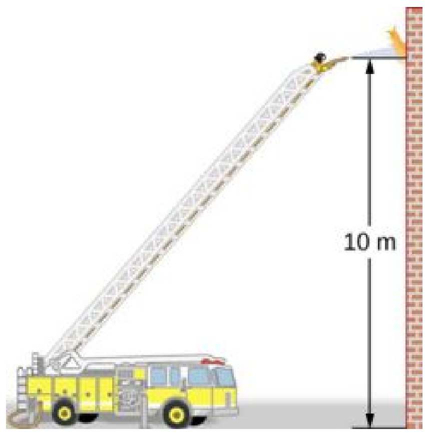


FIGURE 14.33 Pressure in the nozzle of this fire hose is less than at ground level for two reasons: The water has to go uphill to get to the nozzle, and speed increases in the nozzle. In spite of its lowered pressure, the water can exert a large force on anything it strikes by virtue of its kinetic energy. Pressure in the water stream becomes equal to atmospheric pressure once it emerges into the air.

Strategy

We must use Bernoulli's equation to solve for the pressure, since depth is not constant.

Solution

Bernoulli's equation is

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

where subscripts 1 and 2 refer to the initial conditions at ground level and the final conditions inside the nozzle, respectively. We must first find the speeds v_1 and v_2. Since $Q = A_1 v_1$, we get

$$v_1 = \frac{Q}{A_1} = \frac{40.0 \times 10^{-3} \text{m}^3/\text{s}}{\pi(3.20 \times 10^{-2} \text{m})^2} = 12.4 \text{m/s}.$$

Similarly, we find

$$v_2 = 56.6 \text{ m/s}.$$

This rather large speed is helpful in reaching the fire. Now, taking h_1 to be zero, we solve Bernoulli's equation for p_2:

$$p_2 = p_1 + \frac{1}{2}\rho(v_1^2 - v_2^2) - \rho g h_2.$$

Substituting known values yields

$$\begin{aligned} p_2 &= 1.62 \times 10^6 \text{ N/m}^2 + \tfrac{1}{2}(1000 \text{ kg/m}^3)[(12.4 \text{ m/s})^2 - (56.6 \text{ m/s})^2] \\ &\quad -(1000 \text{ kg/m}^3)(9.80 \text{ m/s}^2)(10.0 \text{ m}) \\ &= -2.9 \text{ kPa} \approx 0 \text{ kPa (when compared to air pressure).} \end{aligned}$$

Significance

This value is a gauge pressure, since the initial pressure was given as a gauge pressure. Thus, the nozzle pressure equals atmospheric pressure because the water exits into the atmosphere without changes in its conditions.

14.7 Viscosity and Turbulence

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain what viscosity is
- Calculate flow and resistance with Poiseuille's law
- Explain how pressure drops due to resistance
- Calculate the Reynolds number for an object moving through a fluid
- Use the Reynolds number for a system to determine whether it is laminar or turbulent
- Describe the conditions under which an object has a terminal speed

In Applications of Newton’s Laws, which introduced the concept of friction, we saw that an object sliding across the floor with an initial velocity and no applied force comes to rest due to the force of friction. Friction depends on the types of materials in contact and is proportional to the normal force. We also discussed drag and air resistance in that same chapter. We explained that at low speeds, the drag is proportional to the velocity, whereas at high speeds, drag is proportional to the velocity squared. In this section, we introduce the forces of friction that act on fluids in motion. For example, a fluid flowing through a pipe is subject to resistance, a type of friction, between the fluid and the walls. Friction also occurs between the different layers of fluid. These resistive forces affect the way the fluid flows through the pipe.

Viscosity and Laminar Flow

When you pour yourself a glass of juice, the liquid flows freely and quickly. But if you pour maple syrup on your pancakes, that liquid flows slowly and sticks to the pitcher. The difference is fluid friction, both within the fluid itself and between the fluid and its surroundings. We call this property of fluids *viscosity*. Juice has low viscosity, whereas syrup has high viscosity.

The precise definition of viscosity is based on laminar, or nonturbulent, flow. Figure 14.34 shows schematically how laminar and turbulent flow differ. When flow is laminar, layers flow without mixing. When flow is turbulent, the layers mix, and significant velocities occur in directions other than the overall direction of flow.

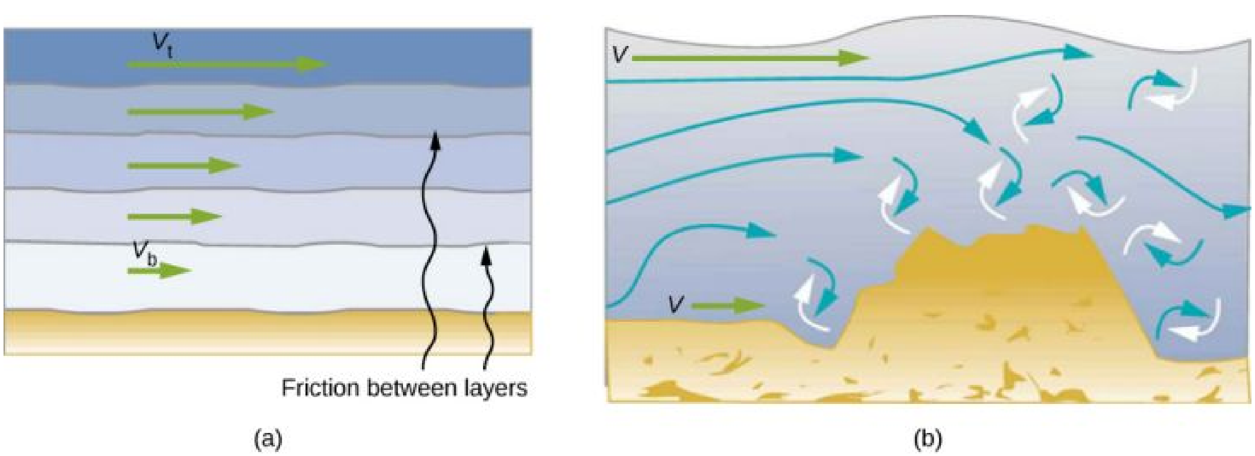


FIGURE 14.34 (a) Laminar flow occurs in layers without mixing. Notice that viscosity causes drag between layers as well as with the fixed surface. The speed near the bottom of the flow (v_b) is less than speed near the top (v_t) because in this case, the surface of the containing vessel is at the bottom. (b) An obstruction in the vessel causes turbulent flow. Turbulent flow mixes the fluid. There is more interaction, greater heating, and more resistance than in laminar flow.

Turbulence is a fluid flow in which layers mix together via eddies and swirls. It has two main causes. First, any obstruction or sharp corner, such as in a faucet, creates turbulence by imparting velocities perpendicular to the flow. Second, high speeds cause turbulence. The drag between adjacent layers of fluid and between the fluid and its surroundings can form swirls and eddies if the speed is great enough. In Figure 14.35, the speed of the accelerating smoke reaches the point that it begins to swirl due to the drag between the smoke and the surrounding air.

FIGURE 14.35 Smoke rises smoothly for a while and then begins to form swirls and eddies. The smooth flow is called laminar flow, whereas the swirls and eddies typify turbulent flow. Smoke rises more rapidly when flowing smoothly than after it becomes turbulent, suggesting that turbulence poses more resistance to flow. (credit: “Creativity103”/Flickr)

Figure 14.36 shows how viscosity is measured for a fluid. The fluid to be measured is placed between two parallel plates. The bottom plate is held fixed, while the top plate is moved to the right, dragging fluid with it. The layer (or lamina) of fluid in contact with either plate does not move relative to the plate, so the top layer moves at speed v while the bottom layer remains at rest. Each successive layer from the top down exerts a force on the one below it, trying to drag it along, producing a continuous variation in speed from v to 0 as shown. Care is taken to ensure that the flow is laminar, that is, the layers do not mix. The motion in the figure is like a continuous shearing motion. Fluids have zero shear strength, but the rate at which they are sheared is related to the same geometrical factors A and L as is shear deformation for solids. In the diagram, the fluid is initially at rest. The layer of fluid in contact with the moving plate is accelerated and starts to move due to the internal friction between moving plate and the fluid. The next layer is in contact with the moving layer; since there is internal friction between the two layers, it also accelerates, and so on through the depth of the fluid. There is also internal friction between the stationary plate and the lowest layer of fluid, next to the station plate. The force is required to keep the plate moving at a constant velocity due to the internal friction.

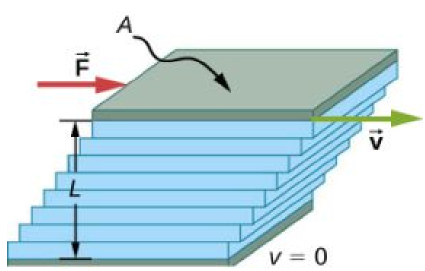


FIGURE 14.36 Measurement of viscosity for laminar flow of fluid between two plates of area A. The bottom plate is fixed. When the top plate is pushed to the right, it drags the fluid along with it.

A force F is required to keep the top plate in Figure 14.36 moving at a constant velocity v, and experiments have shown that this force depends on four factors. First, F is directly proportional to v (until the speed is so high that turbulence occurs—then a much larger force is needed, and it has a more complicated dependence on v). Second, F is proportional to the area A of the plate. This relationship seems reasonable, since A is directly proportional to the amount of fluid being moved. Third, F is inversely proportional to the distance between the plates L. This relationship is also reasonable; *L* is like a lever arm, and the greater the lever arm, the less the force that is needed. Fourth, *F* is directly proportional to the coefficient of viscosity, η. The greater the viscosity, the greater the force required. These dependencies are combined into the equation

$$F = \eta \frac{vA}{L}.$$

This equation gives us a working definition of fluid viscosity η. Solving for η gives

$$\eta = \frac{FL}{vA} \tag{14.17}$$

which defines viscosity in terms of how it is measured.

The SI unit of viscosity is $\mathrm{N \cdot m / \left[(m/s)\, m^2\right] = \left(N/m^2\right) s}$ or $\mathrm{Pa \cdot s}$. Table 14.4 lists the coefficients of viscosity for various fluids. Viscosity varies from one fluid to another by several orders of magnitude. As you might expect, the viscosities of gases are much less than those of liquids, and these viscosities often depend on temperature.

Fluid	Temperature (°C)	Viscosity $\eta \times 10^{-3}$ Pa · s
Air	0	0.0171
	20	0.0181
	40	0.0190
	100	0.0218
Ammonia	20	0.00974
Carbon dioxide	20	0.0147
Helium	20	0.0196
Hydrogen	0	0.0090
Mercury	20	0.0450
Oxygen	20	0.0203
Steam	100	0.0130
Liquid water	0	1.792
	20	1.002
	37	0.6947
	40	0.653
	100	0.282

TABLE 14.4 **Coefficients of Viscosity of Various Fluids**

Fluid	Temperature (°C)	Viscosity $\eta \times 10^{-3}$ Pa · s
Whole blood	20	3.015
	37	2.084
Blood plasma	20	1.810
	37	1.257
Ethyl alcohol	20	1.20
Methanol	20	0.584
Oil (heavy machine)	20	660
Oil (motor, SAE 10)	30	200
Oil (olive)	20	138
Glycerin	20	1500
Honey	20	2000–10000
Maple syrup	20	2000–3000
Milk	20	3.0
Oil (corn)	20	65

TABLE 14.4 **Coefficients of Viscosity of Various Fluids**

Laminar Flow Confined to Tubes: Poiseuille's Law

What causes flow? The answer, not surprisingly, is a pressure difference. In fact, there is a very simple relationship between horizontal flow and pressure. Flow rate *Q* is in the direction from high to low pressure. The greater the pressure differential between two points, the greater the flow rate. This relationship can be stated as

$$Q = \frac{p_2 - p_1}{R}$$

where p_1 and p_2 are the pressures at two points, such as at either end of a tube, and *R* is the resistance to flow. The resistance *R* includes everything, except pressure, that affects flow rate. For example, *R* is greater for a long tube than for a short one. The greater the viscosity of a fluid, the greater the value of *R*. Turbulence greatly increases *R*, whereas increasing the diameter of a tube decreases *R*.

If viscosity is zero, the fluid is frictionless and the resistance to flow is also zero. Comparing frictionless flow in a tube to viscous flow, as in Figure 14.37, we see that for a viscous fluid, speed is greatest at midstream because of drag at the boundaries. We can see the effect of viscosity in a Bunsen burner flame [part (c)], even though the viscosity of natural gas is small.

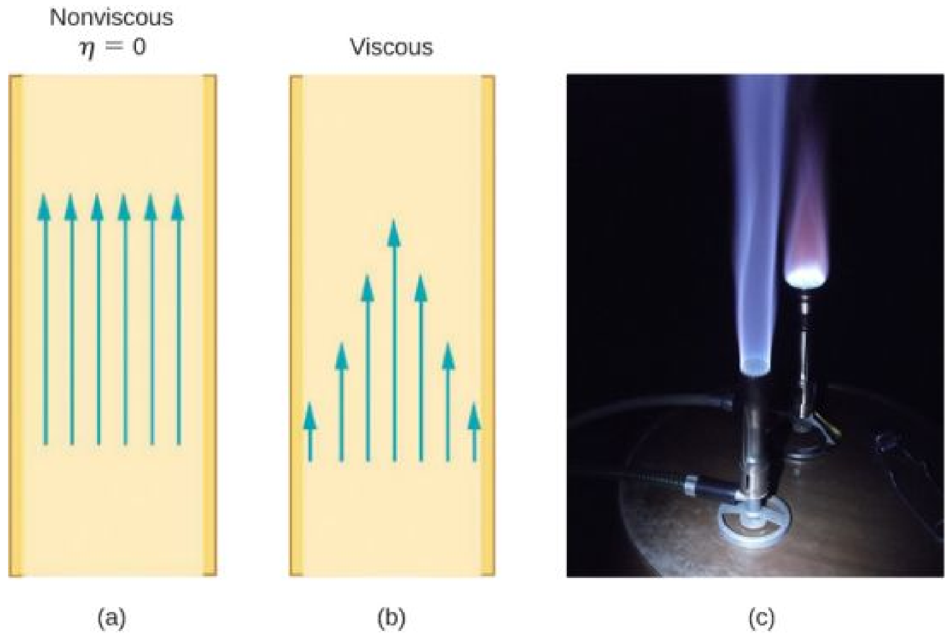


FIGURE 14.37 (a) If fluid flow in a tube has negligible resistance, the speed is the same all across the tube. (b) When a viscous fluid flows through a tube, its speed at the walls is zero, increasing steadily to its maximum at the center of the tube. (c) The shape of a Bunsen burner flame is due to the velocity profile across the tube. (credit c: modification of work by "jasonwoodhead23"/Flickr)

The resistance R to laminar flow of an incompressible fluid with viscosity η through a horizontal tube of uniform radius r and length l, is given by

$$R = \frac{8\eta l}{\pi r^4}. \tag{14.18}$$

This equation is called **Poiseuille's law for resistance**, named after the French scientist J. L. Poiseuille (1799–1869), who derived it in an attempt to understand the flow of blood through the body.

Let us examine Poiseuille's expression for R to see if it makes good intuitive sense. We see that resistance is directly proportional to both fluid viscosity η and the length l of a tube. After all, both of these directly affect the amount of friction encountered—the greater either is, the greater the resistance and the smaller the flow. The radius r of a tube affects the resistance, which again makes sense, because the greater the radius, the greater the flow (all other factors remaining the same). But it is surprising that r is raised to the fourth power in Poiseuille's law. This exponent means that any change in the radius of a tube has a very large effect on resistance. For example, doubling the radius of a tube decreases resistance by a factor of $2^4 = 16$.

Taken together, $Q = \frac{p_2 - p_1}{R}$ and $R = \frac{8\eta l}{\pi r^4}$ give the following expression for flow rate:

$$Q = \frac{(p_2 - p_1)\pi r^4}{8\eta l}. \tag{14.19}$$

This equation describes laminar flow through a tube. It is sometimes called Poiseuille's law for laminar flow, or simply **Poiseuille's law** (Figure 14.38).

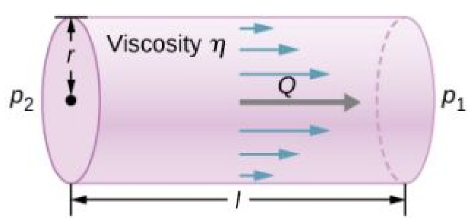


FIGURE 14.38 Poiseuille's law applies to laminar flow of an incompressible fluid of viscosity η through a tube of length l and radius r. The direction of flow is from greater to lower pressure. Flow rate Q is directly proportional to the pressure difference $p_2 - p_1$, and inversely

proportional to the length l of the tube and viscosity η of the fluid. Flow rate increases with radius by a factor of r^4.

EXAMPLE 14.8

Using Flow Rate: Air Conditioning Systems

An air conditioning system is being designed to supply air at a gauge pressure of 0.054 Pa at a temperature of $20\ °\text{C}$. The air is sent through an insulated, round conduit with a diameter of 18.00 cm. The conduit is 20-meters long and is open to a room at atmospheric pressure 101.30 kPa. The room has a length of 12 meters, a width of 6 meters, and a height of 3 meters. (a) What is the volume flow rate through the pipe, assuming laminar flow? (b) Estimate the length of time to completely replace the air in the room. (c) The builders decide to save money by using a conduit with a diameter of 9.00 cm. What is the new flow rate?

Strategy

Assuming laminar flow, Poiseuille's law states that

$$Q = \frac{(p_2 - p_1)\pi r^4}{8\eta l} = \frac{dV}{dt}.$$

We need to compare the conduit radius before and after the flow rate reduction. Note that we are given the diameter of the conduit, so we must divide by two to get the radius.

Solution

a. Assuming a constant pressure difference and using the viscosity $\eta = 0.0181\ \text{mPa} \cdot \text{s}$,
$$Q = \frac{(0.054\ \text{Pa})\,(3.14)\,(0.09\ \text{m})^4}{8\left(0.0181 \times 10^{-3}\ \text{Pa} \cdot \text{s}\right)(20\ \text{m})} = 3.84 \times 10^{-3}\ \frac{\text{m}^3}{\text{s}}.$$

b. Assuming constant flow $Q = \frac{dV}{dt} \approx \frac{\Delta V}{\Delta t}$
$$\Delta t = \frac{\Delta V}{Q} = \frac{(12\ \text{m})\,(6\ \text{m})\,(3\ \text{m})}{3.84 \times 10^{-3}\ \frac{\text{m}^3}{\text{s}}} = 5.63 \times 10^4\ \text{s} = 15.63\ \text{hr}.$$

c. Using laminar flow, Poiseuille's law yields
$$Q = \frac{(0.054\ \text{Pa})\,(3.14)\,(0.045\ \text{m})^4}{8\left(0.0181 \times 10^{-3}\ \text{Pa} \cdot \text{s}\right)(20\ \text{m})} = 2.40 \times 10^{-4}\ \frac{\text{m}^3}{\text{s}}.$$

Thus, the radius of the conduit decreases by half reduces the flow rate to 6.25% of the original value.

Significance

In general, assuming laminar flow, decreasing the radius has a more dramatic effect than changing the length. If the length is increased and all other variables remain constant, the flow rate is decreased:

$$\begin{aligned} \frac{Q_A}{Q_B} &= \frac{\frac{(p_2 - p_1)\pi r_A^4}{8\eta l_A}}{\frac{(p_2 - p_1)\pi r_B^4}{8\eta l_B}} = \frac{l_B}{l_A} \\ Q_B &= \frac{l_A}{l_B} Q_A. \end{aligned}$$

Doubling the length cuts the flow rate to one-half the original flow rate.

If the radius is decreased and all other variables remain constant, the volume flow rate decreases by a much larger factor.

$$\begin{aligned} \frac{Q_A}{Q_B} &= \frac{\frac{(p_2-p_1)\pi r_A^4}{8\eta l_A}}{\frac{(p_2-p_1)\pi r_B^4}{8\eta l_B}} = \left(\frac{r_A}{r_B}\right)^4 \\ Q_B &= \left(\frac{r_B}{r_A}\right)^4 Q_A \end{aligned}$$

Cutting the radius in half decreases the flow rate to one-sixteenth the original flow rate.

Flow and Resistance as Causes of Pressure Drops

Water pressure in homes is sometimes lower than normal during times of heavy use, such as hot summer days. The drop in pressure occurs in the water main before it reaches individual homes. Let us consider flow through the water main as illustrated in Figure 14.39. We can understand why the pressure p_1 to the home drops during times of heavy use by rearranging the equation for flow rate:

$$\begin{aligned} Q &= \frac{p_2-p_1}{R} \\ p_2-p_1 &= RQ. \end{aligned}$$

In this case, p_2 is the pressure at the water works and *R* is the resistance of the water main. During times of heavy use, the flow rate *Q* is large. This means that $p_2 - p_1$ must also be large. Thus p_1 must decrease. It is correct to think of flow and resistance as causing the pressure to drop from p_2 to p_1. The equation $p_2 - p_1 = RQ$ is valid for both laminar and turbulent flows.

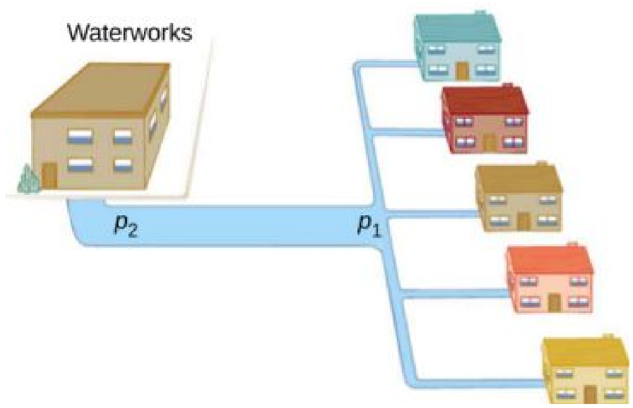


FIGURE 14.39 During times of heavy use, there is a significant pressure drop in a water main, and p_1 supplied to users is significantly less than p_2 created at the water works. If the flow is very small, then the pressure drop is negligible, and $p_2 \approx p_1$.

We can also use $p_2 - p_1 = RQ$ to analyze pressure drops occurring in more complex systems in which the tube radius is not the same everywhere. Resistance is much greater in narrow places, such as in an obstructed coronary artery. For a given flow rate *Q*, the pressure drop is greatest where the tube is most narrow. This is how water faucets control flow. Additionally, *R* is greatly increased by turbulence, and a constriction that creates turbulence greatly reduces the pressure downstream. Plaque in an artery reduces pressure and hence flow, both by its resistance and by the turbulence it creates.

Measuring Turbulence

An indicator called the **Reynolds number** N_R can reveal whether flow is laminar or turbulent. For flow in a tube of uniform diameter, the Reynolds number is defined as

$$N_R = \frac{2\rho v r}{\eta}\text{(flow in tube)} \quad 14.20$$

where ρ is the fluid density, *v* its speed, η its viscosity, and *r* the tube radius. The Reynolds number is a dimensionless quantity. Experiments have revealed that N_R is related to the onset of turbulence. For N_R below about 2000, flow is laminar. For N_R above about 3000, flow is turbulent.

For values of N_R between about 2000 and 3000, flow is unstable—that is, it can be laminar, but small obstructions and surface roughness can make it turbulent, and it may oscillate randomly between being laminar and turbulent. In fact, the flow of a fluid with a Reynolds number between 2000 and 3000 is a good example of chaotic behavior. A system is defined to be chaotic when its behavior is so sensitive to some factor that it is extremely difficult to predict. It is difficult, but not impossible, to predict whether flow is turbulent or not when a fluid's Reynold's number falls in this range due to extremely sensitive dependence on factors like roughness and obstructions on the nature of the flow. A tiny variation in one factor has an exaggerated (or nonlinear) effect on the flow.

EXAMPLE 14.9

Using Flow Rate: Turbulent Flow or Laminar Flow

In Example 14.8, we found the volume flow rate of an air conditioning system to be $Q = 3.84 \times 10^{-3}\ \text{m}^3/\text{s}$. This calculation assumed laminar flow. (a) Was this a good assumption? (b) At what velocity would the flow become turbulent?

Strategy

To determine if the flow of air through the air conditioning system is laminar, we first need to find the velocity, which can be found by

$$Q = Av = \pi r^2 v.$$

Then we can calculate the Reynold's number, using the equation below, and determine if it falls in the range for laminar flow

$$N_R = \frac{2\rho v r}{\eta}.$$

Solution

a. Using the values given:

$$\begin{aligned} v &= \frac{Q}{\pi r^2} = \frac{3.84 \times 10^{-3}\ \frac{\text{m}^3}{\text{s}}}{3.14(0.09\ \text{m})^2} = 0.15\frac{\text{m}}{\text{s}} \\ N_R &= \frac{2\rho v r}{\eta} = \frac{2\left(1.23\frac{\text{kg}}{\text{m}^3}\right)\left(0.15\ \frac{\text{m}}{\text{s}}\right)(0.09\ \text{m})}{0.0181 \times 10^{-3}\ \text{Pa·s}} = 1835. \end{aligned}$$

Since the Reynolds number is 1835 < 2000, the flow is laminar and not turbulent. The assumption that the flow was laminar is valid.

b. To find the maximum speed of the air to keep the flow laminar, consider the Reynold's number.

$$\begin{aligned} N_R &= \frac{2\rho v r}{\eta} \leq 2000 \\ v &= \frac{2000\left(0.0181 \times 10^{-3}\text{Pa·s}\right)}{2\left(1.23\frac{\text{kg}}{\text{m}^3}\right)(0.09\ \text{m})} = 0.16\frac{\text{m}}{\text{s}}. \end{aligned}$$

Significance

When transferring a fluid from one point to another, it desirable to limit turbulence. Turbulence results in wasted energy, as some of the energy intended to move the fluid is dissipated when eddies are formed. In this case, the air conditioning system will become less efficient once the velocity exceeds 0.16 m/s, since this is the point at which turbulence will begin to occur.

Chapter Review

Key Terms

absolute pressure sum of gauge pressure and atmospheric pressure

Archimedes' principle buoyant force on an object equals the weight of the fluid it displaces

Bernoulli's equation equation resulting from applying conservation of energy to an incompressible frictionless fluid: $p + \frac{1}{2}\rho v^2 + \rho gh = \text{constant, throughout the fluid}$

Bernoulli's principle Bernoulli's equation applied at constant depth:

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

buoyant force net upward force on any object in any fluid due to the pressure difference at different depths

density mass per unit volume of a substance or object

flow rate abbreviated *Q*, it is the volume *V* that flows past a particular point during a time *t*, or $Q = dV/dt$

fluids liquids and gases; a fluid is a state of matter that yields to shearing forces

gauge pressure pressure relative to atmospheric pressure

hydraulic jack simple machine that uses cylinders of different diameters to distribute force

hydrostatic equilibrium state at which water is not flowing, or is static

ideal fluid fluid with negligible viscosity

laminar flow type of fluid flow in which layers do not mix

Pascal's principle change in pressure applied to an enclosed fluid is transmitted undiminished to all portions of the fluid and to the walls of its container

Poiseuille's law rate of laminar flow of an incompressible fluid in a tube: $Q = \frac{(p_2 - p_1)\pi r^4}{8\eta l}$.

Poiseuille's law for resistance resistance to laminar flow of an incompressible fluid in a tube: $R = \frac{8\eta l}{\pi r^4}$

pressure force per unit area exerted perpendicular to the area over which the force acts

Reynolds number dimensionless parameter that can reveal whether a particular flow is laminar or turbulent

specific gravity ratio of the density of an object to a fluid (usually water)

turbulence fluid flow in which layers mix together via eddies and swirls

turbulent flow type of fluid flow in which layers mix together via eddies and swirls

viscosity measure of the internal friction in a fluid

Key Equations

Density of a sample at constant density	$\rho = \frac{m}{V}$
Pressure	$p = \frac{F}{A}$
Pressure at a depth *h* in a fluid of constant density	$p = p_0 + \rho gh$
Change of pressure with height in a constant-density fluid	$\frac{dp}{dy} = -\rho g$
Absolute pressure	$p_{\text{abs}} = p_{\text{g}} + p_{\text{atm}}$
Pascal's principle	$\frac{F_1}{A_1} = \frac{F_2}{A_2}$
Volume flow rate	$Q = \frac{dV}{dt}$
Continuity equation (constant density)	$A_1 v_1 = A_2 v_2$
Continuity equation (general form)	$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$
Bernoulli's equation	$p + \frac{1}{2}\rho v^2 + \rho gy = \text{constant}$
Viscosity	$\eta = \frac{FL}{vA}$
Poiseuille's law for resistance	$R = \frac{8\eta l}{\pi r^4}$
Poiseuille's law	$Q = \frac{(p_2 - p_1)\pi r^4}{8\eta l}$

Summary

14.1 Fluids, Density, and Pressure

- A fluid is a state of matter that yields to sideways or shearing forces. Liquids and gases are both fluids. Fluid statics is the physics of stationary fluids.
- Density is the mass per unit volume of a substance or object, defined as $\rho = m/V$. The SI unit of density is kg/m^3.
- Pressure is the force per unit perpendicular area over which the force is applied, $p = F/A$. The SI unit of pressure is the pascal: $1 \text{ Pa} = 1 \text{ N/m}^2$.
- Pressure due to the weight of a liquid of constant density is given by $p = \rho gh$, where *p* is the pressure, *h* is the depth of the liquid, ρ is the density of the liquid, and *g* is the acceleration due to gravity.

14.2 Measuring Pressure

- Gauge pressure is the pressure relative to atmospheric pressure.
- Absolute pressure is the sum of gauge pressure and atmospheric pressure.
- Open-tube manometers have U-shaped tubes and one end is always open. They are used to measure pressure. A mercury barometer is a device that measures atmospheric pressure.
- The SI unit of pressure is the pascal (Pa), but several other units are commonly used.

14.3 Pascal's Principle and Hydraulics

- Pressure is force per unit area.
- A change in pressure applied to an enclosed fluid is transmitted undiminished to all portions of the fluid and to the walls of its container.
- A hydraulic system is an enclosed fluid system used to exert forces.

14.4 Archimedes' Principle and Buoyancy

- Buoyant force is the net upward force on any object in any fluid. If the buoyant force is greater than the object's weight, the object will rise to the surface and float. If the buoyant force is less than the object's weight, the object will sink. If the buoyant force equals the object's weight, the object can remain suspended at its present depth. The buoyant force is always present and acting on any object immersed either partially or entirely in a fluid.
- Archimedes' principle states that the buoyant force on an object equals the weight of the fluid it displaces.

14.5 Fluid Dynamics

- Flow rate *Q* is defined as the volume *V* flowing past a point in time *t*, or $Q = \frac{dV}{dt}$ where *V* is volume and *t* is time. The SI unit of flow rate is m^3/s, but other rates can be used, such as L/min.
- Flow rate and velocity are related by $Q = Av$ where *A* is the cross-sectional area of the flow and *v* is its average velocity.
- The equation of continuity states that for an incompressible fluid, the mass flowing into a pipe must equal the mass flowing out of the pipe.

14.6 Bernoulli's Equation

- Bernoulli's equation states that the sum on each side of the following equation is constant, or the same at any two points in an incompressible frictionless fluid:
$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2.$$
- Bernoulli's principle is Bernoulli's equation applied to situations in which the height of the fluid is constant. The terms involving depth (or height *h*) subtract out, yielding
$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2.$$
- Bernoulli's principle has many applications, including entrainment and velocity measurement.

14.7 Viscosity and Turbulence

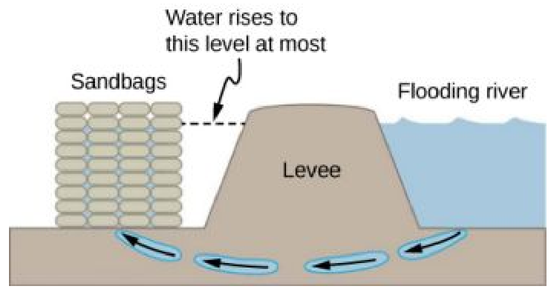
- Laminar flow is characterized by smooth flow of the fluid in layers that do not mix.
- Turbulence is characterized by eddies and swirls that mix layers of fluid together.
- Fluid viscosity η is due to friction within a fluid.
- Flow is proportional to pressure difference and inversely proportional to resistance:
$$Q = \frac{p - 2p_1}{R}.$$
- The pressure drop caused by flow and resistance is given by $p_2 - p_1 = RQ$.
- The Reynolds number N_R can reveal whether flow is laminar or turbulent. It is $N_\text{R} = \frac{2\rho vr}{\eta}$.
- For N_R below about 2000, flow is laminar. For N_R above about 3000, flow is turbulent. For values of N_R between 2000 and 3000, it may be either or both.

Conceptual Questions

14.1 Fluids, Density, and Pressure

1. Which of the following substances are fluids at room temperature and atmospheric pressure: air, mercury, water, glass?
2. Why are gases easier to compress than liquids and solids?
3. Explain how the density of air varies with altitude.
4. The image shows a glass of ice water filled to the brim. Will the water overflow when the ice melts? Explain your answer.

5. How is pressure related to the sharpness of a knife and its ability to cut?
6. Why is a force exerted by a static fluid on a surface always perpendicular to the surface?
7. Imagine that in a remote location near the North Pole, a chunk of ice floats in a lake. Next to the lake, a glacier with the same volume as the floating ice sits on land. If both chunks of ice should melt due to rising global temperatures, and the melted ice all goes into the lake, which one would cause the level of the lake to rise the most? Explain.
8. In ballet, dancing *en pointe* (on the tips of the toes) is much harder on the toes than normal dancing or walking. Explain why, in terms of pressure.
9. Atmospheric pressure exerts a large force (equal to the weight of the atmosphere above your body—about 10 tons) on the top of your body when you are lying on the beach sunbathing. Why are you able to get up?
10. Why does atmospheric pressure decrease more rapidly than linearly with altitude?
11. The image shows how sandbags placed around a leak outside a river levee can effectively stop the flow of water under the levee. Explain how the small amount of water inside the column of sandbags is able to balance the much larger body of water behind the levee.



12. Is there a net force on a dam due to atmospheric pressure? Explain your answer.
13. Does atmospheric pressure add to the gas pressure in a rigid tank? In a toy balloon? When, in general, does atmospheric pressure not affect the total pressure in a fluid?
14. You can break a strong wine bottle by pounding a cork into it with your fist, but the cork must press directly against the liquid filling the bottle—there can be no air between the cork and liquid. Explain why the bottle breaks only if there is no air between the cork and liquid.

14.2 Measuring Pressure

15. Explain why the fluid reaches equal levels on either side of a manometer if both sides are open to the atmosphere, even if the tubes are of different diameters.

14.3 Pascal's Principle and Hydraulics

16. Suppose the master cylinder in a hydraulic system is at a greater height than the cylinder it is controlling. Explain how this will affect the force produced at the cylinder that is being controlled.

14.4 Archimedes' Principle and Buoyancy

17. More force is required to pull the plug in a full bathtub than when it is empty. Does this contradict Pascal's principle? Explain your answer.
18. Do fluids exert buoyant forces in a "weightless" environment, such as in the space shuttle? Explain your answer.

19. Will the same ship float higher in salt water than in freshwater? Explain your answer.

20. Marbles dropped into a partially filled bathtub sink to the bottom. Part of their weight is supported by buoyant force, yet the downward force on the bottom of the tub increases by exactly the weight of the marbles. Explain why.

14.5 Fluid Dynamics

21. Many figures in the text show streamlines. Explain why fluid velocity is greatest where streamlines are closest together. (*Hint:* Consider the relationship between fluid velocity and the cross-sectional area through which the fluid flows.)

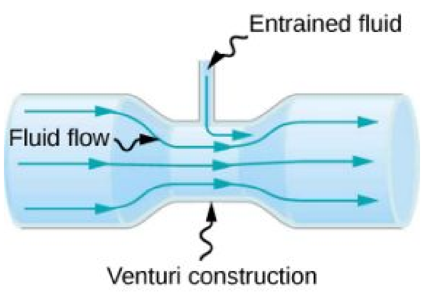
14.6 Bernoulli's Equation

22. You can squirt water from a garden hose a considerably greater distance by partially covering the opening with your thumb. Explain how this works.

23. Water is shot nearly vertically upward in a decorative fountain and the stream is observed to broaden as it rises. Conversely, a stream of water falling straight down from a faucet narrows. Explain why.

24. Look back to Figure 14.29. Answer the following two questions. Why is p_o less than atmospheric? Why is p_o greater than p_i?

25. A tube with a narrow segment designed to enhance entrainment is called a Venturi, such as shown below. Venturis are very commonly used in carburetors and aspirators. How does this structure bolster entrainment?



26. Some chimney pipes have a T-shape, with a crosspiece on top that helps draw up gases whenever there is even a slight breeze. Explain how this works in terms of Bernoulli's principle.

27. Is there a limit to the height to which an entrainment device can raise a fluid? Explain your answer.

28. Why is it preferable for airplanes to take off into the wind rather than with the wind?

29. Roofs are sometimes pushed off vertically during a tropical cyclone, and buildings sometimes explode outward when hit by a tornado. Use Bernoulli's principle to explain these phenomena.

30. It is dangerous to stand close to railroad tracks when a rapidly moving commuter train passes. Explain why atmospheric pressure would push you toward the moving train.

31. Water pressure inside a hose nozzle can be less than atmospheric pressure due to the Bernoulli effect. Explain in terms of energy how the water can emerge from the nozzle against the opposing atmospheric pressure.

32. David rolled down the window on his car while driving on the freeway. An empty plastic bag on the floor promptly flew out the window. Explain why.

33. Based on Bernoulli's equation, what are three forms of energy in a fluid? (Note that these forms are conservative, unlike heat transfer and other dissipative forms not included in Bernoulli's equation.)

34. The old rubber boot shown below has two leaks. To what maximum height can the water squirt from Leak 1? How does the velocity of water emerging from Leak 2 differ from that of Leak 1? Explain your responses in terms of energy.

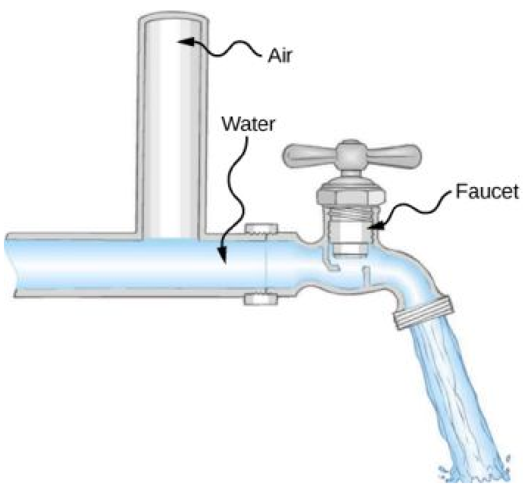
35. Water pressure inside a hose nozzle can be less than atmospheric pressure due to the Bernoulli effect. Explain in terms of energy how the water can emerge from the nozzle against the opposing atmospheric pressure.

14.7 Viscosity and Turbulence

36. Explain why the viscosity of a liquid decreases with temperature, that is, how might an increase in temperature reduce the effects of cohesive forces in a liquid? Also explain why the viscosity of a gas increases with temperature, that is, how does increased gas temperature create more collisions between atoms and molecules?

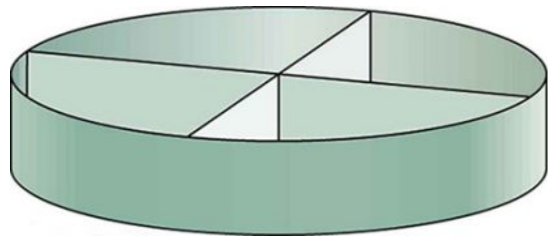
37. When paddling a canoe upstream, it is wisest to travel as near to the shore as possible. When canoeing downstream, it is generally better to stay near the middle. Explain why.

38. Plumbing usually includes air-filled tubes near water faucets (see the following figure). Explain why they are needed and how they work.



39. Doppler ultrasound can be used to measure the speed of blood in the body. If there is a partial constriction of an artery, where would you expect blood speed to be greatest: at or after the constriction? What are the two distinct causes of higher resistance in the constriction?

40. Sink drains often have a device such as that shown below to help speed the flow of water. How does this work?



Problems

14.1 Fluids, Density, and Pressure

41. Gold is sold by the troy ounce (31.103 g). What is the volume of 1 troy ounce of pure gold?

42. Mercury is commonly supplied in flasks containing 34.5 kg (about 76 lb.). What is the volume in liters of this much mercury?

43. What is the mass of a deep breath of air having a volume of 2.00 L? Discuss the effect taking such a breath has on your body's volume and density.

44. A straightforward method of finding the density of an object is to measure its mass and then measure its volume by submerging it in a graduated cylinder. What is the density of a 240-g rock that displaces 89.0 cm^3 of water? (Note that the accuracy and practical applications of this technique are more limited than a variety of others that are based on Archimedes' principle.)

45. Suppose you have a coffee mug with a circular cross-section and vertical sides (uniform radius). What is its inside radius if it holds 375 g of coffee when filled to a depth of 7.50 cm? Assume coffee has the same density as water.

46. A rectangular gasoline tank can hold 50.0 kg of gasoline when full. What is the depth of the tank if it is 0.500-m wide by 0.900-m long? (b) Discuss whether this gas tank has a reasonable volume for a passenger car.

47. A trash compactor can compress its contents to 0.350 times their original volume. Neglecting the mass of air expelled, by what factor is the density of the rubbish increased?

48. A 2.50-kg steel gasoline can holds 20.0 L of gasoline when full. What is the average density of the full gas can, taking into account the volume occupied by steel as well as by gasoline?

49. What is the density of 18.0-karat gold that is a mixture of 18 parts gold, 5 parts silver, and 1 part copper? (These values are parts by mass, not volume.) Assume that this is a simple mixture having an average density equal to the weighted densities of its constituents.

50. The tip of a nail exerts tremendous pressure when hit by a hammer because it exerts a large force over a small area. What force must be exerted on a nail with a circular tip of 1.00-mm diameter to create a pressure of $3.00 \times 10^9 \text{ N/m}^2$? (This high pressure is possible because the hammer striking the nail is brought to rest in such a short distance.)

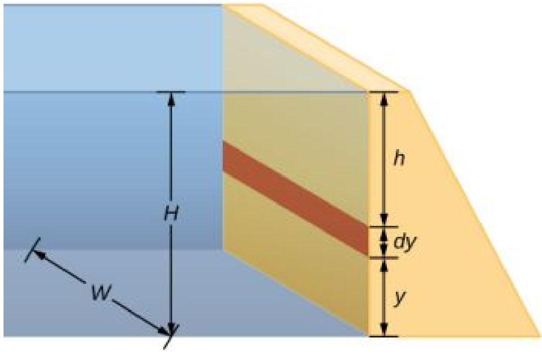
51. A glass tube contains mercury. What would be the height of the column of mercury which would create pressure equal to 1.00 atm?

52. The greatest ocean depths on Earth are found in the Marianas Trench near the Philippines. Calculate the pressure due to the ocean at the bottom of this trench, given its depth is 11.0 km and assuming the density of seawater is constant all the way down.

53. Verify that the SI unit of $h\rho g$ is N/m^2.

54. What pressure is exerted on the bottom of a gas tank that is 0.500-m wide and 0.900-m long and can hold 50.0 kg of gasoline when full?

55. A dam is used to hold back a river. The dam has a height $H = 12 \text{ m}$ and a width $W = 10 \text{ m}$. Assume that the density of the water is $\rho = 1000 \text{ kg/m}^3$. (a) Determine the net force on the dam. (b) Why does the thickness of the dam increase with depth?



14.2 Measuring Pressure

56. Find the gauge and absolute pressures in the balloon and peanut jar shown in Figure 14.12, assuming the manometer connected to the balloon uses water and the manometer connected to the jar contains mercury. Express in units of centimeters of water for the balloon and millimeters of mercury for the jar, taking $h = 0.0500\text{m}$ for each.

57. How tall must a water-filled manometer be to measure blood pressure as high as 300 mm Hg?

58. Assuming bicycle tires are perfectly flexible and support the weight of bicycle and rider by pressure alone, calculate the total area of the tires in contact with the ground if a bicycle and rider have a total mass of 80.0 kg, and the gauge pressure in the tires is $3.50 \times 10^5 \text{ Pa}$.

14.3 Pascal's Principle and Hydraulics

59. How much pressure is transmitted in the hydraulic system considered in Example 14.3? Express your answer in atmospheres.

60. What force must be exerted on the master cylinder of a hydraulic lift to support the weight of a 2000-kg car (a large car) resting on a second cylinder? The master cylinder has a 2.00-cm diameter and the second cylinder has a 24.0-cm diameter.

61. A host pours the remnants of several bottles of wine into a jug after a party. The host then inserts a cork with a 2.00-cm diameter into the bottle, placing it in direct contact with the wine. The host is amazed when the host pounds the cork into place and the bottom of the jug (with a 14.0-cm diameter) breaks away. Calculate the extra force exerted against the bottom if he pounded the cork with a 120-N force.

62. A certain hydraulic system is designed to exert a force 100 times as large as the one put into it. (a) What must be the ratio of the area of the cylinder that is being controlled to the area of the master cylinder? (b) What must be the ratio of their diameters? (c) By what factor is the distance through which the output force moves reduced relative to the distance through which the input force moves? Assume no losses due to friction.

63. Verify that work input equals work output for a hydraulic system assuming no losses due to friction. Do this by showing that the distance the output force moves is reduced by the same factor that the output force is increased. Assume the volume of the fluid is constant. What effect would friction within the fluid and between components in the system have on the output force? How would this depend on whether or not the fluid is moving?

14.4 Archimedes' Principle and Buoyancy

64. What fraction of ice is submerged when it floats in freshwater, given the density of water at $0\ °\text{C}$ is very close to $1000\ \text{kg/m}^3$?

65. If a person's body has a density of $995\ \text{kg/m}^3$, what fraction of the body will be submerged when floating gently in (a) freshwater? (b) In salt water with a density of $1027\ \text{kg/m}^3$?

66. A rock with a mass of 540 g in air is found to have an apparent mass of 342 g when submerged in water. (a) What mass of water is displaced? (b) What is the volume of the rock? (c) What is its average density? Is this consistent with the value for granite?

67. Archimedes' principle can be used to calculate the density of a fluid as well as that of a solid. Suppose a chunk of iron with a mass of 390.0 g in air is found to have an apparent mass of 350.5 g when completely submerged in an unknown liquid. (a) What mass of fluid does the iron displace? (b) What is the volume of iron, using its density as given in Table 14.1? (c) Calculate the fluid's density and identify it.

68. Calculate the buoyant force on a 2.00-L helium balloon. (b) Given the mass of the rubber in the balloon is 1.50 g, what is the net vertical force on the balloon if it is let go? Neglect the volume of the rubber.

69. What is the density of a woman who floats in fresh water with 4.00% of her volume above the surface? (This could be measured by placing her in a tank with marks on the side to measure how much water she displaces when floating and when held under water.) (b) What percent of her volume is above the surface when she floats in seawater?

70. A man has a mass of 80 kg and a density of 955kg/m^3 (excluding the air in his lungs). (a) Calculate his volume. (b) Find the buoyant force air exerts on him. (c) What is the ratio of the buoyant force to his weight?

71. A simple compass can be made by placing a small bar magnet on a cork floating in water. (a) What fraction of a plain cork will be submerged when floating in water? (b) If the cork has a mass of 10.0 g and a 20.0-g magnet is placed on it, what fraction of the cork will be submerged? (c) Will the bar magnet and cork float in ethyl alcohol?

72. What percentage of an iron anchor's weight will be supported by buoyant force when submerged in salt water?

73. Referring to Figure 14.20, prove that the buoyant force on the cylinder is equal to the weight of the fluid displaced (Archimedes' principle). You may assume that the buoyant force is $F_2 - F_1$ and that the ends of the cylinder have equal areas A. Note that the volume of the cylinder (and that of the fluid it displaces) equals $(h_2 - h_1)A$.

74. A 75.0-kg man floats in freshwater with 3.00% of his volume above water when his lungs are empty, and 5.00% of his volume above water when his lungs are full. Calculate the volume of air he inhales—called his lung capacity—in liters. (b) Does this lung volume seem reasonable?

14.5 Fluid Dynamics

75. What is the average flow rate in cm^3/s of gasoline to the engine of a car traveling at 100 km/h if it averages 10.0 km/L?

76. The heart of a resting adult pumps blood at a rate of 5.00 L/min. (a) Convert this to cm^3/s. (b) What is this rate in m^3/s?

77. The Huka Falls on the Waikato River is one of New Zealand's most visited natural tourist attractions. On average, the river has a flow rate of about 300,000 L/s. At the gorge, the river narrows to 20-m wide and averages 20-m deep. (a) What is the average speed of the river in the gorge? (b) What is the average speed of the water in the river downstream of the falls when it widens to 60 m and its depth increases to an average of 40 m?

78. (a) Estimate the time it would take to fill a private swimming pool with a capacity of 80,000 L using a garden hose delivering 60 L/min. (b) How long would it take if you could divert a moderate size river, flowing at $5000\ \text{m}^3/\text{s}$ into the pool?

79. What is the fluid speed in a fire hose with a 9.00-cm diameter carrying 80.0 L of water per second? (b) What is the flow rate in cubic meters per second? (c) Would your answers be different if salt water replaced the fresh water in the fire hose?

80. Water is moving at a velocity of 2.00 m/s through a hose with an internal diameter of 1.60 cm. (a) What is the flow rate in liters per second? (b) The fluid velocity in this hose's nozzle is 15.0 m/s. What is the nozzle's inside diameter?

81. Prove that the speed of an incompressible fluid through a constriction, such as in a Venturi tube, increases by a factor equal to the square of the factor by which the diameter decreases. (The converse applies for flow out of a constriction into a larger-diameter region.)

82. Water emerges straight down from a faucet with a 1.80-cm diameter at a speed of 0.500 m/s. (Because of the construction of the faucet, there is no variation in speed across the stream.) (a) What is the flow rate in cm^3/s? (b) What is the diameter of the stream 0.200 m below the faucet? Neglect any effects due to surface tension.

14.6 Bernoulli's Equation

83. Verify that pressure has units of energy per unit volume.

84. Suppose you have a wind speed gauge like the pitot tube shown in Figure 14.32. By what factor must wind speed increase to double the value of *h* in the manometer? Is this independent of the moving fluid and the fluid in the manometer?

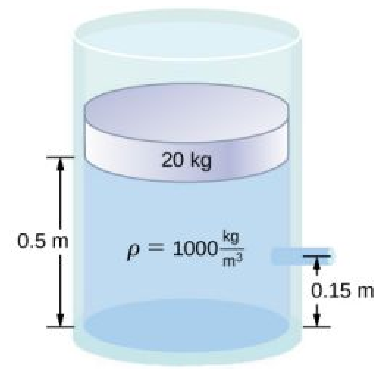
85. If the pressure reading of your pitot tube is 15.0 mm Hg at a speed of 200 km/h, what will it be at 700 km/h at the same altitude?

86. Every few years, winds in Boulder, Colorado, attain sustained speeds of 45.0 m/s (about 100 mph) when the jet stream descends during early spring. Approximately what is the force due to the Bernoulli equation on a roof having an area of $220\mathrm{m}^2$? Typical air density in Boulder is $1.14\mathrm{kg/m}^3$, and the corresponding atmospheric pressure is $8.89 \times 10^4\mathrm{N/m}^2$. (Bernoulli's principle as stated in the text assumes laminar flow. Using the principle here produces only an approximate result, because there is significant turbulence.)

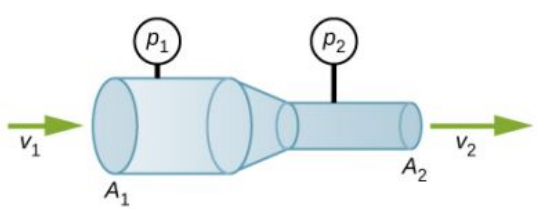
87. What is the pressure drop due to the Bernoulli Effect as water goes into a 3.00-cm-diameter nozzle from a 9.00-cm-diameter fire hose while carrying a flow of 40.0 L/s? (b) To what maximum height above the nozzle can this water rise? (The actual height will be significantly smaller due to air resistance.)

88. (a) Using Bernoulli's equation, show that the measured fluid speed *v* for a pitot tube, like the one in Figure 14.32(b), is given by $v = \left(\frac{2\rho' gh}{\rho}\right)^{1/2}$, where *h* is the height of the manometer fluid, ρ' is the density of the manometer fluid, ρ is the density of the moving fluid, and *g* is the acceleration due to gravity. (Note that *v* is indeed proportional to the square root of *h*, as stated in the text.) (b) Calculate *v* for moving air if a mercury manometer's *h* is 0.200 m.

89. A container of water has a cross-sectional area of $A = 0.1\ \mathrm{m}^2$. A piston sits on top of the water (see the following figure). There is a spout located 0.15 m from the bottom of the tank, open to the atmosphere, and a stream of water exits the spout. The cross sectional area of the spout is $A_\mathrm{s} = 7.0 \times 10^{-4}\mathrm{m}^2$. (a) What is the velocity of the water as it leaves the spout? (b) If the opening of the spout is located 1.5 m above the ground, how far from the spout does the water hit the floor? Ignore all friction and dissipative forces.



90. A fluid of a constant density flows through a reduction in a pipe. Find an equation for the change in pressure, in terms of v_1, A_1, A_2, and the density.



14.7 Viscosity and Turbulence

91. (a) Calculate the retarding force due to the viscosity of the air layer between a cart and a level air track given the following information: air temperature is $20\ ^\circ\mathrm{C}$, the cart is moving at 0.400 m/s, its surface area is $2.50 \times 10^{-2}\ \mathrm{m}^2$, and the thickness of the air layer is $6.00 \times 10^{-5}\,\mathrm{m}$. (b) What is the ratio of this force to the weight of the 0.300-kg cart?

92. The arterioles (small arteries) leading to an organ constrict in order to decrease flow to the organ. To shut down an organ, blood flow is reduced naturally to 1.00% of its original value. By what factor do the radii of the arterioles constrict?

93. A spherical particle falling at a terminal speed in a liquid must have the gravitational force balanced by the drag force and the buoyant force. The buoyant force is equal to the weight of the displaced fluid, while the drag force is assumed to be given by Stokes Law, $F_\mathrm{s} = 6\pi r\eta v$. Show that the terminal speed is given by $v = \frac{2R^2 g}{9\eta}(\rho_\mathrm{s} - \rho_1)$, where *R* is the radius of the sphere, ρ_s is its density, and ρ_1 is the density of the fluid, and η the coefficient of viscosity.

94. Using the equation of the previous problem, find the viscosity of motor oil in which a steel ball of radius 0.8 mm falls with a terminal speed of 4.32 cm/s. The densities of the ball and the oil are 7.86 and 0.88 g/mL, respectively.

95. A skydiver will reach a terminal velocity when the air drag equals their weight. For a skydiver with a large body, turbulence is a factor at high speeds. The drag force then is approximately proportional to the square of the velocity. Taking the drag force to be $F_\mathrm{D} = \frac{1}{2}\rho A v^2$, and setting this equal to the skydiver's weight, find the terminal speed for a person falling "spread eagle."

96. (a) Verify that a 19.0% decrease in laminar flow through a tube is caused by a 5.00% decrease in radius, assuming that all other factors remain constant. (b) What increase in flow is obtained from a 5.00% increase in radius, again assuming all other factors remain constant?

97. When physicians diagnose arterial blockages, they quote the reduction in flow rate. If the flow rate in an artery has been reduced to 10.0% of its normal value by a blood clot and the average pressure difference has increased by 20.0%, by what factor has the clot reduced the radius of the artery?

98. An oil gusher shoots crude oil 25.0 m into the air through a pipe with a 0.100-m diameter. Neglecting air resistance but not the resistance of the pipe, and assuming laminar flow, calculate the pressure at the entrance of the 50.0-m-long vertical pipe. Take the density of the oil to be $900\ \mathrm{kg/m^3}$ and its viscosity to be $1.00(\mathrm{N/m^2})\cdot\mathrm{s}$ (or $1.00\ \mathrm{Pa}\cdot\mathrm{s}$). Note that you must take into account the pressure due to the 50.0-m column of oil in the pipe.

99. Concrete is pumped from a cement mixer to the place it is being laid, instead of being carried in wheelbarrows. The flow rate is 200 L/min through a 50.0-m-long, 8.00-cm-diameter hose, and the pressure at the pump is $8.00 \times 10^6\ \mathrm{N/m^2}$. (a) Calculate the resistance of the hose. (b) What is the viscosity of the concrete, assuming the flow is laminar? (c) How much power is being supplied, assuming the point of use is at the same level as the pump? You may neglect the power supplied to increase the concrete's velocity.

100. Verify that the flow of oil is laminar for an oil gusher that shoots crude oil 25.0 m into the air through a pipe with a 0.100-m diameter. The vertical pipe is 50 m long. Take the density of the oil to be $900\ \mathrm{kg/m^3}$ and its viscosity to be $1.00(\mathrm{N/m^2})\cdot\mathrm{s}$ (or $1.00\ \mathrm{Pa}\cdot\mathrm{s}$).

101. Calculate the Reynolds numbers for the flow of water through (a) a nozzle with a radius of 0.250 cm and (b) a garden hose with a radius of 0.900 cm, when the nozzle is attached to the hose. The flow rate through hose and nozzle is 0.500 L/s. Can the flow in either possibly be laminar?

102. A fire hose has an inside diameter of 6.40 cm. Suppose such a hose carries a flow of 40.0 L/s starting at a gauge pressure of $1.62 \times 10^6\ \mathrm{N/m^2}$. The hose goes 10.0 m up a ladder to a nozzle having an inside diameter of 3.00 cm. Calculate the Reynolds numbers for flow in the fire hose and nozzle to show that the flow in each must be turbulent.

103. At what flow rate might turbulence begin to develop in a water main with a 0.200-m diameter? Assume a $20\ ^\circ\mathrm{C}$ temperature.

Additional Problems

104. Before digital storage devices, such as the memory in your cell phone, music was stored on vinyl disks with grooves with varying depths cut into the disk. A phonograph used a needle, which moved over the grooves, measuring the depth of the grooves. The pressure exerted by a phonograph needle on a record is surprisingly large. If the equivalent of 1.00 g is supported by a needle, the tip of which is a circle with a 0.200-mm radius, what pressure is exerted on the record in Pa?

105. Water towers store water above the level of consumers for times of heavy use, eliminating the need for high-speed pumps. How high above a user must the water level be to create a gauge pressure of $3.00 \times 10^5\ \text{N/m}^2$?

106. The aqueous humor in a person's eye is exerting a force of 0.300 N on the 1.10-cm^2 area of the cornea. What pressure is this in mm Hg?

107. (a) Convert normal blood pressure readings of 120 over 80 mm Hg to newtons per meter squared using the relationship for pressure due to the weight of a fluid ($p = h\rho g$) rather than a conversion factor. (b) Explain why the blood pressure of an infant would likely be smaller than that of an adult. Specifically, consider the smaller height to which blood must be pumped.

108. Pressure cookers have been around for more than 300 years, although their use has greatly declined in recent years (early models had a nasty habit of exploding). How much force must the latches holding the lid onto a pressure cooker be able to withstand if the circular lid is $25.0\ \text{cm}$ in diameter and the gauge pressure inside is 300 atm? Neglect the weight of the lid.

109. Bird bones have air pockets in them to reduce their weight—this also gives them an average density significantly less than that of the bones of other animals. Suppose an ornithologist weighs a bird bone in air and in water and finds its mass is 45.0 g and its apparent mass when submerged is 3.60 g (assume the bone is watertight). (a) What mass of water is displaced? (b) What is the volume of the bone? (c) What is its average density?

110. In an immersion measurement of a woman's density, she is found to have a mass of 62.0 kg in air and an apparent mass of 0.0850 kg when completely submerged with lungs empty. (a) What mass of water does she displace? (b) What is her volume? (c) Calculate her density. (d) If her lung capacity is 1.75 L, is she able to float without treading water with her lungs filled with air?

111. Some fish have a density slightly less than that of water and must exert a force (swim) to stay submerged. What force must an 85.0-kg grouper exert to stay submerged in salt water if its body density is $1015\ \text{kg/m}^3$?

112. The human circulation system has approximately 1×10^9 capillary vessels. Each vessel has a diameter of about $8\,\mu\text{m}$. Assuming cardiac output is 5 L/min, determine the average velocity of blood flow through each capillary vessel.

113. The flow rate of blood through a 2.00×10^{-6} m-radius capillary is $3.80 \times 10^{-9}\ \text{cm}^3/\text{s}$. (a) What is the speed of the blood flow? (b) Assuming all the blood in the body passes through capillaries, how many of them must there be to carry a total flow of $90.0\ \text{cm}^3/\text{s}$?

114. The left ventricle of a resting adult's heart pumps blood at a flow rate of $83.0\ \text{cm}^3/\text{s}$, increasing its pressure by 110 mm Hg, its speed from zero to 30.0 cm/s, and its height by 5.00 cm. (All numbers are averaged over the entire heartbeat.) Calculate the total power output of the left ventricle. Note that most of the power is used to increase blood pressure.

115. A sump pump (used to drain water from the basement of houses built below the water table) is draining a flooded basement at the rate of 0.750 L/s, with an output pressure of $3.00 \times 10^5\ \text{N/m}^2$. (a) The water enters a hose with a 3.00-cm inside diameter and rises 2.50 m above the pump. What is its pressure at this point? (b) The hose goes over the foundation wall, losing 0.500 m in height, and widens to 4.00 cm in diameter. What is the pressure now? You may neglect frictional losses in both parts of the problem.

116. A glucose solution being administered with an IV has a flow rate of$4.00\ \text{cm}^3/\text{min}$. What will the new flow rate be if the glucose is replaced by whole blood having the same density but a viscosity 2.50 times that of the glucose? All other factors remain constant.

117. A small artery has a length of 1.1×10^{-3} m and a radius of 2.5×10^{-5} m. If the pressure drop across the artery is 1.3 kPa, what is the flow rate through the artery? (Assume that the temperature is $37\ °\text{C}$.)

118. Angioplasty is a technique in which arteries partially blocked with plaque are dilated to increase blood flow. By what factor must the radius of an artery be increased in order to increase blood flow by a factor of 10?

119. Suppose a blood vessel's radius is decreased to 90.0% of its original value by plaque deposits and the body compensates by increasing the pressure difference along the vessel to keep the flow rate constant. By what factor must the pressure difference increase? (b) If turbulence is created by the obstruction, what additional effect would it have on the flow rate?

Challenge Problems

120. The pressure on the dam shown early in the problems section increases with depth. Therefore, there is a net torque on the dam. Find the net torque.

121. The temperature of the atmosphere is not always constant and can increase or decrease with height. In a neutral atmosphere, where there is not a significant amount of vertical mixing, the temperature decreases at a rate of approximately 6.5 K per km. The magnitude of the decrease in temperature as height increases is known as the lapse rate (Γ). (The symbol is the upper case Greek letter gamma.) Assume that the surface pressure is $p_0 = 1.013 \times 10^5$ Pa where $\text{T} = 293$ K and the lapse rate is $\left(-\Gamma = 6.5 \frac{\text{K}}{\text{km}}\right)$. Estimate the pressure 3.0 km above the surface of Earth.

122. A submarine is stranded on the bottom of the ocean with its hatch 25.0 m below the surface. Calculate the force needed to open the hatch from the inside, given it is circular and 0.450 m in diameter. Air pressure inside the submarine is 1.00 atm.

123. Logs sometimes float vertically in a lake because one end has become water-logged and denser than the other. What is the average density of a uniform-diameter log that floats with 20.0% of its length above water?

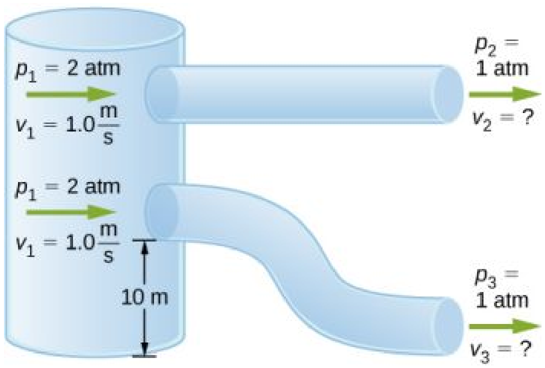
124. Scurrilous con artists have been known to represent gold-plated tungsten ingots as pure gold and sell them at prices much below gold value but high above the cost of tungsten. With what accuracy must you be able to measure the mass of such an ingot in and out of water to tell that it is almost pure tungsten rather than pure gold?

125. The inside volume of a house is equivalent to that of a rectangular solid 13.0 m wide by 20.0 m long by 2.75 m high. The house is heated by a forced air gas heater. The main uptake air duct of the heater is 0.300 m in diameter. What is the average speed of air in the duct if it carries a volume equal to that of the house's interior every 15 minutes?

126. A garden hose with a diameter of 2.0 cm is used to fill a bucket, which has a volume of 0.10 cubic meters. It takes 1.2 minutes to fill. An adjustable nozzle is attached to the hose to decrease the diameter of the opening, which increases the speed of the water. The hose is held level to the ground at a height of 1.0 meters and the diameter is decreased until a flower bed 3.0 meters away is reached. (a) What is the volume flow rate of the water through the nozzle when the diameter is 2.0 cm? (b) What is the speed of the water coming out of the hose? (c) What does the speed of the water coming out of the hose need to be to reach the flower bed 3.0 meters away? (d) What is the diameter of the nozzle needed to reach the flower bed?

127. A frequently quoted rule of thumb in aircraft design is that wings should produce about 1000 N of lift per square meter of wing. (The fact that a wing has a top and bottom surface does not double its area.) (a) At takeoff, an aircraft travels at 60.0 m/s, so that the air speed relative to the bottom of the wing is 60.0 m/s. Given the sea level density of air as $1.29\ \text{kg/m}^3$, how fast must it move over the upper surface to create the ideal lift? (b) How fast must air move over the upper surface at a cruising speed of 245 m/s and at an altitude where air density is one-fourth that at sea level? (Note that this is not all of the aircraft's lift—some comes from the body of the plane, some from engine thrust, and so on. Furthermore, Bernoulli's principle gives an approximate answer because flow over the wing creates turbulence.)

128. Two pipes of equal and constant diameter leave a water pumping station and dump water out of an open end that is open to the atmosphere (see the following figure). The water enters at a pressure of two atmospheres and a speed of $(v_1 = 1.0\,\text{m/s})$. One pipe drops a height of 10 m. What is the velocity of the water as the water leaves each pipe?



129. Fluid originally flows through a tube at a rate of $100\,\text{cm}^3/\text{s}$. To illustrate the sensitivity of flow rate to various factors, calculate the new flow rate for the following changes with all other factors remaining the same as in the original conditions. (a) Pressure difference increases by a factor of 1.50. (b) A new fluid with 3.00 times greater viscosity is substituted. (c) The tube is replaced by one having 4.00 times the length. (d) Another tube is used with a radius 0.100 times the original. (e) Yet another tube is substituted with a radius 0.100 times the original and half the length, and the pressure difference is increased by a factor of 1.50.

130. During a marathon race, a runner's blood flow increases to 10.0 times her resting rate. Her blood's viscosity has dropped to 95.0% of its normal value, and the blood pressure difference across the circulatory system has increased by 50.0%. By what factor has the average radii of her blood vessels increased?

131. Water supplied to a house by a water main has a pressure of $3.00 \times 10^5\,\text{N/m}^2$ early on a summer day when neighborhood use is low. This pressure produces a flow of 20.0 L/min through a garden hose. Later in the day, pressure at the exit of the water main and entrance to the house drops, and a flow of only 8.00 L/min is obtained through the same hose. (a) What pressure is now being supplied to the house, assuming resistance is constant? (b) By what factor did the flow rate in the water main increase in order to cause this decrease in delivered pressure? The pressure at the entrance of the water main is $5.00 \times 10^5\,\text{N/m}^2$, and the original flow rate was 200 L/min. (c) How many more users are there, assuming each would consume 20.0 L/min in the morning?

132. Gasoline is piped underground from refineries to major users. The flow rate is $3.00 \times 10^{-2}\,\text{m}^3/\text{s}$ (about 500 gal/min), the viscosity of gasoline is $1.00 \times 10^{-3}\,(\text{N/m}^2)\cdot\text{s}$, and its density is $680\,\text{kg/m}^3$. (a) What minimum diameter must the pipe have if the Reynolds number is to be less than 2000? (b) What pressure difference must be maintained along each kilometer of the pipe to maintain this flow rate?

CHAPTER 15
Oscillations

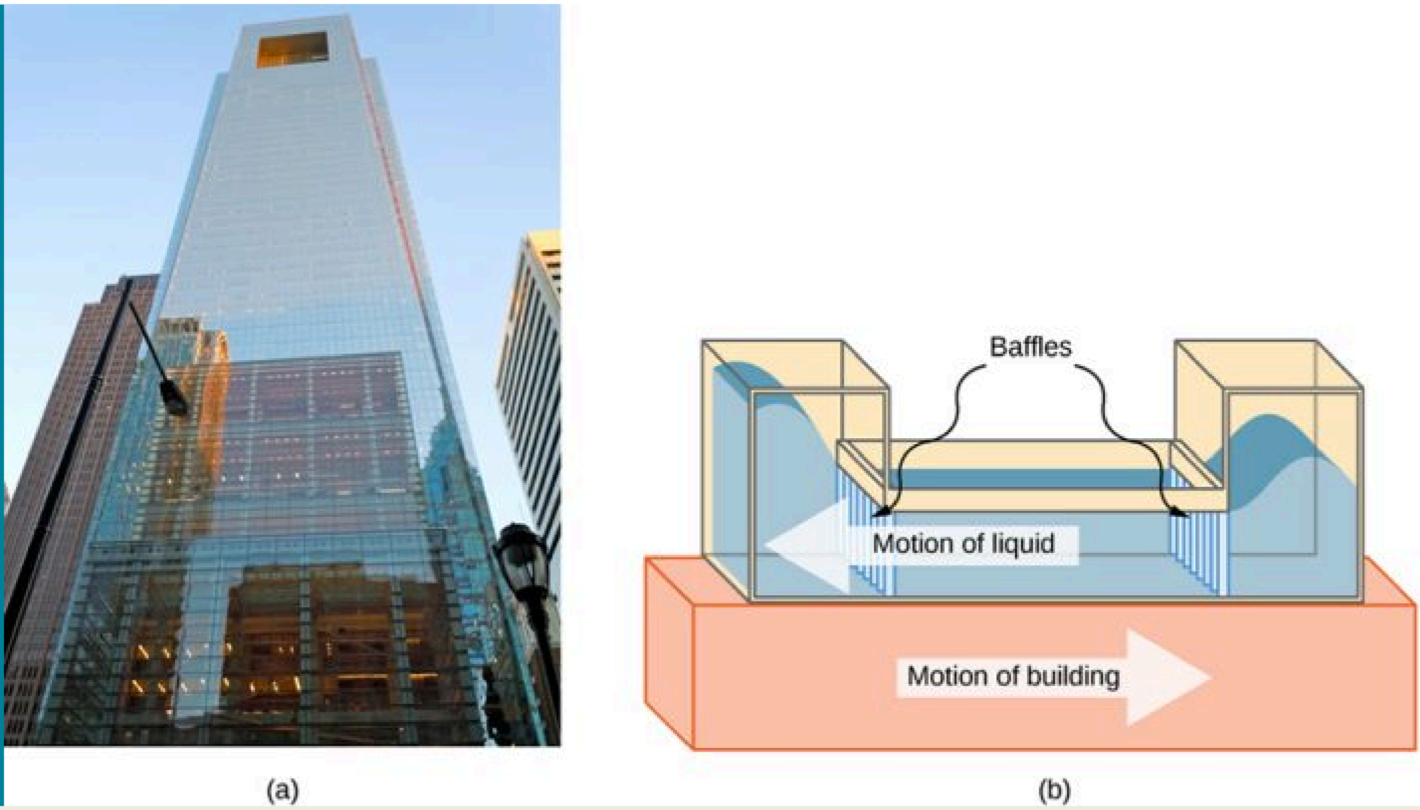


FIGURE 15.1 (a) The Comcast Building in Philadelphia, Pennsylvania, looming high above the skyline, is approximately 305 meters (1000 feet) tall. At this height, the top floors can oscillate back and forth due to seismic activity and fluctuating winds. (b) Shown above is a schematic drawing of a tuned, liquid-column mass damper, installed at the top of the Comcast, consisting of a 300,000-gallon reservoir of water to reduce oscillations.

CHAPTER OUTLINE

INTRODUCTION We begin the study of oscillations with simple systems of pendulums and springs. Although these systems may seem quite basic, the concepts involved have many real-life applications. For example, the Comcast Building in Philadelphia, Pennsylvania, stands approximately 305 meters (1000 feet) tall. As buildings are built taller, they can act as inverted, physical pendulums, with the top floors oscillating due to seismic activity and fluctuating winds. In the Comcast Building, a tuned-mass damper is used to reduce the oscillations. Installed at the top of the building is a tuned, liquid-column mass damper, consisting of a 300,000-gallon reservoir of water. This U-shaped tank allows the water to oscillate freely at a frequency that matches the natural frequency of the building. Damping is provided by tuning the turbulence levels in the moving water using baffles.

15.1 Simple Harmonic Motion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define the terms period and frequency
- List the characteristics of simple harmonic motion
- Explain the concept of phase shift
- Write the equations of motion for the system of a mass and spring undergoing simple harmonic motion
- Describe the motion of a mass oscillating on a vertical spring

When you pluck a guitar string, the resulting sound has a steady tone and lasts a long time (Figure 15.2). The string vibrates around an equilibrium position, and one oscillation is completed when the string starts from the initial position, travels to one of the extreme positions, then to the other extreme position, and returns to its initial position. We define **periodic motion** to be any motion that repeats itself at regular time intervals, such as exhibited by the guitar string or by a child swinging on a swing. In this section, we study the basic characteristics of oscillations and their mathematical description.

FIGURE 15.2 When a guitar string is plucked, the string oscillates up and down in periodic motion. The vibrating string causes the surrounding air molecules to oscillate, producing sound waves. (credit: Yutaka Tsutano)

Period and Frequency in Oscillations

In the absence of friction, the time to complete one oscillation remains constant and is called the **period (T)**. Its units are usually seconds, but may be any convenient unit of time. The word 'period' refers to the time for some event whether repetitive or not, but in this chapter, we shall deal primarily in periodic motion, which is by definition repetitive.

A concept closely related to period is the frequency of an event. **Frequency (f)** is defined to be the number of events per unit time. For periodic motion, frequency is the number of oscillations per unit time. The relationship between frequency and period is

$$f = \frac{1}{T}. \tag{15.1}$$

The SI unit for frequency is the *hertz* (Hz) and is defined as one *cycle per second*:

$$1\ \text{Hz} = 1\frac{\text{cycle}}{\text{s}} \quad \text{or} \quad 1\ \text{Hz} = \frac{1}{\text{s}} = 1\ \text{s}^{-1}.$$

A cycle is one complete **oscillation**.

EXAMPLE 15.1

Determining the Frequency of Medical Ultrasound

Ultrasound machines are used by medical professionals to make images for examining internal organs of the body. An ultrasound machine emits high-frequency sound waves, which reflect off the organs, and a computer receives the waves, using them to create a picture. We can use the formulas presented in this module to determine the frequency, based on what we know about oscillations. Consider a medical imaging device that produces ultrasound by oscillating with a period of $0.400\ \mu\text{s}$. What is the frequency of this oscillation?

Strategy

The period (*T*) is given and we are asked to find frequency (*f*).

Solution

Substitute $0.400\ \mu\text{s}$ for *T* in $f = \frac{1}{T}$:

$$f = \frac{1}{T} = \frac{1}{0.400 \times 10^{-6}\ \text{s}}.$$

Solve to find

$$f = 2.50 \times 10^{6}\ \text{Hz}.$$

Significance

This frequency of sound is much higher than the highest frequency that humans can hear (the range of human hearing is 20 Hz to 20,000 Hz); therefore, it is called ultrasound. Appropriate oscillations at this frequency generate ultrasound used for noninvasive medical diagnoses, such as observations of a fetus in the womb.

Characteristics of Simple Harmonic Motion

A very common type of periodic motion is called **simple harmonic motion (SHM)**. A system that oscillates with SHM is called a **simple harmonic oscillator**.

SIMPLE HARMONIC MOTION

In simple harmonic motion, the acceleration of the system, and therefore the net force, is proportional to the displacement and acts in the opposite direction of the displacement.

A good example of SHM is an object with mass *m* attached to a spring on a frictionless surface, as shown in Figure 15.3. The object oscillates around the equilibrium position, and the net force on the object is equal to the force provided by the spring. This force obeys Hooke's law $F_s = -kx$, as discussed in a previous chapter.

If the net force can be described by Hooke's law and there is no *damping* (slowing down due to friction or other nonconservative forces), then a simple harmonic oscillator oscillates with equal displacement on either side of the equilibrium position, as shown for an object on a spring in Figure 15.3. The maximum displacement from equilibrium is called the **amplitude (*A*)**. The units for amplitude and displacement are the same but depend on the type of oscillation. For the object on the spring, the units of amplitude and displacement are meters.

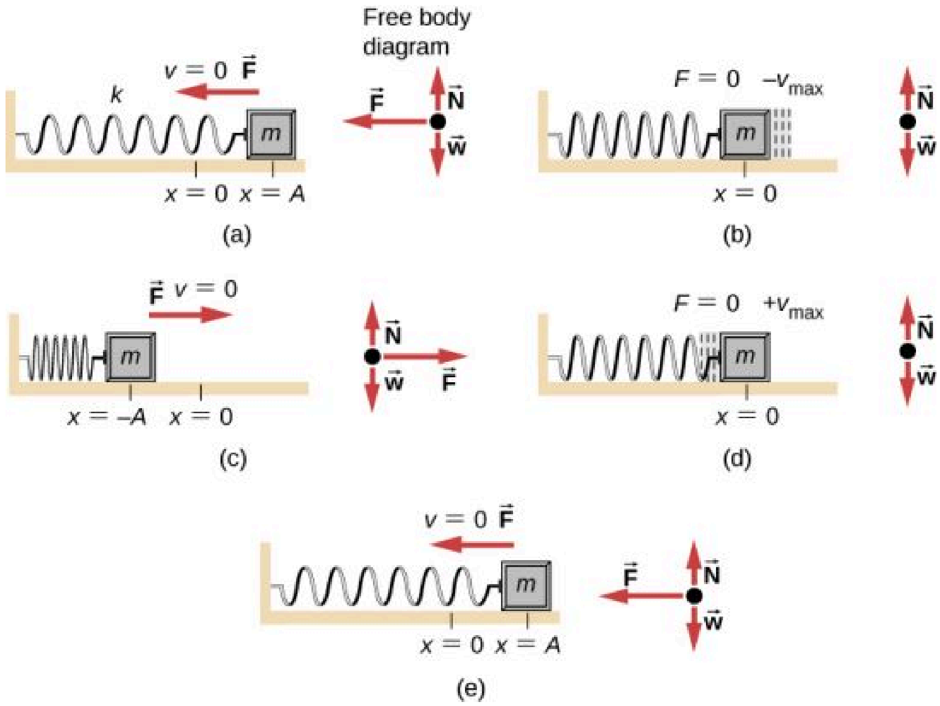


FIGURE 15.3 An object attached to a spring sliding on a frictionless surface is an uncomplicated simple harmonic oscillator. In the above set of figures, a mass is attached to a spring and placed on a frictionless table. The other end of the spring is attached to the wall. The position of the mass, when the spring is neither stretched nor compressed, is marked as $x = 0$ and is the equilibrium position. (a) The mass is displaced to a position $x = A$ and released from rest. (b) The mass accelerates as it moves in the negative *x*-direction, reaching a maximum negative velocity at $x = 0$. (c) The mass continues to move in the negative *x*-direction, slowing until it comes to a stop at $x = -A$. (d) The mass now begins to accelerate in the positive *x*-direction, reaching a positive maximum velocity at $x = 0$. (e) The mass then continues to move in the positive direction until it stops at $x = A$. The mass continues in SHM that has an amplitude *A* and a period *T*. The object's maximum speed occurs as it passes through equilibrium. The stiffer the spring is, the smaller the period *T*. The greater the mass of the object is, the greater the period *T*.

What is so significant about SHM? For one thing, the period *T* and frequency *f* of a simple harmonic oscillator are independent of amplitude. The string of a guitar, for example, oscillates with the same frequency whether plucked gently or hard.

Two important factors do affect the period of a simple harmonic oscillator. The period is related to how stiff the system is. A very stiff object has a large **force constant (*k*)**, which causes the system to have a smaller period. For example, you can adjust a diving board's stiffness—the stiffer it is, the faster it vibrates, and the shorter its period. Period also depends on the mass of the oscillating system. The more massive the system is, the longer the period. For example, a heavy person on a diving board bounces up and down more slowly than a light one. In fact, the mass *m* and the force constant *k* are the *only* factors that affect the period and frequency of SHM. To derive an equation for the period and the frequency, we must first define and analyze the equations of motion. Note that the force constant is sometimes referred to as the *spring constant.*

Equations of SHM

Consider a block attached to a spring on a frictionless table (Figure 15.4). The **equilibrium position** (the position where the spring is neither stretched nor compressed) is marked as $x = 0$. At the equilibrium position, the net force is zero.

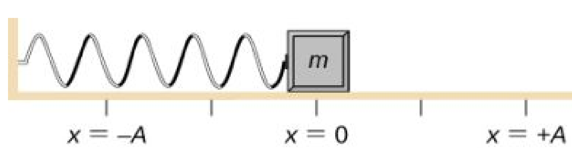


FIGURE 15.4 A block is attached to a spring and placed on a frictionless table. The equilibrium position, where the spring is neither extended nor compressed, is marked as $x = 0$.

Work is done on the block to pull it out to a position of $x = +A$, and it is then released from rest. The maximum *x*-position (*A*) is called the amplitude of the motion. The block begins to oscillate in SHM between $x = +A$ and

$x = -A$, where *A* is the amplitude of the motion and *T* is the period of the oscillation. The period is the time for one oscillation. Figure 15.5 shows the motion of the block as it completes one and a half oscillations after release. Figure 15.6 shows a plot of the position of the block versus time. When the position is plotted versus time, it is clear that the data can be modeled by a cosine function with an amplitude *A* and a period *T*. The cosine function $\cos\theta$ repeats every multiple of 2π, whereas the motion of the block repeats every period *T*. However, the function $\cos\left(\frac{2\pi}{T}t\right)$ repeats every integer multiple of the period. The maximum of the cosine function is one, so it is necessary to multiply the cosine function by the amplitude *A*.

$$x(t) = A\cos\left(\frac{2\pi}{T}t\right) = A\cos(\omega t). \tag{15.2}$$

Recall from the chapter on rotation that the angular frequency equals $\omega = \frac{d\theta}{dt}$. In this case, the period is constant, so the angular frequency is defined as 2π divided by the period, $\omega = \frac{2\pi}{T}$.

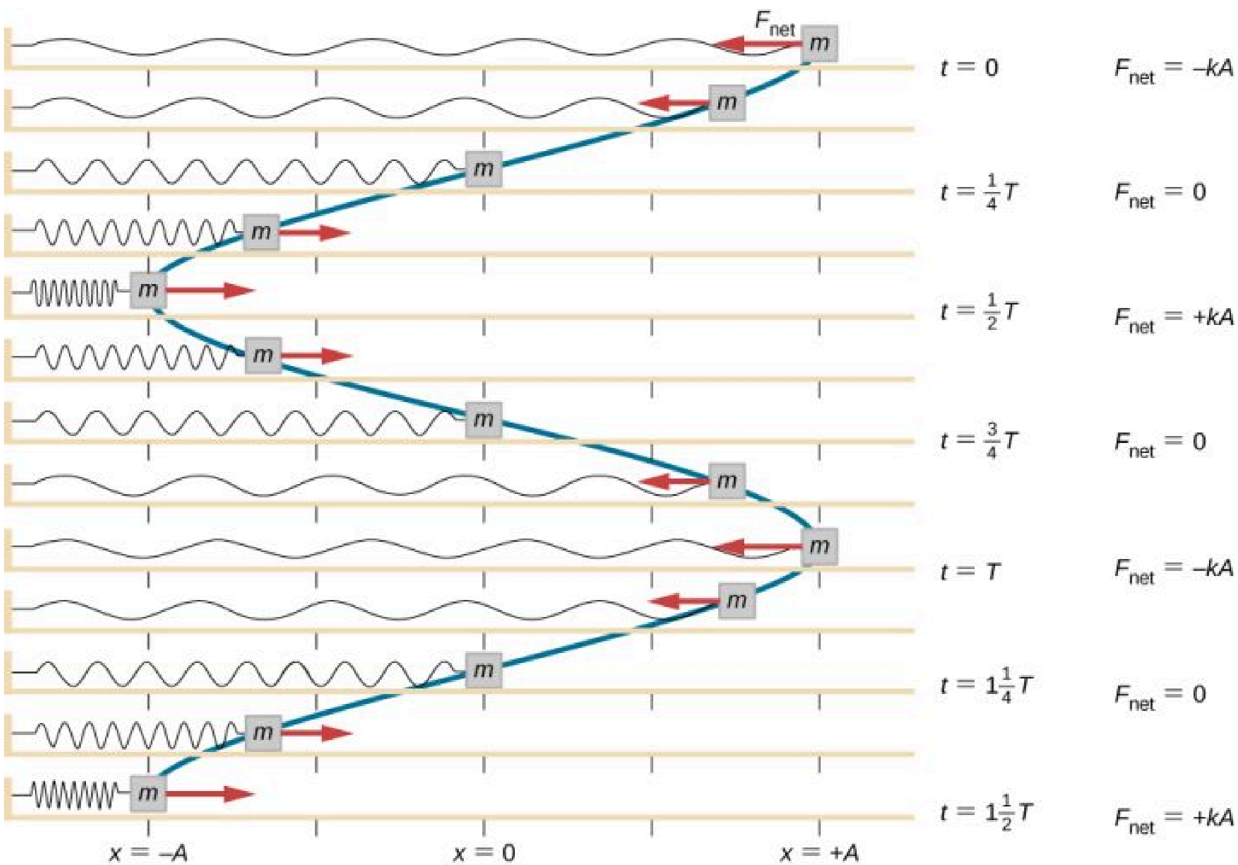


FIGURE 15.5 A block is attached to one end of a spring and placed on a frictionless table. The other end of the spring is anchored to the wall. The equilibrium position, where the net force equals zero, is marked as $x = 0\ \text{m}$. Work is done on the block, pulling it out to $x = +A$, and the block is released from rest. The block oscillates between $x = +A$ and $x = -A$. The force is also shown as a vector.

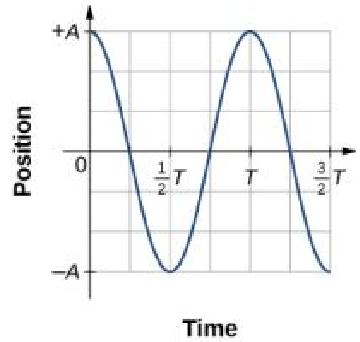


FIGURE 15.6 A graph of the position of the block shown in Figure 15.5 as a function of time. The position can be modeled as a periodic function, such as a cosine or sine function.

The equation for the position as a function of time $x(t) = A\cos(\omega t)$ is good for modeling data, where the position of the block at the initial time $t = 0.00\text{ s}$ is at the amplitude *A* and the initial velocity is zero. Often when taking experimental data, the position of the mass at the initial time $t = 0.00\text{ s}$ is not equal to the amplitude and the initial velocity is not zero. Consider 10 seconds of data collected by a student in lab, shown in Figure 15.7.

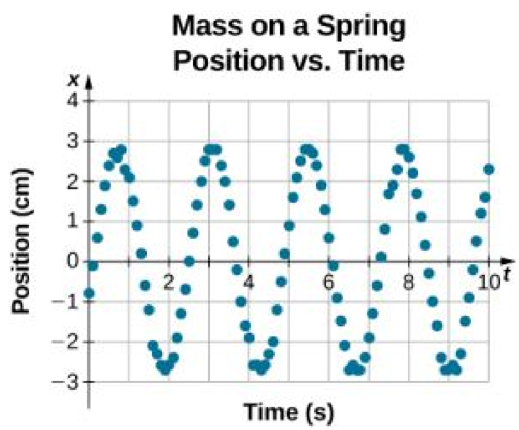


FIGURE 15.7 Data collected by a student in lab indicate the position of a block attached to a spring, measured with a sonic range finder. The data are collected starting at time $t = 0.00\text{s}$, but the initial position is near position $x \approx -0.80\text{ cm} \neq 3.00\text{ cm}$, so the initial position does not equal the amplitude $x_0 = +A$. The velocity is the time derivative of the position, which is the slope at a point on the graph of position versus time. The velocity is not $v = 0.00\text{ m/s}$ at time $t = 0.00\text{ s}$, as evident by the slope of the graph of position versus time, which is not zero at the initial time.

The data in Figure 15.7 can still be modeled with a periodic function, like a cosine function, but the function is shifted to the right. This shift is known as a **phase shift** and is usually represented by the Greek letter phi (ϕ). The equation of the position as a function of time for a block on a spring becomes

$$x(t) = A\cos(\omega t + \phi).$$

This is the generalized equation for SHM where *t* is the time measured in seconds, ω is the angular frequency with units of inverse seconds, *A* is the amplitude measured in meters or centimeters, and ϕ is the phase shift measured in radians (Figure 15.8). It should be noted that because sine and cosine functions differ only by a phase shift, this motion could be modeled using either the cosine or sine function.

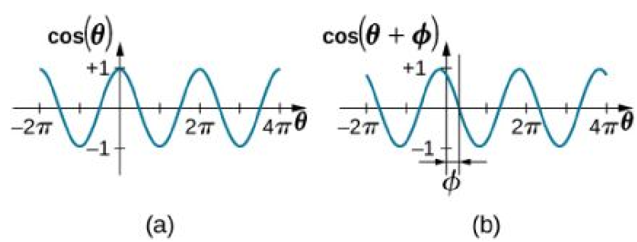


FIGURE 15.8 (a) A cosine function. (b) A cosine function shifted to the left by an angle ϕ. The angle ϕ is known as the phase shift of the function.

The velocity of the mass on a spring, oscillating in SHM, can be found by taking the derivative of the position equation:

$$v(t) = \frac{dx}{dt} = \frac{d}{dt}(A\cos(\omega t + \phi)) = -A\omega\sin(\omega t + \phi) = -v_{max}\sin(\omega t + \phi).$$

Because the sine function oscillates between –1 and +1, the maximum velocity is the amplitude times the angular frequency, $v_{max} = A\omega$. The maximum velocity occurs at the equilibrium position $(x = 0)$ when the mass is moving toward $x = +A$. The maximum velocity in the negative direction is attained at the equilibrium position $(x = 0)$ when the mass is moving toward $x = -A$ and is equal to $-v_{max}$.

The acceleration of the mass on the spring can be found by taking the time derivative of the velocity:

$$a(t) = \frac{dv}{dt} = \frac{d}{dt}(-A\omega\sin(\omega t + \phi)) = -A\omega^2\cos(\omega t + \varphi) = -a_{max}\cos(\omega t + \phi).$$

The maximum acceleration is $a_{max} = A\omega^2$. The maximum acceleration occurs at the position $(x = -A)$, and the acceleration at the position $(x = -A)$ and is equal to a_{max}.

Summary of Equations of Motion for SHM

In summary, the oscillatory motion of a block on a spring can be modeled with the following equations of motion:

$$x(t) = A\cos(\omega t + \phi) \tag{15.3}$$
$$v(t) = -v_{max}\sin(\omega t + \phi) \tag{15.4}$$
$$a(t) = -a_{max}\cos(\omega t + \phi) \tag{15.5}$$
$$x_{max} = A \tag{15.6}$$
$$v_{max} = A\omega \tag{15.7}$$
$$a_{max} = A\omega^2. \tag{15.8}$$

Here, *A* is the amplitude of the motion, *T* is the period, ϕ is the phase shift, and $\omega = \frac{2\pi}{T} = 2\pi f$ is the angular frequency of the motion of the block.

EXAMPLE 15.2

Determining the Equations of Motion for a Block and a Spring

A 2.00-kg block is placed on a frictionless surface. A spring with a force constant of $k = 32.00\text{ N/m}$ is attached to the block, and the opposite end of the spring is attached to the wall. The spring can be compressed or extended. The equilibrium position is marked as $x = 0.00\text{ m}$.

Work is done on the block, pulling it out to $x = +0.02\text{ m}$. The block is released from rest and oscillates between $x = +0.02\text{ m}$ and $x = -0.02\text{ m}$. The period of the motion is 1.57 s. Determine the equations of motion.

Strategy

We first find the angular frequency. The phase shift is zero, $\phi = 0.00\text{ rad}$, because the block is released from rest at $x = A = +0.02\text{ m}$. Once the angular frequency is found, we can determine the maximum velocity and maximum acceleration.

Solution

The angular frequency can be found and used to find the maximum velocity and maximum acceleration:

$$\begin{aligned}
\omega &= \frac{2\pi}{1.57\text{ s}} = 4.00\text{ s}^{-1}; \\
v_{max} &= A\omega = 0.02\text{m}\left(4.00\text{ s}^{-1}\right) = 0.08\text{ m/s}; \\
a_{max} &= A\omega^2 = 0.02\text{ m}\left(4.00\text{ s}^{-1}\right)^2 = 0.32\text{ m/s}^2.
\end{aligned}$$

All that is left is to fill in the equations of motion:

$$\begin{aligned} x(t) &= A\cos(\omega t + \phi) = (0.02\text{ m})\cos\left(4.00\text{ s}^{-1}t\right); \\ v(t) &= -v_{\max}\sin(\omega t + \phi) = (-0.08\text{ m/s})\sin\left(4.00\text{ s}^{-1}t\right); \\ a(t) &= -a_{\max}\cos(\omega t + \phi) = \left(-0.32\text{ m/s}^2\right)\cos\left(4.00\text{ s}^{-1}t\right). \end{aligned}$$

Significance

The position, velocity, and acceleration can be found for any time. It is important to remember that when using these equations, your calculator must be in radians mode.

The Period and Frequency of a Mass on a Spring

One interesting characteristic of the SHM of an object attached to a spring is that the angular frequency, and therefore the period and frequency of the motion, depend on only the mass and the force constant, and not on other factors such as the amplitude of the motion. We can use the equations of motion and Newton's second law ($\vec{\mathbf{F}}_{\text{net}} = m\vec{\mathbf{a}}$) to find equations for the angular frequency, frequency, and period.

Consider the block on a spring on a frictionless surface. There are three forces on the mass: the weight, the normal force, and the force due to the spring. The only two forces that act perpendicular to the surface are the weight and the normal force, which have equal magnitudes and opposite directions, and thus sum to zero. The only force that acts parallel to the surface is the force due to the spring, so the net force must be equal to the force of the spring:

$$\begin{aligned} F_x &= -kx; \\ ma &= -kx; \\ m\frac{d^2x}{dt^2} &= -kx; \\ \frac{d^2x}{dt^2} &= -\frac{k}{m}x. \end{aligned}$$

Substituting the equations of motion for *x* and *a* gives us

$$-A\omega^2\cos(\omega t + \phi) = -\frac{k}{m}A\cos(\omega t + \phi).$$

Cancelling out like terms and solving for the angular frequency yields

$$\omega = \sqrt{\frac{k}{m}}. \tag{15.9}$$

The angular frequency depends only on the force constant and the mass, and not the amplitude. The angular frequency is defined as $\omega = 2\pi/T$, which yields an equation for the period of the motion:

$$T = 2\pi\sqrt{\frac{m}{k}}. \tag{15.10}$$

The period also depends only on the mass and the force constant. The greater the mass, the longer the period. The stiffer the spring, the shorter the period. The frequency is

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}. \tag{15.11}$$

Vertical Motion and a Horizontal Spring

When a spring is hung vertically and a block is attached and set in motion, the block oscillates in SHM. In this case, there is no normal force, and the net effect of the force of gravity is to change the equilibrium position. Consider Figure 15.9. Two forces act on the block: the weight and the force of the spring. The weight is constant and the force of the spring changes as the length of the spring changes.

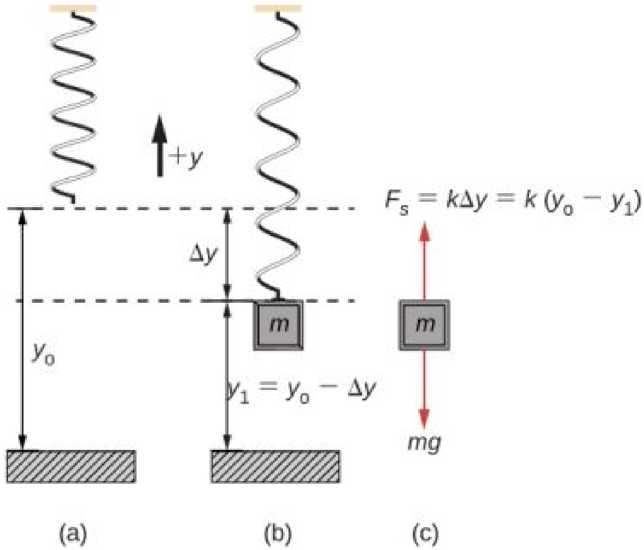


FIGURE 15.9 A spring is hung from the ceiling. When a block is attached, the block is at the equilibrium position where the weight of the block is equal to the force of the spring. (a) The spring is hung from the ceiling and the equilibrium position is marked as y_o. (b) A mass is attached to the spring and a new equilibrium position is reached ($y_1 = y_o - \Delta y$) when the force provided by the spring equals the weight of the mass. (c) The free-body diagram of the mass shows the two forces acting on the mass: the weight and the force of the spring.

When the block reaches the equilibrium position, as seen in Figure 15.9, the force of the spring equals the weight of the block, $F_{\text{net}} = F_{\text{s}} - mg = 0$, where

$$-k(-\Delta y) = mg.$$

From the figure, the change in the position is $\Delta y = y_0 - y_1$ and since $-k(-\Delta y) = mg$, we have

$$k(y_0 - y_1) - mg = 0.$$

If the block is displaced and released, it will oscillate around the new equilibrium position. As shown in Figure 15.10, if the position of the block is recorded as a function of time, the recording is a periodic function.

If the block is displaced to a position *y*, the net force becomes $F_{\text{net}} = k(y_0 - y) - mg = 0$. But we found that at the equilibrium position, $mg = k\Delta y = ky_0 - ky_1$. Substituting for the weight in the equation yields

$$F_{\text{net}} = ky_0 - ky - (ky_0 - ky_1) = -k(y - ky_1).$$

Recall that y_1 is just the equilibrium position and any position can be set to be the point $y = 0.00\text{m}$. So let's set y_1 to $y = 0.00\text{ m}$. The net force then becomes

$$F_{\text{net}} = -ky;$$

$$m\frac{d^2y}{dt^2} = -ky.$$

This is just what we found previously for a horizontally sliding mass on a spring. The constant force of gravity only served to shift the equilibrium location of the mass. Therefore, the solution should be the same form as for a block on a horizontal spring, $y(t) = A\cos(\omega t + \phi)$. The equations for the velocity and the acceleration also have the same form as for the horizontal case. Note that the inclusion of the phase shift means that the motion can actually be modeled using either a cosine or a sine function, since these two functions only differ by a phase shift.

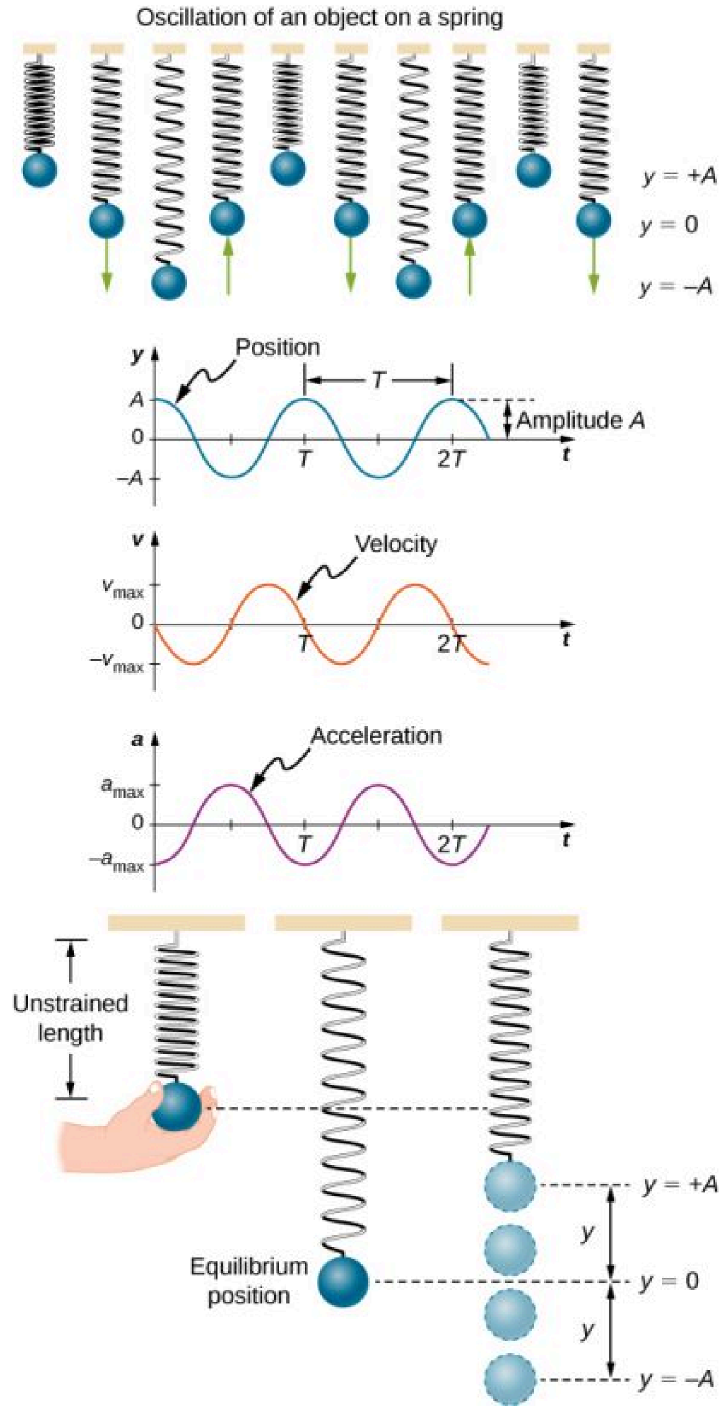


FIGURE 15.10 Graphs of $y(t)$, $v(t)$, and $a(t)$ versus t for the motion of an object on a vertical spring. The net force on the object can be described by Hooke's law, so the object undergoes SHM. Note that the initial position has the vertical displacement at its maximum value A; v is initially zero and then negative as the object moves down; the initial acceleration is negative, back toward the equilibrium position and becomes zero at that point.

15.2 Energy in Simple Harmonic Motion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the energy conservation of the system of a mass and a spring
- Explain the concepts of stable and unstable equilibrium points

To produce a deformation in an object, we must do work. That is, whether you pluck a guitar string or compress a car's shock absorber, a force must be exerted through a distance. If the only result is deformation, and no work goes into thermal, sound, or kinetic energy, then all the work is initially stored in the deformed object as some form of potential energy.

Consider the example of a block attached to a spring on a frictionless table, oscillating in SHM. The force of the spring is a conservative force (which you studied in the chapter on potential energy and conservation of energy), and we can define a potential energy for it. This potential energy is the energy stored in the spring when the spring is extended or compressed. In this case, the block oscillates in one dimension with the force of the spring acting parallel to the motion:

$$W = \int_{x_i}^{x_f} F_x dx = \int_{x_i}^{x_f} -kxdx = \left[-\frac{1}{2}kx^2\right]_{x_i}^{x_f} = -\left[\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right] = -\left[U_f - U_i\right] = -\Delta U.$$

When considering the energy stored in a spring, the equilibrium position, marked as $x_i = 0.00 \text{ m}$, is the position at which the energy stored in the spring is equal to zero. When the spring is stretched or compressed a distance *x*, the potential energy stored in the spring is

$$U = \frac{1}{2}kx^2.$$

Energy and the Simple Harmonic Oscillator

To study the energy of a simple harmonic oscillator, we need to consider all the forms of energy. Consider the example of a block attached to a spring, placed on a frictionless surface, oscillating in SHM. The potential energy stored in the deformation of the spring is

$$U = \frac{1}{2}kx^2.$$

In a simple harmonic oscillator, the energy oscillates between kinetic energy of the mass $K = \frac{1}{2}mv^2$ and potential energy $U = \frac{1}{2}kx^2$ stored in the spring. In the SHM of the mass and spring system, there are no dissipative forces, so the total energy is the sum of the potential energy and kinetic energy. In this section, we consider the conservation of energy of the system. The concepts examined are valid for all simple harmonic oscillators, including those where the gravitational force plays a role.

Consider Figure 15.11, which shows an oscillating block attached to a spring. In the case of undamped SHM, the energy oscillates back and forth between kinetic and potential, going completely from one form of energy to the other as the system oscillates. So for the simple example of an object on a frictionless surface attached to a spring, the motion starts with all of the energy stored in the spring as **elastic potential energy**. As the object starts to move, the elastic potential energy is converted into kinetic energy, becoming entirely kinetic energy at the equilibrium position. The energy is then converted back into elastic potential energy by the spring as it is stretched or compressed. The velocity becomes zero when the kinetic energy is completely converted, and this cycle then repeats. Understanding the conservation of energy in these cycles will provide extra insight here and in later applications of SHM, such as alternating circuits.

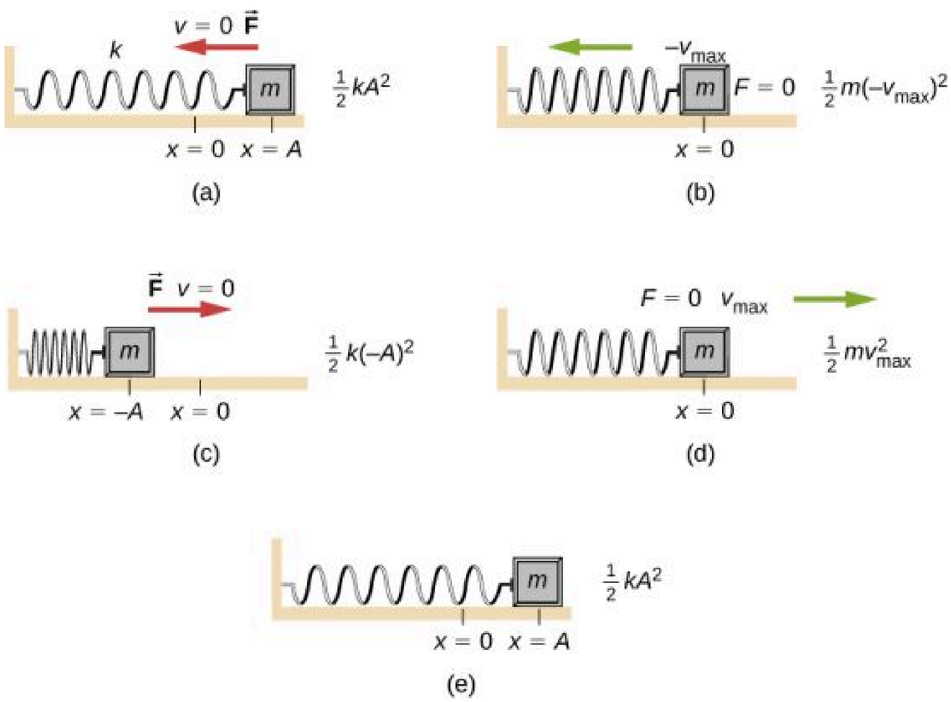


FIGURE 15.11 The transformation of energy in SHM for an object attached to a spring on a frictionless surface. (a) When the mass is at the position $x = +A$, all the energy is stored as potential energy in the spring $U = \frac{1}{2}kA^2$. The kinetic energy is equal to zero because the velocity of the mass is zero. (b) As the mass moves toward $x = -A$, the mass crosses the position $x = 0$. At this point, the spring is neither extended nor compressed, so the potential energy stored in the spring is zero. At $x = 0$, the total energy is all kinetic energy where $K = \frac{1}{2}m(-v_{\text{max}})^2$. (c) The mass continues to move until it reaches $x = -A$ where the mass stops and starts moving toward $x = +A$. At the position $x = -A$, the total energy is stored as potential energy in the compressed $U = \frac{1}{2}k(-A)^2$ and the kinetic energy is zero. (d) As the mass passes through the position $x = 0$, the kinetic energy is $K = \frac{1}{2}mv_{\text{max}}^2$ and the potential energy stored in the spring is zero. (e) The mass returns to the position $x = +A$, where $K = 0$ and $U = \frac{1}{2}kA^2$.

Consider Figure 15.11, which shows the energy at specific points on the periodic motion. While staying constant, the energy oscillates between the kinetic energy of the block and the potential energy stored in the spring:

$$E_{\text{Total}} = U + K = \frac{1}{2}kx^2 + \frac{1}{2}mv^2.$$

The motion of the block on a spring in SHM is defined by the position $x(t) = A\cos(\omega t + \phi)$ with a velocity of $v(t) = -A\omega\sin(\omega t + \phi)$. Using these equations, the trigonometric identity $\cos^2\theta + \sin^2\theta = 1$ and $\omega = \sqrt{\frac{k}{m}}$, we can find the total energy of the system:

$$\begin{aligned} E_{\text{Total}} &= \frac{1}{2}kA^2\cos^2(\omega t + \phi) + \frac{1}{2}mA^2\omega^2\sin^2(\omega t + \phi) \\ &= \frac{1}{2}kA^2\cos^2(\omega t + \phi) + \frac{1}{2}mA^2\left(\frac{k}{m}\right)\sin^2(\omega t + \phi) \\ &= \frac{1}{2}kA^2\cos^2(\omega t + \phi) + \frac{1}{2}kA^2\sin^2(\omega t + \phi) \\ &= \frac{1}{2}kA^2\left(\cos^2(\omega t + \phi) + \sin^2(\omega t + \phi)\right) \\ &= \frac{1}{2}kA^2. \end{aligned}$$

The total energy of the system of a block and a spring is equal to the sum of the potential energy stored in the spring plus the kinetic energy of the block and is proportional to the square of the amplitude $E_{\text{Total}} = (1/2)kA^2$. The total energy of the system is constant.

A closer look at the energy of the system shows that the kinetic energy oscillates like a sine-squared function, while the potential energy oscillates like a cosine-squared function. However, the total energy for the system is constant and is proportional to the amplitude squared. Figure 15.12 shows a plot of the potential, kinetic, and total energies of the block and spring system as a function of time. Also plotted are the position and velocity as a function of time. Before time $t = 0.0\text{ s}$, the block is attached to the spring and placed at the equilibrium position. Work is done on the block by applying an external force, pulling it out to a position of $x = +A$. The system now has potential energy stored in the spring. At time $t = 0.00\ \text{s}$, the position of the block is equal to the amplitude, the potential energy stored in the spring is equal to $U = \frac{1}{2}kA^2$, and the force on the block is maximum and points in the negative x-direction $(F_S = -kA)$. The velocity and kinetic energy of the block are zero at time $t = 0.00\ \text{s}$. At time $t = 0.00\ \text{s}$, the block is released from rest.

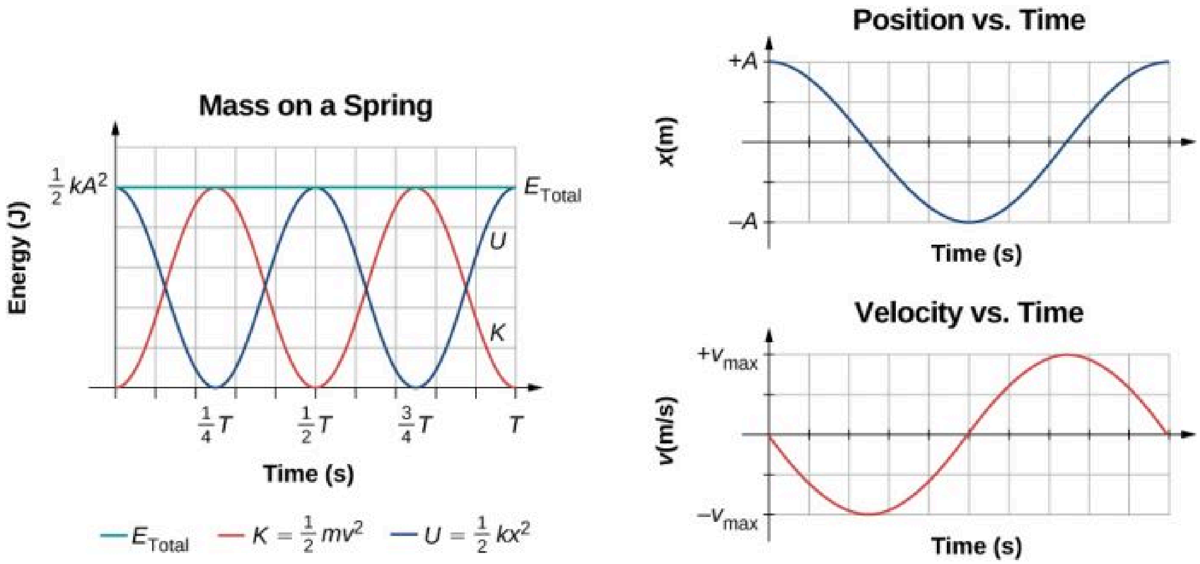


FIGURE 15.12 Graph of the kinetic energy, potential energy, and total energy of a block oscillating on a spring in SHM. Also shown are the graphs of position versus time and velocity versus time. The total energy remains constant, but the energy oscillates between kinetic energy and potential energy. When the kinetic energy is maximum, the potential energy is zero. This occurs when the velocity is maximum and the mass is at the equilibrium position. The potential energy is maximum when the speed is zero. The total energy is the sum of the kinetic energy plus the potential energy and it is constant.

Oscillations About an Equilibrium Position

We have just considered the energy of SHM as a function of time. Another interesting view of the simple harmonic oscillator is to consider the energy as a function of position. Figure 15.13 shows a graph of the energy versus position of a system undergoing SHM.

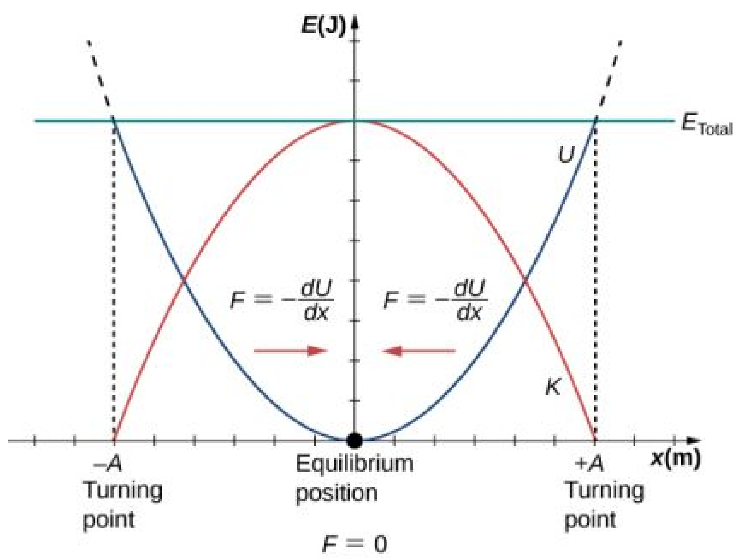


FIGURE 15.13 A graph of the kinetic energy (red), potential energy (blue), and total energy (green) of a simple harmonic oscillator. The force is equal to $F = -\frac{dU}{dx}$. The equilibrium position is shown as a black dot and is the point where the force is equal to zero. The force is positive when $x < 0$, negative when $x > 0$, and equal to zero when $x = 0$.

The potential energy curve in Figure 15.13 resembles a bowl. When a marble is placed in a bowl, it settles to the equilibrium position at the lowest point of the bowl $(x = 0)$. This happens because a **restoring force** points toward the equilibrium point. This equilibrium point is sometimes referred to as a *fixed point*. When the marble is disturbed to a different position $(x = +A)$, the marble oscillates around the equilibrium position. Looking back at the graph of potential energy, the force can be found by looking at the slope of the potential energy graph $\left(F = -\frac{dU}{dx}\right)$. Since the force on either side of the fixed point points back toward the equilibrium point, the equilibrium point is called a **stable equilibrium point**. The points $x = A$ and $x = -A$ are called the turning points. (See Potential Energy and Conservation of Energy.)

Stability is an important concept. If an equilibrium point is stable, a slight disturbance of an object that is initially at the stable equilibrium point will cause the object to oscillate around that point. The stable equilibrium point occurs because the force on either side is directed toward it. For an unstable equilibrium point, if the object is disturbed slightly, it does not return to the equilibrium point.

Consider the marble in the bowl example. If the bowl is right-side up, the marble, if disturbed slightly, will oscillate around the stable equilibrium point. If the bowl is turned upside down, the marble can be balanced on the top, at the equilibrium point where the net force is zero. However, if the marble is disturbed slightly, it will not return to the equilibrium point, but will instead roll off the bowl. The reason is that the force on either side of the equilibrium point is directed away from that point. This point is an unstable equilibrium point.

Figure 15.14 shows three conditions. The first is a stable equilibrium point (a), the second is an unstable equilibrium point (b), and the last is also an unstable equilibrium point (c), because the force on only one side points toward the equilibrium point.

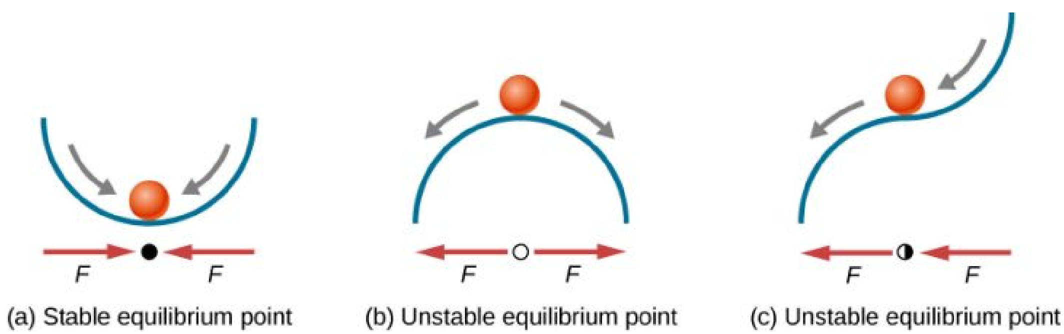


FIGURE 15.14 Examples of equilibrium points. (a) Stable equilibrium point; (b) unstable equilibrium point; (c) unstable equilibrium point (sometimes referred to as a half-stable equilibrium point).

The process of determining whether an equilibrium point is stable or unstable can be formalized. Consider the potential energy curves shown in Figure 15.15. The force can be found by analyzing the slope of the graph. The force is $F = -\frac{dU}{dx}$. In (a), the fixed point is at $x = 0.00\ \text{m}$. When $x < 0.00\ \text{m}$, the force is positive. When $x > 0.00\ \text{m}$, the force is negative. This is a stable point. In (b), the fixed point is at $x = 0.00\ \text{m}$. When $x < 0.00\ \text{m}$, the force is negative. When $x > 0.00\ \text{m}$, the force is also negative. This is an unstable point.

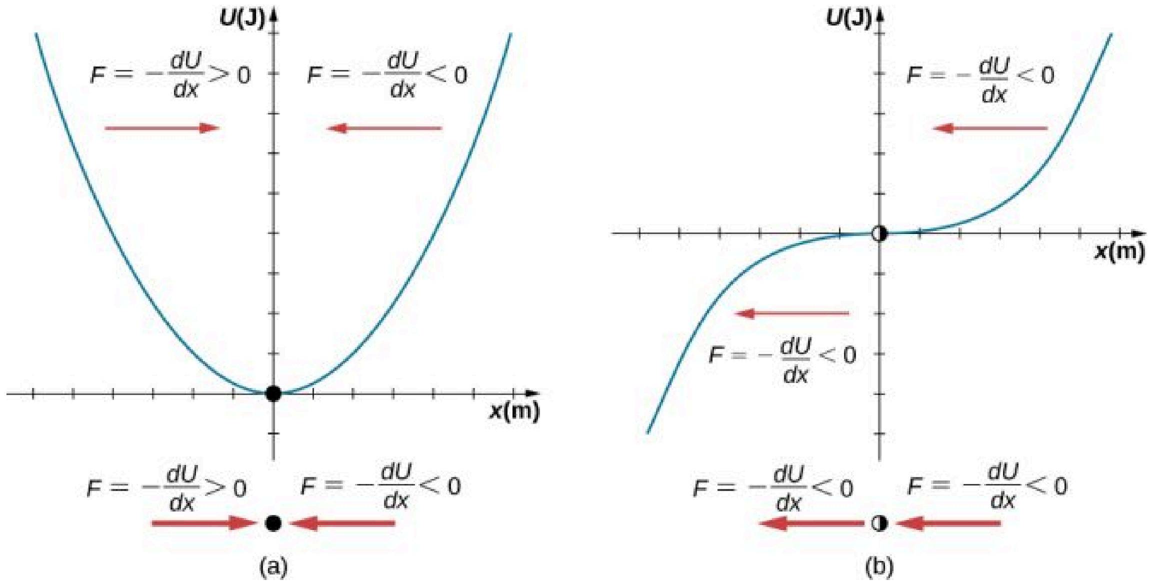


FIGURE 15.15 Two examples of a potential energy function. The force at a position is equal to the negative of the slope of the graph at that position. (a) A potential energy function with a stable equilibrium point. (b) A potential energy function with an unstable equilibrium point. This point is sometimes called half-stable because the force on one side points toward the fixed point.

A practical application of the concept of stable equilibrium points is the force between two neutral atoms in a molecule. If two molecules are in close proximity, separated by a few atomic diameters, they can experience an attractive force. If the molecules move close enough so that the electron shells of the other electrons overlap, the force between the molecules becomes repulsive. The attractive force between the two atoms may cause the atoms to form a molecule. The force between the two molecules is not a linear force and cannot be modeled simply as two masses separated by a spring, but the atoms of the molecule can oscillate around an equilibrium point when displaced a small amount from the equilibrium position. The atoms oscillate due the attractive force and repulsive force between the two atoms.

Consider one example of the interaction between two atoms known as the van Der Waals interaction. It is beyond the scope of this chapter to discuss in depth the interactions of the two atoms, but the oscillations of the atoms can be examined by considering one example of a model of the potential energy of the system. One suggestion to model the potential energy of this molecule is with the Lennard-Jones 6-12 potential:

$$U(x) = 4\varepsilon \left[\left(\frac{\sigma}{x} \right)^{12} - \left(\frac{\sigma}{x} \right)^{6} \right].$$

A graph of this function is shown in Figure 15.16. The two parameters ε and σ are found experimentally.

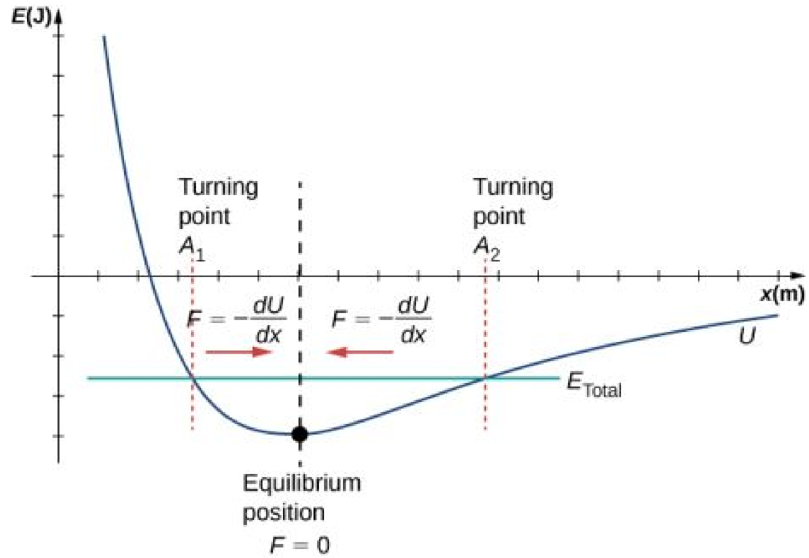


FIGURE 15.16 The Lennard-Jones potential energy function for a system of two neutral atoms. If the energy is below some maximum energy, the system oscillates near the equilibrium position between the two turning points.

From the graph, you can see that there is a potential energy well, which has some similarities to the potential energy well of the potential energy function of the simple harmonic oscillator discussed in Figure 15.13. The Lennard-Jones potential has a stable equilibrium point where the potential energy is minimum and the force on either side of the equilibrium point points toward equilibrium point. Note that unlike the simple harmonic oscillator, the potential well of the Lennard-Jones potential is not symmetric. This is due to the fact that the force between the atoms is not a Hooke's law force and is not linear. The atoms can still oscillate around the equilibrium position x_{min} because when $x < x_{\text{min}}$, the force is positive; when $x > x_{\text{min}}$, the force is negative. Notice that as *x* approaches zero, the slope is quite steep and negative, which means that the force is large and positive. This suggests that it takes a large force to try to push the atoms close together. As *x* becomes increasingly large, the slope becomes less steep and the force is smaller and negative. This suggests that if given a large enough energy, the atoms can be separated.

If you are interested in this interaction, find the force between the molecules by taking the derivative of the potential energy function. You will see immediately that the force does not resemble a Hooke's law force $(F = -kx)$, but if you are familiar with the binomial theorem:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots,$$

the force can be approximated by a Hooke's law force.

Velocity and Energy Conservation

Getting back to the system of a block and a spring in Figure 15.11, once the block is released from rest, it begins to move in the negative direction toward the equilibrium position. The potential energy decreases and the magnitude of the velocity and the kinetic energy increase. At time $t = T/4$, the block reaches the equilibrium position $x = 0.00 \text{ m}$, where the force on the block and the potential energy are zero. At the equilibrium position, the block reaches a negative velocity with a magnitude equal to the maximum velocity $v = -A\omega$. The kinetic energy is maximum and equal to $K = \frac{1}{2}mv^2 = \frac{1}{2}mA^2\omega^2 = \frac{1}{2}kA^2$. At this point, the force on the block is zero, but momentum carries the block, and it continues in the negative direction toward $x = -A$. As the block continues to move, the force on it acts in the positive direction and the magnitude of the velocity and kinetic energy decrease. The potential energy increases as the spring compresses. At time $t = T/2$, the block reaches $x = -A$. Here the velocity and kinetic energy are equal to zero. The force on the block is $F = +kA$ and the potential energy stored in the spring is $U = \frac{1}{2}kA^2$. During the oscillations, the total energy is constant and equal to the sum of the potential energy and the kinetic energy of the system,

$$E_{\text{Total}} = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kA^2. \tag{15.12}$$

The equation for the energy associated with SHM can be solved to find the magnitude of the velocity at any position:

$$|v| = \sqrt{\frac{k}{m}\left(A^2 - x^2\right)}. \tag{15.13}$$

The energy in a simple harmonic oscillator is proportional to the square of the amplitude. When considering many forms of oscillations, you will find the energy proportional to the amplitude squared.

CHECK YOUR UNDERSTANDING 15.1

Why would it hurt more if you snapped your hand with a ruler than with a loose spring, even if the displacement of each system is equal?

CHECK YOUR UNDERSTANDING 15.2

Identify one way you could decrease the maximum velocity of a simple harmonic oscillator.

15.3 Comparing Simple Harmonic Motion and Circular Motion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe how the sine and cosine functions relate to the concepts of circular motion
- Describe the connection between simple harmonic motion and circular motion

An easy way to model SHM is by considering uniform circular motion. Figure 15.17 shows one way of using this method. A peg (a cylinder of wood) is attached to a vertical disk, rotating with a constant angular frequency. Figure 15.18 shows a side view of the disk and peg. If a lamp is placed above the disk and peg, the peg produces a shadow. Let the disk have a radius of $r = A$ and define the position of the shadow that coincides with the center line of the disk to be $x = 0.00\ m$. As the disk rotates at a constant rate, the shadow oscillates between $x = +A$ and $x = -A$. Now imagine a block on a spring beneath the floor as shown in Figure 15.18.

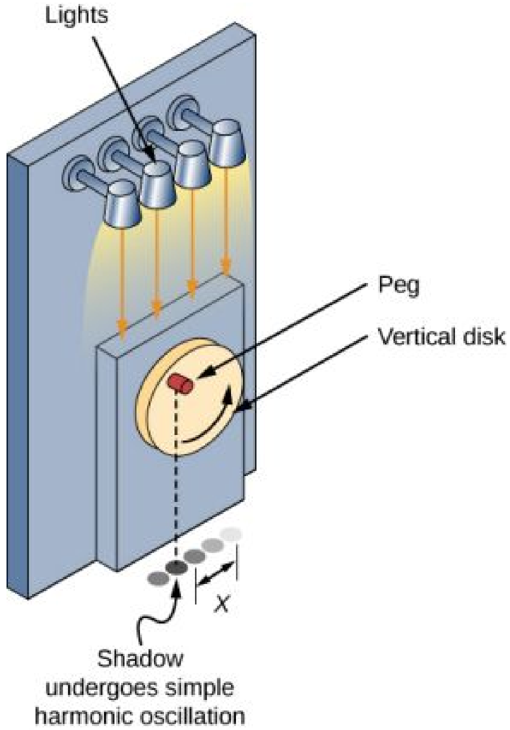


FIGURE 15.17 SHM can be modeled as rotational motion by looking at the shadow of a peg on a wheel rotating at a constant angular frequency.

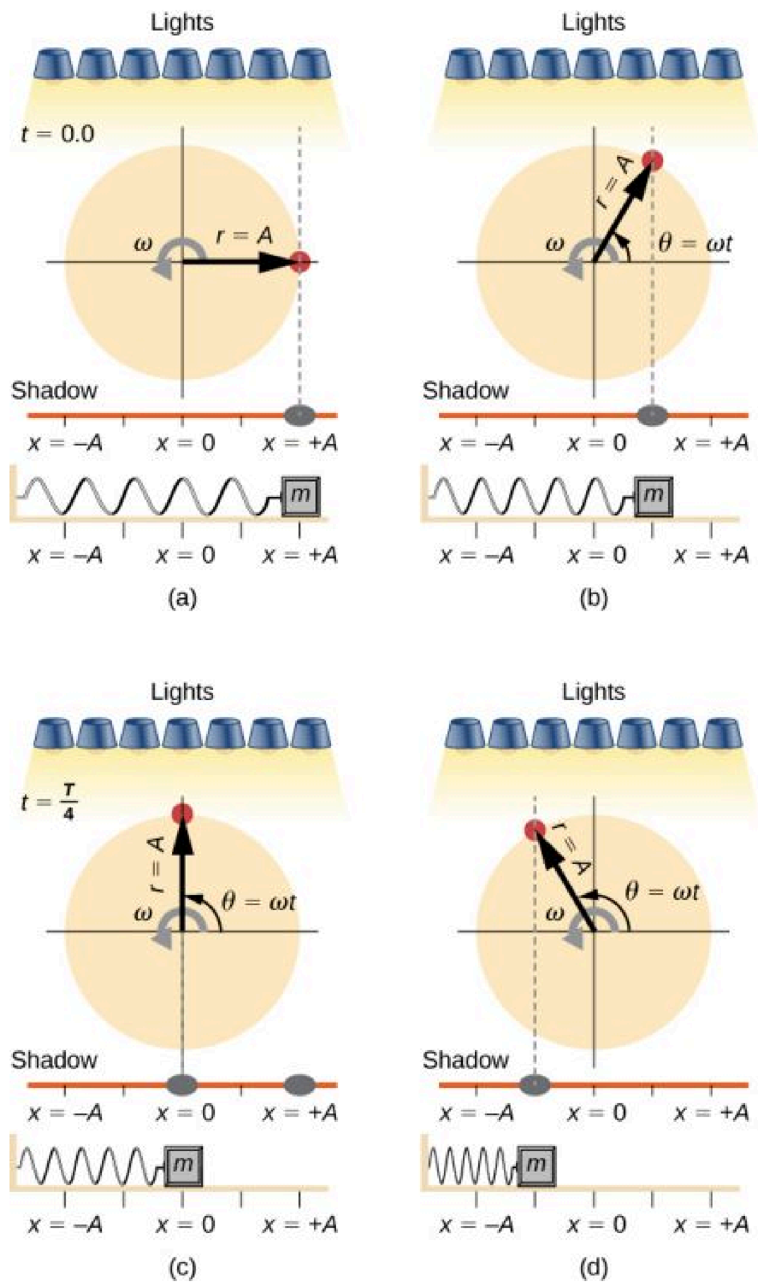


FIGURE 15.18 Light shines down on the disk so that the peg makes a shadow. If the disk rotates at just the right angular frequency, the shadow follows the motion of the block on a spring. If there is no energy dissipated due to nonconservative forces, the block and the shadow will oscillate back and forth in unison. In this figure, four snapshots are taken at four different times. (a) The wheel starts at $\theta = 0^o$ and the shadow of the peg is at $x = +A$, representing the mass at position $x = +A$. (b) As the disk rotates through an angle $\theta = \omega t$, the shadow of the peg is between $x = +A$ and $x = 0$. (c) The disk continues to rotate until $\theta = 90^0$, at which the shadow follows the mass to $x = 0$. (d) The disk continues to rotate, the shadow follows the position of the mass.

If the disk turns at the proper angular frequency, the shadow follows along with the block. The position of the shadow can be modeled with the equation

$$x(t) = A\cos(\omega t). \tag{15.14}$$

Recall that the block attached to the spring does not move at a constant velocity. How often does the wheel have to turn to have the peg's shadow always on the block? The disk must turn at a constant angular frequency equal to 2π times the frequency of oscillation $(\omega = 2\pi f)$.

Figure 15.19 shows the basic relationship between uniform circular motion and SHM. The peg lies at the tip of the radius, a distance *A* from the center of the disk. The *x*-axis is defined by a line drawn parallel to the ground, cutting the disk in half. The *y*-axis (not shown) is defined by a line perpendicular to the ground, cutting the disk into a left half and a right half. The center of the disk is the point $(x = 0, y = 0)$. The projection of the position of the peg onto the fixed *x*-axis gives the position of the shadow, which undergoes SHM analogous to the system of the block and spring. At the time shown in the figure, the projection has position *x* and moves to the left with velocity *v*. The tangential velocity of the peg around the circle equals $\bar{v}_{\max}$ of the block on the spring. The *x*-component of the velocity is equal to the velocity of the block on the spring.

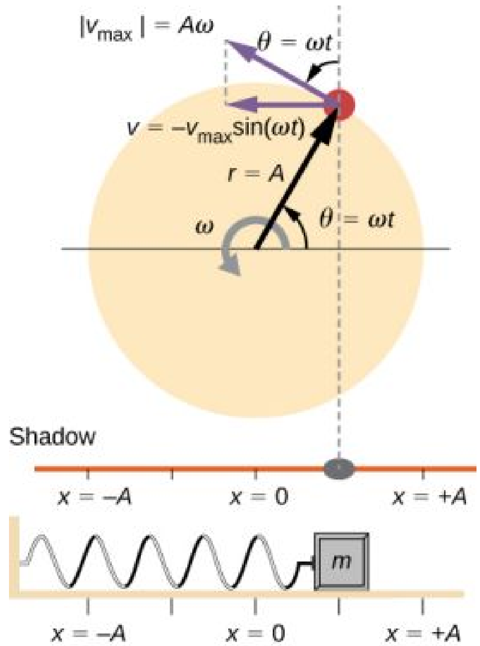


FIGURE 15.19 A peg moving on a circular path with a constant angular velocity ω is undergoing uniform circular motion. Its projection on the *x*-axis undergoes SHM. Also shown is the velocity of the peg around the circle, $v_{\max}$, and its projection, which is *v*. Note that these velocities form a similar triangle to the displacement triangle.

We can use Figure 15.19 to analyze the velocity of the shadow as the disk rotates. The peg moves in a circle with a speed of $v_{\max} = A\omega$. The shadow moves with a velocity equal to the component of the peg's velocity that is parallel to the surface where the shadow is being produced:

$$v = -v_{\max}\sin(\omega t). \quad (15.15)$$

It follows that the acceleration is

$$a = -a_{\max}\cos(\omega t). \quad (15.16)$$

CHECK YOUR UNDERSTANDING 15.3

Identify an object that undergoes uniform circular motion. Describe how you could trace the SHM of this object.

15.4 Pendulums

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- State the forces that act on a simple pendulum
- Determine the angular frequency, frequency, and period of a simple pendulum in terms of the length of the pendulum and the acceleration due to gravity
- Define the period for a physical pendulum
- Define the period for a torsional pendulum

Pendulums are in common usage. Grandfather clocks use a pendulum to keep time and a pendulum can be used to measure the acceleration due to gravity. For small displacements, a pendulum is a simple harmonic oscillator.

The Simple Pendulum

A **simple pendulum** is defined to have a point mass, also known as the pendulum bob, which is suspended from a string of length *L* with negligible mass (Figure 15.20). Here, the only forces acting on the bob are the force of gravity (i.e., the weight of the bob) and tension from the string. The mass of the string is assumed to be negligible as compared to the mass of the bob.

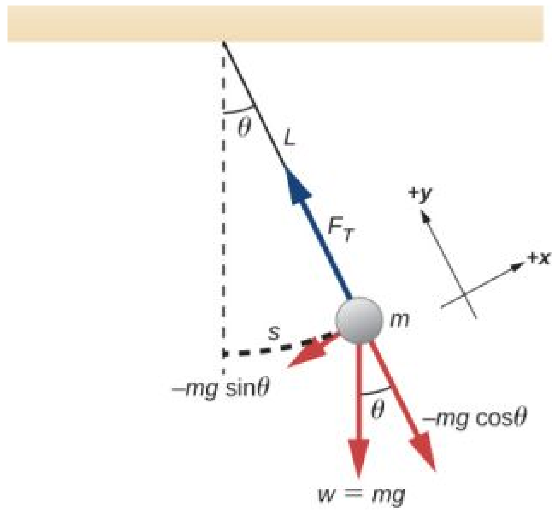


FIGURE 15.20 A simple pendulum has a small-diameter bob and a string that has a very small mass but is strong enough not to stretch appreciably. The linear displacement from equilibrium is *s*, the length of the arc. Also shown are the forces on the bob, which result in a net force of $-mg\sin\theta$ toward the equilibrium position—that is, a restoring force.

Consider the torque on the pendulum. The force providing the restoring torque is the component of the weight of the pendulum bob that acts along the arc length. The torque is the length of the string *L* times the component of the net force that is perpendicular to the radius of the arc. The minus sign indicates the torque acts in the opposite direction of the angular displacement:

$$\begin{aligned}\tau &= -L\,(mg\sin\theta)\,;\\ I\alpha &= -L\,(mg\sin\theta)\,;\\ I\frac{d^2\theta}{dt^2} &= -L\,(mg\sin\theta)\,;\\ mL^2\frac{d^2\theta}{dt^2} &= -L\,(mg\sin\theta)\,;\\ \frac{d^2\theta}{dt^2} &= -\tfrac{g}{L}\sin\theta.\end{aligned}$$

The solution to this differential equation involves advanced calculus, and is beyond the scope of this text. But note that for small angles (less than 15 degrees or about 0.26 radians), $\sin\theta$ and θ differ by less than 1%, if θ is measured in radians. We can then use the small angle approximation $\sin\theta \approx \theta$. The angle θ describes the position of the pendulum. Using the small angle approximation gives an approximate solution for small angles,

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta. \quad (15.17)$$

Because this equation has the same form as the equation for SHM, the solution is easy to find. The angular frequency is

$$\omega = \sqrt{\frac{g}{L}} \quad (15.18)$$

and the period is

$$T = 2\pi\sqrt{\frac{L}{g}}. \quad (15.19)$$

The period of a simple pendulum depends on its length and the acceleration due to gravity. The period is completely independent of other factors, such as mass and the maximum displacement. As with simple harmonic oscillators, the period T for a pendulum is nearly independent of amplitude, especially if θ is less than about $15°$. Even simple pendulum clocks can be finely adjusted and remain accurate.

Note the dependence of T on g. If the length of a pendulum is precisely known, it can actually be used to measure the acceleration due to gravity, as in the following example.

EXAMPLE 15.3

Measuring Acceleration due to Gravity by the Period of a Pendulum

What is the acceleration due to gravity in a region where a simple pendulum having a length 75.000 cm has a period of 1.7357 s?

Strategy

We are asked to find g given the period T and the length L of a pendulum. We can solve $T = 2\pi\sqrt{\frac{L}{g}}$ for g, assuming only that the angle of deflection is less than $15°$.

Solution

1. Square $T = 2\pi\sqrt{\frac{L}{g}}$ and solve for g:
$$g = 4\pi^2 \frac{L}{T^2}.$$
2. Substitute known values into the new equation:
$$g = 4\pi^2 \frac{0.75000 \text{ m}}{(1.7357 \text{ s})^2}.$$
3. Calculate to find g:
$$g = 9.8281 \text{ m/s}^2.$$

Significance

This method for determining g can be very accurate, which is why length and period are given to five digits in this example. For the precision of the approximation $\sin\theta \approx \theta$ to be better than the precision of the pendulum length and period, the maximum displacement angle should be kept below about $0.5°$.

CHECK YOUR UNDERSTANDING 15.4

An engineer builds two simple pendulums. Both are suspended from small wires secured to the ceiling of a room. Each pendulum hovers 2 cm above the floor. Pendulum 1 has a bob with a mass of 10 kg. Pendulum 2 has a bob with a mass of 100 kg. Describe how the motion of the pendulums will differ if the bobs are both displaced by $12°$.

Physical Pendulum

Any object can oscillate like a pendulum. Consider a coffee mug hanging on a hook in the pantry. If the mug gets knocked, it oscillates back and forth like a pendulum until the oscillations die out. We have described a simple pendulum as a point mass and a string. A **physical pendulum** is any object whose oscillations are similar to those of the simple pendulum, but cannot be modeled as a point mass on a string, and the mass distribution must be included into the equation of motion.

As for the simple pendulum, the restoring force of the physical pendulum is the force of gravity. With the simple pendulum, the force of gravity acts on the center of the pendulum bob. In the case of the physical pendulum, the force of gravity acts on the center of mass (CM) of an object. The object oscillates about a point *O*. Consider an object of a generic shape as shown in Figure 15.21.

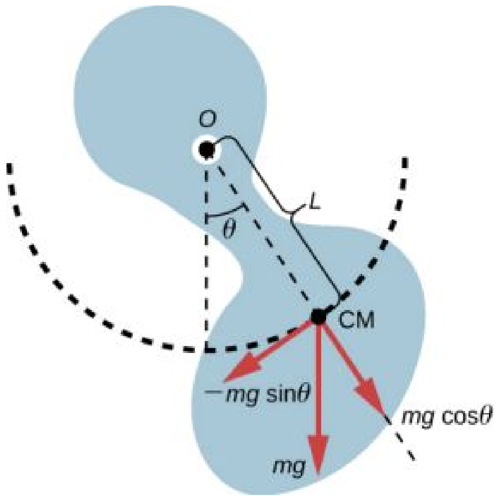


FIGURE 15.21 A physical pendulum is any object that oscillates as a pendulum, but cannot be modeled as a point mass on a string. The force of gravity acts on the center of mass (CM) and provides the restoring force that causes the object to oscillate. The minus sign on the component of the weight that provides the restoring force is present because the force acts in the opposite direction of the increasing angle θ.

When a physical pendulum is hanging from a point but is free to rotate, it rotates because of the torque applied at the CM, produced by the component of the object's weight that acts tangent to the motion of the CM. Taking the counterclockwise direction to be positive, the component of the gravitational force that acts tangent to the motion is $-mg\sin\theta$. The minus sign is the result of the restoring force acting in the opposite direction of the increasing angle. Recall that the torque is equal to $\vec{\tau} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}$. The magnitude of the torque is equal to the length of the radius arm times the tangential component of the force applied, $|\tau| = rF\sin\theta$. Here, the length *L* of the radius arm is the distance between the point of rotation and the CM. To analyze the motion, start with the net torque. Like the simple pendulum, consider only small angles so that $\sin\theta \approx \theta$. Recall from Fixed-Axis Rotation on rotation that the net torque is equal to the moment of inertia $I = \int r^2 dm$ times the angular acceleration α, where $\alpha = \frac{d^2\theta}{dt^2}$:

$$I\alpha = \tau_{\text{net}} = L(-mg)\sin\theta.$$

Using the small angle approximation and rearranging:

$$\begin{aligned} I\alpha &= -L(mg)\theta; \\ I\frac{d^2\theta}{dt^2} &= -L(mg)\theta; \\ \frac{d^2\theta}{dt^2} &= -\left(\frac{mgL}{I}\right)\theta. \end{aligned}$$

Once again, the equation says that the second time derivative of the position (in this case, the angle) equals minus a constant $\left(-\frac{mgL}{I}\right)$ times the position. The solution is

$$\theta(t) = \Theta\cos(\omega t + \phi),$$

where Θ is the maximum angular displacement. The angular frequency is

$$\omega = \sqrt{\frac{mgL}{I}}. \quad (15.20)$$

The period is therefore

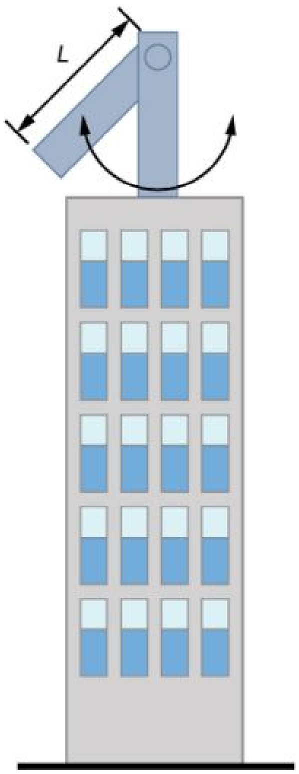
$$T = 2\pi\sqrt{\frac{I}{mgL}}. \tag{15.21}$$

Note that for a simple pendulum, the moment of inertia is $I = \int r^2\, dm = mL^2$ and the period reduces to $T = 2\pi\sqrt{\frac{L}{g}}$.

EXAMPLE 15.4

Reducing the Swaying of a Skyscraper

In extreme conditions, skyscrapers can sway up to two meters with a frequency of up to 20.00 Hz due to high winds or seismic activity. Several companies have developed physical pendulums that are placed on the top of the skyscrapers. As the skyscraper sways to the right, the pendulum swings to the left, reducing the sway. Assuming the oscillations have a frequency of 0.50 Hz, design a pendulum that consists of a long beam, of constant density, with a mass of 100 metric tons and a pivot point at one end of the beam. What should be the length of the beam?



Strategy

We are asked to find the length of the physical pendulum with a known mass. We first need to find the moment of inertia of the beam. We can then use the equation for the period of a physical pendulum to find the length.

Solution

1. Find the moment of inertia for the CM:
   Use the parallel axis theorem to find the moment of inertia about the point of rotation for a rod length *L*:
   $$I = I_{\text{CM}} + \frac{L}{4}^2 M = \frac{1}{12}ML^2 + \frac{1}{4}ML^2 = \frac{1}{3}ML^2.$$
2. The period of a physical pendulum with its center of mass a distance *l* from the pivot point has a period of $T = 2\pi\sqrt{\frac{I}{mgl}}$. In this problem, *L* has been defined as the total length of the rod, so $l = L/2$. Use the moment

of inertia to solve for the length *L*:

$$T = \sqrt{\frac{\frac{1}{3}ML^2}{Mg\frac{L}{2}}} = 2\pi\sqrt{\frac{2L}{3g}};$$

$$L = \frac{3T^2 g}{8\pi^2} = 1.49 \text{ m}.$$

Significance

There are many ways to reduce the oscillations, including modifying the shape of the skyscrapers, using multiple physical pendulums, and using tuned-mass dampers.

Torsional Pendulum

A **torsional pendulum** consists of a rigid body suspended by a light wire or spring (Figure 15.22). When the body is twisted some small maximum angle (Θ) and released from rest, the body oscillates between $(\theta = +\Theta)$ and $(\theta = -\Theta)$. The restoring torque is supplied by the shearing of the string or wire.

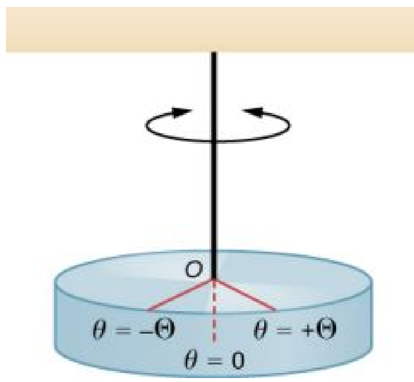


FIGURE 15.22 A torsional pendulum consists of a rigid body suspended by a string or wire. The rigid body oscillates between $\theta = +\Theta$ and $\theta = -\Theta$.

The restoring torque can be modeled as being proportional to the angle:

$$\tau = -\kappa\theta.$$

The variable kappa (κ) is known as the torsion constant of the wire or string. The minus sign shows that the restoring torque acts in the opposite direction to increasing angular displacement. The net torque is equal to the moment of inertia times the angular acceleration:

$$I\frac{d^2\theta}{dt^2} = -\kappa\theta;$$

$$\frac{d^2\theta}{dt^2} = -\frac{\kappa}{I}\theta.$$

This equation says that the second time derivative of the position (in this case, the angle) equals a negative constant times the position. This looks very similar to the equation of motion for the SHM $\frac{d^2x}{dt^2} = -\frac{k}{m}x$, where the period was found to be $T = 2\pi\sqrt{\frac{m}{k}}$. Therefore, the period of the torsional pendulum can be found using

$$T = 2\pi\sqrt{\frac{I}{\kappa}}. \quad (15.22)$$

The units for the torsion constant are $[\kappa] = \text{N-m} = \left(\text{kg}\frac{\text{m}}{\text{s}^2}\right)\text{m} = \text{kg}\ \frac{\text{m}^2}{\text{s}^2}$ and the units for the moment of inertial are $[I] = \text{kg-m}^2$, which show that the unit for the period is the second.

EXAMPLE 15.5

Measuring the Torsion Constant of a String

A rod has a length of $L = 0.30$ m and a mass of 4.00 kg. A string is attached to the CM of the rod and the system is hung from the ceiling (Figure 15.23). The rod is displaced 10 degrees from the equilibrium position and released from rest. The rod oscillates with a period of 0.5 s. What is the torsion constant κ?

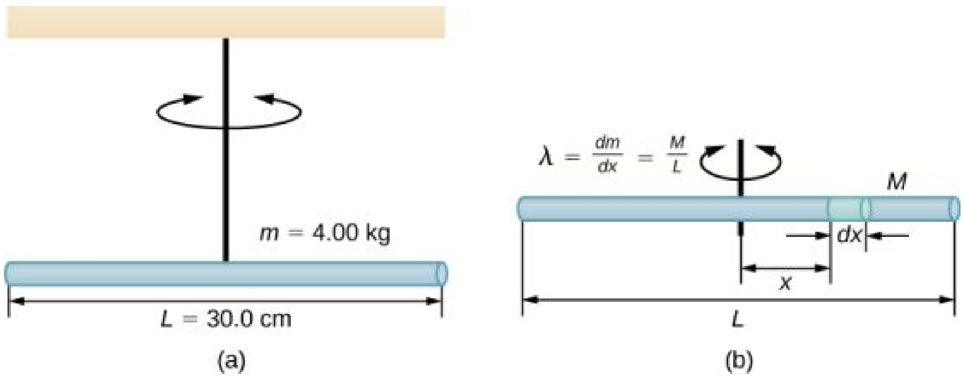


FIGURE 15.23 (a) A rod suspended by a string from the ceiling. (b) Finding the rod's moment of inertia.

Strategy

We are asked to find the torsion constant of the string. We first need to find the moment of inertia.

Solution

1. Find the moment of inertia for the CM:

$$I_{\text{CM}} = \int x^2 dm = \int_{-L/2}^{+L/2} x^2 \lambda dx = \lambda \left[\frac{x^3}{3}\right]_{-L/2}^{+L/2} = \lambda \frac{2L^3}{24} = \left(\frac{M}{L}\right)\frac{2L^3}{24} = \frac{1}{12}ML^2.$$

2. Calculate the torsion constant using the equation for the period:

$$\begin{aligned} T &= 2\pi\sqrt{\frac{I}{\kappa}}; \\ \kappa &= I\left(\frac{2\pi}{T}\right)^2 = \left(\frac{1}{12}ML^2\right)\left(\frac{2\pi}{T}\right)^2; \\ &= \left(\frac{1}{12}(4.00\text{ kg})(0.30\text{ m})^2\right)\left(\frac{2\pi}{0.50\text{ s}}\right)^2 = 4.73\text{ N}\cdot\text{m}. \end{aligned}$$

Significance

Like the force constant of the system of a block and a spring, the larger the torsion constant, the shorter the period.

15.5 Damped Oscillations

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the motion of damped harmonic motion
- Write the equations of motion for damped harmonic oscillations
- Describe the motion of driven, or forced, damped harmonic motion
- Write the equations of motion for forced, damped harmonic motion

In the real world, oscillations seldom follow true SHM. Friction of some sort usually acts to dampen the motion so it dies away, or needs more force to continue. In this section, we examine some examples of damped harmonic motion and see how to modify the equations of motion to describe this more general case.

A guitar string stops oscillating a few seconds after being plucked. To keep swinging on a playground swing, you must keep pushing (Figure 15.24). Although we can often make friction and other nonconservative forces small or negligible, completely undamped motion is rare. In fact, we may even want to damp oscillations, such as with car shock absorbers.

FIGURE 15.24 To counteract dampening forces, you need to keep pumping a swing. (credit: Bob Mical)

Figure 15.25 shows a mass *m* attached to a spring with a force constant k. The mass is raised to a position A_0, the initial amplitude, and then released. The mass oscillates around the equilibrium position in a fluid with viscosity but the amplitude decreases for each oscillation. For a system that has a small amount of damping, the period and frequency are constant and are nearly the same as for SHM, but the amplitude gradually decreases as shown. This occurs because the non-conservative damping force removes energy from the system, usually in the form of thermal energy.

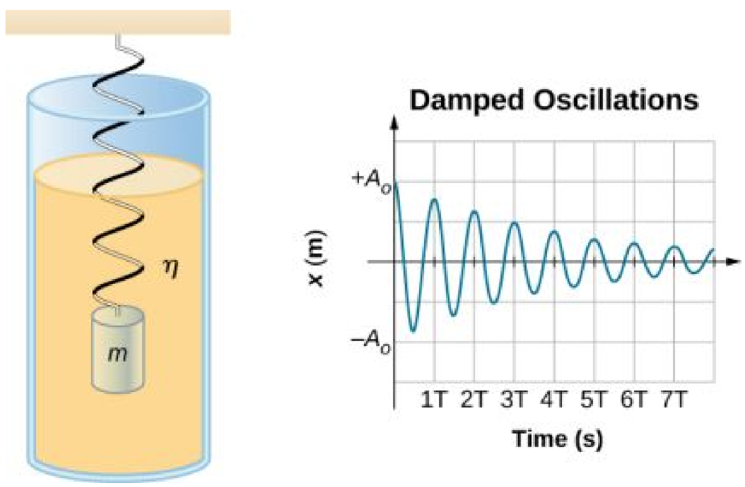


FIGURE 15.25 For a mass on a spring oscillating in a viscous fluid, the period remains constant, but the amplitudes of the oscillations decrease due to the damping caused by the fluid.

Consider the forces acting on the mass. Note that the only contribution of the weight is to change the equilibrium position, as discussed earlier in the chapter. Therefore, the net force is equal to the force of the spring and the damping force (F_D). If the magnitude of the velocity is small, meaning the mass oscillates slowly, the damping force is proportional to the velocity and acts against the direction of motion $(F_D = -bv)$. The net force on the mass is therefore

$$ma = -bv - kx.$$

Writing this as a differential equation in *x*, we obtain

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0. \tag{15.23}$$

To determine the solution to this equation, consider the plot of position versus time shown in Figure 15.26. The curve resembles a cosine curve oscillating in the envelope of an exponential function $A_0 e^{-\alpha t}$ where $\alpha = \frac{b}{2m}$. The solution is

$$x(t) = A_0 e^{-\frac{b}{2m}t} \cos(\omega t + \phi). \quad (15.24)$$

It is left as an exercise to prove that this is, in fact, the solution. To prove that it is the right solution, take the first and second derivatives with respect to time and substitute them into Equation 15.23. It is found that Equation 15.24 is the solution if

$$\omega = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}.$$

Recall that the angular frequency of a mass undergoing SHM is equal to the square root of the force constant divided by the mass. This is often referred to as the **natural angular frequency**, which is represented as

$$\omega_0 = \sqrt{\frac{k}{m}}. \quad (15.25)$$

The angular frequency for damped harmonic motion becomes

$$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}. \quad (15.26)$$

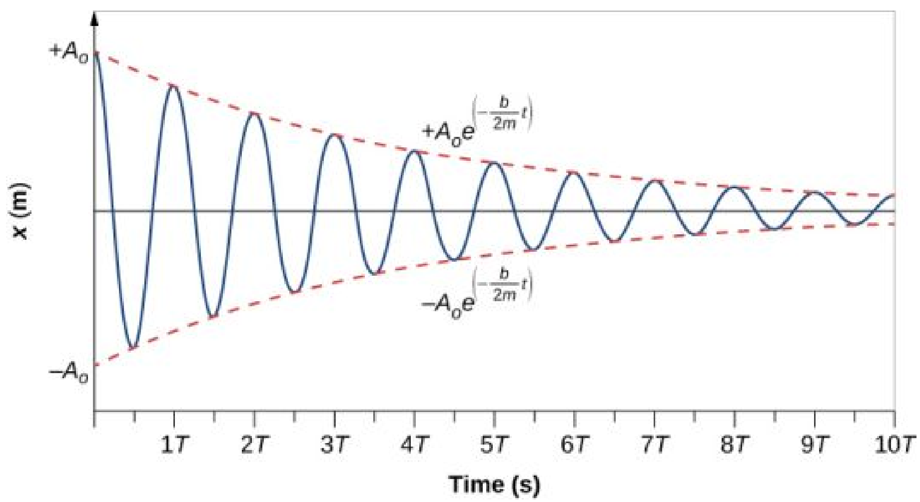


FIGURE 15.26 Position versus time for the mass oscillating on a spring in a viscous fluid. Notice that the curve appears to be a cosine function inside an exponential envelope.

Recall that when we began this description of damped harmonic motion, we stated that the damping must be small. Two questions come to mind. Why must the damping be small? And how small is small? If you gradually *increase* the amount of damping in a system, the period and frequency begin to be affected, because damping opposes and hence slows the back and forth motion. (The net force is smaller in both directions.) If there is very large damping, the system does not even oscillate—it slowly moves toward equilibrium. The angular frequency is equal to

$$\omega = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}.$$

As b increases, $\frac{k}{m} - \left(\frac{b}{2m}\right)^2$ becomes smaller and eventually reaches zero when $b = \sqrt{4mk}$. If b becomes any larger, $\frac{k}{m} - \left(\frac{b}{2m}\right)^2$ becomes a negative number and $\sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$ is a complex number.

Figure 15.27 shows the displacement of a harmonic oscillator for different amounts of damping. When the damping constant is small, $b < \sqrt{4mk}$, the system oscillates while the amplitude of the motion decays exponentially. This system is said to be **underdamped**, as in curve (a). Many systems are underdamped, and oscillate while the amplitude decreases exponentially, such as the mass oscillating on a spring. The damping may be quite small, but eventually the mass comes to rest. If the damping constant is $b = \sqrt{4mk}$, the system is said to be **critically damped**, as in curve (b). An example of a critically damped system is the shock absorbers in a car. It is advantageous to have the oscillations decay as fast as possible. Here, the system does not oscillate, but asymptotically approaches the equilibrium condition as quickly as possible. Curve (c) in Figure 15.27 represents an **overdamped** system where $b > \sqrt{4mk}$. An overdamped system will approach equilibrium over a longer period of time.

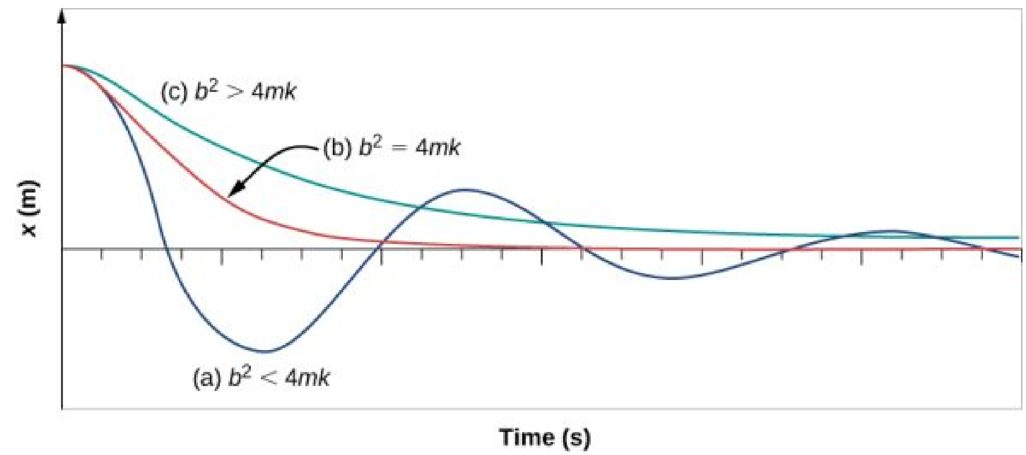


FIGURE 15.27 The position versus time for three systems consisting of a mass and a spring in a viscous fluid. (a) If the damping is small $\left(b < \sqrt{4mk}\right)$, the mass oscillates, slowly losing amplitude as the energy is dissipated by the non-conservative force(s). The limiting case is (b) where the damping is $\left(b = \sqrt{4mk}\right)$. (c) If the damping is very large $\left(b > \sqrt{4mk}\right)$, the mass does not oscillate when displaced, but attempts to return to the equilibrium position.

Critical damping is often desired, because such a system returns to equilibrium rapidly and remains at equilibrium as well. In addition, a constant force applied to a critically damped system moves the system to a new equilibrium position in the shortest time possible without overshooting or oscillating about the new position.

CHECK YOUR UNDERSTANDING 15.5

Why are completely undamped harmonic oscillators so rare?

15.6 Forced Oscillations

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define forced oscillations
- List the equations of motion associated with forced oscillations
- Explain the concept of resonance and its impact on the amplitude of an oscillator
- List the characteristics of a system oscillating in resonance

Sit in front of a piano sometime and sing a loud brief note at it with the dampers off its strings (Figure 15.28). It will sing the same note back at you—the strings, having the same frequencies as your voice, are resonating in response to the forces from the sound waves that you sent to them. This is a good example of the fact that objects—in this case, piano strings—can be forced to oscillate, and oscillate most easily at their natural frequency. In this section, we briefly explore applying a periodic driving force acting on a simple harmonic oscillator. The driving force puts energy into the system at a certain frequency, not necessarily the same as the natural frequency of the system. Recall that the natural frequency is the frequency at which a system would oscillate if there were no driving and no damping force.

FIGURE 15.28 You can cause the strings in a piano to vibrate simply by producing sound waves from your voice. (credit: Matt Billings)

Most of us have played with toys involving an object supported on an elastic band, something like the paddle ball suspended from a finger in Figure 15.29. Imagine the finger in the figure is your finger. At first, you hold your finger steady, and the ball bounces up and down with a small amount of damping. If you move your finger up and down slowly, the ball follows along without bouncing much on its own. As you increase the frequency at which you move your finger up and down, the ball responds by oscillating with increasing amplitude. When you drive the ball at its natural frequency, the ball's oscillations increase in amplitude with each oscillation for as long as you drive it. The phenomenon of driving a system with a frequency equal to its natural frequency is called **resonance**. A system being driven at its natural frequency is said to *resonate*. As the driving frequency gets progressively higher than the resonant or natural frequency, the amplitude of the oscillations becomes smaller until the oscillations nearly disappear, and your finger simply moves up and down with little effect on the ball.

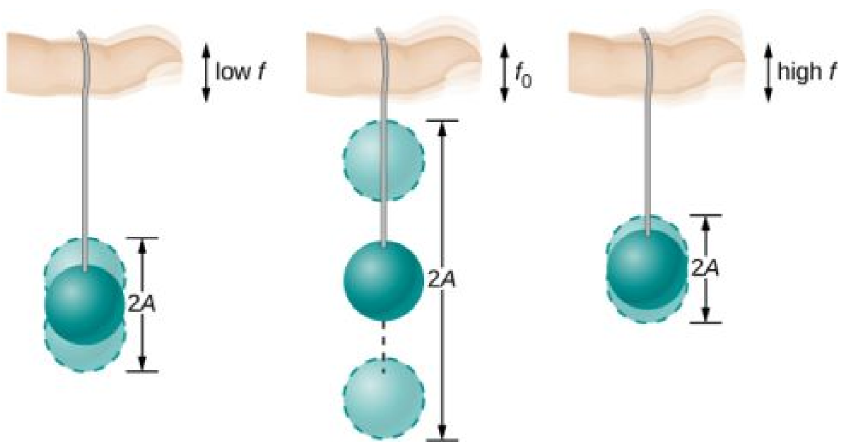


FIGURE 15.29 The paddle ball on its rubber band moves in response to the finger supporting it. If the finger moves with the natural frequency f_0 of the ball on the rubber band, then a resonance is achieved, and the amplitude of the ball's oscillations increases dramatically. At higher and lower driving frequencies, energy is transferred to the ball less efficiently, and it responds with lower-amplitude oscillations.

Consider a simple experiment. Attach a mass *m* to a spring in a viscous fluid, similar to the apparatus discussed in the damped harmonic oscillator. This time, instead of fixing the free end of the spring, attach the free end to a disk that is driven by a variable-speed motor. The motor turns with an angular driving frequency of ω. The rotating disk provides energy to the system by the work done by the driving force $(F_d = F_0 \sin(\omega t))$. The experimental apparatus is shown in Figure 15.30.

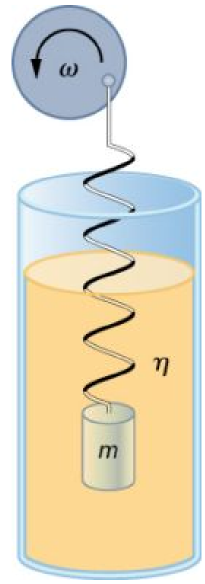


FIGURE 15.30 Forced, damped harmonic motion produced by driving a spring and mass with a disk driven by a variable-speed motor.

Using Newton's second law ($\vec{\mathbf{F}}_{\text{net}} = m\vec{\mathbf{a}}$), we can analyze the motion of the mass. The resulting equation is similar to the force equation for the damped harmonic oscillator, with the addition of the driving force:

$$-kx - b\frac{dx}{dt} + F_0 \sin(\omega t) = m\frac{d^2x}{dt^2}. \tag{15.27}$$

When an oscillator is forced with a periodic driving force, the motion may seem chaotic. The motions of the oscillator is known as transients. After the transients die out, the oscillator reaches a steady state, where the motion is periodic. After some time, the steady state solution to this differential equation is

$$x(t) = A\cos(\omega t + \phi). \tag{15.28}$$

Once again, it is left as an exercise to prove that this equation is a solution. Taking the first and second time derivative of $x(t)$ and substituting them into the force equation shows that $x(t) = A\sin(\omega t + \phi)$ is a solution as long as the amplitude is equal to

$$A = \frac{F_0}{\sqrt{m^2\left(\omega^2 - \omega_0^2\right)^2 + b^2\omega^2}} \tag{15.29}$$

where $\omega_0 = \sqrt{\frac{k}{m}}$ is the natural frequency of the mass/spring system. Recall that the angular frequency, and therefore the frequency, of the motor can be adjusted. Looking at the denominator of the equation for the amplitude, when the driving frequency is much smaller, or much larger, than the natural frequency, the square of the difference of the two angular frequencies $\left(\omega^2 - \omega_0^2\right)^2$ is positive and large, making the denominator large, and the result is a small amplitude for the oscillations of the mass. As the frequency of the driving force approaches the natural frequency of the system, the denominator becomes small and the amplitude of the oscillations becomes large. The maximum amplitude results when the frequency of the driving force equals the natural frequency of the system $\left(A_{\max} = \frac{F_0}{b\omega}\right)$.

Figure 15.31 shows a graph of the amplitude of a damped harmonic oscillator as a function of the frequency of the periodic force driving it. Each of the three curves on the graph represents a different amount of damping. All three curves peak at the point where the frequency of the driving force equals the natural frequency of the harmonic oscillator. The highest peak, or greatest response, is for the least amount of damping, because less energy is removed by the damping force. Note that since the amplitude grows as the damping decreases, taking this to the limit where there is no damping ($b = 0$), the amplitude becomes infinite.

Note that a small-amplitude driving force can produce a large-amplitude response. This phenomenon is known as resonance. A common example of resonance is a parent pushing a small child on a swing. When the child wants to go higher, the parent does not move back and then, getting a running start, slam into the child, applying a great force in a short interval. Instead, the parent applies small pushes to the child at just the right frequency, and the amplitude of the child's swings increases.

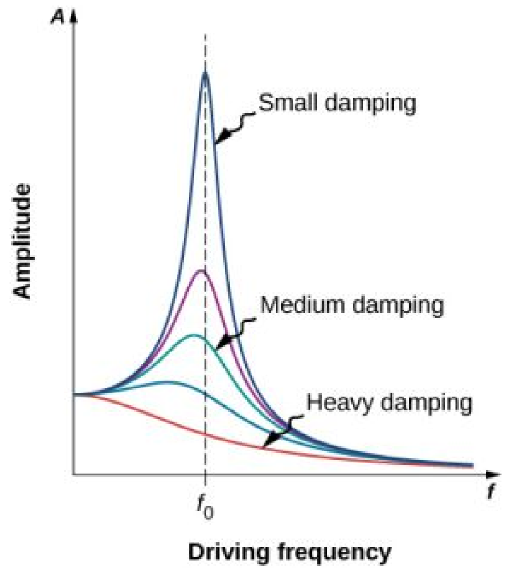


FIGURE 15.31 Amplitude of a harmonic oscillator as a function of the frequency of the driving force. The curves represent the same oscillator with the same natural frequency but with different amounts of damping. Resonance occurs when the driving frequency equals the natural frequency, and the greatest response is for the least amount of damping. The narrowest response is also for the least damping.

It is interesting to note that the widths of the resonance curves shown in Figure 15.31 depend on damping: the less the damping, the narrower the resonance. The consequence is that if you want a driven oscillator to resonate at a very specific frequency, you need as little damping as possible. For instance, a radio has a circuit that is used to choose a particular radio station. In this case, the forced damped oscillator consists of a resistor, capacitor, and inductor, which will be discussed later in this course. The circuit is "tuned" to pick a particular radio station. Here it is desirable to have the resonance curve be very narrow, to pick out the exact frequency of the radio station chosen. The narrowness of the graph, and the ability to pick out a certain frequency, is known as the quality of the system. The quality, Q, is defined as the spread of the angular frequency, or equivalently, as inverse of the fractional bandwidth, i.e., the natural frequency divided by the spread in the frequency at half the maximum amplitude $\left(Q = \frac{\omega_0}{\Delta\omega}\right)$ as shown in Figure 15.32. For a small damping, the quality is approximately equal to $Q = \frac{m\omega_0}{b}$.

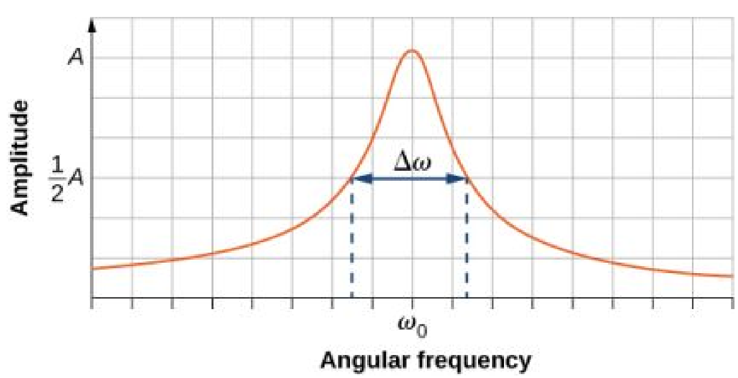


FIGURE 15.32 The quality of a system is defined as the spread in the frequencies at half the amplitude divided by the natural frequency.

These features of driven harmonic oscillators apply to a huge variety of systems. For instance, magnetic resonance imaging (MRI) is a widely used medical diagnostic tool in which atomic nuclei (mostly hydrogen nuclei or protons) are made to resonate by incoming radio waves (on the order of 100 MHz). In all of these cases, the efficiency of energy transfer from the driving force into the oscillator is best at resonance. Figure 15.33 shows the London Millennium Footbridge that allows pedestrians to cross the River Thames in London. This bridge was nicknamed “Wobbly Bridge” when pedestrians experienced swaying motion while crossing it. The bridge was closed for roughly two years to get rid of this motion.

FIGURE 15.33 Initially when people crossed the London Millennium Footbridge, they experienced a swaying motion. People continuing to cross reinforced the oscillation’s amplitude, thereby increasing the problematic swaying. (credit: Adrian Pingstone/Wikimedia Commons)

CHECK YOUR UNDERSTANDING 15.6

A famous magic trick involves a performer singing a note toward a crystal glass until the glass shatters. Explain why the trick works in terms of resonance and natural frequency.

Chapter Review

Key Terms

amplitude (*A*) maximum displacement from the equilibrium position of an object oscillating around the equilibrium position

critically damped condition in which the damping of an oscillator causes it to return as quickly as possible to its equilibrium position without oscillating back and forth about this position

elastic potential energy potential energy stored as a result of deformation of an elastic object, such as the stretching of a spring

equilibrium position position where the spring is neither stretched nor compressed

force constant (*k*) characteristic of a spring which is defined as the ratio of the force applied to the spring to the displacement caused by the force

frequency (*f*) number of events per unit of time

natural angular frequency angular frequency of a system oscillating in SHM

oscillation single fluctuation of a quantity, or repeated and regular fluctuations of a quantity, between two extreme values around an equilibrium or average value

overdamped condition in which damping of an oscillator causes it to return to equilibrium without oscillating; oscillator moves more slowly toward equilibrium than in the critically damped system

period (*T*) time taken to complete one oscillation

periodic motion motion that repeats itself at regular time intervals

phase shift angle, in radians, that is used in a cosine or sine function to shift the function left or right, used to match up the function with the initial conditions of data

physical pendulum any extended object that swings like a pendulum

resonance large amplitude oscillations in a system produced by a small amplitude driving force, which has a frequency equal to the natural frequency

restoring force force acting in opposition to the force caused by a deformation

simple harmonic motion (SHM) oscillatory motion in a system where the restoring force is proportional to the displacement, which acts in the direction opposite to the displacement

simple harmonic oscillator a device that oscillates in SHM where the restoring force is proportional to the displacement and acts in the direction opposite to the displacement

simple pendulum point mass, called a pendulum bob, attached to a near massless string

stable equilibrium point point where the net force on a system is zero, but a small displacement of the mass will cause a restoring force that points toward the equilibrium point

torsional pendulum any suspended object that oscillates by twisting its suspension

underdamped condition in which damping of an oscillator causes the amplitude of oscillations of a damped harmonic oscillator to decrease over time, eventually approaching zero

Key Equations

Relationship between frequency and period	$f = \frac{1}{T}$
Position in SHM with $\phi = 0.00$	$x(t) = A\cos(\omega t)$
General position in SHM	$x(t) = A\cos(\omega t + \phi)$
General velocity in SHM	$v(t) = -A\omega\sin(\omega t + \phi)$
General acceleration in SHM	$a(t) = -A\omega^2\cos(\omega t + \phi)$
Maximum displacement (amplitude) of SHM	$x_{\text{max}} = A$
Maximum velocity of SHM	$\lvert v_{\text{max}}\rvert = A\omega$
Maximum acceleration of SHM	$\lvert a_{\text{max}}\rvert = A\omega^2$
Angular frequency of a mass-spring system in SHM	$\omega = \sqrt{\frac{k}{m}}$
Period of a mass-spring system in SHM	$T = 2\pi\sqrt{\frac{m}{k}}$
Frequency of a mass-spring system in SHM	$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$
Energy in a mass-spring system in SHM	$E_{\text{Total}} = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kA^2$
The velocity of the mass in a spring-mass system in SHM	$v = \pm\sqrt{\frac{k}{m}\left(A^2 - x^2\right)}$

The x-component of the radius of a rotating disk	$x(t) = A\cos(\omega t + \phi)$
The x-component of the velocity of the edge of a rotating disk	$v(t) = -v_{\max}\sin(\omega t + \phi)$
The x-component of the acceleration of the edge of a rotating disk	$a(t) = -a_{\max}\cos(\omega t + \phi)$
Force equation for a simple pendulum	$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$
Angular frequency for a simple pendulum	$\omega = \sqrt{\frac{g}{L}}$
Period of a simple pendulum	$T = 2\pi\sqrt{\frac{L}{g}}$
Angular frequency of a physical pendulum	$\omega = \sqrt{\frac{mgL}{I}}$
Period of a physical pendulum	$T = 2\pi\sqrt{\frac{I}{mgL}}$
Period of a torsional pendulum	$T = 2\pi\sqrt{\frac{I}{\kappa}}$
Newton's second law for harmonic motion	$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$
Solution for underdamped harmonic motion	$x(t) = A_0 e^{-\frac{b}{2m}t}\cos(\omega t + \phi)$
Natural angular frequency of a mass-spring system	$\omega_0 = \sqrt{\frac{k}{m}}$
Angular frequency of underdamped harmonic motion	$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$
Newton's second law for forced, damped oscillation	$-kx - b\frac{dx}{dt} + F_o\sin(\omega t) = m\frac{d^2x}{dt^2}$
Solution to Newton's second law for forced, damped oscillations	$x(t) = A\cos(\omega t + \phi)$
Amplitude of system undergoing forced, damped oscillations	$A = \frac{F_o}{\sqrt{m^2\left(\omega^2 - \omega_o^2\right)^2 + b^2\omega^2}}$

Summary

15.1 Simple Harmonic Motion

- Periodic motion is a repeating oscillation. The time for one oscillation is the period T and the number of oscillations per unit time is the frequency f. These quantities are related by $f = \frac{1}{T}$.
- Simple harmonic motion (SHM) is oscillatory motion for a system where the restoring force is proportional to the displacement and acts in the direction opposite to the displacement.
- Maximum displacement is the amplitude A. The angular frequency ω, period T, and frequency f of a simple harmonic oscillator are given by $\omega = \sqrt{\frac{k}{m}}, T = 2\pi\sqrt{\frac{m}{k}}$, and $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$, where m is the mass of the system and k is the force constant.
- Displacement as a function of time in SHM is given by $x(t) = A\cos\left(\frac{2\pi}{T}t + \phi\right) = A\cos(\omega t + \phi)$.
- The velocity is given by $v(t) = -A\omega\sin(\omega t + \phi) = -v_{\max}\sin(\omega t + \phi)$, where $v_{\max} = A\omega = A\sqrt{\frac{k}{m}}$.
- The acceleration is $a(t) = -A\omega^2\cos(\omega t + \phi) = -a_{\max}\cos(\omega t + \phi)$, where $a_{\max} = A\omega^2 = A\frac{k}{m}$.

15.2 Energy in Simple Harmonic Motion

- The simplest type of oscillations are related to systems that can be described by Hooke's law, $F = -kx$, where F is the restoring force, x is the displacement from equilibrium or deformation, and k is the force constant of the system.
- Elastic potential energy U stored in the deformation of a system that can be described by Hooke's law is given by $U = \frac{1}{2}kx^2$.
- Energy in the simple harmonic oscillator is shared between elastic potential energy and kinetic energy, with the total being constant:

$$E_{\text{Total}} = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \text{constant}.$$

- The magnitude of the velocity as a function of position for the simple harmonic oscillator can be found by using
$$|v| = \sqrt{\frac{k}{m}\left(A^2 - x^2\right)}.$$

15.3 Comparing Simple Harmonic Motion and Circular Motion

- A projection of uniform circular motion undergoes simple harmonic oscillation.
- Consider a circle with a radius *A*, moving at a constant angular speed ω. A point on the edge of the circle moves at a constant tangential speed of $v_{\text{max}} = A\omega$. The projection of the radius onto the *x*-axis is $x\,(t) = A\cos\,(\omega t + \phi)$, where (ϕ) is the phase shift. The *x*-component of the tangential velocity is $v\,(t) = -A\omega\sin\,(\omega t + \phi)$.

15.4 Pendulums

- A mass *m* suspended by a wire of length *L* and negligible mass is a simple pendulum and undergoes SHM for amplitudes less than about $15°$. The period of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$, where *L* is the length of the string and *g* is the acceleration due to gravity.
- The period of a physical pendulum $T = 2\pi\sqrt{\frac{I}{mgL}}$ can be found if the moment of inertia is known. The length between the point of rotation and the center of mass is *L*.
- The period of a torsional pendulum $T = 2\pi\sqrt{\frac{I}{\kappa}}$ can be found if the moment of inertia and torsion constant are known.

15.5 Damped Oscillations

- Damped harmonic oscillators have non-conservative forces that dissipate their energy.
- Critical damping returns the system to equilibrium as fast as possible without overshooting.
- An underdamped system will oscillate through the equilibrium position.
- An overdamped system moves more slowly toward equilibrium than one that is critically damped.

15.6 Forced Oscillations

- A system's natural frequency is the frequency at which the system oscillates if not affected by driving or damping forces.
- A periodic force driving a harmonic oscillator at its natural frequency produces resonance. The system is said to resonate.
- The less damping a system has, the higher the amplitude of the forced oscillations near resonance. The more damping a system has, the broader response it has to varying driving frequencies.

Conceptual Questions

15.1 Simple Harmonic Motion

1. What conditions must be met to produce SHM?

2. (a) If frequency is not constant for some oscillation, can the oscillation be SHM? (b) Can you think of any examples of harmonic motion where the frequency may depend on the amplitude?

3. Give an example of a simple harmonic oscillator, specifically noting how its frequency is independent of amplitude.

4. Explain why you expect an object made of a stiff material to vibrate at a higher frequency than a similar object made of a more pliable material.

5. As you pass a freight truck with a trailer on a highway, you notice that its trailer is bouncing up and down slowly. Is it more likely that the trailer is heavily loaded or nearly empty? Explain your answer.

6. Some people modify cars to be much closer to the ground than when manufactured. Should they install stiffer springs? Explain your answer.

15.2 Energy in Simple Harmonic Motion

7. Describe a system in which elastic potential energy is stored.

8. Explain in terms of energy how dissipative forces such as friction reduce the amplitude of a harmonic oscillator. Also explain how a driving mechanism can compensate. (A pendulum clock is such a system.)

9. The temperature of the atmosphere oscillates from a maximum near noontime and a minimum near sunrise. Would you consider the atmosphere to be in stable or unstable equilibrium?

15.3 Comparing Simple Harmonic Motion and Circular Motion

10. Can this analogy of SHM to circular motion be carried out with an object oscillating on a spring vertically hung from the ceiling? Why or why not? If given the choice, would you prefer to use a sine function or a cosine function to model the motion?

11. If the maximum speed of the mass attached to a spring, oscillating on a frictionless table, was increased, what characteristics of the rotating disk would need to be changed?

15.4 Pendulums

12. Pendulum clocks are made to run at the correct rate by adjusting the pendulum's length. Suppose you move from one city to another where the acceleration due to gravity is slightly greater, taking your pendulum clock with you, will you have to lengthen or shorten the pendulum to keep the correct time, other factors remaining constant? Explain your answer.

13. A pendulum clock works by measuring the period of a pendulum. In the springtime the clock runs with perfect time, but in the summer and winter the length of the pendulum changes. When most materials are heated, they expand. Does the clock run too fast or too slow in the summer? What about the winter?

14. With the use of a phase shift, the position of an object may be modeled as a cosine or sine function. If given the option, which function would you choose? Assuming that the phase shift is zero, what are the initial conditions of function; that is, the initial position, velocity, and acceleration, when using a sine function? How about when a cosine function is used?

15.5 Damped Oscillations

15. Give an example of a damped harmonic oscillator. (They are more common than undamped or simple harmonic oscillators.)

16. How would a car bounce after a bump under each of these conditions?
(a) overdamping
(b) underdamping
(c) critical damping

17. Most harmonic oscillators are damped and, if undriven, eventually come to a stop. Why?

15.6 Forced Oscillations

18. Why are soldiers in general ordered to "route step" (walk out of step) across a bridge?

19. Do you think there is any harmonic motion in the physical world that is not damped harmonic motion? Try to make a list of five examples of undamped harmonic motion and damped harmonic motion. Which list was easier to make?

20. Some engineers use sound to diagnose performance problems with car engines. Occasionally, a part of the engine is designed that resonates at the frequency of the engine. The unwanted oscillations can cause noise that irritates the driver or could lead to the part failing prematurely. In one case, a part was located that had a length *L* made of a material with a mass *M*. What can be done to correct this problem?

Problems

15.1 Simple Harmonic Motion

21. Prove that using $x(t) = A\sin(\omega t + \phi)$ will produce the same results for the period for the oscillations of a mass and a spring. Why do you think the cosine function was chosen?

22. What is the period of 60.0 Hz of electrical power?

23. If your heart rate is 150 beats per minute during strenuous exercise, what is the time per beat in units of seconds?

24. Find the frequency of a tuning fork that takes $2.50 \times 10^{-3}\,\text{s}$ to complete one oscillation.

25. A stroboscope is set to flash every $8.00 \times 10^{-5}\,\text{s}$. What is the frequency of the flashes?

26. A tire has a tread pattern with a crevice every 2.00 cm. Each crevice makes a single vibration as the tire moves. What is the frequency of these vibrations if the car moves at 30.0 m/s?

27. Each piston of an engine makes a sharp sound every other revolution of the engine. (a) How fast is a race car going if its eight-cylinder engine emits a sound of frequency 750 Hz, given that the engine makes 2000 revolutions per kilometer? (b) At how many revolutions per minute is the engine rotating?

28. A type of cuckoo clock keeps time by having a mass bouncing on a spring, usually something cute like a cherub in a chair. What force constant is needed to produce a period of 0.500 s for a 0.0150-kg mass?

29. A mass m_0 is attached to a spring and hung vertically. The mass is raised a short distance in the vertical direction and released. The mass oscillates with a frequency f_0. If the mass is replaced with a mass nine times as large, and the experiment was repeated, what would be the frequency of the oscillations in terms of f_0 ?

30. A 0.500-kg mass suspended from a spring oscillates with a period of 1.50 s. How much mass must be added to the object to change the period to 2.00 s?

31. How much leeway (both percentage and mass) would you have in the selection of the mass of the object in the previous problem if you did not wish the new period to be greater than 2.01 s or less than 1.99 s?

15.2 Energy in Simple Harmonic Motion

32. Fish are hung on a spring scale to determine their mass. (a) What is the force constant of the spring in such a scale if it the spring stretches 8.00 cm for a 10.0 kg load? (b) What is the mass of a fish that stretches the spring 5.50 cm? (c) How far apart are the half-kilogram marks on the scale?

33. It is weigh-in time for the local under-85-kg rugby team. The bathroom scale used to assess eligibility can be described by Hooke's law and is depressed 0.75 cm by its maximum load of 120 kg. (a) What is the spring's effective force constant? (b) A player stands on the scales and depresses it by 0.48 cm. Is he eligible to play on this under-85-kg team?

34. One type of BB gun uses a spring-driven plunger to blow the BB from its barrel. (a) Calculate the force constant of its plunger's spring if you must compress it 0.150 m to drive the 0.0500-kg plunger to a top speed of 20.0 m/s. (b) What force must be exerted to compress the spring?

35. When an 80.0-kg man stands on a pogo stick, the spring is compressed 0.120 m. (a) What is the force constant of the spring? (b) Will the spring be compressed more when he hops down the road?

36. A spring has a length of 0.200 m when a 0.300-kg mass hangs from it, and a length of 0.750 m when a 1.95-kg mass hangs from it. (a) What is the force constant of the spring? (b) What is the unloaded length of the spring?

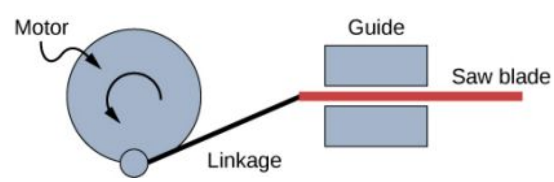
37. The length of nylon rope from which a mountain climber is suspended has an effective force constant of 1.40×10^4 N/m. (a) What is the frequency at which he bounces, given his mass plus and the mass of his equipment are 90.0 kg? (b) How much would this rope stretch to break the climber's fall if he free-falls 2.00 m before the rope runs out of slack? (*Hint:* Use conservation of energy.) (c) Repeat both parts of this problem in the situation where twice this length of nylon rope is used.

15.3 Comparing Simple Harmonic Motion and Circular Motion

38. The motion of a mass on a spring hung vertically, where the mass oscillates up and down, can also be modeled using the rotating disk. Instead of the lights being placed horizontally along the top and pointing down, place the lights vertically and have the lights shine on the side of the rotating disk. A shadow will be produced on a nearby wall, and will move up and down. Write the equations of motion for the shadow taking the position at $t = 0.0$ s to be $y = 0.0$ m with the mass moving in the positive *y*-direction.

39. (a) A novelty clock has a 0.0100-kg-mass object bouncing on a spring that has a force constant of 1.25 N/m. What is the maximum velocity of the object if the object bounces 3.00 cm above and below its equilibrium position? (b) How many joules of kinetic energy does the object have at its maximum velocity?

40. Reciprocating motion uses the rotation of a motor to produce linear motion up and down or back and forth. This is how a reciprocating saw operates, as shown below.



If the motor rotates at 60 Hz and has a radius of 3.0 cm, estimate the maximum speed of the saw blade as it moves left and right. This design is known as a scotch yoke.

41. A student stands on the edge of a merry-go-round which rotates five times a minute and has a radius of two meters one evening as the sun is setting. The student produces a shadow on the nearby building. (a) Write an equation for the position of the shadow. (b) Write an equation for the velocity of the shadow.

15.4 Pendulums

42. What is the length of a pendulum that has a period of 0.500 s?

43. Some people think a pendulum with a period of 1.00 s can be driven with “mental energy” or psycho kinetically, because its period is the same as an average heartbeat. True or not, what is the length of such a pendulum?

44. What is the period of a 1.00-m-long pendulum?

45. How long does it take a child on a swing to complete one swing if her center of gravity is 4.00 m below the pivot?

46. The pendulum on a cuckoo clock is 5.00-cm long. What is its frequency?

47. Two parakeets sit on a swing with their combined CMs 10.0 cm below the pivot. At what frequency do they swing?

48. (a) A pendulum that has a period of 3.00000 s and that is located where the acceleration due to gravity is $9.79\ \mathrm{m/s^2}$ is moved to a location where the acceleration due to gravity is $9.82\ \mathrm{m/s^2}$. What is its new period? (b) Explain why so many digits are needed in the value for the period, based on the relation between the period and the acceleration due to gravity.

49. A pendulum with a period of 2.00000 s in one location ($g = 9.80\mathrm{m/s^2}$) is moved to a new location where the period is now 1.99796 s. What is the acceleration due to gravity at its new location?

50. (a) What is the effect on the period of a pendulum if you double its length? (b) What is the effect on the period of a pendulum if you decrease its length by 5.00%?

15.5 Damped Oscillations

51. The amplitude of a lightly damped oscillator decreases by 3.0% during each cycle. What percentage of the mechanical energy of the oscillator is lost in each cycle?

15.6 Forced Oscillations

52. How much energy must the shock absorbers of a 1200-kg car dissipate in order to damp a bounce that initially has a velocity of 0.800 m/s at the equilibrium position? Assume the car returns to its original vertical position.

53. If a car has a suspension system with a force constant of $5.00 \times 10^4\ \mathrm{N/m}$, how much energy must the car’s shocks remove to dampen an oscillation starting with a maximum displacement of 0.0750 m?

54. (a) How much will a spring that has a force constant of 40.0 N/m be stretched by an object with a mass of 0.500 kg when hung motionless from the spring? (b) Calculate the decrease in gravitational potential energy of the 0.500-kg object when it descends this distance. (c) Part of this gravitational energy goes into the spring. Calculate the energy stored in the spring by this stretch, and compare it with the gravitational potential energy. Explain where the rest of the energy might go.

55. Suppose you have a 0.750-kg object on a horizontal surface connected to a spring that has a force constant of 150 N/m. There is simple friction between the object and surface with a static coefficient of friction $\mu_\mathrm{s} = 0.100$. (a) How far can the spring be stretched without moving the mass? (b) If the object is set into oscillation with an amplitude twice the distance found in part (a), and the kinetic coefficient of friction is $\mu_\mathrm{k} = 0.0850$, what total distance does it travel before stopping? Assume it starts at the maximum amplitude.

Additional Problems

56. Suppose you attach an object with mass *m* to a vertical spring originally at rest, and let it bounce up and down. You release the object from rest at the spring’s original rest length, the length of the spring in equilibrium, without the mass attached. The amplitude of the motion is the distance between the equilibrium position of the spring without the mass attached and the equilibrium position of the spring with the mass attached. (a) Show that the spring exerts an upward force of 2.00*mg* on the object at its lowest point. (b) If the spring has a force constant of 10.0 N/m, is hung horizontally, and the position of the free end of the spring is marked as $y = 0.00\text{ m}$, where is the new equilibrium position if a 0.25-kg-mass object is hung from the spring? (c) If the spring has a force constant of 10.0 N/m and a 0.25-kg-mass object is set in motion as described, find the amplitude of the oscillations. (d) Find the maximum velocity.

57. A diver on a diving board is undergoing SHM. Her mass is 55.0 kg and the period of her motion is 0.800 s. The next diver's period of simple harmonic oscillation is 1.05 s. What is the second diver's mass if the mass of the board is negligible?

58. Suppose a diving board with no one on it bounces up and down in a SHM with a frequency of 4.00 Hz. The board has an effective mass of 10.0 kg. What is the frequency of the SHM of a 75.0-kg diver on the board?

59. The device pictured in the following figure entertains infants while keeping them from wandering. The child bounces in a harness suspended from a door frame by a spring. (a) If the spring stretches 0.250 m while supporting an 8.0-kg child, what is its force constant? (b) What is the time for one complete bounce of this child? (c) What is the child's maximum velocity if the amplitude of her bounce is 0.200 m?

FIGURE 15.34 (credit: Lisa Doehnert)

60. A mass is placed on a frictionless, horizontal table. A spring ($k = 100\text{ N/m}$), which can be stretched or compressed, is placed on the table. A 5.00-kg mass is attached to one end of the spring, the other end is anchored to the wall. The equilibrium position is marked at zero. A student moves the mass out to $x = 4.00\text{cm}$ and releases it from rest. The mass oscillates in SHM. (a) Determine the equations of motion. (b) Find the position, velocity, and acceleration of the mass at time $t = 3.00\text{ s}$.

61. Find the ratio of the new/old periods of a pendulum if the pendulum were transported from Earth to the Moon, where the acceleration due to gravity is 1.63 m/s^2.

62. At what rate will a pendulum clock run on the Moon, where the acceleration due to gravity is 1.63 m/s^2, if it keeps time accurately on Earth? That is, find the time (in hours) it takes the clock's hour hand to make one revolution on the Moon.

63. If a pendulum-driven clock gains 5.00 s/day, what fractional change in pendulum length must be made for it to keep perfect time?

64. A 2.00-kg object hangs, at rest, on a 1.00-m-long string attached to the ceiling. A 100-g mass is fired with a speed of 20 m/s at the 2.00-kg mass, and the 100.00-g mass collides perfectly elastically with the 2.00-kg mass. Write an equation for the motion of the hanging mass after the collision. Assume air resistance is negligible.

65. A 2.00-kg object hangs, at rest, on a 1.00-m-long string attached to the ceiling. A 100-g object is fired with a speed of 20 m/s at the 2.00-kg object, and the two objects collide and stick together in a totally inelastic collision. Write an equation for the motion of the system after the collision. Assume air resistance is negligible.

66. Assume that a pendulum used to drive a grandfather clock has a length $L_0 = 1.00\text{ m}$ and a mass *M* at temperature $T = 20.00°\text{C}$. It can be modeled as a physical pendulum as a rod oscillating around one end. By what percentage will the period change if the temperature increases by $10°\text{C}$? Assume the length of the rod changes linearly with temperature, where $L = L_0\,(1 + \alpha\Delta T)$ and the rod is made of brass $\left(\alpha = 18 \times 10^{-6}\,°\text{C}^{-1}\right)$.

67. A 2.00-kg block lies at rest on a frictionless table. A spring, with a spring constant of 100 N/m is attached to the wall and to the block. A second block of 0.50 kg is placed on top of the first block. The 2.00-kg block is gently pulled to a position $x = +A$ and released from rest. There is a coefficient of friction of 0.45 between the two blocks. (a) What is the period of the oscillations? (b) What is the largest amplitude of motion that will allow the blocks to oscillate without the 0.50-kg block sliding off?

Challenge Problems

68. A suspension bridge oscillates with an effective force constant of 1.00×10^8 N/m. (a) How much energy is needed to make it oscillate with an amplitude of 0.100 m? (b) If soldiers march across the bridge with a cadence equal to the bridge's natural frequency and impart 1.00×10^4 J of energy each second, how long does it take for the bridge's oscillations to go from 0.100 m to 0.500 m amplitude.

69. Near the top of the Citigroup Center building in New York City, there is an object with mass of 4.00×10^5 kg on springs that have adjustable force constants. Its function is to dampen wind-driven oscillations of the building by oscillating at the same frequency as the building is being driven—the driving force is transferred to the object, which oscillates instead of the entire building. (a) What effective force constant should the springs have to make the object oscillate with a period of 2.00 s? (b) What energy is stored in the springs for a 2.00-m displacement from equilibrium?

70. Parcels of air (small volumes of air) in a stable atmosphere (where the temperature increases with height) can oscillate up and down, due to the restoring force provided by the buoyancy of the air parcel. The frequency of the oscillations are a measure of the stability of the atmosphere. Assuming that the acceleration of an air parcel can be modeled as $\frac{\partial^2 z'}{\partial t^2} = \frac{g}{\rho_o} \frac{\partial \rho(z)}{\partial z} z'$, prove that $z' = z_0' e^{t\sqrt{-N^2}}$ is a solution, where N is known as the Brunt-Väisälä frequency. Note that in a stable atmosphere, the density decreases with height and parcel oscillates up and down.

71. Consider the van der Waals potential $U(r) = U_o\left[\left(\frac{R_o}{r}\right)^{12} - 2\left(\frac{R_o}{r}\right)^6\right]$, used to model the potential energy function of two molecules, where the minimum potential is at $r = R_o$. Find the force as a function of *r*. Consider a small displacement $r = R_o + r'$ and use the binomial theorem: $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$, to show that the force does approximate a Hooke's law force.

72. Suppose the length of a clock's pendulum is changed by 1.000%, exactly at noon one day. What time will the clock read 24.00 hours later, assuming it the pendulum has kept perfect time before the change? Note that there are two answers, and perform the calculation to four-digit precision.

73. (a) The springs of a pickup truck act like a single spring with a force constant of 1.30×10^5 N/m. By how much will the truck be depressed by its maximum load of 1000 kg? (b) If the pickup truck has four identical springs, what is the force constant of each?

CHAPTER 16
Waves

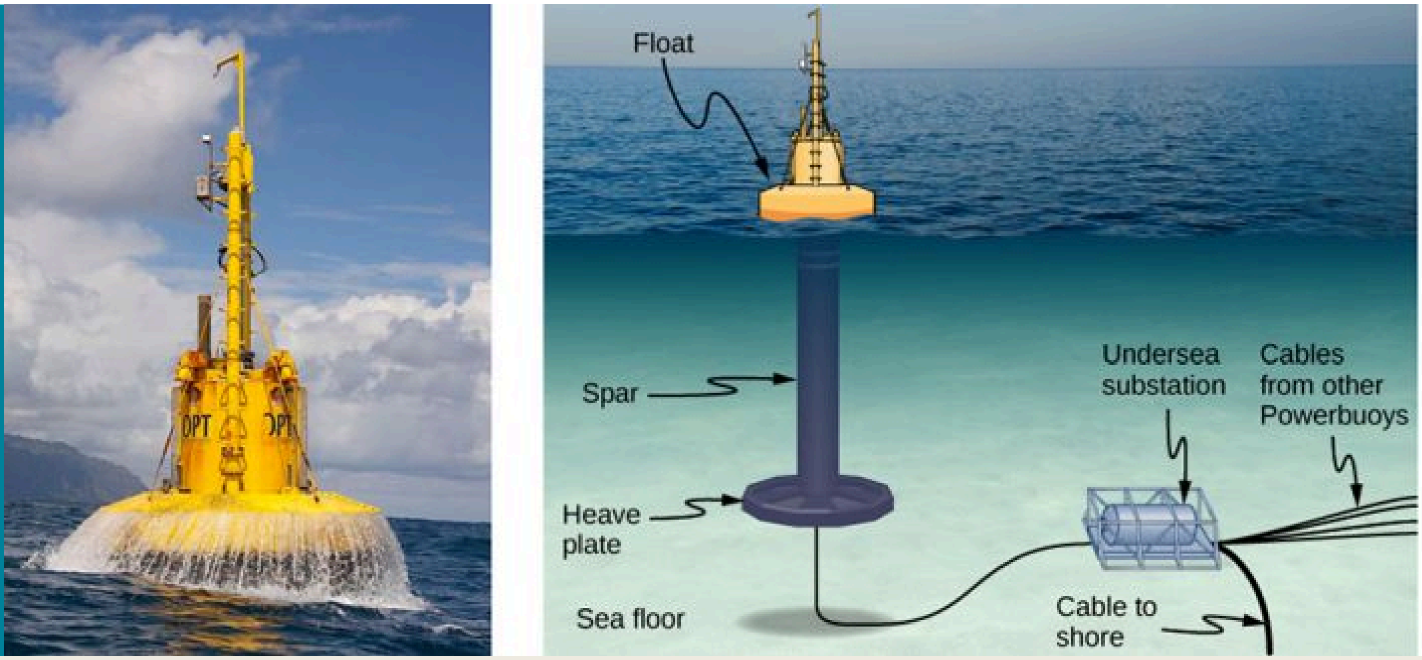


FIGURE 16.1 From the world of renewable energy sources comes the electric power-generating buoy. Although there are many versions, this one converts the up-and-down motion, as well as side-to-side motion, of the buoy into rotational motion in order to turn an electric generator, which stores the energy in batteries.

CHAPTER OUTLINE

INTRODUCTION In this chapter, we study the physics of wave motion. We concentrate on mechanical waves, which are disturbances that move through a medium such as air or water. Like simple harmonic motion studied in the preceding chapter, the energy transferred through the medium is proportional to the amplitude squared. Surface water waves in the ocean are transverse waves in which the energy of the wave travels horizontally while the water oscillates up and down due to some restoring force. In the picture above, a buoy is used to convert the awesome power of ocean waves into electricity. The up-and-down motion of the buoy generated as the waves pass is converted into rotational motion that turns a rotor in an electric generator. The generator charges batteries, which are in turn used to provide a consistent energy source for the end user. This model was successfully tested by the US Navy in a project to provide power to coastal security networks and was able to provide an average power of 350 W. The buoy survived the difficult ocean environment, including operation off the New Jersey coast through Hurricane Irene in 2011.

The concepts presented in this chapter will be the foundation for many interesting topics, from the transmission of information to the concepts of quantum mechanics.

16.1 Traveling Waves

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the basic characteristics of wave motion
- Define the terms wavelength, amplitude, period, frequency, and wave speed
- Explain the difference between longitudinal and transverse waves, and give examples of each type
- List the different types of waves

We saw in Oscillations that oscillatory motion is an important type of behavior that can be used to model a wide range of physical phenomena. Oscillatory motion is also important because oscillations can generate waves, which are of fundamental importance in physics. Many of the terms and equations we studied in the chapter on oscillations apply equally well to wave motion (Figure 16.2).

FIGURE 16.2 An ocean wave is probably the first picture that comes to mind when you hear the word "wave." Although this breaking wave, and ocean waves in general, have apparent similarities to the basic wave characteristics we will discuss, the mechanisms driving ocean waves are highly complex and beyond the scope of this chapter. It may seem natural, and even advantageous, to apply the concepts in this chapter to ocean waves, but ocean waves are nonlinear, and the simple models presented in this chapter do not fully explain them. (credit: Steve Jurvetson)

Types of Waves

A **wave** is a disturbance that propagates, or moves from the place it was created. There are three basic types of waves: mechanical waves, electromagnetic waves, and matter waves.

Basic **mechanical waves** are governed by Newton's laws and require a medium. A medium is the substance mechanical waves propagate through, and the medium produces an elastic restoring force when it is deformed. Mechanical waves transfer energy and momentum, without transferring mass. Some examples of mechanical waves are water waves, sound waves, and seismic waves. The medium for water waves is water; for sound waves, the medium is usually air. (Sound waves can travel in other media as well; we will look at that in more detail in Sound.) For surface water waves, the disturbance occurs on the surface of the water, perhaps created by a rock thrown into a pond or by a swimmer splashing the surface repeatedly. For sound waves, the disturbance is a change in air pressure, perhaps created by the oscillating cone inside a speaker or a vibrating tuning fork. In both cases, the disturbance is the oscillation of the molecules of the fluid. In mechanical waves, energy and momentum transfer with the motion of the wave, whereas the mass oscillates around an equilibrium point. (We discuss this in Energy and Power of a Wave.) Earthquakes generate seismic waves from several types of disturbances, including the disturbance of Earth's surface and pressure disturbances under the surface. Seismic waves travel through the solids and liquids that form Earth. In this chapter, we focus on mechanical waves.

Electromagnetic waves are associated with oscillations in electric and magnetic fields and do not require a medium. Examples include gamma rays, X-rays, ultraviolet waves, visible light, infrared waves, microwaves, and radio waves. Electromagnetic waves can travel through a vacuum at the speed of light, $v = c = 2.99792458 \times 10^8 \text{ m/s}$. For example, light from distant stars travels through the vacuum of space and reaches Earth. Electromagnetic waves have some characteristics that are similar to mechanical waves; they are covered in more detail in Electromagnetic Waves (http://openstax.org/books/university-physics-volume-2/pages/16-introduction).

Matter waves are a central part of the branch of physics known as quantum mechanics. These waves are associated with protons, electrons, neutrons, and other fundamental particles found in nature. The theory that all types of matter have wave-like properties was first proposed by Louis de Broglie in 1924. Matter waves are discussed in Photons and Matter Waves (http://openstax.org/books/university-physics-volume-3/pages/6-introduction).

Mechanical Waves

Mechanical waves exhibit characteristics common to all waves, such as amplitude, wavelength, period, frequency, and energy. All wave characteristics can be described by a small set of underlying principles.

The simplest mechanical waves repeat themselves for several cycles and are associated with simple harmonic motion. These simple harmonic waves can be modeled using some combination of sine and cosine functions. For example, consider the simplified surface water wave that moves across the surface of water as illustrated in Figure 16.3. Unlike complex ocean waves, in surface water waves, the medium, in this case water, moves vertically, oscillating up and down, whereas the disturbance of the wave moves horizontally through the medium. In Figure 16.3, the waves causes a seagull to move up and down in simple harmonic motion as the wave crests and troughs (peaks and valleys) pass under the bird. The crest is the highest point of the wave, and the trough is the lowest part of the wave. The time for one complete oscillation of the up-and-down motion is the wave's period *T*. The wave's frequency is the number of waves that pass through a point per unit time and is equal to $f = 1/T$. The period can be expressed using any convenient unit of time but is usually measured in seconds; frequency is usually measured in hertz (Hz), where $1 \text{ Hz} = 1 \text{ s}^{-1}$.

The length of the wave is called the **wavelength** and is represented by the Greek letter lambda (λ), which is measured in any convenient unit of length, such as a centimeter or meter. The wavelength can be measured between any two similar points along the medium that have the same height and the same slope. In Figure 16.3, the wavelength is shown measured between two crests. As stated above, the period of the wave is equal to the time for one oscillation, but it is also equal to the time for one wavelength to pass through a point along the wave's path.

The amplitude of the wave (*A*) is a measure of the maximum displacement of the medium from its equilibrium position. In the figure, the equilibrium position is indicated by the dotted line, which is the height of the water if there were no waves moving through it. In this case, the wave is symmetrical, the crest of the wave is a distance $+A$ above the equilibrium position, and the trough is a distance $-A$ below the equilibrium position. The units for the amplitude can be centimeters or meters, or any convenient unit of distance.

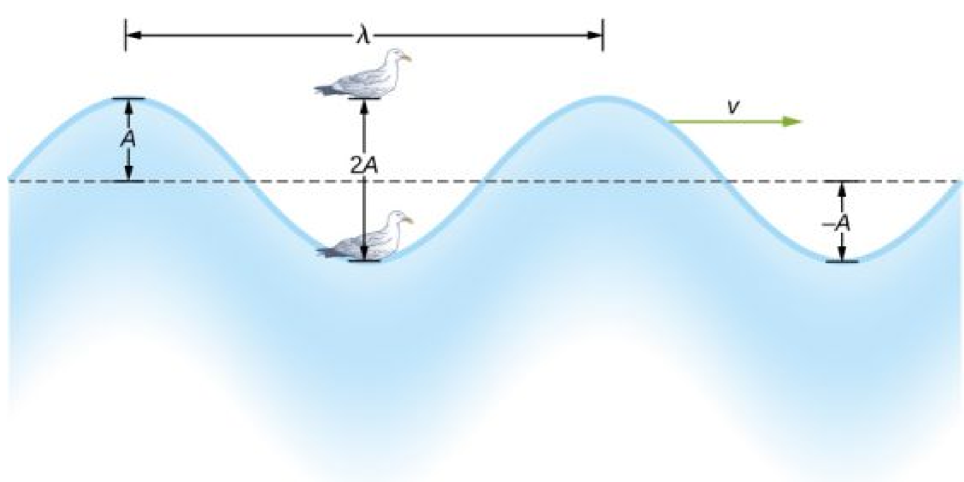


FIGURE 16.3 An idealized surface water wave passes under a seagull that bobs up and down in simple harmonic motion. The wave has a

wavelength λ, which is the distance between adjacent identical parts of the wave. The amplitude *A* of the wave is the maximum displacement of the wave from the equilibrium position, which is indicated by the dotted line. In this example, the medium moves up and down, whereas the disturbance of the surface propagates parallel to the surface at a speed *v*.

The water wave in the figure moves through the medium with a propagation velocity $\vec{\mathbf{v}}$. The magnitude of the **wave velocity** is the distance the wave travels in a given time, which is one wavelength in the time of one period, and the **wave speed** is the magnitude of wave velocity. In equation form, this is

$$v = \frac{\lambda}{T} = \lambda f. \tag{16.1}$$

This fundamental relationship holds for all types of waves. For water waves, *v* is the speed of a surface wave; for sound, *v* is the speed of sound; and for visible light, *v* is the speed of light.

Transverse and Longitudinal Waves

We have seen that a simple mechanical wave consists of a periodic disturbance that propagates from one place to another through a medium. In Figure 16.4(a), the wave propagates in the horizontal direction, whereas the medium is disturbed in the vertical direction. Such a wave is called a **transverse wave**. In a transverse wave, the wave may propagate in any direction, but the disturbance of the medium is perpendicular to the direction of propagation. In contrast, in a **longitudinal wave** or compressional wave, the disturbance is parallel to the direction of propagation. Figure 16.4(b) shows an example of a longitudinal wave. The size of the disturbance is its amplitude *A* and is completely independent of the speed of propagation *v*.

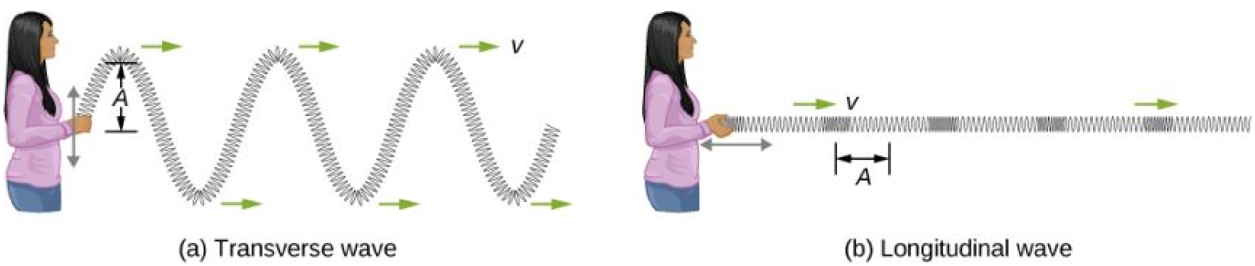


FIGURE 16.4 (a) In a transverse wave, the medium oscillates perpendicular to the wave velocity. Here, the spring moves vertically up and down, while the wave propagates horizontally to the right. (b) In a longitudinal wave, the medium oscillates parallel to the propagation of the wave. In this case, the spring oscillates back and forth, while the wave propagates to the right.

A simple graphical representation of a section of the spring shown in Figure 16.4(b) is shown in Figure 16.5. Figure 16.5(a) shows the equilibrium position of the spring before any waves move down it. A point on the spring is marked with a blue dot. Figure 16.5(b) through (g) show snapshots of the spring taken one-quarter of a period apart, sometime after the end of` the spring is oscillated back and forth in the *x*-direction at a constant frequency. The disturbance of the wave is seen as the compressions and the expansions of the spring. Note that the blue dot oscillates around its equilibrium position a distance *A*, as the longitudinal wave moves in the positive *x*-direction with a constant speed. The distance *A* is the amplitude of the wave. The *y*-position of the dot does not change as the wave moves through the spring. The wavelength of the wave is measured in part (d). The wavelength depends on the speed of the wave and the frequency of the driving force.

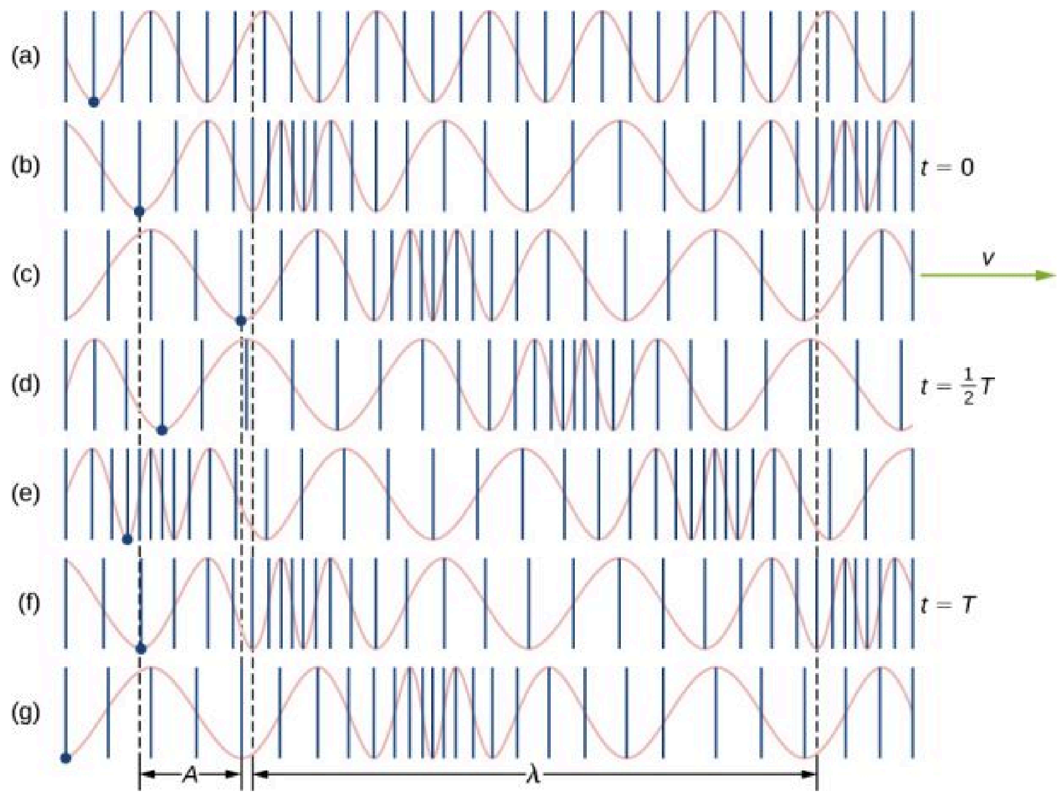


FIGURE 16.5 (a) This is a simple, graphical representation of a section of the stretched spring shown in Figure 16.4(b), representing the spring's equilibrium position before any waves are induced on the spring. A point on the spring is marked by a blue dot. (b–g) Longitudinal waves are created by oscillating the end of the spring (not shown) back and forth along the *x*-axis. The longitudinal wave, with a wavelength λ, moves along the spring in the +*x*-direction with a wave speed *v*. For convenience, the wavelength is measured in (d). Note that the point on the spring that was marked with the blue dot moves back and forth a distance *A* from the equilibrium position, oscillating around the equilibrium position of the point.

Waves may be transverse, longitudinal, or a combination of the two. Examples of transverse waves are the waves on stringed instruments or surface waves on water, such as ripples moving on a pond. Sound waves in air and water are longitudinal. With sound waves, the disturbances are periodic variations in pressure that are transmitted in fluids. Fluids do not have appreciable shear strength, and for this reason, the sound waves in them are longitudinal waves. Sound in solids can have both longitudinal and transverse components, such as those in a seismic wave. Earthquakes generate seismic waves under Earth's surface with both longitudinal and transverse components (called compressional or P-waves and shear or S-waves, respectively). The components of seismic waves have important individual characteristics—they propagate at different speeds, for example. Earthquakes also have surface waves that are similar to surface waves on water. Ocean waves also have both transverse and longitudinal components.

EXAMPLE 16.1

Wave on a String

A student takes a 30.00-m-long string and attaches one end to the wall in the physics lab. The student then holds the free end of the rope, keeping the tension constant in the rope. The student then begins to send waves down the string by moving the end of the string up and down with a frequency of 2.00 Hz. The maximum displacement of the end of the string is 20.00 cm. The first wave hits the lab wall 6.00 s after it was created. (a) What is the speed of the wave? (b) What is the period of the wave? (c) What is the wavelength of the wave?

Strategy

a. The speed of the wave can be derived by dividing the distance traveled by the time.
b. The period of the wave is the inverse of the frequency of the driving force.
c. The wavelength can be found from the speed and the period $v = \lambda/T$.

Solution

a. The first wave traveled 30.00 m in 6.00 s:

$$v = \frac{30.00\text{ m}}{6.00\text{ s}} = 5.00\frac{\text{m}}{\text{s}}.$$

b. The period is equal to the inverse of the frequency:

$$T = \frac{1}{f} = \frac{1}{2.00\text{ s}^{-1}} = 0.50\text{ s}.$$

c. The wavelength is equal to the velocity times the period:

$$\lambda = vT = 5.00\frac{\text{m}}{\text{s}}(0.50\text{ s}) = 2.50\text{ m}.$$

Significance

The frequency of the wave produced by an oscillating driving force is equal to the frequency of the driving force.

CHECK YOUR UNDERSTANDING 16.1

When a guitar string is plucked, the guitar string oscillates as a result of waves moving through the string. The vibrations of the string cause the air molecules to oscillate, forming sound waves. The frequency of the sound waves is equal to the frequency of the vibrating string. Is the wavelength of the sound wave always equal to the wavelength of the waves on the string?

EXAMPLE 16.2

Characteristics of a Wave

A transverse mechanical wave propagates in the positive *x*-direction through a spring (as shown in Figure 16.4(a)) with a constant wave speed, and the medium oscillates between $+A$ and $-A$ around an equilibrium position. The graph in Figure 16.6 shows the height of the spring (*y*) versus the position (*x*), where the *x*-axis points in the direction of propagation. The figure shows the height of the spring versus the *x*-position at $t = 0.00$ s as a dotted line and the wave at $t = 3.00$ s as a solid line. Assume the wave has not traveled more than 1 wavelength in this time. (a) Determine the wavelength and amplitude of the wave. (b) Find the propagation velocity of the wave. (c) Calculate the period and frequency of the wave.

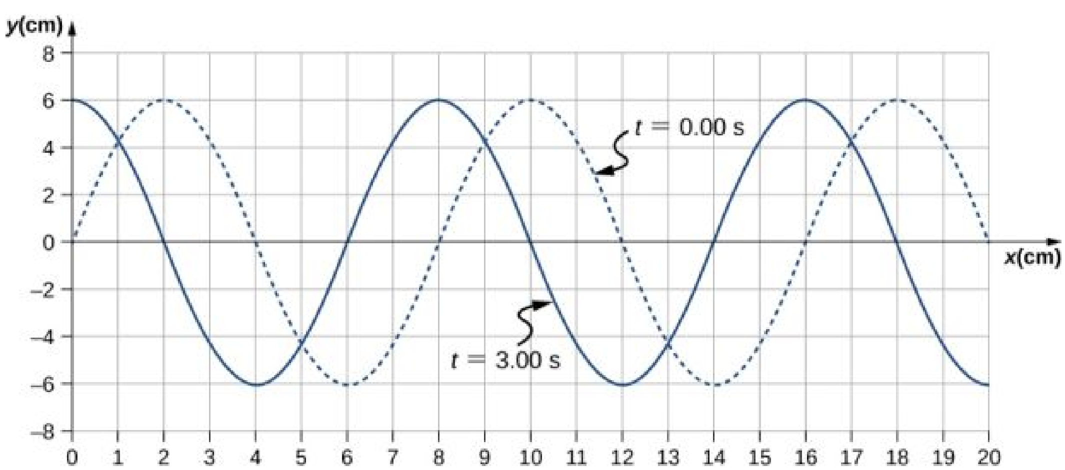


FIGURE 16.6 A transverse wave shown at two instants of time.

Strategy

a. The amplitude and wavelength can be determined from the graph.
b. Since the velocity is constant, the velocity of the wave can be found by dividing the distance traveled by the

wave by the time it took the wave to travel the distance.

c. The period can be found from $v = \frac{\lambda}{T}$ and the frequency from $f = \frac{1}{T}$.

Solution

a. Read the wavelength from the graph, looking at the purple arrow in Figure 16.7. Read the amplitude by looking at the green arrow. The wavelength is $\lambda = 8.00\ \text{cm}$ and the amplitude is $A = 6.00\ \text{cm}$.

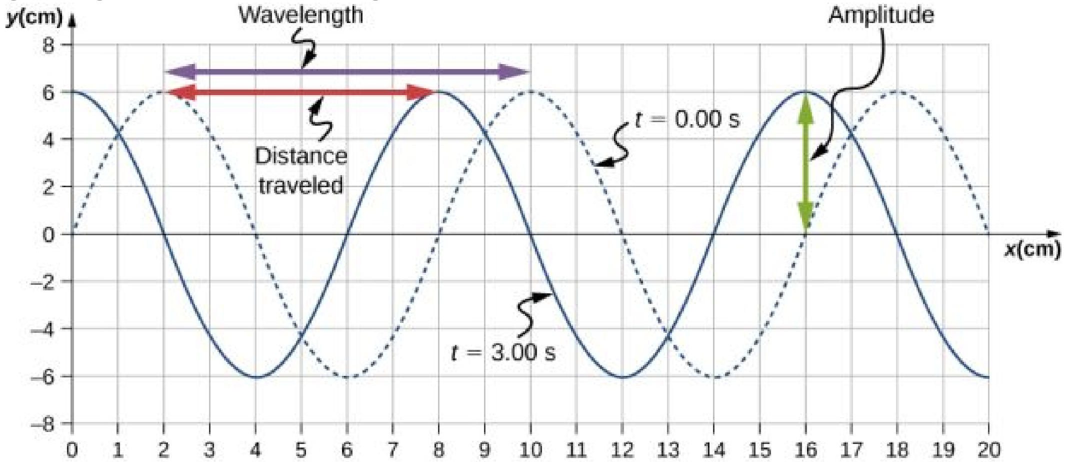


FIGURE 16.7 Characteristics of the wave marked on a graph of its displacement.

b. The distance the wave traveled from time $t = 0.00\ \text{s}$ to time $t = 3.00\ \text{s}$ can be seen in the graph. Consider the red arrow, which shows the distance the crest has moved in 3 s. The distance is $8.00\ \text{cm} - 2.00\ \text{cm} = 6.00\ \text{cm}$. The velocity is

$$v = \frac{\Delta x}{\Delta t} = \frac{8.00\ \text{cm} - 2.00\ \text{cm}}{3.00\ \text{s} - 0.00\ \text{s}} = 2.00\ \text{cm/s}.$$

c. The period is $T = \frac{\lambda}{v} = \frac{8.00\ \text{cm}}{2.00\ \text{cm/s}} = 4.00\ \text{s}$ and the frequency is $f = \frac{1}{T} = \frac{1}{4.00\ \text{s}} = 0.25\ \text{Hz}.$

Significance

Note that the wavelength can be found using any two successive identical points that repeat, having the same height and slope. You should choose two points that are most convenient. The displacement can also be found using any convenient point.

CHECK YOUR UNDERSTANDING 16.2

The propagation velocity of a transverse or longitudinal mechanical wave may be constant as the wave disturbance moves through the medium. Consider a transverse mechanical wave: Is the velocity of the medium also constant?

16.2 Mathematics of Waves

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Model a wave, moving with a constant wave velocity, with a mathematical expression
- Calculate the velocity and acceleration of the medium
- Show how the velocity of the medium differs from the wave velocity (propagation velocity)

In the previous section, we described periodic waves by their characteristics of wavelength, period, amplitude, and wave speed of the wave. Waves can also be described by the motion of the particles of the medium through which the waves move. The position of particles of the medium can be mathematically modeled as **wave functions**, which can be used to find the position, velocity, and acceleration of the particles of the medium of the wave at any time.

Pulses

A **pulse** can be described as wave consisting of a single disturbance that moves through the medium with a constant amplitude. The pulse moves as a pattern that maintains its shape as it propagates with a constant wave speed. Because the wave speed is constant, the distance the pulse moves in a time Δt is equal to $\Delta x = v\Delta t$ (Figure 16.8).

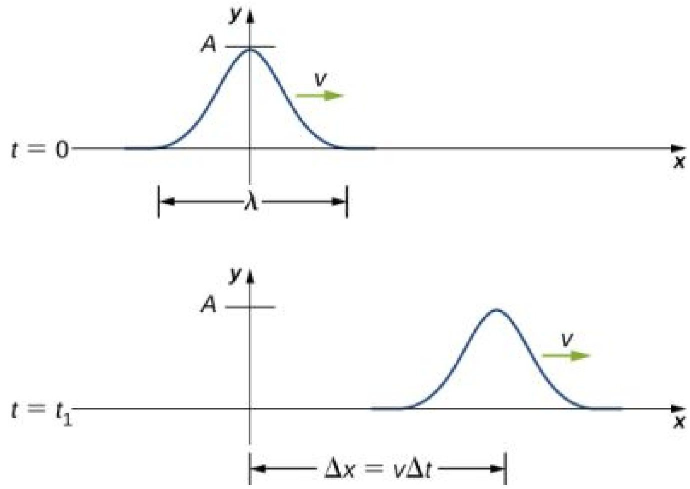


FIGURE 16.8 The pulse at time $t = 0$ is centered on $x = 0$ with amplitude *A*. The pulse moves as a pattern with a constant shape, with a constant maximum value *A*. The velocity is constant and the pulse moves a distance $\Delta x = v\Delta t$ in a time Δt. The distance traveled is measured with any convenient point on the pulse. In this figure, the crest is used.

Modeling a One-Dimensional Sinusoidal Wave using a Wave Function

Consider a string kept at a constant tension F_T where one end is fixed and the free end is oscillated between $y = +A$ and $y = -A$ by a mechanical device at a constant frequency. Figure 16.9 shows snapshots of the wave at an interval of an eighth of a period, beginning after one period $(t = T)$.

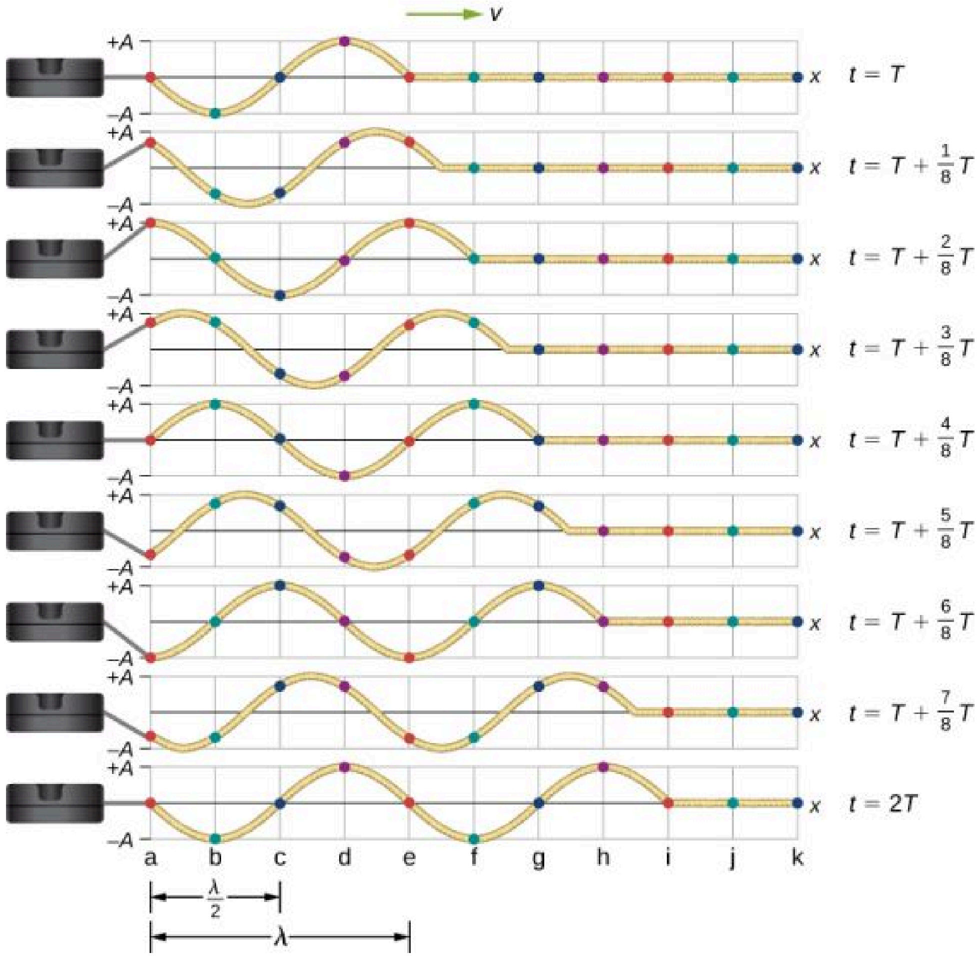


FIGURE 16.9 Snapshots of a transverse wave moving through a string under tension, beginning at time $t = T$ and taken at intervals of $\frac{1}{8}T$. Colored dots are used to highlight points on the string. Points that are a wavelength apart in the *x*-direction are highlighted with the same color dots.

Notice that each select point on the string (marked by colored dots) oscillates up and down in simple harmonic motion, between $y = +A$ and $y = -A$, with a period *T*. The wave on the string is sinusoidal and is translating in the positive *x*-direction as time progresses.

At this point, it is useful to recall from your study of algebra that if *f*(*x*) is some function, then $f\,(x - d)$ is the same function translated in the positive *x*-direction by a distance *d*. The function $f\,(x + d)$ is the same function translated in the negative *x*-direction by a distance *d*. We want to define a wave function that will give the *y*-position of each segment of the string for every position *x* along the string for every time *t*.

Looking at the first snapshot in Figure 16.9, the *y*-position of the string between $x = 0$ and $x = \lambda$ can be modeled as a sine function. This wave propagates down the string one wavelength in one period, as seen in the last snapshot. The wave therefore moves with a constant wave speed of $v = \lambda/T$.

Recall that a sine function is a function of the angle θ, oscillating between $+1$ and -1, and repeating every 2π radians (Figure 16.10). However, the *y*-position of the medium, or the wave function, oscillates between $+A$ and $-A$, and repeats every wavelength λ.

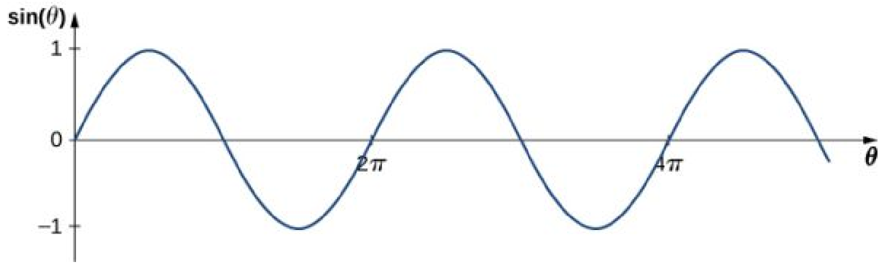


FIGURE 16.10 A sine function oscillates between $+1$ and -1 every 2π radians.

To construct our model of the wave using a periodic function, consider the ratio of the angle and the position,

$$\begin{aligned}\frac{\theta}{x} &= \frac{2\pi}{\lambda},\\ \theta &= \frac{2\pi}{\lambda}x.\end{aligned}$$

Using $\theta = \frac{2\pi}{\lambda}x$ and multiplying the sine function by the amplitude *A*, we can now model the *y*-position of the string as a function of the position *x*:

$$y(x) = A\sin\left(\frac{2\pi}{\lambda}x\right).$$

The wave on the string travels in the positive *x*-direction with a constant velocity *v*, and moves a distance *vt* in a time *t*. The wave function can now be defined by

$$y(x,t) = A\sin\left(\frac{2\pi}{\lambda}(x - vt)\right).$$

It is often convenient to rewrite this wave function in a more compact form. Multiplying through by the ratio $\frac{2\pi}{\lambda}$ leads to the equation

$$y(x,t) = A\sin\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{\lambda}vt\right).$$

The value $\frac{2\pi}{\lambda}$ is defined as the **wave number**. The symbol for the wave number is *k* and has units of inverse meters, m^{-1} :

$$k \equiv \frac{2\pi}{\lambda} \tag{16.2}$$

Recall from Oscillations that the angular frequency is defined as $\omega \equiv \frac{2\pi}{T}$. The second term of the wave function becomes

$$\frac{2\pi}{\lambda}vt = \frac{2\pi}{\lambda}\left(\frac{\lambda}{T}\right)t = \frac{2\pi}{T}t = \omega t.$$

The wave function for a simple harmonic wave on a string reduces to

$$y(x,t) = A\sin(kx \mp \omega t),$$

where *A* is the amplitude, $k = \frac{2\pi}{\lambda}$ is the wave number, $\omega = \frac{2\pi}{T}$ is the angular frequency, the minus sign is for waves moving in the positive *x*-direction, and the plus sign is for waves moving in the negative *x*-direction. The velocity of the wave is equal to

$$v = \frac{\lambda}{T} = \frac{\lambda}{T}\left(\frac{2\pi}{2\pi}\right) = \frac{\omega}{k}. \tag{16.3}$$

Think back to our discussion of a mass on a spring, when the position of the mass was modeled as $x(t) = A\cos(\omega t + \phi)$. The angle ϕ is a phase shift, added to allow for the fact that the mass may have initial conditions other than $x = +A$ and $v = 0$. For similar reasons, the initial phase is added to the wave function. The wave function modeling a sinusoidal wave, allowing for an initial phase shift ϕ, is

$$y(x,t) = A\sin(kx \mp \omega t + \phi) \tag{16.4}$$

The value

$$(kx \mp \omega t + \phi) \tag{16.5}$$

is known as the phase of the wave, where ϕ is the initial phase of the wave function. Whether the temporal term ωt is negative or positive depends on the direction of the wave. First consider the minus sign for a wave with an initial phase equal to zero $(\phi = 0)$. The phase of the wave would be $(kx - \omega t)$. Consider following a point on a wave, such as a crest. A crest will occur when $\sin\,(kx - \omega t) = 1.00$, that is, when $kx - \omega t = n\pi + \frac{\pi}{2}$, for any integral value of *n*. For instance, one particular crest occurs at $kx - \omega t = \frac{\pi}{2}$. As the wave moves, time increases and *x* must also increase to keep the phase equal to $\frac{\pi}{2}$. Therefore, the minus sign is for a wave moving in the positive *x*-direction. Using the plus sign, $kx + \omega t = \frac{\pi}{2}$. As time increases, *x* must decrease to keep the phase equal to $\frac{\pi}{2}$. The plus sign is used for waves moving in the negative *x*-direction. In summary, $y(x,t) = A\sin(kx - \omega t + \phi)$ models a wave moving in the positive *x*-direction and $y(x,t) = A\sin(kx + \omega t + \phi)$ models a wave moving in the negative *x*-direction.

Equation 16.4 is known as a simple harmonic wave function. A wave function is any function such that $f(x,t) = f(x - vt)$. Later in this chapter, we will see that it is a solution to the linear wave equation. Note that $y(x,t) = A\cos(kx + \omega t + \phi')$ works equally well because it corresponds to a different phase shift $\phi' = \phi - \frac{\pi}{2}$.

PROBLEM-SOLVING STRATEGY

Finding the Characteristics of a Sinusoidal Wave

1. To find the amplitude, wavelength, period, and frequency of a sinusoidal wave, write down the wave function in the form $y(x,t) = A\sin(kx - \omega t + \phi)$.
2. The amplitude can be read straight from the equation and is equal to *A*.
3. The period of the wave can be derived from the angular frequency $\left(T = \frac{2\pi}{\omega}\right)$.
4. The frequency can be found using $f = \frac{1}{T}$.
5. The wavelength can be found using the wave number $\left(\lambda = \frac{2\pi}{k}\right)$.

EXAMPLE 16.3

Characteristics of a Traveling Wave on a String

A transverse wave on a taut string is modeled with the wave function

$$y(x,t) = A\sin(kx - \omega t) = 0.2\text{ m}\sin\left(6.28\text{ m}^{-1}x - 1.57\text{ s}^{-1}t\right).$$

Find the amplitude, wavelength, period, and speed of the wave.

Strategy

All these characteristics of the wave can be found from the constants included in the equation or from simple combinations of these constants.

Solution

1. The amplitude, wave number, and angular frequency can be read directly from the wave equation:
$$y(x,t) = A\sin(kx - \omega t) = 0.2\text{ m}\sin\left(6.28\text{ m}^{-1}x - 1.57\text{ s}^{-1}t\right).$$
$$(A = 0.2\text{ m};\ k = 6.28\text{ m}^{-1};\ \omega = 1.57\text{ s}^{-1})$$

2. The wave number can be used to find the wavelength:

$$k = \frac{2\pi}{\lambda}.$$
$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{6.28\text{ m}^{-1}} = 1.0\text{ m}.$$

3. The period of the wave can be found using the angular frequency:

$$\omega = \frac{2\pi}{T}.$$
$$T = \frac{2\pi}{\omega} = \frac{2\pi}{1.57\text{ s}^{-1}} = 4\text{ s}.$$

4. The speed of the wave can be found using the wave number and the angular frequency. The direction of the wave can be determined by considering the sign of $kx \mp \omega t$: A negative sign suggests that the wave is moving in the positive *x*-direction:

$$|v| = \frac{\omega}{k} = \frac{1.57\text{ s}^{-1}}{6.28\text{ m}^{-1}} = 0.25\text{ m/s}.$$

Significance

All of the characteristics of the wave are contained in the wave function. Note that the wave speed is the speed of the wave in the direction parallel to the motion of the wave. Plotting the height of the medium *y* versus the position *x* for two times $t = 0.00\text{ s}$ and $t = 0.80\text{ s}$ can provide a graphical visualization of the wave (Figure 16.11).

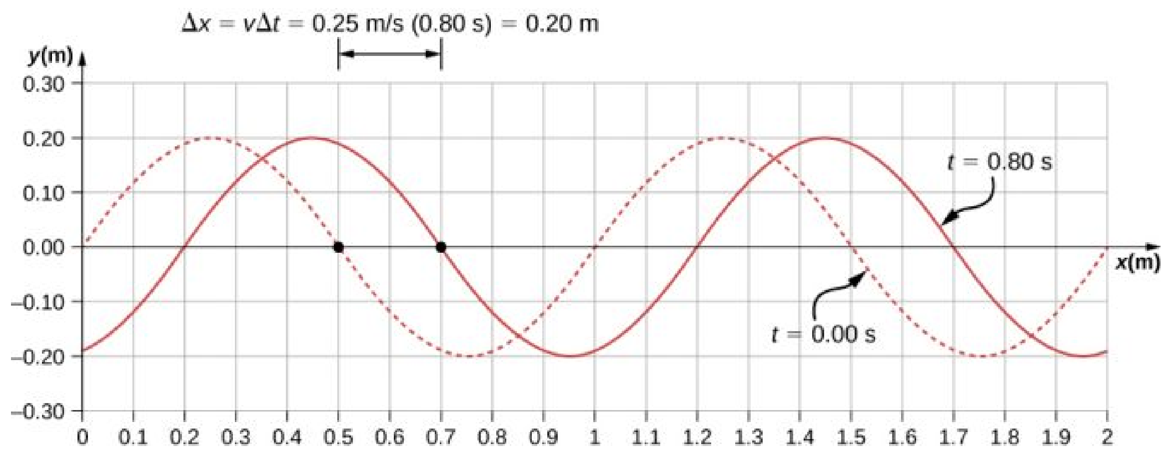


FIGURE 16.11 A graph of height of the wave *y* as a function of position *x* for snapshots of the wave at two times. The dotted line represents the wave at time $t = 0.00\text{ s}$ and the solid line represents the wave at $t = 0.80\text{ s}$. Since the wave velocity is constant, the distance the wave travels is the wave velocity times the time interval. The black dots indicate the points used to measure the displacement of the wave. The medium moves up and down, whereas the wave moves to the right.

There is a second velocity to the motion. In this example, the wave is transverse, moving horizontally as the medium oscillates up and down perpendicular to the direction of motion. The graph in Figure 16.12 shows the motion of the medium at point $x = 0.60\text{ m}$ as a function of time. Notice that the medium of the wave oscillates up and down between $y = +0.20\text{ m}$ and $y = -0.20\text{ m}$ every period of 4.0 seconds.

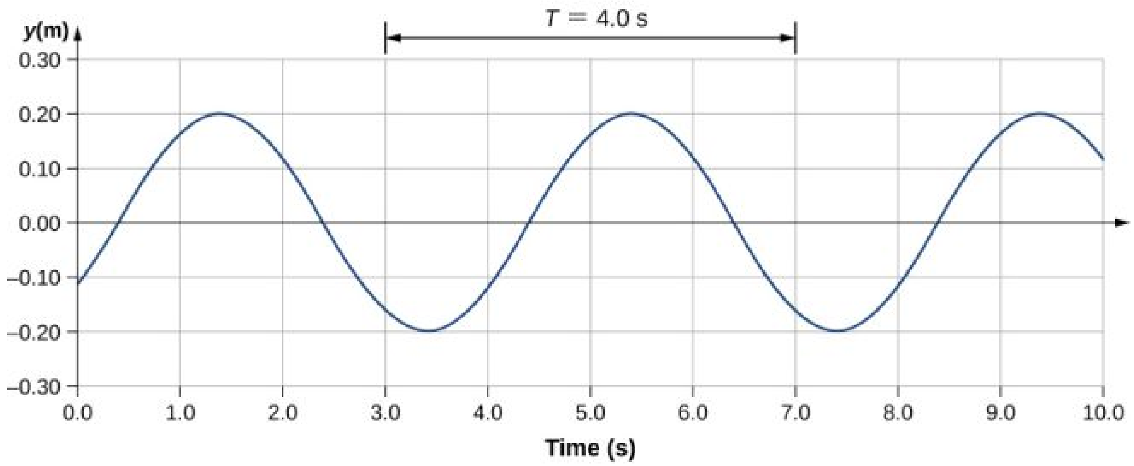


FIGURE 16.12 A graph of height of the wave y as a function of time t for the position $x = 0.6\ \mathrm{m}$. The medium oscillates between $y = +0.20\ \mathrm{m}$ and $y = -0.20\ \mathrm{m}$ every period. The period represented picks two convenient points in the oscillations to measure the period. The period can be measured between any two adjacent points with the same amplitude and the same velocity, $(\partial y/\partial t)$. The velocity can be found by looking at the slope tangent to the point on a y-versus-t plot. Notice that at times $t = 3.00\ s$ and $t = 7.00s$, the heights and the velocities are the same and the period of the oscillation is 4.00 s.

CHECK YOUR UNDERSTANDING 16.3

The wave function above is derived using a sine function. Can a cosine function be used instead?

Velocity and Acceleration of the Medium

As seen in Example 16.4, the wave speed is constant and represents the speed of the wave as it propagates through the medium, not the speed of the particles that make up the medium. The particles of the medium oscillate around an equilibrium position as the wave propagates through the medium. In the case of the transverse wave propagating in the x-direction, the particles oscillate up and down in the y-direction, perpendicular to the motion of the wave. The velocity of the particles of the medium is not constant, which means there is an acceleration. The velocity of the medium, which is perpendicular to the wave velocity in a transverse wave, can be found by taking the partial derivative of the position equation with respect to time. The partial derivative is found by taking the derivative of the function, treating all variables as constants, except for the variable in question. In the case of the partial derivative with respect to time t, the position x is treated as a constant. Although this may sound strange if you haven't seen it before, the object of this exercise is to find the transverse velocity at a point, so in this sense, the x-position is not changing. We have

$$\begin{aligned} y(x,t) &= A\sin(kx - \omega t + \phi) \\ v_y(x,t) &= \frac{\partial y(x,t)}{\partial t} = \frac{\partial}{\partial t}\left(A\sin(kx - \omega t + \phi)\right) \\ &= -A\omega\cos(kx - \omega t + \phi) \\ &= -v_{y\,\max}\cos(kx - \omega t + \phi). \end{aligned}$$

The magnitude of the maximum velocity of the medium is $\left|v_{y_{\max}}\right| = A\omega$. This may look familiar from the Oscillations and a mass on a spring.

We can find the acceleration of the medium by taking the partial derivative of the velocity equation with respect to time,

$$\begin{aligned} a_y(x,t) &= \frac{\partial v_y}{\partial t} = \frac{\partial}{\partial t}\left(-A\omega\cos(kx - \omega t + \phi)\right) \\ &= -A\omega^2\sin(kx - \omega t + \phi) \\ &= -a_{y\,\max}\sin(kx - \omega t + \phi). \end{aligned}$$

The magnitude of the maximum acceleration is $\left|a_{y_{\max}}\right| = A\omega^2$. The particles of the medium, or the mass elements, oscillate in simple harmonic motion for a mechanical wave.

The Linear Wave Equation

We have just determined the velocity of the medium at a position *x* by taking the partial derivative, with respect to time, of the position *y*. For a transverse wave, this velocity is perpendicular to the direction of propagation of the wave. We found the acceleration by taking the partial derivative, with respect to time, of the velocity, which is the second time derivative of the position:

$$a_y(x,t) = \frac{\partial^2 y(x,t)}{\partial t^2} = \frac{\partial^2}{\partial t^2}\left(A\sin(kx-\omega t+\phi)\right) = -A\omega^2\sin(kx-\omega t+\phi).$$

Now consider the partial derivatives with respect to the other variable, the position *x*, holding the time constant. The first derivative is the slope of the wave at a point *x* at a time *t*,

$$\text{slope} = \frac{\partial y(x,t)}{\partial x} = \frac{\partial}{\partial x}\left(A\sin(kx-\omega t+\phi)\right) = Ak\cos(kx-\omega t+\phi).$$

The second partial derivative expresses how the slope of the wave changes with respect to position—in other words, the curvature of the wave, where

$$\text{curvature} = \frac{\partial^2 y(x,t)}{\partial x^2} = \frac{\partial^2}{\partial x^2}\left(A\sin(kx-\omega t+\phi)\right) = -Ak^2\sin(kx-\omega t+\phi).$$

The ratio of the acceleration and the curvature leads to a very important relationship in physics known as the **linear wave equation**. Taking the ratio and using the equation $v = \omega/k$ yields the linear wave equation (also known simply as the wave equation or the equation of a vibrating string),

$$\begin{aligned} \frac{\frac{\partial^2 y(x,t)}{\partial t^2}}{\frac{\partial^2 y(x,t)}{\partial x^2}} &= \frac{-A\omega^2\sin(kx-\omega t+\phi)}{-Ak^2\sin(kx-\omega t+\phi)} \\ &= \frac{\omega^2}{k^2} = v^2, \end{aligned}$$

$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}. \qquad \textbf{16.6}$$

Equation 16.6 is the linear wave equation, which is one of the most important equations in physics and engineering. We derived it here for a transverse wave, but it is equally important when investigating longitudinal waves. This relationship was also derived using a sinusoidal wave, but it successfully describes any wave or pulse that has the form $y(x,t) = f(x \mp vt)$. These waves result due to a linear restoring force of the medium—thus, the name linear wave equation. Any wave function that satisfies this equation is a linear wave function.

An interesting aspect of the linear wave equation is that if two wave functions are individually solutions to the linear wave equation, then the sum of the two linear wave functions is also a solution to the wave equation. Consider two transverse waves that propagate along the *x*-axis, occupying the same medium. Assume that the individual waves can be modeled with the wave functions $y_1(x,t) = f(x \mp vt)$ and $y_2(x,t) = g(x \mp vt)$, which are solutions to the linear wave equations and are therefore linear wave functions. The sum of the wave functions is the wave function

$$y_1(x,t) + y_2(x,t) = f(x \mp vt) + g(x \mp vt).$$

Consider the linear wave equation:

$$\begin{aligned} \frac{\partial^2(f+g)}{\partial x^2} &= \frac{1}{v^2}\frac{\partial^2(f+g)}{\partial t^2} \\ \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 g}{\partial x^2} &= \frac{1}{v^2}\left[\frac{\partial^2 f}{\partial t^2} + \frac{\partial^2 g}{\partial t^2}\right]. \end{aligned}$$

This has shown that if two linear wave functions are added algebraically, the resulting wave function is also linear. This wave function models the displacement of the medium of the resulting wave at each position along the *x*-axis.

If two linear waves occupy the same medium, they are said to interfere. If these waves can be modeled with a linear wave function, these wave functions add to form the wave equation of the wave resulting from the interference of the individual waves. The displacement of the medium at every point of the resulting wave is the algebraic sum of the displacements due to the individual waves.

Taking this analysis a step further, if wave functions $y_1\,(x,t) = f\,(x \mp vt)$ and $y_2\,(x,t) = g\,(x \mp vt)$ are solutions to the linear wave equation, then $Ay_1\,(x,t) + By_2\,(x,t)$, where *A* and *B* are constants, is also a solution to the linear wave equation. This property is known as the principle of superposition. Interference and superposition are covered in more detail in Interference of Waves.

EXAMPLE 16.4

Interference of Waves on a String

Consider a very long string held taut by two students, one on each end. Student A oscillates the end of the string producing a wave modeled with the wave function $y_1\,(x,t) = A\,\sin\,(kx - \omega t)$ and student B oscillates the string producing at twice the frequency, moving in the opposite direction. Both waves move at the same speed $v = \frac{\omega}{k}$. The two waves interfere to form a resulting wave whose wave function is $y_R\,(x,t) = y_1\,(x,t) + y_2\,(x,t)$. Find the velocity of the resulting wave using the linear wave equation $\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}$.

Strategy

First, write the wave function for the wave created by the second student. Note that the angular frequency of the second wave is twice the frequency of the first wave (2ω), and since the velocity of the two waves are the same, the wave number of the second wave is twice that of the first wave $(2k)$. Next, write the wave equation for the resulting wave function, which is the sum of the two individual wave functions. Then find the second partial derivative with respect to position and the second partial derivative with respect to time. Use the linear wave equation to find the velocity of the resulting wave.

Solution

1. Write the wave function of the second wave: $y_2\,(x,t) = A\,\sin\,(2kx + 2\omega t)$.
2. Write the resulting wave function:
$$y_R\,(x,t) = y_1\,(x,t) + y\,(x,t) = A\,\sin\,(kx - \omega t) + A\,\sin\,(2kx + 2\omega t).$$
3. Find the partial derivatives:
$$\begin{aligned}
\frac{\partial y_R(x,t)}{\partial x} &= +Ak\,\cos\,(kx - \omega t) + 2Ak\,\cos\,(2kx + 2\omega t), \\
\frac{\partial^2 y_R(x,t)}{\partial x^2} &= -Ak^2\,\sin\,(kx - \omega t) - 4Ak^2\,\sin\,(2kx + 2\omega t), \\
\frac{\partial y_R(x,t)}{\partial t} &= -A\omega\,\cos\,(kx - \omega t) + 2A\omega\,\cos\,(2kx + 2\omega t), \\
\frac{\partial^2 y_R(x,t)}{\partial t^2} &= -A\omega^2\,\sin\,(kx - \omega t) - 4A\omega^2\,\sin\,(2kx + 2\omega t).
\end{aligned}$$
4. Use the wave equation to find the velocity of the resulting wave:
$$\begin{aligned}
\frac{\partial^2 y(x,t)}{\partial x^2} &= \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}, \\
-Ak^2\,\sin\,(kx - \omega t) - 4Ak^2\,\sin\,(2kx + 2\omega t) &= \frac{1}{v^2}\left(-A\omega^2\,\sin\,(kx - \omega t) - 4A\omega^2\,\sin\,(2kx + 2\omega t)\right), \\
k^2\,(-A\,\sin\,(kx - \omega t) - 4A\,\sin\,(2kx + 2\omega t)) &= \frac{\omega^2}{v^2}(-A\,\sin\,(kx - \omega t) - 4A\,\sin\,(2kx + 2\omega t)), \\
k^2 &= \frac{\omega^2}{v^2},\ |v| = \frac{\omega}{k}.
\end{aligned}$$

Significance

The speed of the resulting wave is equal to the speed of the original waves $\left(v = \frac{\omega}{k}\right)$. We will show in the next section that the speed of a simple harmonic wave on a string depends on the tension in the string and the mass per length of the string. For this reason, it is not surprising that the component waves as well as the resultant wave all travel at the same speed.

CHECK YOUR UNDERSTANDING 16.4

The wave equation $\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}$ works for any wave of the form $y(x,t) = f(x \mp vt)$. In the previous section, we stated that a cosine function could also be used to model a simple harmonic mechanical wave. Check if the wave

$$y(x,t) = 0.50\,\text{m}\cos\left(0.20\pi\,\text{m}^{-1}x - 4.00\pi\,\text{s}^{-1}t + \frac{\pi}{10}\right)$$

is a solution to the wave equation.

Any disturbance that complies with the wave equation can propagate as a wave moving along the *x*-axis with a wave speed *v*. It works equally well for waves on a string, sound waves, and electromagnetic waves. This equation is extremely useful. For example, it can be used to show that electromagnetic waves move at the speed of light.

16.3 Wave Speed on a Stretched String

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Determine the factors that affect the speed of a wave on a string
- Write a mathematical expression for the speed of a wave on a string and generalize these concepts for other media

The speed of a wave depends on the characteristics of the medium. For example, in the case of a guitar, the strings vibrate to produce the sound. The speed of the waves on the strings, and the wavelength, determine the frequency of the sound produced. The strings on a guitar have different thickness but may be made of similar material. They have different *linear densities*, where the linear density is defined as the mass per length,

$$\mu = \frac{\text{mass of string}}{\text{length of string}} = \frac{m}{l}. \tag{16.7}$$

In this chapter, we consider only strings with a constant linear density. If the linear density is constant, then the mass (Δm) of a small length of string (Δx) is $\Delta m = \mu\Delta x$. For example, if the string has a length of 2.00 m and a mass of 0.06 kg, then the linear density is $\mu = \frac{0.06\text{ kg}}{2.00\text{ m}} = 0.03\frac{\text{kg}}{\text{m}}$. If a 1.00-mm section is cut from the string, the mass of the 1.00-mm length is $\Delta m = \mu\Delta x = \left(0.03\frac{\text{kg}}{\text{m}}\right)0.001\text{ m} = 3.00 \times 10^{-5}\text{ kg}$. The guitar also has a method to change the tension of the strings. The tension of the strings is adjusted by turning spindles, called the tuning pegs, around which the strings are wrapped. For the guitar, the linear density of the string and the tension in the string determine the speed of the waves in the string and the frequency of the sound produced is proportional to the wave speed.

Wave Speed on a String under Tension

To see how the speed of a wave on a string depends on the tension and the linear density, consider a pulse sent down a taut string (Figure 16.13). When the taut string is at rest at the equilibrium position, the tension in the string F_T is constant. Consider a small element of the string with a mass equal to $\Delta m = \mu\Delta x$. The mass element is at rest and in equilibrium and the force of tension of either side of the mass element is equal and opposite.

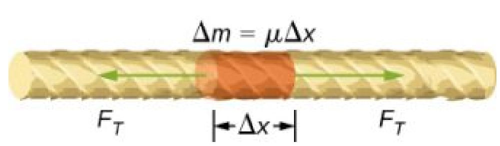


FIGURE 16.13 Mass element of a string kept taut with a tension F_T. The mass element is in static equilibrium, and the force of tension

acting on either side of the mass element is equal in magnitude and opposite in direction.

If you pluck a string under tension, a transverse wave moves in the positive *x*-direction, as shown in Figure 16.14. The mass element is small but is enlarged in the figure to make it visible. The small mass element oscillates perpendicular to the wave motion as a result of the restoring force provided by the string and does not move in the *x*-direction. The tension F_T in the string, which acts in the positive and negative *x*-direction, is approximately constant and is independent of position and time.

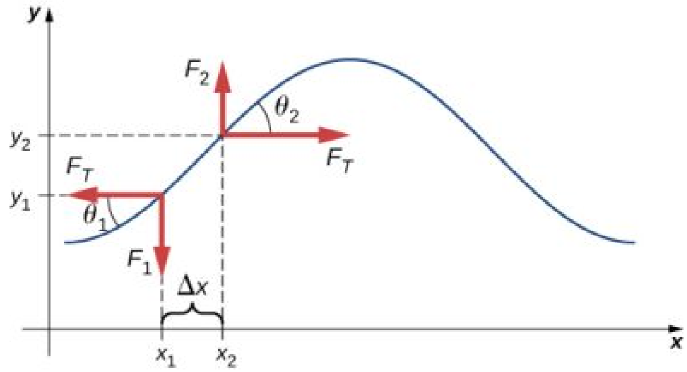


FIGURE 16.14 A string under tension is plucked, causing a pulse to move along the string in the positive *x*-direction.

Assume that the inclination of the displaced string with respect to the horizontal axis is small. The net force on the element of the string, acting parallel to the string, is the sum of the tension in the string and the restoring force. The *x*-components of the force of tension cancel, so the net force is equal to the sum of the *y*-components of the force. The magnitude of the *x*-component of the force is equal to the horizontal force of tension of the string F_T as shown in Figure 16.14. To obtain the *y*-components of the force, note that $\tan\theta_1 = \frac{-F_1}{F_T}$ and $\tan\theta_2 = \frac{F_2}{F_T}$. The $\tan\theta$ is equal to the slope of a function at a point, which is equal to the partial derivative of *y* with respect to *x* at that point. Therefore, $\frac{F_1}{F_T}$ is equal to the negative slope of the string at x_1 and $\frac{F_2}{F_T}$ is equal to the slope of the string at x_2 :

$$\frac{F_1}{F_T} = -\left(\frac{\partial y}{\partial x}\right)_{x_1} \text{ and } \frac{F_2}{F_T} = \left(\frac{\partial y}{\partial x}\right)_{x_2}.$$

The net force on the small mass element can be written as

$$F_{\text{net}} = F_1 + F_2 = F_T\left[\left(\frac{\partial y}{\partial x}\right)_{x_2} - \left(\frac{\partial y}{\partial x}\right)_{x_1}\right].$$

Using Newton’s second law, the net force is equal to the mass times the acceleration. The linear density of the string μ is the mass per length of the string, and the mass of the portion of the string is $\mu\Delta x$,

$$F_T\left[\left(\frac{\partial y}{\partial x}\right)_{x_2} - \left(\frac{\partial y}{\partial x}\right)_{x_1}\right] = \Delta m a,$$
$$F_T\left[\left(\frac{\partial y}{\partial x}\right)_{x_2} - \left(\frac{\partial y}{\partial x}\right)_{x_1}\right] = \mu\Delta x\frac{\partial^2 y}{\partial t^2}.$$

Dividing by $F_T\Delta x$ and taking the limit as Δx approaches zero,

$$\frac{\left[\left(\frac{\partial y}{\partial x}\right)_{x_2} - \left(\frac{\partial y}{\partial x}\right)_{x_1}\right]}{\Delta x} = \frac{\mu}{F_T}\frac{\partial^2 y}{\partial t^2}$$

$$\lim_{\Delta x \to 0} \frac{\left[\left(\frac{\partial y}{\partial x}\right)_{x_2} - \left(\frac{\partial y}{\partial x}\right)_{x_1}\right]}{\Delta x} = \frac{\mu}{F_T}\frac{\partial^2 y}{\partial t^2}$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{F_T}\frac{\partial^2 y}{\partial t^2}.$$

Recall that the linear wave equation is

$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}.$$

Therefore,

$$\frac{1}{v^2} = \frac{\mu}{F_T}.$$

Solving for v, we see that the speed of the wave on a string depends on the tension and the linear density.

SPEED OF A WAVE ON A STRING UNDER TENSION

The speed of a pulse or wave on a string under tension can be found with the equation

$$|v| = \sqrt{\frac{F_T}{\mu}} \tag{16.8}$$

where F_T is the tension in the string and μ is the mass per length of the string.

EXAMPLE 16.5

The Wave Speed of a Guitar String

On a six-string guitar, the high E string has a linear density of $\mu_{\text{High E}} = 3.09 \times 10^{-4}$ kg/m and the low E string has a linear density of $\mu_{\text{Low E}} = 5.78 \times 10^{-3}$ kg/m. (a) If the high E string is plucked, producing a wave in the string, what is the speed of the wave if the tension of the string is 56.40 N? (b) The linear density of the low E string is approximately 20 times greater than that of the high E string. For waves to travel through the low E string at the same wave speed as the high E, would the tension need to be larger or smaller than the high E string? What would be the approximate tension? (c) Calculate the tension of the low E string needed for the same wave speed.

Strategy

a. The speed of the wave can be found from the linear density and the tension $v = \sqrt{\frac{F_T}{\mu}}$.
b. From the equation $v = \sqrt{\frac{F_T}{\mu}}$, if the linear density is increased by a factor of almost 20, the tension would need to be increased by a factor of 20.
c. Knowing the velocity and the linear density, the velocity equation can be solved for the force of tension $F_T = \mu v^2$.

Solution

a. Use the velocity equation to find the speed:

$$v = \sqrt{\frac{F_T}{\mu}} = \sqrt{\frac{56.40\text{ N}}{3.09 \times 10^{-4}\text{ kg/m}}} = 427.23\text{ m/s}.$$

b. The tension would need to be increased by a factor of approximately 20. The tension would be slightly less than 1128 N.

c. Use the velocity equation to find the actual tension:

$$F_T = \mu v^2 = 5.78 \times 10^{-3}\,\text{kg/m}(427.23\,\text{m/s})^2 = 1055.00\,\text{N}.$$

This solution is within 7% of the approximation.

Significance

The standard notes of the six string (high E, B, G, D, A, low E) are tuned to vibrate at the fundamental frequencies (329.63 Hz, 246.94Hz, 196.00Hz, 146.83Hz, 110.00Hz, and 82.41Hz) when plucked. The frequencies depend on the speed of the waves on the string and the wavelength of the waves. The six strings have different linear densities and are "tuned" by changing the tensions in the strings. We will see in Interference of Waves that the wavelength depends on the length of the strings and the boundary conditions. To play notes other than the fundamental notes, the lengths of the strings are changed by pressing down on the strings.

CHECK YOUR UNDERSTANDING 16.5

The wave speed of a wave on a string depends on the tension and the linear mass density. If the tension is doubled, what happens to the speed of the waves on the string?

Speed of Compression Waves in a Fluid

The speed of a wave on a string depends on the square root of the tension divided by the mass per length, the linear density. In general, the speed of a wave through a medium depends on the elastic property of the medium and the inertial property of the medium.

$$|v| = \sqrt{\frac{\text{elastic property}}{\text{inertial property}}}$$

The elastic property describes the tendency of the particles of the medium to return to their initial position when perturbed. The inertial property describes the tendency of the particle to resist changes in velocity.

The speed of a longitudinal wave through a liquid or gas depends on the density of the fluid and the bulk modulus of the fluid,

$$v = \sqrt{\frac{B}{\rho}}. \tag{16.9}$$

Here the bulk modulus is defined as $B = -\frac{\Delta P}{\Delta V/V_0}$, where ΔP is the change in the pressure and the denominator is the ratio of the change in volume to the initial volume, and $\rho \equiv \frac{m}{V}$ is the mass per unit volume. For example, sound is a mechanical wave that travels through a fluid or a solid. The speed of sound in air with an atmospheric pressure of 1.013×10^5 Pa and a temperature of $20°\text{C}$ is $v_\text{s} \approx 343.00\,\text{m/s}$. Because the density depends on temperature, the speed of sound in air depends on the temperature of the air. This will be discussed in detail in Sound.

16.4 Energy and Power of a Wave

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain how energy travels with a pulse or wave
- Describe, using a mathematical expression, how the energy in a wave depends on the amplitude of the wave

All waves carry energy, and sometimes this can be directly observed. Earthquakes can shake whole cities to the ground, performing the work of thousands of wrecking balls (Figure 16.15). Loud sounds can pulverize nerve cells in the inner ear, causing permanent hearing loss. Ultrasound is used for deep-heat treatment of muscle strains. A laser beam can burn away a malignancy. Water waves chew up beaches.

FIGURE 16.15 The destructive effect of an earthquake is observable evidence of the energy carried in these waves. The Richter scale rating of earthquakes is a logarithmic scale related to both their amplitude and the energy they carry.

In this section, we examine the quantitative expression of energy in waves. This will be of fundamental importance in later discussions of waves, from sound to light to quantum mechanics.

Energy in Waves

The amount of energy in a wave is related to its amplitude and its frequency. Large-amplitude earthquakes produce large ground displacements. Loud sounds have high-pressure amplitudes and come from larger-amplitude source vibrations than soft sounds. Large ocean breakers churn up the shore more than small ones. Consider the example of the seagull and the water wave earlier in the chapter (Figure 16.3). Work is done on the seagull by the wave as the seagull is moved up, changing its potential energy. The larger the amplitude, the higher the seagull is lifted by the wave and the larger the change in potential energy.

The energy of the wave depends on both the amplitude and the frequency. If the energy of each wavelength is considered to be a discrete packet of energy, a high-frequency wave will deliver more of these packets per unit time than a low-frequency wave. We will see that the average rate of energy transfer in mechanical waves is proportional to both the square of the amplitude and the square of the frequency. If two mechanical waves have equal amplitudes, but one wave has a frequency equal to twice the frequency of the other, the higher-frequency wave will have a rate of energy transfer a factor of four times as great as the rate of energy transfer of the lower-frequency wave. It should be noted that although the rate of energy transport is proportional to both the square of the amplitude and square of the frequency in mechanical waves, the rate of energy transfer in electromagnetic waves is proportional to the square of the amplitude, but independent of the frequency.

Power in Waves

Consider a sinusoidal wave on a string that is produced by a string vibrator, as shown in Figure 16.16. The string vibrator is a device that vibrates a rod up and down. A string of uniform linear mass density is attached to the rod, and the rod oscillates the string, producing a sinusoidal wave. The rod does work on the string, producing energy that propagates along the string. Consider a mass element of the string with a mass Δm, as seen in Figure 16.16. As the energy propagates along the string, each mass element of the string is driven up and down at the same frequency as the wave. Each mass element of the string can be modeled as a simple harmonic oscillator. Since the string has a constant linear density $\mu = \frac{\Delta m}{\Delta x}$, each mass element of the string has the mass $\Delta m = \mu \Delta x$.

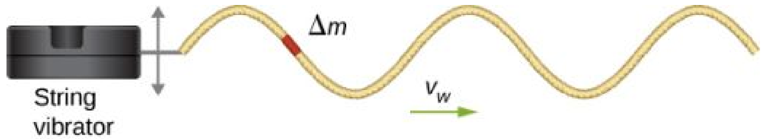


FIGURE 16.16 A string vibrator is a device that vibrates a rod. A string is attached to the rod, and the rod does work on the string, driving the string up and down. This produces a sinusoidal wave in the string, which moves with a wave velocity *v*. The wave speed depends on the tension in the string and the linear mass density of the string. A section of the string with mass Δm oscillates at the same frequency as the wave.

The total mechanical energy of the wave is the sum of its kinetic energy and potential energy. The kinetic energy $K = \frac{1}{2}mv^2$ of each mass element of the string of length Δx is $\Delta K = \frac{1}{2}(\Delta m)\, v_y^2$, as the mass element oscillates perpendicular to the direction of the motion of the wave. Using the constant linear mass density, the kinetic energy of each mass element of the string with length Δx is

$$\Delta K = \frac{1}{2}(\mu \Delta x)\, v_y^2.$$

A differential equation can be formed by letting the length of the mass element of the string approach zero,

$$dK = \lim_{\Delta x \to 0} \frac{1}{2}(\mu \Delta x)\, v_y^2 = \frac{1}{2}(\mu dx)\, v_y^2.$$

Since the wave is a sinusoidal wave with an angular frequency ω, the position of each mass element may be modeled as $y\,(x, t) = A \sin\,(kx - \omega t)$. Each mass element of the string oscillates with a velocity $v_y = \frac{\partial y(x,t)}{\partial t} = -A\omega \cos\,(kx - \omega t)$. The kinetic energy of each mass element of the string becomes

$$\begin{aligned} dK &= \tfrac{1}{2}(\mu dx)\,(-A\omega \cos\,(kx - \omega t))^2, \\ &= \tfrac{1}{2}(\mu dx)\, A^2\omega^2 \cos^2\,(kx - \omega t)\,. \end{aligned}$$

The wave can be very long, consisting of many wavelengths. To standardize the energy, consider the kinetic energy associated with a wavelength of the wave. This kinetic energy can be integrated over the wavelength to find the energy associated with each wavelength of the wave:

$$\begin{aligned} dK &= \tfrac{1}{2}(\mu dx)\, A^2\omega^2 \cos^2\,(kx)\,, \\ \int_0^{K_\lambda} dK &= \int_0^{\lambda} \frac{1}{2}\mu A^2\omega^2 \cos^2\,(kx)\, dx = \tfrac{1}{2}\mu A^2\omega^2 \int_0^{\lambda} \cos^2\,(kx)\, dx, \\ K_\lambda &= \tfrac{1}{2}\mu A^2\omega^2 \left[\tfrac{1}{2}x + \tfrac{1}{4k}\sin\,(2kx)\right]_0^{\lambda} = \tfrac{1}{2}\mu A^2\omega^2 \left[\tfrac{1}{2}\lambda + \tfrac{1}{4k}\sin\,(2k\lambda) - \tfrac{1}{4k}\sin\,(0)\right], \\ K_\lambda &= \tfrac{1}{4}\mu A^2\omega^2 \lambda. \end{aligned}$$

There is also potential energy associated with the wave. Much like the mass oscillating on a spring, there is a conservative restoring force that, when the mass element is displaced from the equilibrium position, drives the mass element back to the equilibrium position. The potential energy of the mass element can be found by considering the linear restoring force of the string, In Oscillations, we saw that the potential energy stored in a spring with a linear restoring force is equal to $U = \frac{1}{2}k_s x^2$, where the equilibrium position is defined as $x = 0.00\ \text{m}$. When a mass attached to the spring oscillates in simple harmonic motion, the angular frequency is equal to $\omega = \sqrt{\frac{k_s}{m}}$. As each mass element oscillates in simple harmonic motion, the spring constant is equal to $k_s = \Delta m\omega^2$. The potential energy of the mass element is equal to

$$\Delta U = \frac{1}{2}k_s x^2 = \frac{1}{2}\Delta m\omega^2 x^2.$$

Note that k_s is the spring constant and not the wave number $k = \frac{2\pi}{\lambda}$. This equation can be used to find the energy over a wavelength. Integrating over the wavelength, we can compute the potential energy over a wavelength:

$$\begin{aligned} dU &= \tfrac{1}{2}k_s x^2 = \tfrac{1}{2}\mu\omega^2 x^2\,dx, \\ U_\lambda &= \tfrac{1}{2}\mu\omega^2 A^2 \int_0^\lambda \sin^2(kx)\,dx = \tfrac{1}{4}\mu A^2\omega^2\lambda. \end{aligned}$$

The potential energy associated with a wavelength of the wave is equal to the kinetic energy associated with a wavelength.

The total energy associated with a wavelength is the sum of the potential energy and the kinetic energy:

$$\begin{aligned} E_\lambda &= U_\lambda + K_\lambda, \\ E_\lambda &= \tfrac{1}{4}\mu A^2\omega^2\lambda + \tfrac{1}{4}\mu A^2\omega^2\lambda = \tfrac{1}{2}\mu A^2\omega^2\lambda. \end{aligned}$$

The time-averaged power of a sinusoidal mechanical wave, which is the average rate of energy transfer associated with a wave as it passes a point, can be found by taking the total energy associated with the wave divided by the time it takes to transfer the energy. If the velocity of the sinusoidal wave is constant, the time for one wavelength to pass by a point is equal to the period of the wave, which is also constant. For a sinusoidal mechanical wave, the time-averaged power is therefore the energy associated with a wavelength divided by the period of the wave. The wavelength of the wave divided by the period is equal to the velocity of the wave,

$$P_{\text{ave}} = \frac{E_\lambda}{T} = \frac{1}{2}\mu A^2\omega^2\frac{\lambda}{T} = \frac{1}{2}\mu A^2\omega^2 v. \tag{16.10}$$

Note that this equation for the time-averaged power of a sinusoidal mechanical wave shows that the power is proportional to the square of the amplitude of the wave and to the square of the angular frequency of the wave. Recall that the angular frequency is equal to $\omega = 2\pi f$, so the power of a mechanical wave is equal to the square of the amplitude and the square of the frequency of the wave.

EXAMPLE 16.6

Power Supplied by a String Vibrator

Consider a two-meter-long string with a mass of 70.00 g attached to a string vibrator as illustrated in Figure 16.16. The tension in the string is 90.0 N. When the string vibrator is turned on, it oscillates with a frequency of 60 Hz and produces a sinusoidal wave on the string with an amplitude of 4.00 cm and a constant wave speed. What is the time-averaged power supplied to the wave by the string vibrator?

Strategy

The power supplied to the wave should equal the time-averaged power of the wave on the string. We know the mass of the string (m_s), the length of the string (L_s), and the tension (F_T) in the string. The speed of the wave on the string can be derived from the linear mass density and the tension. The string oscillates with the same frequency as the string vibrator, from which we can find the angular frequency.

Solution

1. Begin with the equation of the time-averaged power of a sinusoidal wave on a string:
$$P = \frac{1}{2}\mu A^2\omega^2 v.$$
The amplitude is given, so we need to calculate the linear mass density of the string, the angular frequency of the wave on the string, and the speed of the wave on the string.
2. We need to calculate the linear density to find the wave speed:
$$\mu = \frac{m_s}{L_s} = \frac{0.070\ \text{kg}}{2.00\ \text{m}} = 0.035\ \text{kg/m}.$$
3. The wave speed can be found using the linear mass density and the tension of the string:

$$v = \sqrt{\frac{F_T}{\mu}} = \sqrt{\frac{90.00\text{ N}}{0.035\text{ kg/m}}} = 50.71\text{ m/s}.$$

4. The angular frequency can be found from the frequency:
$$\omega = 2\pi f = 2\pi\left(60\text{ s}^{-1}\right) = 376.80\text{ s}^{-1}.$$

5. Calculate the time-averaged power:
$$P = \frac{1}{2}\mu A^2\omega^2 v = \frac{1}{2}\left(0.035\frac{\text{kg}}{\text{m}}\right)(0.040\text{ m})^2\left(376.80\text{ s}^{-1}\right)^2\left(50.71\frac{\text{m}}{\text{s}}\right) = 201.59\text{ W}.$$

Significance

The time-averaged power of a sinusoidal wave is proportional to the square of the amplitude of the wave and the square of the angular frequency of the wave. This is true for most mechanical waves. If either the angular frequency or the amplitude of the wave were doubled, the power would increase by a factor of four. The time-averaged power of the wave on a string is also proportional to the speed of the sinusoidal wave on the string. If the speed were doubled, by increasing the tension by a factor of four, the power would also be doubled.

CHECK YOUR UNDERSTANDING 16.6

Is the time-averaged power of a sinusoidal wave on a string proportional to the linear density of the string?

The equations for the energy of the wave and the time-averaged power were derived for a sinusoidal wave on a string. In general, the energy of a mechanical wave and the power are proportional to the amplitude squared and to the angular frequency squared (and therefore the frequency squared).

Another important characteristic of waves is the intensity of the waves. Waves can also be concentrated or spread out. Waves from an earthquake, for example, spread out over a larger area as they move away from a source, so they do less damage the farther they get from the source. Changing the area the waves cover has important effects. All these pertinent factors are included in the definition of **intensity (*I*)** as power per unit area:

$$I = \frac{P}{A}, \tag{16.11}$$

where *P* is the power carried by the wave through area *A*. The definition of intensity is valid for any energy in transit, including that carried by waves. The SI unit for intensity is watts per square meter (W/m^2). Many waves are spherical waves that move out from a source as a sphere. For example, a sound speaker mounted on a post above the ground may produce sound waves that move away from the source as a spherical wave. Sound waves are discussed in more detail in the next chapter, but in general, the farther you are from the speaker, the less intense the sound you hear. As a spherical wave moves out from a source, the surface area of the wave increases as the radius increases $\left(A = 4\pi r^2\right)$. The intensity for a spherical wave is therefore

$$I = \frac{P}{4\pi r^2}. \tag{16.12}$$

If there are no dissipative forces, the energy will remain constant as the spherical wave moves away from the source, but the intensity will decrease as the surface area increases.

In the case of the two-dimensional circular wave, the wave moves out, increasing the circumference of the wave as the radius of the circle increases. If you toss a pebble in a pond, the surface ripple moves out as a circular wave. As the ripple moves away from the source, the amplitude decreases. The energy of the wave spreads around a larger circumference and the intensity decreases proportional to $\frac{1}{r}$, which is also the same in the case of a spherical wave, since intensity is proportional to the amplitude squared.

16.5 Interference of Waves

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain how mechanical waves are reflected and transmitted at the boundaries of a medium
- Define the terms interference and superposition
- Find the resultant wave of two identical sinusoidal waves that differ only by a phase shift

Up to now, we have been studying mechanical waves that propagate continuously through a medium, but we have not discussed what happens when waves encounter the boundary of the medium or what happens when a wave encounters another wave propagating through the same medium. Waves do interact with boundaries of the medium, and all or part of the wave can be reflected. For example, when you stand some distance from a rigid cliff face and yell, you can hear the sound waves reflect off the rigid surface as an echo. Waves can also interact with other waves propagating in the same medium. If you throw two rocks into a pond some distance from one another, the circular ripples that result from the two stones seem to pass through one another as they propagate out from where the stones entered the water. This phenomenon is known as interference. In this section, we examine what happens to waves encountering a boundary of a medium or another wave propagating in the same medium. We will see that their behavior is quite different from the behavior of particles and rigid bodies. Later, when we study modern physics, we will see that only at the scale of atoms do we see similarities in the properties of waves and particles.

Reflection and Transmission

When a wave propagates through a medium, it reflects when it encounters the boundary of the medium. The wave before hitting the boundary is known as the incident wave. The wave after encountering the boundary is known as the reflected wave. How the wave is reflected at the boundary of the medium depends on the boundary conditions; waves will react differently if the boundary of the medium is fixed in place or free to move (Figure 16.17). A **fixed boundary condition** exists when the medium at a boundary is fixed in place so it cannot move. A **free boundary condition** exists when the medium at the boundary is free to move.

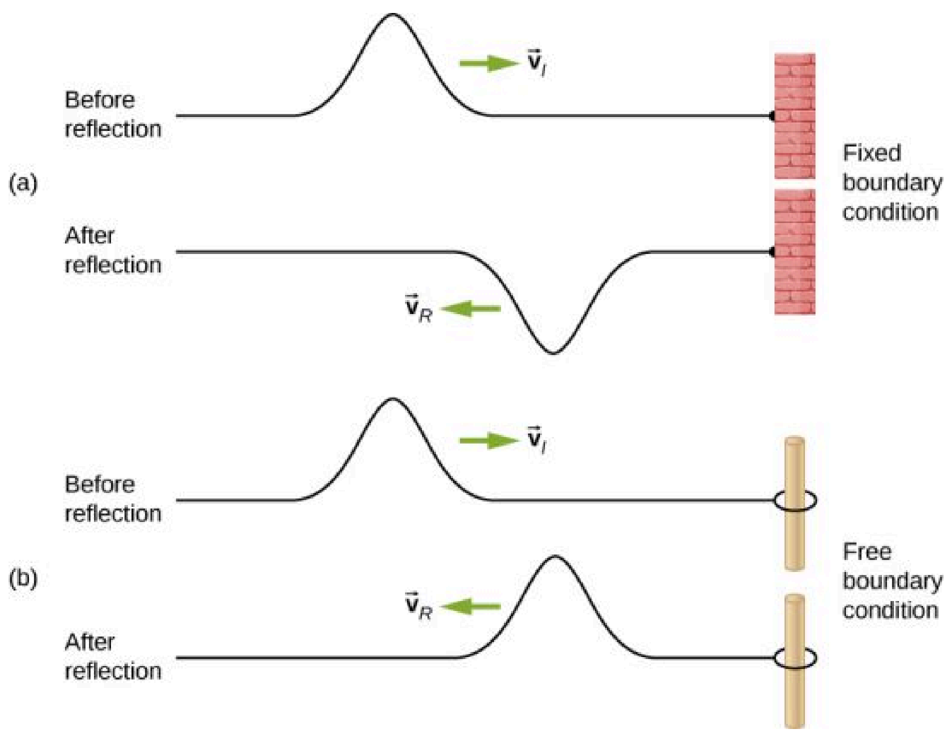


FIGURE 16.17 (a) One end of a string is fixed so that it cannot move. A wave propagating on the string, encountering this *fixed boundary condition*, is reflected $180°(\pi \text{ rad})$ out of phase with respect to the incident wave. (b) One end of a string is tied to a solid ring of negligible mass on a frictionless lab pole, where the ring is free to move. A wave propagating on the string, encountering this *free boundary condition*, is reflected in phase $0°(0 \text{ rad})$ with respect to the wave.

Part (a) of the Figure 16.17 shows a fixed boundary condition. Here, one end of the string is fixed to a wall so the end of the string is fixed in place and the medium (the string) at the boundary cannot move. When the wave is reflected, the amplitude of the reflected wave is exactly the same as the amplitude of the incident wave, but the reflected wave is reflected $180°$ (π rad) out of phase with respect to the incident wave. The phase change can be explained using Newton's third law: Recall that Newton's third law states that when object *A* exerts a force on object *B*, then object *B* exerts an equal and opposite force on object *A*. As the incident wave encounters the wall, the string exerts an upward force on the wall and the wall reacts by exerting an equal and opposite force on the string. The reflection at a fixed boundary is inverted. Note that the figure shows a crest of the incident wave reflected as a trough. If the incident wave were a trough, the reflected wave would be a crest.

Part (b) of the figure shows a free boundary condition. Here, one end of the string is tied to a solid ring of negligible mass on a frictionless pole, so the end of the string is free to move up and down. As the incident wave encounters the boundary of the medium, it is also reflected. In the case of a free boundary condition, the reflected wave is in phase with respect to the incident wave. In this case, the wave encounters the free boundary applying an upward force on the ring, accelerating the ring up. The ring travels up to the maximum height equal to the amplitude of the wave and then accelerates down towards the equilibrium position due to the tension in the string. The figure shows the crest of an incident wave being reflected in phase with respect to the incident wave as a crest. If the incident wave were a trough, the reflected wave would also be a trough. The amplitude of the reflected wave would be equal to the amplitude of the incident wave.

In some situations, the boundary of the medium is neither fixed nor free. Consider Figure 16.18(a), where a low-linear mass density string is attached to a string of a higher linear mass density. In this case, the reflected wave is out of phase with respect to the incident wave. There is also a transmitted wave that is in phase with respect to the incident wave. Both the transmitted and the reflected waves have amplitudes less than the amplitude of the incident wave. If the tension is the same in both strings, the wave speed is higher in the string with the lower linear mass density.

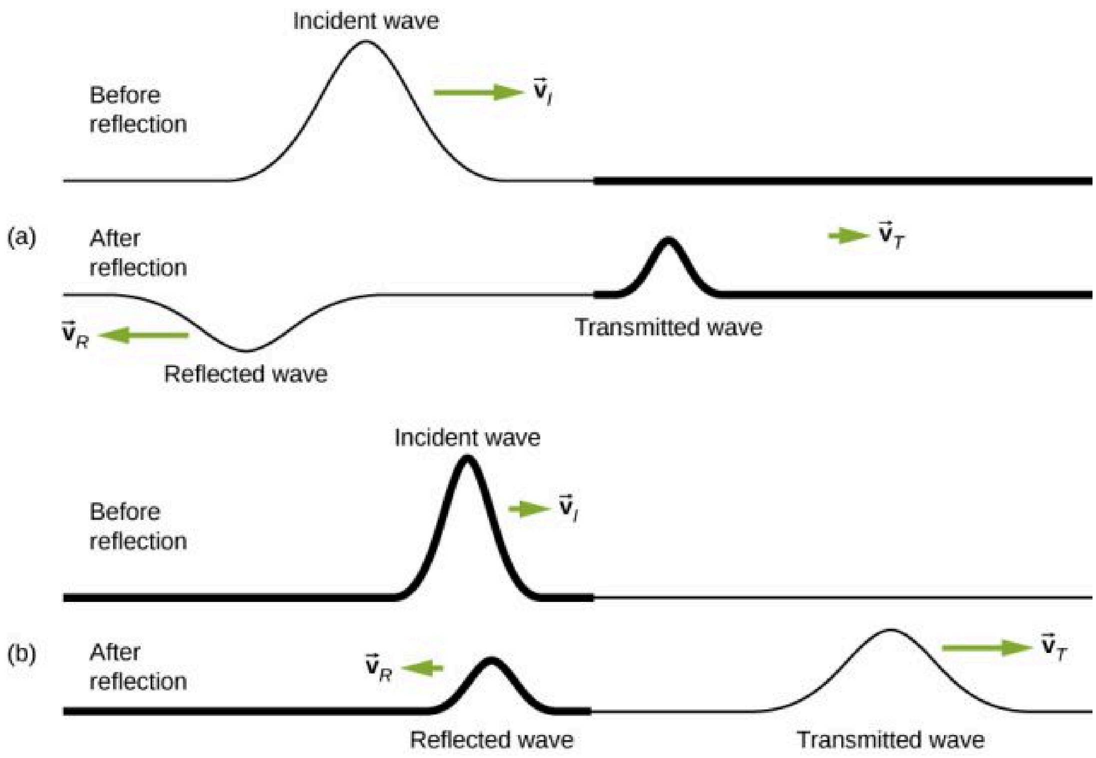


FIGURE 16.18 Waves traveling along two types of strings: a thick string with a high linear density and a thin string with a low linear density. Both strings are under the same tension, so a wave moves faster on the low-density string than on the high-density string. (a) A wave moving from a low-density to a high-density medium results in a reflected wave that is $180°$(π rad) out of phase with respect to the incident pulse (or wave) and a transmitted wave that is in phase with the incident wave. (b) When a wave moves from a high-density medium to a low-density medium, both the reflected and transmitted wave are in phase with respect to the incident wave.

Part (b) of the figure shows a high-linear mass density string is attached to a string of a lower linear density. In this case, the reflected wave is in phase with respect to the incident wave. There is also a transmitted wave that is in phase with respect to the incident wave. Both the incident and the reflected waves have amplitudes less than the amplitude of the incident wave. Here you may notice that if the tension is the same in both strings, the wave speed is higher in the string with the lower linear mass density.

Superposition and Interference

Most waves do not look very simple. Complex waves are more interesting, even beautiful, but they look formidable. Most interesting mechanical waves consist of a combination of two or more traveling waves propagating in the same medium. The principle of superposition can be used to analyze the combination of waves.

Consider two simple pulses of the same amplitude moving toward one another in the same medium, as shown in Figure 16.19. Eventually, the waves overlap, producing a wave that has twice the amplitude, and then continue on unaffected by the encounter. The pulses are said to interfere, and this phenomenon is known as **interference**.

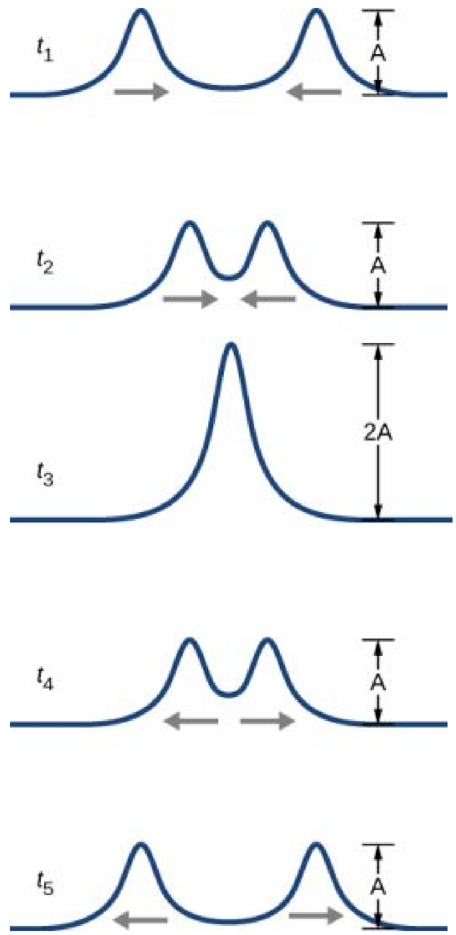


FIGURE 16.19 Two pulses moving toward one another experience interference. The term interference refers to what happens when two waves overlap.

To analyze the interference of two or more waves, we use the principle of superposition. For mechanical waves, the principle of **superposition** states that if two or more traveling waves combine at the same point, the resulting position of the mass element of the medium, at that point, is the algebraic sum of the position due to the individual waves. This property is exhibited by many waves observed, such as waves on a string, sound waves, and surface water waves. Electromagnetic waves also obey the superposition principle, but the electric and magnetic fields of the combined wave are added instead of the displacement of the medium. Waves that obey the superposition principle are linear waves; waves that do not obey the superposition principle are said to be nonlinear waves. In this chapter, we deal with linear waves, in particular, sinusoidal waves.

The superposition principle can be understood by considering the linear wave equation. In Mathematics of a Wave, we defined a linear wave as a wave whose mathematical representation obeys the linear wave equation. For a transverse wave on a string with an elastic restoring force, the linear wave equation is

$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}.$$

Any wave function $y(x,t) = y(x \mp vt)$, where the argument of the function is linear $(x \mp vt)$ is a solution to the linear wave equation and is a linear wave function. If wave functions $y_1(x,t)$ and $y_2(x,t)$ are solutions to the linear wave equation, the sum of the two functions $y_1(x,t) + y_2(x,t)$ is also a solution to the linear wave equation. Mechanical waves that obey superposition are normally restricted to waves with amplitudes that are small with respect to their wavelengths. If the amplitude is too large, the medium is distorted past the region where the restoring force of the medium is linear.

Waves can interfere constructively or destructively. Figure 16.20 shows two identical sinusoidal waves that arrive at the same point exactly in phase. Figure 16.20(a) and (b) show the two individual waves, Figure 16.20(c) shows the resultant wave that results from the algebraic sum of the two linear waves. The crests of the two waves are precisely aligned, as are the troughs. This superposition produces **constructive interference**. Because the disturbances add, constructive interference produces a wave that has twice the amplitude of the individual waves, but has the same wavelength.

Figure 16.21 shows two identical waves that arrive exactly $180°$ out of phase, producing **destructive interference**. Figure 16.21(a) and (b) show the individual waves, and Figure 16.21(c) shows the superposition of the two waves. Because the troughs of one wave add the crest of the other wave, the resulting amplitude is zero for destructive interference—the waves completely cancel.

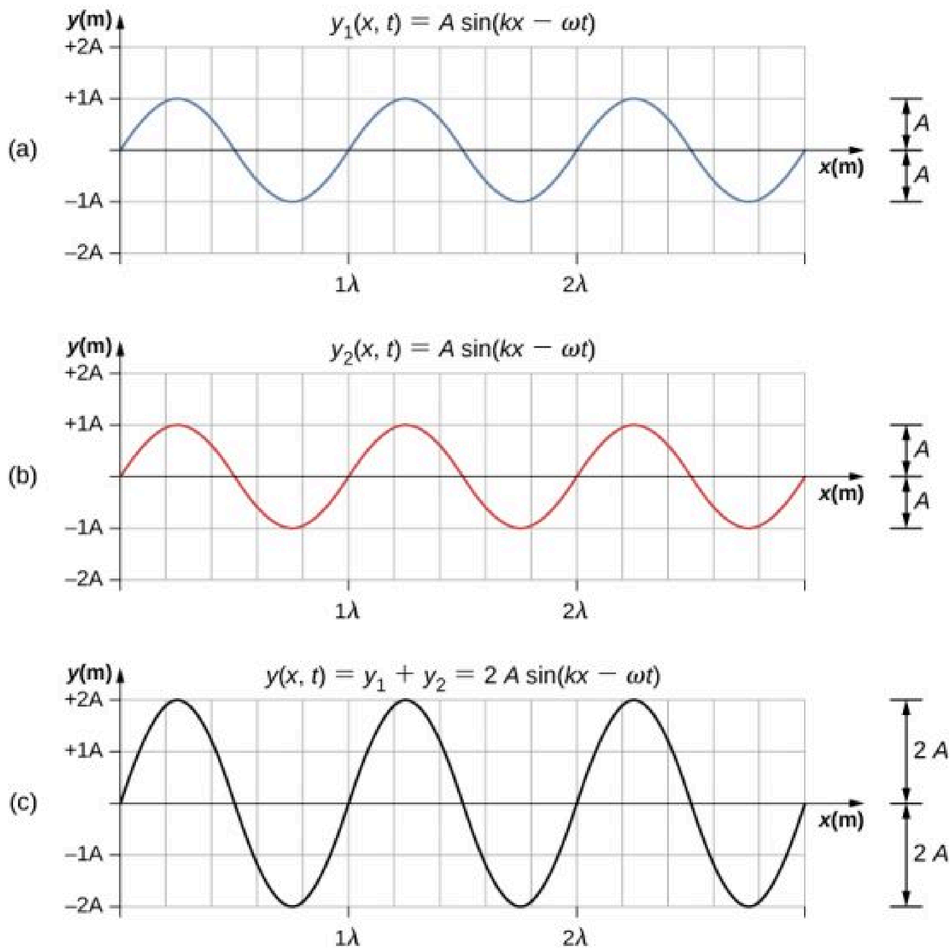


FIGURE 16.20 Constructive interference of two identical waves produces a wave with twice the amplitude, but the same wavelength.

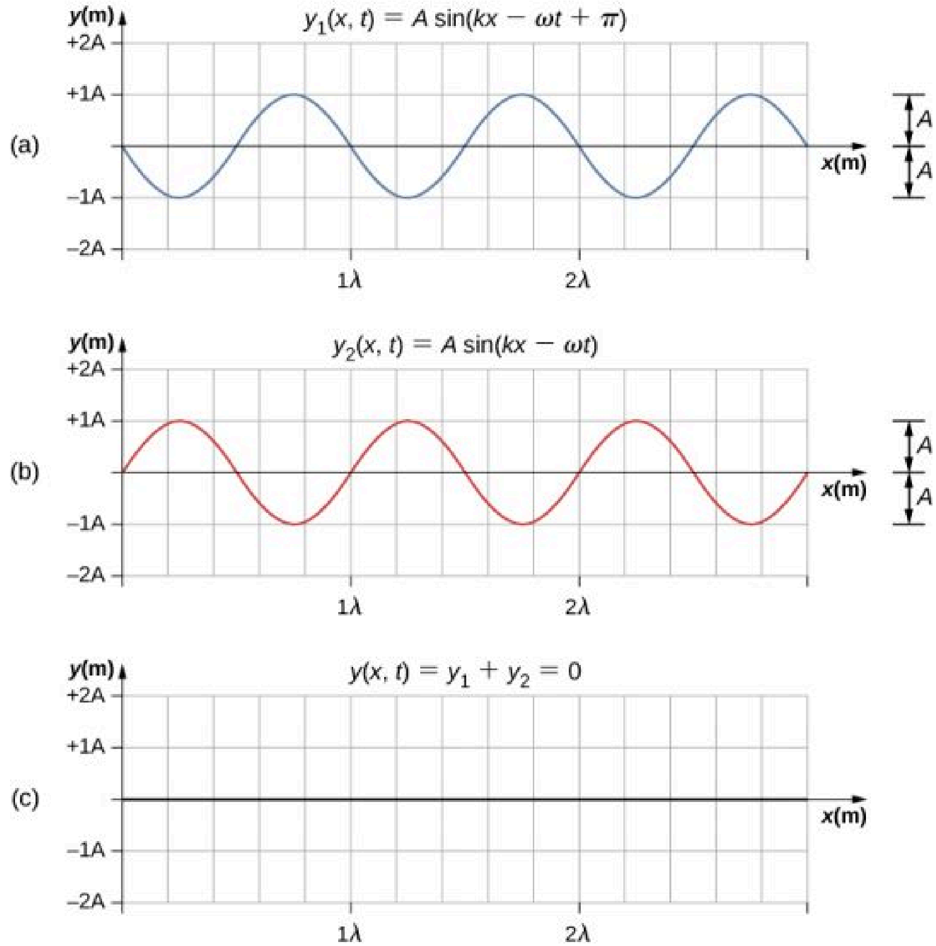


FIGURE 16.21 Destructive interference of two identical waves, one with a phase shift of $180°$ (π rad), produces zero amplitude, or complete cancellation.

When linear waves interfere, the resultant wave is just the algebraic sum of the individual waves as stated in the principle of superposition. Figure 16.22 shows two waves (red and blue) and the resultant wave (black). The resultant wave is the algebraic sum of the two individual waves.

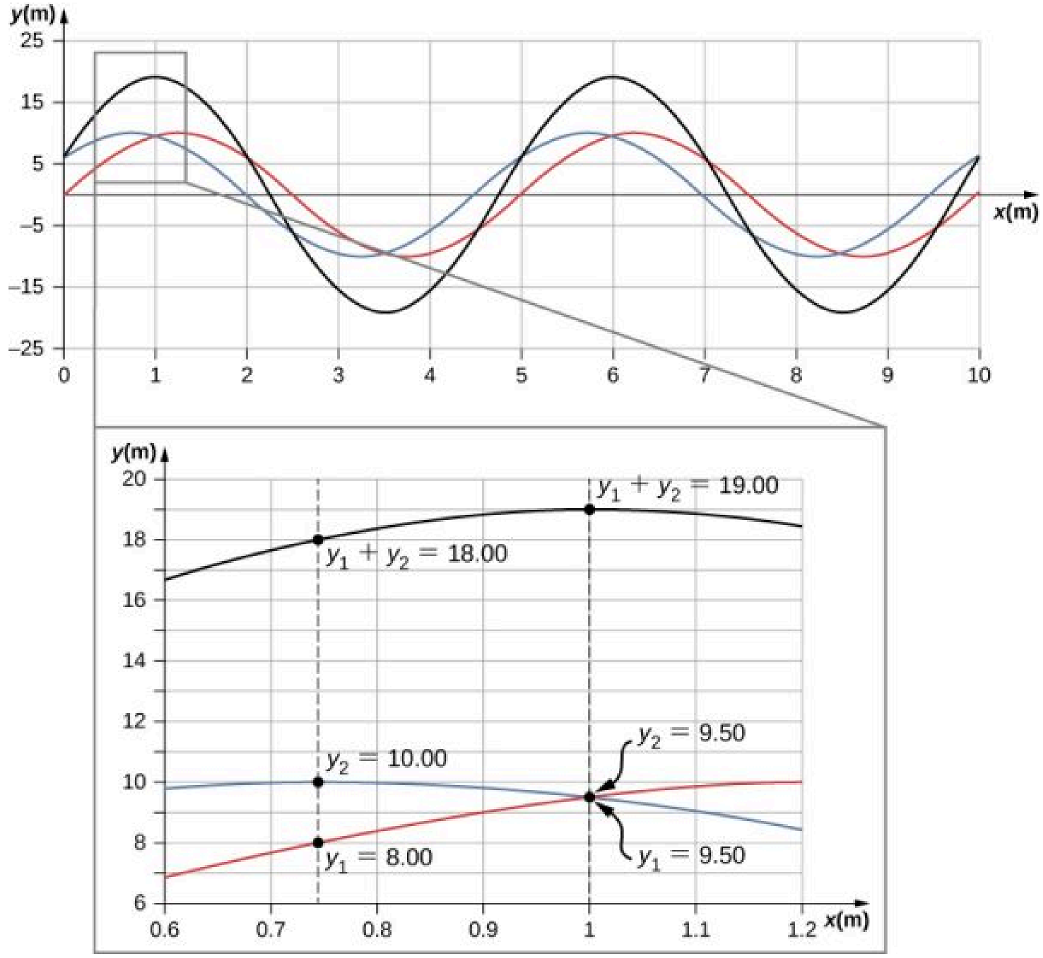


FIGURE 16.22 When two linear waves in the same medium interfere, the height of resulting wave is the sum of the heights of the individual waves, taken point by point. This plot shows two waves (red and blue) added together, along with the resulting wave (black). These graphs represent the height of the wave at each point. The waves may be any linear wave, including ripples on a pond, disturbances on a string, sound, or electromagnetic waves.

The superposition of most waves produces a combination of constructive and destructive interference, and can vary from place to place and time to time. Sound from a stereo, for example, can be loud in one spot and quiet in another. Varying loudness means the sound waves add partially constructively and partially destructively at different locations. A stereo has at least two speakers creating sound waves, and waves can reflect from walls. All these waves interfere, and the resulting wave is the superposition of the waves.

We have shown several examples of the superposition of waves that are similar. Figure 16.23 illustrates an example of the superposition of two dissimilar waves. Here again, the disturbances add, producing a resultant wave.

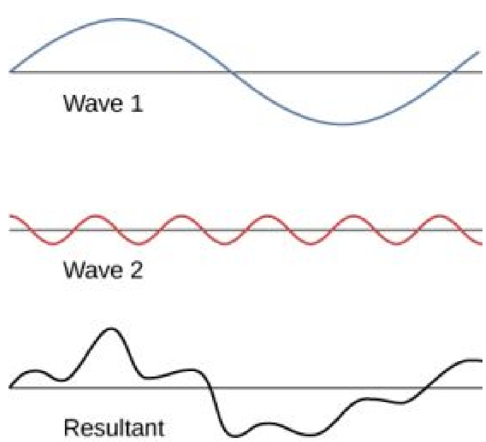


FIGURE 16.23 Superposition of nonidentical waves exhibits both constructive and destructive interference.

At times, when two or more mechanical waves interfere, the pattern produced by the resulting wave can be rich in complexity, some without any readily discernable patterns. For example, plotting the sound wave of your favorite music can look quite complex and is the superposition of the individual sound waves from many instruments; it is the complexity that makes the music interesting and worth listening to. At other times, waves can interfere and produce interesting phenomena, which are complex in their appearance and yet beautiful in simplicity of the physical principle of superposition, which formed the resulting wave. One example is the phenomenon known as standing waves, produced by two identical waves moving in different directions. We will look more closely at this phenomenon in the next section.

INTERACTIVE

Engage the Phet simulation below to make waves with a dripping faucet, audio speaker, or laser! Add a second source or a pair of slits to create an interference pattern. You can observe one source or two sources. Using two sources, you can observe the interference patterns that result from varying the frequencies and the amplitudes of the sources.

[Access multimedia content (https://openstax.org/books/university-physics-volume-1/pages/16-5-interference-of-waves)](https://openstax.org/books/university-physics-volume-1/pages/16-5-interference-of-waves)

Superposition of Sinusoidal Waves that Differ by a Phase Shift

Many examples in physics consist of two sinusoidal waves that are identical in amplitude, wave number, and angular frequency, but differ by a phase shift:

$$\begin{aligned} y_1(x,t) &= A\sin(kx-\omega t+\phi), \\ y_2(x,t) &= A\sin(kx-\omega t). \end{aligned}$$

When these two waves exist in the same medium, the resultant wave resulting from the superposition of the two individual waves is the sum of the two individual waves:

$$y_R(x,t) = y_1(x,t) + y_2(x,t) = A\sin(kx-\omega t+\phi) + A\sin(kx-\omega t).$$

The resultant wave can be better understood by using the trigonometric identity:

$$\sin u + \sin v = 2\sin\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right),$$

where $u = kx - \omega t + \phi$ and $v = kx - \omega t$. The resulting wave becomes

$$\begin{aligned} y_R(x,t) &= y_1(x,t) + y_2(x,t) = A\sin(kx-\omega t+\phi) + A\sin(kx-\omega t) \\ &= 2A\sin\left(\tfrac{(kx-\omega t+\phi)+(kx-\omega t)}{2}\right)\cos\left(\tfrac{(kx-\omega t+\phi)-(kx-\omega t)}{2}\right) \\ &= 2A\sin\left(kx-\omega t+\tfrac{\phi}{2}\right)\cos\left(\tfrac{\phi}{2}\right). \end{aligned}$$

This equation is usually written as

$$y_R(x,t) = \left[2A\cos\left(\frac{\phi}{2}\right)\right]\sin\left(kx-\omega t+\frac{\phi}{2}\right). \tag{16.13}$$

The resultant wave has the same wave number and angular frequency, an amplitude of $A_R = \left[2A\cos\left(\frac{\phi}{2}\right)\right]$, and a phase shift equal to half the original phase shift. Examples of waves that differ only in a phase shift are shown in Figure 16.24. The red and blue waves each have the same amplitude, wave number, and angular frequency, and differ only in a phase shift. They therefore have the same period, wavelength, and frequency. The green wave is the result of the superposition of the two waves. When the two waves have a phase difference of zero, the waves are in phase, and the resultant wave has the same wave number and angular frequency, and an amplitude equal to twice the individual amplitudes (part (a)). This is constructive interference. If the phase difference is $180°$, the waves interfere in destructive interference (part (c)). The resultant wave has an amplitude of zero. Any other phase difference results in a wave with the same wave number and angular frequency as the two incident waves but with a phase shift of $\phi/2$ and an amplitude equal to $2A\cos(\phi/2)$. Examples are shown in parts (b) and (d).

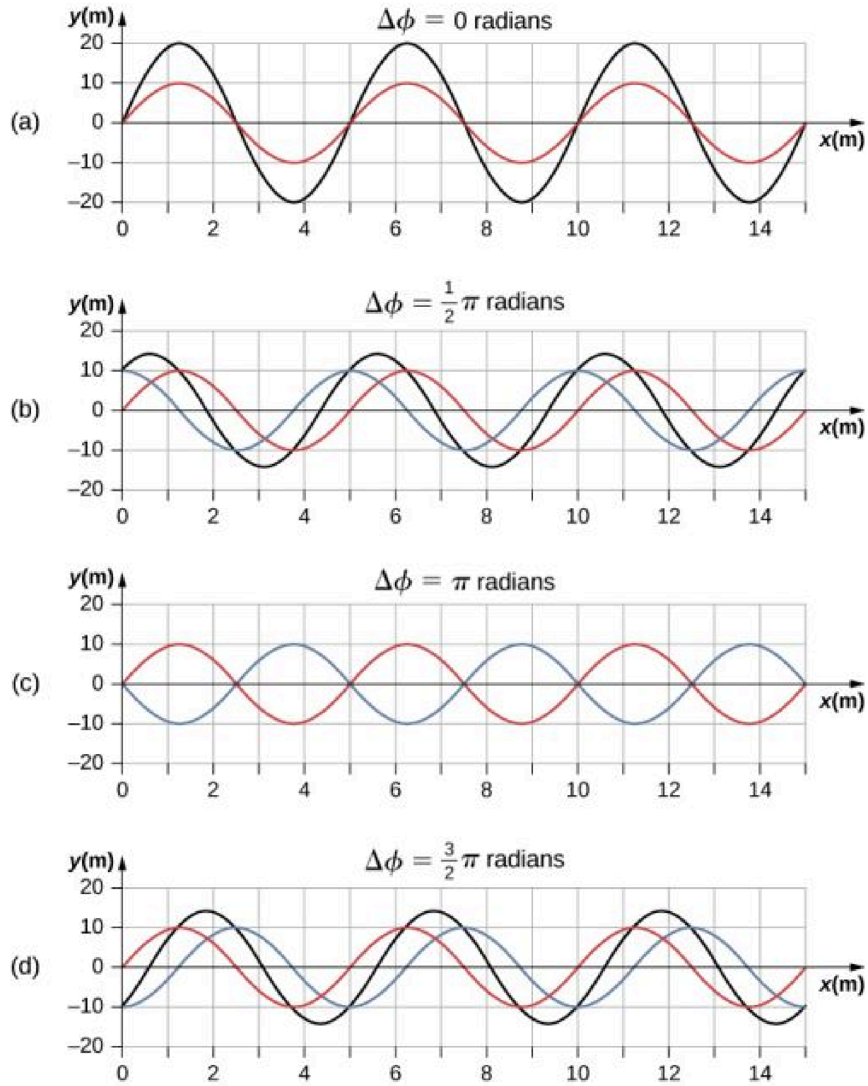


FIGURE 16.24 Superposition of two waves with identical amplitudes, wavelengths, and frequency, but that differ in a phase shift. The red wave is defined by the wave function $y_1(x, t) = A\sin(kx - \omega t)$ and the blue wave is defined by the wave function $y_2(x, t) = A\sin(kx - \omega t + \phi)$. The black line shows the result of adding the two waves. The phase difference between the two waves are (a) 0.00 rad, (b) $\pi/2$ rad, (c) π rad, and (d) $3\pi/2$ rad.

16.6 Standing Waves and Resonance

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe standing waves and explain how they are produced
- Describe the modes of a standing wave on a string
- Provide examples of standing waves beyond the waves on a string

Throughout this chapter, we have been studying traveling waves, or waves that transport energy from one place to another. Under certain conditions, waves can bounce back and forth through a particular region, effectively becoming stationary. These are called **standing waves**.

Another related effect is known as resonance. In Oscillations, we defined resonance as a phenomenon in which a small-amplitude driving force could produce large-amplitude motion. Think of a child on a swing, which can be modeled as a physical pendulum. Relatively small-amplitude pushes by a parent can produce large-amplitude swings. Sometimes this resonance is good—for example, when producing music with a stringed instrument. At other times, the effects can be devastating, such as the collapse of a building during an earthquake. In the case of standing waves, the relatively large amplitude standing waves are produced by the superposition of smaller amplitude component waves.

Standing Waves

Sometimes waves do not seem to move; rather, they just vibrate in place. You can see unmoving waves on the surface of a glass of milk in a refrigerator, for example. Vibrations from the refrigerator motor create waves on the milk that oscillate up and down but do not seem to move across the surface. Figure 16.25 shows an experiment you can try at home. Take a bowl of milk and place it on a common box fan. Vibrations from the fan will produce circular standing waves in the milk. The waves are visible in the photo due to the reflection from a lamp. These waves are formed by the superposition of two or more traveling waves, such as illustrated in Figure 16.26 for two identical waves moving in opposite directions. The waves move through each other with their disturbances adding as they go by. If the two waves have the same amplitude and wavelength, then they alternate between constructive and destructive interference. The resultant looks like a wave standing in place and, thus, is called a standing wave.

FIGURE 16.25 Standing waves are formed on the surface of a bowl of milk sitting on a box fan. The vibrations from the fan causes the surface of the milk to oscillate. The waves are visible due to the reflection of light from a lamp. (credit: David Chelton)

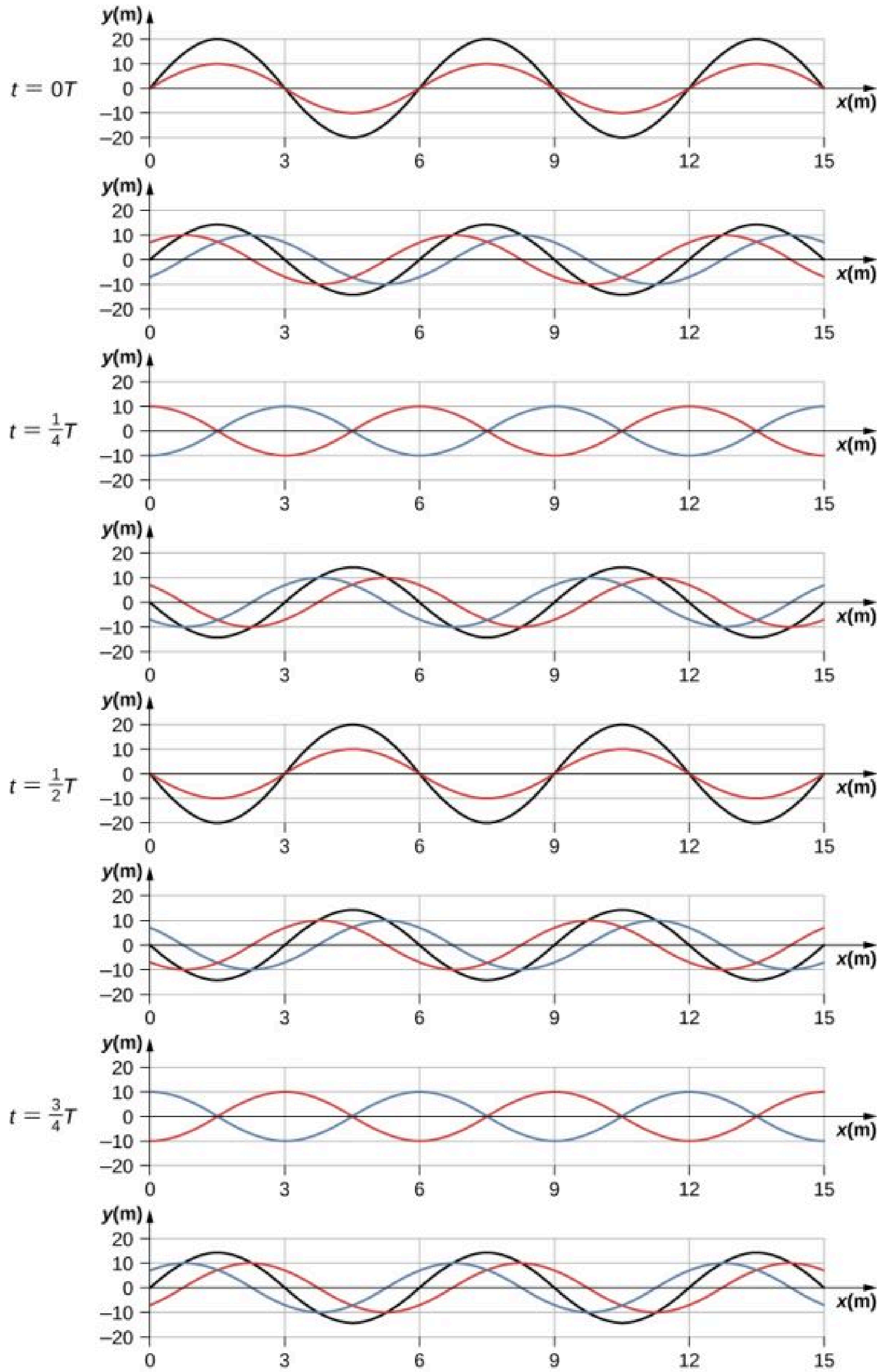


FIGURE 16.26 Time snapshots of two sine waves. The red wave is moving in the −*x*-direction and the blue wave is moving in the +*x*-direction. The resulting wave is shown in black. Consider the resultant wave at the points $x = 0\,\text{m}, 3\,\text{m}, 6\,\text{m}, 9\,\text{m}, 12\,\text{m}, 15\,\text{m}$ and notice that the resultant wave always equals zero at these points, no matter what the time is. These points are known as fixed points (nodes). In between each two nodes is an antinode, a place where the medium oscillates with an amplitude equal to the sum of the amplitudes of the individual waves.

Consider two identical waves that move in opposite directions. The first wave has a wave function of $y_1\,(x, t) = A\,\sin\,(kx - \omega t)$ and the second wave has a wave function $y_2\,(x, t) = A\,\sin\,(kx + \omega t)$. The waves interfere and form a resultant wave

$$y(x,t) = y_1(x,t) + y_2(x,t),$$
$$y(x,t) = A\sin(kx - \omega t) + A\sin(kx + \omega t).$$

This can be simplified using the trigonometric identity

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta,$$

where $\alpha = kx$ and $\beta = \omega t$, giving us

$$y(x,t) = A[\sin(kx)\cos(\omega t) - \cos(kx)\sin(\omega t) + \sin(kx)\cos(\omega t) + \cos(kx)\sin(\omega t)],$$

which simplifies to

$$y(x,t) = [2A\sin(kx)]\cos(\omega t). \tag{16.14}$$

Notice that the resultant wave is a sine wave that is a function only of position, multiplied by a cosine function that is a function only of time. Graphs of *y*(*x*,*t*) as a function of *x* for various times are shown in Figure 16.26. The red wave moves in the negative *x*-direction, the blue wave moves in the positive *x*-direction, and the black wave is the sum of the two waves. As the red and blue waves move through each other, they move in and out of constructive interference and destructive interference.

Initially, at time $t = 0$, the two waves are in phase, and the result is a wave that is twice the amplitude of the individual waves. The waves are also in phase at the time $t = \frac{T}{2}$. In fact, the waves are in phase at any integer multiple of half of a period:

$$t = n\frac{T}{2} \text{ where } n = 0, 1, 2, 3.... \text{ (in phase)}.$$

At other times, the two waves are $180°(\pi \text{ radians})$ out of phase, and the resulting wave is equal to zero. This happens at

$$t = \frac{1}{4}T, \frac{3}{4}T, \frac{5}{4}T,..., \frac{n}{4}T \text{ where } n = 1, 3, 5.... \text{ (out of phase)}.$$

Notice that some *x*-positions of the resultant wave are always zero no matter what the phase relationship is. These positions are called **nodes**. Where do the nodes occur? Consider the solution to the sum of the two waves

$$y(x,t) = [2A\sin(kx)]\cos(\omega t).$$

Finding the positions where the sine function equals zero provides the positions of the nodes.

$$\begin{aligned} \sin(kx) &= 0 \\ kx &= 0, \pi, 2\pi, 3\pi,... \\ \tfrac{2\pi}{\lambda}x &= 0, \pi, 2\pi, 3\pi,... \\ x &= 0, \tfrac{\lambda}{2}, \lambda, \tfrac{3\lambda}{2},... = n\tfrac{\lambda}{2} \quad n = 0, 1, 2, 3,.... \end{aligned}$$

There are also positions where *y* oscillates between $y = \pm A$. These are the **antinodes**. We can find them by considering which values of *x* result in $\sin(kx) = \pm 1$.

$$\begin{aligned} \sin(kx) &= \pm 1 \\ kx &= \tfrac{\pi}{2}, \tfrac{3\pi}{2}, \tfrac{5\pi}{2},... \\ \tfrac{2\pi}{\lambda}x &= \tfrac{\pi}{2}, \tfrac{3\pi}{2}, \tfrac{5\pi}{2},... \\ x &= \tfrac{\lambda}{4}, \tfrac{3\lambda}{4}, \tfrac{5\lambda}{4},... = n\tfrac{\lambda}{4} \quad n = 1, 3, 5,.... \end{aligned}$$

What results is a standing wave as shown in Figure 16.27, which shows snapshots of the resulting wave of two identical waves moving in opposite directions. The resulting wave appears to be a sine wave with nodes at integer multiples of half wavelengths. The antinodes oscillate between $y = \pm 2A$ due to the cosine term, $\cos(\omega t)$, which oscillates between ± 1.

The resultant wave appears to be standing still, with no apparent movement in the *x*-direction, although it is composed of one wave function moving in the positive, whereas the second wave is moving in the negative x-direction. Figure 16.27 shows various snapshots of the resulting wave. The nodes are marked with red dots while the antinodes are marked with blue dots.

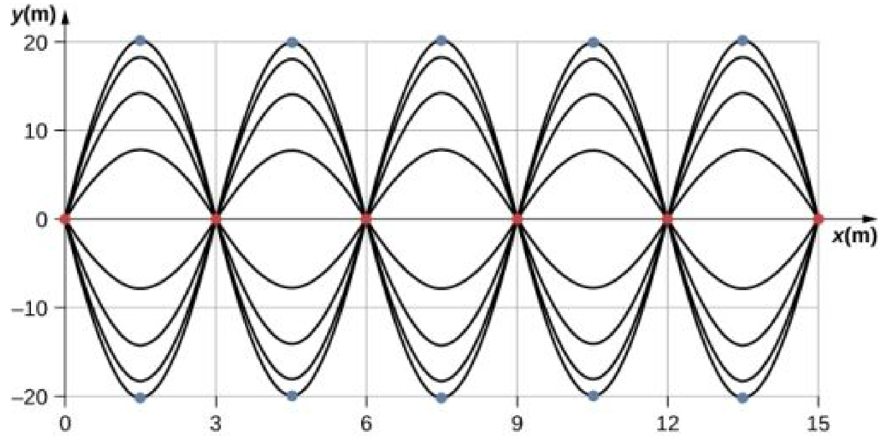


FIGURE 16.27 When two identical waves are moving in opposite directions, the resultant wave is a standing wave. Nodes appear at integer multiples of half wavelengths. Antinodes appear at odd multiples of quarter wavelengths, where they oscillate between $y = \pm A$. The nodes are marked with red dots and the antinodes are marked with blue dots.

A common example of standing waves are the waves produced by stringed musical instruments. When the string is plucked, pulses travel along the string in opposite directions. The ends of the strings are fixed in place, so nodes appear at the ends of the strings—the boundary conditions of the system, regulating the resonant frequencies in the strings. The resonance produced on a string instrument can be modeled in a physics lab using the apparatus shown in Figure 16.28.

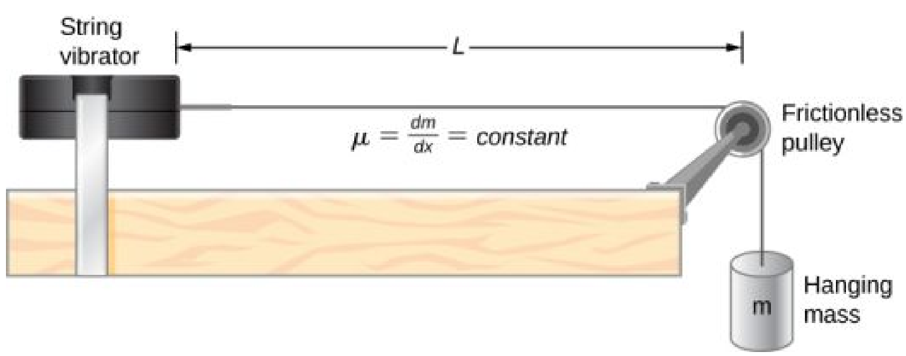


FIGURE 16.28 A lab setup for creating standing waves on a string. The string has a node on each end and a constant linear density. The length between the fixed boundary conditions is *L*. The hanging mass provides the tension in the string, and the speed of the waves on the string is proportional to the square root of the tension divided by the linear mass density.

The lab setup shows a string attached to a string vibrator, which oscillates the string with an adjustable frequency *f*. The other end of the string passes over a frictionless pulley and is tied to a hanging mass. The magnitude of the tension in the string is equal to the weight of the hanging mass. The string has a constant linear density (mass per length) μ and the speed at which a wave travels down the string equals $v = \sqrt{\frac{F_T}{\mu}} = \sqrt{\frac{mg}{\mu}}$ Equation 16.7. The symmetrical boundary conditions (a node at each end) dictate the possible frequencies that can excite standing waves. Starting from a frequency of zero and slowly increasing the frequency, the first mode $n = 1$ appears as shown in Figure 16.29. The first mode, also called the fundamental mode or the first harmonic, shows half of a wavelength has formed, so the wavelength is equal to twice the length between the nodes $\lambda_1 = 2L$. The **fundamental frequency**, or first harmonic frequency, that drives this mode is

$$f_1 = \frac{v}{\lambda_1} = \frac{v}{2L},$$

where the speed of the wave is $v = \sqrt{\frac{F_T}{\mu}}$. Keeping the tension constant and increasing the frequency leads to the second harmonic or the $n = 2$ mode. This mode is a full wavelength $\lambda_2 = L$ and the frequency is twice the fundamental frequency:

$$f_2 = \frac{v}{\lambda_2} = \frac{v}{L} = 2f_1.$$

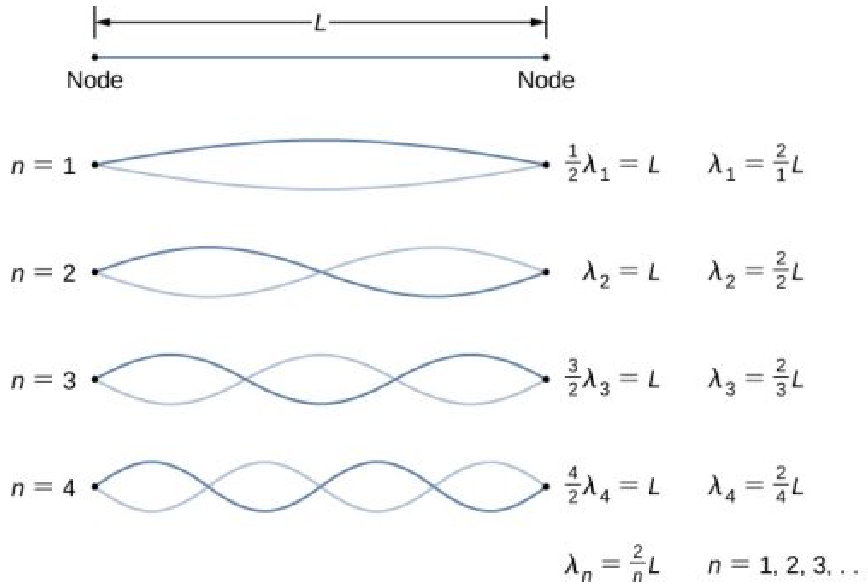


FIGURE 16.29 Standing waves created on a string of length *L*. A node occurs at each end of the string. The nodes are boundary conditions that limit the possible frequencies that excite standing waves. (Note that the amplitudes of the oscillations have been kept constant for visualization. The standing wave patterns possible on the string are known as the normal modes. Conducting this experiment in the lab would result in a decrease in amplitude as the frequency increases.)

The next two modes, or the third and fourth harmonics, have wavelengths of $\lambda_3 = \frac{2}{3}L$ and $\lambda_4 = \frac{2}{4}L$, driven by frequencies of $f_3 = \frac{3v}{2L} = 3f_1$ and $f_4 = \frac{4v}{2L} = 4f_1$. All frequencies above the frequency f_1 are known as the **overtones**. The equations for the wavelength and the frequency can be summarized as:

$$\lambda_n = \frac{2}{n}L \quad n = 1, 2, 3, 4, 5... \tag{16.15}$$

$$f_n = n\frac{v}{2L} = nf_1 \quad n = 1, 2, 3, 4, 5... \tag{16.16}$$

The standing wave patterns that are possible for a string, the first four of which are shown in Figure 16.29, are known as the **normal modes**, with frequencies known as the normal frequencies. In summary, the first frequency to produce a normal mode is called the fundamental frequency (or first harmonic). Any frequencies above the fundamental frequency are overtones. The second frequency of the $n = 2$ normal mode of the string is the first overtone (or second harmonic). The frequency of the $n = 3$ normal mode is the second overtone (or third harmonic) and so on.

The solutions shown as Equation 16.15 and Equation 16.16 are for a string with the boundary condition of a node on each end. When the boundary condition on either side is the same, the system is said to have symmetric boundary conditions. Equation 16.15 and Equation 16.16 are good for any symmetric boundary conditions, that is, nodes at both ends or antinodes at both ends.

EXAMPLE 16.7

Standing Waves on a String

Consider a string of $L = 2.00\ \text{m}$ attached to an adjustable-frequency string vibrator as shown in Figure 16.30. The waves produced by the vibrator travel down the string and are reflected by the fixed boundary condition at the pulley. The string, which has a linear mass density of $\mu = 0.006\ \text{kg/m}$, is passed over a frictionless pulley of a negligible mass, and the tension is provided by a 2.00-kg hanging mass. (a) What is the velocity of the waves on the string? (b) Draw a sketch of the first three normal modes of the standing waves that can be produced on the string and label each with the wavelength. (c) List the frequencies that the string vibrator must be tuned to in order to produce the first three normal modes of the standing waves.

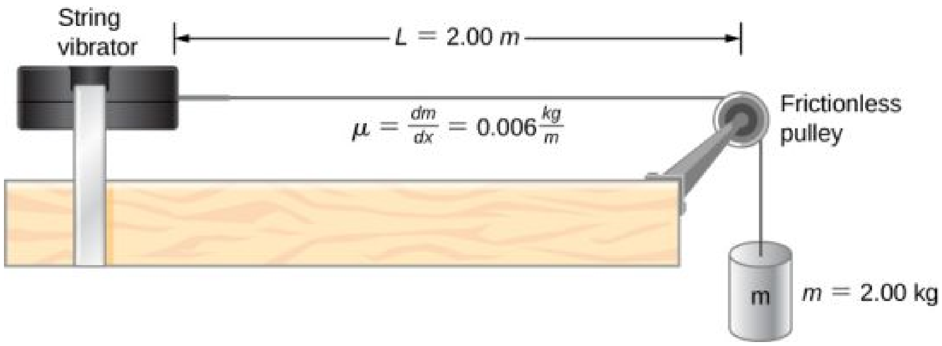


FIGURE 16.30 A string attached to an adjustable-frequency string vibrator.

Strategy

a. The velocity of the wave can be found using $v = \sqrt{\frac{F_T}{\mu}}$. The tension is provided by the weight of the hanging mass.
b. The standing waves will depend on the boundary conditions. There must be a node at each end. The first mode will be one half of a wave. The second can be found by adding a half wavelength. That is the shortest length that will result in a node at the boundaries. For example, adding one quarter of a wavelength will result in an antinode at the boundary and is not a mode which would satisfy the boundary conditions. This is shown in Figure 16.31.
c. Since the wave speed velocity is the wavelength times the frequency, the frequency is wave speed divided by the wavelength.

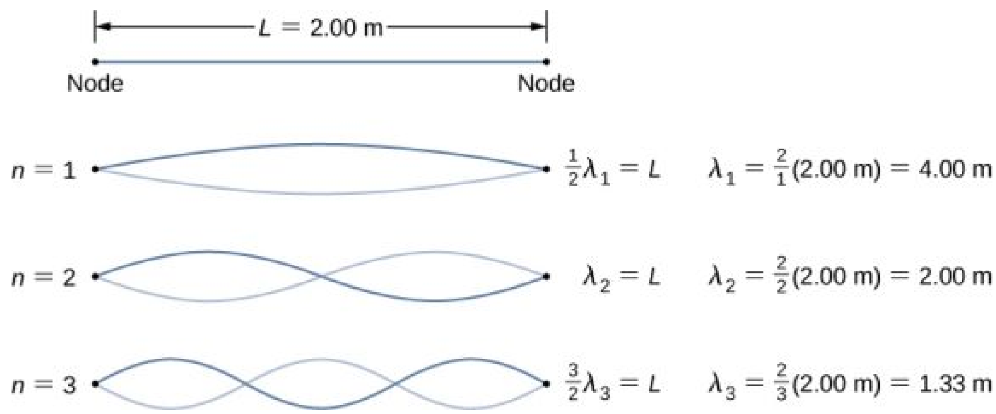
FIGURE 16.31 (a) The figure represents the second mode of the string that satisfies the boundary conditions of a node at each end of the string. (b)This figure could not possibly be a normal mode on the string because it does not satisfy the boundary conditions. There is a node on one end, but an antinode on the other.

Solution

a. Begin with the velocity of a wave on a string. The tension is equal to the weight of the hanging mass. The linear mass density and mass of the hanging mass are given:

$$v = \sqrt{\frac{F_T}{\mu}} = \sqrt{\frac{mg}{\mu}} = \sqrt{\frac{2\text{ kg}\left(9.8\frac{\text{m}}{\text{s}}\right)}{0.006\frac{\text{kg}}{\text{m}}}} = 57.15\text{ m/s}.$$

b. The first normal mode that has a node on each end is a half wavelength. The next two modes are found by adding a half of a wavelength.



c. The frequencies of the first three modes are found by using $f = \frac{v_w}{\lambda}$.

$$f_1 = \frac{v_w}{\lambda_1} = \frac{57.15 \text{ m/s}}{4.00 \text{ m}} = 14.29 \text{ Hz}$$
$$f_2 = \frac{v_w}{\lambda_2} = \frac{57.15 \text{ m/s}}{2.00 \text{ m}} = 28.58 \text{ Hz}$$
$$f_3 = \frac{v_w}{\lambda_3} = \frac{57.15 \text{ m/s}}{1.333 \text{ m}} = 42.87 \text{ Hz}$$

Significance

The three standing modes in this example were produced by maintaining the tension in the string and adjusting the driving frequency. Keeping the tension in the string constant results in a constant velocity. The same modes could have been produced by keeping the frequency constant and adjusting the speed of the wave in the string (by changing the hanging mass.)

INTERACTIVE

Engage the Phet simulation below to play with a 1D or 2D system of coupled mass-spring oscillators. Vary the number of masses, set the initial conditions, and watch the system evolve. See the spectrum of normal modes for arbitrary motion. See longitudinal or transverse modes in the 1D system.

[Access multimedia content (https://openstax.org/books/university-physics-volume-1/pages/16-6-standing-waves-and-resonance)](https://openstax.org/books/university-physics-volume-1/pages/16-6-standing-waves-and-resonance)

CHECK YOUR UNDERSTANDING 16.7

The equations for the wavelengths and the frequencies of the modes of a wave produced on a string:

$$\lambda_n = \frac{2}{n}L \quad n = 1, 2, 3, 4, 5\ldots \text{ and}$$
$$f_n = n\frac{v}{2L} = nf_1 \quad n = 1, 2, 3, 4, 5\ldots$$

were derived by considering a wave on a string where there were symmetric boundary conditions of a node at each end. These modes resulted from two sinusoidal waves with identical characteristics except they were moving in opposite directions, confined to a region L with nodes required at both ends. Will the same equations work if there were symmetric boundary conditions with antinodes at each end? What would the normal modes look like for a medium that was free to oscillate on each end? Don't worry for now if you cannot imagine such a medium, just consider two sinusoidal wave functions in a region of length L, with antinodes on each end.

The free boundary conditions shown in the last Check Your Understanding may seem hard to visualize. How can there be a system that is free to oscillate on each end? In Figure 16.32 are shown two possible configuration of a metallic rods (shown in red) attached to two supports (shown in blue). In part (a), the rod is supported at the ends, and there are fixed boundary conditions at both ends. Given the proper frequency, the rod can be driven into resonance with a wavelength equal to length of the rod, with nodes at each end. In part (b), the rod is supported at positions one quarter of the length from each end of the rod, and there are free boundary conditions at both ends. Given the proper frequency, this rod can also be driven into resonance with a wavelength equal to the length of the rod, but there are antinodes at each end. If you are having trouble visualizing the wavelength in this figure, remember that the wavelength may be measured between any two nearest identical points and consider Figure 16.33.

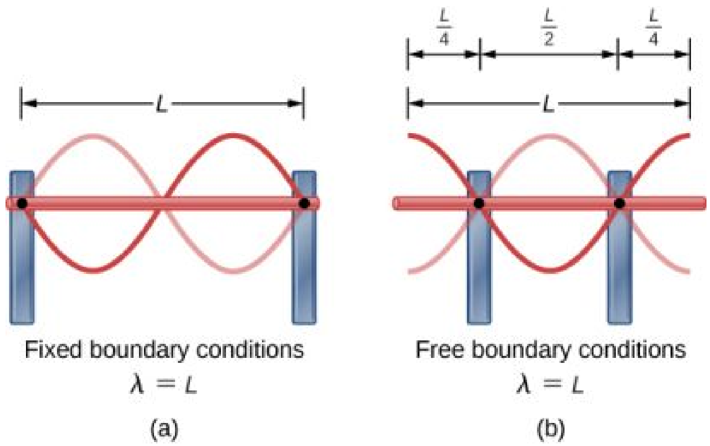


FIGURE 16.32 (a) A metallic rod of length L (red) supported by two supports (blue) on each end. When driven at the proper frequency, the rod can resonate with a wavelength equal to the length of the rod with a node on each end. (b) The same metallic rod of length L (red) supported by two supports (blue) at a position a quarter of the length of the rod from each end. When driven at the proper frequency, the rod can resonate with a wavelength equal to the length of the rod with an antinode on each end.

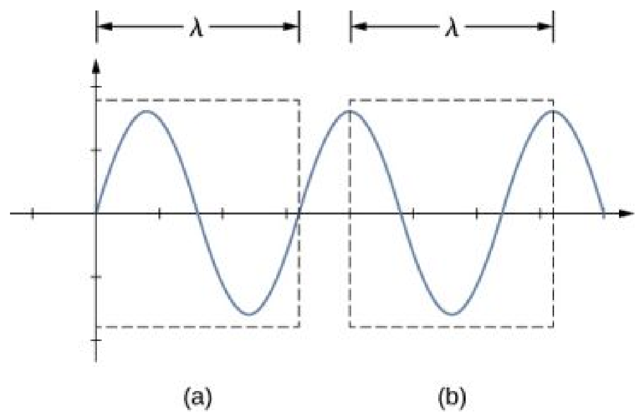


FIGURE 16.33 A wavelength may be measure between the nearest two repeating points. On the wave on a string, this means the same height and slope. (a) The wavelength is measured between the two nearest points where the height is zero and the slope is maximum and positive. (b) The wavelength is measured between two identical points where the height is maximum and the slope is zero.

Note that the study of standing waves can become quite complex. In Figure 16.32(a), the $n = 2$ mode of the standing wave is shown, and it results in a wavelength equal to L. In this configuration, the $n = 1$ mode would also have been possible with a standing wave equal to $2L$. Is it possible to get the $n = 1$ mode for the configuration shown in part (b)? The answer is no. In this configuration, there are additional conditions set beyond the boundary conditions. Since the rod is mounted at a point one quarter of the length from each side, a node must exist there, and this limits the possible modes of standing waves that can be created. We leave it as an exercise for the reader to consider if other modes of standing waves are possible. It should be noted that when a system is driven at a frequency that does not cause the system to resonate, vibrations may still occur, but the amplitude of the vibrations will be much smaller than the amplitude at resonance.

A field of mechanical engineering uses the sound produced by the vibrating parts of complex mechanical systems to troubleshoot problems with the systems. Suppose a part in an automobile is resonating at the frequency of the car's engine, causing unwanted vibrations in the automobile. This may cause the engine to fail prematurely. The engineers use microphones to record the sound produced by the engine, then use a technique called Fourier analysis to find frequencies of sound produced with large amplitudes and then look at the parts list of the automobile to find a part that would resonate at that frequency. The solution may be as simple as changing the composition of the material used or changing the length of the part in question.

There are other numerous examples of resonance in standing waves in the physical world. The air in a tube, such as found in a musical instrument like a flute, can be forced into resonance and produce a pleasant sound, as we discuss in Sound.

At other times, resonance can cause serious problems. A closer look at earthquakes provides evidence for conditions appropriate for resonance, standing waves, and constructive and destructive interference. A building may vibrate for several seconds with a driving frequency matching that of the natural frequency of vibration of the building—producing a resonance resulting in one building collapsing while neighboring buildings do not. Often, buildings of a certain height are devastated while other taller buildings remain intact. The building height matches the condition for setting up a standing wave for that particular height. The span of the roof is also important. Often it is seen that gymnasiums, supermarkets, and churches suffer damage when individual homes suffer far less damage. The roofs with large surface areas supported only at the edges resonate at the frequencies of the earthquakes, causing them to collapse. As the earthquake waves travel along the surface of Earth and reflect off denser rocks, constructive interference occurs at certain points. Often areas closer to the epicenter are not damaged, while areas farther away are damaged.

Chapter Review

Key Terms

antinode location of maximum amplitude in standing waves

constructive interference when two waves arrive at the same point exactly in phase; that is, the crests of the two waves are precisely aligned, as are the troughs

destructive interference when two identical waves arrive at the same point exactly out of phase; that is, precisely aligned crest to trough

fixed boundary condition when the medium at a boundary is fixed in place so it cannot move

free boundary condition exists when the medium at the boundary is free to move

fundamental frequency lowest frequency that will produce a standing wave

intensity (I) power per unit area

interference overlap of two or more waves at the same point and time

linear wave equation equation describing waves that result from a linear restoring force of the medium; any function that is a solution to the wave equation describes a wave moving in the positive *x*-direction or the negative *x*-direction with a constant wave speed *v*

longitudinal wave wave in which the disturbance is parallel to the direction of propagation

mechanical wave wave that is governed by Newton's laws and requires a medium

node point where the string does not move; more generally, nodes are where the wave disturbance is zero in a standing wave

normal mode possible standing wave pattern for a standing wave on a string

overtone frequency that produces standing waves and is higher than the fundamental frequency

pulse single disturbance that moves through a medium, transferring energy but not mass

standing wave wave that can bounce back and forth through a particular region, effectively becoming stationary

superposition phenomenon that occurs when two or more waves arrive at the same point

transverse wave wave in which the disturbance is perpendicular to the direction of propagation

wave disturbance that moves from its source and carries energy

wave function mathematical model of the position of particles of the medium

wave number $\frac{2\pi}{\lambda}$

wave speed magnitude of the wave velocity

wave velocity velocity at which the disturbance moves; also called the propagation velocity

wavelength distance between adjacent identical parts of a wave

Key Equations

Wave speed	$v = \frac{\lambda}{T} = \lambda f$
Linear mass density	$\mu = \frac{\text{mass of the string}}{\text{length of the string}}$
Speed of a wave or pulse on a string under tension	$\lvert v \rvert = \sqrt{\frac{F_T}{\mu}}$
Speed of a compression wave in a fluid	$v = \sqrt{\frac{B}{\rho}}$
Resultant wave from superposition of two sinusoidal waves that are identical except for a phase shift	$y_R(x,t) = \left[2A\cos\left(\frac{\phi}{2}\right)\right]\sin\left(kx - \omega t + \frac{\phi}{2}\right)$
Wave number	$k \equiv \frac{2\pi}{\lambda}$
Wave speed	$v = \frac{\omega}{k}$
A periodic wave	$y(x,t) = A\sin(kx \mp \omega t + \phi)$
Phase of a wave	$kx \mp \omega t + \phi$
The linear wave equation	$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v_w^2}\frac{\partial^2 y(x,t)}{\partial t^2}$
Power averaged over a wavelength	$P_{\text{ave}} = \frac{E_\lambda}{T} = \frac{1}{2}\mu A^2\omega^2\frac{\lambda}{T} = \frac{1}{2}\mu A^2\omega^2 v$
Intensity	$I = \frac{P}{A}$

Intensity for a spherical wave	$I = \frac{P}{4\pi r^2}$
Equation of a standing wave	$y(x, t) = [2A \sin(kx)] \cos(\omega t)$
Wavelength for symmetric boundary conditions	$\lambda_n = \frac{2}{n} L, \quad n = 1, 2, 3, 4, 5...$
Frequency for symmetric boundary conditions	$f_n = n\frac{v}{2L} = nf_1, \quad n = 1, 2, 3, 4, 5...$

Summary

16.1 Traveling Waves

- A wave is a disturbance that moves from the point of origin with a wave velocity *v*.
- A wave has a wavelength λ, which is the distance between adjacent identical parts of the wave. Wave velocity and wavelength are related to the wave's frequency and period by $v = \frac{\lambda}{T} = \lambda f$.
- Mechanical waves are disturbances that move through a medium and are governed by Newton's laws.
- Electromagnetic waves are disturbances in the electric and magnetic fields, and do not require a medium.
- Matter waves are a central part of quantum mechanics and are associated with protons, electrons, neutrons, and other fundamental particles found in nature.
- A transverse wave has a disturbance perpendicular to the wave's direction of propagation, whereas a longitudinal wave has a disturbance parallel to its direction of propagation.

16.2 Mathematics of Waves

- A wave is an oscillation (of a physical quantity) that travels through a medium, accompanied by a transfer of energy. Energy transfers from one point to another in the direction of the wave motion. The particles of the medium oscillate up and down, back and forth, or both up and down and back and forth, around an equilibrium position.
- A snapshot of a sinusoidal wave at time $t = 0.00$ s can be modeled as a function of position. Two examples of such functions are $y(x) = A \sin(kx + \phi)$ and $y(x) = A \cos(kx + \phi)$.
- Given a function of a wave that is a snapshot of the wave, and is only a function of the position *x*, the motion of the pulse or wave moving at a constant velocity can be modeled with the function, replacing *x* with $x \mp vt$. The minus sign is for motion in the positive direction and the plus sign for the negative direction.
- The wave function is given by $y(x, t) = A \sin(kx - \omega t + \phi)$ where $k = 2\pi/\lambda$ is defined as the wave number, $\omega = 2\pi/T$ is the angular frequency, and ϕ is the phase shift.
- The wave moves with a constant velocity v_w, where the particles of the medium oscillate about an equilibrium position. The constant velocity of a wave can be found by $v = \frac{\lambda}{T} = \frac{\omega}{k}$.

16.3 Wave Speed on a Stretched String

- The speed of a wave on a string depends on the linear density of the string and the tension in the string. The linear density is mass per unit length of the string.
- In general, the speed of a wave depends on the square root of the ratio of the elastic property to the inertial property of the medium.
- The speed of a wave through a fluid is equal to the square root of the ratio of the bulk modulus of the fluid to the density of the fluid.
- The speed of sound through air at $T = 20°\text{C}$ is approximately $v_s = 343.00$ m/s.

16.4 Energy and Power of a Wave

- The energy and power of a wave are proportional to the square of the amplitude of the wave and the square of the angular frequency of the wave.
- The time-averaged power of a sinusoidal wave on a string is found by $P_{\text{ave}} = \frac{1}{2}\mu A^2 \omega^2 v$, where μ is the linear mass density of the string, *A* is the amplitude of the wave, ω is the angular frequency of the wave, and *v* is the speed of the wave.
- Intensity is defined as the power divided by the area. In a spherical wave, the area is $A = 4\pi r^2$ and the intensity is $I = \frac{P}{4\pi r^2}$. As the wave moves out from a source, the energy is conserved, but the intensity decreases as the area increases.

16.5 Interference of Waves

- Superposition is the combination of two waves at the same location.
- Constructive interference occurs from the

superposition of two identical waves that are in phase.

- Destructive interference occurs from the superposition of two identical waves that are $180°$ (π radians) out of phase.
- The wave that results from the superposition of two sine waves that differ only by a phase shift is a wave with an amplitude that depends on the value of the phase difference.

16.6 Standing Waves and Resonance

- A standing wave is the superposition of two waves which produces a wave that varies in amplitude but does not propagate.
- Nodes are points of no motion in standing waves.
- An antinode is the location of maximum amplitude of a standing wave.
- Normal modes of a wave on a string are the possible standing wave patterns. The lowest frequency that will produce a standing wave is known as the fundamental frequency. The higher frequencies which produce standing waves are called overtones.

Conceptual Questions

16.1 Traveling Waves

1. Give one example of a transverse wave and one example of a longitudinal wave, being careful to note the relative directions of the disturbance and wave propagation in each.

2. A sinusoidal transverse wave has a wavelength of 2.80 m. It takes 0.10 s for a portion of the string at a position *x* to move from a maximum position of $y = 0.03$ m to the equilibrium position $y = 0$. What are the period, frequency, and wave speed of the wave?

3. What is the difference between propagation speed and the frequency of a mechanical wave? Does one or both affect wavelength? If so, how?

4. Consider a stretched spring, such as a slinky. The stretched spring can support longitudinal waves and transverse waves. How can you produce transverse waves on the spring? How can you produce longitudinal waves on the spring?

5. Consider a wave produced on a stretched spring by holding one end and shaking it up and down. Does the wavelength depend on the distance you move your hand up and down?

6. A sinusoidal, transverse wave is produced on a stretched spring, having a period *T*. Each section of the spring moves perpendicular to the direction of propagation of the wave, in simple harmonic motion with an amplitude *A*. Does each section oscillate with the same period as the wave or a different period? If the amplitude of the transverse wave were doubled but the period stays the same, would your answer be the same?

7. An electromagnetic wave, such as light, does not require a medium. Can you think of an example that would support this claim?

16.2 Mathematics of Waves

8. If you were to shake the end of a taut spring up and down 10 times a second, what would be the frequency and the period of the sinusoidal wave produced on the spring?

9. If you shake the end of a stretched spring up and down with a frequency *f*, you can produce a sinusoidal, transverse wave propagating down the spring. Does the wave number depend on the frequency you are shaking the spring?

10. Does the vertical speed of a segment of a horizontal taut string through which a sinusoidal, transverse wave is propagating depend on the wave speed of the transverse wave?

11. In this section, we have considered waves that move at a constant wave speed. Does the medium accelerate?

12. If you drop a pebble in a pond you may notice that several concentric ripples are produced, not just a single ripple. Why do you think that is?

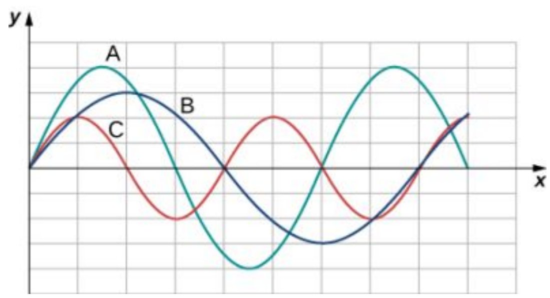
16.3 Wave Speed on a Stretched String

13. If the tension in a string were increased by a factor of four, by what factor would the wave speed of a wave on the string increase?

14. Does a sound wave move faster in seawater or fresh water, if both the sea water and fresh water are at the same temperature and the sound wave moves near the surface? $\left(\rho_{\text{w}} \approx 1000 \frac{\text{kg}}{\text{m}^3}, \rho_{\text{s}} \approx 1030 \frac{\text{kg}}{\text{m}^3}, B_{\text{w}} = 2.15 \times 10^9 \text{ Pa}, \ B_{\text{s}} = 2.34 \times 10^9 \text{ Pa}\right)$

15. Guitars have strings of different linear mass density. If the lowest density string and the highest density string are under the same tension, which string would support waves with the higher wave speed?

16. Shown below are three waves that were sent down a string at different times. The tension in the string remains constant. (a) Rank the waves from the smallest wavelength to the largest wavelength. (b) Rank the waves from the lowest frequency to the highest frequency.



17. Electrical power lines connected by two utility poles are sometimes heard to hum when driven into oscillation by the wind. The speed of the waves on the power lines depend on the tension. What provides the tension in the power lines?

18. Two strings, one with a low mass density and one with a high linear density are spliced together. The higher density end is tied to a lab post and a student holds the free end of the low-mass density string. The student gives the string a flip and sends a pulse down the strings. If the tension is the same in both strings, does the pulse travel at the same wave velocity in both strings? If not, where does it travel faster, in the low density string or the high density string?

16.4 Energy and Power of a Wave

19. Consider a string with under tension with a constant linear mass density. A sinusoidal wave is produced by some external driving force. If the frequency, keeping the wave amplitude as it was originally, how is the time-averaged power of the wave affected? If the amplitude of the wave is decreased by half, keeping the wave frequency as it was originally, how is the time-averaged power affected? Explain your answer.

20. Circular water waves decrease in amplitude as they move away from where a rock is dropped. Explain why.

21. In a transverse wave on a string, the motion of the string is perpendicular to the motion of the wave. If this is so, how is possible to move energy along the length of the string?

22. The energy from the sun warms the portion of the earth facing the sun during the daylight hours. Why are the North and South Poles cold while the equator is quite warm?

23. The intensity of a spherical waves decreases as the wave moves away from the source. If the intensity of the wave at the source is I_0, how far from the source will the intensity decrease by a factor of nine?

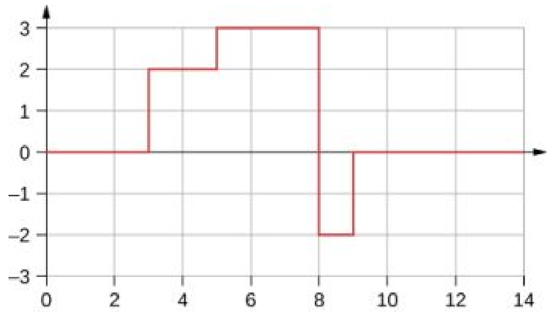
16.5 Interference of Waves

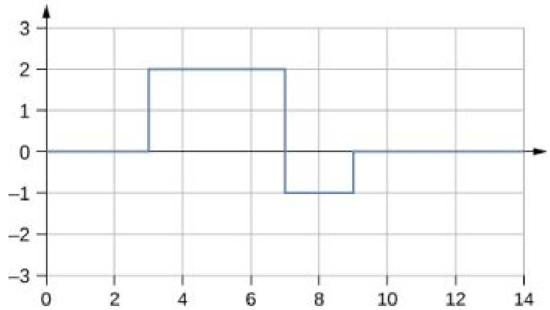
24. An incident sinusoidal wave is sent along a string that is fixed to the wall with a wave speed of *v*. The wave reflects off the end of the string. Describe the reflected wave.

25. A string of a length of 2.00 m with a linear mass density of $\mu = 0.006\ \text{kg/m}$ is attached to the end of a 2.00-m-long string with a linear mass density of $\mu = 0.012\ \text{kg/m}$. The free end of the higher-density string is fixed to the wall, and a student holds the free end of the low-density string, keeping the tension constant in both strings. The student sends a pulse down the string. Describe what happens at the interface between the two strings.

26. A long, tight spring is held by two students, one student holding each end. Each student gives the end a flip sending one wavelength of a sinusoidal wave down the spring in opposite directions. When the waves meet in the middle, what does the wave look like?

27. Many of the topics discussed in this chapter are useful beyond the topics of mechanical waves. It is hard to conceive of a mechanical wave with sharp corners, but you could encounter such a wave form in your digital electronics class, as shown below. This could be a signal from a device known as an analog to digital converter, in which a continuous voltage signal is converted into a discrete signal or a digital recording of sound. What is the result of the superposition of the two signals?





28. A string of a constant linear mass density is held taut by two students, each holding one end. The tension in the string is constant. The students each send waves down the string by wiggling the string. (a) Is it possible for the waves to have different wave speeds? (b) Is it possible for the waves to have different frequencies? (c) Is it possible for the waves to have different wavelengths?

16.6 Standing Waves and Resonance

29. A truck manufacturer finds that a strut in the engine is failing prematurely. A sound engineer determines that the strut resonates at the frequency of the engine and suspects that this could be the problem. What are two possible characteristics of the strut can be modified to correct the problem?

30. Why do roofs of gymnasiums and churches seem to fail more than family homes when an earthquake occurs?

31. Wine glasses can be set into resonance by moistening your finger and rubbing it around the rim of the glass. Why?

32. Air conditioning units are sometimes placed on the roof of homes in the city. Occasionally, the air conditioners cause an undesirable hum throughout the upper floors of the homes. Why does this happen? What can be done to reduce the hum?

33. Consider a standing wave modeled as $y(x,t) = 4.00\text{ cm}\sin\left(3\text{ m}^{-1}x\right)\cos\left(4\text{ s}^{-1}t\right)$. Is there a node or an antinode at $x = 0.00\text{ m}$? What about a standing wave modeled as $y(x,t) = 4.00\text{ cm}\sin\left(3\text{ m}^{-1}x + \frac{\pi}{2}\right)\cos\left(4\text{ s}^{-1}t\right)$? Is there a node or an antinode at the $x = 0.00\text{ m}$ position?

Problems

16.1 Traveling Waves

34. Storms in the South Pacific can create waves that travel all the way to the California coast, 12,000 km away. How long does it take them to travel this distance if they travel at 15.0 m/s?

35. Waves on a swimming pool propagate at 0.75 m/s. You splash the water at one end of the pool and observe the wave go to the opposite end, reflect, and return in 30.00 s. How far away is the other end of the pool?

36. Wind gusts create ripples on the ocean that have a wavelength of 5.00 cm and propagate at 2.00 m/s. What is their frequency?

37. How many times a minute does a boat bob up and down on ocean waves that have a wavelength of 40.0 m and a propagation speed of 5.00 m/s?

38. Scouts at a camp shake the rope bridge they have just crossed and observe the wave crests to be 8.00 m apart. If they shake the bridge twice per second, what is the propagation speed of the waves?

39. What is the wavelength of the waves you create in a swimming pool if you splash your hand at a rate of 2.00 Hz and the waves propagate at a wave speed of 0.800 m/s?

40. What is the wavelength of an earthquake that shakes you with a frequency of 10.0 Hz and gets to another city 84.0 km away in 12.0 s?

41. Radio waves transmitted through empty space at the speed of light $\left(v = c = 3.00 \times 10^8\text{ m/s}\right)$ by the *Voyager* spacecraft have a wavelength of 0.120 m. What is their frequency?

42. Your ear is capable of differentiating sounds that arrive at each ear just 0.34 ms apart, which is useful in determining where low frequency sound is originating from. (a) Suppose a low-frequency sound source is placed to the right of a person, whose ears are approximately 18 cm apart, and the speed of sound generated is 340 m/s. How long is the interval between when the sound arrives at the right ear and the sound arrives at the left ear? (b) Assume the same person was scuba diving and a low-frequency sound source was to the right of the scuba diver. How long is the interval between when the sound arrives at the right ear and the sound arrives at the left ear, if the speed of sound in water is 1500 m/s? (c) What is significant about the time interval of the two situations?

43. (a) Seismographs measure the arrival times of earthquakes with a precision of 0.100 s. To get the distance to the epicenter of the quake, geologists compare the arrival times of S- and P-waves, which travel at different speeds. If S- and P-waves arrive at 4.00 and 7.20 km/s, respectively, in the region considered, how precisely can the distance to the source of the earthquake be determined? (b) Seismic waves from underground detonations of nuclear bombs can be used to locate the test site and detect violations of test bans. Discuss whether your answer to (a) implies a serious limit to such detection. (Note also that the uncertainty is greater if there is an uncertainty in the propagation speeds of the S- and P-waves.)

44. A Girl Scout is taking a 10.00-km hike to earn a merit badge. While on the hike, she sees a cliff some distance away. She wishes to estimate the time required to walk to the cliff. She knows that the speed of sound is approximately 343 meters per second. She yells and finds that the echo returns after approximately 2.00 seconds. If she can hike 1.00 km in 10 minutes, how long would it take her to reach the cliff?

45. A quality assurance engineer at a frying pan company is asked to qualify a new line of nonstick-coated frying pans. The coating needs to be 1.00 mm thick. One method to test the thickness is for the engineer to pick a percentage of the pans manufactured, strip off the coating, and measure the thickness using a micrometer. This method is a destructive testing method. Instead, the engineer decides that every frying pan will be tested using a nondestructive method. An ultrasonic transducer is used that produces sound waves with a frequency of $f = 25\ \text{kHz}$. The sound waves are sent through the coating and are reflected by the interface between the coating and the metal pan, and the time is recorded. The wavelength of the ultrasonic waves in the coating is 0.076 m. What should be the time recorded if the coating is the correct thickness (1.00 mm)?

16.2 Mathematics of Waves

46. A pulse can be described as a single wave disturbance that moves through a medium. Consider a pulse that is defined at time $t = 0.00\ \text{s}$ by the equation $y(x) = \frac{6.00\ \text{m}^3}{x^2 + 2.00\ \text{m}^2}$ centered around $x = 0.00\ \text{m}$. The pulse moves with a velocity of $v = 3.00\ \text{m/s}$ in the positive x-direction. (a) What is the amplitude of the pulse? (b) What is the equation of the pulse as a function of position and time? (c) Where is the pulse centered at time $t = 5.00\ \text{s}$?

47. A transverse wave on a string is modeled with the wave function $y(x,t) = (0.20\ \text{cm})\sin\left(2.00\ \text{m}^{-1}x - 3.00\ \text{s}^{-1}t + \frac{\pi}{16}\right)$. What is the height of the string with respect to the equilibrium position at a position $x = 4.00\ \text{m}$ and a time $t = 10.00\ \text{s}$?

48. Consider the wave function $y(x,t) = (3.00\ \text{cm})\sin\left(0.4\ \text{m}^{-1}x + 2.00\ \text{s}^{-1}t + \frac{\pi}{10}\right)$. What are the period, wavelength, speed, and initial phase shift of the wave modeled by the wave function?

49. A pulse is defined as $y(x,t) = e^{-2.77\left(\frac{2.00(x - 2.00\ \text{m/s}(t))}{5.00\ \text{m}}\right)^2}$. Use a spreadsheet, or other computer program, to plot the pulse as the height of medium *y* as a function of position *x*. Plot the pulse at times $t = 0.00\ \text{s}$ and $t = 3.00\ \text{s}$ on the same graph. Where is the pulse centered at time $t = 3.00\ \text{s}$? Use your spreadsheet to check your answer.

50. A wave is modeled at time $t = 0.00\ \text{s}$ with a wave function that depends on position. The equation is $y(x) = (0.30\ \text{m})\sin\left(6.28\ \text{m}^{-1}x\right)$. The wave travels a distance of 4.00 meters in 0.50 s in the positive *x*-direction. Write an equation for the wave as a function of position and time.

51. A wave is modeled with the function $y(x,t) = (0.25\ \text{m})\cos\left(0.30\ \text{m}^{-1}x - 0.90\ \text{s}^{-1}t + \frac{\pi}{3}\right)$. Find the (a) amplitude, (b) wave number, (c) angular frequency, (d) wave speed, (e) initial phase shift, (f) wavelength, and (g) period of the wave.

52. A surface ocean wave has an amplitude of 0.60

m and the distance from trough to trough is 8.00 m. It moves at a constant wave speed of 1.50 m/s propagating in the positive x-direction. At $t = 0$, the water displacement at $x = 0$ is zero, and v_y is positive. (a) Assuming the wave can be modeled as a sine wave, write a wave function to model the wave. (b) Use a spreadsheet to plot the wave function at times $t = 0.00\text{ s}$ and $t = 2.00\text{ s}$ on the same graph. Verify that the wave moves 3.00 m in those 2.00 s.

53. A wave is modeled by the wave function $y(x, t) = (0.30\text{ m})\sin\left[\frac{2\pi}{4.50\text{ m}}\left(x - 18.00\frac{\text{m}}{\text{s}}t\right)\right]$. What are the amplitude, wavelength, wave speed, period, and frequency of the wave?

54. A transverse wave on a string is described with the wave function $y(x, t) = (0.50\text{ cm})\sin\left(1.57\text{ m}^{-1}x - 6.28\text{ s}^{-1}t\right)$. (a) What is the wave velocity of the wave? (b) What is the magnitude of the maximum velocity of the string perpendicular to the direction of the motion?

55. A swimmer in the ocean observes one day that the ocean surface waves are periodic and resemble a sine wave. The swimmer estimates that the vertical distance between the crest and the trough of each wave is approximately 0.45 m, and the distance between each crest is approximately 1.8 m. The swimmer counts that 12 waves pass every two minutes. Determine the simple harmonic wave function that would describes these waves.

56. Consider a wave described by the wave function $y(x, t) = 0.3\text{ m}\sin\left(2.00\text{ m}^{-1}x - 628.00\text{ s}^{-1}t\right)$. (a) How many crests pass by an observer at a fixed location in 2.00 minutes? (b) How far has the wave traveled in that time?

57. Consider two waves defined by the wave functions $y_1(x, t) = 0.50\text{ m}\sin\left(\frac{2\pi}{3.00\text{ m}}x + \frac{2\pi}{4.00\text{ s}}t\right)$ and $y_2(x, t) = 0.50\text{ m}\sin\left(\frac{2\pi}{6.00\text{ m}}x - \frac{2\pi}{4.00\text{ s}}t\right)$. What are the similarities and differences between the two waves?

58. Consider two waves defined by the wave functions $y_1(x, t) = 0.20\text{ m}\sin\left(\frac{2\pi}{6.00\text{ m}}x - \frac{2\pi}{4.00\text{ s}}t\right)$ and $y_2(x, t) = 0.20\text{ m}\cos\left(\frac{2\pi}{6.00\text{ m}}x - \frac{2\pi}{4.00\text{ s}}t\right)$. What are the similarities and differences between the two waves?

59. The speed of a transverse wave on a string is 300.00 m/s, its wavelength is 0.50 m, and the amplitude is 20.00 cm. How much time is required for a particle on the string to move through a distance of 5.00 km?

16.3 Wave Speed on a Stretched String

60. Transverse waves are sent along a 5.00-m-long string with a speed of 30.00 m/s. The string is under a tension of 10.00 N. What is the mass of the string?

61. A copper wire has a density of $\rho = 8920\text{ kg/m}^3$, a radius of 1.20 mm, and a length *L*. The wire is held under a tension of 10.00 N. Transverse waves are sent down the wire. (a) What is the linear mass density of the wire? (b) What is the speed of the waves through the wire?

62. A piano wire has a linear mass density of $\mu = 4.95 \times 10^{-3}\text{ kg/m}$. Under what tension must the string be kept to produce waves with a wave speed of 500.00 m/s?

63. A string with a linear mass density of $\mu = 0.0060\text{ kg/m}$ is tied to the ceiling. A 20-kg mass is tied to the free end of the string. The string is plucked, sending a pulse down the string. Estimate the speed of the pulse as it moves down the string.

64. A cord has a linear mass density of $\mu = 0.0075\text{ kg/m}$ and a length of three meters. The cord is plucked and it takes 0.20 s for the pulse to reach the end of the string. What is the tension of the string?

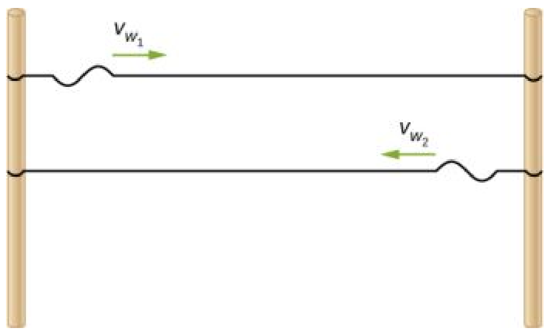
65. A string is 3.00 m long with a mass of 5.00 g. The string is held taut with a tension of 500.00 N applied to the string. A pulse is sent down the string. How long does it take the pulse to travel the 3.00 m of the string?

66. Two strings are attached to poles, however the first string is twice as long as the second. If both strings have the same tension and mu, what is the ratio of the speed of the pulse of the wave from the first string to the second string?

67. Two strings are attached to poles, however the first string is twice the linear mass density mu of the second. If both strings have the same tension, what is the ratio of the speed of the pulse of the wave from the first string to the second string?

68. Transverse waves travel through a string where the tension equals 7.00 N with a speed of 20.00 m/s. What tension would be required for a wave speed of 25.00 m/s?

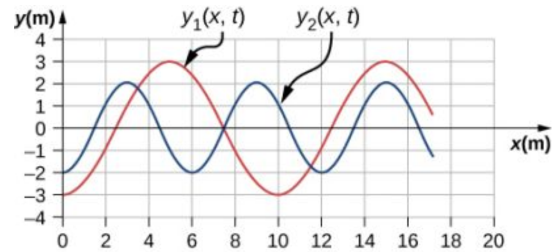
69. Two strings are attached between two poles separated by a distance of 2.00 m as shown below, both under the same tension of 600.00 N. String 1 has a linear density of $\mu_1 = 0.0025\ \text{kg/m}$ and string 2 has a linear mass density of $\mu_2 = 0.0035\ \text{kg/m}$. Transverse wave pulses are generated simultaneously at opposite ends of the strings. How much time passes before the pulses pass one another?



70. Two strings are attached between two poles separated by a distance of 2.00 meters as shown in the preceding figure, both strings have a linear density of $\mu_1 = 0.0025\ \text{kg/m}$, the tension in string 1 is 600.00 N and the tension in string 2 is 700.00 N. Transverse wave pulses are generated simultaneously at opposite ends of the strings. How much time passes before the pulses pass one another?

71. The note E_4 is played on a piano and has a frequency of $f = 393.88$. If the linear mass density of this string of the piano is $\mu = 0.012\ \text{kg/m}$ and the string is under a tension of 1000.00 N, what is the speed of the wave on the string and the wavelength of the wave?

72. Two transverse waves travel through a taut string. The speed of each wave is $v = 30.00\ \text{m/s}$. A plot of the vertical position as a function of the horizontal position is shown below for the time $t = 0.00\ \text{s}$. (a) What is the wavelength of each wave? (b) What is the frequency of each wave? (c) What is the maximum vertical speed of each string?



73. A sinusoidal wave travels down a taut, horizontal string with a linear mass density of $\mu = 0.060\ \text{kg/m}$. The maximum vertical speed of the wave is $v_{y\ \text{max}} = 0.30\ \text{cm/s}$. The wave is modeled with the wave equation $y(x, t) = A \sin\left(6.00\ \text{m}^{-1} x - 24.00\ \text{s}^{-1} t\right)$. (a) What is the amplitude of the wave? (b) What is the tension in the string?

74. The speed of a transverse wave on a string is $v = 60.00\ \text{m/s}$ and the tension in the string is $F_T = 100.00\ \text{N}$. What must the tension be to increase the speed of the wave to $v = 120.00\ \text{m/s}$?

16.4 Energy and Power of a Wave

75. A string of length 5 m and a mass of 90 g is held under a tension of 100 N. A wave travels down the string that is modeled as $y(x, t) = 0.01\ \text{m} \sin\left(15.7\ \text{m}^{-1} x - 1170.12\ \text{s}^{-1} t\right)$. What is the power over one wavelength?

76. Ultrasound of intensity $1.50 \times 10^2\ \text{W/m}^2$ is produced by the rectangular head of a medical imaging device measuring 3.00 cm by 5.00 cm. What is its power output?

77. The low-frequency speaker of a stereo set has a surface area of $A = 0.05\ \text{m}^2$ and produces 1 W of acoustical power. (a) What is the intensity at the speaker? (b) If the speaker projects sound uniformly in all directions, at what distance from the speaker is the intensity $0.1\ \text{W/m}^2$?

78. To increase the intensity of a wave by a factor of 50, by what factor should the amplitude be increased?

79. A device called an insolation meter is used to measure the intensity of sunlight. It has an area of $100\ \text{cm}^2$ and registers 6.50 W. What is the intensity in W/m^2?

80. Energy from the Sun arrives at the top of Earth's atmosphere with an intensity of $1400\ \text{W/m}^2$. How long does it take for $1.80 \times 10^9\ \text{J}$ to arrive on an area of $1.00\ \text{m}^2$?

81. Suppose you have a device that extracts energy from ocean breakers in direct proportion to their intensity. If the device produces 10.0 kW of power on a day when the breakers are 1.20 m high, how much will it produce when they are 0.600 m high?

82. A photovoltaic array of (solar cells) is 10.0% efficient in gathering solar energy and converting it to electricity. If the average intensity of sunlight on one day is $70.00\ \mathrm{W/m^2}$, what area should your array have to gather energy at the rate of 100 W? (b) What is the maximum cost of the array if it must pay for itself in two years of operation averaging 10.0 hours per day? Assume that it earns money at the rate of 9.00 cents per kilowatt-hour.

83. A microphone receiving a pure sound tone feeds an oscilloscope, producing a wave on its screen. If the sound intensity is originally $2.00 \times 10^{-5}\ \mathrm{W/m^2}$, but is turned up until the amplitude increases by 30.0%, what is the new intensity?

84. A string with a mass of 0.30 kg has a length of 4.00 m. If the tension in the string is 50.00 N, and a sinusoidal wave with an amplitude of 2.00 cm is induced on the string, what must the frequency be for an average power of 100.00 W?

85. The power versus time for a point on a string $(\mu = 0.05\ \mathrm{kg/m})$ in which a sinusoidal traveling wave is induced. The wave is modeled with the wave equation $y(x, t) = A \sin\left(20.93\ \mathrm{m^{-1}}x - \omega t\right)$. What is the frequency and amplitude of the wave?

86. A string is under tension F_{T1}. Energy is transmitted by a wave on the string at rate P_1 by a wave of frequency f_1. What is the ratio of the new energy transmission rate P_2 to P_1 if the tension is doubled?

87. A 250-Hz tuning fork is struck and the intensity at the source is I_1 at a distance of one meter from the source. (a) What is the intensity at a distance of 4.00 m from the source? (b) How far from the tuning fork is the intensity a tenth of the intensity at the source?

88. A sound speaker is rated at a voltage of $V = 120.00\ \mathrm{V}$ and a current of $I = 10.00\ \mathrm{A}$. Electrical power consumption is $P = IV$. To test the speaker, a signal of a sine wave is applied to the speaker. Assuming that the sound wave moves as a spherical wave and that all of the energy applied to the speaker is converted to sound energy, how far from the speaker is the intensity equal to $3.82\ \mathrm{W/m^2}$?

89. The energy of a ripple on a pond is proportional to the amplitude squared. If the amplitude of the ripple is 0.1 cm at a distance from the source of 6.00 meters, what was the amplitude at a distance of 2.00 meters from the source?

16.5 Interference of Waves

90. Consider two sinusoidal waves traveling along a string, modeled as $y_1(x, t) = 0.3\ \mathrm{m} \sin\left(4\ \mathrm{m^{-1}}x + 3\ \mathrm{s^{-1}}t\right)$ and $y_2(x, t) = 0.6\ \mathrm{m} \sin\left(8\ \mathrm{m^{-1}}x - 6\ \mathrm{s^{-1}}t\right)$. What is the height of the resultant wave formed by the interference of the two waves at the position $x = 0.5\ \mathrm{m}$ at time $t = 0.2\ \mathrm{s}$?

91. Consider two sinusoidal sine waves traveling along a string, modeled as $y_1(x, t) = 0.3\ \mathrm{m} \sin\left(4\ \mathrm{m^{-1}}x + 3\ \mathrm{s^{-1}}t + \frac{\pi}{3}\right)$ and $y_2(x, t) = 0.6\ \mathrm{m} \sin\left(8\ \mathrm{m^{-1}}x - 6\ \mathrm{s^{-1}}t\right)$. What is the height of the resultant wave formed by the interference of the two waves at the position $x = 1.0\ \mathrm{m}$ at time $t = 3.0\ \mathrm{s}$?

92. Consider two sinusoidal sine waves traveling along a string, modeled as $y_1(x, t) = 0.3\ \mathrm{m} \sin\left(4\ \mathrm{m^{-1}}x - 3\ \mathrm{s^{-1}}t\right)$ and $y_2(x, t) = 0.3\ \mathrm{m} \sin\left(4\ \mathrm{m^{-1}}x + 3\ \mathrm{s^{-1}}t\right)$. What is the wave function of the resulting wave? [*Hint:* Use the trig identity $\sin(u \pm v) = \sin u \cos v \pm \cos u \sin v$

93. Two sinusoidal waves are moving through a medium in the same direction, both having amplitudes of 3.00 cm, a wavelength of 5.20 m, and a period of 6.52 s, but one has a phase shift of an angle ϕ. What is the phase shift if the resultant wave has an amplitude of 5.00 cm? [*Hint:* Use the trig identity $\sin u + \sin v = 2 \sin\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right)$

94. Two sinusoidal waves are moving through a medium in the positive *x*-direction, both having amplitudes of 6.00 cm, a wavelength of 4.3 m, and a period of 6.00 s, but one has a phase shift of an angle $\phi = 0.50\ \mathrm{rad}$. What is the height of the resultant wave at a time $t = 3.15\ \mathrm{s}$ and a position $x = 0.45\ \mathrm{m}$?

95. Two sinusoidal waves are moving through a medium in the positive *x*-direction, both having amplitudes of 7.00 cm, a wave number of $k = 3.00\ \text{m}^{-1}$, an angular frequency of $\omega = 2.50\ \text{s}^{-1}$, and a period of 6.00 s, but one has a phase shift of an angle $\phi = \frac{\pi}{12}$ rad. What is the height of the resultant wave at a time $t = 2.00$ s and a position $x = 0.53$ m?

96. Consider two waves $y_1(x, t)$ and $y_2(x, t)$ that are identical except for a phase shift propagating in the same medium. (a)What is the phase shift, in radians, if the amplitude of the resulting wave is 1.75 times the amplitude of the individual waves? (b) What is the phase shift in degrees? (c) What is the phase shift as a percentage of the individual wavelength?

97. Two sinusoidal waves, which are identical except for a phase shift, travel along in the same direction. The wave equation of the resultant wave is $y_R(x, t) = 0.70\ \text{m}\sin\left(3.00\ \text{m}^{-1}x - 6.28\ \text{s}^{-1}t + \pi/16\ \text{rad}\right)$. What are the angular frequency, wave number, amplitude, and phase shift of the individual waves?

98. Two sinusoidal waves, which are identical except for a phase shift, travel along in the same direction. The wave equation of the resultant wave is $y_R(x, t) = 0.35\ \text{cm}\sin\left(6.28\ \text{m}^{-1}x - 1.57\ \text{s}^{-1}t + \frac{\pi}{4}\right)$. What are the period, wavelength, amplitude, and phase shift of the individual waves?

99. Consider two wave functions, $y_1(x, t) = 4.00\ \text{m}\sin\left(\pi\ \text{m}^{-1}x - \pi\ \text{s}^{-1}t\right)$ and $y_2(x, t) = 4.00\ \text{m}\sin\left(\pi\ \text{m}^{-1}x - \pi\ \text{s}^{-1}t + \frac{\pi}{3}\right)$. (a) Using a spreadsheet, plot the two wave functions and the wave that results from the superposition of the two wave functions as a function of position $(0.00 \le x \le 6.00\ \text{m})$ for the time $t = 0.00$ s. (b) What are the wavelength and amplitude of the two original waves? (c) What are the wavelength and amplitude of the resulting wave?

100. Consider two wave functions, $y_2(x, t) = 2.00\ \text{m}\sin\left(\frac{\pi}{2}\text{m}^{-1}x - \frac{\pi}{3}\text{s}^{-1}t\right)$ and $y_2(x, t) = 2.00\ \text{m}\sin\left(\frac{\pi}{2}\text{m}^{-1}x - \frac{\pi}{3}\text{s}^{-1}t + \frac{\pi}{6}\right)$. (a) Verify that $y_R = 2A\cos\left(\frac{\phi}{2}\right)\sin\left(kx - \omega t + \frac{\phi}{2}\right)$ is the solution for the wave that results from a superposition of the two waves. Make a column for x, y_1, y_2, $y_1 + y_2$, and $y_R = 2A\cos\left(\frac{\phi}{2}\right)\sin\left(kx - \omega t + \frac{\phi}{2}\right)$. Plot four waves as a function of position where the range of x is from 0 to 12 m.

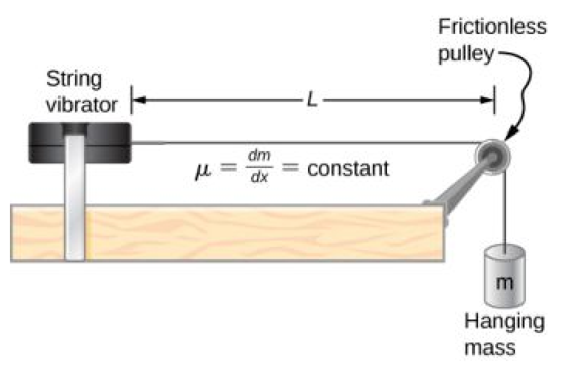
101. Consider two wave functions that differ only by a phase shift, $y_1(x, t) = A\cos(kx - \omega t)$ and $y_2(x, t) = A\cos(kx - \omega t + \phi)$. Use the trigonometric identities $\cos u + \cos v = 2\cos\left(\frac{u-v}{2}\right)\cos\left(\frac{u+v}{2}\right)$ and $\cos(-\theta) = \cos(\theta)$ to find a wave equation for the wave resulting from the superposition of the two waves. Does the resulting wave function come as a surprise to you?

16.6 Standing Waves and Resonance

102. A wave traveling on a Slinky® that is stretched to 4 m takes 2.4 s to travel the length of the Slinky and back again. (a) What is the speed of the wave? (b) Using the same Slinky stretched to the same length, a standing wave is created which consists of three antinodes and four nodes. At what frequency must the Slinky be oscillating?

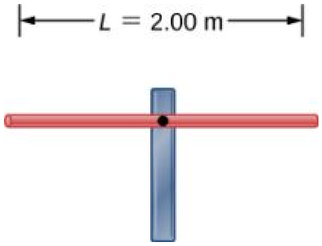
103. A 2-m long string is stretched between two supports with a tension that produces a wave speed equal to $v_w = 50.00$ m/s. What are the wavelength and frequency of the first three modes that resonate on the string?

104. Consider the experimental setup shown below. The length of the string between the string vibrator and the pulley is $L = 1.20$ m. The linear density of the string is $\mu = 0.006$ kg/m. The string vibrator can oscillate at any frequency. The hanging mass is 2.00 kg. (a)What are the wavelength and frequency of $n = 6$ mode? (b) The string oscillates the air around the string. What is the wavelength of the sound if the speed of the sound is $v_s = 343.00$ m/s?



105. A cable with a linear density of $\mu = 0.02$ kg/m is hung from telephone poles. The tension in the cable is 500.00 N. The distance between poles is 20 meters. The wind blows across the line, causing the cable resonate. A standing waves pattern is produced that has 4.5 wavelengths between the two poles. The speed of sound at the current temperature $T = 20°\text{C}$ is 343.00 m/s. What are the frequency and wavelength of the hum?

106. Consider a rod of length *L*, mounted in the center to a support. A node must exist where the rod is mounted on a support, as shown below. Draw the first two normal modes of the rod as it is driven into resonance. Label the wavelength and the frequency required to drive the rod into resonance.



107. Consider two wave functions $y(x, t) = 0.30\text{ cm}\sin\left(3\text{ m}^{-1}x - 4\text{ s}^{-1}t\right)$ and $y(x, t) = 0.30\text{ cm}\sin\left(3\text{ m}^{-1}x + 4\text{ s}^{-1}t\right)$. Write a wave function for the resulting standing wave.

108. A 2.40-m wire has a mass of 7.50 g and is under a tension of 160 N. The wire is held rigidly at both ends and set into oscillation. (a) What is the speed of waves on the wire? The string is driven into resonance by a frequency that produces a standing wave with a wavelength equal to 1.20 m. (b) What is the frequency used to drive the string into resonance?

109. A string with a linear mass density of 0.0062 kg/m and a length of 3.00 m is set into the $n = 100$ mode of resonance. The tension in the string is 20.00 N. What is the wavelength and frequency of the wave?

110. A string with a linear mass density of 0.0075 kg/m and a length of 6.00 m is set into the $n = 4$ mode of resonance by driving with a frequency of 100.00 Hz. What is the tension in the string?

111. Two sinusoidal waves with identical wavelengths and amplitudes travel in opposite directions along a string producing a standing wave. The linear mass density of the string is $\mu = 0.075\text{ kg/m}$ and the tension in the string is $F_T = 5.00\text{ N}$. The time interval between instances of total destructive interference is $\Delta t = 0.13\text{ s}$. What is the wavelength of the waves?

112. A string, fixed on both ends, is 5.00 m long and has a mass of 0.15 kg. The tension if the string is 90 N. The string is vibrating to produce a standing wave at the fundamental frequency of the string. (a) What is the speed of the waves on the string? (b) What is the wavelength of the standing wave produced? (c) What is the period of the standing wave?

113. A string is fixed at both end. The mass of the string is 0.0090 kg and the length is 3.00 m. The string is under a tension of 200.00 N. The string is driven by a variable frequency source to produce standing waves on the string. Find the wavelengths and frequency of the first four modes of standing waves.

114. The frequencies of two successive modes of standing waves on a string are 258.36 Hz and 301.42 Hz. What is the next frequency above 100.00 Hz that would produce a standing wave?

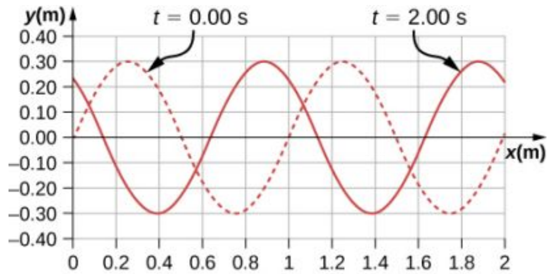
115. A string is fixed at both ends to supports 3.50 m apart and has a linear mass density of $\mu = 0.005\text{ kg/m}$. The string is under a tension of 90.00 N. A standing wave is produced on the string with six nodes and five antinodes. What are the wave speed, wavelength, frequency, and period of the standing wave?

116. Sine waves are sent down a 1.5-m-long string fixed at both ends. The waves reflect back in the opposite direction. The amplitude of the wave is 4.00 cm. The propagation velocity of the waves is 175 m/s. The $n = 6$ resonance mode of the string is produced. Write an equation for the resulting standing wave.

Additional Problems

117. Ultrasound equipment used in the medical profession uses sound waves of a frequency above the range of human hearing. If the frequency of the sound produced by the ultrasound machine is $f = 30\text{ kHz}$, what is the wavelength of the ultrasound in bone, if the speed of sound in bone is $v = 3000\text{ m/s}$?

118. Shown below is the plot of a wave function that models a wave at time $t = 0.00\text{ s}$ and $t = 2.00\text{ s}$. The dotted line is the wave function at time $t = 0.00\text{ s}$ and the solid line is the function at time $t = 2.00\text{ s}$. Estimate the amplitude, wavelength, velocity, and period of the wave.



119. The speed of light in air is approximately $v = 3.00 \times 10^8$ m/s and the speed of light in glass is $v = 2.00 \times 10^8$ m/s. A red laser with a wavelength of $\lambda = 633.00$ nm shines light incident of the glass, and some of the red light is transmitted to the glass. The frequency of the light is the same for the air and the glass. (a) What is the frequency of the light? (b) What is the wavelength of the light in the glass?

120. A radio station broadcasts radio waves at a frequency of 101.7 MHz. The radio waves move through the air at approximately the speed of light in a vacuum. What is the wavelength of the radio waves?

121. A sunbather stands waist deep in the ocean and observes that six crests of periodic surface waves pass each minute. The crests are 16.00 meters apart. What is the wavelength, frequency, period, and speed of the waves?

122. A tuning fork vibrates producing sound at a frequency of 512 Hz. The speed of sound of sound in air is $v = 343.00$ m/s if the air is at a temperature of $20.00°\text{C}$. What is the wavelength of the sound?

123. A motorboat is traveling across a lake at a speed of $v_b = 15.00$ m/s. The boat bounces up and down every 0.50 s as it travels in the same direction as a wave. It bounces up and down every 0.30 s as it travels in a direction opposite the direction of the waves. What is the speed and wavelength of the wave?

124. Use the linear wave equation to show that the wave speed of a wave modeled with the wave function $y(x,t) = 0.20\text{ m}\sin\left(3.00\text{ m}^{-1}x + 6.00\text{ s}^{-1}t\right)$ is $v = 2.00$ m/s. What are the wavelength and the speed of the wave?

125. Given the wave functions $y_1(x,t) = A\sin(kx - \omega t)$ and $y_2(x,t) = A\sin(kx - \omega t + \phi)$ with $\phi \neq \frac{\pi}{2}$, show that $y_1(x,t) + y_2(x,t)$ is a solution to the linear wave equation with a wave velocity of $v = \frac{\omega}{k}$.

126. A transverse wave on a string is modeled with the wave function $y(x,t) = 0.10\text{ m}\sin\left(0.15\text{ m}^{-1}x + 1.50\text{ s}^{-1}t + 0.20\right)$. (a) Find the wave velocity. (b) Find the position in the *y*-direction, the velocity perpendicular to the motion of the wave, and the acceleration perpendicular to the motion of the wave, of a small segment of the string centered at $x = 0.40$ m at time $t = 5.00$ s.

127. A sinusoidal wave travels down a taut, horizontal string with a linear mass density of $\mu = 0.060$ kg/m. The magnitude of maximum vertical acceleration of the wave is $a_{y\text{ max}} = 0.90\text{ m/s}^2$ and the amplitude of the wave is 0.40 m. The string is under a tension of $F_T = 600.00$ N. The wave moves in the negative *x*-direction. Write an equation to model the wave.

128. A transverse wave on a string ($\mu = 0.0030$ kg/m) is described with the equation $y(x,t) = 0.30\text{ m}\sin\left(\frac{2\pi}{4.00\text{ m}}\left(x - 16.00\frac{\text{m}}{\text{s}}t\right)\right)$. What is the tension under which the string is held taut?

129. A transverse wave on a horizontal string ($\mu = 0.0060$ kg/m) is described with the equation $y(x,t) = 0.30\text{ m}\sin\left(\frac{2\pi}{4.00\text{ m}}(x - v_w t)\right)$. The string is under a tension of 300.00 N. What are the wave speed, wave number, and angular frequency of the wave?

130. A student holds an inexpensive sonic range finder and uses the range finder to find the distance to the wall. The sonic range finder emits a sound wave. The sound wave reflects off the wall and returns to the range finder. The round trip takes 0.012 s. The range finder was calibrated for use at room temperature $T = 20°\text{C}$, but the temperature in the room is actually $T = 23°\text{C}$. Assuming that the timing mechanism is perfect, what percentage of error can the student expect due to the calibration?

131. A wave on a string is driven by a string vibrator, which oscillates at a frequency of 100.00 Hz and an amplitude of 1.00 cm. The string vibrator operates at a voltage of 12.00 V and a current of 0.20 A. The power consumed by the string vibrator is $P = IV$. Assume that the string vibrator is 90% efficient at converting electrical energy into the energy associated with the vibrations of the string. The string is 3.00 m long, and is under a tension of 60.00 N. What is the linear mass density of the string?

132. A traveling wave on a string is modeled by the wave equation $y(x, t) = 3.00\,\text{cm}\sin\left(8.00\,\text{m}^{-1}x + 100.00\,\text{s}^{-1}t\right)$. The string has a linear mass density of $\mu = 0.00800\,\text{kg/m}$. What is the average power transferred by the wave on the string?

133. A transverse wave on a string has a wavelength of 5.0 m, a period of 0.02 s, and an amplitude of 1.5 cm. The average power transferred by the wave is 5.00 W. What is the tension in the string?

134. (a) What is the intensity of a laser beam used to burn away cancerous tissue that, when 90.0% absorbed, puts 500.0 J of energy into a circular spot 2.00 mm in diameter in 4.00 s? (b) Discuss how this intensity compares to the average intensity of sunlight (about 1kW/m^2) and the implications that would have if the laser beam entered your eye. Note how your answer depends on the time duration of the exposure.

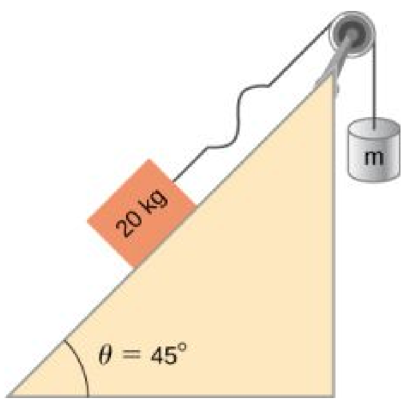
135. Consider two periodic wave functions, $y_1(x, t) = A\sin(kx - \omega t)$ and $y_2(x, t) = A\sin(kx - \omega t + \phi)$. (a) For what values of ϕ will the wave that results from a superposition of the wave functions have an amplitude of 2*A*? (b) For what values of ϕ will the wave that results from a superposition of the wave functions have an amplitude of zero?

136. Consider two periodic wave functions, $y_1(x, t) = A\sin(kx - \omega t)$ and $y_2(x, t) = A\cos(kx - \omega t + \phi)$. (a) For what values of ϕ will the wave that results from a superposition of the wave functions have an amplitude of 2*A*? (b) For what values of ϕ will the wave that results from a superposition of the wave functions have an amplitude of zero?

137. A trough with dimensions 10.00 meters by 0.10 meters by 0.10 meters is partially filled with water. Small-amplitude surface water waves are produced from both ends of the trough by paddles oscillating in simple harmonic motion. The height of the water waves are modeled with two sinusoidal wave equations, $y_1(x, t) = 0.3\,\text{m}\sin\left(4\,\text{m}^{-1}x - 3\,\text{s}^{-1}t\right)$ and $y_2(x, t) = 0.3\,\text{m}\cos\left(4\,\text{m}^{-1}x + 3\,\text{s}^{-1}t - \frac{\pi}{2}\right)$. What is the wave function of the resulting wave after the waves reach one another and before they reach the end of the trough (i.e., assume that there are only two waves in the trough and ignore reflections)? Use a spreadsheet to check your results. (*Hint:* Use the trig identities $\sin(u \pm v) = \sin u\cos v \pm \cos u\sin v$ and $\cos(u \pm v) = \cos u\cos v \mp \sin u\sin v$)

138. A seismograph records the S- and P-waves from an earthquake 20.00 s apart. If they traveled the same path at constant wave speeds of $v_S = 4.00\,\text{km/s}$ and $v_P = 7.50\,\text{km/s}$, how far away is the epicenter of the earthquake?

139. Consider what is shown below. A 20.00-kg mass rests on a frictionless ramp inclined at $45°$. A string with a linear mass density of $\mu = 0.025\,\text{kg/m}$ is attached to the 20.00-kg mass. The string passes over a frictionless pulley of negligible mass and is attached to a hanging mass (*m*). The system is in static equilibrium. A wave is induced on the string and travels up the ramp. (a) What is the mass of the hanging mass (*m*)? (b) At what wave speed does the wave travel up the string?



140. Consider the superposition of three wave functions $y(x, t) = 3.00\,\text{cm}\sin\left(2\,\text{m}^{-1}x - 3\,\text{s}^{-1}t\right)$, $y(x, t) = 3.00\,\text{cm}\sin\left(6\,\text{m}^{-1}x + 3\,\text{s}^{-1}t\right)$, and $y(x, t) = 3.00\,\text{cm}\sin\left(2\,\text{m}^{-1}x - 4\,\text{s}^{-1}t\right)$. What is the height of the resulting wave at position $x = 3.00\,\text{m}$ at time $t = 10.0\,\text{s}$?

141. A string has a mass of 150 g and a length of 3.4 m. One end of the string is fixed to a lab stand and the other is attached to a spring with a spring constant of $k_s = 100\,\text{N/m}$. The free end of the spring is attached to another lab pole. The tension in the string is maintained by the spring. The lab poles are separated by a distance that stretches the spring 2.00 cm. The string is plucked and a pulse travels along the string. What is the propagation speed of the pulse?

142. A standing wave is produced on a string under a tension of 70.0 N by two sinusoidal transverse waves that are identical, but moving in opposite directions. The string is fixed at $x = 0.00\text{ m}$ and $x = 10.00\text{ m}$. Nodes appear at $x = 0.00\text{ m}$, 2.00 m, 4.00 m, 6.00 m, 8.00 m, and 10.00 m. The amplitude of the standing wave is 3.00 cm. It takes 0.10 s for the antinodes to make one complete oscillation. (a) What are the wave functions of the two sine waves that produce the standing wave? (b) What are the maximum velocity and acceleration of the string, perpendicular to the direction of motion of the transverse waves, at the antinodes?

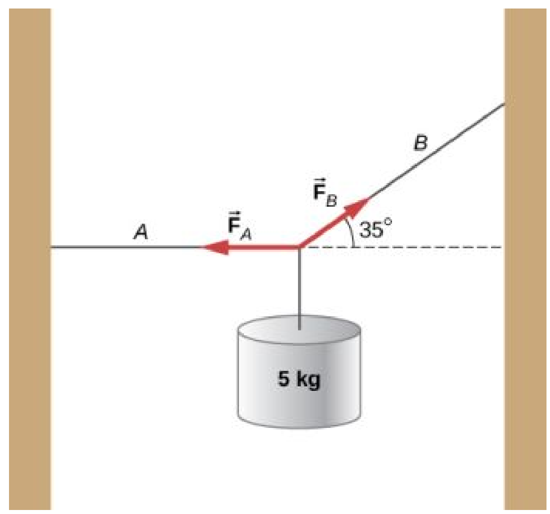
143. A string with a length of 4 m is held under a constant tension. The string has a linear mass density of $\mu = 0.006\text{ kg/m}$. Two resonant frequencies of the string are 400 Hz and 480 Hz. There are no resonant frequencies between the two frequencies. (a) What are the wavelengths of the two resonant modes? (b) What is the tension in the string?

Challenge Problems

144. A copper wire has a radius of $200\ \mu\text{m}$ and a length of 5.0 m. The wire is placed under a tension of 3000 N and the wire stretches by a small amount. The wire is plucked and a pulse travels down the wire. What is the propagation speed of the pulse? (Assume the temperature does not change: $\left(\rho = 8.96 \frac{\text{g}}{\text{cm}^3}, Y = 1.1\, \times\, 10^{11}\, \frac{\text{N}}{\text{m}}\right).$)

145. A pulse moving along the *x* axis can be modeled as the wave function $y\,(x, t) = 4.00\text{ m}e^{-\left(\frac{x + (2.00\text{ m/s})t}{1.00\text{ m}}\right)^2}$. (a)What are the direction and propagation speed of the pulse? (b) How far has the wave moved in 3.00 s? (c) Plot the pulse using a spreadsheet at time $t = 0.00\text{ s}$ and $t = 3.00\text{ s}$ to verify your answer in part (b).

146. A string with a linear mass density of $\mu = 0.0085\text{ kg/m}$ is fixed at both ends. A 5.0-kg mass is hung from the string, as shown below. If a pulse is sent along section *A*, what is the wave speed in section *A* and the wave speed in section *B*?



147. Consider two wave functions $y_1\,(x, t) = A\sin\,(kx - \omega t)$ and $y_2\,(x, t) = A\sin\,(kx + \omega t + \phi)$. What is the wave function resulting from the interference of the two wave? (*Hint:* $\sin\,(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$ and $\phi = \frac{\phi}{2} + \frac{\phi}{2}$.)

148. The wave function that models a standing wave is given as $y_R\,(x, t) = 6.00\text{ cm}\sin\left(3.00\text{ m}^{-1}x + 1.20\text{ rad}\right)$ $\cos\left(6.00\text{ s}^{-1}t + 1.20\text{ rad}\right)$. What are two wave functions that interfere to form this wave function? Plot the two wave functions and the sum of the sum of the two wave functions at $t = 1.00\text{ s}$ to verify your answer.

149. Consider two wave functions $y_1\,(x, t) = A\sin\,(kx - \omega t)$ and $y_2\,(x, t) = A\sin\,(kx + \omega t + \phi)$. The resultant wave form when you add the two functions is

$y_R = 2A \sin\left(kx + \frac{\phi}{2}\right) \cos\left(\omega t + \frac{\phi}{2}\right).$

Consider the case where $A = 0.03 \text{ m}^{-1}$, $k = 1.26 \text{ m}^{-1}$, $\omega = \pi \text{ s}^{-1}$, and $\phi = \frac{\pi}{10}$. (a) Where are the first three nodes of the standing wave function starting at zero and moving in the positive *x* direction? (b) Using a spreadsheet, plot the two wave functions and the resulting function at time $t = 1.00 \text{ s}$ to verify your answer.

CHAPTER 17
Sound

FIGURE 17.1 Hearing is an important human sense that can detect frequencies of sound, ranging between 20 Hz and 20 kHz. However, other species have very different ranges of hearing. Bats, for example, emit clicks in ultrasound, using frequencies beyond 20 kHz. They can detect nearby insects by hearing the echo of these ultrasonic clicks. Ultrasound is important in several human applications, including probing the interior structures of human bodies, Earth, and the Sun. Ultrasound is also useful in industry for nondestructive testing. (credit: modification of work by Derek Keats/flickr)

CHAPTER OUTLINE

INTRODUCTION Sound is an example of a mechanical wave, specifically, a pressure wave: Sound waves travel through the air and other media as oscillations of molecules. Normal human hearing encompasses an impressive range of frequencies from 20 Hz to 20 kHz. Sounds below 20 Hz are called infrasound, whereas those above 20 kHz are called ultrasound. Some animals, like the bat shown in Figure 17.1, can hear sounds in the ultrasonic range.

Many of the concepts covered in Waves also have applications in the study of sound. For example, when a sound wave encounters an interface between two media with different wave speeds, reflection and transmission of the wave occur.

Ultrasound has many uses in science, engineering, and medicine. Ultrasound is used for nondestructive testing in engineering, such as testing the thickness of coating on metal. In medicine, sound waves are far less destructive than X-rays and can be used to image the fetus in a mother's womb without danger to the fetus or the mother. Later in this chapter, we discuss the Doppler effect, which can be used to determine the velocity of blood in the arteries or wind speed in weather systems.

17.1 Sound Waves

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the difference between sound and hearing
- Describe sound as a wave
- List the equations used to model sound waves
- Describe compression and rarefactions as they relate to sound

The physical phenomenon of **sound** is a disturbance of matter that is transmitted from its source outward. **Hearing** is the perception of sound, just as seeing is the perception of visible light. On the atomic scale, sound is a disturbance of atoms that is far more ordered than their thermal motions. In many instances, sound is a periodic wave, and the atoms undergo simple harmonic motion. Thus, sound waves can induce oscillations and resonance effects (Figure 17.2).

FIGURE 17.2 This glass has been shattered by a high-intensity sound wave of the same frequency as the resonant frequency of the glass. (credit: "||read||"/Flickr)

INTERACTIVE

This video (https://openstax.org/l/21waveswineglas) shows waves on the surface of a wine glass, being driven by sound waves from a speaker. As the frequency of the sound wave approaches the resonant frequency of the wine glass, the amplitude and frequency of the waves on the wine glass increase. When the resonant frequency is reached, the glass shatters.

A speaker produces a sound wave by oscillating a cone, causing vibrations of air molecules. In Figure 17.3, a speaker vibrates at a constant frequency and amplitude, producing vibrations in the surrounding air molecules. As the speaker oscillates back and forth, it transfers energy to the air, mostly as thermal energy. But a small part of the speaker's energy goes into compressing and expanding the surrounding air, creating slightly higher and lower local pressures. These compressions (high-pressure regions) and rarefactions (low-pressure regions) move out as longitudinal pressure waves having the same frequency as the speaker—they are the disturbance that is a sound wave. (Sound waves in air and most fluids are longitudinal, because fluids have almost no shear strength. In solids, sound waves can be both transverse and longitudinal.)

Figure 17.3(a) shows the compressions and rarefactions, and also shows a graph of gauge pressure versus distance from a speaker. As the speaker moves in the positive *x*-direction, it pushes air molecules, displacing them from their equilibrium positions. As the speaker moves in the negative *x*-direction, the air molecules move back toward their equilibrium positions due to a restoring force. The air molecules oscillate in simple harmonic motion about their equilibrium positions, as shown in part (b). Note that sound waves in air are longitudinal, and in the figure, the wave propagates in the positive *x*-direction and the molecules oscillate parallel to the direction in which the wave propagates.

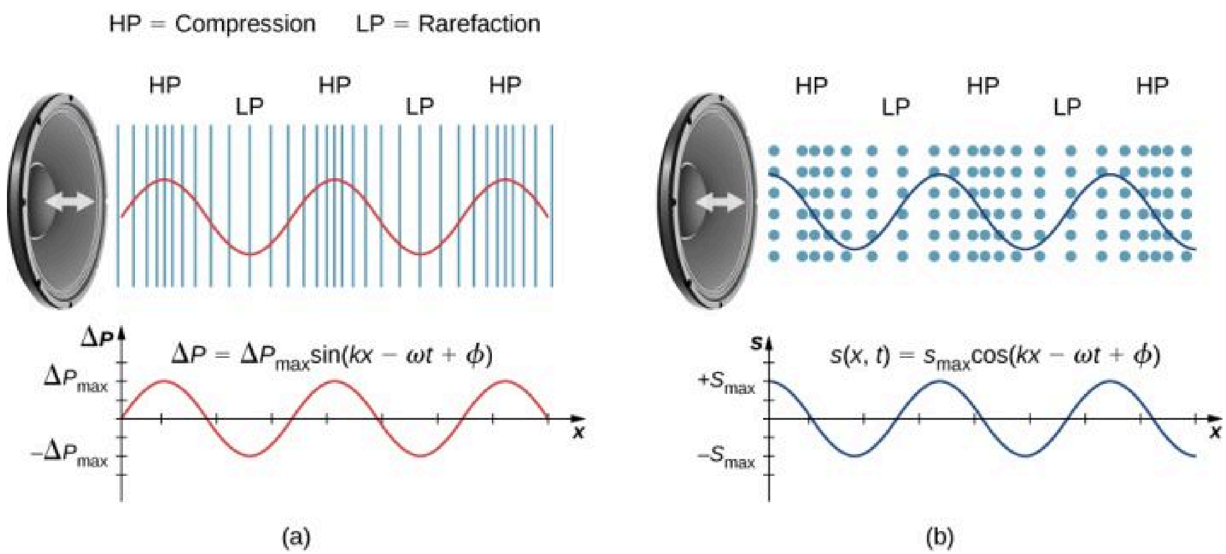


FIGURE 17.3 (a) A vibrating cone of a speaker, moving in the positive *x*-direction, compresses the air in front of it and expands the air behind it. As the speaker oscillates, it creates another compression and rarefaction as those on the right move away from the speaker. After many vibrations, a series of compressions and rarefactions moves out from the speaker as a sound wave. The red graph shows the gauge pressure of the air versus the distance from the speaker. Pressures vary only slightly from atmospheric pressure for ordinary sounds. Note that gauge pressure is modeled with a sine function, where the crests of the function line up with the compressions and the troughs line up with the rarefactions. (b) Sound waves can also be modeled using the displacement of the air molecules. The blue graph shows the displacement of the air molecules versus the position from the speaker and is modeled with a cosine function. Notice that the displacement is zero for the molecules in their equilibrium position and are centered at the compressions and rarefactions. Compressions are formed when molecules on either side of the equilibrium molecules are displaced toward the equilibrium position. Rarefactions are formed when the molecules are displaced away from the equilibrium position.

MODELS DESCRIBING SOUND

Sound can be modeled as a pressure wave by considering the change in pressure from average pressure,

$$\Delta P = \Delta P_{\max} \sin (kx \mp \omega t + \phi) . \tag{17.1}$$

This equation is similar to the periodic wave equations seen in Waves, where ΔP is the change in pressure, $\Delta P_{\max}$ is the maximum change in pressure, $k = \frac{2\pi}{\lambda}$ is the wave number, $\omega = \frac{2\pi}{T} = 2\pi f$ is the angular frequency, and ϕ is the initial phase. The wave speed can be determined from $v = \frac{\omega}{k} = \frac{\lambda}{T}$. Sound waves can also be modeled in terms of the displacement of the air molecules. The displacement of the air molecules can be modeled using a cosine function:

$$s (x, t) = s_{\max} \cos (kx \mp \omega t + \phi) . \tag{17.2}$$

In this equation, s is the displacement and s_{max} is the maximum displacement.

Not shown in the figure is the amplitude of a sound wave as it decreases with distance from its source, because the energy of the wave is spread over a larger and larger area. The intensity decreases as it moves away from the speaker, as discussed in Waves. The energy is also absorbed by objects and converted into thermal energy by the viscosity of the air. In addition, during each compression, a little heat transfers to the air; during each rarefaction, even less heat transfers from the air, and these heat transfers reduce the organized disturbance into random thermal motions. Whether the heat transfer from compression to rarefaction is significant depends on how far apart they are—that is, it depends on wavelength. Wavelength, frequency, amplitude, and speed of propagation are important characteristics for sound, as they are for all waves.

17.2 Speed of Sound

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the relationship between wavelength and frequency of sound
- Determine the speed of sound in different media
- Derive the equation for the speed of sound in air
- Determine the speed of sound in air for a given temperature

Sound, like all waves, travels at a certain speed and has the properties of frequency and wavelength. You can observe direct evidence of the speed of sound while watching a fireworks display (Figure 17.4). You see the flash of an explosion well before you hear its sound and possibly feel the pressure wave, implying both that sound travels at a finite speed and that it is much slower than light.

FIGURE 17.4 When a firework shell explodes, we perceive the light energy before the sound energy because sound travels more slowly than light does.

The difference between the speed of light and the speed of sound can also be experienced during an electrical storm. The flash of lighting is often seen before the clap of thunder. You may have heard that if you count the number of seconds between the flash and the sound, you can estimate the distance to the source. Every five seconds converts to about one mile. The velocity of any wave is related to its frequency and wavelength by

$$v = f\lambda, \tag{17.3}$$

where v is the speed of the wave, f is its frequency, and λ is its wavelength. Recall from Waves that the wavelength is the length of the wave as measured between sequential identical points. For example, for a surface water wave or sinusoidal wave on a string, the wavelength can be measured between any two convenient sequential points with the same height and slope, such as between two sequential crests or two sequential troughs. Similarly, the wavelength of a sound wave is the distance between sequential identical parts of a wave—for example, between sequential compressions (Figure 17.5). The frequency is the same as that of the source and is the number of waves that pass a point per unit time.

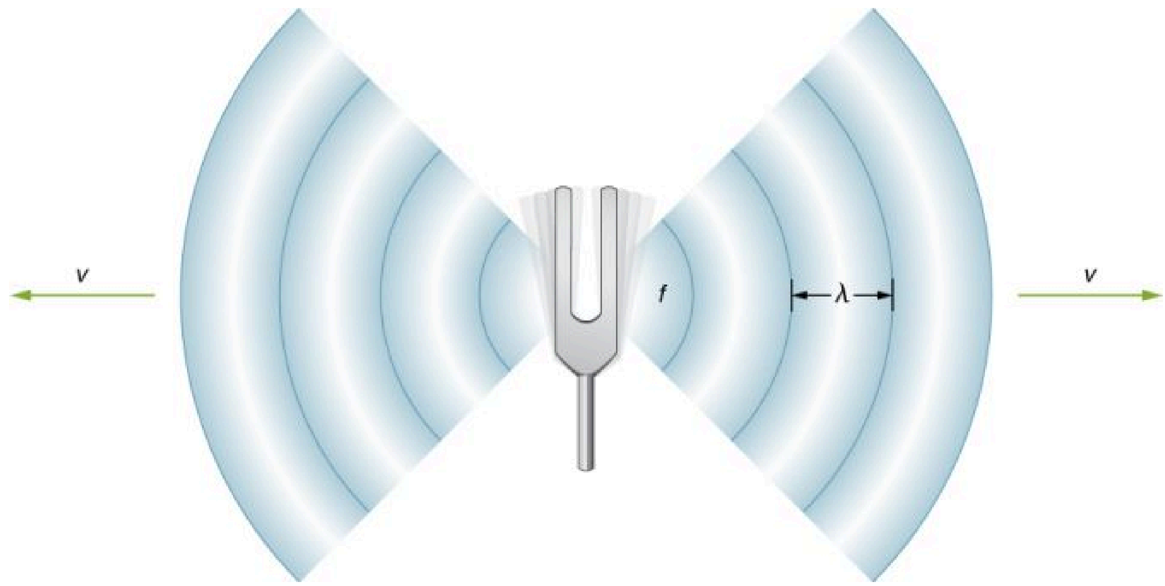


FIGURE 17.5 A sound wave emanates from a source, such as a tuning fork, vibrating at a frequency f. It propagates at speed v and has a wavelength λ.

Speed of Sound in Various Media

Table 17.1 shows that the speed of sound varies greatly in different media. The speed of sound in a medium depends on how quickly vibrational energy can be transferred through the medium. For this reason, the derivation of the speed of sound in a medium depends on the medium and on the state of the medium. In general, the equation for the speed of a mechanical wave in a medium depends on the square root of the restoring force, or the elastic property, divided by the inertial property,

$$v = \sqrt{\frac{\text{elastic property}}{\text{inertial property}}}.$$

Also, sound waves satisfy the wave equation derived in Waves,

$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}.$$

Recall from Waves that the speed of a wave on a string is equal to $v = \sqrt{\frac{F_T}{\mu}}$, where the restoring force is the tension in the string F_T and the linear density μ is the inertial property. In a fluid, the speed of sound depends on the bulk modulus and the density,

$$v = \sqrt{\frac{B}{\rho}}. \tag{17.4}$$

The speed of sound in a solid depends on the Young's modulus of the medium and the density,

$$v = \sqrt{\frac{Y}{\rho}}. \tag{17.5}$$

In an ideal gas (see The Kinetic Theory of Gases (http://openstax.org/books/university-physics-volume-2/pages/2-introduction)), the equation for the speed of sound is

$$v = \sqrt{\frac{\gamma R T_K}{M}}, \tag{17.6}$$

where γ is the adiabatic index, $R = 8.31\ \text{J/mol} \cdot \text{K}$ is the gas constant, T_K is the absolute temperature in kelvins, and *M* is the molar mass. In general, the more rigid (or less compressible) the medium, the faster the speed of sound. This observation is analogous to the fact that the frequency of simple harmonic motion is directly proportional to the stiffness of the oscillating object as measured by *k*, the spring constant. The greater the density of a medium, the slower the speed of sound. This observation is analogous to the fact that the frequency of a simple harmonic motion is inversely proportional to *m*, the mass of the oscillating object. The speed of sound in air is low, because air is easily compressible. Because liquids and solids are relatively rigid and very difficult to compress, the speed of sound in such media is generally greater than in gases.

Medium	*v* (m/s)
Gases at $0°C$	
Air	331
Carbon dioxide	259
Oxygen	316
Helium	965
Hydrogen	1290
Liquids at $20°C$	
Ethanol	1160
Mercury	1450
Water, fresh	1480
Sea Water	1540
Human tissue	1540
Solids (longitudinal or bulk)	
Vulcanized rubber	54
Polyethylene	920
Marble	3810
Glass, Pyrex	5640

TABLE 17.1 Speed of Sound in Various Media

Medium	v (m/s)
Lead	1960
Aluminum	5120
Steel	5960

TABLE 17.1 Speed of Sound in Various Media

Because the speed of sound depends on the density of the material, and the density depends on the temperature, there is a relationship between the temperature in a given medium and the speed of sound in the medium. For air at sea level, the speed of sound is given by

$$v = 331\frac{\text{m}}{\text{s}}\sqrt{1 + \frac{T_\text{C}}{273°\text{C}}} = 331\frac{\text{m}}{\text{s}}\sqrt{\frac{T_\text{K}}{273\text{ K}}} \tag{17.7}$$

where the temperature in the first equation (denoted as T_C) is in degrees Celsius and the temperature in the second equation (denoted as T_K) is in kelvins. The speed of sound in gases is related to the average speed of particles in the gas, $v_\text{rms} = \sqrt{\frac{3k_\text{B}T}{m}}$, where k_B is the Boltzmann constant $(1.38 \times 10^{-23}\text{ J/K})$ and *m* is the mass of each (identical) particle in the gas. Note that *v* refers to the speed of the coherent propagation of a disturbance (the wave), whereas v_rms describes the speeds of particles in random directions. Thus, it is reasonable that the speed of sound in air and other gases should depend on the square root of temperature. While not negligible, this is not a strong dependence. At $0°\text{C}$, the speed of sound is 331 m/s, whereas at $20.0°\text{C}$, it is 343 m/s, less than a 4% increase. Figure 17.6 shows how a bat uses the speed of sound to sense distances.

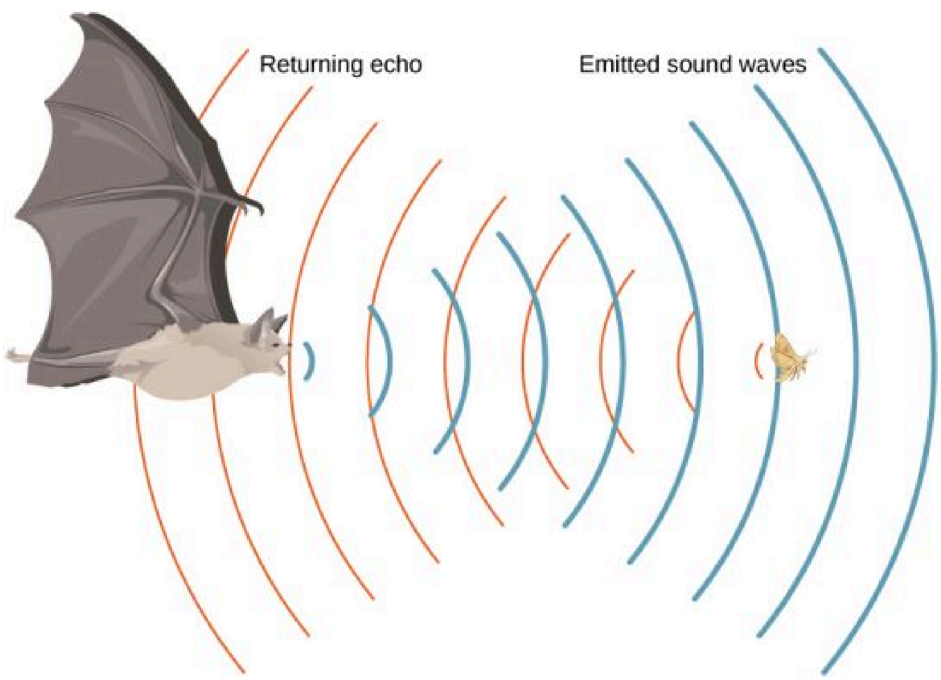


FIGURE 17.6 A bat uses sound echoes to find its way about and to catch prey. The time for the echo to return is directly proportional to the distance.

Derivation of the Speed of Sound in Air

As stated earlier, the speed of sound in a medium depends on the medium and the state of the medium. The derivation of the equation for the speed of sound in air starts with the mass flow rate and continuity equation discussed in Fluid Mechanics.

Consider fluid flow through a pipe with cross-sectional area *A* (Figure 17.7). The mass in a small volume of length *x* of the pipe is equal to the density times the volume, or $m = \rho V = \rho A x$. The mass flow rate is

$$\frac{dm}{dt} = \frac{d}{dt}(\rho V) = \frac{d}{dt}(\rho A x) = \rho A \frac{dx}{dt} = \rho A v.$$

The continuity equation from Fluid Mechanics states that the mass flow rate into a volume has to equal the mass flow rate out of the volume, $\rho_{\text{in}} A_{\text{in}} v_{\text{in}} = \rho_{\text{out}} A_{\text{out}} v_{\text{out}}$.

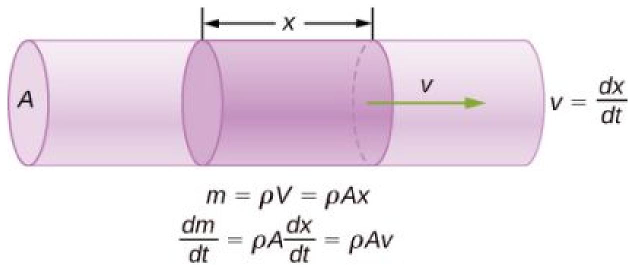


FIGURE 17.7 The mass of a fluid in a volume is equal to the density times the volume, $m = \rho V = \rho A x$. The mass flow rate is the time derivative of the mass.

Now consider a sound wave moving through a parcel of air. A parcel of air is a small volume of air with imaginary boundaries (Figure 17.8). The density, temperature, and velocity on one side of the volume of the fluid are given as ρ, T, v, and on the other side are $\rho + d\rho, T + dT, v + dv$.

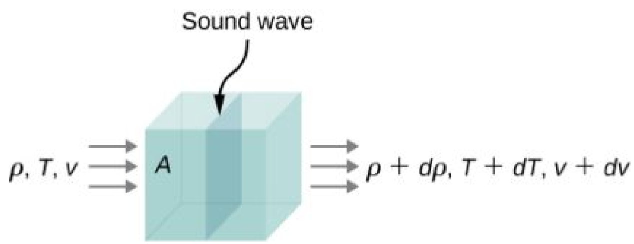


FIGURE 17.8 A sound wave moves through a volume of fluid. The density, temperature, and velocity of the fluid change from one side to the other.

The continuity equation states that the mass flow rate entering the volume is equal to the mass flow rate leaving the volume, so

$$\rho A v = (\rho + d\rho)\, A\, (v + dv)\,.$$

This equation can be simplified, noting that the area cancels and considering that the multiplication of two infinitesimals is approximately equal to zero: $d\rho\,(dv) \approx 0$,

$$\begin{aligned}
\rho v &= (\rho + d\rho)(v + dv) \\
\rho v &= \rho v + \rho\,(dv) + (d\rho)\, v + (d\rho)(dv) \\
0 &= \rho\,(dv) + (d\rho)\, v \\
\rho\, dv &= -v d\rho.
\end{aligned}$$

The net force on the volume of fluid (Figure 17.9) equals the sum of the forces on the left face and the right face:

$$\begin{aligned}
F_{\text{net}} &= p\, dy\, dz - (p + dp)\, dy\, dz \\
&= p\, dy\, dz - p dy\, dz - dp\, dy\, dz \\
&= -dp\, dy\, dz \\
ma &= -dp\, dy\, dz.
\end{aligned}$$

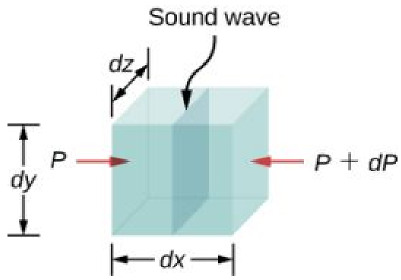


FIGURE 17.9 A sound wave moves through a volume of fluid. The force on each face can be found by the pressure times the area.

The acceleration is the force divided by the mass and the mass is equal to the density times the volume, $m = \rho V = \rho\, dx\, dy\, dz$. We have

$$\begin{aligned} ma &= -dp\, dy\, dz \\ a &= -\frac{dp\, dy\, dz}{m} = -\frac{dp\, dy\, dz}{\rho\, dx\, dy\, dz} = -\frac{dp}{(\rho\, dx)} \\ \frac{dv}{dt} &= -\frac{dp}{(\rho\, dx)} \\ dv &= -\frac{dp}{(\rho\, dx)} dt = -\frac{dp}{\rho}\frac{1}{v} \\ \rho v\, dv &= -dp. \end{aligned}$$

From the continuity equation $\rho\, dv = -v d\rho$, we obtain

$$\begin{aligned} \rho v dv &= -dp \\ (-v d\rho)\, v &= -dp \\ v &= \sqrt{\frac{dp}{d\rho}}. \end{aligned}$$

Consider a sound wave moving through air. During the process of compression and expansion of the gas, no heat is added or removed from the system. A process where heat is not added or removed from the system is known as an adiabatic system. Adiabatic processes are covered in detail in [The First Law of Thermodynamics (http://openstax.org/books/university-physics-volume-2/pages/3-introduction)](http://openstax.org/books/university-physics-volume-2/pages/3-introduction), but for now it is sufficient to say that for an adiabatic process, $pV^{\gamma} = \text{constant}$, where p is the pressure, V is the volume, and gamma (γ) is a constant that depends on the gas. For air, $\gamma = 1.40$. The density equals the number of moles times the molar mass divided by the volume, so the volume is equal to $V = \frac{nM}{\rho}$. The number of moles and the molar mass are constant and can be absorbed into the constant $p\left(\frac{1}{\rho}\right)^{\gamma} = \text{constant}$. Taking the natural logarithm of both sides yields $\ln p - \gamma \ln \rho = \text{constant}$. Differentiating with respect to the density, the equation becomes

$$\begin{aligned} \ln p - \gamma \ln \rho &= \text{constant} \\ \frac{d}{d\rho}(\ln p - \gamma \ln \rho) &= \frac{d}{d\rho}(\text{constant}) \\ \frac{1}{p}\frac{dp}{d\rho} - \frac{\gamma}{\rho} &= 0 \\ \frac{dp}{d\rho} &= \frac{\gamma p}{\rho}. \end{aligned}$$

If the air can be considered an ideal gas, we can use the ideal gas law:

$$\begin{aligned} pV &= nRT = \frac{m}{M} RT \\ p &= \frac{m}{V}\frac{RT}{M} = \rho \frac{RT}{M}. \end{aligned}$$

Here M is the molar mass of air:

$$\frac{dp}{d\rho} = \frac{\gamma p}{\rho} = \frac{\gamma\left(\rho \frac{RT}{M}\right)}{\rho} = \frac{\gamma RT}{M}.$$

Since the speed of sound is equal to $v = \sqrt{\frac{dp}{d\rho}}$, the speed is equal to

$$v = \sqrt{\frac{\gamma RT}{M}}.$$

Note that the velocity is faster at higher temperatures and slower for heavier gases. For air, $\gamma = 1.4$, $M = 0.02897 \frac{\text{kg}}{\text{mol}}$, and $R = 8.31 \frac{\text{J}}{\text{mol·K}}$. If the temperature is $T_\text{C} = 20°\text{C}\,(T = 293\text{ K})$, the speed of sound is $v = 343\text{ m/s}.$

The equation for the speed of sound in air $v = \sqrt{\frac{\gamma RT}{M}}$ can be simplified to give the equation for the speed of sound in air as a function of absolute temperature:

$$\begin{aligned} v &= \sqrt{\frac{\gamma RT}{M}} \\ &= \sqrt{\frac{\gamma RT}{M}\left(\frac{273\text{ K}}{273\text{ K}}\right)} = \sqrt{\frac{(273\text{ K})\gamma R}{M}}\sqrt{\frac{T}{273\text{ K}}} \\ &\approx 331\frac{\text{m}}{\text{s}}\sqrt{\frac{T}{273\text{ K}}}. \end{aligned}$$

One of the more important properties of sound is that its speed is nearly independent of the frequency. This independence is certainly true in open air for sounds in the audible range. If this independence were not true, you would certainly notice it for music played by a marching band in a football stadium, for example. Suppose that high-frequency sounds traveled faster—then the farther you were from the band, the more the sound from the low-pitch instruments would lag that from the high-pitch ones. But the music from all instruments arrives in cadence independent of distance, so all frequencies must travel at nearly the same speed. Recall that

$$v = f\lambda.$$

In a given medium under fixed conditions, *v* is constant, so there is a relationship between *f* and λ; the higher the frequency, the smaller the wavelength (Figure 17.10).

FIGURE 17.10 Because they travel at the same speed in a given medium, low-frequency sounds must have a greater wavelength than high-frequency sounds. Here, the lower-frequency sounds are emitted by the large speaker, called a woofer, whereas the higher-frequency sounds are emitted by the small speaker, called a tweeter. (credit: modification of work by Jane Whitney)

EXAMPLE 17.1

Calculating Wavelengths

Calculate the wavelengths of sounds at the extremes of the audible range, 20 and 20,000 Hz, in $30.0°\text{C}$ air. (Assume that the frequency values are accurate to two significant figures.)

Strategy

To find wavelength from frequency, we can use $v = f\lambda$.

Solution

1. Identify knowns. The value for *v* is given by
$$v = (331\text{ m/s})\sqrt{\frac{T}{273\text{ K}}}.$$
2. Convert the temperature into kelvins and then enter the temperature into the equation
$$v = (331\text{ m/s})\sqrt{\frac{303\text{ K}}{273\text{ K}}} = 348.7\text{ m/s}.$$
3. Solve the relationship between speed and wavelength for λ:
$$\lambda = \frac{v}{f}.$$
4. Enter the speed and the minimum frequency to give the maximum wavelength:
$$\lambda_{\text{max}} = \frac{348.7\text{ m/s}}{20\text{ Hz}} = 17\text{ m}.$$
5. Enter the speed and the maximum frequency to give the minimum wavelength:
$$\lambda_{\text{min}} = \frac{348.7\text{ m/s}}{20{,}000\text{ Hz}} = 0.017\text{ m} = 1.7\text{ cm}.$$

Significance

Because the product of *f* multiplied by λ equals a constant, the smaller *f* is, the larger λ must be, and vice versa.

The speed of sound can change when sound travels from one medium to another, but the frequency usually remains the same. This is similar to the frequency of a wave on a string being equal to the frequency of the force oscillating the string. If *v* changes and *f* remains the same, then the wavelength λ must change. That is, because $v = f\lambda$, the higher the speed of a sound, the greater its wavelength for a given frequency.

CHECK YOUR UNDERSTANDING 17.1

Imagine you observe two firework shells explode. You hear the explosion of one as soon as you see it. However, you see the other shell for several milliseconds before you hear the explosion. Explain why this is so.

Although sound waves in a fluid are longitudinal, sound waves in a solid travel both as longitudinal waves and transverse waves. Seismic waves, which are essentially sound waves in Earth's crust produced by earthquakes, are an interesting example of how the speed of sound depends on the rigidity of the medium. Earthquakes produce both longitudinal and transverse waves, and these travel at different speeds. The bulk modulus of granite is greater than its shear modulus. For that reason, the speed of longitudinal or pressure waves (P-waves) in earthquakes in granite is significantly higher than the speed of transverse or shear waves (S-waves). Both types of earthquake waves travel slower in less rigid material, such as sediments. P-waves have speeds of 4 to 7 km/s, and S-waves range in speed from 2 to 5 km/s, both being faster in more rigid material. The P-wave gets progressively farther ahead of the S-wave as they travel through Earth's crust. The time between the P- and S-waves is routinely used to determine the distance to their source, the epicenter of the earthquake. Because S-waves do not pass through the liquid core, two shadow regions are produced (Figure 17.11).

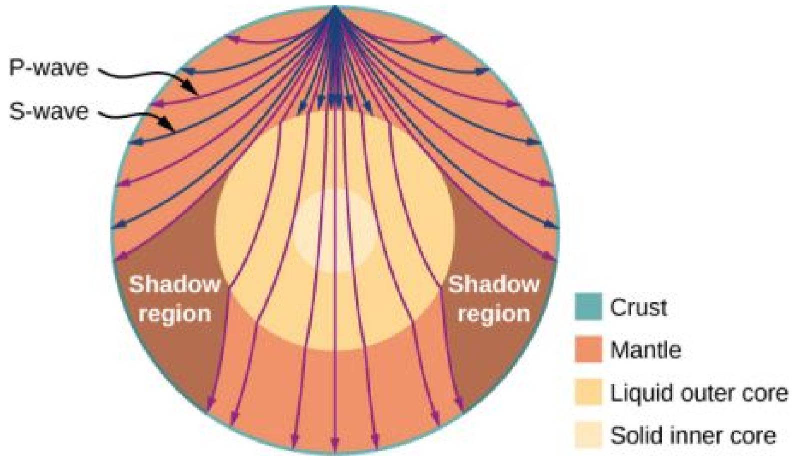


FIGURE 17.11 Earthquakes produce both longitudinal waves (P-waves) and transverse waves (S-waves), and these travel at different speeds. Both waves travel at different speeds in the different regions of Earth, but in general, P-waves travel faster than S-waves. S-waves cannot be supported by the liquid core, producing shadow regions.

Seismologists and geophysicists use properties and velocities of earthquake waves to study the Earth's interior, which due to it's depth and pressure is not observable through many other means. In fact, the discoveries of the structure of the Earth, illustrated in the figure above, resulted from earthquake observations. In 1914, Beno Gutenberg used differences in wave speeds to determine that there must be a liquid core within the mantle. In 1936, Inge Lehmann began investigating P-waves from a New Zealand earthquake that had unexpectedly reached Europe, which should have been in the shadow region. Up until that point, seismologists had explained such shadow waves as being caused by some type of diffraction (as Gutenberg himself assumed) or a result of faulty seismometers. However, Lehmann had installed the European instruments herself, and so trusted their accuracy. She calculated that the amplitude of the waves must be caused by the existence of a solid inner core within the liquid core. This model has been accepted and reinforced by decades of subsequent calculations, including those from nuclear test explosions, which can be measured very precisely.

As sound waves move away from a speaker, or away from the epicenter of an earthquake, their power per unit area decreases. This is why the sound is very loud near a speaker and becomes less loud as you move away from the speaker. This also explains why there can be an extreme amount of damage at the epicenter of an earthquake but only tremors are felt in areas far from the epicenter. The power per unit area is known as the intensity, and in the next section, we will discuss how the intensity depends on the distance from the source.

17.3 Sound Intensity

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define the term intensity
- Explain the concept of sound intensity level
- Describe how the human ear translates sound

In a quiet forest, you can sometimes hear a single leaf fall to the ground. But when a passing motorist has their stereo turned up, you cannot even hear what the person next to you in your car is saying (Figure 17.12). We are all very familiar with the loudness of sounds and are aware that loudness is related to how energetically the source is vibrating. High noise exposure is hazardous to hearing, which is why it is important for people working in industrial settings to wear ear protection. The relevant physical quantity is sound intensity, a concept that is valid for all sounds whether or not they are in the audible range.

FIGURE 17.12 Noise on crowded roadways, like this one in Delhi, makes it hard to hear others unless they shout. (credit: “Lingaraj G J”/Flickr)

In Waves, we defined intensity as the power per unit area carried by a wave. Power is the rate at which energy is transferred by the wave. In equation form, intensity I is

$$I = \frac{P}{A}, \tag{17.8}$$

where P is the power through an area A. The SI unit for I is $\mathrm{W/m^2}$. If we assume that the sound wave is spherical, and that no energy is lost to thermal processes, the energy of the sound wave is spread over a larger area as distance increases, so the intensity decreases. The area of a sphere is $A = 4\pi r^2$. As the wave spreads out from r_1 to r_2, the energy also spreads out over a larger area:

$$\begin{aligned} P_1 &= P_2 \\ I_1 4\pi r_1^2 &= I_2 4\pi r_2^2; \end{aligned}$$

$$I_2 = I_1 \left(\frac{r_1}{r_2}\right)^2. \tag{17.9}$$

The intensity decreases as the wave moves out from the source. In an inverse square relationship, such as the intensity, when you double the distance, the intensity decreases to one quarter,

$$I_2 = I_1 \left(\frac{r_1}{r_2}\right)^2 = I_1 \left(\frac{r_1}{2r_1}\right)^2 = \frac{1}{4} I_1.$$

Generally, when considering the intensity of a sound wave, we take the intensity to be the time-averaged value of the power, denoted by $\langle P \rangle$, divided by the area,

$$I = \frac{\langle P \rangle}{A}. \tag{17.10}$$

The intensity of a sound wave is proportional to the change in the pressure squared and inversely proportional to the density and the speed. Consider a parcel of a medium initially undisturbed and then influenced by a sound wave at time t, as shown in Figure 17.13.

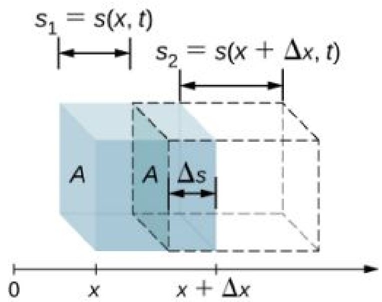


FIGURE 17.13 An undisturbed parcel of a medium with a volume $V = A\Delta x$ shown in blue. A sound wave moves through the medium at time *t*, and the parcel is displaced and expands, as shown by dotted lines. The change in volume is $\Delta V = A\Delta s = A\,(s_2 - s_1)$, where s_1 is the displacement of the leading edge of the parcel and s_2 is the displacement of the trailing edge of the parcel. In the figure, $s_2 > s_1$ and the parcel expands, but the parcel can either expand or compress ($s_2 < s_1$), depending on which part of the sound wave (compression or rarefaction) is moving through the parcel.

As the sound wave moves through the parcel, the parcel is displaced and may expand or contract. If $s_2 > s_1$, the volume has increased and the pressure decreases. If $s_2 < s_1$, the volume has decreased and the pressure increases. The change in the volume is

$$\Delta V = A\Delta s = A\,(s_2 - s_1) = A\,(s\,(x + \Delta x, t) - s\,(x, t))\,.$$

The fractional change in the volume is the change in volume divided by the original volume:

$$\frac{dV}{V} = \lim_{\Delta x \to 0} \frac{A\,[s\,(x + \Delta x, t) - s\,(x, t)]}{A\Delta x} = \frac{\partial s\,(x, t)}{\partial x}.$$

The fractional change in volume is related to the pressure fluctuation by the bulk modulus$\beta = -\frac{\Delta p(x,t)}{dV/V}$. Recall that the minus sign is required because the volume is *inversely* related to the pressure. (We use lowercase *p* for pressure to distinguish it from power, denoted by *P*.) The change in pressure is therefore $\Delta p\,(x, t) = -\beta\frac{dV}{V} = -\beta\frac{\partial s(x,t)}{\partial x}$. If the sound wave is sinusoidal, then the displacement as shown in Equation 17.2 is $s\,(x, t) = s_{\max}\cos\,(kx \mp \omega t + \phi)$ and the pressure is found to be

$$\Delta p\,(x, t) = -\beta\frac{dV}{V} = -\beta\frac{\partial s\,(x, t)}{\partial x} = \beta k s_{\max}\sin\,(kx - \omega t + \phi) = \Delta p_{\max}\sin\,(kx - \omega t + \phi)\,.$$

The intensity of the sound wave is the power per unit area, and the power is the force times the velocity, $I = \frac{P}{A} = \frac{Fv}{A} = pv$. Here, the velocity is the velocity of the oscillations of the medium, and not the velocity of the sound wave. The velocity of the medium is the time rate of change in the displacement:

$$v\,(x, t) = \frac{\partial}{\partial y}s\,(x, t) = \frac{\partial}{\partial y}\,(s_{\max}\cos\,(kx - \omega t + \phi)) = s_{\max}\omega\,\sin\,(kx - \omega t + \phi)\,.$$

Thus, the intensity becomes

$$\begin{aligned} I &= \Delta p\,(x, t)\,v\,(x, t) \\ &= \beta k s_{\max}\sin\,(kx - \omega t + \phi)\,[s_{\max}\omega\,\sin\,(kx - \omega t + \phi)] \\ &= \beta k \omega s_{\max}^2\sin^2\,(kx - \omega t + \phi)\,. \end{aligned}$$

To find the time-averaged intensity over one period $T = \frac{2\pi}{\omega}$ for a position *x*, we integrate over the period, $I = \frac{\beta k \omega s_{\max}^2}{2}$. Using $\Delta p_{\max} = \beta k s_{\max}$, $v = \sqrt{\frac{\beta}{\rho}}$, and $v = \frac{\omega}{k}$, we obtain

$$I = \frac{\beta k \omega s_{\max}^2}{2} = \frac{\beta^2 k^2 \omega s_{\max}^2}{2\beta k} = \frac{\omega(\Delta p_{\max})^2}{2\left(\rho v^2\right)k} = \frac{v(\Delta p_{\max})^2}{2\left(\rho v^2\right)} = \frac{(\Delta p_{\max})^2}{2\rho v}.$$

That is, the intensity of a sound wave is related to its amplitude squared by

$$I = \frac{(\Delta p_{\max})^2}{2\rho v}. \qquad \textbf{17.11}$$

Here, Δp_{max} is the pressure variation or pressure amplitude in units of pascals (Pa) or N/m^2. The energy (as kinetic energy $\frac{1}{2}mv^2$) of an oscillating element of air due to a traveling sound wave is proportional to its amplitude squared. In this equation, ρ is the density of the material in which the sound wave travels, in units of kg/m^3, and v is the speed of sound in the medium, in units of m/s. The pressure variation is proportional to the amplitude of the oscillation, so I varies as $(\Delta p)^2$. This relationship is consistent with the fact that the sound wave is produced by some vibration; the greater its pressure amplitude, the more the air is compressed in the sound it creates.

Human Hearing and Sound Intensity Levels

As stated earlier in this chapter, hearing is the perception of sound. The hearing mechanism involves some interesting physics. The sound wave that impinges upon our ear is a pressure wave. The ear is a **transducer** that converts sound waves into electrical nerve impulses in a manner much more sophisticated than, but analogous to, a microphone. Figure 17.14 shows the anatomy of the ear.

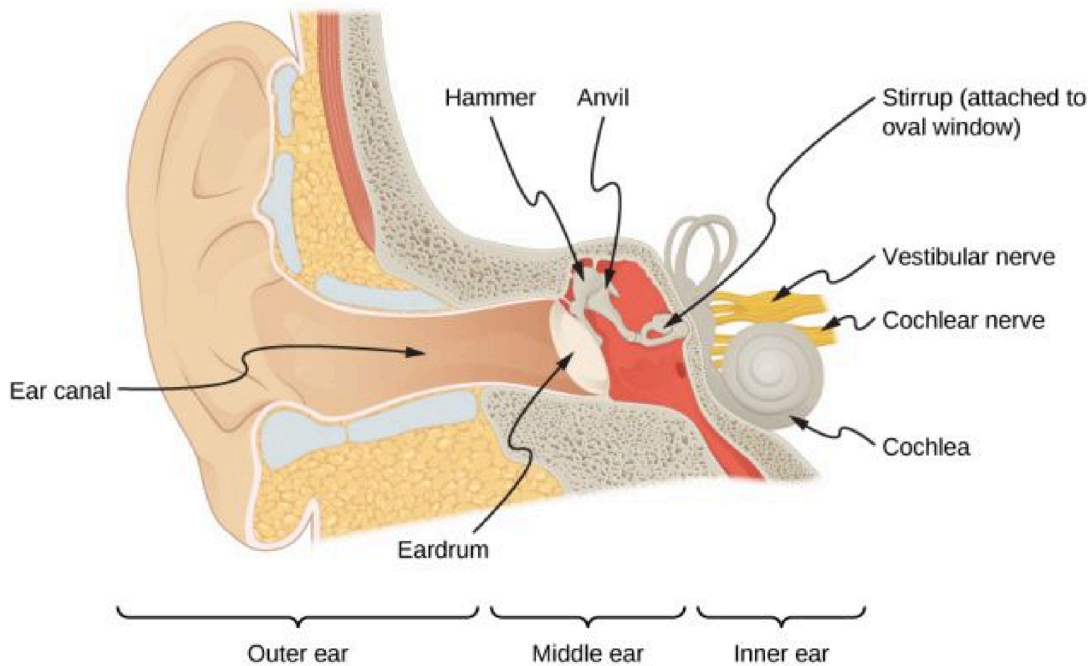


FIGURE 17.14 The anatomy of the human ear.

The outer ear, or ear canal, carries sound to the recessed, protected eardrum. The air column in the ear canal resonates and is partially responsible for the sensitivity of the ear to sounds in the 2000–5000-Hz range. The middle ear converts sound into mechanical vibrations and applies these vibrations to the cochlea.

INTERACTIVE

Watch this video (https://openstax.org/l/21humanear) for a more detailed discussion of the workings of the human ear.

The range of intensities that the human ear can hear depends on the frequency of the sound, but, in general, the range is quite large. The minimum threshold intensity that can be heard is $I_0 = 10^{-12}\ \text{W/m}^2$. Pain is experienced at intensities of $I_{\text{pain}} = 1\ \text{W/m}^2$. Measurements of sound intensity (in units of W/m^2) are very cumbersome due to this large range in values. For this reason, as well as for other reasons, the concept of sound intensity level was proposed.

The **sound intensity level** β of a sound, measured in decibels, having an intensity I in watts per meter squared, is defined as

$$\beta(\text{dB}) = 10 \log_{10}\left(\frac{I}{I_0}\right), \qquad (17.12)$$

where $I_0 = 10^{-12}\ \text{W/m}^2$ is a reference intensity, corresponding to the threshold intensity of sound that a person with normal hearing can perceive at a frequency of 1.00 kHz. It is more common to consider sound intensity levels in dB than in W/m^2. How human ears perceive sound can be more accurately described by the logarithm of the intensity rather than directly by the intensity. Because β is defined in terms of a ratio, it is a unitless quantity, telling you the *level* of the sound relative to a fixed standard $(10^{-12}\ \text{W/m}^2)$. The units of decibels (dB) are used to indicate this ratio is multiplied by 10 in its definition. The bel, upon which the decibel is based, is named for Alexander Graham Bell, the inventor of the telephone.

The decibel level of a sound having the threshold intensity of $10^{-12}\ \text{W/m}^2$ is $\beta = 0\ \text{dB}$, because $\log_{10} 1 = 0$. Table 17.2 gives levels in decibels and intensities in watts per meter squared for some familiar sounds. The ear is sensitive to as little as a trillionth of a watt per meter squared—even more impressive when you realize that the area of the eardrum is only about $1\ \text{cm}^2$, so that only $10^{-16}\ \text{W}$ falls on it at the threshold of hearing. Air molecules in a sound wave of this intensity vibrate over a distance of less than one molecular diameter, and the gauge pressures involved are less than $10^{-9}\ \text{atm}$.

Sound intensity level β (dB)	**Intensity I (W/m^2)**	**Example/effect**
0	1×10^{-12}	Threshold of hearing at 1000 Hz
10	1×10^{-11}	Rustle of leaves
20	1×10^{-10}	Whisper at 1-m distance
30	1×10^{-9}	Quiet home
40	1×10^{-8}	Average home
50	1×10^{-7}	Average office, soft music
60	1×10^{-6}	Normal conversation
70	1×10^{-5}	Noisy office, busy traffic
80	1×10^{-4}	Loud radio, classroom lecture
90	1×10^{-3}	Inside a heavy truck; damage from prolonged exposure[1]
100	1×10^{-2}	Noisy factory, siren at 30 m; damage from 8 h per day exposure
110	1×10^{-1}	Damage from 30 min per day exposure
120	1	Loud rock concert; pneumatic chipper at 2 m; threshold of pain

TABLE 17.2 Sound Intensity Levels and Intensities [1] Several government agencies and health-related professional associations recommend that 85 dB not be exceeded for 8-hour daily exposures in the absence of hearing protection.

1 Several government agencies and health-related professional associations recommend that 85 dB not be exceeded for 8-hour daily exposures in the absence of hearing protection.

Sound intensity level β (dB)	Intensity I $\left(\mathrm{W/m^2}\right)$	Example/effect
140	1×10^2	Jet airplane at 30 m; severe pain, damage in seconds
160	1×10^4	Bursting of eardrums

TABLE 17.2 **Sound Intensity Levels and Intensities** [1] Several government agencies and health-related professional associations recommend that 85 dB not be exceeded for 8-hour daily exposures in the absence of hearing protection.

An observation readily verified by examining Table 17.2 or by using Equation 17.12 is that each factor of 10 in intensity corresponds to 10 dB. For example, a 90-dB sound compared with a 60-dB sound is 30 dB greater, or three factors of 10 (that is, 10^3 times) as intense. Another example is that if one sound is 10^7 as intense as another, it is 70 dB higher (Table 17.3).

I_2/I_1	$\beta_2 - \beta_1$
2.0	3.0 dB
5.0	7.0 dB
10.0	10.0 dB
100.0	20.0 dB
1000.0	30.0 dB

TABLE 17.3 **Ratios of Intensities and Corresponding Differences in Sound Intensity Levels**

EXAMPLE 17.2

Calculating Sound Intensity Levels

Calculate the sound intensity level in decibels for a sound wave traveling in air at $0°\mathrm{C}$ and having a pressure amplitude of 0.656 Pa.

Strategy

We are given Δp, so we can calculate I using the equation $I = \frac{(\Delta p)^2}{2\rho v_\mathrm{W}}$. Using I, we can calculate β straight from its definition in $\beta(dB) = 10 \log_{10}\left(\frac{I}{I_0}\right)$.

Solution

1. Identify knowns:
   Sound travels at 331 m/s in air at $0°\mathrm{C}$.
   Air has a density of $1.29\ \mathrm{kg/m^3}$ at atmospheric pressure and $0°\mathrm{C}$.
2. Enter these values and the pressure amplitude into $I = \frac{(\Delta p)^2}{2\rho v}$.
$$I = \frac{(\Delta p)^2}{2\rho v} = \frac{(0.656\ \mathrm{Pa})^2}{2(1.29\ \mathrm{kg/m^3})(331\ \mathrm{m/s})} = 5.04 \times 10^{-4}\ \mathrm{W/m^2}.$$
3. Enter the value for I and the known value for I_0 into $\beta(\mathrm{dB}) = 10 \log_{10}(I/I_0)$. Calculate to find the sound intensity level in decibels:

$$10 \log_{10}(5.04 \times 10^8) = 10(8.70)\text{dB} = 87 \text{ dB}.$$

Significance

This 87-dB sound has an intensity five times as great as an 80-dB sound. So a factor of five in intensity corresponds to a difference of 7 dB in sound intensity level. This value is true for any intensities differing by a factor of five.

EXAMPLE 17.3

Changing Intensity Levels of a Sound

Show that if one sound is twice as intense as another, it has a sound level about 3 dB higher.

Strategy

We are given that the ratio of two intensities is 2 to 1, and are then asked to find the difference in their sound levels in decibels. We can solve this problem by using of the properties of logarithms.

Solution

1. Identify knowns:
   The ratio of the two intensities is 2 to 1, or
   $$\frac{I_2}{I_1} = 2.00.$$
   We wish to show that the difference in sound levels is about 3 dB. That is, we want to show:
   $$\beta_2 - \beta_1 = 3 \text{ dB}.$$
   Note that
   $$\log_{10} b - \log_{10} a = \log_{10}\left(\frac{b}{a}\right).$$
2. Use the definition of β to obtain
   $$\beta_2 - \beta_1 = 10 \log_{10}\left(\frac{I_2}{I_1}\right) = 10 \log_{10} 2.00 = 10(0.301) \text{ dB}.$$
   Thus,
   $$\beta_2 - \beta_1 = 3.01 \text{ dB}.$$

Significance

This means that the two sound intensity levels differ by 3.01 dB, or about 3 dB, as advertised. Note that because only the ratio I_2/I_1 is given (and not the actual intensities), this result is true for any intensities that differ by a factor of two. For example, a 56.0-dB sound is twice as intense as a 53.0-dB sound, a 97.0-dB sound is half as intense as a 100-dB sound, and so on.

CHECK YOUR UNDERSTANDING 17.2

Identify common sounds at the levels of 10 dB, 50 dB, and 100 dB.

Another decibel scale is also in use, called the **sound pressure level**, based on the ratio of the pressure amplitude to a reference pressure. This scale is used particularly in applications where sound travels in water. It is beyond the scope of this text to treat this scale because it is not commonly used for sounds in air, but it is important to note that very different decibel levels may be encountered when sound pressure levels are quoted.

Hearing and Pitch

The human ear has a tremendous range and sensitivity. It can give us a wealth of simple information—such as pitch, loudness, and direction.

The perception of frequency is called **pitch**. Typically, humans have excellent relative pitch and can discriminate between two sounds if their frequencies differ by 0.3% or more. For example, 500.0 and 501.5 Hz are noticeably different. Musical **notes** are sounds of a particular frequency that can be produced by most instruments and in Western music have particular names, such as A-sharp, C, or E-flat.

The perception of intensity is called **loudness**. At a given frequency, it is possible to discern differences of about 1 dB, and a change of 3 dB is easily noticed. But loudness is not related to intensity alone. Frequency has a major effect on how loud a sound seems. Sounds near the high- and low-frequency extremes of the hearing range seem even less loud, because the ear is less sensitive at those frequencies. When a violin plays middle C, there is no mistaking it for a piano playing the same note. The reason is that each instrument produces a distinctive set of frequencies and intensities. We call our perception of these combinations of frequencies and intensities tone quality or, more commonly, the **timbre** of the sound. Timbre is the shape of the wave that arises from the many reflections, resonances, and superposition in an instrument.

A unit called a **phon** is used to express loudness numerically. Phons differ from decibels because the phon is a unit of loudness perception, whereas the decibel is a unit of physical intensity. Figure 17.15 shows the relationship of loudness to intensity (or intensity level) and frequency for persons with normal hearing. The curved lines are equal-loudness curves. Each curve is labeled with its loudness in phons. Any sound along a given curve is perceived as equally loud by the average person. The curves were determined by having large numbers of people compare the loudness of sounds at different frequencies and sound intensity levels. At a frequency of 1000 Hz, phons are taken to be numerically equal to decibels.

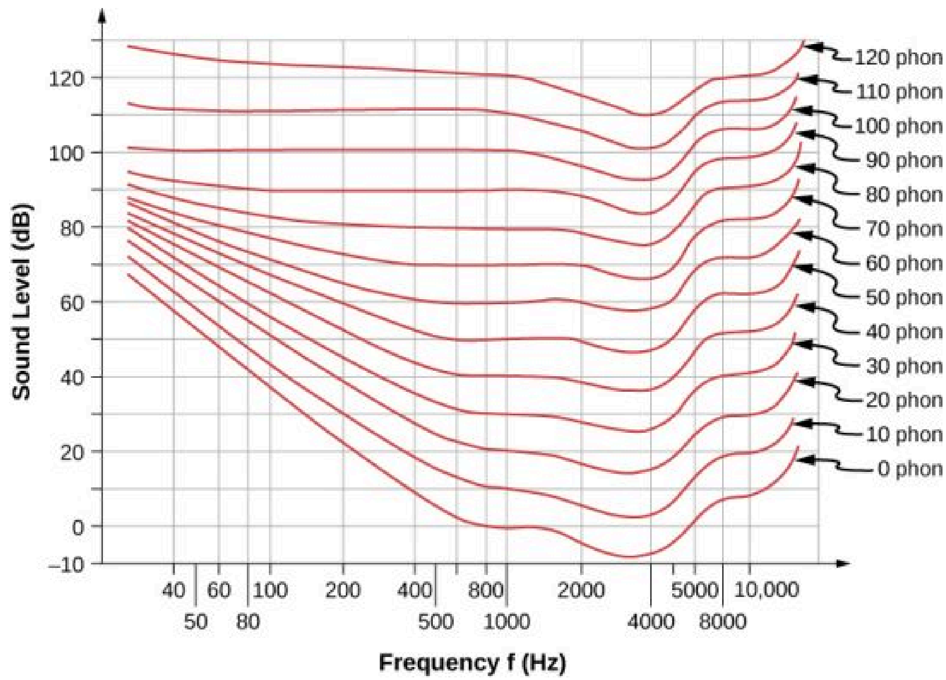


FIGURE 17.15 The relationship of loudness in phons to intensity level (in decibels) and intensity (in watts per meter squared) for persons with normal hearing. The curved lines are equal-loudness curves—all sounds on a given curve are perceived as equally loud. Phons and decibels are defined to be the same at 1000 Hz.

EXAMPLE 17.4

Measuring Loudness

(a) What is the loudness in phons of a 100-Hz sound that has an intensity level of 80 dB? (b) What is the intensity level in decibels of a 4000-Hz sound having a loudness of 70 phons? (c) At what intensity level will an 8000-Hz sound have the same loudness as a 200-Hz sound at 60 dB?

Strategy

The graph in Figure 17.15 should be referenced to solve this example. To find the loudness of a given sound, you must know its frequency and intensity level, locate that point on the square grid, and then interpolate between loudness curves to get the loudness in phons. Once that point is located, the intensity level can be determined from the vertical axis.

Solution

1. Identify knowns: The square grid of the graph relating phons and decibels is a plot of intensity level versus frequency—both physical quantities: 100 Hz at 80 dB lies halfway between the curves marked 70 and 80 phons.
   Find the loudness: 75 phons.
2. Identify knowns: Values are given to be 4000 Hz at 70 phons.
   Follow the 70-phon curve until it reaches 4000 Hz. At that point, it is below the 70 dB line at about 67 dB.
   Find the intensity level: 67 dB.
3. Locate the point for a 200 Hz and 60 dB sound.
   Find the loudness: This point lies just slightly above the 50-phon curve, and so its loudness is 51 phons.
   Look for the 51-phon level is at 8000 Hz: 63 dB.

Significance

These answers, like all information extracted from Figure 17.15, have uncertainties of several phons or several decibels, partly due to difficulties in interpolation, but mostly related to uncertainties in the equal-loudness curves.

CHECK YOUR UNDERSTANDING 17.3

Describe how amplitude is related to the loudness of a sound.

In this section, we discussed the characteristics of sound and how we hear, but how are the sounds we hear produced? Interesting sources of sound are musical instruments and the human voice, and we will discuss these sources. But before we can understand how musical instruments produce sound, we need to look at the basic mechanisms behind these instruments. The theories behind the mechanisms used by musical instruments involve interference, superposition, and standing waves, which we discuss in the next section.

17.4 Normal Modes of a Standing Sound Wave

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the mechanism behind sound-reducing headphones
- Describe resonance in a tube closed at one end and open at the other end
- Describe resonance in a tube open at both ends

Interference is the hallmark of waves, all of which exhibit constructive and destructive interference exactly analogous to that seen for water waves. In fact, one way to prove something "is a wave" is to observe interference effects. Since sound is a wave, we expect it to exhibit interference.

Interference of Sound Waves

In Waves, we discussed the interference of wave functions that differ only in a phase shift. We found that the wave function resulting from the superposition of $y_1\,(x, t) = A\sin(kx - \omega t + \phi)$ and $y_2\,(x, t) = A\sin(kx - \omega t)$ is

$$y(x, t) = \left[2A\cos\left(\frac{\phi}{2}\right)\right]\sin\left(kx - \omega t + \frac{\phi}{2}\right).$$

One way for two identical waves that are initially in phase to become out of phase with one another is to have the waves travel different distances; that is, they have different path lengths. Sound waves provide an excellent example of a phase shift due to a path difference. As we have discussed, sound waves can basically be modeled as longitudinal waves, where the molecules of the medium oscillate around an equilibrium position, or as pressure waves.

When the waves leave the speakers, they move out as spherical waves (Figure 17.16). The waves interfere; constructive inference is produced by the combination of two crests or two troughs, as shown. Destructive interference is produced by the combination of a trough and a crest.

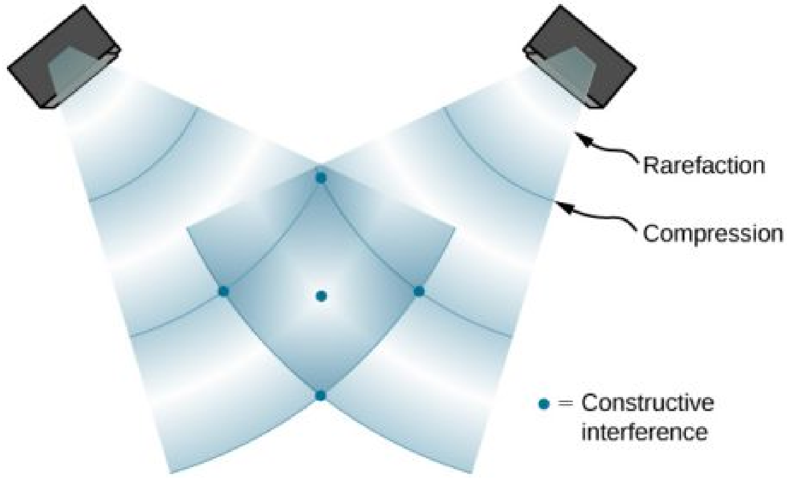
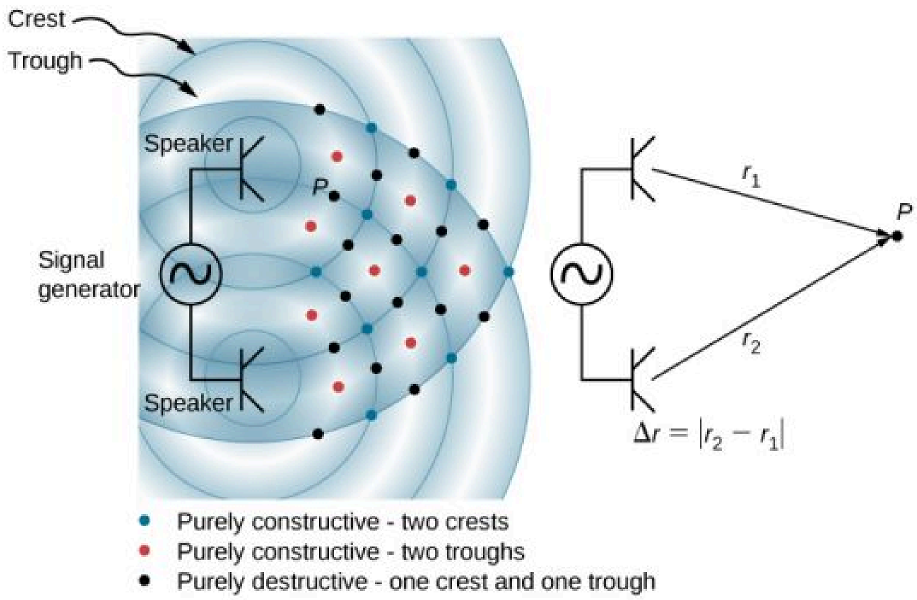


FIGURE 17.16 When sound waves are produced by a speaker, they travel at the speed of sound and move out as spherical waves. Here, two speakers produce the same steady tone (frequency). The result is points of high-intensity sound (highlighted), which result from two crests (compression) or two troughs (rarefaction) overlapping. Destructive interference results from a crest and trough overlapping. The points where there is constructive interference in the figure occur because the two waves are in phase at those points. Points of destructive interference (Figure 17.17) are the result of the two waves being out of phase.



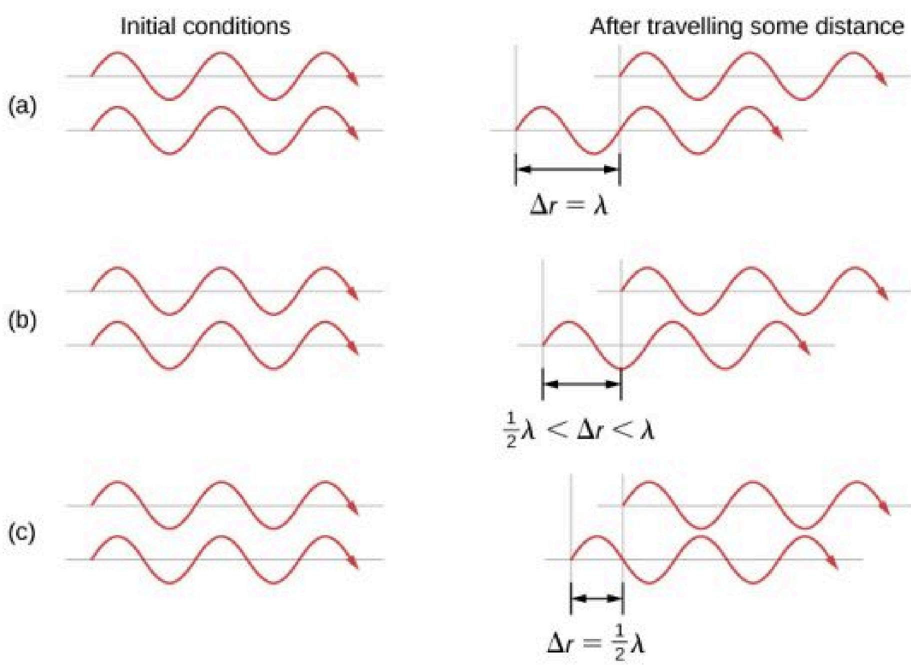


FIGURE 17.17 Two speakers being driven by a single signal generator. The sound waves produced by the speakers are in phase and are of a single frequency. The sound waves interfere with each other. When two crests or two troughs coincide, there is constructive interference, marked by the red and blue dots. When a trough and a crest coincide, destructive interference occurs, marked by black dots. The phase difference is due to the path lengths traveled by the individual waves. Two identical waves travel two different path lengths to a point *P*. (a) The difference in the path lengths is one wavelength, resulting in total constructive interference and a resulting amplitude equal to twice the original amplitude. (b) The difference in the path lengths is less than one wavelength but greater than one half a wavelength, resulting in an amplitude greater than zero and less than twice the original amplitude. (c) The difference in the path lengths is one half of a wavelength, resulting in total destructive interference and a resulting amplitude of zero.

The phase difference at each point is due to the different path lengths traveled by each wave. When the difference in the path lengths is an integer multiple of a wavelength,

$$\Delta r = |r_2 - r_1| = n\lambda, \text{ where } n = 0, 1, 2, 3, \ldots,$$

the waves are in phase and there is constructive interference. When the difference in path lengths is an odd multiple of a half wavelength,

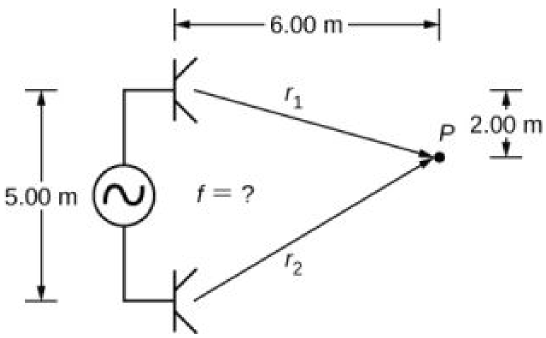
$$\Delta r = |r_2 - r_1| = n\frac{\lambda}{2},\ \text{where } n = 1, 3, 5, \ldots,$$

the waves are $180°(\pi\ \text{rad})$ out of phase and the result is destructive interference. These points can be located with a sound-level intensity meter.

EXAMPLE 17.5

Interference of Sound Waves

Two speakers are separated by 5.00 m and are being driven by a signal generator at an unknown frequency. A student with a sound-level meter walks out 6.00 m and down 2.00 m, and finds the first minimum intensity, as shown below. What is the frequency supplied by the signal generator? Assume the wave speed of sound is $v = 343.00\ \text{m/s}$.

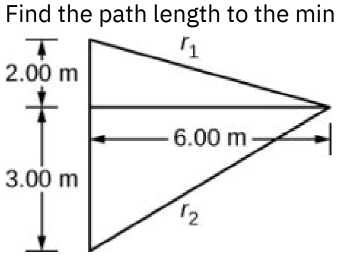


Strategy

The wave velocity is equal to $v = \frac{\lambda}{T} = \lambda f$. The frequency is then $f = \frac{v}{\lambda}$. A minimum intensity indicates destructive interference and the first such point occurs where there is path difference of $\Delta r = \lambda/2$, which can be found from the geometry.

Solution

1. Find the path length to the minimum point from each speaker.



$$r_1 = \sqrt{(6.00\ \text{m})^2 + (2.00\ \text{m})^2} = 6.32\ \text{m},\ \ r_2 = \sqrt{(6.00\ \text{m})^2 + (3.00\ \text{m})^2} = 6.71\ \text{m}$$

2. Use the difference in the path length to find the wavelength.

$$\Delta r = |r_2 - r_1| = |6.71\ \text{m} - 6.32\ \text{m}| = 0.39\ \text{m}$$

$$\lambda = 2\Delta r = 2\,(0.39\ \text{m}) = 0.78\ \text{m}$$

3. Find the frequency.

$$f = \frac{v}{\lambda} = \frac{343.00\ \text{m/s}}{0.78\ \text{m}} = 439.74\ \text{Hz}$$

Significance

If point *P* were a point of maximum intensity, then the path length would be an integer multiple of the wavelength.

CHECK YOUR UNDERSTANDING 17.4

If you walk around two speakers playing music, how come you do not notice places where the music is very loud or very soft, that is, where there is constructive and destructive interference?

The concept of a phase shift due to a difference in path length is very important. You will use this concept again in Interference (http://openstax.org/books/university-physics-volume-3/pages/3-introduction) and Photons and Matter Waves (http://openstax.org/books/university-physics-volume-3/pages/6-introduction), where we discuss how Thomas Young used this method in his famous double-slit experiment to provide evidence that light has wavelike properties.

Noise Reduction through Destructive Interference

Figure 17.18 shows a clever use of sound interference to cancel noise. Larger-scale applications of active noise reduction by destructive interference have been proposed for entire passenger compartments in commercial aircraft. To obtain destructive interference, a fast electronic analysis is performed, and a second sound is introduced $180°$ out of phase with the original sound, with its maxima and minima exactly reversed from the incoming noise. Sound waves in fluids are pressure waves and are consistent with Pascal's principle; that is, pressures from two different sources add and subtract like simple numbers. Therefore, positive and negative gauge pressures add to a much smaller pressure, producing a lower-intensity sound. Although completely destructive interference is possible only under the simplest conditions, it is possible to reduce noise levels by 30 dB or more using this technique.

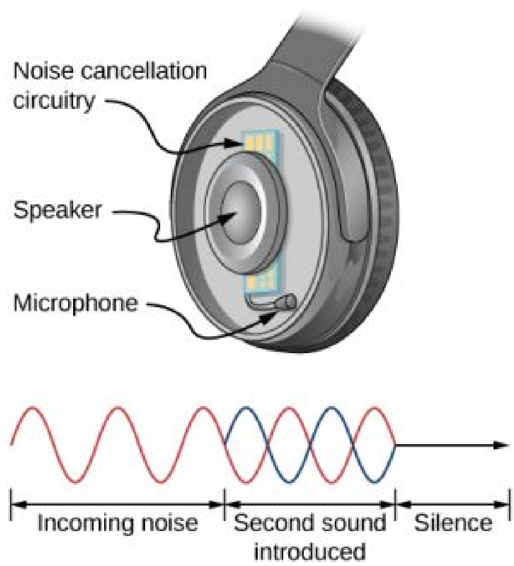


FIGURE 17.18 Headphones designed to cancel noise with destructive interference create a sound wave exactly opposite to the incoming sound. These headphones can be more effective than the simple passive attenuation used in most ear protection. Such headphones were used on the record-setting, around-the-world nonstop flight of the *Voyager* aircraft in 1986 to protect the pilots' hearing from engine noise.

CHECK YOUR UNDERSTANDING 17.5

Describe how noise-canceling headphones differ from standard headphones used to block outside sounds.

Where else can we observe sound interference? All sound resonances, such as in musical instruments, are due to constructive and destructive interference. Only the resonant frequencies interfere constructively to form standing waves, whereas others interfere destructively and are absent.

Resonance in a Tube Closed at one End

As we discussed in Waves, *standing waves* are formed by two waves moving in opposite directions. When two identical sinusoidal waves move in opposite directions, the waves may be modeled as

$$y_1(x,t) = A\sin(kx - \omega t) \text{ and } y_2(x,t) = A\sin(kx + \omega t).$$

When these two waves interfere, the resultant wave is a standing wave:

$$y_R(x,t) = [2A\sin(kx)]\cos(\omega t).$$

Resonance can be produced due to the boundary conditions imposed on a wave. In Waves, we showed that resonance could be produced in a string under tension that had symmetrical boundary conditions, specifically, a node at each end. We defined a node as a fixed point where the string did not move. We found that the symmetrical boundary conditions resulted in some frequencies resonating and producing standing waves, also called normal modes, while other frequencies interfere destructively. Sound waves can resonate in a hollow tube, and the frequencies of the sound waves that resonate depend on the boundary conditions.

Suppose we have a tube that is closed at one end and open at the other. If we hold a vibrating tuning fork near the open end of the tube, an incident sound wave travels through the tube and reflects off the closed end. The reflected sound has the same frequency and wavelength as the incident sound wave, but is traveling in the opposite direction. At the closed end of the tube, the molecules of air have very little freedom to oscillate, and a node arises. At the open end, the molecules are free to move, and at the right frequency, an antinode occurs. Unlike the symmetrical boundary conditions for the standing waves on the string, the boundary conditions for a tube open at one end and closed at the other end are anti-symmetrical: a node at the closed end and an antinode at the open end.

If the tuning fork has just the right frequency, the air column in the tube resonates loudly, but at most frequencies it vibrates very little. This observation just means that the air column has only certain natural frequencies. Consider the lowest frequency that will cause the tube to resonate, producing a loud sound. There will be a node at the closed end and an antinode at the open end, as shown in Figure 17.19.

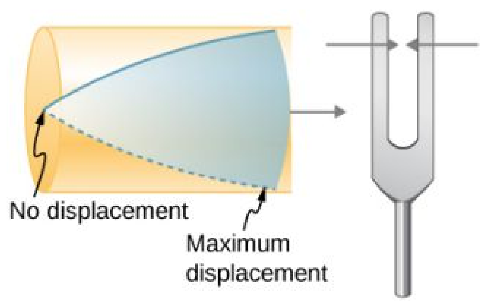


FIGURE 17.19 Resonance of air in a tube closed at one end, caused by a tuning fork that vibrates at the lowest frequency that can produce resonance (the fundamental frequency). A node exists at the closed end and an antinode at the open end.

The standing wave formed in the tube has an antinode at the open end and a node at the closed end. The distance from a node to an antinode is one-fourth of a wavelength, and this equals the length of the tube; thus, $\lambda_1 = 4L$. This same resonance can be produced by a vibration introduced at or near the closed end of the tube (Figure 17.20). It is best to consider this a natural vibration of the air column, independently of how it is induced.

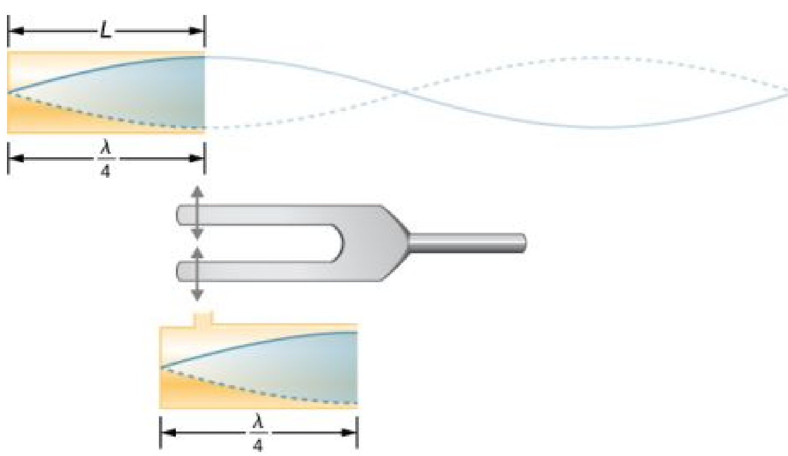


FIGURE 17.20 The same standing wave is created in the tube by a vibration introduced near its closed end.

Given that maximum air displacements are possible at the open end and none at the closed end, other shorter wavelengths can resonate in the tube, such as the one shown in Figure 17.21. Here the standing wave has three-fourths of its wavelength in the tube, or $\frac{3}{4}\lambda_3 = L$, so that $\lambda_3 = \frac{4}{3}L$. Continuing this process reveals a whole series of shorter-wavelength and higher-frequency sounds that resonate in the tube. We use specific terms for the resonances in any system. The lowest resonant frequency is called the **fundamental**, while all higher resonant frequencies are called **overtones**. The resonant frequencies that are integral multiples of the fundamental are collectively called **harmonics**. The fundamental is the first harmonic, the second harmonic is twice the frequency of the first harmonic, and so on. Some of these harmonics may not exist for a given scenario. Figure 17.22 shows the fundamental and the first three overtones (or the first, third, fifth, and seventh harmonics) in a tube closed at one end.

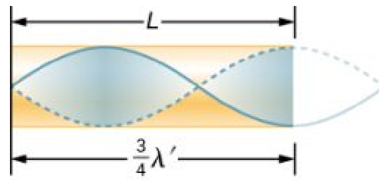


FIGURE 17.21 Another resonance for a tube closed at one end. This standing wave has maximum air displacement at the open end and none at the closed end. The wavelength is shorter, with three-fourths λ' equaling the length of the tube, so that $\lambda' = 4L/3$. This higher-frequency vibration is the first overtone.

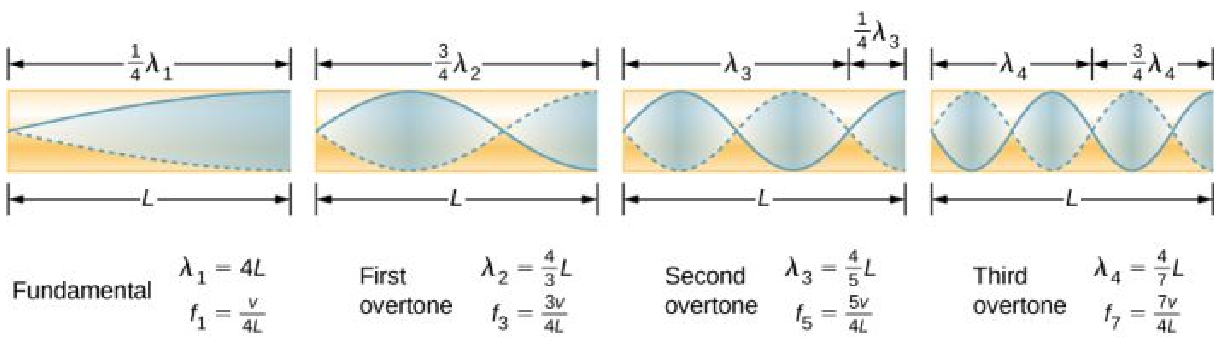


FIGURE 17.22 The fundamental and three lowest overtones for a tube closed at one end. All have maximum air displacements at the open end and none at the closed end.

The relationship for the resonant wavelengths of a tube closed at one end is

$$\lambda_n = \frac{4}{n}L \quad n = 1, 3, 5, ... \tag{17.13}$$

Now let us look for a pattern in the resonant frequencies for a simple tube that is closed at one end. The fundamental has $\lambda = 4L$, and frequency is related to wavelength and the speed of sound as given by

$v = f\lambda.$

Solving for f in this equation gives

$$f = \frac{v}{\lambda} = \frac{v}{4L},$$

where v is the speed of sound in air. Similarly, the first overtone has $\lambda = 4L/3$ (see Figure 17.22), so that

$$f_3 = 3\frac{v}{4L} = 3f_1.$$

Because $f_3 = 3f_1$, we call the first overtone the third harmonic. Continuing this process, we see a pattern that can be generalized in a single expression. The resonant frequencies of a tube closed at one end are

$$f_n = n\frac{v}{4L}, \quad n = 1, 3, 5, ..., \tag{17.14}$$

where f_1 is the fundamental, f_3 is the first overtone, and so on. It is interesting that the resonant frequencies depend on the speed of sound and, hence, on temperature. This dependence poses a noticeable problem for organs in old unheated cathedrals, and it is also the reason why musicians commonly bring their wind instruments to room temperature before playing them.

Resonance in a Tube Open at Both Ends

Another source of standing waves is a tube that is open at both ends. In this case, the boundary conditions are symmetrical: an antinode at each end. The resonances of tubes open at both ends can be analyzed in a very similar fashion to those for tubes closed at one end. The air columns in tubes open at both ends have maximum air displacements at both ends (Figure 17.23). Standing waves form as shown.

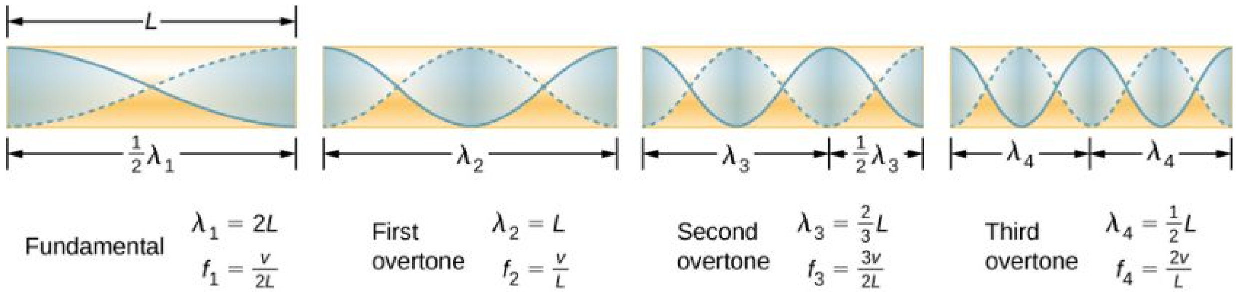


FIGURE 17.23 The resonant frequencies of a tube open at both ends, including the fundamental and the first three overtones. In all cases, the maximum air displacements occur at both ends of the tube, giving it different natural frequencies than a tube closed at one end.

The relationship for the resonant wavelengths of a tube open at both ends is

$$\lambda_n = \frac{2}{n}L, \quad n = 1, 2, 3, \ldots. \tag{17.15}$$

Based on the fact that a tube open at both ends has maximum air displacements at both ends, and using Figure 17.23 as a guide, we can see that the resonant frequencies of a tube open at both ends are

$$f_n = n\frac{v}{2L}, \quad n = 1, 2, 3\ldots, \tag{17.16}$$

where f_1 is the fundamental, f_2 is the first overtone, f_3 is the second overtone, and so on. Note that a tube open at both ends has a fundamental frequency twice what it would have if closed at one end. It also has a different spectrum of overtones than a tube closed at one end.

Note that a tube open at both ends has symmetrical boundary conditions, similar to the string fixed at both ends discussed in Waves. The relationships for the wavelengths and frequencies of a stringed instrument are the same as given in Equation 17.15 and Equation 17.16. The speed of the wave on the string (from Waves) is $v = \sqrt{\frac{F_T}{\mu}}$. The air around the string vibrates at the same frequency as the string, producing sound of the same frequency. The sound wave moves at the speed of sound and the wavelength can be found using $v = \lambda f$.

CHECK YOUR UNDERSTANDING 17.6

How is it possible to use a standing wave's node and antinode to determine the length of a closed-end tube?

INTERACTIVE

This video (https://openstax.org/l/21soundwaves) lets you visualize sound waves.

CHECK YOUR UNDERSTANDING 17.7

You observe two musical instruments that you cannot identify. One plays high-pitched sounds and the other plays low-pitched sounds. How could you determine which is which without hearing either of them play?

17.5 Sources of Musical Sound

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the resonant frequencies in instruments that can be modeled as a tube with symmetrical boundary conditions
- Describe the resonant frequencies in instruments that can be modeled as a tube with anti-symmetrical boundary conditions

Some musical instruments, such as woodwinds, brass, and pipe organs, can be modeled as tubes with symmetrical boundary conditions, that is, either open at both ends or closed at both ends (Figure 17.24). Other instruments can be modeled as tubes with anti-symmetrical boundary conditions, such as a tube with one end open and the other end closed (Figure 17.25).

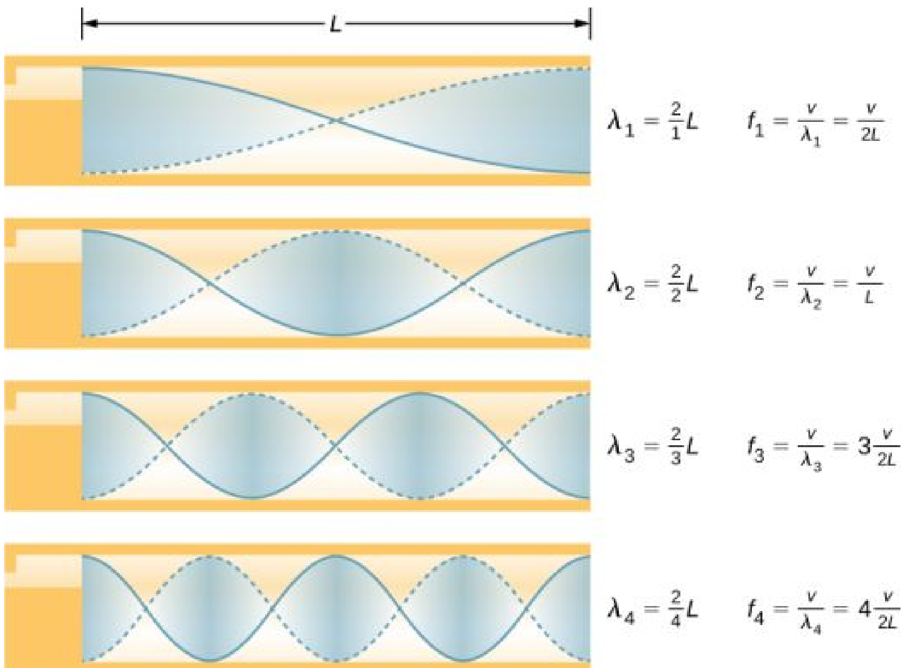


FIGURE 17.24 Some musical instruments can be modeled as a pipe open at both ends.

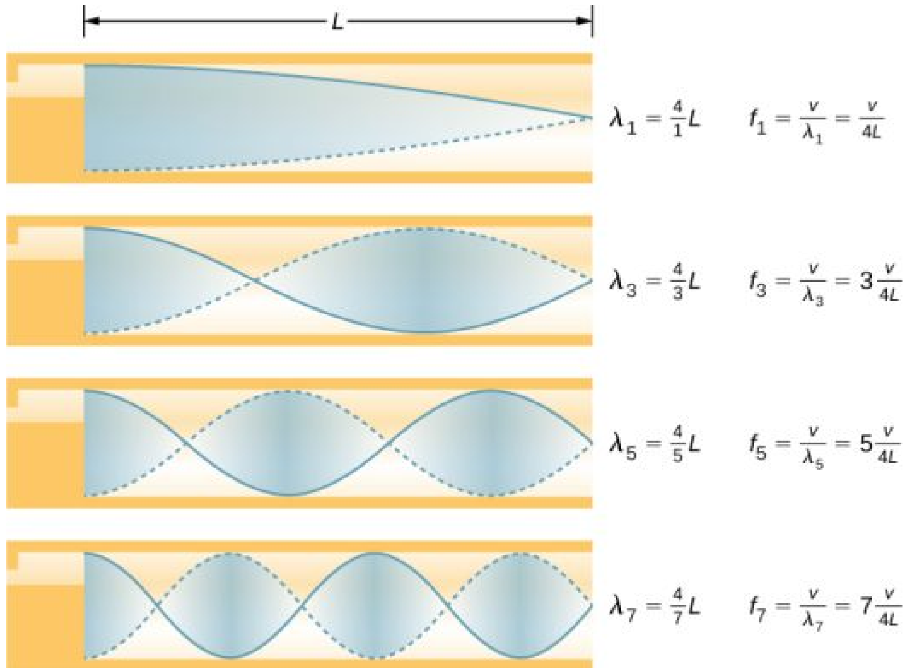


FIGURE 17.25 Some musical instruments can be modeled as a pipe closed at one end.

Resonant frequencies are produced by longitudinal waves that travel down the tubes and interfere with the reflected waves traveling in the opposite direction. A pipe organ is manufactured with various tubes of fixed lengths to produce different frequencies. The waves are the result of compressed air allowed to expand in the tubes. Even in open tubes, some reflection occurs due to the constraints of the sides of the tubes and the atmospheric pressure outside the open tube.

The antinodes do not occur at the opening of the tube, but rather depend on the radius of the tube. The waves do not fully expand until they are outside the open end of a tube, and for a thin-walled tube, an *end correction* should be added. This end correction is approximately 0.6 times the radius of the tube and should be added to the length of the tube.

Players of instruments such as the flute or oboe vary the length of the tube by opening and closing finger holes. On a trombone, you change the tube length by using a sliding tube. Bugles have a fixed length and can produce only a limited range of frequencies.

The fundamental and overtones can be present simultaneously in a variety of combinations. For example, middle C on a trumpet sounds distinctively different from middle C on a clarinet, although both instruments are modified versions of a tube closed at one end. The fundamental frequency is the same (and usually the most intense), but the overtones and their mix of intensities are different and subject to shading by the musician. This mix is what gives various musical instruments (and human voices) their distinctive characteristics, whether they have air columns, strings, sounding boxes, or drumheads. In fact, much of our speech is determined by shaping the cavity formed by the throat and mouth, and positioning the tongue to adjust the fundamental and combination of overtones. For example, simple resonant cavities can be made to resonate with the sound of the vowels (Figure 17.26). In boys at puberty, the larynx grows and the shape of the resonant cavity changes, giving rise to the difference in predominant frequencies in speech between men and women.

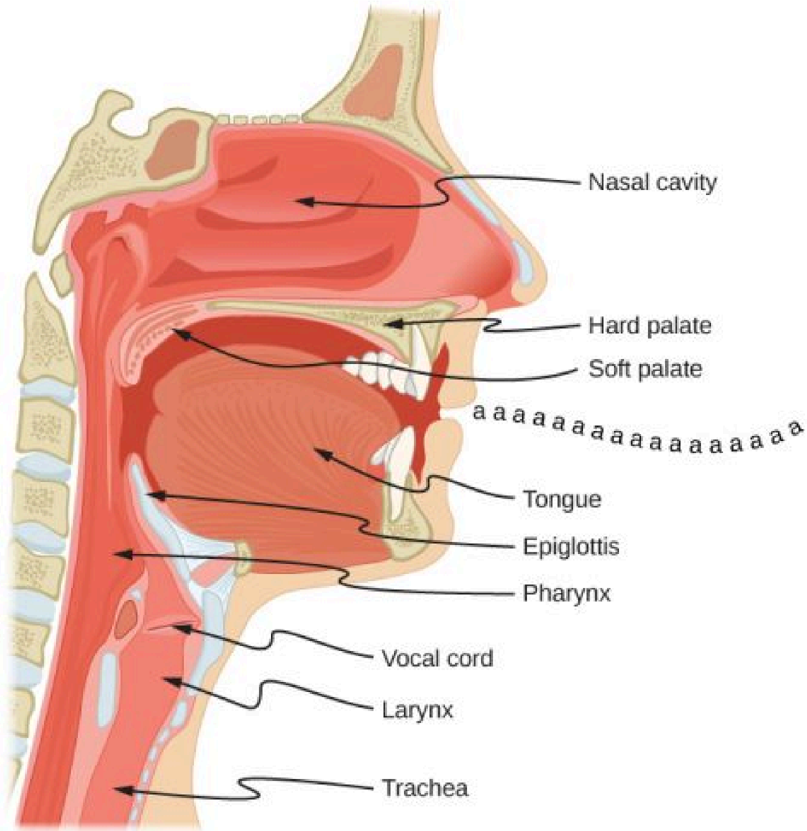


FIGURE 17.26 The throat and mouth form an air column closed at one end that resonates in response to vibrations in the voice box. The spectrum of overtones and their intensities vary with mouth shaping and tongue position to form different sounds. The voice box can be replaced with a mechanical vibrator, and understandable speech is still possible. Variations in basic shapes make different voices recognizable.

EXAMPLE 17.6

Finding the Length of a Tube with a 128-Hz Fundamental

(a) What length should a tube closed at one end have on a day when the air temperature is

$22.0°\text{C}$ if its fundamental frequency is to be 128 Hz (C below middle C)?

(b) What is the frequency of its fourth overtone?

Strategy

The length *L* can be found from the relationship $f_n = n\frac{v}{4L}$, but we first need to find the speed of sound *v*.

Solution

a. Identify knowns: The fundamental frequency is 128 Hz, and the air temperature is $22.0°\text{C}$. Use $f_n = n\frac{v}{4L}$ to find the fundamental frequency ($n = 1$),

$$f_1 = \frac{v}{4L}.$$

Solve this equation for length,

$$L = \frac{v}{4f_1}.$$

Find the speed of sound using $v = (331\text{ m/s})\sqrt{\frac{T}{273\text{ K}}}$,

$$v = (331\text{ m/s})\sqrt{\frac{295\text{ K}}{273\text{ K}}} = 344\text{ m/s}.$$

Enter the values of the speed of sound and frequency into the expression for *L*.

$$L = \frac{v}{4f_1} = \frac{344 \text{ m/s}}{4(128 \text{ Hz})} = 0.672 \text{ m}$$

b. Identify knowns: The first overtone has $n = 3$, the second overtone has $n = 5$, the third overtone has $n = 7$, and the fourth overtone has $n = 9$.
Enter the value for the fourth overtone into $f_n = n\frac{v}{4L}$,

$$f_9 = 9\frac{v}{4L} = 9f_1 = 1.15 \text{ kHz}.$$

Significance

Many wind instruments are modified tubes that have finger holes, valves, and other devices for changing the length of the resonating air column and hence, the frequency of the note played. Horns producing very low frequencies require tubes so long that they are coiled into loops. An example is the tuba. Whether an overtone occurs in a simple tube or a musical instrument depends on how it is stimulated to vibrate and the details of its shape. The trombone, for example, does not produce its fundamental frequency and only makes overtones.

If you have two tubes with the same fundamental frequency, but one is open at both ends and the other is closed at one end, they would sound different when played because they have different overtones. Middle C, for example, would sound richer played on an open tube, because it has even multiples of the fundamental as well as odd. A closed tube has only odd multiples.

Resonance

Resonance occurs in many different systems, including strings, air columns, and atoms. As we discussed in earlier chapters, resonance is the driven or forced oscillation of a system at its natural frequency. At resonance, energy is transferred rapidly to the oscillating system, and the amplitude of its oscillations grows until the system can no longer be described by Hooke's law. An example of this is the distorted sound intentionally produced in certain types of rock music.

Wind instruments use resonance in air columns to amplify tones made by lips or vibrating reeds. Other instruments also use air resonance in clever ways to amplify sound. Figure 17.27 shows a violin and a guitar, both of which have sounding boxes but with different shapes, resulting in different overtone structures. The vibrating string creates a sound that resonates in the sounding box, greatly amplifying the sound and creating overtones that give the instrument its characteristic timbre. The more complex the shape of the sounding box, the greater its ability to resonate over a wide range of frequencies. The marimba, like the one shown in Figure 17.28, uses pots or gourds below the wooden slats to amplify their tones. The resonance of the pot can be adjusted by adding water.

(a) (b)

FIGURE 17.27 String instruments such as (a) violins and (b) guitars use resonance in their sounding boxes to amplify and enrich the sound created by their vibrating strings. The bridge and supports couple the string vibrations to the sounding boxes and air within. (credit a: modification of work by Feliciano Guimarães; credit b: modification of work by Steve Snodgrass)

FIGURE 17.28 This marimba uses gourds as resonance chambers to amplify its sound. (credit: “APC Events”/Flickr)

We have emphasized sound applications in our discussions of resonance and standing waves, but these ideas apply to any system that has wave characteristics. Vibrating strings, for example, are actually resonating and have fundamentals and overtones similar to those for air columns. More subtle are the resonances in atoms due to the wave character of their electrons. Their orbitals can be viewed as standing waves, which have a fundamental (ground state) and overtones (excited states). It is fascinating that wave characteristics apply to such a wide range of physical systems.

17.6 Beats

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Determine the beat frequency produced by two sound waves that differ in frequency
- Describe how beats are produced by musical instruments

The study of music provides many examples of the superposition of waves and the constructive and destructive interference that occurs. Very few examples of music being performed consist of a single source playing a single frequency for an extended period of time. You will probably agree that a single frequency of sound for an extended period might be boring to the point of irritation, similar to the unwanted drone of an aircraft engine or a loud fan. Music is pleasant and interesting due to mixing the changing frequencies of various instruments and voices.

An interesting phenomenon that occurs due to the constructive and destructive interference of two or more frequencies of sound is the phenomenon of **beats**. If two sounds differ in frequencies, the sound waves can be modeled as

$$y_1 = A\cos(k_1 x - 2\pi f_1 t) \text{ and } y_2 = A\cos(k_2 x - 2\pi f_2 t).$$

Using the trigonometric identity $\cos u + \cos v = 2\cos\left(\frac{u+v}{2}\right)\cos\left(\frac{u-v}{2}\right)$ and considering the point in space as $x = 0.0\text{ m}$, we find the resulting sound at a point in space, from the superposition of the two sound waves, is equal to Figure 17.29:

$$y(t) = 2A\cos\left(2\pi f_{avg} t\right)\cos\left(2\pi\left(\frac{|f_2 - f_1|}{2}\right)t\right),$$

where the **beat frequency** is

$$f_{\text{beat}} = |f_2 - f_1|. \tag{17.17}$$

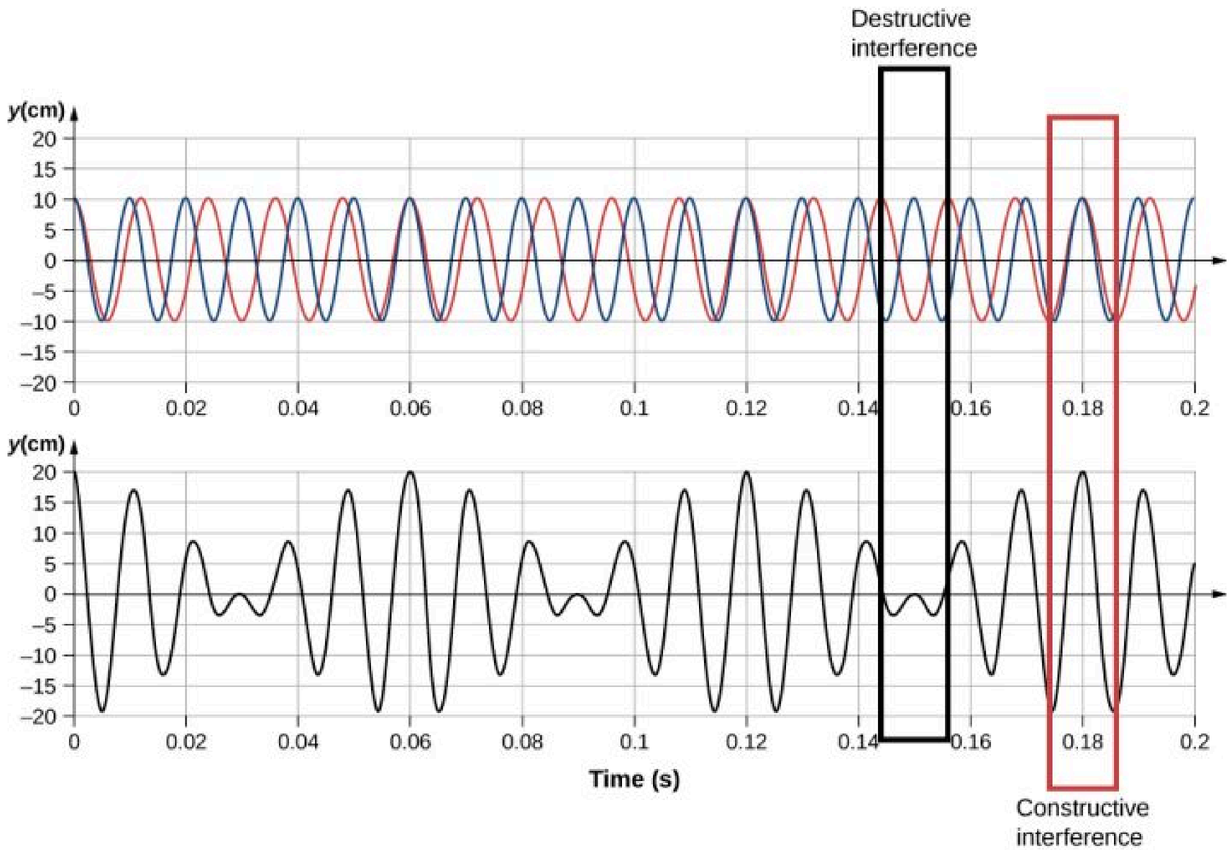


FIGURE 17.29 Beats produced by the constructive and destructive interference of two sound waves that differ in frequency.

These beats can be used by piano tuners to tune a piano. A tuning fork is struck and a note is played on the piano. As the piano tuner tunes the string, the beats have a lower frequency as the frequency of the note played approaches the frequency of the tuning fork.

EXAMPLE 17.7

Find the Beat Frequency Between Two Tuning Forks

What is the beat frequency produced when a tuning fork of a frequency of 256 Hz and a tuning fork of a frequency of 512 Hz are struck simultaneously?

Strategy

The beat frequency is the difference of the two frequencies.

Solution

We use $f_{\text{beat}} = |f_2 - f_1|$:

$$|f_2 - f_1| = (512 - 256)\,\text{Hz} = 256\,\text{Hz}.$$

Significance

The beat frequency is the absolute value of the difference between the two frequencies. A negative frequency would not make sense.

CHECK YOUR UNDERSTANDING 17.8

What would happen if more than two frequencies interacted? Consider three frequencies.

The study of the superposition of various waves has many interesting applications beyond the study of sound. In later chapters, we will discuss the wave properties of particles. The particles can be modeled as a “wave packet” that results from the superposition of various waves, where the particle moves at the “group velocity” of the wave packet.

17.7 The Doppler Effect

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the change in observed frequency as a moving source of sound approaches or departs from a stationary observer
- Explain the change in observed frequency as an observer moves toward or away from a stationary source of sound

The characteristic sound of a motorcycle buzzing by is an example of the **Doppler effect**. Specifically, if you are standing on a street corner and observe an ambulance with a siren sounding passing at a constant speed, you notice two characteristic changes in the sound of the siren. First, the sound increases in loudness as the ambulance approaches and decreases in loudness as it moves away, which is expected. But in addition, the high-pitched siren shifts dramatically to a lower-pitched sound. As the ambulance passes, the frequency of the sound heard by a stationary observer changes from a constant high frequency to a constant lower frequency, even though the siren is producing a constant source frequency. The closer the ambulance brushes by, the more abrupt the shift. Also, the faster the ambulance moves, the greater the shift. We also hear this characteristic shift in frequency for passing cars, airplanes, and trains.

The Doppler effect is an alteration in the observed frequency of a sound due to motion of either the source or the observer. Although less familiar, this effect is easily noticed for a stationary source and moving observer. For example, if you ride a train past a stationary warning horn, you will hear the horn’s frequency shift from high to low as you pass by. The actual change in frequency due to relative motion of source and observer is called a **Doppler shift**. The Doppler effect and Doppler shift are named for the Austrian physicist and mathematician Christian Johann Doppler (1803–1853), who did experiments with both moving sources and moving observers. Doppler, for example, had musicians play on a moving open train car and also play standing next to the train tracks as a train passed by. Their music was observed both on and off the train, and changes in frequency were measured.

What causes the Doppler shift? Figure 17.30 illustrates sound waves emitted by stationary and moving sources in a stationary air mass. Each disturbance spreads out spherically from the point at which the sound is emitted. If the source is stationary, then all of the spheres representing the air compressions in the sound wave are centered on the same point, and the stationary observers on either side hear the same wavelength and frequency as emitted by the source (case a). If the source is moving, the situation is different. Each compression of the air moves out in a sphere from the point at which it was emitted, but the point of emission moves. This moving emission point causes the air compressions to be closer together on one side and farther apart on the other. Thus, the wavelength is shorter in the direction the source is moving (on the right in case b), and longer in the opposite direction (on the left in case b). Finally, if the observers move, as in case (c), the frequency at which they receive the compressions changes. The observer moving toward the source receives them at a higher frequency, and the person moving away from the source receives them at a lower frequency.

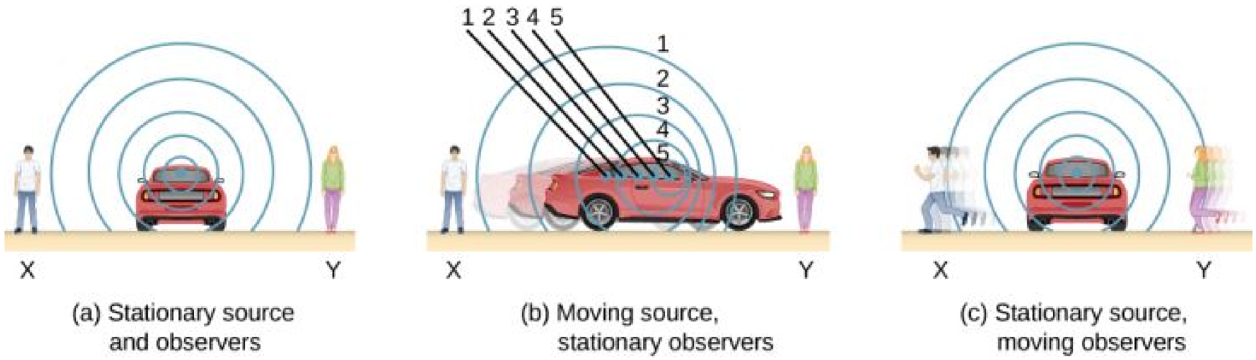


FIGURE 17.30 Sounds emitted by a source spread out in spherical waves. (a) When the source, observers, and air are stationary, the wavelength and frequency are the same in all directions and to all observers. (b) Sounds emitted by a source moving to the right spread out from the points at which they were emitted. The wavelength is reduced, and consequently, the frequency is increased in the direction of motion, so that the observer on the right hears a higher-pitched sound. The opposite is true for the observer on the left, where the wavelength is increased and the frequency is reduced. (c) The same effect is produced when the observers move relative to the source. Motion toward the source increases frequency as the observer on the right passes through more wave crests than she would if stationary. Motion away from the source decreases frequency as the observer on the left passes through fewer wave crests than he would if stationary.

We know that wavelength and frequency are related by $v = f\lambda$, where *v* is the fixed speed of sound. The sound moves in a medium and has the same speed *v* in that medium whether the source is moving or not. Thus, *f* multiplied by λ is a constant. Because the observer on the right in case (b) receives a shorter wavelength, the frequency she receives must be higher. Similarly, the observer on the left receives a longer wavelength, and hence he hears a lower frequency. The same thing happens in case (c). A higher frequency is received by the observer moving toward the source, and a lower frequency is received by an observer moving away from the source. In general, then, relative motion of source and observer toward one another increases the received frequency. Relative motion apart decreases frequency. The greater the relative speed, the greater the effect.

The Doppler effect occurs not only for sound, but for any wave when there is relative motion between the observer and the source. Doppler shifts occur in the frequency of sound, light, and water waves, for example. Doppler shifts can be used to determine velocity, such as when ultrasound is reflected from blood in a medical diagnostic. The relative velocities of stars and galaxies is determined by the shift in the frequencies of light received from them and has implied much about the origins of the universe. Modern physics has been profoundly affected by observations of Doppler shifts.

Derivation of the Observed Frequency due to the Doppler Shift

Consider two stationary observers *X* and *Y* in Figure 17.31, located on either side of a stationary source. Each observer hears the same frequency, and that frequency is the frequency produced by the stationary source.

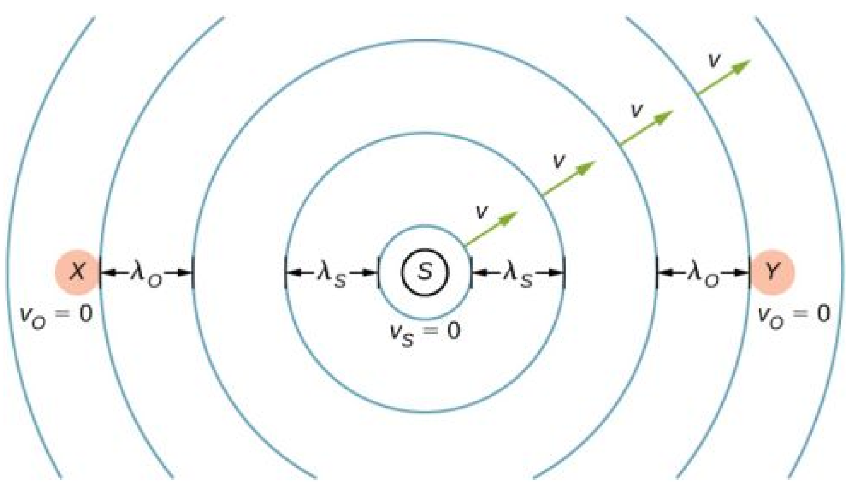


FIGURE 17.31 A stationary source sends out sound waves at a constant frequency f_s, with a constant wavelength λ_s, at the speed of sound *v*. Two stationary observers *X* and *Y*, on either side of the source, observe a frequency $f_o = f_s$, with a wavelength $\lambda_o = \lambda_s$.

Now consider a stationary observer *X* with a source moving away from the observer with a constant speed $v_s < v$ (Figure 17.32). At time $t = 0$, the source sends out a sound wave, indicated in black. This wave moves out at the speed of sound *v*. The position of the sound wave at each time interval of period T_s is shown as dotted lines. After one period, the source has moved $\Delta x = v_s T_s$ and emits a second sound wave, which moves out at the speed of sound. The source continues to move and produce sound waves, as indicated by the circles numbered 3 and 4. Notice that as the waves move out, they remained centered at their respective point of origin.

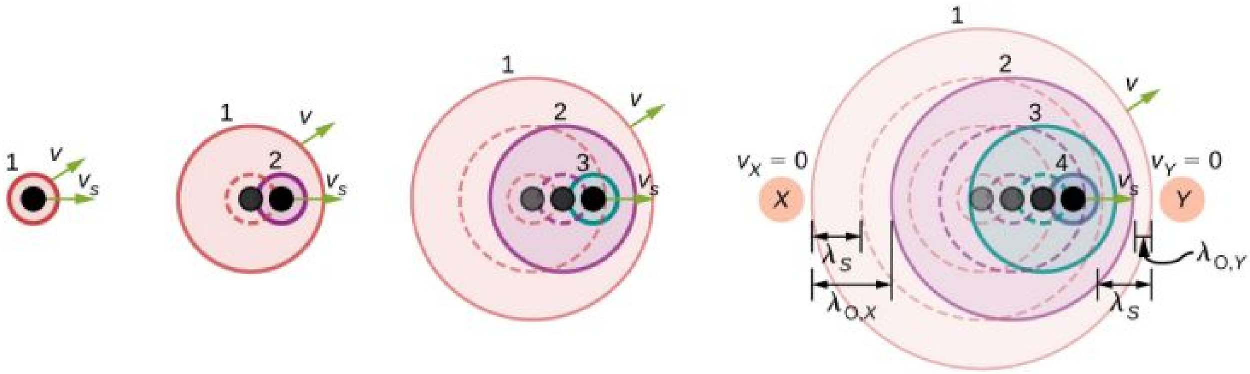


FIGURE 17.32 A source moving at a constant speed v_s away from an observer *X*. The moving source sends out sound waves at a constant frequency f_s, with a constant wavelength λ_s, at the speed of sound *v*. Snapshots of the source at an interval of T_s are shown as the source moves away from the stationary observer *X*. The solid lines represent the position of the sound waves after four periods from the initial time. The dotted lines are used to show the positions of the waves at each time period. The observer hears a wavelength of $\lambda_o = \lambda_s + \Delta x = \lambda_s + v_s T_s$.

Using the fact that the wavelength is equal to the speed times the period, and the period is the inverse of the frequency, we can derive the observed frequency:

$$\begin{aligned} \lambda_o &= \lambda_s + \Delta x \\ vT_o &= vT_s + v_s T_s \\ \frac{v}{f_o} &= \frac{v}{f_s} + \frac{v_s}{f_s} = \frac{v+v_s}{f_s} \\ f_o &= f_s\left(\frac{v}{v+v_s}\right). \end{aligned}$$

As the source moves away from the observer, the observed frequency is lower than the source frequency.

Now consider a source moving at a constant velocity v_s, moving toward a stationary observer *Y*, also shown in Figure 17.32. The wavelength is observed by *Y* as $\lambda_o = \lambda_s - \Delta x = \lambda_s - v_s T_s$. Once again, using the fact that the wavelength is equal to the speed times the period, and the period is the inverse of the frequency, we can derive the observed frequency:

$$\begin{aligned} \lambda_o &= \lambda_s - \Delta x \\ vT_o &= vT_s - v_s T_s \\ \frac{v}{f_o} &= \frac{v}{f_s} - \frac{v_s}{f_s} = \frac{v-v_s}{f_s} \\ f_o &= f_s\left(\frac{v}{v-v_s}\right). \end{aligned}$$

When a source is moving and the observer is stationary, the observed frequency is

$$f_o = f_s\left(\frac{v}{v \mp v_s}\right), \qquad (17.18)$$

where f_o is the frequency observed by the stationary observer, f_s is the frequency produced by the moving source, *v* is the speed of sound, v_s is the constant speed of the source, and the top sign is for the source approaching the observer and the bottom sign is for the source departing from the observer.

What happens if the observer is moving and the source is stationary? If the observer moves toward the stationary source, the observed frequency is higher than the source frequency. If the observer is moving away from the stationary source, the observed frequency is lower than the source frequency. Consider observer *X* in Figure 17.33 as the observer moves toward a stationary source with a speed v_o. The source emits a tone with a constant frequency f_s and constant period T_s. The observer hears the first wave emitted by the source. If the observer were stationary, the time for one wavelength of sound to pass should be equal to the period of the source T_{S}. Since the observer is moving toward the source, the time for one wavelength to pass is less than T_{S} and is equal to the observed period $T_{\mathrm{O}} = T_{\mathrm{S}} - \Delta t$. At time $t = 0$, the observer starts at the beginning of a wavelength and moves toward the second wavelength as the wavelength moves out from the source. The wavelength is equal to the distance the observer traveled plus the distance the sound wave traveled until it is met by the observer:

$$\begin{aligned} \lambda_{\mathrm{S}} &= vT_{\mathrm{O}} + v_{\mathrm{O}}T_{\mathrm{O}} \\ vT_{\mathrm{S}} &= (v + v_{\mathrm{O}})T_{\mathrm{O}} \\ v\left(\frac{1}{f_{\mathrm{S}}}\right) &= (v + v_{\mathrm{O}})\left(\frac{1}{f_{\mathrm{O}}}\right) \\ f_{\mathrm{O}} &= f_{\mathrm{S}}\left(\frac{v+v_{\mathrm{O}}}{v}\right). \end{aligned}$$

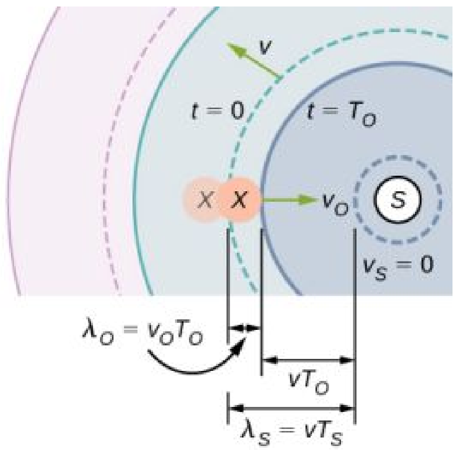


FIGURE 17.33 A stationary source emits a sound wave with a constant frequency f_{S}, with a constant wavelength λ_{S} moving at the speed of sound *v*. Observer *X* moves toward the source with a constant speed v_{O}, and the figure shows the initial and final position of observer *X*. Observer *X* observes a frequency higher than the source frequency. The dotted lines show the position of the waves at $t = 0$. The solid lines show the position of the waves at $t = T_{\mathrm{O}}$.

If the observer is moving away from the source (Figure 17.34), the observed frequency can be found:

$$\begin{aligned} \lambda_{\mathrm{S}} &= vT_{\mathrm{O}} - v_{\mathrm{O}}T_{\mathrm{O}} \\ vT_{\mathrm{S}} &= (v - v_{\mathrm{O}})T_{\mathrm{O}} \\ v\left(\frac{1}{f_{\mathrm{S}}}\right) &= (v - v_{\mathrm{O}})\left(\frac{1}{f_{\mathrm{O}}}\right) \\ f_{\mathrm{O}} &= f_{\mathrm{S}}\left(\frac{v-v_{\mathrm{O}}}{v}\right). \end{aligned}$$

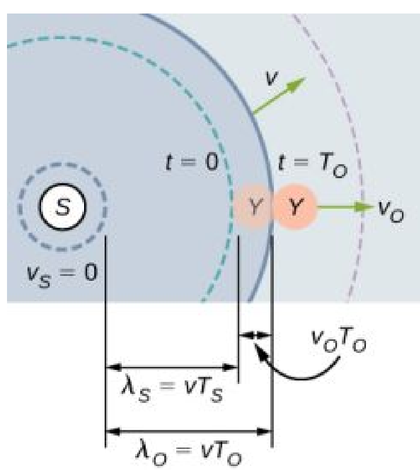


FIGURE 17.34 A stationary source emits a sound wave with a constant frequency f_{S}, with a constant wavelength λ_{S} moving at the speed of sound *v*. Observer *Y* moves away from the source with a constant speed v_{O}, and the figure shows initial and final position of the observer *Y*. Observer *Y* observes a frequency lower than the source frequency. The dotted lines show the position of the waves at $t = 0$. The solid lines show the position of the waves at $t = T_{\mathrm{O}}$.

The equations for an observer moving toward or away from a stationary source can be combined into one equation:

$$f_o = f_s \left(\frac{v \pm v_o}{v} \right), \tag{17.19}$$

where f_o is the observed frequency, f_s is the source frequency, v is the speed of sound, v_o is the speed of the observer, the top sign is for the observer approaching the source and the bottom sign is for the observer departing from the source.

Equation 17.18 and Equation 17.19 can be summarized in one equation (the top sign is for approaching) and is further illustrated in Table 17.4:

$$f_o = f_s \left(\frac{v \pm v_o}{v \mp v_s} \right), \tag{17.20}$$

Doppler shift $f_o = f_s \left(\frac{v \pm v_o}{v \mp v_s} \right)$	Stationary observer	Observer moving towards source	Observer moving away from source
Stationary source	$f_o = f_s$	$f_o = f_s \left(\frac{v+v_o}{v} \right)$	$f_o = f_s \left(\frac{v-v_o}{v} \right)$
Source moving towards observer	$f_o = f_s \left(\frac{v}{v-v_s} \right)$	$f_o = f_s \left(\frac{v+v_o}{v-v_s} \right)$	$f_o = f_s \left(\frac{v-v_o}{v-v_s} \right)$
Source moving away from observer	$f_o = f_s \left(\frac{v}{v+v_s} \right)$	$f_o = f_s \left(\frac{v+v_o}{v+v_s} \right)$	$f_o = f_s \left(\frac{v-v_o}{v+v_s} \right)$

TABLE 17.4

where f_o is the observed frequency, f_s is the source frequency, v is the speed of sound, v_o is the speed of the observer, v_s is the speed of the source, the top sign is for approaching and the bottom sign is for departing.

INTERACTIVE

The Doppler effect involves motion and a video (https://openstax.org/l/21doppler) will help visualize the effects of a moving observer or source. This video shows a moving source and a stationary observer, and a moving observer and a stationary source. It also discusses the Doppler effect and its application to light.

EXAMPLE 17.8

Calculating a Doppler Shift

Suppose a train that has a 150-Hz horn is moving at 35.0 m/s in still air on a day when the speed of sound is 340 m/s.

(a) What frequencies are observed by a stationary person at the side of the tracks as the train approaches and after it passes?

(b) What frequency is observed by the train's engineer traveling on the train?

Strategy

To find the observed frequency in (a), we must use $f_{obs} = f_s \left(\frac{v}{v \mp v_s} \right)$ because the source is moving. The minus sign is used for the approaching train, and the plus sign for the receding train. In (b), there are two Doppler shifts—one for a moving source and the other for a moving observer.

Solution

a. Enter known values into $f_o = f_s \left(\frac{v}{v - v_s}\right)$:

$$f_o = f_s \left(\frac{v}{v - v_s}\right) = (150\,\text{Hz}) \left(\frac{340\,\text{m/s}}{340\,\text{m/s} - 35.0\,\text{m/s}}\right).$$

Calculate the frequency observed by a stationary person as the train approaches:

$$f_o = (150\,\text{Hz})\,(1.11) = 167\,\text{Hz}.$$

Use the same equation with the plus sign to find the frequency heard by a stationary person as the train recedes:

$$f_o = f_s \left(\frac{v}{v + v_s}\right) = (150\,\text{Hz}) \left(\frac{340\,\text{m/s}}{340\,\text{m/s} + 35.0\,\text{m/s}}\right).$$

Calculate the second frequency:

$$f_o = (150\,\text{Hz})\,(0.907) = 136\,\text{Hz}.$$

b. Identify knowns:
   - It seems reasonable that the engineer would receive the same frequency as emitted by the horn, because the relative velocity between them is zero.
   - Relative to the medium (air), the speeds are $v_s = v_o = 35.0\,\text{m/s}$.
   - The first Doppler shift is for the moving observer; the second is for the moving source.

Use the following equation:

$$f_o = \left[f_s \left(\frac{v \pm v_o}{v}\right)\right] \left(\frac{v}{v \mp v_s}\right).$$

The quantity in the square brackets is the Doppler-shifted frequency due to a moving observer. The factor on the right is the effect of the moving source.
Because the train engineer is moving in the direction toward the horn, we must use the plus sign for v_{obs}; however, because the horn is also moving in the direction away from the engineer, we also use the plus sign for v_s. But the train is carrying both the engineer and the horn at the same velocity, so $v_s = v_o$. As a result, everything but f_s cancels, yielding

$$f_o = f_s.$$

Significance

For the case where the source and the observer are not moving together, the numbers calculated are valid when the source (in this case, the train) is far enough away that the motion is nearly along the line joining source and observer. In both cases, the shift is significant and easily noticed. Note that the shift is 17.0 Hz for motion toward and 14.0 Hz for motion away. The shifts are not symmetric.

For the engineer riding in the train, we may expect that there is no change in frequency because the source and observer move together. This matches your experience. For example, there is no Doppler shift in the frequency of conversations between driver and passenger on a motorcycle. People talking when a wind moves the air between them also observe no Doppler shift in their conversation. The crucial point is that source and observer are not moving relative to each other.

CHECK YOUR UNDERSTANDING 17.9

Describe a situation in your life when you might rely on the Doppler shift to help you either while driving a car or walking near traffic.

The Doppler effect and the Doppler shift have many important applications in science and engineering. For example, the Doppler shift in ultrasound can be used to measure blood velocity, and police use the Doppler shift in radar (a microwave) to measure car velocities. In meteorology, the Doppler shift is used to track the motion of storm clouds;

such "Doppler Radar" can give the velocity and direction of rain or snow in weather fronts. In astronomy, we can examine the light emitted from distant galaxies and determine their speed relative to ours. As galaxies move away from us, their light is shifted to a lower frequency, and so to a longer wavelength—the so-called red shift. Such information from galaxies far, far away has allowed us to estimate the age of the universe (from the Big Bang) as about 14 billion years.

17.8 Shock Waves

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the mechanism behind sonic booms
- Describe the difference between sonic booms and shock waves
- Describe a bow wake

When discussing the Doppler effect of a moving source and a stationary observer, the only cases we considered were cases where the source was moving at speeds that were less than the speed of sound. Recall that the observed frequency for a moving source approaching a stationary observer is $f_{\mathrm{O}} = f_{\mathrm{S}}\left(\frac{v}{v - v_{\mathrm{S}}}\right)$. As the source approaches the speed of sound, the observed frequency increases. According to the equation, if the source moves at the speed of sound, the denominator is equal to zero, implying the observed frequency is infinite. If the source moves at speeds greater than the speed of sound, the observed frequency is negative.

What could this mean? What happens when a source approaches the speed of sound? It was once argued by some scientists that such a large pressure wave would result from the constructive interference of the sound waves, that it would be impossible for a plane to exceed the speed of sound because the pressures would be great enough to destroy the airplane. But now planes routinely fly faster than the speed of sound. On July 28, 1976, Captain Eldon W. Joersz and Major George T. Morgan flew a Lockheed SR-71 Blackbird #61-7958 at 3529.60 km/h (2193.20 mi/h), which is Mach 2.85. The Mach number is the speed of the source divided by the speed of sound:

$$M = \frac{v_{\mathrm{S}}}{v}. \tag{17.21}$$

You will see that interesting phenomena occur when a source approaches and exceeds the speed of sound.

Doppler Effect and High Velocity

What happens to the sound produced by a moving source, such as a jet airplane, that approaches or even exceeds the speed of sound? The answer to this question applies not only to sound but to all other waves as well. Suppose a jet plane is coming nearly straight at you, emitting a sound of frequency f_{S}. The greater the plane's speed v_{S}, the greater the Doppler shift and the greater the value observed for f_{O} (Figure 17.35).

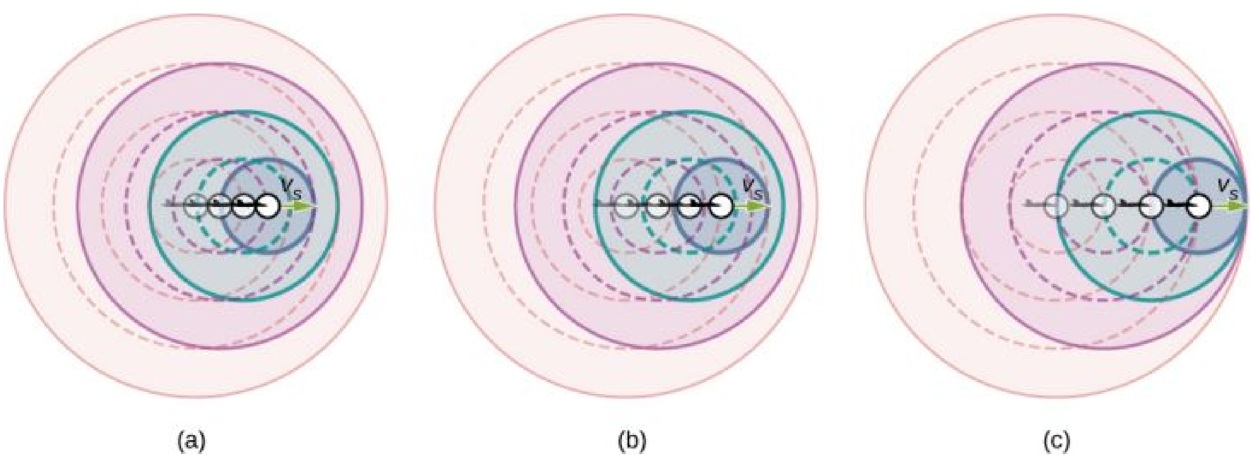


FIGURE 17.35 Because of the Doppler shift, as a moving source approaches a stationary observer, the observed frequency is higher than the source frequency. The faster the source is moving, the higher the observed frequency. In this figure, the source in (b) is moving faster than the source in (a). Shown are four time steps, the first three shown as dotted lines. (c) If a source moves at the speed of sound, each successive wave interfere with the previous one and the observer observes them all at the same instant.

Now, as v_s approaches the speed of sound, f_o approaches infinity, because the denominator in $f_o = f_s\left(\frac{v}{v \mp v_s}\right)$ approaches zero. At the speed of sound, this result means that in front of the source, each successive wave interferes with the previous one because the source moves forward at the speed of sound. The observer gets them all at the same instant, so the frequency is infinite [part (c) of the figure].

Shock Waves and Sonic Booms

If the source exceeds the speed of sound, no sound is received by the observer until the source has passed, so that the sounds from the approaching source are mixed with those from it when receding. This mixing appears messy, but something interesting happens—a shock wave is created (Figure 17.36).

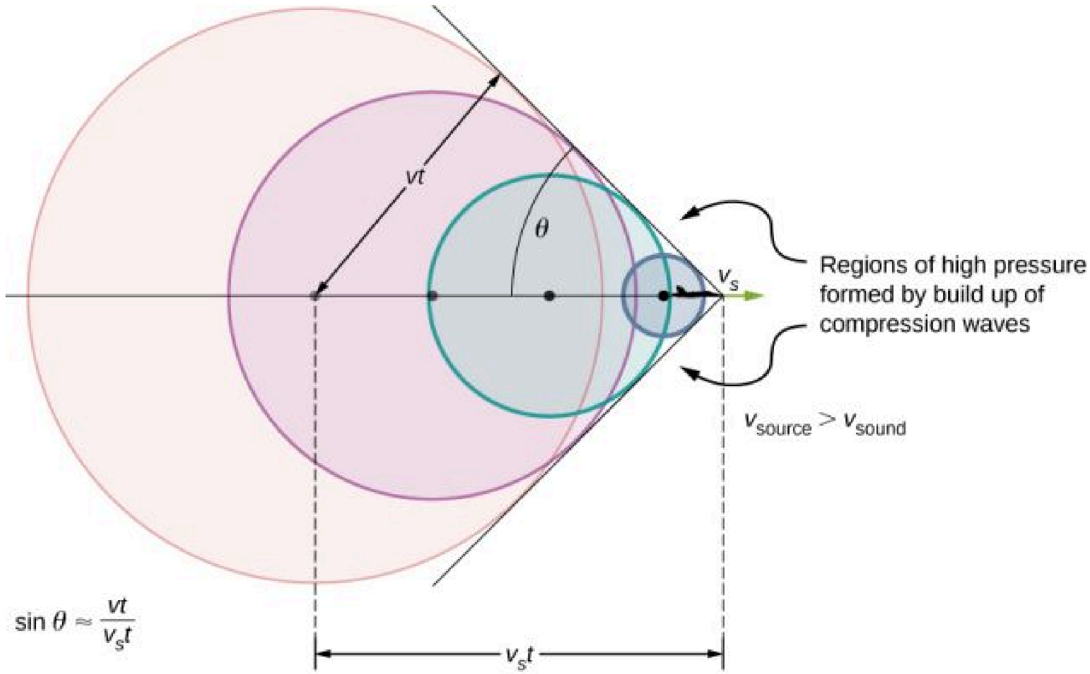


FIGURE 17.36 Sound waves from a source that moves faster than the speed of sound spread spherically from the point where they are emitted, but the source moves ahead of each wave. Constructive interference along the lines shown (actually a cone in three dimensions) creates a shock wave called a sonic boom. The faster the speed of the source, the smaller the angle θ.

Constructive interference along the lines shown (a cone in three dimensions) from similar sound waves arriving there simultaneously. This superposition forms a disturbance called a **shock wave**, a constructive interference of sound created by an object moving faster than sound. Inside the cone, the interference is mostly destructive, so the sound intensity there is much less than on the shock wave. The angle of the shock wave can be found from the geometry. In time t the source has moved $v_s t$ and the sound wave has moved a distance vt and the angle can be found using $\sin\theta = \frac{vt}{v_s t} = \frac{v}{v_s}$. Note that the Mach number is defined as $\frac{v_s}{v}$ so the sine of the angle equals the inverse of the Mach number,

$$\sin\theta = \frac{v}{v_s} = \frac{1}{M}. \qquad (17.22)$$

You may have heard of the common term '**sonic boom**.' A common misconception is that the sonic boom occurs as the plane breaks the sound barrier; that is, accelerates to a speed higher than the speed of sound. Actually, the sonic boom occurs as the shock wave sweeps along the ground.

An aircraft creates two shock waves, one from its nose and one from its tail (Figure 17.37). During television coverage of space shuttle landings, two distinct booms could often be heard. These were separated by exactly the time it would take the shuttle to pass by a point. Observers on the ground often do not see the aircraft creating the sonic boom, because it has passed by before the shock wave reaches them, as seen in the figure. If the aircraft flies close by at low altitude, pressures in the sonic boom can be destructive and break windows as well as rattle nerves. Because of how destructive sonic booms can be, researchers have developed experimental aircraft and components to reduce their impact. Many of these efforts were led by Christine Darden, a "human computer" who advanced to become the head of a NASA program dedicated to predicting and reducing sonic booms. From the 1960's until the late 1990's, Darden employed new materials, innovative designs, and advanced computer simulations. She led experiments involving panels placed on the aircraft that dispersed the shock waves. In recent years, these principles have been revisited and advanced, including in an experimental plane, the X-59, designed for quieter and more efficient flight.

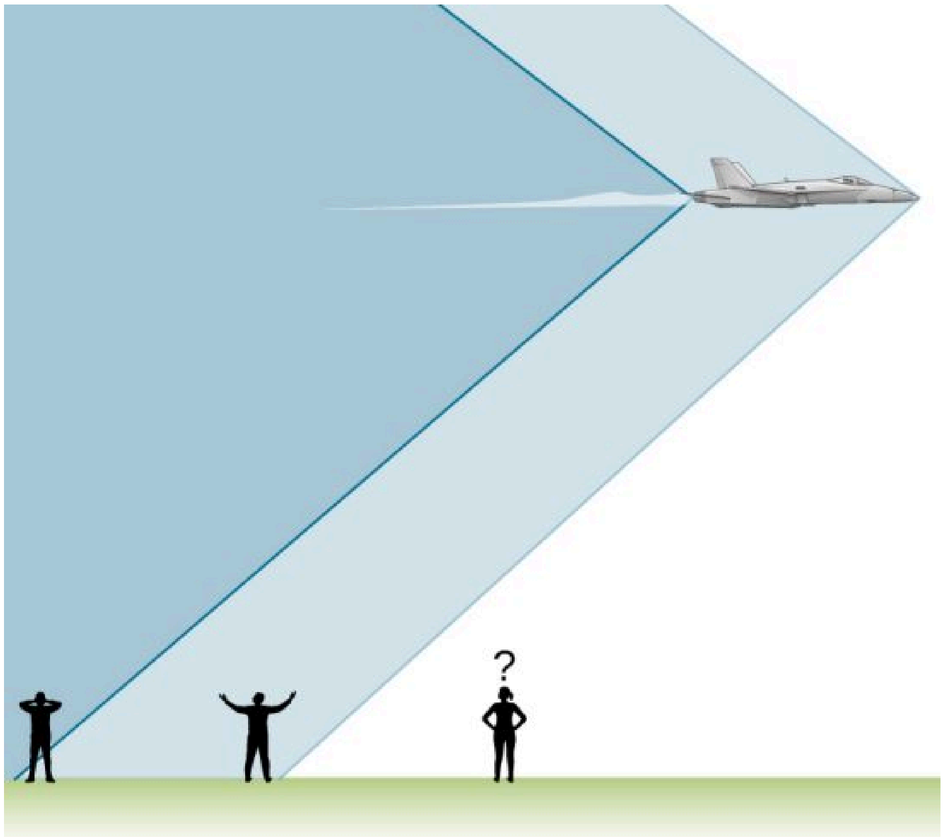


FIGURE 17.37 Two sonic booms experienced by observers, created by the nose and tail of an aircraft as the shock wave sweeps along the ground, are observed on the ground after the plane has passed by.

Shock waves are one example of a broader phenomenon called bow wakes. A **bow wake**, such as the one in Figure 17.38, is created when the wave source moves faster than the wave propagation speed. Water waves spread out in circles from the point where created, and the bow wake is the familiar V-shaped wake, trailing the source. A more exotic bow wake is created when a subatomic particle travels through a medium faster than the speed of light travels in that medium. (In a vacuum, the maximum speed of light is $c = 3.00 \times 10^8$ m/s; in the medium of water, the speed of light is closer to 0.75*c*.) If the particle creates light in its passage, that light spreads on a cone with an angle indicative of the speed of the particle, as illustrated in Figure 17.39. Such a bow wake is called Cerenkov radiation and is commonly observed in particle physics.

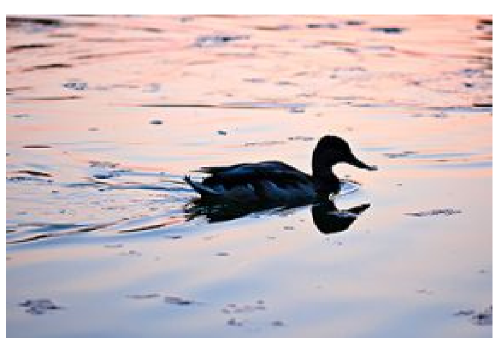


FIGURE 17.38 Bow wake created by a duck. Constructive interference produces the rather structured wake, whereas relatively little wave

action occurs inside the wake, where interference is mostly destructive. (credit: Horia Varlan)

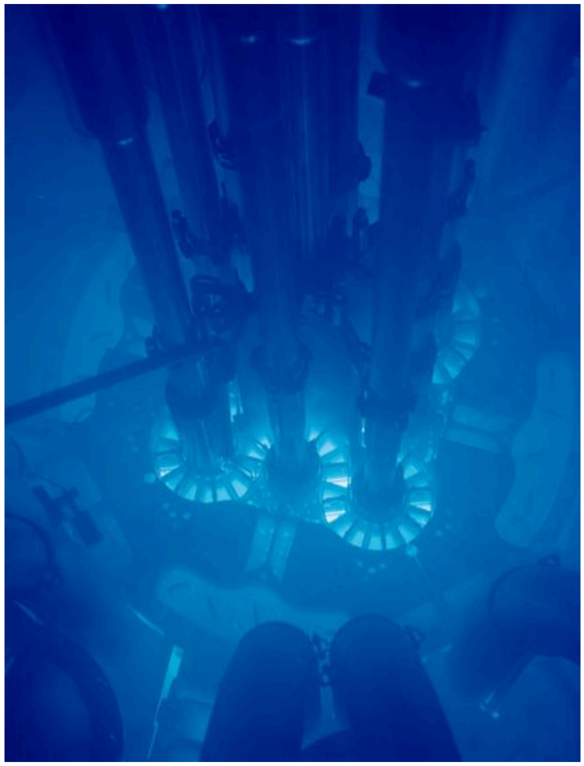


FIGURE 17.39 The blue glow in this research reactor pool is Cerenkov radiation caused by subatomic particles traveling faster than the speed of light in water. (credit: Idaho National Laboratory)

Chapter Review

Key Terms

beat frequency frequency of beats produced by sound waves that differ in frequency

beats constructive and destructive interference of two or more frequencies of sound

bow wake v-shaped disturbance created when the wave source moves faster than the wave propagation speed

Doppler effect alteration in the observed frequency of a sound due to motion of either the source or the observer

Doppler shift actual change in frequency due to relative motion of source and observer

fundamental the lowest-frequency resonance

harmonics the term used to refer collectively to the fundamental and its overtones

hearing perception of sound

loudness perception of sound intensity

notes basic unit of music with specific names, combined to generate tunes

overtones all resonant frequencies higher than the fundamental

phon numerical unit of loudness

pitch perception of the frequency of a sound

shock wave wave front that is produced when a sound source moves faster than the speed of sound

sonic boom loud noise that occurs as a shock wave as it sweeps along the ground

sound traveling pressure wave that may be periodic; the wave can be modeled as a pressure wave or as an oscillation of molecules

sound intensity level unitless quantity telling you the level of the sound relative to a fixed standard

sound pressure level ratio of the pressure amplitude to a reference pressure

timbre number and relative intensity of multiple sound frequencies

transducer device that converts energy of a signal into measurable energy form, for example, a microphone converts sound waves into an electrical signal

Key Equations

Pressure of a sound wave	$\Delta P = \Delta P_{\max} \sin(kx \mp \omega t + \phi)$
Displacement of the oscillating molecules of a sound wave	$s(x,t) = s_{\max} \cos(kx \mp \omega t + \phi)$
Velocity of a wave	$v = f\lambda$
Speed of sound in a fluid	$v = \sqrt{\frac{B}{\rho}}$
Speed of sound in a solid	$v = \sqrt{\frac{Y}{\rho}}$
Speed of sound in an ideal gas	$v = \sqrt{\frac{\gamma RT}{M}}$
Speed of sound in air as a function of temperature	$v = 331 \frac{\text{m}}{\text{s}} \sqrt{\frac{T_{\text{K}}}{273\text{ K}}} = 331 \frac{\text{m}}{\text{s}} \sqrt{1 + \frac{T_{\text{C}}}{273°\text{C}}}$
Decrease in intensity as a spherical wave expands	$I_2 = I_1 \left(\frac{r_1}{r_2}\right)^2$
Intensity averaged over a period	$I = \frac{\langle P \rangle}{A}$
Intensity of sound	$I = \frac{(\Delta p_{\max})^2}{2\rho v}$
Sound intensity level	$\beta(dB) = 10 \log_{10}\left(\frac{I}{I_0}\right)$
Resonant wavelengths of a tube closed at one end	$\lambda_n = \frac{4}{n}L, \quad n = 1, 3, 5, \ldots$
Resonant frequencies of a tube closed at one end	$f_n = n\frac{v}{4L}, \quad n = 1, 3, 5, \ldots$
Resonant wavelengths of a tube open at both ends	$\lambda_n = \frac{2}{n}L, \quad n = 1, 2, 3, \ldots$
Resonant frequencies of a tube open at both ends	$f_n = n\frac{v}{2L}, \quad n = 1, 2, 3, \ldots$
Beat frequency produced by two waves that differ in frequency	$f_{\text{beat}} = \lvert f_2 - f_1 \rvert$
Observed frequency for a stationary observer and a moving source	$f_{\text{o}} = f_{\text{s}} \left(\frac{v}{v \mp v_{\text{s}}}\right)$

Observed frequency for a moving observer and a stationary source	$f_o = f_s\left(\frac{v \pm v_o}{v}\right)$
Doppler shift for the observed frequency	$f_o = f_s\left(\frac{v \pm v_o}{v \mp v_s}\right)$
Mach number	$M = \frac{v_s}{v}$
Sine of angle formed by shock wave	$\sin\theta = \frac{v}{v_s} = \frac{1}{M}$

Summary

17.1 Sound Waves

- Sound is a disturbance of matter (a pressure wave) that is transmitted from its source outward. Hearing is the perception of sound.
- Sound can be modeled in terms of pressure or in terms of displacement of molecules.
- The human ear is sensitive to frequencies between 20 Hz and 20 kHz.

17.2 Speed of Sound

- The speed of sound depends on the medium and the state of the medium.
- In a fluid, because the absence of shear forces, sound waves are longitudinal. A solid can support both longitudinal and transverse sound waves.
- In air, the speed of sound is related to air temperature *T* by $v = 331\frac{\text{m}}{\text{s}}\sqrt{\frac{T_\text{K}}{273\text{ K}}} = 331\frac{\text{m}}{\text{s}}\sqrt{1 + \frac{T_\text{C}}{273°\text{C}}}.$
- *v* is the same for all frequencies and wavelengths of sound in air.

17.3 Sound Intensity

- Intensity $I = P/A$ is the same for a sound wave as was defined for all waves, where *P* is the power crossing area *A*. The SI unit for *I* is watts per meter squared. The intensity of a sound wave is also related to the pressure amplitude Δp:
$$I = \frac{(\Delta p)^2}{2\,\rho v},$$
where ρ is the density of the medium in which the sound wave travels and v_w is the speed of sound in the medium.
- Sound intensity level in units of decibels (dB) is
$$\beta(\text{dB}) = 10\log_{10}\left(\frac{I}{I_0}\right),$$
where $I_0 = 10^{-12}\text{ W/m}^2$ is the threshold intensity of hearing.
- The perception of frequency is pitch. The perception of intensity is loudness and loudness has units of phons.

17.4 Normal Modes of a Standing Sound Wave

- Unwanted sound can be reduced using destructive interference.
- Sound has the same properties of interference and resonance as defined for all waves.
- In air columns, the lowest-frequency resonance is called the fundamental, whereas all higher resonant frequencies are called overtones. Collectively, they are called harmonics.

17.5 Sources of Musical Sound

- Some musical instruments can be modeled as pipes that have symmetrical boundary conditions: open at both ends or closed at both ends. Other musical instruments can be modeled as pipes that have anti-symmetrical boundary conditions: closed at one end and open at the other.
- Some instruments, such as the pipe organ, have several tubes with different lengths. Instruments such as the flute vary the length of the tube by closing the holes along the tube. The trombone varies the length of the tube using a sliding bar.
- String instruments produce sound using a vibrating string with nodes at each end. The air around the string oscillates at the frequency of the string. The relationship for the frequencies for the string is the same as for the symmetrical boundary conditions of the pipe, with the length of the pipe replaced by the length of the string and the velocity replaced by $v = \sqrt{\frac{F_T}{\mu}}$.

17.6 Beats

- When two sound waves that differ in frequency interfere, beats are created with a beat frequency that is equal to the absolute value of the difference in the frequencies.

17.7 The Doppler Effect

- The Doppler effect is an alteration in the observed frequency of a sound due to motion of either the source or the observer.
- The actual change in frequency is called the

Doppler shift.

17.8 Shock Waves

- The Mach number is the velocity of a source divided by the speed of sound, $M = \frac{v_S}{v}$.
- When a sound source moves faster than the speed of sound, a shock wave is produced as the sound waves interfere.
- A sonic boom is the intense sound that occurs as the shock wave moves along the ground.
- The angle the shock wave produces can be found as $\sin\theta = \frac{v}{v_S} = \frac{1}{M}$.
- A bow wake is produced when an object moves faster than the speed of a mechanical wave in the medium, such as a boat moving through the water.

Conceptual Questions

17.1 Sound Waves

1. What is the difference between sound and hearing?

2. You will learn that light is an electromagnetic wave that can travel through a vacuum. Can sound waves travel through a vacuum?

3. Sound waves can be modeled as a change in pressure. Why is the change in pressure used and not the actual pressure?

17.2 Speed of Sound

4. How do sound vibrations of atoms differ from thermal motion?

5. When sound passes from one medium to another where its propagation speed is different, does its frequency or wavelength change? Explain your answer briefly.

6. A popular party trick is to inhale helium and speak in a high-frequency, funny voice. Explain this phenomenon.

7. You may have used a sonic range finder in lab to measure the distance of an object using a clicking sound from a sound transducer. What is the principle used in this device?

8. The sonic range finder discussed in the preceding question often needs to be calibrated. During the calibration, the software asks for the room temperature. Why do you suppose the room temperature is required?

17.3 Sound Intensity

9. Six members of a synchronized swim team wear earplugs to protect themselves against water pressure at depths, but they can still hear the music and perform the combinations in the water perfectly. One day, they were asked to leave the pool so the dive team could practice a few dives, and they tried to practice on a mat, but seemed to have a lot more difficulty. Why might this be?

10. A community is concerned about a plan to bring train service to their downtown from the town's outskirts. The current sound intensity level, even though the rail yard is blocks away, is 70 dB downtown. The mayor assures the public that there will be a difference of only 30 dB in sound in the downtown area. Should the townspeople be concerned? Why?

17.4 Normal Modes of a Standing Sound Wave

11. You are given two wind instruments of identical length. One is open at both ends, whereas the other is closed at one end. Which is able to produce the lowest frequency?

12. What is the difference between an overtone and a harmonic? Are all harmonics overtones? Are all overtones harmonics?

13. Two identical columns, open at both ends, are in separate rooms. In room *A*, the temperature is $T = 20°\text{C}$ and in room *B*, the temperature is $T = 25°\text{C}$. A speaker is attached to the end of each tube, causing the tubes to resonate at the fundamental frequency. Is the frequency the same for both tubes? Which has the higher frequency?

17.5 Sources of Musical Sound

14. How does an unamplified guitar produce sounds so much more intense than those of a plucked string held taut by a simple stick?

15. Consider three pipes of the same length (*L*). Pipe *A* is open at both ends, pipe *B* is closed at both ends, and pipe *C* has one open end and one closed end. If the velocity of sound is the same in each of the three tubes, in which of the tubes could the lowest fundamental frequency be produced? In which of the tubes could the highest fundamental frequency be produced?

16. Pipe *A* has a length *L* and is open at both ends.

Pipe *B* has a length *L*/2 and has one open end and one closed end. Assume the speed of sound to be the same in both tubes. Which of the harmonics in each tube would be equal?

17. A string is tied between two lab posts a distance *L* apart. The tension in the string and the linear mass density is such that the speed of a wave on the string is $v = 343 \text{ m/s}$. A tube with symmetric boundary conditions has a length *L* and the speed of sound in the tube is $v = 343 \text{ m/s}$. What could be said about the frequencies of the harmonics in the string and the tube? What if the velocity in the string were $v = 686 \text{ m/s}$?

17.6 Beats

18. Two speakers are attached to variable-frequency signal generator. Speaker *A* produces a constant-frequency sound wave of 1.00 kHz, and speaker *B* produces a tone of 1.10 kHz. The beat frequency is 0.10 kHz. If the frequency of each speaker is doubled, what is the beat frequency produced?

19. The label has been scratched off a tuning fork and you need to know its frequency. From its size, you suspect that it is somewhere around 250 Hz. You find a 250-Hz tuning fork and a 270-Hz tuning fork. When you strike the 250-Hz fork and the fork of unknown frequency, a beat frequency of 5 Hz is produced. When you strike the unknown with the 270-Hz fork, the beat frequency is 15 Hz. What is the unknown frequency? Could you have deduced the frequency using just the 250-Hz fork?

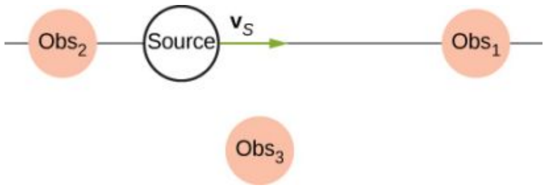
20. Referring to the preceding question, if you had only the 250-Hz fork, could you come up with a solution to the problem of finding the unknown frequency?

21. A "showy" custom-built car has two brass horns that are supposed to produce the same frequency but actually emit 263.8 and 264.5 Hz. What beat frequency is produced?

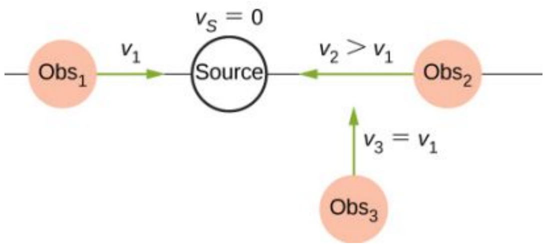
17.7 The Doppler Effect

22. Is the Doppler shift real or just a sensory illusion?

23. Three stationary observers observe the Doppler shift from a source moving at a constant velocity. The observers are stationed as shown below. Which observer will observe the highest frequency? Which observer will observe the lowest frequency? What can be said about the frequency observed by observer 3?



24. Shown below is a stationary source and moving observers. Describe the frequencies observed by the observers for this configuration.



25. Prior to 1980, conventional radar was used by weather forecasters. In the 1960s, weather forecasters began to experiment with Doppler radar. What do you think is the advantage of using Doppler radar?

17.8 Shock Waves

26. What is the difference between a sonic boom and a shock wave?

27. Due to efficiency considerations related to its bow wake, the supersonic transport aircraft must maintain a cruising speed that is a constant ratio to the speed of sound (a constant Mach number). If the aircraft flies from warm air into colder air, should it increase or decrease its speed? Explain your answer.

28. When you hear a sonic boom, you often cannot see the plane that made it. Why is that?

Problems

17.1 Sound Waves

29. Consider a sound wave modeled with the equation $s(x, t) = 4.00 \text{ nm} \cos\left(3.66 \text{ m}^{-1} x - 1256 \text{ s}^{-1} t\right)$. What is the maximum displacement, the wavelength, the frequency, and the speed of the sound wave?

30. Consider a sound wave moving through the air modeled with the equation $s(x, t) = 6.00 \text{ nm}$

$\cos\left(54.93\ \text{m}^{-1}x - 18.84\ \times\ 10^3\ \text{s}^{-1}t\right)$. What is the shortest time required for an air molecule to move between 3.00 nm and –3.00 nm?

31. Consider a diagnostic ultrasound of frequency 5.00 MHz that is used to examine an irregularity in soft tissue. (a) What is the wavelength in air of such a sound wave if the speed of sound is 343 m/s? (b) If the speed of sound in tissue is 1800 m/s, what is the wavelength of this wave in tissue?

32. A sound wave is modeled as $\Delta P = 1.80\ \text{Pa}\sin\left(55.41\ \text{m}^{-1}\ x - 18{,}840\ \text{s}^{-1}t\right)$. What is the maximum change in pressure, the wavelength, the frequency, and the speed of the sound wave?

33. A sound wave is modeled with the wave function $\Delta P = 1.20\ \text{Pa}\sin\left(kx - 6.28\ \times\ 10^4\text{s}^{-1}t\right)$ and the sound wave travels in air at a speed of $v = 343.00\ \text{m/s}$. (a) What is the wave number of the sound wave? (b) What is the value for $\Delta P\,(3.00\ \text{m}, 20.00\ \text{s})$?

34. The displacement of the air molecules in sound wave is modeled with the wave function $s\,(x, t) = 5.00\ \text{nm}$ $\cos\left(91.54\ \text{m}^{-1}x - 3.14\ \times\ 10^4\ \text{s}^{-1}t\right)$. (a) What is the wave speed of the sound wave? (b) What is the maximum speed of the air molecules as they oscillate in simple harmonic motion? (c) What is the magnitude of the maximum acceleration of the air molecules as they oscillate in simple harmonic motion?

35. A speaker is placed at the opening of a long horizontal tube. The speaker oscillates at a frequency *f*, creating a sound wave that moves down the tube. The wave moves through the tube at a speed of $v = 340.00\ \text{m/s}$. The sound wave is modeled with the wave function $s\,(x, t) = s_{\text{max}}\cos\,(kx - \omega t + \phi)$. At time $t = 0.00\ \text{s}$, an air molecule at $x = 3.5\ \text{m}$ is at the maximum displacement of 7.00 nm. At the same time, another molecule at $x = 3.7\ \text{m}$ has a displacement of 3.00 nm. What is the frequency at which the speaker is oscillating?

36. A 250-Hz tuning fork is struck and begins to vibrate. A sound-level meter is located 34.00 m away. It takes the sound $\Delta t = 0.10\ \text{s}$ to reach the meter. The maximum displacement of the tuning fork is 1.00 mm. Write a wave function for the sound.

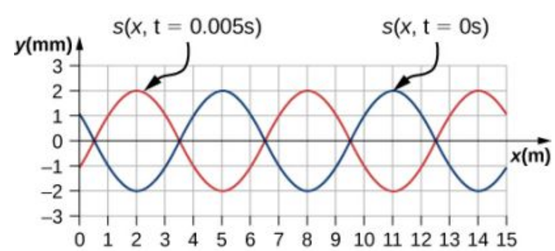
37. A sound wave produced by an ultrasonic transducer, moving in air, is modeled with the wave equation $s\,(x, t) = 4.50\ \text{nm}\cos\left(9.15\ \times\ 10^4\ \text{m}^{-1}x - 2\pi\,(5.00\ \text{MHz})\,t\right)$. The transducer is to be used in nondestructive testing to test for fractures in steel beams. The speed of sound in the steel beam is $v = 5950\ \text{m/s}$. Find the wave function for the sound wave in the steel beam.

38. Porpoises emit sound waves that they use for navigation. If the wavelength of the sound wave emitted is 4.5 cm, and the speed of sound in the water is $v = 1530\ \text{m/s}$, what is the period of the sound?

39. Bats use sound waves to catch insects. Bats can detect frequencies up to 100 kHz. If the sound waves travel through air at a speed of $v = 343\ \text{m/s}$, what is the wavelength of the sound waves?

40. A bat sends of a sound wave 100 kHz and the sound waves travel through air at a speed of $v = 343\ \text{m/s}$. (a) If the maximum pressure difference is 1.30 Pa, what is a wave function that would model the sound wave, assuming the wave is sinusoidal? (Assume the phase shift is zero.) (b) What are the period and wavelength of the sound wave?

41. Consider the graph shown below of a compression wave. Shown are snapshots of the wave function for $t = 0.000\ \text{s}$ (blue) and $t = 0.005\ \text{s}$ (orange). What are the wavelength, maximum displacement, velocity, and period of the compression wave?



42. Consider the graph in the preceding problem of a compression wave. Shown are snapshots of the wave function for $t = 0.000\ \text{s}$ (blue) and $t = 0.005\ \text{s}$ (orange). Given that the displacement of the molecule at time $t = 0.00\ \text{s}$ and position $x = 0.00\ \text{m}$ is $s\,(0.00\ \text{m}, 0.00\ \text{s}) = 1.08\ \text{mm}$, derive a wave function to model the compression wave.

43. A guitar string oscillates at a frequency of 100 Hz and produces a sound wave. (a) What do you think the frequency of the sound wave is that the vibrating string produces? (b) If the speed of the sound wave is $v = 343\text{ m/s}$, what is the wavelength of the sound wave?

17.2 Speed of Sound

44. When poked by a spear, an operatic soprano lets out a 1200-Hz shriek. What is its wavelength if the speed of sound is 345 m/s?

45. What frequency sound has a 0.10-m wavelength when the speed of sound is 340 m/s?

46. Calculate the speed of sound on a day when a 1500-Hz frequency has a wavelength of 0.221 m.

47. (a) What is the speed of sound in a medium where a 100-kHz frequency produces a 5.96-cm wavelength? (b) Which substance in Table 17.1 is this likely to be?

48. Show that the speed of sound in $20.0°\text{C}$ air is 343 m/s, as claimed in the text.

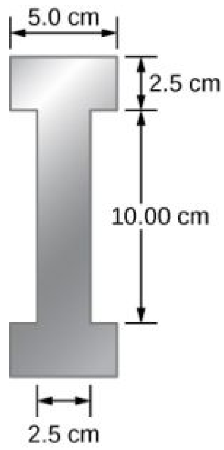
49. Air temperature in the Sahara Desert can reach $56.0°\text{C}$ (about $134°\text{F}$). What is the speed of sound in air at that temperature?

50. Dolphins make sounds in air and water. What is the ratio of the wavelength of a sound in air to its wavelength in seawater? Assume air temperature is $20.0°\text{C}$.

51. A sonar echo returns to a submarine 1.20 s after being emitted. What is the distance to the object creating the echo? (Assume that the submarine is in the ocean, not in fresh water.)

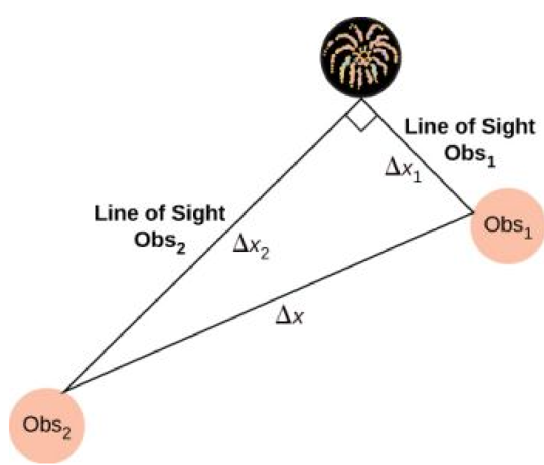
52. (a) If a submarine's sonar can measure echo times with a precision of 0.0100 s, what is the smallest difference in distances it can detect? (Assume that the submarine is in the ocean, not in fresh water.) (b) Discuss the limits this time resolution imposes on the ability of the sonar system to detect the size and shape of the object creating the echo.

53. Ultrasonic sound waves are often used in methods of nondestructive testing. For example, this method can be used to find structural faults in a steel I-beams used in building. Consider a 10.00 meter long, steel I-beam with a cross-section shown below. The weight of the I-beam is 3846.50 N. What would be the speed of sound through in the I-beam? $(Y_{\text{steel}} = 200\text{ GPa}, \beta_{\text{steel}} = 159\text{ GPa})$.



54. A physicist at a fireworks display times the lag between seeing an explosion and hearing its sound, and finds it to be 0.400 s. (a) How far away is the explosion if air temperature is $24.0°\text{C}$ and if you neglect the time taken for light to reach the physicist? (b) Calculate the distance to the explosion taking the speed of light into account. Note that this distance is negligibly greater.

55. During a 4th of July celebration, an M80 firework explodes on the ground, producing a bright flash and a loud bang. The air temperature of the night air is $T_{\text{F}} = 95.00°\text{F}$. Two observers see the flash and hear the bang. The first observer notes the time between the flash and the bang as 0.10 second. The second observer notes the difference as 0.15 seconds. The line of sight between the two observers meet at a right angle as shown below. What is the distance Δx between the two observers?



56. The density of a sample of water is $\rho = 998.00\text{ kg/m}^3$ and the bulk modulus is $\beta = 2.15\text{ GPa}$. What is the speed of sound through the sample?

57. Suppose a bat uses sound echoes to locate its insect prey, 3.00 m away. (See Figure 17.6.) (a) Calculate the echo times for temperatures of $5.00°C$ and $35.0°C$. (b) What percent uncertainty does this cause for the bat in locating the insect? (c) Discuss the significance of this uncertainty and whether it could cause difficulties for the bat. (In practice, the bat continues to use sound as it closes in, eliminating most of any difficulties imposed by this and other effects, such as motion of the prey.)

17.3 Sound Intensity

58. What is the intensity in watts per meter squared of a 85.0-dB sound?

59. The warning tag on a lawn mower states that it produces noise at a level of 91.0 dB. What is the intensity of this sound in watts per meter squared?

60. A sound wave traveling in air has a pressure amplitude of 0.5 Pa. What is the intensity of the wave?

61. What intensity level does the sound in the preceding problem correspond to?

62. What sound intensity level in dB is produced by earphones that create an intensity of $4.00 \times 10^{-2}\ \text{W/m}^2$?

63. What is the decibel level of a sound that is twice as intense as a 90.0-dB sound? (b) What is the decibel level of a sound that is one-fifth as intense as a 90.0-dB sound?

64. What is the intensity of a sound that has a level 7.00 dB lower than a $4.00 \times 10^{-9}\text{-W/m}^2$ sound? (b) What is the intensity of a sound that is 3.00 dB higher than a $4.00 \times 10^{-9}\text{-W/m}^2$ sound?

65. People with good hearing can perceive sounds as low as −8.00 dB at a frequency of 3000 Hz. What is the intensity of this sound in watts per meter squared?

66. If a large housefly 3.0 m away from you makes a noise of 40.0 dB, what is the noise level of 1000 flies at that distance, assuming interference has a negligible effect?

67. Ten cars in a circle at a boom box competition produce a 120-dB sound intensity level at the center of the circle. What is the average sound intensity level produced there by each stereo, assuming interference effects can be neglected?

68. The amplitude of a sound wave is measured in terms of its maximum gauge pressure. By what factor does the amplitude of a sound wave increase if the sound intensity level goes up by 40.0 dB?

69. If a sound intensity level of 0 dB at 1000 Hz corresponds to a maximum gauge pressure (sound amplitude) of 10^{-9} atm, what is the maximum gauge pressure in a 60-dB sound? What is the maximum gauge pressure in a 120-dB sound?

70. An 8-hour exposure to a sound intensity level of 90.0 dB may cause hearing damage. What energy in joules falls on a 0.800-cm-diameter eardrum so exposed?

71. Sound is more effectively transmitted into a stethoscope by direct contact rather than through the air, and it is further intensified by being concentrated on the smaller area of the eardrum. It is reasonable to assume that sound is transmitted into a stethoscope 100 times as effectively compared with transmission though the air. What, then, is the gain in decibels produced by a stethoscope that has a sound gathering area of $15.0\ \text{cm}^2$, and concentrates the sound onto two eardrums with a total area of $0.900\ \text{cm}^2$ with an efficiency of 40.0%?

72. Loudspeakers can produce intense sounds with surprisingly small energy input in spite of their low efficiencies. Calculate the power input needed to produce a 90.0-dB sound intensity level for a 12.0-cm-diameter speaker that has an efficiency of 1.00%. (This value is the sound intensity level right at the speaker.)

73. The factor of 10^{-12} in the range of intensities to which the ear can respond, from threshold to that causing damage after brief exposure, is truly remarkable. If you could measure distances over the same range with a single instrument and the smallest distance you could measure was 1 mm, what would the largest be?

74. What are the closest frequencies to 500 Hz that an average person can clearly distinguish as being different in frequency from 500 Hz? The sounds are not present simultaneously.

75. Can you tell that your roommate turned up the sound on the TV if its average sound intensity level goes from 70 to 73 dB?

76. If a woman needs an amplification of 5.0×10^5 times the threshold intensity to enable her to hear at all frequencies, what is her overall hearing loss in dB? Note that smaller amplification is appropriate for more intense sounds to avoid further damage to her hearing from levels above 90 dB.

77. A person has a hearing threshold 10 dB above normal at 100 Hz and 50 dB above normal at 4000 Hz. How much more intense must a 100-Hz tone be than a 4000-Hz tone if they are both barely audible to this person?

17.4 Normal Modes of a Standing Sound Wave

78. (a) What is the fundamental frequency of a 0.672-m-long tube, open at both ends, on a day when the speed of sound is 344 m/s? (b) What is the frequency of its second harmonic?

79. What is the length of a tube that has a fundamental frequency of 176 Hz and a first overtone of 352 Hz if the speed of sound is 343 m/s?

80. The ear canal resonates like a tube closed at one end. (See Figure 17.14.) If ear canals range in length from 1.80 to 2.60 cm in an average population, what is the range of fundamental resonant frequencies? Take air temperature to be $37.0°C$, which is the same as body temperature.

81. Calculate the first overtone in an ear canal, which resonates like a 2.40-cm-long tube closed at one end, by taking air temperature to be $37.0°C$. Is the ear particularly sensitive to such a frequency? (The resonances of the ear canal are complicated by its nonuniform shape, which we shall ignore.)

82. A crude approximation of voice production is to consider the breathing passages and mouth to be a resonating tube closed at one end. (a) What is the fundamental frequency if the tube is 0.240 m long, by taking air temperature to be $37.0°C$? (b) What would this frequency become if the person replaced the air with helium? Assume the same temperature dependence for helium as for air.

83. A 4.0-m-long pipe, open at one end and closed at one end, is in a room where the temperature is $T = 22°C$. A speaker capable of producing variable frequencies is placed at the open end and is used to cause the tube to resonate. (a) What is the wavelength and the frequency of the fundamental frequency? (b) What is the frequency and wavelength of the first overtone?

84. A 4.0-m-long pipe, open at both ends, is placed in a room where the temperature is $T = 25°C$. A speaker capable of producing variable frequencies is placed at the open end and is used to cause the tube to resonate. (a) What are the wavelength and the frequency of the fundamental frequency? (b) What are the frequency and wavelength of the first overtone?

85. A nylon guitar string is fixed between two lab posts 2.00 m apart. The string has a linear mass density of $\mu = 7.20 \text{ g/m}$ and is placed under a tension of 160.00 N. The string is placed next to a tube, open at both ends, of length *L*. The string is plucked and resonates at the fundamental frequency while the tube resonates at the $n = 3$ mode. The speed of sound is 343 m/s. What is the length of the tube?

86. A 512-Hz tuning fork is struck and placed next to a tube with a movable piston, creating a tube with a variable length. The piston is slid down the pipe and resonance is reached when the piston is 115.50 cm from the open end. The next resonance is reached when the piston is 82.50 cm from the open end. (a) What is the speed of sound in the tube? (b) How far from the open end will the piston cause the next mode of resonance?

87. Students in a physics lab are asked to find the length of an air column. They obtain a long tube that is closed at one end that has a fundamental frequency of 256 Hz. They hold the tube vertically and fill it with water to the top, then lower the water while a 256-Hz tuning fork is rung and listen for the first resonance. (a) What is the air temperature if the resonance occurs for a length of 0.336 m? (b) At what length will they observe the second resonance (first overtone)?

17.5 Sources of Musical Sound

88. If a wind instrument, such as a tuba, has a fundamental frequency of 32.0 Hz, what are its first three overtones? It is closed at one end. (The overtones of a real tuba are more complex than this example, because it is a tapered tube.)

89. What are the first three overtones of a bassoon that has a fundamental frequency of 90.0 Hz? It is open at both ends. (The overtones of a real bassoon are more complex than this example, because its double reed makes it act more like a tube closed at one end.)

90. How long must a flute be in order to have a fundamental frequency of 262 Hz (this frequency corresponds to middle C on the evenly tempered chromatic scale) on a day when air temperature is $20.0°C$? It is open at both ends.

91. What length should an oboe have to produce a fundamental frequency of 110 Hz on a day when the speed of sound is 343 m/s? It is open at both ends.

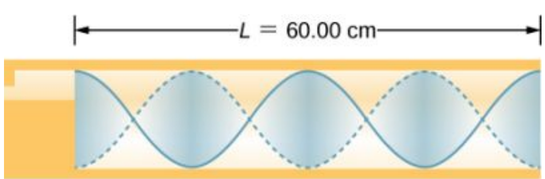
92. (a) Find the length of an organ pipe closed at one end that produces a fundamental frequency of 256 Hz when air temperature is $18.0°C$. (b) What is its fundamental frequency at $25.0°C$?

93. An organ pipe $(L = 3.00 \text{ m})$ is closed at both ends. Compute the wavelengths and frequencies of the first three modes of resonance. Assume the speed of sound is $v = 343.00 \text{ m/s}$.

94. An organ pipe $(L = 3.00 \text{ m})$ is closed at one end. Compute the wavelengths and frequencies of the first three modes of resonance. Assume the speed of sound is $v = 343.00 \text{ m/s}$.

95. A sound wave of a frequency of 2.00 kHz is produced by a string oscillating in the $n = 6$ mode. The linear mass density of the string is $\mu = 0.0065 \text{ kg/m}$ and the length of the string is 1.50 m. What is the tension in the string?

96. Consider the sound created by resonating the tube shown below. The air temperature is $T_C = 30.00°C$. What are the wavelength, wave speed, and frequency of the sound produced?



97. A student holds an 80.00-cm lab pole one quarter of the length from the end of the pole. The lab pole is made of aluminum. The student strikes the lab pole with a hammer. The pole resonates at the lowest possible frequency. What is that frequency?

98. A string on the violin has a length of 24.00 cm and a mass of 0.860 g. The fundamental frequency of the string is 1.00 kHz. (a) What is the speed of the wave on the string? (b) What is the tension in the string?

99. By what fraction will the frequencies produced by a wind instrument change when air temperature goes from $10.0°C$ to $30.0°C$? That is, find the ratio of the frequencies at those temperatures.

17.6 Beats

100. What beat frequencies are present: (a) If the musical notes A and C are played together (frequencies of 220 and 264 Hz)? (b) If D and F are played together (frequencies of 297 and 352 Hz)? (c) If all four are played together?

101. What beat frequencies result if a piano hammer hits three strings that emit frequencies of 127.8, 128.1, and 128.3 Hz?

102. A piano tuner hears a beat every 2.00 s when listening to a 264.0-Hz tuning fork and a single piano string. What are the two possible frequencies of the string?

103. Two identical strings, of identical lengths of 2.00 m and linear mass density of $\mu = 0.0065 \text{ kg/m}$, are fixed on both ends. String *A* is under a tension of 120.00 N. String *B* is under a tension of 130.00 N. They are each plucked and produce sound at the $n = 10$ mode. What is the beat frequency?

104. A piano tuner uses a 512-Hz tuning fork to tune a piano. He strikes the fork and hits a key on the piano and hears a beat frequency of 5 Hz. He tightens the string of the piano, and repeats the procedure. Once again he hears a beat frequency of 5 Hz. What happened?

105. A string with a linear mass density of $\mu = 0.0062 \text{ kg/m}$ is stretched between two posts 1.30 m apart. The tension in the string is 150.00 N. The string oscillates and produces a sound wave. A 1024-Hz tuning fork is struck and the beat frequency between the two sources is 52.83 Hz. What are the possible frequency and wavelength of the wave on the string?

106. A car has two horns, one emitting a frequency of 199 Hz and the other emitting a frequency of 203 Hz. What beat frequency do they produce?

107. The middle C hammer of a piano hits two strings, producing beats of 1.50 Hz. One of the strings is tuned to 260.00 Hz. What frequencies could the other string have?

108. Two tuning forks having frequencies of 460 and 464 Hz are struck simultaneously. What average frequency will you hear, and what will the beat frequency be?

109. Twin jet engines on an airplane are producing an average sound frequency of 4100 Hz with a beat frequency of 0.500 Hz. What are their individual frequencies?

110. Three adjacent keys on a piano (F, F-sharp, and G) are struck simultaneously, producing frequencies of 349, 370, and 392 Hz. What beat frequencies are produced by this discordant combination?

17.7 The Doppler Effect

111. (a) What frequency is received by a person watching an oncoming ambulance moving at 110 km/h and emitting a steady 800-Hz sound from its siren? The speed of sound on this day is 345 m/s. (b) What frequency does she receive after the ambulance has passed?

112. (a) At an air show a jet flies directly toward the stands at a speed of 1200 km/h, emitting a frequency of 3500 Hz, on a day when the speed of sound is 342 m/s. What frequency is received by the observers? (b) What frequency do they receive as the plane flies directly away from them?

113. What frequency is received by a mouse just before being dispatched by a hawk flying at it at 25.0 m/s and emitting a screech of frequency 3500 Hz? Take the speed of sound to be 331 m/s.

114. A spectator at a parade receives an 888-Hz tone from an oncoming trumpeter who is playing an 880-Hz note. At what speed is the musician approaching if the speed of sound is 338 m/s?

115. A commuter train blows its 200-Hz horn as it approaches a crossing. The speed of sound is 335 m/s. (a) An observer waiting at the crossing receives a frequency of 208 Hz. What is the speed of the train? (b) What frequency does the observer receive as the train moves away?

116. Can you perceive the shift in frequency produced when you pull a tuning fork toward you at 10.0 m/s on a day when the speed of sound is 344 m/s? To answer this question, calculate the factor by which the frequency shifts and see if it is greater than 0.300%.

117. Two eagles fly directly toward one another, the first at 15.0 m/s and the second at 20.0 m/s. Both screech, the first one emitting a frequency of 3200 Hz and the second one emitting a frequency of 3800 Hz. What frequencies do they receive if the speed of sound is 330 m/s?

118. Student *A* runs down the hallway of the school at a speed of $v_0 = 5.00 \text{ m/s}$, carrying a ringing 1024.00-Hz tuning fork toward a concrete wall. The speed of sound is $v = 343.00 \text{ m/s}$. Student *B* stands at rest at the wall. (a) What is the frequency heard by student *B*? (b) What is the beat frequency heard by student *A*?

119. An ambulance with a siren $(f = 1.00 \text{kHz})$ blaring is approaching an accident scene. The ambulance is moving at 70.00 mph. A nurse is approaching the scene from the opposite direction, running at $v_o = 7.00 \text{ m/s}$. What frequency does the nurse observe? Assume the speed of sound is $v = 343.00 \text{ m/s}$.

120. The frequency of the siren of an ambulance is 900 Hz and is approaching you. You are standing on a corner and observe a frequency of 960 Hz. What is the speed of the ambulance (in mph) if the speed of sound is $v = 340.00 \text{ m/s}$?

121. What is the minimum speed at which a source must travel toward you for you to be able to hear that its frequency is Doppler shifted? That is, what speed produces a shift of 0.300% on a day when the speed of sound is 331 m/s?

17.8 Shock Waves

122. An airplane is flying at Mach 1.50 at an altitude of 7500.00 meters, where the speed of sound is $v = 343.00 \text{ m/s}$. How far away from a stationary observer will the plane be when the observer hears the sonic boom?

123. A jet flying at an altitude of 8.50 km has a speed of Mach 2.00, where the speed of sound is $v = 340.00 \text{ m/s}$. How long after the jet is directly overhead, will a stationary observer hear a sonic boom?

124. The shock wave off the front of a fighter jet has an angle of $\theta = 70.00°$. The jet is flying at 1200 km/h. What is the speed of sound?

125. A plane is flying at Mach 1.2, and an observer on the ground hears the sonic boom 15.00 seconds after the plane is directly overhead. What is the altitude of the plane? Assume the speed of sound is $v_{\text{w}} = 343.00 \text{ m/s}$.

126. A bullet is fired and moves at a speed of 1342 mph. Assume the speed of sound is $v = 340.00 \text{ m/s}$. What is the angle of the shock wave produced?

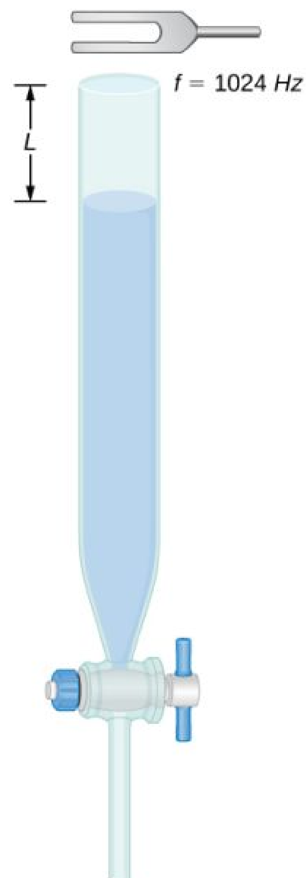
127. A speaker is placed at the opening of a long horizontal tube. The speaker oscillates at a frequency of *f*, creating a sound wave that moves down the tube. The wave moves through the tube at a speed of $v = 340.00 \text{ m/s}$. The sound wave is modeled with the wave function $s(x, t) = s_{\text{max}} \cos(kx - \omega t + \phi)$. At time $t = 0.00 \text{ s}$, an air molecule at $x = 2.3 \text{ m}$ is at the maximum displacement of 6.34 nm. At the same time, another molecule at $x = 2.7 \text{ m}$ has a displacement of 2.30 nm. What is the wave function of the sound wave, that is, find the wave number, angular frequency, and the initial phase shift?

128. An airplane moves at Mach 1.2 and produces a shock wave. (a) What is the speed of the plane in meters per second? (b) What is the angle that the shock wave moves?

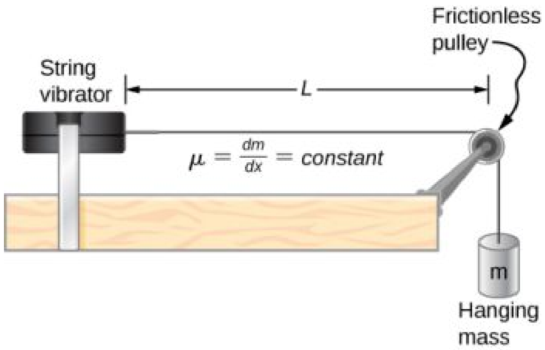
Additional Problems

129. A 0.80-m-long tube is opened at both ends. The air temperature is $26°\text{C}$. The air in the tube is oscillated using a speaker attached to a signal generator. What are the wavelengths and frequencies of first two modes of sound waves that resonate in the tube?

130. A tube filled with water has a valve at the bottom to allow the water to flow out of the tube. As the water is emptied from the tube, the length *L* of the air column changes. A 1024-Hz tuning fork is placed at the opening of the tube. Water is removed from the tube until the $n = 5$ mode of a sound wave resonates. What is the length of the air column if the temperature of the air in the room is $18°\text{C}$?



131. Consider the following figure. The length of the string between the string vibrator and the pulley is $L = 1.00 \text{ m}$. The linear density of the string is $\mu = 0.006 \text{ kg/m}$. The string vibrator can oscillate at any frequency. The hanging mass is 2.00 kg. (a)What are the wavelength and frequency of $n = 6$ mode? (b) The string oscillates the air around the string. What is the wavelength of the sound if the speed of the sound is $v_s = 343.00 \text{ m/s}$?



132. Early Doppler shift experiments were conducted using a band playing music on a train. A trumpet player on a moving railroad flatcar plays a 320-Hz note. The sound waves heard by a stationary observer on a train platform hears a frequency of 350 Hz. What is the flatcar's speed in mph? The temperature of the air is $T_C = 22°C$.

133. Two cars move toward one another, both sounding their horns ($f_s = 800\text{ Hz}$). Car A is moving at 65 mph and Car B is at 75 mph. What is the beat frequency heard by each driver? The air temperature is $T_C = 22.00°C$.

134. Student A runs after Student B. Student A carries a tuning fork ringing at 1024 Hz, and student B carries a tuning fork ringing at 1000 Hz. Student A is running at a speed of $v_A = 5.00\text{ m/s}$ and Student B is running at $v_B = 6.00\text{ m/s}$. What is the beat frequency heard by each student? The speed of sound is $v = 343.00\text{ m/s}$.

135. Suppose that the sound level from a source is 75 dB and then drops to 52 dB, with a frequency of 600 Hz. Determine the (a) initial and (b) final sound intensities and the (c) initial and (d) final sound wave amplitudes. The air temperature is $T_C = 24.00°C$ and the air density is $\rho = 1.184\text{ kg/m}^3$.

136. The Doppler shift for a Doppler radar is found by $f = f_R\left(\frac{1+\frac{v}{c}}{1-\frac{v}{c}}\right)$, where f_R is the frequency of the radar, *f* is the frequency observed by the radar, *c* is the speed of light, and *v* is the speed of the target. What is the beat frequency observed at the radar, assuming the speed of the target is much slower than the speed of light?

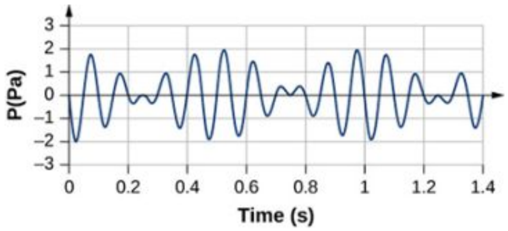
137. A stationary observer hears a frequency of 1000.00 Hz as a source approaches and a frequency of 850.00 Hz as a source departs. The source moves at a constant velocity of 75 mph. What is the temperature of the air?

138. A flute plays a note with a frequency of 600 Hz. The flute can be modeled as a pipe open at both ends, where the flute player changes the length with his finger positions. What is the length of the tube if this is the fundamental frequency?

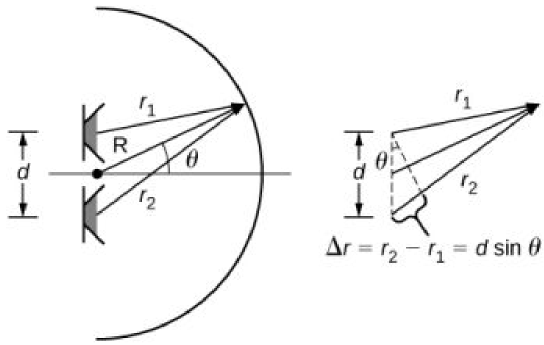
Challenge Problems

139. Two sound speakers are separated by a distance *d*, each sounding a frequency *f*. An observer stands at one speaker and walks in a straight line a distance *x*, perpendicular to the the two speakers, until he comes to the first maximum intensity of sound. The speed of sound is *v*. How far is he from the speaker? Assume the distance, *d*, between the slits is larger than the wavelength of the sound.

140. Consider the beats shown below. This is a graph of the gauge pressure versus time for the position $x = 0.00\text{ m}$. The wave moves with a speed of $v = 343.00\text{ m/s}$. (a) How many beats are there per second? (b) How many times does the wave oscillate per second? (c) Write a wave function for the gauge pressure as a function of time.



141. Two speakers producing the same frequency of sound are a distance of *d* apart. Consider an arc along a circle of radius *R*, centered at the midpoint of the speakers, as shown below. (a) At what angles will there be maxima? (b) At what angle will there be minima?



142. A string has a length of 1.5 m, a linear mass density $\mu = 0.008 \text{ kg/m}$,, and a tension of 120 N. If the air temperature is $T = 22°\text{C}$, what should the length of a pipe open at both ends for it to have the same frequency for the $n = 3$ mode?

143. A string $\left(\mu = 0.006 \frac{\text{kg}}{\text{m}}, L = 1.50 \text{ m}\right)$ is fixed at both ends and is under a tension of 155 N. It oscillates in the $n = 10$ mode and produces sound. A tuning fork is ringing nearby, producing a beat frequency of 23.76 Hz. (a) What is the frequency of the sound from the string? (b) What is the frequency of the tuning fork if the tuning fork frequency is lower? (c) What should be the tension of the string for the beat frequency to be zero?

144. A string has a linear mass density μ, a length *L*, and a tension of F_T, and oscillates in a mode *n* at a frequency *f*. Find the ratio of $\frac{\Delta f}{f}$ for a small change in tension.

145. A string has a linear mass density $\mu = 0.007 \text{ kg/m}$, a length $L = 0.70 \text{ m}$, a tension of $F_T = 110 \text{ N}$, and oscillates in a mode $n = 3$. (a) What is the frequency of the oscillations? (b) Use the result in the preceding problem to find the change in the frequency when the tension is increased by 1.00%.

146. A speaker powered by a signal generator is used to study resonance in a tube. The signal generator can be adjusted from a frequency of 1000 Hz to 1800 Hz. First, a 0.75-m-long tube, open at both ends, is studied. The temperature in the room is $T_{\text{F}} = 85.00°\text{F}$. (a) Which normal modes of the pipe can be studied? What are the frequencies and wavelengths? Next a cap is place on one end of the 0.75-meter-long pipe. (b) Which normal modes of the pipe can be studied? What are the frequencies and wavelengths?

147. A string on the violin has a length of 23.00 cm and a mass of 0.900 grams. The tension in the string 850.00 N. The temperature in the room is $T_C = 24.00°\text{C}$. The string is plucked and oscillates in the $n = 9$ mode. (a) What is the speed of the wave on the string? (b) What is the wavelength of the sounding wave produced? (c) What is the frequency of the oscillating string? (d) What is the frequency of the sound produced? (e) What is the wavelength of the sound produced?

APPENDIX A

Units

Quantity	Common Symbol	Unit	Unit in Terms of Base SI Units
Acceleration	$\vec{\mathbf{a}}$	m/s^2	m/s^2
Amount of substance	n	**mole**	mol
Angle	θ, ϕ	radian (rad)	
Angular acceleration	$\vec{\alpha}$	rad/s^2	s^{-2}
Angular frequency	ω	rad/s	s^{-1}
Angular momentum	$\vec{\mathbf{L}}$	$\mathrm{kg \cdot m^2/s}$	$\mathrm{kg \cdot m^2/s}$
Angular velocity	$\vec{\omega}$	rad/s	s^{-1}
Area	A	m^2	m^2
Atomic number	Z		
Capacitance	C	farad (F)	$\mathrm{A^2 \cdot s^4/kg \cdot m^2}$
Charge	q, Q, e	coulomb (C)	$\mathrm{A \cdot s}$
Charge density:			
Line	λ	C/m	$\mathrm{A \cdot s/m}$
Surface	σ	C/m^2	$\mathrm{A \cdot s/m^2}$
Volume	ρ	C/m^3	$\mathrm{A \cdot s/m^3}$
Conductivity	σ	$1/\Omega \cdot \mathrm{m}$	$\mathrm{A^2 \cdot s^3/kg \cdot m^3}$
Current	I	**ampere**	A
Current density	$\vec{\mathbf{J}}$	A/m^2	A/m^2
Density	ρ	kg/m^3	kg/m^3
Dielectric constant	κ		
Electric dipole moment	$\vec{\mathbf{p}}$	$\mathrm{C \cdot m}$	$\mathrm{A \cdot s \cdot m}$

TABLE A1 **Units Used in Physics (Fundamental units in bold)**

Quantity	Common Symbol	Unit	Unit in Terms of Base SI Units
Electric field	$\vec{\mathbf{E}}$	N/C	$\text{kg} \cdot \text{m/A} \cdot \text{s}^3$
Electric flux	Φ	$\text{N} \cdot \text{m}^2/\text{C}$	$\text{kg} \cdot \text{m}^3/\text{A} \cdot \text{s}^3$
Electromotive force	ε	volt (V)	$\text{kg} \cdot \text{m}^2/\text{A} \cdot \text{s}^3$
Energy	*E,U,K*	joule (J)	$\text{kg} \cdot \text{m}^2/\text{s}^2$
Entropy	*S*	J/K	$\text{kg} \cdot \text{m}^2/\text{s}^2 \cdot \text{K}$
Force	$\vec{\mathbf{F}}$	newton (N)	$\text{kg} \cdot \text{m/s}^2$
Frequency	*f*	hertz (Hz)	s^{-1}
Heat	*Q*	joule (J)	$\text{kg} \cdot \text{m}^2/\text{s}^2$
Inductance	*L*	henry (H)	$\text{kg} \cdot \text{m}^2/\text{A}^2 \cdot \text{s}^2$
Length:	ℓ, L	**meter**	m
Displacement	$\Delta x, \Delta\vec{\mathbf{r}}$		
Distance	*d, h*		
Position	$x, y, z, \vec{\mathbf{r}}$		
Magnetic dipole moment	$\vec{\mu}$	$\text{N} \cdot \text{J/T}$	$\text{A} \cdot \text{m}^2$
Magnetic field	$\vec{\mathbf{B}}$	tesla (T) $= \left(\text{Wb/m}^2\right)$	$\text{kg/A} \cdot \text{s}^2$
Magnetic flux	Φ_m	weber (Wb)	$\text{kg} \cdot \text{m}^2/\text{A} \cdot \text{s}^2$
Mass	*m, M*	**kilogram**	kg
Molar specific heat	*C*	$\text{J/mol} \cdot \text{K}$	$\text{kg} \cdot \text{m}^2/\text{s}^2 \cdot \text{mol} \cdot \text{K}$
Moment of inertia	*I*	$\text{kg} \cdot \text{m}^2$	$\text{kg} \cdot \text{m}^2$
Momentum	$\vec{\mathbf{p}}$	$\text{kg} \cdot \text{m/s}$	$\text{kg} \cdot \text{m/s}$
Period	*T*	s	s
Permeability of free space	μ_0	$\text{N/A}^2 = (\text{H/m})$	$\text{kg} \cdot \text{m/A}^2 \cdot \text{s}^2$
Permittivity of free space	ε_0	$\text{C}^2/\text{N} \cdot \text{m}^2 = (\text{F/m})$	$\text{A}^2 \cdot \text{s}^4/\text{kg} \cdot \text{m}^3$
Potential	*V*	volt (V) = (J/C)	$\text{kg} \cdot \text{m}^2/\text{A} \cdot \text{s}^3$

TABLE A1 **Units Used in Physics (Fundamental units in bold)**

Quantity	Common Symbol	Unit	Unit in Terms of Base SI Units
Power	P	watt (W) $= (\mathrm{J/s})$	$\mathrm{kg \cdot m^2/s^3}$
Pressure	p	pascal (P) $= \left(\mathrm{N/m^2}\right)$	$\mathrm{kg/m \cdot s^2}$
Resistance	R	ohm (Ω) $= (\mathrm{V/A})$	$\mathrm{kg \cdot m^2/A^2 \cdot s^3}$
Specific heat	c	$\mathrm{J/kg \cdot K}$	$\mathrm{m^2/s^2 \cdot K}$
Speed	v	m/s	m/s
Temperature	T	**kelvin**	K
Time	t	**second**	s
Torque	$\vec{\tau}$	$\mathrm{N \cdot m}$	$\mathrm{kg \cdot m^2/s^2}$
Velocity	$\vec{\mathbf{v}}$	m/s	m/s
Volume	V	$\mathrm{m^3}$	$\mathrm{m^3}$
Wavelength	λ	m	m
Work	W	joule (J) $= (\mathrm{N \cdot m})$	$\mathrm{kg \cdot m^2/s^2}$

TABLE A1 **Units Used in Physics (Fundamental units in bold)**

APPENDIX B

Conversion Factors

	m	cm	km
1 meter	1	10^2	10^{-3}
1 centimeter	10^{-2}	1	10^{-5}
1 kilometer	10^3	10^5	1
1 inch	2.540×10^{-2}	2.540	2.540×10^{-5}
1 foot	0.3048	30.48	3.048×10^{-4}
1 mile	1609	1.609×10^5	1.609
1 angstrom	10^{-10}		
1 fermi	10^{-15}		
1 light-year			9.460×10^{12}
	in.	**ft**	**mi**
1 meter	39.37	3.281	6.214×10^{-4}
1 centimeter	0.3937	3.281×10^{-2}	6.214×10^{-6}
1 kilometer	3.937×10^4	3.281×10^3	0.6214
1 inch	1	8.333×10^{-2}	1.578×10^{-5}
1 foot	12	1	1.894×10^{-4}
1 mile	6.336×10^4	5280	1

TABLE B1 **Length**

Area

$1\ \text{cm}^2 = 0.155\ \text{in.}^2$

$1\ \text{m}^2 = 10^4\ \text{cm}^2 = 10.76\ \text{ft}^2$

$1\ \text{in.}^2 = 6.452\ \text{cm}^2$

$1\ \text{ft}^2 = 144\ \text{in.}^2 = 0.0929\ \text{m}^2$

Volume

$1\ \text{liter} = 1000\ \text{cm}^3 = 10^{-3}\ \text{m}^3 = 0.03531\ \text{ft}^3 = 61.02\ \text{in.}^3$

$1\ \text{ft}^3 = 0.02832\ \text{m}^3 = 28.32\ \text{liters} = 7.477\ \text{gallons}$

$1\ \text{gallon} = 3.788\ \text{liters}$

	s	min	h	day	yr
1 second	1	1.667×10^{-2}	2.778×10^{-4}	1.157×10^{-5}	3.169×10^{-8}
1 minute	60	1	1.667×10^{-2}	6.944×10^{-4}	1.901×10^{-6}
1 hour	3600	60	1	4.167×10^{-2}	1.141×10^{-4}
1 day	8.640×10^{4}	1440	24	1	2.738×10^{-3}
1 year	3.156×10^{7}	5.259×10^{5}	8.766×10^{3}	365.25	1

TABLE B2 Time

	m/s	cm/s	ft/s	mi/h
1 meter/second	1	10^2	3.281	2.237
1 centimeter/second	10^{-2}	1	3.281×10^{-2}	2.237×10^{-2}
1 foot/second	0.3048	30.48	1	0.6818
1 mile/hour	0.4470	44.70	1.467	1

TABLE B3 Speed

Acceleration

$1\ \text{m/s}^2 = 100\ \text{cm/s}^2 = 3.281\ \text{ft/s}^2$

$1\ \text{cm/s}^2 = 0.01\ \text{m/s}^2 = 0.03281\ \text{ft/s}^2$

$1\ \text{ft/s}^2 = 0.3048\ \text{m/s}^2 = 30.48\ \text{cm/s}^2$

$1\ \text{mi/h} \cdot \text{s} = 1.467\ \text{ft/s}^2$

	kg	g	slug	u
1 kilogram	1	10^3	6.852×10^{-2}	6.024×10^{26}
1 gram	10^{-3}	1	6.852×10^{-5}	6.024×10^{23}
1 slug	14.59	1.459×10^{4}	1	8.789×10^{27}
1 atomic mass unit	1.661×10^{-27}	1.661×10^{-24}	1.138×10^{-28}	1
1 metric ton	1000			

TABLE B4 Mass

	N	dyne	lb
1 newton	1	10^5	0.2248
1 dyne	10^{-5}	1	2.248×10^{-6}
1 pound	4.448	4.448×10^5	1

TABLE B5 Force

	Pa	dyne/cm^2	atm	cmHg	lb/in.^{2}
1 pascal	1	10	9.869×10^{-6}	7.501×10^{-4}	1.450×10^{-4}
1 dyne/centimeter^2	10^{-1}	1	9.869×10^{-7}	7.501×10^{-5}	1.450×10^{-5}
1 atmosphere	1.013×10^5	1.013×10^6	1	76	14.70
1 centimeter mercury*	1.333×10^3	1.333×10^4	1.316×10^{-2}	1	0.1934
1 pound/inch^2	6.895×10^3	6.895×10^4	6.805×10^{-2}	5.171	1
1 bar	10^5				
1 torr				1 (mmHg)	

*Where the acceleration due to gravity is $9.80665\ \mathrm{m/s^2}$ and the temperature is $0°\mathrm{C}$

TABLE B6 Pressure

	J	erg	ft.lb
1 joule	1	10^7	0.7376
1 erg	10^{-7}	1	7.376×10^{-8}
1 foot-pound	1.356	1.356×10^7	1
1 electron-volt	1.602×10^{-19}	1.602×10^{-12}	1.182×10^{-19}
1 calorie	4.186	4.186×10^7	3.088
1 British thermal unit	1.055×10^3	1.055×10^{10}	7.779×10^2
1 kilowatt-hour	3.600×10^6		
	eV	**cal**	**Btu**
1 joule	6.242×10^{18}	0.2389	9.481×10^{-4}
1 erg	6.242×10^{11}	2.389×10^{-8}	9.481×10^{-11}

TABLE B7 Work, Energy, Heat

	J	erg	ft.lb
1 foot-pound	8.464×10^{18}	0.3239	1.285×10^{-3}
1 electron-volt	1	3.827×10^{-20}	1.519×10^{-22}
1 calorie	2.613×10^{19}	1	3.968×10^{-3}
1 British thermal unit	6.585×10^{21}	2.520×10^{2}	1

TABLE B7 Work, Energy, Heat

Power

$1\ \text{W} = 1\ \text{J/s}$

$1\ \text{hp} = 746\ \text{W} = 550\ \text{ft} \cdot \text{lb/s}$

$1\ \text{Btu/h} = 0.293\ \text{W}$

Angle

$1\ \text{rad} = 57.30^\circ = 180^\circ/\pi$

$1^\circ = 0.01745\ \text{rad} = \pi/180\ \text{rad}$

$1\ \text{revolution} = 360^\circ = 2\pi\ \text{rad}$

$1\ \text{rev/min (rpm)} = 0.1047\ \text{rad/s}$

APPENDIX C

Fundamental Constants

Quantity	Symbol	Value
Atomic mass unit	u	$1.660\ 538\ 782\ (83) \times 10^{-27}$ kg $931.494\ 028\ (23)$ MeV/c^2
Avogadro's number	N_A	$6.02214076 \times 10^{23}$ reciprocal mole$\left(\mathrm{mol}^{-1}\right)$
Bohr magneton	$\mu_B = \frac{e\hbar}{2m_e}$	$9.274\ 009\ 15\ (23) \times 10^{-24}$ J/T
Bohr radius	$a_0 = \frac{\hbar^2}{m_e e^2 k_e}$	$5.291\ 772\ 085\ 9\ (36) \times 10^{-11}$ m
Boltzmann's constant	$k_B = \frac{R}{N_A}$	1.380649×10^{-23} joule per kelvin$\left(\mathrm{J \cdot K^{-1}}\right)$
Compton wavelength	$\lambda_C = \frac{h}{m_e c}$	$2.426\ 310\ 217\ 5\ (33) \times 10^{-12}$ m
Coulomb constant	$k_e = \frac{1}{4\pi\varepsilon_0}$	$8.987\ 551\ 788... \times 10^9$ N · m^2/C^2 (exact)
Deuteron mass	m_d	$3.343\ 583\ 20\ (17) \times 10^{-27}$ kg $2.013\ 553\ 212\ 724\,(78)$ u $1875.612\ 859$ MeV/c^2
Electron mass	m_e	$9.109\ 382\ 15\ (45) \times 10^{-31}$ kg $5.485\ 799\ 094\ 3\,(23) \times 10^{-4}$ u $0.510\ 998\ 910\ (13)$ MeV/c^2
Electron volt	eV	$1.602\ 176\ 487\ (40) \times 10^{-19}$ J
Elementary charge	e	$1.602176634 \times 10^{-19}$ C
Gas constant	R	$8.314\ 472\ (15)$ J/mol · K
Gravitational constant	G	$6.674\ 28\ (67) \times 10^{-11}$ N · m^2/kg^2
Neutron mass	m_n	$1.674\ 927\ 211\ (84) \times 10^{-27}$ kg $1.008\ 664\ 915\ 97\ (43)$ u $939.565\ 346\ (23)$ MeV/c^2

TABLE C1 **Fundamental Constants** *Note:* These constants are the values recommended in 2006 by CODATA, based on a least-squares adjustment of data from different measurements. The numbers in parentheses for the values represent the uncertainties of the last two digits.

Quantity	Symbol	Value
Nuclear magneton	$\mu_n = \frac{e\hbar}{2m_p}$	$5.050\,783\,24\ (13) \times 10^{-27}$ J/T
Permeability of free space	μ_0	$4\pi \times 10^{-7}$ T · m/A (exact)
Permittivity of free space	$\varepsilon_0 = \frac{1}{\mu_0 c^2}$	$8.854\,187\,817... \times 10^{-12}\ C^2/N \cdot m^2$ (exact)
Planck's constant	h $\hbar = \frac{h}{2\pi}$	$6.62607015 \times 10^{-34}\ kg \cdot m^2 \cdot s^{-1}$ $1.05457182 \times 10^{-34}\ kg \cdot m^2 \cdot s^{-1}$
Proton mass	m_p	$1.672\,621\,637\ (83) \times 10^{-27}$ kg $1.007\,276\,466\,77\ (10)$ u $938.272\,013\ (23)\ MeV/c^2$
Rydberg constant	R_H	$1.097\,373\,156\,852\,7\ (73) \times 10^7\ m^{-1}$
Speed of light in vacuum	c	$2.997\,924\,58 \times 10^8$ m/s (exact)

TABLE C1 Fundamental Constants *Note:* These constants are the values recommended in 2006 by CODATA, based on a least-squares adjustment of data from different measurements. The numbers in parentheses for the values represent the uncertainties of the last two digits.

Useful combinations of constants for calculations:

$$hc = 12{,}400\ \text{eV} \cdot \text{Å} = 1240\ \text{eV} \cdot \text{nm} = 1240\ \text{MeV} \cdot \text{fm}$$

$$\hbar c = 1973\ \text{eV} \cdot \text{Å} = 197.3\ \text{eV} \cdot \text{nm} = 197.3\ \text{MeV} \cdot \text{fm}$$

$$k_e e^2 = 14.40\ \text{eV} \cdot \text{Å} = 1.440\ \text{eV} \cdot \text{nm} = 1.440\ \text{MeV} \cdot \text{fm}$$

$$k_B T = 0.02585\ \text{eV at } T = 300\ \text{K}$$

APPENDIX D

Astronomical Data

Celestial Object	Mean Distance from Sun (million km)	Period of Revolution (d = days) (y = years)	Period of Rotation at Equator	Eccentricity of Orbit
Sun	–	–	27 d	–
Mercury	57.9	88 d	59 d	0.206
Venus	108.2	224.7 d	243 d	0.007
Earth	149.6	365.25 d	23 h 56 min 4 s	0.017
Mars	227.9	687 d	24 h 37 min 23 s	0.093
Jupiter	778.4	11.9 y	9 h 50 min 30 s	0.048
Saturn	1426.7	29.5 y	10 h 14 min	0.054
Uranus	2871.0	84.0 y	17 h 14 min	0.047
Neptune	4498.3	164.8 y	16 h	0.009
Earth's Moon	149.6 (0.386 from Earth)	27.3 d	27.3 d	0.055
Celestial Object	**Equatorial Diameter (km)**	**Mass (Earth = 1)**	**Density (g/cm^3)**	
Sun	1,392,000	333,000.00	1.4	
Mercury	4879	0.06	5.4	
Venus	12,104	0.82	5.2	
Earth	12,756	1.00	5.5	
Mars	6794	0.11	3.9	
Jupiter	142,984	317.83	1.3	
Saturn	120,536	95.16	0.7	
Uranus	51,118	14.54	1.3	

TABLE D1 Astronomical Data

Other Data:

Mass of Earth: 5.97×10^{24} kg

Mass of the Moon: 7.36×10^{22} kg

Mass of the Sun: 1.99×10^{30} kg

APPENDIX E

Mathematical Formulas

Quadratic formula

If $ax^2 + bx + c = 0$, then $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Triangle of base b and height h	Area $= \frac{1}{2}bh$	
Circle of radius r	Circumference $= 2\pi r$	Area $= \pi r^2$
Sphere of radius r	Surface area $= 4\pi r^2$	Volume $= \frac{4}{3}\pi r^3$
Cylinder of radius r and height h	Area of curved surface $= 2\pi rh$	Volume $= \pi r^2 h$

TABLE E1 Geometry

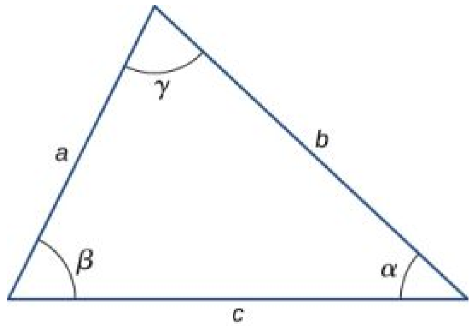
Trigonometry

Trigonometric Identities

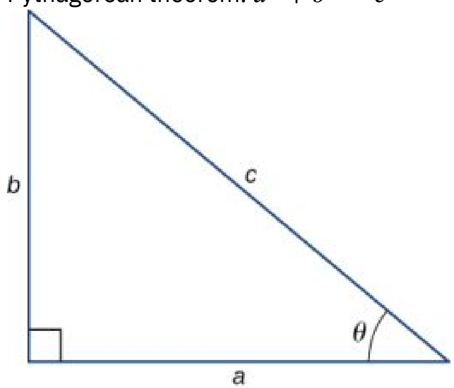
1. $\sin\theta = 1/\csc\theta$
2. $\cos\theta = 1/\sec\theta$
3. $\tan\theta = 1/\cot\theta$
4. $\sin\left(90^0 - \theta\right) = \cos\theta$
5. $\cos\left(90^0 - \theta\right) = \sin\theta$
6. $\tan\left(90^0 - \theta\right) = \cot\theta$
7. $\sin^2\theta + \cos^2\theta = 1$
8. $\sec^2\theta - \tan^2\theta = 1$
9. $\tan\theta = \sin\theta/\cos\theta$
10. $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$
11. $\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$
12. $\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$
13. $\sin 2\theta = 2\sin\theta\cos\theta$
14. $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$
15. $\sin\alpha + \sin\beta = 2\sin\frac{1}{2}(\alpha + \beta)\cos\frac{1}{2}(\alpha - \beta)$
16. $\cos\alpha + \cos\beta = 2\cos\frac{1}{2}(\alpha + \beta)\cos\frac{1}{2}(\alpha - \beta)$
17. $\mathrm{s} = r\theta$

Triangles

1. Law of sines: $\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$
2. Law of cosines: $c^2 = a^2 + b^2 - 2ab\cos\gamma$



3. Pythagorean theorem: $a^2 + b^2 = c^2$



Series expansions

1. Binomial theorem: $(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)a^{n-2}b^2}{2!} + \frac{n(n-1)(n-2)a^{n-3}b^3}{3!} + \cdots$
2. $(1 \pm x)^n = 1 \pm \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} \pm \cdots \left(x^2 < 1\right)$
3. $(1 \pm x)^{-n} = 1 \mp \frac{nx}{1!} + \frac{n(n+1)x^2}{2!} \mp \cdots \left(x^2 < 1\right)$
4. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$
5. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$
6. $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$
7. $e^x = 1 + x + \frac{x^2}{2!} + \cdots$
8. $\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots (|x| < 1)$

Derivatives

1. $\frac{d}{dx}[af(x)] = a\frac{d}{dx}f(x)$
2. $\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$
3. $\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}g(x) + g(x)\frac{d}{dx}f(x)$
4. $\frac{d}{dx}f(u) = \left[\frac{d}{du}f(u)\right]\frac{du}{dx}$
5. $\frac{d}{dx}x^m = mx^{m-1}$
6. $\frac{d}{dx}\sin x = \cos x$
7. $\frac{d}{dx}\cos x = -\sin x$
8. $\frac{d}{dx}\tan x = \sec^2 x$
9. $\frac{d}{dx}\cot x = -\csc^2 x$
10. $\frac{d}{dx}\sec x = \tan x \sec x$
11. $\frac{d}{dx}\csc x = -\cot x \csc x$
12. $\frac{d}{dx}e^x = e^x$

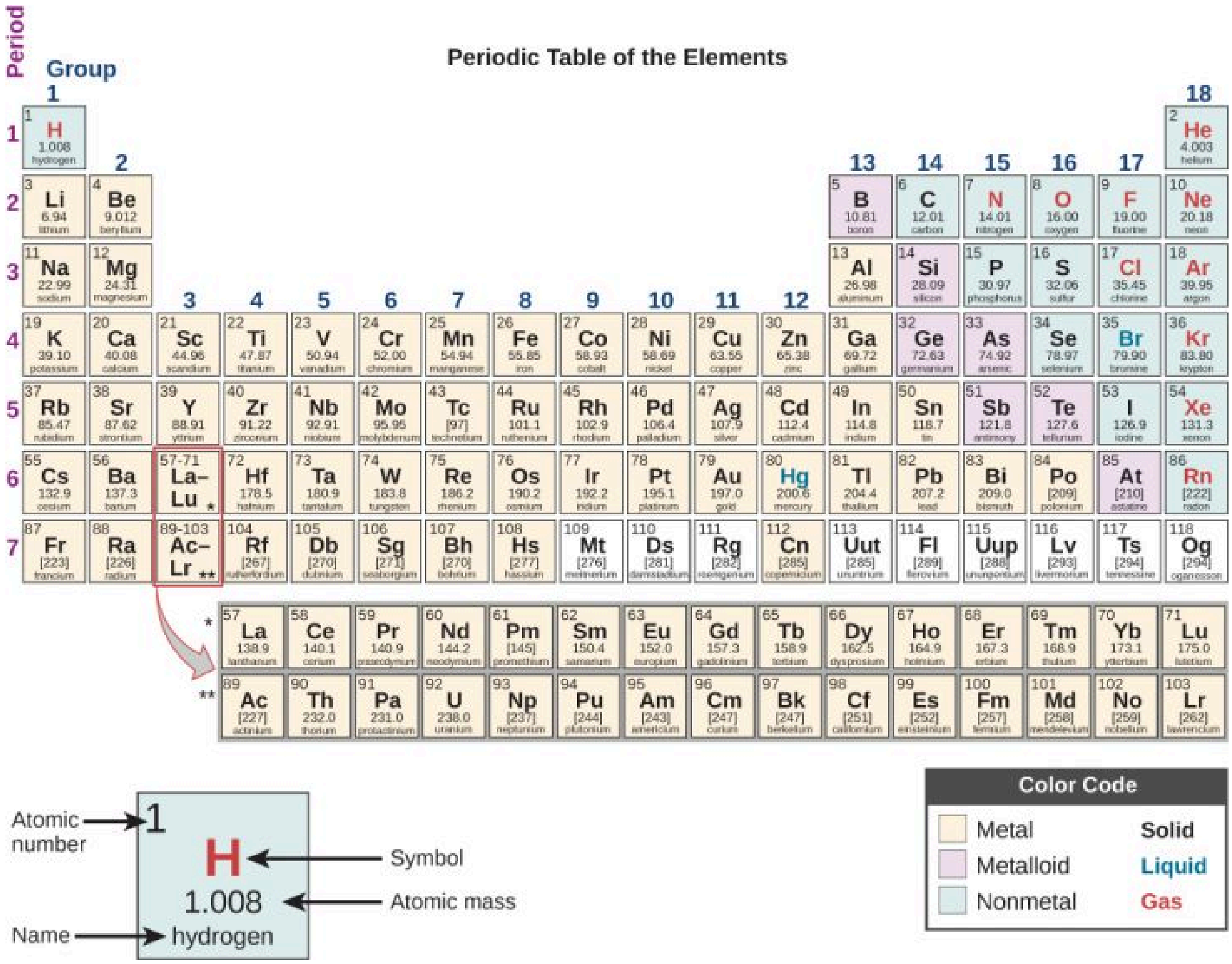
13. $\frac{d}{dx}\ln x = \frac{1}{x}$
14. $\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$
15. $\frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$
16. $\frac{d}{dx}\tan^{-1} x = \frac{1}{1+x^2}$

Integrals

1. $\int af(x)\,dx = a\int f(x)\,dx$
2. $\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$
3. $$\begin{aligned}\int x^m\,dx &= \frac{x^{m+1}}{m+1}\ (m \neq -1)\\ &= \ln x (m = -1)\end{aligned}$$
4. $\int \sin x\,dx = -\cos x$
5. $\int \cos x\,dx = \sin x$
6. $\int \tan x\,dx = \ln|\sec x|$
7. $\int \sin^2 ax\,dx = \frac{x}{2} - \frac{\sin 2ax}{4a}$
8. $\int \cos^2 ax\,dx = \frac{x}{2} + \frac{\sin 2ax}{4a}$
9. $\int \sin ax \cos ax\,dx = -\frac{\cos 2ax}{4a}$
10. $\int e^{ax}\,dx = \frac{1}{a}e^{ax}$
11. $\int xe^{ax}\,dx = \frac{e^{ax}}{a^2}(ax - 1)$
12. $\int \ln ax\,dx = x\ln ax - x$
13. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$
14. $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a}\ln\left|\frac{x+a}{x-a}\right|$
15. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\frac{x}{a}$
16. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a}$
17. $\int \sqrt{a^2 + x^2}\,dx = \frac{x}{2}\sqrt{a^2 + x^2} + \frac{a^2}{2}\sinh^{-1}\frac{x}{a}$
18. $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}$
19. $\int \frac{1}{(x^2 + a^2)^{3/2}}\,dx = \frac{x}{a^2\sqrt{x^2+a^2}}$
20. $\int \frac{x}{(x^2 + a^2)^{3/2}}\,dx = -\frac{1}{\sqrt{x^2+a^2}}$

APPENDIX F

Chemistry



APPENDIX G

The Greek Alphabet

Name	Capital	Lowercase	Name	Capital	Lowercase
Alpha	A	α	Nu	N	ν
Beta	B	β	Xi	Ξ	ξ
Gamma	Γ	γ	Omicron	O	o
Delta	Δ	δ	Pi	Π	π
Epsilon	E	ε	Rho	P	ρ
Zeta	Z	ζ	Sigma	Σ	σ
Eta	H	η	Tau	T	τ
Theta	Θ	θ	Upsilon	Υ	υ
Iota	I	ι	Phi	Φ	ϕ
Kappa	K	κ	Chi	X	χ
Lambda	Λ	λ	Psi	Ψ	ψ
Mu	M	μ	Omega	Ω	ω

TABLE G1 The Greek Alphabet

ANSWER KEY

Chapter 1

Check Your Understanding

1.1 4.79×10^2 Mg or 479 Mg
1.2 $3 \times 10^8 \mathrm{m/s}$
1.3 $10^8\ \mathrm{km}^2$
1.4 The numbers were too small, by a factor of 4.45.
1.5 $4\pi r^3/3$
1.6 yes
1.7 $3 \times 10^4 \mathrm{m}$ or 30 km. It is probably an underestimate because the density of the atmosphere decreases with altitude. (In fact, 30 km does not even get us out of the stratosphere.)
1.8 No, the coach's new stopwatch will not be helpful. The uncertainty in the stopwatch is too great to differentiate between the sprint times effectively.

Conceptual Questions

1. Physics is the science concerned with describing the interactions of energy, matter, space, and time to uncover the fundamental mechanisms that underlie every phenomenon.
3. No, neither of these two theories is more valid than the other. Experimentation is the ultimate decider. If experimental evidence does not suggest one theory over the other, then both are equally valid. A given physicist might prefer one theory over another on the grounds that one seems more simple, more natural, or more beautiful than the other, but that physicist would quickly acknowledge that he or she cannot say the other theory is invalid. Rather, he or she would be honest about the fact that more experimental evidence is needed to determine which theory is a better description of nature.
5. Probably not. As the saying goes, "Extraordinary claims require extraordinary evidence."
7. Conversions between units require factors of 10 only, which simplifies calculations. Also, the same basic units can be scaled up or down using metric prefixes to sizes appropriate for the problem at hand.
9. a. Base units are defined by a particular process of measuring a base quantity whereas derived units are defined as algebraic combinations of base units. b. A base quantity is chosen by convention and practical considerations. Derived quantities are expressed as algebraic combinations of base quantities. c. A base unit is a standard for expressing the measurement of a base quantity within a particular system of units. So, a measurement of a base quantity could be expressed in terms of a base unit in any system of units using the same base quantities. For example, length is a base quantity in both SI and the English system, but the meter is a base unit in the SI system only.
11. a. Uncertainty is a quantitative measure of precision. b. Discrepancy is a quantitative measure of accuracy.
13. Check to make sure it makes sense and assess its significance.

Problems

15. a. 10^3; b. 10^5; c. 10^2; d. 10^{15}; e. 10^2; f. 10^{57}
17. 10^2 generations
19. 10^{11} atoms
21. 10^3 nerve impulses/s
23. 10^{26} floating-point operations per human lifetime
25. a. 957 ks; b. 4.5 cs or 45 ms; c. 550 ns; d. 31.6 Ms
27. a. 75.9 Mm; b. 7.4 mm; c. 88 pm; d. 16.3 Tm
29. a. 3.8 cg or 38 mg; b. 230 Eg; c. 24 ng; d. 8 Eg e. 4.2 g
31. a. 27.8 m/s; b. 62 mi/h
33. a. 3.6 km/h; b. 2.2 mi/h

35. $1.05 \times 10^5 \text{ ft}^2$
37. 8.847 km
39. a. 1.3×10^{-9}m; b. 40 km/My
41. $10^6 \text{Mg}/\mu\text{L}$
43. 62.4 lbm/ft^3
45. 0.017 rad
47. 1 light-nanosecond
49. $3.6 \times 10^{-4}\text{m}^3$
51. a. Yes, both terms have dimension L^2T^{-2} b. No. c. Yes, both terms have dimension LT^{-1} d. Yes, both terms have dimension LT^{-2}
53. a. [v] = LT^{-1}; b. [a] = LT^{-2}; c. $\left[\int v dt\right] = \text{L}$; d. $\left[\int a dt\right] = \text{LT}^{-1}$; e. $\left[\frac{da}{dt}\right] = \text{LT}^{-3}$
55. a. L; b. L; c. L^0 = 1 (that is, it is dimensionless)
57. 10^{28} atoms
59. 10^{51} molecules
61. 10^{16} solar systems
63. a. Volume = 10^{27} m^3, diameter is 10^9 m.; b. 10^{11} m
65. a. A reasonable estimate might be one operation per second for a total of 10^9 in a lifetime.; b. about $(10^9)(10^{-17}$ s) = 10^{-8} s, or about 10 ns
67. 2 kg
69. 4%
71. 67 mL
73. a. The number 99 has 2 significant figures; 100. has 3 significant figures. b. 1.00%; c. percent uncertainties
75. a. 2%; b. 1 mm Hg
77. 7.557 cm^2
79. a. 37.2 lb; because the number of bags is an exact value, it is not considered in the significant figures; b. 1.4 N; because the value 55 kg has only two significant figures, the final value must also contain two significant figures

Additional Problems

81. a. $[s_0] = \text{L}$ and units are meters (m); b. $[v_0] = \text{LT}^{-1}$ and units are meters per second (m/s); c. $[a_0] = \text{LT}^{-2}$ and units are meters per second squared (m/s^2); d. $[j_0] = \text{LT}^{-3}$ and units are meters per second cubed (m/s^3); e. $[S_0] = \text{LT}^{-4}$ and units are m/s^4; f. $[c] = \text{LT}^{-5}$ and units are m/s^5.
83. a. 0.059%; b. 0.01%; c. 4.681 m/s; d. 0.07%, 0.003 m/s
85. a. 0.02%; b. 1×10^4 lbm
87. a. 143.6 cm^3; b. 0.1 cm^3 or 0.084%

Challenge Problems

89. Since each term in the power series involves the argument raised to a different power, the only way that every term in the power series can have the same dimension is if the argument is dimensionless. To see this explicitly, suppose $[x] = L^aM^bT^c$. Then, $[x^n] = [x]^n = L^{an}M^{bn}T^{cn}$. If we want $[x] = [x^n]$, then an = a, bn = b, and cn = c for all n. The only way this can happen is if a = b = c = 0.

Chapter 2

Check Your Understanding

2.1 a. not equal because they are orthogonal; b. not equal because they have different magnitudes; c. not equal because they have different magnitudes and directions; d. not equal because they are antiparallel; e. equal.
2.2 16 m; $\vec{\mathbf{D}} = -16 \text{ m}\hat{\mathbf{u}}$
2.3 G = 28.2 cm, $\theta_G = 291°$
2.4 $\vec{\mathbf{D}} = (-5.0\hat{\mathbf{i}} - 3.0\hat{\mathbf{j}})$cm; the fly moved 5.0 cm to the left and 3.0 cm down from its landing site.
2.5 5.83 cm, $211°$

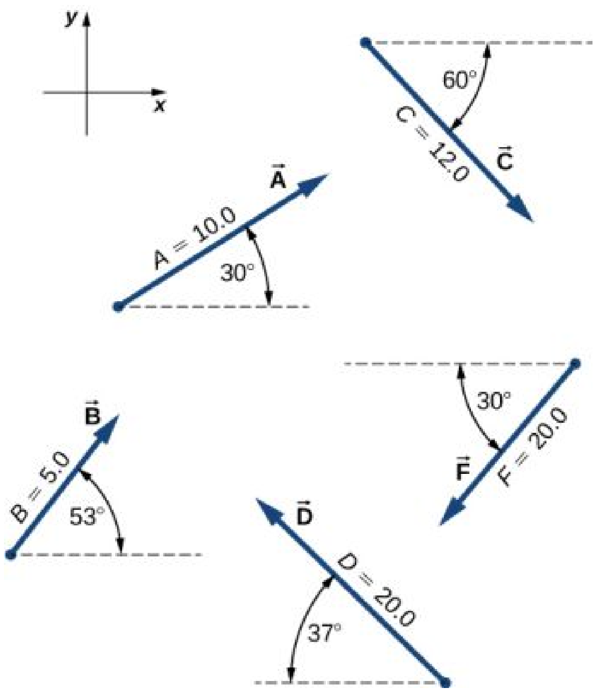
2.6 $\vec{\mathbf{D}} = (-20\text{ m})\hat{\mathbf{i}}$
2.7 35.2 m/s = 126.7 km/h
2.8 $\vec{\mathbf{G}} = (10.25\hat{\mathbf{i}} - 26.22\hat{\mathbf{j}})\text{cm}$
2.9 D = 55.7 N; direction $65.7°$ north of east
2.10 $\hat{\mathbf{v}} = 0.8\hat{\mathbf{i}} + 0.6\hat{\mathbf{j}}$, $36.87°$ north of east
2.11 $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = -57.3$, $\vec{\mathbf{F}} \cdot \vec{\mathbf{C}} = 27.8$
2.13 $131.9°$
2.14 $W_1 = 1.5\text{ J}$, $W_2 = 0.3\text{ J}$
2.15 $\vec{\mathbf{A}} \times \vec{\mathbf{B}} = -40.1\hat{\mathbf{k}}$ or, equivalently, $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right| = 40.1$, and the direction is into the page; $\vec{\mathbf{C}} \times \vec{\mathbf{F}} = +157.6\hat{\mathbf{k}}$ or, equivalently, $\left|\vec{\mathbf{C}} \times \vec{\mathbf{F}}\right| = 157.6$, and the direction is out of the page.
2.16 a. $-2\hat{\mathbf{k}}$, b. 2, c. $153.4°$, d. $135°$

Conceptual Questions

1. scalar
3. answers may vary
5. parallel, sum of magnitudes, antiparallel, zero
7. no, yes
9. zero, yes
11. no
13. equal, equal, the same
15. a unit vector of the x-axis
17. They are equal.
19. yes
21. a. $C = \vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$, b. $\vec{\mathbf{C}} = \vec{\mathbf{A}} \times \vec{\mathbf{B}}$ or $\vec{\mathbf{C}} = \vec{\mathbf{A}} - \vec{\mathbf{B}}$, c. $\vec{\mathbf{C}} = \vec{\mathbf{A}} \times \vec{\mathbf{B}}$, d. $\vec{\mathbf{C}} = A\vec{\mathbf{B}}$, e. $\vec{\mathbf{C}} + 2\vec{\mathbf{A}} = \vec{\mathbf{B}}$, f. $\vec{\mathbf{C}} = \vec{\mathbf{A}} \times \vec{\mathbf{B}}$, g. left side is a scalar and right side is a vector, h. $\vec{\mathbf{C}} = 2\vec{\mathbf{A}} \times \vec{\mathbf{B}}$, i. $\vec{\mathbf{C}} = \vec{\mathbf{A}}/B$, j. $\vec{\mathbf{C}} = \vec{\mathbf{A}}/B$
23. They are orthogonal.

Problems

25. $\vec{\mathbf{h}} = -49\text{ m}\hat{\mathbf{u}}$, 49 m
27. 30.8 m, $35.7°$ west of north (equivalently, $54.2°$ north of west or $125.7°$ from east)
29. 134 km, $80°$
31. 7.34 km, $63.5°$ south of east
33. 3.8 km east, 3.2 km north, 7.0 km
35. 14.3 km, $65°$
37. a. $\vec{\mathbf{A}} = +8.66\hat{\mathbf{i}} + 5.00\hat{\mathbf{j}}$, b. $\vec{\mathbf{B}} = +3.01\hat{\mathbf{i}} + 3.99\hat{\mathbf{j}}$, c. $\vec{\mathbf{C}} = +6.00\hat{\mathbf{i}} - 10.39\hat{\mathbf{j}}$, d. $\vec{\mathbf{D}} = -15.97\hat{\mathbf{i}} + 12.04\hat{\mathbf{j}}$, f. $\vec{\mathbf{F}} = -17.32\hat{\mathbf{i}} - 10.00\hat{\mathbf{j}}$



39. a. 1.94 km, 7.24 km; b. proof

41. 3.8 km east, 3.2 km north, 2.0 km, $\vec{\mathbf{D}} = (3.8\hat{\mathbf{i}} + 3.2\hat{\mathbf{j}})\text{km}$

43. $P_1(2.165\text{ m}, 1.250\text{ m})$, $P_2(-1.900\text{ m}, 3.290\text{ m})$, 4.55 m

45. 8.60 m, $A(2\sqrt{5}\text{ m}, 0.647\pi)$, $B(3\sqrt{2}\text{ m}, 0.75\pi)$

47. a. $\vec{\mathbf{A}} + \vec{\mathbf{B}} = -4\hat{\mathbf{i}} - 6\hat{\mathbf{j}}$, $\left|\vec{\mathbf{A}} + \vec{\mathbf{B}}\right| = 7.211, \theta = 236°$; b. $\vec{\mathbf{A}} - \vec{\mathbf{B}} = -2\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$, $\left|\vec{\mathbf{A}} - \vec{\mathbf{B}}\right| = 2\sqrt{2}, \theta = 135°$

49. a. $\vec{\mathbf{C}} = (5.0\hat{\mathbf{i}} - 1.0\hat{\mathbf{j}} - 3.0\hat{\mathbf{k}})\text{m}, C = 5.92\text{ m}$;
b. $\vec{\mathbf{D}} = (4.0\hat{\mathbf{i}} - 11.0\hat{\mathbf{j}} + 15.0\hat{\mathbf{k}})\text{m}, D = 19.03\text{ m}$

51. $\vec{\mathbf{D}} = (3.3\hat{\mathbf{i}} - 6.6\hat{\mathbf{j}})\text{km}$, $\hat{\mathbf{i}}$ is to the east, 7.34 km, $-63.5°$

53. a. $\vec{\mathbf{R}} = -1.35\hat{\mathbf{i}} - 22.04\hat{\mathbf{j}}$, b. $\vec{\mathbf{R}} = -17.98\hat{\mathbf{i}} + 0.89\hat{\mathbf{j}}$

55. $\vec{\mathbf{D}} = (200\hat{\mathbf{i}} + 300\hat{\mathbf{j}})\text{yd}$, D = 360.5 yd, $56.3°$ north of east; The numerical answers would stay the same but the physical unit would be meters. The physical meaning and distances would be about the same because 1 yd is comparable with 1 m.

57. $\vec{\mathbf{R}} = (-3\hat{\mathbf{i}} - 16\hat{\mathbf{j}})\text{m}$

59. $\vec{\mathbf{E}} = E\hat{\mathbf{E}}$, $E_x = +178.9\text{V/m}$, $E_y = -357.8\text{V/m}$, $E_z = 0.0\text{V/m}$, $\theta_E = -\tan^{-1}(2)$

61. a. $-34.290\vec{\mathbf{R}}_B = (-12.278\hat{\mathbf{i}} + 7.089\hat{\mathbf{j}} + 2.500\hat{\mathbf{k}})\text{km}$, $\vec{\mathbf{R}}_D = (-34.290\hat{\mathbf{i}} + 3.000\hat{\mathbf{k}})\text{km}$; b. $\left|\vec{\mathbf{R}}_B - \vec{\mathbf{R}}_D\right| = 23.131\text{ km}$

63. a. 0, b. 0, c. −0.866, d. −17.32

65. $\theta_i = 64.12°, \theta_j = 150.79°, \theta_k = 77.39°$

67. a. $-120\hat{\mathbf{k}}$, b. $0\hat{\mathbf{k}}$, c. $94\hat{\mathbf{k}}$, d. $-240\hat{\mathbf{k}}$, e. $4.0\hat{\mathbf{k}}$, f. $-3.0\hat{\mathbf{k}}$, g. $15\hat{\mathbf{k}}$, h. 0

69. a. 0, b. 0, c. $20{,}000\hat{\mathbf{k}}$

Additional Problems

71. a. 18.4 km and 26.2 km, b. 31.5 km and 5.56 km

73. a. $(r, \pi - \varphi)$, b. $(2r, \varphi + 2\pi)$, c. $(3r, -\varphi)$

75. $d_{\text{PM}} = 6.2\text{ nmi} = 11.4\text{ km}$, $d_{\text{NP}} = 7.2\text{ nmi} = 13.3\text{ km}$

77. proof

79. a. 10.00 m, b. 5π m, c. 0

81. 22.2 km/h, $35.8°$ south of west

83. 270 m, $4.2°$ north of west

85. $\vec{\mathbf{B}} = -4.0\hat{\mathbf{i}} + 3.0\hat{\mathbf{j}}$ or $\vec{\mathbf{B}} = 4.0\hat{\mathbf{i}} - 3.0\hat{\mathbf{j}}$

87. proof

Challenge Problems

89. $G_H = 19\,\text{N}/\sqrt{17} \approx 4.6\,\text{N}$

91. proof

Chapter 3

Check Your Understanding

3.1 (a) The rider's displacement is $\Delta x = x_f - x_0 = -1\text{ km}$. (The displacement is negative because we take east to be positive and west to be negative.) (b) The distance traveled is 3 km + 2 km = 5 km. (c) The magnitude of the displacement is 1 km.

3.2 (a) Taking the derivative of *x*(*t*) gives *v*(*t*) = −6*t* m/s. (b) No, because time can never be negative. (c) The velocity is *v*(1.0 s) = −6 m/s and the speed is $|v(1.0\text{ s})| = 6\text{ m/s}$.

3.3 Inserting the knowns, we have

$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{2.0\times10^7\text{ m/s}-0}{10^{-4}\text{ s}-0} = 2.0\times10^{11}\text{m/s}^2.$

3.4 If we take east to be positive, then the airplane has negative acceleration because it is accelerating toward the west. It is also acceleration opposite to the motion; its acceleration is opposite in direction to its velocity.

3.5 To answer this, choose an equation that allows us to solve for time *t*, given only *a* , v_0 , and *v*:

$v = v_0 + at.$

Rearrange to solve for *t*:

$t = \frac{v-v_0}{a} = \frac{400\text{ m/s}-0\text{ m/s}}{20\text{ m/s}^2} = 20\text{ s}.$

3.6 $a = \frac{2}{3}\text{ m/s}^2.$

3.7 It takes 2.47 s to hit the water. The quantity distance traveled increases faster.

3.8
a. The velocity function is the integral of the acceleration function plus a constant of integration. By Equation 3.18,

$$v(t) = \int a(t)dt + C_1 = \int (5-10t)dt + C_1 = 5t - 5t^2 + C_1.$$

Since *v*(0) = 0, we have C_1 = 0; so,

$v(t) = 5t - 5t^2.$

b. By Equation 3.19,

$$x(t) = \int v(t)dt + C_2 = \int (5t - 5t^2)dt + C_2 = \frac{5}{2}t^2 - \frac{5}{3}t^3 + C_2.$$

Since *x*(0) = 0, we have C_2 = 0, and

$x(t) = \frac{5}{2}t^2 - \frac{5}{3}t^3.$

c. The velocity can be written as *v*(*t*) = 5*t*(1 – *t*), which equals zero at *t* = 0, and *t* = 1 s.

Conceptual Questions

1. You drive your car into town and return to drive past your house to a friend's house.

3. If the bacteria are moving back and forth, then the displacements are canceling each other and the final displacement is small.

5. Distance traveled

7. Average speed is the total distance traveled divided by the elapsed time. If you go for a walk, leaving and returning to your home, your average speed is a positive number. Since Average velocity = Displacement/Elapsed time, your average velocity is zero.

9. Average speed. They are the same if the car doesn't reverse direction.

11. No, in one dimension constant speed requires zero acceleration.

13. A ball is thrown into the air and its velocity is zero at the apex of the throw, but acceleration is not zero.

15. Plus, minus

17. If the acceleration, time, and displacement are the knowns, and the initial and final velocities are the unknowns, then two kinematic equations must be solved simultaneously. Also if the final velocity, time, and displacement are the knowns then two kinematic equations must be solved for the initial velocity and acceleration.

19. a. at the top of its trajectory; b. yes, at the top of its trajectory; c. yes

21. Earth $v = v_0 - gt = -gt$; Moon $v' = -\frac{g}{6}t'$ $v = v'$ $-gt = -\frac{g}{6}t'$ $t' = 6t$; Earth $y = -\frac{1}{2}gt^2$ Moon $y' = -\frac{1}{2}\frac{g}{6}(6t)^2 = -\frac{1}{2}g6t^2 = -6\left(\frac{1}{2}gt^2\right) = -6y$

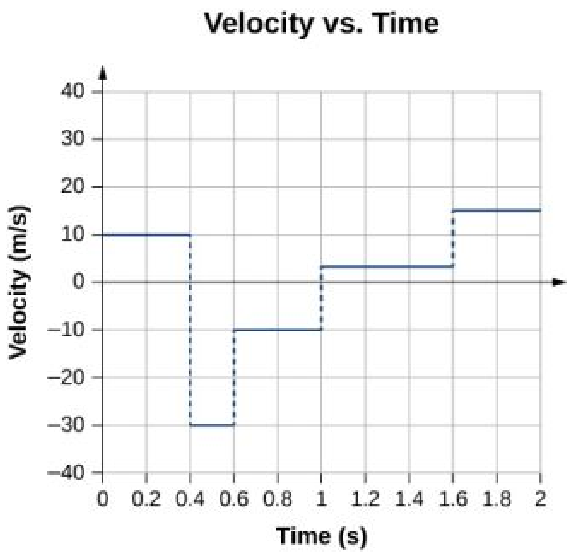
Problems

25. a. $\vec{\mathbf{x}}_1 = (-2.0\,\text{km})\hat{\mathbf{i}}$, $\vec{\mathbf{x}}_2 = (5.0\,\text{km})\hat{\mathbf{i}}$; b. 7.0 km east

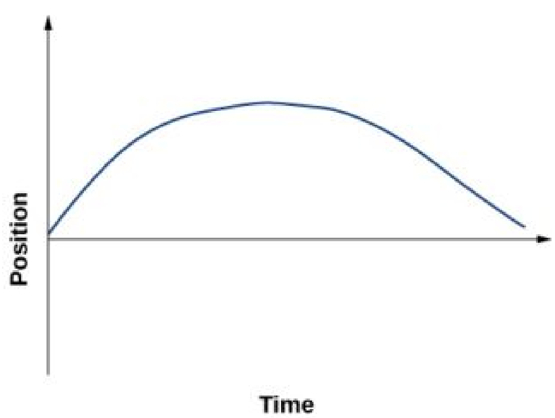
27. a. $t = 2.0$ s; b. $x(6.0) - x(3.0) = -8.0 - (-2.0) = -6.0\,\text{m}$

29. a. 150.0 s, $\bar{v} = 902\,\text{m/s}$; b. 263% the speed of sound at sea level or about Mach 2.

31.



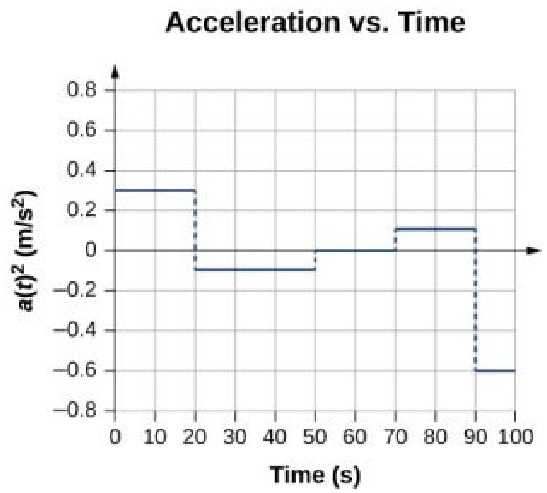
33.



35. a. $v(t) = (10 - 4t)\text{m/s}$; $v(2\text{ s}) = 2$ m/s, $v(3\text{ s}) = -2$ m/s; b. $|v(2\text{ s})| = 2\,\text{m/s}$, $|v(3\text{ s})| = 2\,\text{m/s}$; (c) $\bar{v} = 0\,\text{m/s}$

37. $a = 4.29\text{m/s}^2$

39.

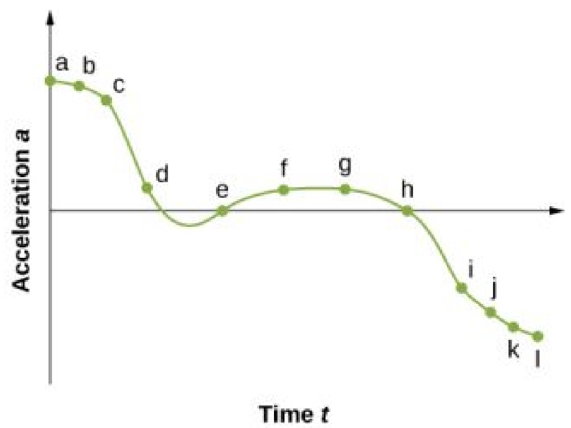


41. $a = 11.1g$

43. 150 m

45. a. 525 m;
b. $v = 180\ \text{m/s}$

47. a.



b. The acceleration has the greatest positive value at t_a
c. The acceleration is zero at t_e and t_h
d. The acceleration is negative at t_i,t_j,t_k,t_l

49. a. $a = -1.3\ \text{m/s}^2$;
b. $v_0 = 18\ \text{m/s}$;
c. $t = 13.8\ \text{s}$

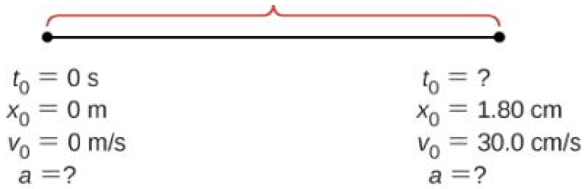
51. $v = 502.20\ \text{m/s}$

53. a.

b. Knowns: $a = 2.40\ \text{m/s}^2$, $t = 12.0\ \text{s}$, $v_0 = 0\ \text{m/s}$, and $x_0 = 0\ \text{m}$;
c. $x = x_0 + v_0 t + \frac{1}{2}at^2 = \frac{1}{2}at^2 = 2.40\ \text{m/s}^2 (12.0\ \text{s})^2 = 172.80\ \text{m}$, the answer seems reasonable at about

172.8 m; d. $v = 28.8\ \text{m/s}$

55. a.



b. Knowns: $v = 30.0\ \text{cm/s}$, $x = 1.80\ \text{cm}$;
c. $a = 250\ \text{cm/s}^2,\quad t = 0.12\ \text{s}$;
d. yes

57. a. 6.87 m/s^2; b. $x = 52.26\ \text{m}$

59. a. $a = 8450\ \text{m/s}^2$;
b. $t = 0.0077\ \text{s}$

61. a. $a = 9.18\ g$;
b. $t = 6.67 \times 10^{-3}\ \text{s}$;

c.
$$a = -40.0\ \text{m/s}^2$$
$$a = 4.08\ g$$

63. Knowns: $x = 3\ \text{m}$, $v = 0\ \text{m/s}$, $v_0 = 54\ \text{m/s}$. We want *a*, so we can use this equation: $a = -486\ \text{m/s}^2$.

65. a. $a = 32.58\ \text{m/s}^2$;
b. $v = 161.85\ \text{m/s}$;
c. $v > v_{\text{max}}$, because the assumption of constant acceleration is not valid for a dragster. A dragster changes gears and would have a greater acceleration in first gear than second gear than third gear, and so on. The acceleration would be greatest at the beginning, so it would not be accelerating at $32.6\ \text{m/s}^2$ during the last few meters, but substantially less, and the final velocity would be less than $162\ \text{m/s}$.

67. a.
$$y = -8.23\ \text{m}$$
$$v_1 = -18.9\ \text{m/s}$$
;

b.
$$y = -18.9\ \text{m}$$
$$v_2 = -23.8\ \text{m/s}$$
;

c.
$$y = -32.0\ \text{m}$$
$$v_3 = -28.7\ \text{m/s}$$
;

d.
$$y = -47.6\ \text{m}$$
$$v_4 = -33.6\ \text{m/s}$$
;

e.
$$y = -65.6\ \text{m}$$
$$v_5 = -38.5\ \text{m/s}$$

69. a. Knowns: $a = -9.8\ \text{m/s}^2\quad v_0 = -1.4\ \text{m/s}\quad t = 1.8\ \text{s}\quad y_0 = 0\ \text{m}$;

b. $y = y_0 + v_0 t - \frac{1}{2}gt^2 \quad y = v_0 t - \frac{1}{2}gt = -1.4\text{ m/s}(1.8\text{ sec}) - \frac{1}{2}(9.8)(1.8\text{ s})^2 = -18.4\text{ m}$ and the origin is at the rescuers, who are 18.4 m above the water.

71. a. $v^2 = v_0^2 - 2g(y - y_0) \quad y_0 = 0 \quad v = 0 \quad y = \frac{v_0^2}{2g} = \frac{(4.0\text{ m/s})^2}{2(9.80)} = 0.82\text{ m}$; b. to the apex $v = 0.41\text{ s}$ times 2 to the board = 0.82 s from the board to the water

$y = y_0 + v_0 t - \frac{1}{2}gt^2 \quad y = -1.80\text{ m} \quad y_0 = 0 \quad v_0 = 4.0\text{ m/s}-1.8 = 4.0t - 4.9t^2 \quad 4.9t^2 - 4.0t - 1.80 = 0,$

solution to quadratic equation gives 1.13 s; c. $v^2 = v_0^2 - 2g(y - y_0) \quad y_0 = 0 \quad v_0 = 4.0\text{ m/s} \quad y = -1.80\text{ m}$

$v = 7.16\text{ m/s}$

73. Time to the apex: $t = 1.12\text{ s}$ times 2 equals 2.24 s to a height of 2.20 m. To 1.80 m in height is an additional 0.40 m. $y = y_0 + v_0 t - \frac{1}{2}gt^2 \quad y = -0.40\text{ m} \quad y_0 = 0 \quad v_0 = -11.0\text{ m/s}$.

$$y = y_0 + v_0 t - \frac{1}{2}gt^2 \quad y = -0.40\text{ m} \quad y_0 = 0 \quad v_0 = -11.0\text{ m/s}$$
$$-0.40 = -11.0t - 4.9t^2 \quad \text{or} \quad 4.9t^2 + 11.0t - 0.40 = 0$$

Take the positive root, so the time to go the additional 0.4 m is 0.04 s. Total time is $2.24\text{ s} + 0.04\text{ s} = 2.28\text{ s}$.

75. a. $v^2 = v_0^2 - 2g(y - y_0) \quad y_0 = 0 \quad v = 0 \quad y = 2.50\text{ m}$; b. $t = 0.72\text{ s}$ times 2 gives 1.44 s in the air

$$v_0^2 = 2gy \Rightarrow v_0 = \sqrt{2(9.80)(2.50)} = 7.0\text{ m/s}$$

77. a. $v = 70.0\text{ m/s}$; b. time heard after rock begins to fall: 0.75 s, time to reach the ground: 6.09 s

79. a. $A = \text{m/s}^2 \quad B = \text{m/s}^{5/2}$;

b. $v(t) = \int a(t)dt + C_1 = \int \left(A - Bt^{1/2}\right) dt + C_1 = At - \frac{2}{3}Bt^{3/2} + C_1$;

$$v(0) = 0 = C_1 \quad \text{so} \quad v(t_0) = At_0 - \frac{2}{3}Bt_0^{3/2}$$

c. $x(t) = \int v(t)dt + C_2 = \int \left(At - \frac{2}{3}Bt^{3/2}\right) dt + C_2 = \frac{1}{2}At^2 - \frac{4}{15}Bt^{5/2} + C_2$

$$x(0) = 0 = C_2 \quad \text{so} \quad x(t_0) = \frac{1}{2}At_0^2 - \frac{4}{15}Bt_0^{5/2}$$

81. a. $a(t) = 3.2\text{m/s}^2 \quad t \le 5.0\text{ s}$;

$$a(t) = 1.5\text{m/s}^2 \quad 5.0\text{ s} \le t \le 11.0\text{ s}$$
$$a(t) = 0\text{m/s}^2 \quad t > 11.0\text{ s}$$

$$x(t) = \int v(t)dt + C_2 = \int 3.2tdt + C_2 = 1.6t^2 + C_2$$

$t \le 5.0\text{ s}$

$x(0) = 0 \Rightarrow C_2 = 0$ therefore, $x(2.0\text{ s}) = 6.4\text{ m}$

$$x(t) = \int v(t)dt + C_2 = \int [16.0 - 1.5(t - 5.0)]\,dt + C_2 = 16t - 1.5\left(\frac{t^2}{2} - 5.0t\right) + C_2$$

$5.0 \le t \le 11.0\text{ s}$

b. $x(5\text{ s}) = 1.6(5.0)^2 = 40\text{ m} = 16(5.0\text{ s}) - 1.5\left(\frac{5^2}{2} - 5.0(5.0)\right) + C_2$

$40 = 98.75 + C_2 \Rightarrow C_2 = -58.75$

$x(7.0\text{ s}) = 16(7.0) - 1.5\left(\frac{7^2}{2} - 5.0(7)\right) - 58.75 = 69\text{ m}$

$$x(t) = \int 7.0dt + C_2 = 7t + C_2$$

$t \ge 11.0\text{ s}$

$x(11.0\text{ s}) = 16(11) - 1.5\left(\frac{11^2}{2} - 5.0(11)\right) - 58.75 = 109 = 7(11.0\text{ s}) + C_2 \Rightarrow C_2 = 32\text{ m}$

$x(t) = 7t + 32\text{ m}$

$x \ge 11.0\text{ s} \Rightarrow x(12.0\text{ s}) = 7(12) + 32 = 116\text{ m}$

Additional Problems

83. Take west to be the positive direction.
1st plane: $\bar{v} = 600\text{ km/h}$
2nd plane $\bar{v} = 667.0\text{ km/h}$

85. $a = \frac{v - v_0}{t - t_0}$, $t = 0$, $a = \frac{-3.4\text{ cm/s} - v_0}{4\text{ s}} = 1.2\text{ cm/s}^2 \Rightarrow v_0 = -8.2\text{ cm/s}$ $v = v_0 + at = -8.2 + 1.2\,t$;
$v = -7.0\text{ cm/s}$ $v = -1.0\text{ cm/s}$

87. $a = -3\text{ m/s}^2$

89. a.
$v = 8.7 \times 10^5\text{ m/s}$;
b. $t = 7.8 \times 10^{-8}\text{ s}$

91. $1\text{ km} = v_0(80.0\text{ s}) + \frac{1}{2}a(80.0)^2$; $2\text{ km} = v_0(200.0) + \frac{1}{2}a(200.0)^2$ solve simultaneously to get $a = -\frac{0.1}{2400.0}\text{km/s}^2$ and $v_0 = 0.014167\text{ km/s}$, which is 51.0 km/h. Velocity at the end of the trip is $v = 21.0\text{ km/h}$.

93. $a = -0.9\text{ m/s}^2$

95. Equation for the speeding car: This car has a constant velocity, which is the average velocity, and is not accelerating, so use the equation for displacement with $x_0 = 0$:$x = x_0 + \bar{v}t = \bar{v}t$; Equation for the police car: This car is accelerating, so use the equation for displacement with $x_0 = 0$ and $v_0 = 0$, since the police car starts from rest: $x = x_0 + v_0 t + \frac{1}{2}at^2 = \frac{1}{2}at^2$; Now we have an equation of motion for each car with a common parameter, which can be eliminated to find the solution. In this case, we solve for t. Step 1, eliminating x: $x = \bar{v}t = \frac{1}{2}at^2$; Step 2, solving for t: $t = \frac{2\bar{v}}{a}$. The speeding car has a constant velocity of 40 m/s, which is its average velocity. The acceleration of the police car is 4 m/s^2. Evaluating t, the time for the police car to reach the speeding car, we have $t = \frac{2\bar{v}}{a} = \frac{2(40)}{4} = 20\text{ s}$.

97. At this acceleration she comes to a full stop in $t = \frac{-v_0}{a} = \frac{8}{0.5} = 16\text{ s}$, but the distance covered is $x = 8\text{ m/s}(16\text{ s}) - \frac{1}{2}(0.5)(16\text{ s})^2 = 64\text{ m}$, which is less than the distance she is away from the finish line, so she never finishes the race.

99. $x_1 = \frac{3}{2}v_0 t$

$x_2 = \frac{5}{3}x_1$

101. $v_0 = 7.9$ m/s velocity at the bottom of the window.
$v = 7.9$ m/s
$v_0 = 14.5$ m/s

103. a. $v = 5.42$ m/s;
b. $v = 4.64$ m/s;
c. $a = 2874.28$ m/s^2;
d. $(x - x_0) = 5.11 \times 10^{-3}$ m

105. Consider the players fall from rest at the height 1.0 m and 0.3 m.
0.9 s
0.5 s

107. a. $t = 6.37$ s taking the positive root;
b. $v = 59.5$ m/s

109. a. $y = 4.9$ m;
b. $v = 38.3$ m/s;
c. -33.3 m

111. $h = \frac{1}{2}gt^2$, h = total height and time to drop to ground
$\frac{2}{3}h = \frac{1}{2}g(t-1)^2$ in $t - 1$ seconds it drops $2/3h$
$\frac{2}{3}\left(\frac{1}{2}gt^2\right) = \frac{1}{2}g(t-1)^2$ or $\frac{t^2}{3} = \frac{1}{2}(t-1)^2$
$0 = t^2 - 6t + 3$ $t = \frac{6 \pm \sqrt{6^2 - 4 \cdot 3}}{2} = 3 \pm \frac{\sqrt{24}}{2}$
t = 5.45 s and h = 145.5 m. Other root is less than 1 s. Check for t = 4.45 s $h = \frac{1}{2}gt^2 = 97.0$ m $= \frac{2}{3}(145.5)$

Challenge Problems

113. a. $v(t) = 10t - 12t^2$ m/s, $a(t) = 10 - 24t$ m/s^2;
b. $v(2\text{ s}) = -28$ m/s, $a(2\text{ s}) = -38$m/s^2; c. The slope of the position function is zero or the velocity is zero. There are two possible solutions: t = 0, which gives x = 0, or t = 10.0/12.0 = 0.83 s, which gives x = 1.16 m. The second answer is the correct choice; d. 0.83 s (e) 1.16 m

115. 96 km/h $= 26.67$ m/s, $a = \frac{26.67\text{ m/s}}{4.0\text{ s}} = 6.67$m/s^2, 295.38 km/h = 82.05 m/s, $t = 12.3$ s time to accelerate to maximum speed
$x = 504.55$ m distance covered during acceleration
7495.44 m at a constant speed
$\frac{7495.44\text{ m}}{82.05\text{ m/s}} = 91.35$ s so total time is $91.35\text{ s} + 12.3\text{ s} = 103.65$ s.

Chapter 4

Check Your Understanding

4.1 (a) Taking the derivative with respect to time of the position function, we have $\vec{\mathbf{v}}(t) = 9.0t^2\hat{\mathbf{i}}$ and $\vec{\mathbf{v}}(3.0\text{s}) = 81.0\hat{\mathbf{i}}$m/s. (b) Since the velocity function is nonlinear, we suspect the average velocity is not equal to the instantaneous velocity. We check this and find
$\vec{\mathbf{v}}_{\text{avg}} = \frac{\vec{\mathbf{r}}(t_2) - \vec{\mathbf{r}}(t_1)}{t_2 - t_1} = \frac{\vec{\mathbf{r}}(4.0\text{ s}) - \vec{\mathbf{r}}(2.0\text{ s})}{4.0\text{ s} - 2.0\text{ s}} = \frac{[(192-24)\hat{\mathbf{i}} + (4-4)\hat{\mathbf{i}}]\text{ m}}{2.0\text{ s}} = 84\hat{\mathbf{j}}$m/s,
which is different from $\vec{\mathbf{v}}(3.0\text{s}) = 81.0\hat{\mathbf{i}}$m/s.

4.2 The acceleration vector is constant and doesn't change with time. If a, b, and c are not zero, then the velocity function must be linear in time. We have $\vec{\mathbf{v}}(t) = \int \vec{\mathbf{a}}dt = \int (a\hat{\mathbf{i}} + b\hat{\mathbf{j}} + c\hat{\mathbf{k}})dt = (a\hat{\mathbf{i}} + b\hat{\mathbf{j}} + c\hat{\mathbf{k}})t$ m/s, since taking the derivative of the velocity function produces $\vec{\mathbf{a}}(t)$. If any of the components of the acceleration are zero, then that component of the velocity would be a constant.

4.3 (a) Choose the top of the cliff where the rock is thrown from the origin of the coordinate system. Although it is arbitrary, we typically choose time t = 0 to correspond to the origin. (b) The equation that describes the horizontal motion is $x = x_0 + v_x t$. With $x_0 = 0$, this equation becomes $x = v_x t$. (c) Equation 4.16 through

Equation 4.18 and Equation 4.19 describe the vertical motion, but since $y_0 = 0$ and $v_{0y} = 0$, these equations simplify greatly to become $y = \frac{1}{2}(v_{0y} + v_y)t = \frac{1}{2}v_y t,\ v_y = -gt,\ y = -\frac{1}{2}gt^2,$ and $v_y^2 = -2gy$. (d) We use the kinematic equations to find the *x* and *y* components of the velocity at the point of impact. Using $v_y^2 = -2gy$ and noting the point of impact is −100.0 m, we find the *y* component of the velocity at impact is $v_y = 44.3\text{ m/s}$. We are given the *x* component, $v_x = 15.0\text{ m/s}$, so we can calculate the total velocity at impact: *v* = 46.8 m/s and $\theta = 71.3°$ below the horizontal.

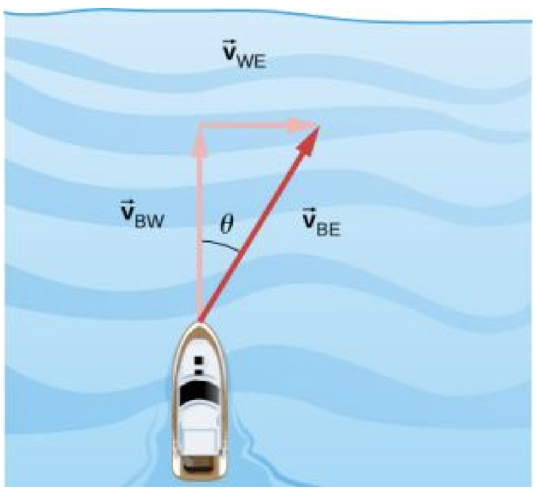
4.4 The golf shot at $30°$.

4.5 134.0 cm/s

4.6 Labeling subscripts for the vector equation, we have B = boat, R = river, and E = Earth. The vector equation becomes $\vec{\mathbf{v}}_{\text{BE}} = \vec{\mathbf{v}}_{\text{BR}} + \vec{\mathbf{v}}_{\text{RE}}$. We have right triangle geometry shown in Figure 04_05_BoatRiv_img. Solving for $\vec{\mathbf{v}}_{\text{BE}}$, we have

$$v_{\text{BE}} = \sqrt{v_{\text{BR}}^2 + v_{\text{RE}}^2} = \sqrt{4.5^2 + 3.0^2}$$

$$v_{\text{BE}} = 5.4\text{ m/s},\quad \theta = \tan^{-1}\left(\frac{3.0}{4.5}\right) = 33.7°.$$



Conceptual Questions

1. straight line

3. The slope must be zero because the velocity vector is tangent to the graph of the position function.

5. No, motions in perpendicular directions are independent.

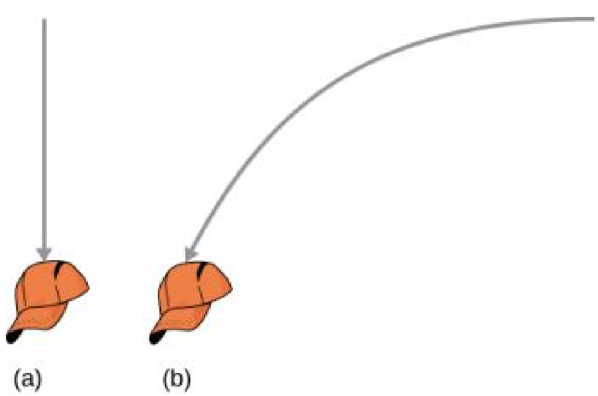
7. a. no; b. minimum at apex of trajectory and maximum at launch and impact; c. no, velocity is a vector; d. yes, where it lands

9. They both hit the ground at the same time.

11. yes

13. If he is going to pass the ball to another player, he needs to keep his eyes on the reference frame in which the other players on the team are located.

15.



Problems

17. $\vec{\mathbf{r}} = 1.0\hat{\mathbf{i}} - 4.0\hat{\mathbf{j}} + 6.0\hat{\mathbf{k}}$

19. $\Delta\vec{\mathbf{r}}_{\text{Total}} = 472.0\ \text{m}\hat{\mathbf{i}} + 80.3\ \text{m}\hat{\mathbf{j}}$

21. Sum of displacements $= -6.4\ \text{km}\hat{\mathbf{i}} + 9.4\ \text{km}\hat{\mathbf{j}}$

23. a. $\vec{\mathbf{v}}(t) = 8.0t\hat{\mathbf{i}} + 6.0t^2\hat{\mathbf{k}}, \quad \vec{\mathbf{v}}(0) = 0, \quad \vec{\mathbf{v}}(1.0) = 8.0\hat{\mathbf{i}} + 6.0\hat{\mathbf{k}}\text{m/s},$

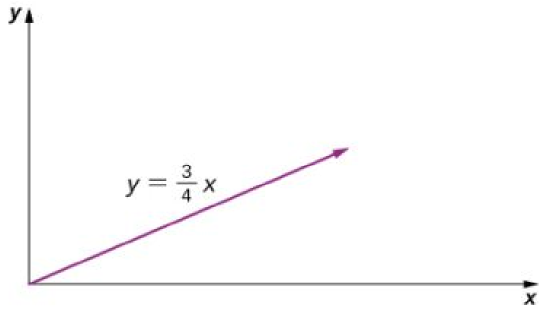
b. $\vec{\mathbf{v}}_{\text{avg}} = 4.0 \quad \hat{\mathbf{i}} + 2.0\hat{\mathbf{k}}\ \text{m/s}$

25. $\Delta\vec{\mathbf{r}}_1 = 20.00\ \text{m}\hat{\mathbf{j}}, \Delta\vec{\mathbf{r}}_2 = (2.000 \times 10^4\ \text{m})(\cos 30°\hat{\mathbf{i}} + \sin 30°\hat{\mathbf{j}})$

$\Delta\vec{\mathbf{r}} = 1.700 \times 10^4\ \text{m}\hat{\mathbf{i}} + 1.002 \times 10^4\ \text{m}\hat{\mathbf{j}}$

27. a. $\vec{\mathbf{v}}(t) = (4.0t\hat{\mathbf{i}} + 3.0t\hat{\mathbf{j}})\text{m/s}, \vec{\mathbf{r}}(t) = (2.0t^2\hat{\mathbf{i}} + \frac{3}{2}t^2\hat{\mathbf{j}})\ \text{m},$

b. $x(t) = 2.0t^2\ \text{m}, y(t) = \frac{3}{2}t^2\ \text{m}, t^2 = \frac{x}{2} \Rightarrow y = \frac{3}{4}x$



29. a. $\vec{\mathbf{v}}(t) = (6.0t\hat{\mathbf{i}} - 21.0t^2\hat{\mathbf{j}} + 10.0t^{-3}\hat{\mathbf{k}})\text{m/s}$

b. $\vec{\mathbf{a}}(t) = (6.0\hat{\mathbf{i}} - 42.0t\hat{\mathbf{j}} - 30t^{-4}\hat{\mathbf{k}})\text{m/s}^2$

c. $\vec{\mathbf{v}}(2.0s) = (12.0\hat{\mathbf{i}} - 84.0\hat{\mathbf{j}} + 1.25\hat{\mathbf{k}})\text{m/s}$

d. $\vec{\mathbf{v}}(1.0\ \text{s}) = 6.0\hat{\mathbf{i}} - 21.0\hat{\mathbf{j}} + 10.0\hat{\mathbf{k}}\text{m/s}, \ |\vec{\mathbf{v}}(1.0\ \text{s})| = 24.0\ \text{m/s}$

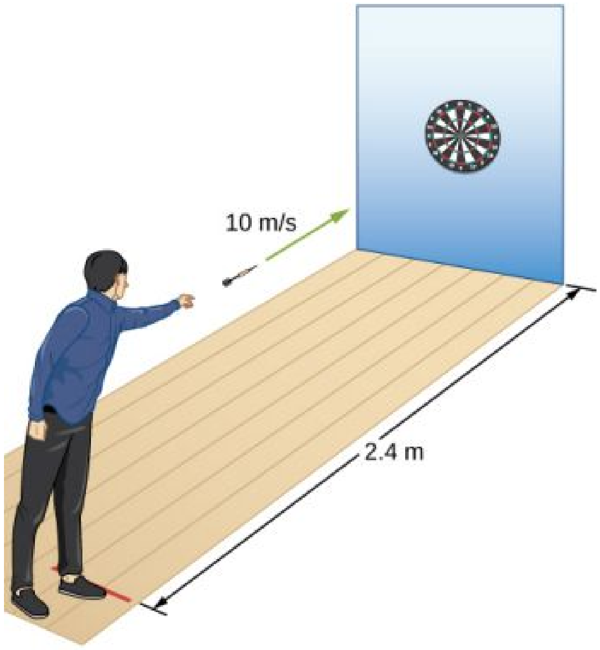
$\vec{\mathbf{v}}(3.0\ \text{s}) = 18.0\hat{\mathbf{i}} - 189.0\hat{\mathbf{j}} + 0.37\hat{\mathbf{k}}\text{m/s},, \ |\vec{\mathbf{v}}(3.0\ \text{s})| = 190\ \text{m/s}$

e. $\vec{\mathbf{r}}(t) = (3.0t^2\hat{\mathbf{i}} - 7.0t^3\hat{\mathbf{j}} - 5.0t^{-2}\hat{\mathbf{k}})\text{m}, \vec{\mathbf{v}}_{\text{avg}} = 9.0\hat{\mathbf{i}} - 49.0\hat{\mathbf{j}} + 3.75\hat{\mathbf{k}}\text{m/s}$

31. a. $\vec{\mathbf{v}}(t) = -\sin(1.0t)\hat{\mathbf{i}} + \cos(1.0t)\hat{\mathbf{j}} + \hat{\mathbf{k}}$, b. $\vec{\mathbf{a}}(t) = -\cos(1.0t)\hat{\mathbf{i}} - \sin(1.0t)\hat{\mathbf{j}}$

33. a. $t = 0.55\ \text{s}$, b. $x = 110\ \text{m}$

35. a. $t = 0.24\text{s}, \ d = 0.28\ \text{m}$, b. They aim high.



37. a., $t = 12.8\ \text{s}, \ x = 5619\ \text{m}$ b. $v_y = 125.0\ \text{m/s}, \ v_x = 439.0\ \text{m/s}, \ |\vec{v}| = 456.0\ \text{m/s}$

39. a. $v_y = v_{0y} - gt$, $t = 10\text{s}$, $v_y = 0$, $v_{0y} = 98.0\text{ m/s}$, $v_0 = 196.0\text{ m/s}$,
b. $h = 490.0\text{ m}$,
c. $v_{0x} = 169.7\text{ m/s}$, $x = 3394.0\text{ m}$,

d. $x = 169.7\text{ m/s}\,(15.0\text{ s}) = 2550\text{ m}$
$y = (98.0\text{ m/s})\,(15.0\text{ s}) – 4.9(15.0\text{s})^2 = 368\text{ m}$
$\overset{\mathbf{r}}{\mathbf{s}} = 2550\text{ m}\hat{\mathbf{i}} + 368\text{ m}\hat{\mathbf{j}}$

41. $-100\text{ m} = (-2.0\text{ m/s})t - (4.9\text{ m/s}^2)t^2$, $t = 4.3\text{ s}$, $x = 86.0\text{ m}$

43. $R_{Moon} = 48\text{ m}$

45. a. $v_{0y} = 24\text{ m/s}v_y^2 = v_{0y}^2 - 2gy \Rightarrow h = 29.3\text{ m}$,
b. $t = 2.4\text{ s}$ $v_{0x} = 18\text{ m/s}$ $x = 43.2\text{ m}$,
c. $y = -100\text{ m}$ $y_0 = 0$ $y - y_0 = v_{0y}t - \frac{1}{2}gt^2$ $-100 = 24t - 4.9t^2 \Rightarrow t = 7.58\text{ s}$,
d. $x = 136.44\text{ m}$,
e. $t = 2.0\text{ s}$ $y = 28.4\text{ m}$ $x = 36\text{ m}$
$t = 4.0\text{ s}$ $y = 17.6\text{ m}$ $x = 72\text{ m}$
$t = 6.0\text{ s}$ $y = -32.4\text{ m}$ $x = 108\text{ m}$

47. $v_{0y} = 12.9\text{ m/s}$ $y - y_0 = v_{0y}t - \frac{1}{2}gt^2$ $-20.0 = 12.9t - 4.9t^2$
$t = 3.7\text{ s}$ $v_{0x} = 15.3\text{ m/s} \Rightarrow x = 56.7\text{ m}$
So the golfer’s shot lands 13.3 m short of the green.

49. a. $R = 60.8\text{ m}$,
b. $R = 137.8\text{ m}$

51. a. $v_y^2 = v_{0y}^2 - 2gy \Rightarrow y = 2.9\text{ m/s}$
$y = 3.3\text{ m/s}$
$y = \frac{v_{0y}^2}{2g} = \frac{(v_0 \sin\theta)^2}{2g} \Rightarrow \sin\theta = 0.91 \Rightarrow \theta = 65.5°$

53. $R = 18.5\text{ m}$

55. $y = (\tan\theta_0)x - \left[\frac{g}{2(v_0\cos\theta_0)^2}\right]x^2 \Rightarrow v_0 = 16.4\text{ m/s}$

57. $R = \frac{v_0^2 \sin 2\theta_0}{g} \Rightarrow \theta_0 = 15.9°$

59. (a) It takes the wide receiver 1.1 s to cover the last 10 m of his run.
$T_{\text{tof}} = \frac{2(v_0 \sin\theta)}{g} \Rightarrow \sin\theta = 0.27 \Rightarrow \theta = 15.6°$
(b) $x = \frac{v_0^2 \sin\left(2\theta\right)}{g} = 21\text{ m}$
Therefore, the ball will be overthrown, and the receiver will not be able to catch it.

61. $a_C = 40\text{ m/s}^2$

63. $a_C = \frac{v^2}{r} \Rightarrow v^2 = r\ a_C = 78.4$, $v = 8.85\text{ m/s}$
$T = 5.68\text{ s}$, which is $0.176\text{ rev/s} = 10.6\text{ rev/min}$

65. Venus is 108.2 million km from the Sun and has an orbital period of 0.6152 y.
$r = 1.082 \times 10^{11}\text{ m}$ $T = 1.94 \times 10^7\text{ s}$
$v = 3.5 \times 10^4\text{ m/s}$, $\text{a}_C = 1.135 \times 10^{-2}\text{ m/s}^2$

67. $360\text{ rev/min} = 6\text{ rev/s}$
$v = 3.8\text{ m/s}$ $a_C = 144.\text{ m/s}^2$

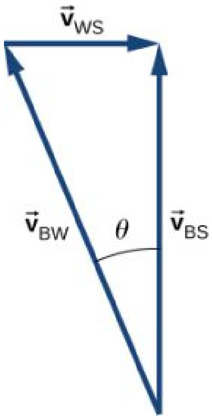
69. a. $O'(t) = (4.0\hat{\mathbf{i}} + 3.0\hat{\mathbf{j}} + 5.0\hat{\mathbf{k}})t\text{ m}$,
b. $\vec{\mathbf{r}}_{PS} = \vec{\mathbf{r}}_{PS'} + \vec{\mathbf{r}}_{S'S}$, $\vec{\mathbf{r}}(t) = \vec{\mathbf{r}}'(t) + (4.0\hat{\mathbf{i}} + 3.0\hat{\mathbf{j}} + 5.0\hat{\mathbf{k}})t\text{ m}$,
c. $\vec{\mathbf{v}}(t) = \vec{\mathbf{v}}'(t) + (4.0\hat{\mathbf{i}} + 3.0\hat{\mathbf{j}} + 5.0\hat{\mathbf{k}})\text{ m/s}$, d. The accelerations are the same.

71. $\vec{\mathbf{v}}_{PC} = (2.0\hat{\mathbf{i}} + 5.0\hat{\mathbf{j}} + 4.0\hat{\mathbf{k}})\text{m/s}$

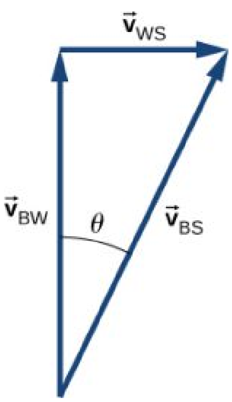
73. a. A = air, S = seagull, G = ground

$\vec{v}_{SA} = 9.0\text{ m/s}$ velocity of seagull with respect to still air
$\vec{v}_{AG} = ?\ \ \vec{v}_{SG} = 5\text{ m/s}\ \vec{v}_{SG} = \vec{v}_{SA} + \vec{v}_{AG} \Rightarrow \vec{v}_{AG} = \vec{v}_{SG} - \vec{v}_{SA}$
$\vec{v}_{AG} = -4.0\text{ m/s}$
b. $\vec{v}_{SG} = \vec{v}_{SA} + \vec{v}_{AG} \Rightarrow \vec{v}_{SG} = -13.0\text{ m/s}$
$\frac{-6000\text{ m}}{-13.0\text{ m/s}} = 7\text{ min }42\text{ s}$

75. Take the positive direction to be the same direction that the river is flowing, which is east. S = shore/Earth, W = water, and B = boat.
a. $\vec{v}_{BS} = 11\text{ km/h}$
$t = 8.2\text{ min}$
b. $\vec{v}_{BS} = -5\text{ km/h}$
$t = 18\text{ min}$
c. $\vec{v}_{BS} = \vec{v}_{BW} + \vec{v}_{WS}\ \theta = 22°$ west of north

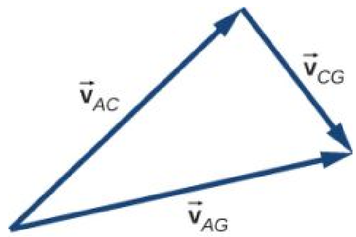


d. $|\vec{v}_{BS}| = 7.4\text{ km/h}\ t = 6.5\text{ min}$
e. $\vec{v}_{BS} = 8.54\text{ km/h}$, but only the component of the velocity straight across the river is used to get the time



$t = 6.0\text{ min}$
Downstream = 0.3 km

77. $\vec{v}_{AG} = \vec{v}_{AC} + \vec{v}_{CG}$
$|\vec{v}_{AC}| = 25\text{ km/h}\ \ |\vec{v}_{CG}| = 15\text{ km/h}\ \ |\vec{v}_{AG}| = 29.15\text{ km/h}\ \vec{v}_{AG} = \vec{v}_{AC} + \vec{v}_{CG}$
The angle between $\vec{v}_{AC}$ and $\vec{v}_{AG}$ is $31°$, so the direction of the wind is $14°$ north of east.



Additional Problems

79. $a_C = 39.6\ \text{m/s}^2$

81. $90.0\ \text{km/h} = 25.0\ \text{m/s},\ 9.0\ \text{km/h} = 2.5\ \text{m/s}, 60.0\ \text{km/h} = 16.7\ \text{m/s}$
$a_T = -2.5\ \text{m/s}^2, a_C = 1.86\ \text{m/s}^2, a = 3.1\ \text{m/s}^2$

83. The radius of the circle of revolution at latitude λ is $R_E \cos\lambda$. The velocity of the body is $\frac{2\pi r}{T}$. $a_C = \frac{4\pi^2 R_E \cos\lambda}{T^2}$ for $\lambda = 40°,\ a_C = 0.26\%\ g$

85. $a_T = 3.00\ \text{m/s}^2$
$v(5\ \text{s}) = 15.00\ \text{m/s}\ \ a_C = 150.00\ \text{m/s}^2\ \ \theta = 88.8°$ with respect to the tangent to the circle of revolution directed inward. $|\vec{\mathbf{a}}| = 150.03\ \text{m/s}^2$

87. $\vec{\mathbf{a}}(t) = -A\omega^2 \cos\omega t\hat{\mathbf{i}} - A\omega^2 \sin\omega t\hat{\mathbf{j}}$
$a_C = 5.0\ \text{m}\omega^2\ \ \omega = 0.89\ \text{rad/s}$
$\vec{\mathbf{v}}(t) = -2.24\ \text{m/s}\ \hat{\mathbf{i}} - 3.87\ \text{m/s}\hat{\mathbf{j}}$

89. $\vec{\mathbf{r}}_1 = 1.5\hat{\mathbf{j}} + 4.0\hat{\mathbf{k}}\ \ \vec{\mathbf{r}}_2 = \Delta\vec{\mathbf{r}} + \vec{\mathbf{r}}_1 = 2.5\hat{\mathbf{i}} + 4.7\hat{\mathbf{j}} + 2.8\hat{\mathbf{k}}$

91. $v_x(t) = 265.0\ \text{m/s}$
$v_y(t) = 20.0\ \text{m/s}$
$\vec{\mathbf{v}}(5.0\ \text{s}) = (265.0\hat{\mathbf{i}} + 20.0\hat{\mathbf{j}})\text{m/s}$

93. $R = 1.07\ \text{m}$

95. $v_0 = 20.1\ \text{m/s}$

97. $v = 3072.5\ \text{m/s}$
$a_C = 0.223\ \text{m/s}^2$

Challenge Problems

99. a. $-400.0\ \text{m} = v_{0y}t - 4.9t^2\ \ 359.0\ \text{m} = v_{0x}t\ \ t = \frac{359.0}{v_{0x}}\ \ -400.0 = 359.0\frac{v_{0y}}{v_{0x}} - 4.9(\frac{359.0}{v_{0x}})^2$
$-400.0 = 359.0\tan 40 - \frac{631{,}516.9}{v_{0x}^2} \Rightarrow v_{0x}^2 = 900.6\ \ v_{0x} = 30.0\ \text{m/s}\ \ v_{0y} = v_{0x}\tan 40 = 25.2\ \text{m/s}$
$v = 39.2\ \text{m/s}$, b. $t = 12.0\ \text{s}$

101. a. $\vec{\mathbf{r}}_{TC} = (-32 + 80t)\hat{\mathbf{i}} + 50t\hat{\mathbf{j}},\ \ |\vec{\mathbf{r}}_{TC}|^2 = (-32 + 80t)^2 + (50t)^2$
$2r\frac{dr}{dt} = 2(-32 + 80t)(80) + 5000t\ \ \frac{dr}{dt} = \frac{160(-32+80t)+5000t}{2r} = 0$
$17800t = 5184 \Rightarrow t = 0.29\ \text{hr}$,
b. $|\vec{\mathbf{r}}_{TC}| = 17\ \text{km}$

Chapter 5

Check Your Understanding

5.1 14 N, 56° measured from the positive x-axis

5.2 a. Their weight acts downward, and the force of air resistance with the parachute acts upward. b. neither; the forces are equal in magnitude

5.3 $0.1\ \text{m/s}^2$

5.4 $40\ \text{m/s}^2$

5.5 a. $F_{\text{net}} = 159.0\hat{\mathbf{i}} + 763.9\hat{\mathbf{j}}\ \text{N}$; b. $a = 0.1590\hat{\mathbf{i}} + 70.639\hat{\mathbf{j}}\ \text{N}$

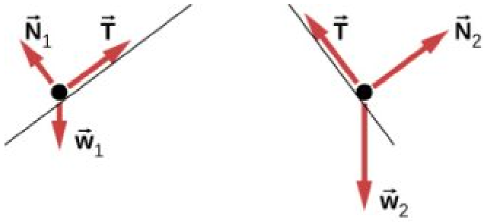
5.6 $a = 2.78\ \text{m/s}^2$

5.7 a. $3.0\ \text{m/s}^2$; b. 18 N

5.8 a. $1.7\ \text{m/s}^2$; b. $1.3\ \text{m/s}^2$

5.9 6.0×10^2 N

5.10



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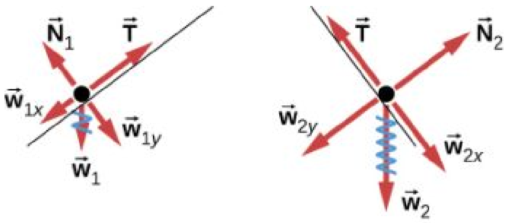


Conceptual Questions

1. Forces are directional and have magnitude.

3. The cupcake velocity before the braking action was the same as that of the car. Therefore, the cupcakes were unrestricted bodies in motion, and when the car suddenly stopped, the cupcakes kept moving forward according to Newton's first law.

5. No. If the force were zero at this point, then there would be nothing to change the object's momentary zero velocity. Since we do not observe the object hanging motionless in the air, the force could not be zero.

7. The astronaut is truly weightless in the location described, because there is no large body (planet or star) nearby to exert a gravitational force. Her mass is 70 kg regardless of where she is located.

9. The force you exert (a contact force equal in magnitude to your weight) is small. Earth is extremely massive by comparison. Thus, the acceleration of Earth would be incredibly small. To see this, use Newton's second law to calculate the acceleration you would cause if your weight is 600.0 N and the mass of Earth is 6.00×10^{24} kg.

11. a. action: Earth pulls on the Moon, reaction: Moon pulls on Earth; b. action: foot applies force to ball, reaction: ball applies force to foot; c. action: rocket pushes on gas, reaction: gas pushes back on rocket; d. action: car tires push backward on road, reaction: road pushes forward on tires; e. action: jumper pushes down on ground, reaction: ground pushes up on jumper; f. action: gun pushes forward on bullet, reaction: bullet pushes backward on gun.

13. a. The rifle (the shell supported by the rifle) exerts a force to expel the bullet; the reaction to this force is the force that the bullet exerts on the rifle (shell) in opposite direction. b. In a recoilless rifle, the shell is not secured in the rifle; hence, as the bullet is pushed to move forward, the shell is pushed to eject from the opposite end of the barrel. c. It is not safe to stand behind a recoilless rifle.

15. a. Yes, the force can be acting to the left; the particle would experience acceleration opposite to the motion and lose speed. B. Yes, the force can be acting downward because its weight acts downward even as it moves to the right.

17. two forces of different types: weight acting downward and normal force acting upward

Problems

19. a. $\vec{\mathbf{F}}_{\text{net}} = 5.0\hat{\mathbf{i}} + 10.0\hat{\mathbf{j}}$ N; b. the magnitude is $F_{\text{net}} = 11$ N, and the direction is $\theta = 63°$

21. a. $\vec{\mathbf{F}}_{\text{net}} = 660.0\hat{\mathbf{i}} + 150.0\hat{\mathbf{j}}$ N; b. $F_{\text{net}} = 676.6$ N at $\theta = 12.8°$ from David's rope

23. a. $\vec{\mathbf{F}}_{\text{net}} = 95.0\hat{\mathbf{i}} + 283\hat{\mathbf{j}}$N; b. 299 N at 71° north of east; c. $\vec{\mathbf{F}}_{\text{DS}} = -\left(95.0\hat{\mathbf{i}} + 283\hat{\mathbf{j}}\right)$ N

25. Running from rest, the sprinter attains a velocity of $v = 12.96$ m/s, at end of acceleration. We find the time for acceleration using $x = 20.00\text{ m} = 0 + 0.5at_1{}^2$, or $t_1 = 3.086\text{ s}$. For maintained velocity, $x_2 = vt_2$, or $t_2 = x_2/v = 80.00\text{ m}/12.96\text{ m/s} = 6.173\text{ s}$. Total time $= 9.259\text{ s}$.

27. a. $m = 56.0\text{ kg}$; b. $a_{\text{meas}} = a_{\text{astro}} + a_{\text{ship}}$, where $a_{\text{ship}} = \frac{m_{\text{astro}}a_{\text{astro}}}{m_{\text{ship}}}$; c. If the force could be exerted on the astronaut by another source (other than the spaceship), then the spaceship would not experience a recoil.

29. $F_{\text{net}} = 4.12 \times 10^5\text{ N}$

31. $a = 253\text{ m/s}^2$

33. $F_{\text{net}} = F - f = ma \Rightarrow F = 1.26 \times 10^3\text{ N}$

35.
$$v^2 = v_0^2 + 2ax \Rightarrow a = -7.80\text{ m/s}^2$$
$$F_{\text{net}} = -7.80 \times 10^3\text{ N}$$

37. a. $\vec{\mathbf{F}}_{\text{net}} = m\vec{\mathbf{a}} \Rightarrow \vec{\mathbf{a}} = 9.0\hat{\mathbf{i}}\text{ m/s}^2$; b. The acceleration has magnitude 9.0 m/s^2, so $x = 110\text{ m}$.

39. $1.6\hat{\mathbf{i}} - 0.8\hat{\mathbf{j}}\text{ m/s}^2$

41. a. $\begin{aligned} w_{\text{Moon}} &= mg_{\text{Moon}} \\ m &= 150\text{ kg} \\ w_{\text{Earth}} &= 1.5 \times 10^3\text{ N} \end{aligned}$; b. Mass does not change, so the suited astronaut's mass on both Earth and the Moon is 150 kg.

43. a. $\begin{aligned} F_{\text{h}} &= 3.68 \times 10^3\text{ N and} \\ w &= 7.35 \times 10^2\text{ N} \\ \frac{F_{\text{h}}}{w} &= 5.00\text{ times greater than weight} \end{aligned}$;
b. $\begin{aligned} F_{\text{net}} &= 3750\text{ N} \\ \theta &= 11.3^\circ\text{ from horizontal} \end{aligned}$

45.
$$\begin{aligned} w &= 19.6\text{ N} \\ F_{\text{net}} &= 5.40\text{ N} \\ F_{\text{net}} &= ma \Rightarrow a = 2.70\text{ m/s}^2 \end{aligned}$$

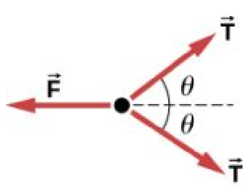
47. 98 N

49. 497 N

51. a. $F_{\text{net}} = 2.64 \times 10^7\text{ N}$; b. The force exerted on the ship is also $2.64 \times 10^7\text{ N}$ because it is opposite the shell's direction of motion.

53. Because the weight of the history book is the force exerted by Earth on the history book, we represent it as $\vec{\mathbf{F}}_{\text{EH}} = -14\hat{\mathbf{j}}\text{ N}$. Aside from this, the history book interacts only with the physics book. Because the acceleration of the history book is zero, the net force on it is zero by Newton's second law: $\vec{\mathbf{F}}_{\text{PH}} + \vec{\mathbf{F}}_{\text{EH}} = \vec{\mathbf{0}}$, where $\vec{\mathbf{F}}_{\text{PH}}$ is the force exerted by the physics book on the history book. Thus, $\vec{\mathbf{F}}_{\text{PH}} = -\vec{\mathbf{F}}_{\text{EH}} = -\left(-14\hat{\mathbf{j}}\right)\text{ N} = 14\hat{\mathbf{j}}\text{ N}$. We find that the physics book exerts an upward force of magnitude 14 N on the history book. The physics book has three forces exerted on it: $\vec{\mathbf{F}}_{\text{EP}}$ due to Earth, $\vec{\mathbf{F}}_{\text{HP}}$ due to the history book, and $\vec{\mathbf{F}}_{\text{DP}}$ due to the desktop. Since the physics book weighs 18 N, $\vec{\mathbf{F}}_{\text{EP}} = -18\hat{\mathbf{j}}\text{ N}$. From Newton's third law, $\vec{\mathbf{F}}_{\text{HP}} = -\vec{\mathbf{F}}_{\text{PH}}$, so $\vec{\mathbf{F}}_{\text{HP}} = -14\hat{\mathbf{j}}\text{ N}$. Newton's second law applied to the physics book gives $\sum\vec{\mathbf{F}} = \vec{\mathbf{0}}$, or $\vec{\mathbf{F}}_{\text{DP}} + \vec{\mathbf{F}}_{\text{EP}} + \vec{\mathbf{F}}_{\text{HP}} = \vec{\mathbf{0}}$, so $\vec{\mathbf{F}}_{\text{DP}} = -\left(-18\hat{\mathbf{j}}\right) - \left(-14\hat{\mathbf{j}}\right) = 32\hat{\mathbf{j}}\text{ N}$. The desk exerts an upward force of 32 N on the physics book. To arrive at this solution, we apply Newton's second law twice and Newton's third law once.

55. a. The free-body diagram of pulley 4:



b. $T = mg, \quad F = 2T\cos\theta = 2mg\cos\theta$

57. a. 65 N

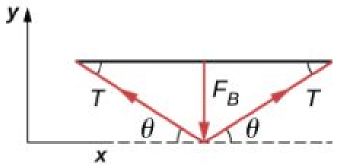
b. $1.22 \times 10^4\ \text{m/s}^2$

59. a. $T = 1.96 \times 10^{-4}\ \text{N}$;

b. $T' = 4.71 \times 10^{-4}\ \text{N}$

$\frac{T'}{T} = 2.40$ times the tension in the vertical strand

61.



$F_{y\ \text{net}} = F_\perp - 2T\sin\theta = 0$

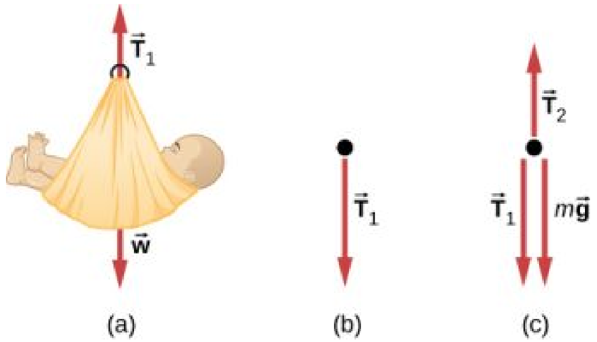
$F_\perp = 2T\sin\theta$

$T = \frac{F_\perp}{2\sin\theta}$

63. a. see Example 5.13; b. 1.5 N; c. 15 N

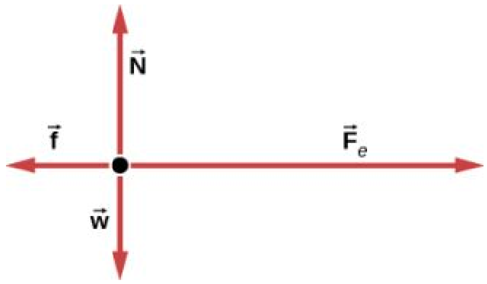
65. a. 5.6 kg; b. 55 N; c. $T_2 = 60\ \text{N}$;

d.

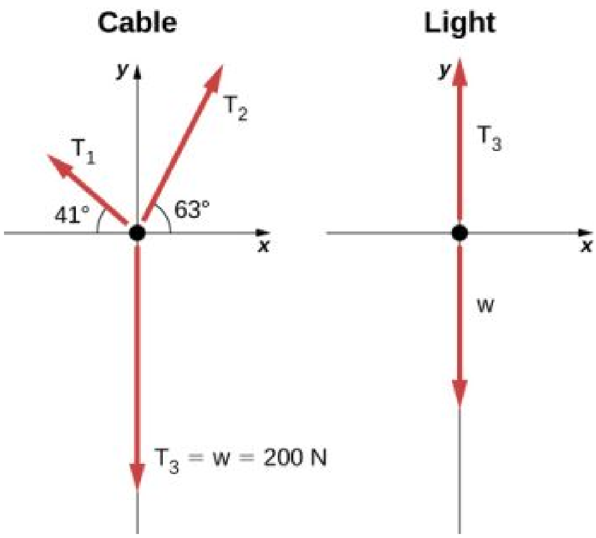


67. a. $4.9\ \text{m/s}^2$, 17 N; b. 9.8 N

69.



71.



Additional Problems

73. 5.90 kg

75.

77. a. $F_{\text{net}} = \frac{m(v^2 - v_0{}^2)}{2x}$; b. 2590 N

79. $\vec{\mathbf{F}}_{\text{net}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 + \vec{\mathbf{F}}_3 = (6.02\hat{\mathbf{i}} + 14.0\hat{\mathbf{j}})\text{N}$

$\vec{\mathbf{F}}_{\text{net}} = m\vec{\mathbf{a}} \Rightarrow \vec{\mathbf{a}} = \frac{\vec{\mathbf{F}}_{\text{net}}}{m} = \frac{6.02\hat{\mathbf{i}} + 14.0\hat{\mathbf{j}}\text{N}}{10.0\text{kg}} = (0.602\hat{\mathbf{i}} + 1.40\hat{\mathbf{j}})\text{m/s}^2$

81.
$\vec{\mathbf{F}}_{\text{net}} = \vec{\mathbf{F}}_A + \vec{\mathbf{F}}_B$

$\vec{\mathbf{F}}_{\text{net}} = A\hat{\mathbf{i}} + \left(-1.41A\hat{\mathbf{i}} - 1.41A\hat{\mathbf{j}}\right)$

$\vec{\mathbf{F}}_{\text{net}} = A\left(-0.41\hat{\mathbf{i}} - 1.41\hat{\mathbf{j}}\right)$

$\theta = 254°$

(We add $180°$, because the angle is in quadrant IV.)

83. $F = 2mk^2x^2$; First, take the derivative of the velocity function to obtain $a = 2kxv = 2kx\left(kx^2\right) = 2k^2x^3$. Then apply Newton's second law $F = ma = 2mk^2x^2$.

85. a. For box A, $N_A = mg$ and $N_B = mg\cos\theta$; b. $N_A > N_B$ because for $\theta < 90°$, $\cos\theta < 1$; c. $N_A > N_B$ when $\theta = 10°$

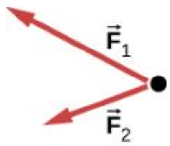
87. a. 8.66 N; b. 0.433 m

89. 0.40 or 40%

91. 16 N

Challenge Problems

93. a.



; b. No; $\vec{\mathbf{F}}_R$ is not shown, because it would replace $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$. (If we want to show it, we could draw it and then place squiggly lines on $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ to show that they are no longer considered.

95. a. 14.1 m/s; b. 601 N

97. $\frac{F}{m}t^2$

99. 936 N

101. $\vec{\mathbf{a}} = -248\hat{\mathbf{i}} - 433\hat{\mathbf{j}}\text{m/s}^2$

103. 0.548 m/s^2

105. a. $T_1 = \frac{2mg}{\sin\theta}, T_2 = \frac{mg}{\sin\left(\arctan\left(\frac{1}{2}\tan\theta\right)\right)}, T_3 = \frac{2mg}{\tan\theta}$; b. $\phi = \arctan\left(\frac{1}{2}\tan\theta\right)$; c. $2.56°$; (d) $x = d\left(2\cos\theta + 2\cos\left(\arctan\left(\frac{1}{2}\tan\theta\right)\right) + 1\right)$

107. a. $\vec{\mathbf{a}} = \left(\frac{5.00}{m}\hat{\mathbf{i}} + \frac{3.00}{m}\hat{\mathbf{j}}\right)\text{ m/s}^2$; b. 1.38 kg; c. 21.2 m/s; d. $\vec{\mathbf{v}} = \left(18.1\hat{\mathbf{i}} + 10.9\hat{\mathbf{j}}\right)\text{ m/s}^2$

109. a. $0.900\hat{\mathbf{i}} + 0.600\hat{\mathbf{j}}\text{ N}$; b. 1.08 N

Chapter 6

Check Your Understanding

6.1 $F_s = 645\text{ N}$

6.2 $a = 3.68\text{ m/s}^2, T = 18.4\text{ N}$

6.3 $T = \frac{2m_1m_2}{m_1+m_2}g$ (This is found by substituting the equation for acceleration in Figure 6.7(a), into the equation for tension in Figure 6.7(b).)

6.4 1.49 s

6.5 49.4 degrees

6.6 128 m; no

6.7 a. 4.9 N; b. 0.98 m/s^2

6.8 -0.23 m/s^2; the negative sign indicates that the snowboarder is slowing down.

6.9 0.40

6.10 34 m/s

6.11 0.27 kg/m

Conceptual Questions

1. The scale is in free fall along with the astronauts, so the reading on the scale would be 0. There is no difference in the apparent weightlessness; in the aircraft and in orbit, free fall is occurring.

3. If you do not let up on the brake pedal, the car's wheels will lock so that they are not rolling; sliding friction is now involved and the sudden change (due to the larger force of static friction) causes the jerk.

5. 5.00 N

7. Centripetal force is defined as any net force causing uniform circular motion. The centripetal force is not a new kind of force. The label "centripetal" refers to *any* force that keeps something turning in a circle. That force could be tension, gravity, friction, electrical attraction, the normal force, or any other force. Any combination of these could be the source of centripetal force, for example, the centripetal force at the top of the path of a tetherball swung through a vertical circle is the result of both tension and gravity.

9. The driver who cuts the corner (on Path 2) has a more gradual curve, with a larger radius. That one will be the better racing line. If the driver goes too fast around a corner using a racing line, he will still slide off the track; the key is to stay at the maximum value of static friction. So, the driver wants maximum possible speed and maximum friction. Consider the equation for centripetal force: $F_c = m\frac{v^2}{r}$ where *v* is speed and *r* is the radius of curvature. So by decreasing the curvature (1/*r*) of the path that the car takes, we reduce the amount of force the tires have to exert on the road, meaning we can now increase the speed, *v*. Looking at this from the point of view of the driver on Path 1, we can reason this way: the sharper the turn, the smaller the turning circle; the smaller the turning circle, the larger is the required centripetal force. If this centripetal force is not exerted, the result is a skid.

11. The barrel of the dryer provides a centripetal force on the clothes (including the water droplets) to keep them moving in a circular path. As a water droplet comes to one of the holes in the barrel, it will move in a path tangent to the circle.

13. (a) If there is no friction, then there is no centripetal force. This means that in the inertial frame of the ground, the lunch box will move along a straight path tangent to the circle, and thus follows path *B*. This is a result of Newton's first law of motion. (b) The dust trail traces the motion of the box relative to the merry-go-round and will be radially out.

15. There must be a centripetal force to maintain the circular motion; this is provided by the nail at the center. Newton's third law explains the phenomenon. The action force is the force of the string on the mass; the reaction force is the force of the mass on the string. This reaction force causes the string to stretch.

17. Since the radial friction with the tires supplies the centripetal force, and friction is nearly 0 when the car encounters the ice, the car will obey Newton's first law and go off the road in a straight line path, tangent to the curve. A common misconception is that the car will follow a curved path off the road.

19. Anna is correct. The satellite is freely falling toward Earth due to gravity, even though gravity is weaker at the altitude of the satellite, and *g* is not $9.80\ \text{m/s}^2$. Free fall does not depend on the value of *g*; that is, you could experience free fall on Mars if you jumped off Olympus Mons (the tallest volcano in the solar system).

21. The pros of wearing body suits include: (1) the body suit reduces the drag force on the swimmer and the athlete can move more easily; (2) the tightness of the suit reduces the surface area of the athlete, and even though this is a small amount, it can make a difference in performance time. The cons of wearing body suits are: (1) The tightness of the suits can induce cramping and breathing problems. (2) Heat will be retained and thus the athlete could overheat during a long period of use.

23. The oil is less dense than the water and so rises to the top when a light rain falls and collects on the road. This creates a dangerous situation in which friction is greatly lowered, and so a car can lose control. In a heavy rain, the oil is dispersed and does not affect the motion of cars as much.

Problems

25. a. 170 N; b. 170 N

27. $\vec{\mathbf{F}}_3 = (-7\hat{\mathbf{i}} + 2\hat{\mathbf{j}})\ \text{N}$

29. 376 N pointing up (along the dashed line in the figure); the force is used to raise the heel of the foot.

31. −68.5 N

33. a. $7.70\ \text{m/s}^2$; b. 4.33 s

35. a. 46.4 m/s; b. $2.40 \times 10^3\ \text{m/s}^2$; c. 5.99×10^3 N; ratio of 245

37. a. $1.87 \times 10^4\ \text{N}$; b. $1.67 \times 10^4\ \text{N}$; c. $1.56 \times 10^4\ \text{N}$; d. 19.4 m, 0 m/s

39. a. 10 kg; b. 140 N; c. 98 N; d. 0

41. a. $3.35\ \text{m/s}^2$; b. 4.2 s

43. a. $2.0\ \text{m/s}^2$; b. 7.8 N; c. 2.0 m/s

45. a. $4.43\ \text{m/s}^2$ (mass 1 accelerates up the ramp as mass 2 falls with the same acceleration); b. 21.5 N

47. a. 10.0 N; b. 97.0 N

49. a. $4.9\ \text{m/s}^2$; b. The cabinet will not slip. c. The cabinet will slip.

51. a. 32.3 N, $35.2°$; b. 0; c. $0.301\ \text{m/s}^2$ in the direction of $\vec{\mathbf{F}}_{\text{tot}}$

53.
$$\begin{aligned}\text{net } F_y &= 0 \Rightarrow N = mg\cos\theta \\ \text{net } F_x &= ma \\ a &= g(\sin\theta - \mu_k \cos\theta)\end{aligned}$$

55. a. $0.737\ \text{m/s}^2$; b. $5.71°$

57. a. $10.8\ \text{m/s}^2$; b. $7.85\ \text{m/s}^2$; c. $2.00\ \text{m/s}^2$

59. a. $9.09\ \text{m/s}^2$; b. $6.16\ \text{m/s}^2$; c. $0.294\ \text{m/s}^2$

61. a. 272 N, 512 N; b. 0.268
63. a. 46.5 N; b. 0.655 m/s^2
65. a. 483 N; b. 17.4 N; c. 2.24, 0.0807
67. $4.14°$
69. a. 24.6 m; b. 36.6 m/s^2; c. 3.73 times g
71. a. 16.2 m/s; b. 0.234
73. a. 179 N; b. 290 N; c. 8.3 m/s
75. 20.7 m/s
77. 21 m/s
79. 115 m/s or 414 km/h
81. $v_{\text{T}} = 11.8 \text{ m/s}; \text{v}_2 = 9.9 \text{ m/s}$
83. $\left(\frac{110}{65}\right)^2 = 2.86$ times
85. Stokes' law is $F_{\text{s}} = 6\pi r\eta v$. Solving for the viscosity, $\eta = \frac{F_{\text{s}}}{6\pi rv}$. Considering only the units, this becomes $[\eta] = \frac{\text{kg}}{\text{m·s}}$.
87. $0.76 \text{ kg/m} \cdot \text{s}$
89. a. 0.049 kg/s; b. 0.57 m

Additional Problems

91. a. 1860 N, 2.53; b. The value (1860 N) is more force than you expect to experience on an elevator. The force of 1860 N is 418 pounds, compared to the force on a typical elevator of 904 N (which is about 203 pounds); this is calculated for a speed from 0 to 10 miles per hour, which is about 4.5 m/s, in 2.00 s). c. The acceleration $a = 1.53 \times g$ is much higher than any standard elevator. The final speed is too large (30.0 m/s is VERY fast)! The time of 2.00 s is not unreasonable for an elevator.
93. 199 N
95. 15 N
97. 12 N
99. $a_x = 0.40 \text{ m/s}^2$ and $T = 11.2 \times 10^3 \text{ N}$
101. $m(6pt + 2q)$
103. $\vec{\mathbf{v}}(t) = \left(\frac{pt}{m} + \frac{nt^2}{2m}\right)\hat{\mathbf{i}} + \left(\frac{qt^2}{2m}\right)\hat{\mathbf{j}}$ and $\vec{\mathbf{r}}(t) = \left(\frac{pt^2}{2m} + \frac{nt^3}{6m}\right)\hat{\mathbf{i}} + \left(\frac{qt^3}{6m}\right)\hat{\mathbf{j}}$
105. 9.2 m/s
107. 1.3 s
109. 3.5 m/s^2
111. a. 0.75; b. 1200 N; c. 1.2 m/s^2 and 1080 N; d. -1.2 m/s^2; e. 120 N
113. 0.789
115. a. 0.186 N; b. 0.774 N; c. 0.48 N
117. 13 m/s
119. 0.21
121. a. 28,300 N; b. 2540 m
123. 25 N
125. $a = \frac{F}{4} - \mu_k g$
127. 11 m

Challenge Problems

129. $v = \sqrt{v_0{}^2 - 2gr_0\left(1 - \frac{r_0}{r}\right)}$
131. 78.7 m
133. a. 98 m/s; b. 490 m; c. 107 m/s; d. 10.1 s
135. a. $v = 20.0(1 - e^{-0.01t})$; b. $v_{\text{limiting}} = 20 \text{ m/s}$

Chapter 7

Check Your Understanding

7.1 No, only its magnitude can be constant; its direction must change, to be always opposite the relative displacement along the surface.
7.2 No, it's only approximately constant near Earth's surface.
7.3 $W = 35\ \text{J}$
7.4 a. The spring force is the opposite direction to a compression (as it is for an extension), so the work it does is negative. b. The work done depends on the square of the displacement, which is the same for $x = \pm 6\ \text{cm}$, so the magnitude is 0.54 J.
7.5 a. the car; b. the truck
7.6 against
7.7 $\sqrt{3}\ \text{m/s}$
7.8 980 W

Conceptual Questions

1. When you push on the wall, this "feels" like work; however, there is no displacement so there is no physical work. Energy is consumed, but no energy is transferred.
3. If you continue to push on a wall without breaking through the wall, you continue to exert a force with no displacement, so no work is done.
5. The total displacement of the ball is zero, so no work is done.
7. Both require the same gravitational work, but the stairs allow Tarzan to take this work over a longer time interval and hence gradually exert his energy, rather than dramatically by climbing a vine.
9. The first particle has a kinetic energy of $4(\frac{1}{2}mv^2)$ whereas the second particle has a kinetic energy of $2(\frac{1}{2}mv^2)$, so the first particle has twice the kinetic energy of the second particle.
11. The mower would gain energy if $-90° < \theta < 90°$. It would lose energy if $90° < \theta < 270°$. The mower may also lose energy due to friction with the grass while pushing; however, we are not concerned with that energy loss for this problem.
13. The second marble has twice the kinetic energy of the first because kinetic energy is directly proportional to mass, like the work done by gravity.
15. Unless the environment is nearly frictionless, you are doing some positive work on the environment to cancel out the frictional work against you, resulting in zero total work producing a constant velocity.
17. Appliances are rated in terms of the energy consumed in a relatively small time interval. It does not matter how long the appliance is on, only the rate of change of energy per unit time.
19. The spark occurs over a relatively short time span, thereby delivering a very low amount of energy to your body.
21. If the force is antiparallel or points in an opposite direction to the velocity, the power expended can be negative.

Problems

23. 3.00 J
25. a. 592 kJ; b. –588 kJ; c. 0 J
27. 3.14 kJ
29. a. –700 J; b. 0 J; c. 700 J; d. 38.6 N; e. 0 J
31. 100 J
33. a. 2.45 J; b. – 2.45 J; c. 0 J
35. a. 2.22 kJ; b. –2.22 kJ; c. 0 J
37. 18.6 kJ
39. a. 2.32 kN; b. 22.0 kJ
41. 835 N
43. 257 J
45. a. 1.47 m/s; b. answers may vary

47. a. 772 kJ; b. 4.0 kJ; c. 1.8×10^{-16} J
49. a. 2.6 kJ; b. 640 J
51. 2.72 kN
53. 102 N
55. 2.8 m/s
57. $W(\text{bullet}) = 20 \times W(\text{crate})$
59. 12.8 kN
61. 0.25
63. a. 24 m/s, −4.8 m/s^2; b. 29.4 m
65. 313 m/s
67. a. 40; b. 8 million
69. $149
71. a. 208 W; b. 141 s
73. a. 3.20 s; b. 4.04 s
75. a. 224 s; b. 24.8 MW; c. 49.7 kN
77. a. 1.57 kW; b. 6.28 kW
79. $6.83 \mu\text{W}$
81. a. 8.51 J; b. 8.51 W
83. 1.7 kW

Additional Problems

85. $15\,\text{N} \cdot \text{m}$
87. $39\,\text{N} \cdot \text{m}$
89. a. $208\,\text{N} \cdot \text{m}$; b. $240\,\text{N} \cdot \text{m}$
91. a. $-0.5\,\text{N} \cdot \text{m}$; b. $-0.83\,\text{N} \cdot \text{m}$
93. a. 10. J; b. −10. J; c. 380 N/m
95. 160 J/s
97. a. 20 N; b. 40 W

Challenge Problems

99. If crate goes up: a. 3.46 kJ; b. −1.89 kJ; c. −1.57 kJ; d. 0; If crate goes down: a. −0.39 kJ; b. −1.18 kJ; c. 1.57 kJ; d. 0
101. 8.0 J
103. 35.7 J
105. 24.3 J
107. a. 40 hp; b. 39.8 MJ, independent of speed; c. 80 hp, 79.6 MJ at 30 m/s; d. If air resistance is proportional to speed, the car gets about 22 mpg at 34 mph and half that at twice the speed, closer to actual driving experience.

Chapter 8

Check Your Understanding

8.1 $(4.63\,\text{J}) - (-2.38\,\text{J}) = 7.00\,\text{J}$
8.2 35.3 kJ, 143 kJ, 0
8.3 22.8 cm. Using 0.02 m for the initial displacement of the spring (see above), we calculate the final displacement of the spring to be 0.028 m; therefore the length of the spring is the unstretched length plus the displacement, or 22.8 cm.
8.4 It increases because you had to exert a downward force, doing positive work, to pull the mass down, and that's equal to the change in the total potential energy.
8.5 2.83 N
8.6 $F = 4.8\,\text{N}$, directed toward the origin
8.7 $0.10\,\text{m}$
8.8 b. At any given height, the gravitational potential energy is the same going up or down, but the kinetic energy

is less going down than going up, since air resistance is dissipative and does negative work. Therefore, at any height, the speed going down is less than the speed going up, so it must take a longer time to go down than to go up.

8.9 constant $U(x) = -1\ \text{J}$

8.10 a. yes, motion confined to $-1.055\ \text{m} \leq x \leq 1.055\ \text{m}$; b. same equilibrium points and types as in example

8.11 $x(t) = \pm\sqrt{(2E/k)}\sin\left[\left(\sqrt{k/m}\right)t\right]$ and $v_0 = \pm\sqrt{(2E/m)}$

Conceptual Questions

1. The potential energy of a system can be negative because its value is relative to a defined point.

3. If the reference point of the ground is zero gravitational potential energy, the javelin first increases its gravitational potential energy, followed by a decrease in its gravitational potential energy as it is thrown until it hits the ground. The overall change in gravitational potential energy of the javelin is zero unless the center of mass of the javelin is lower than from where it is initially thrown, and therefore would have slightly less gravitational potential energy.

5. the vertical height from the ground to the object

7. A force that takes energy away from the system that can't be recovered if we were to reverse the action.

9. The change in kinetic energy is the net work. Since conservative forces are path independent, when you are back to the same point the kinetic and potential energies are exactly the same as the beginning. During the trip the total energy is conserved, but both the potential and kinetic energy change.

11. The car experiences a change in gravitational potential energy as it goes down the hills because the vertical distance is decreasing. Some of this change of gravitational potential energy will be taken away by work done by friction. The rest of the energy results in a kinetic energy increase, making the car go faster. Lastly, the car brakes and will lose its kinetic energy to the work done by braking to a stop.

13. It states that total energy of the system *E* is conserved as long as there are no non-conservative forces acting on the object.

15. He puts energy into the system through his legs compressing and expanding.

17. Four times the original height would double the impact speed.

Problems

19. 40,000

21. a. $-200\ \text{J}$; b. $-200\ \text{J}$; c. $-100\ \text{J}$; d. $-300\ \text{J}$

23. a. $0.068\ \text{J}$; b. $-0.068\ \text{J}$; c. $0.068\ \text{J}$; d. $0.068\ \text{J}$; e. $-0.068\ \text{J}$; f. $46\ \text{cm}$

25. $-120\ \text{J}$

27. a. $\left(\frac{2a}{b}\right)^{1/6}$; b. 0; c. $\sim x^6$

29. $14\ \text{m/s}$

31. $14\ \text{J}$

33. proof

35. $9.7\ \text{m/s}$

37. $39\ \text{m/s}$

39. 1900 J

41. –39 J

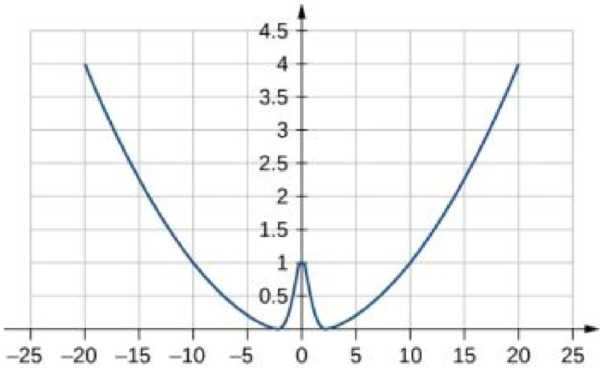
43. 3.5 cm

45. $10x$ with x-axis pointed away from the wall and origin at the wall

47. 4.6 m/s

49. a. 5.6 m/s; b. 5.2 m/s; c. 6.4 m/s; d. no; e. yes

51. a.



where $k = 0.02, A = 1, \alpha = 1$; b. $F(x) = -\frac{dU(x)}{dx} = -kx + 2A\alpha xe^{-\alpha x^2}$; c. The potential energy at $x = 0$ must be less than the kinetic plus potential energy at $x =$ a or $A \leq \frac{1}{2}mv^2 + \frac{1}{2}ka^2 + Ae^{-\alpha a^2}$. Solving this for *A* matches results in the problem.

53. 490 N/m
55. a. 70.6 m/s; b. 69.9 m/s
57. a. 180 N/m; b. 11 m
59. a. 9.8×10^3 J; b. 1.4×10^3 J; c. 14 m/s
61. a. 47.6 m; b. 1.88×10^5 J; c. 373 N
63. 33.9 cm
65. a. Zero, since the total energy of the system is zero and the kinetic energy at the lowest point is zero; b. –0.038 J; c. 0.62 m/s
67. 42 cm

Additional Problems

69. –0.44 J
71. 3.6 m/s
73. $bD^4/4$
75. proof
77. a. $\sqrt{\frac{2m^2gh}{k(m+M)}}$; b. $\frac{mMgh}{m+M}$
79. a. 2.24 m/s; b. 1.94 m/s; c. 1.94 m/s
81. 18 m/s
83. $v_A = 24$ m/s; $v_B = 14$ m/s; $v_C = 31$ m/s
85. a. Loss of energy is $240\ \text{N} \cdot \text{m}$; b. $F = 8\ \text{N}$
87. 89.7 m/s
89. 32 J

Chapter 9

Check Your Understanding

9.1 To reach a final speed of $v_\text{f} = \frac{1}{4}\left(3.0 \times 10^8 \text{ m/s}\right)$ at an acceleration of 10*g*, the time required is

$$\begin{aligned} 10g &= \frac{v_\text{f}}{\Delta t} \\ \Delta t &= \frac{v_\text{f}}{10g}\ \frac{\frac{1}{4}\left(3.0 \times 10^8 \text{ m/s}\right)}{10g} = 7.7 \times 10^5 \text{ s} = 8.9 \text{ d} \end{aligned}$$

9.2 If the phone bounces up with approximately the same initial speed as its impact speed, the change in momentum of the phone will be $\Delta\vec{\mathbf{p}} = m\Delta\vec{\mathbf{v}} - \left(-m\Delta\vec{\mathbf{v}}\right) = 2m\Delta\vec{\mathbf{v}}$. This is twice the momentum change than when the phone does not bounce, so the impulse-momentum theorem tells us that more force must be applied to the phone.

9.3 If the smaller cart were rolling at 1.33 m/s to the left, then conservation of momentum gives

$$\begin{aligned}(m_1+m_2)\vec{\mathbf{v}}_{\mathbf{f}} &= m_1 v_1 \hat{\mathbf{i}} - m_2 v_2 \hat{\mathbf{i}} \\ \vec{\mathbf{v}}_{\mathbf{f}} &= \left(\frac{m_1 v_1 - m_2 v_2}{m_1+m_2}\right)\hat{\mathbf{i}} \\ &= \left[\frac{(0.675\text{ kg})(0.75\text{ m/s}) - (0.500\text{ kg})(1.33\text{ m/s})}{1.175\text{ kg}}\right]\hat{\mathbf{i}} \\ &= -(0.135\text{ m/s})\hat{\mathbf{i}}\end{aligned}$$

Thus, the final velocity is 0.135 m/s to the left.

9.4 If the ball does not bounce, its final momentum $\vec{\mathbf{p}}_{\mathbf{2}}$ is zero, so

$$\begin{aligned}\Delta\vec{\mathbf{p}} &= \vec{\mathbf{p}}_{\mathbf{2}} - \vec{\mathbf{p}}_{\mathbf{1}} \\ &= (0)\hat{\mathbf{j}} - (-1.4\text{ kg}\cdot\text{m/s})\hat{\mathbf{j}} \\ &= +(1.4\text{ kg}\cdot\text{m/s})\hat{\mathbf{j}}\end{aligned}$$

9.5 Consider the impulse momentum theory, which is $\vec{J} = \Delta\vec{p}$. If $\vec{J} = 0$, we have the situation described in the example. If a force acts on the system, then $\vec{J} = \vec{F}_{\text{ave}}\Delta t$. Thus, instead of $\vec{\mathbf{p}}_f = \vec{\mathbf{p}}_i$, we have

$$\vec{F}_{\text{ave}}\Delta t = \Delta\vec{p} = \vec{p}_{\text{f}} - \vec{p}_{\text{i}}$$

where $\vec{\mathbf{F}}_{\text{ave}}$ is the force due to friction.

9.6 The impulse is the change in momentum multiplied by the time required for the change to occur. By conservation of momentum, the changes in momentum of the probe and the comment are of the same magnitude, but in opposite directions, and the interaction time for each is also the same. Therefore, the impulse each receives is of the same magnitude, but in opposite directions. Because they act in opposite directions, the impulses are not the same. As for the impulse, the force on each body acts in opposite directions, so the forces on each are not equal. However, the change in kinetic energy differs for each, because the collision is not elastic.

9.7 This solution represents the case in which no interaction takes place: the first puck misses the second puck and continues on with a velocity of 2.5 m/s to the left. This case offers no meaningful physical insights.

9.8 If zero friction acts on the car, then it will continue to slide indefinitely ($d \to \infty$), so we cannot use the work-energy theorem as is done in the example. Thus, we could not solve the problem from the information given.

9.9 Were the initial velocities not at right angles, then one or both of the velocities would have to be expressed in component form. The mathematical analysis of the problem would be slightly more involved, but the physical result would not change.

9.10 The volume of a scuba tank is about 11 L. Assuming air is an ideal gas, the number of gas molecules in the tank is

$$\begin{aligned}PV &= NRT \\ N &= \frac{PV}{RT} = \frac{(2500\text{ psi})\left(0.011\text{ m}^3\right)}{(8.31\text{ J/mol}\cdot\text{K})(300\text{ K})}\left(\frac{6894.8\text{ Pa}}{1\text{ psi}}\right) \\ &= 7.59\times10^1\text{ mol}\end{aligned}$$

The average molecular mass of air is 29 g/mol, so the mass of air contained in the tank is about 2.2 kg. This is about 10 times less than the mass of the tank, so it is safe to neglect it. Also, the initial force of the air pressure is roughly proportional to the surface area of each piece, which is in turn proportional to the mass of each piece (assuming uniform thickness). Thus, the initial acceleration of each piece would change very little if we explicitly consider the air.

9.11 The average radius of Earth's orbit around the Sun is 1.496×10^9 m. Taking the Sun to be the origin, and noting that the mass of the Sun is approximately the same as the masses of the Sun, Earth, and Moon combined, the center of mass of the Earth + Moon system and the Sun is

$$\begin{aligned}R_{\text{CM}} &= \frac{m_{\text{Sun}}R_{\text{Sun}} + m_{\text{em}}R_{\text{em}}}{m_{\text{Sun}}} \\ &= \frac{\left(1.989\times10^{30}\text{ kg}\right)(0) + \left(5.97\times10^{24}\text{ kg} + 7.36\times10^{22}\text{ kg}\right)\left(1.496\times10^9\text{ m}\right)}{1.989\times10^{30}\text{ kg}} \\ &= 4.6\text{ km}\end{aligned}$$

Thus, the center of mass of the Sun, Earth, Moon system is 4.6 km from the center of the Sun.

9.12 On a macroscopic scale, the size of a unit cell is negligible and the crystal mass may be considered to be distributed homogeneously throughout the crystal. Thus,

$$\vec{\mathbf{r}}_{CM} = \frac{1}{M}\sum_{j=1}^{N} m_j \vec{r}_j = \frac{1}{M}\sum_{j=1}^{N} m\vec{r}_j = \frac{m}{M}\sum_{j=1}^{N} \vec{r}_j = \frac{Nm}{M}\frac{\sum_{j=1}^{N}\vec{r}_j}{N}$$

where we sum over the number *N* of unit cells in the crystal and *m* is the mass of a unit cell. Because *Nm* = *M*, we can write

$$\vec{\mathbf{r}}_{CM} = \frac{m}{M}\sum_{j=1}^{N} \vec{r}_j = \frac{Nm}{M}\frac{\sum_{j=1}^{N}\vec{r}_j}{N} = \frac{1}{N}\sum_{j=1}^{N}\vec{r}_j.$$

This is the definition of the geometric center of the crystal, so the center of mass is at the same point as the geometric center.

9.13 The explosions would essentially be spherically symmetric, because gravity would not act to distort the trajectories of the expanding projectiles.

9.14 The notation m_g stands for the mass of the fuel and *m* stands for the mass of the rocket plus the initial mass of the fuel. Note that m_g changes with time, so we write it as $m_g\,(t)$. Using m_R as the mass of the rocket with no fuel, the total mass of the rocket plus fuel is $m = m_R + m_g\,(t)$. Differentiation with respect to time gives

$$\frac{dm}{dt} = \frac{dm_R}{dt} + \frac{dm_g(t)}{dt} = \frac{dm_g(t)}{dt}$$

where we used $\frac{dm_R}{dt} = 0$ because the mass of the rocket does not change. Thus, time rate of change of the mass of the rocket is the same as that of the fuel.

Conceptual Questions

1. Since $K = p^2/2m$, then if the momentum is fixed, the object with smaller mass has more kinetic energy.

3. Yes; impulse is the force applied multiplied by the time during which it is applied ($J = F\Delta t$), so if a small force acts for a long time, it may result in a larger impulse than a large force acting for a small time.

5. By friction, the road exerts a horizontal force on the tires of the car, which changes the momentum of the car.

7. Momentum is conserved when the mass of the system of interest remains constant during the interaction in question and when no *net* external force acts on the system during the interaction.

9. To accelerate air molecules in the direction of motion of the car, the car must exert a force on these molecules by Newton's second law $\vec{F} = d\vec{p}/dt$. By Newton's third law, the air molecules exert a force of equal magnitude but in the opposite direction on the car. This force acts in the direction opposite the motion of the car and constitutes the force due to air resistance.

11. No, he is not a closed system because a net nonzero external force acts on him in the form of the starting blocks pushing on his feet.

13. Yes, all the kinetic energy can be lost if the two masses come to rest due to the collision (i.e., they stick together).

15. The angle between the directions must be 90°. Any system that has zero net external force in one direction and nonzero net external force in a perpendicular direction will satisfy these conditions.

17. Yes, the rocket speed can exceed the exhaust speed of the gases it ejects. The thrust of the rocket does not depend on the relative speeds of the gases and rocket, it simply depends on conservation of momentum.

Problems

19. a. magnitude: $25\text{ kg}\cdot\text{m/s}$; b. same as a.

21. $1.78 \times 10^{29}\text{ kg}\cdot\text{m/s}$

23. $1.3 \times 10^{9}\text{ kg}\cdot\text{m/s}$

25. a. $1.50 \times 10^{6}\text{ N}$; b. $1.00 \times 10^{5}\text{ N}$

27. $4.69 \times 10^{5}\text{ N}$

29. $2.10 \times 10^{3}\text{ N}$

31. $\vec{\mathbf{p}}(t) = \left(10\hat{\mathbf{i}} + 20t\hat{\mathbf{j}}\right)\text{ kg}\cdot\text{m/s}; \vec{\mathbf{F}} = (20\text{ N})\,\hat{\mathbf{j}}$

33. Let the positive x-axis be in the direction of the original momentum. Then $p_x = 1.5\ \text{kg} \cdot \text{m/s}$ and $p_y = 7.5\ \text{kg} \cdot \text{m/s}$
35. $(0.122\ \text{m/s})\hat{\mathbf{i}}$
37. a. 47 m/s in the bullet to block direction; b. $70.6\ \text{N} \cdot \text{s}$, toward the bullet; c. $70.6\ \text{N} \cdot \text{s}$, toward the block; d. magnitude is $2.35 \times 10^4\ \text{N}$
39. 2:5
41. 5.9 m/s
43. a. 6.80 m/s, 5.33°; b. yes (calculate the ratio of the initial and final kinetic energies)
45. 2.5 cm
47. the speed of the leading bumper car is 6.00 m/s and that of the trailing bumper car is 5.60 m/s
49. 6.6%
51. 1.8 m/s
53. 22.1 m/s at $32.2°$ below the horizontal
55. a. 33 m/s and 110 m/s; b. 57 m; c. 480 m
57. $(732\ \text{m/s})\,\hat{\mathbf{i}} + (-79.6\ \text{m/s})\,\hat{\mathbf{j}}$
59. $-(1.47\ \text{m/s})\,\hat{\mathbf{i}} + (1.75\ \text{m/s})\,\hat{\mathbf{j}}$
61. 341 m/s at 86.8° with respect to the $\hat{\mathbf{i}}$ axis.
63. With the origin defined to be at the position of the 150-g mass, $x_{\text{CM}} = -1.23\text{cm}$ and $y_{\text{CM}} = 0.69\text{cm}$
65. $y_{\text{CM}} = \begin{cases} \frac{h}{2} - \frac{1}{4}gt^2, & t < T \\ h - \frac{1}{2}gt^2 - \frac{1}{4}gT^2 + \frac{1}{2}gtT, & t \geq T \end{cases}$
67. a. $R_1 = 4\ \text{m}$, $R_2 = 2\ \text{m}$; b. $X_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, Y_{\text{CM}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$; c. yes, with $R = \frac{1}{m_1 + m_2}\sqrt{16m_1^2 + 4m_2^2}$
69. $x_{cm} = \frac{3}{4}L\left(\frac{\rho_1 + \rho_0}{\rho_1 + 2\rho_0}\right)$
71. $\left(\frac{2a}{3}, \frac{2b}{3}\right)$
73. $(x_{\text{CM}}, y_{\text{CM}}, z_{\text{CM}}) = (0, 0, h/4)$
75. $(x_{\text{CM}}, y_{\text{CM}}, z_{\text{CM}}) = (0, 4R/(3\pi), 0)$
77. (a) 0.413 m/s, (b) about 0.2 J
79. 1551 kg
81. 4.9 km/s

Additional Problems

84. the elephant has a higher momentum
86. Answers may vary. The first clause is true, but the second clause is not true in general because the velocity of an object with small mass may be large enough so that the momentum of the object is greater than that of a larger-mass object with a smaller velocity.
88. $4.5 \times 10^3\ \text{N}$
90. $\vec{\mathbf{J}} = \int_0^{\tau} \left[m\vec{g} - m\vec{g}\left(1 - e^{-bt/m}\right)\right] dt = \frac{m^2}{b}\vec{g}\left(e^{-b\tau/m} - 1\right)$
92. a. $-\left(2.1 \times 10^3\ \text{kg} \cdot \text{m/s}\right)\hat{\mathbf{i}}$, b. $-\left(24 \times 10^3\ \text{N}\right)\hat{\mathbf{i}}$
94. a. $\left(1.1 \times 10^3\ \text{kg} \cdot \text{m/s}\right)\hat{\mathbf{i}}$, b. $(0.010\ \text{kg} \cdot \text{m/s})\,\hat{\mathbf{i}}$, c. $-(0.00093\ \text{m/s})\,\hat{\mathbf{i}}$, d. $-(0.0012\ \text{m/s})\,\hat{\mathbf{i}}$
96. $-(7.2\ \text{m/s})\,\hat{\mathbf{i}}$
98. $v_{1,\text{f}} = v_{1,\text{i}}\frac{m_1 - m_2}{m_1 + m_2},\ v_{2,\text{f}} = v_{1,\text{i}}\frac{2m_1}{m_1 + m_2}$
100. 2.8 m/s
102. 0.094 m/s
104. final velocity of cue ball is $-(0.76\ \text{m/s})\,\hat{\mathbf{i}}$, final velocities of the other two balls are 2.6 m/s at ±30° with respect to the initial velocity of the cue ball
106. ball 1: $-(1.4\ \text{m/s})\,\hat{\mathbf{i}} - (0.4\ \text{m/s})\,\hat{\mathbf{j}}$, ball 2: $(2.2\ \text{m/s})\,\hat{\mathbf{i}} + (2.4\ \text{m/s})\,\hat{\mathbf{j}}$
108. ball 1: $(1.4\ \text{m/s})\,\hat{\mathbf{i}} - (1.7\ \text{m/s})\,\hat{\mathbf{j}}$, ball 2: $-(2.8\ \text{m/s})\,\hat{\mathbf{i}} + (0.012\ \text{m/s})\,\hat{\mathbf{j}}$

110. $\theta_{cm} = \frac{1}{8}, r_{cm} = \frac{8R}{3\pi}\sqrt{2 - \sqrt{2}}$

112. Answers may vary. The rocket is propelled forward not by the gasses pushing against the surface of Earth, but by conservation of momentum. The momentum of the gas being expelled out the back of the rocket must be compensated by an increase in the forward momentum of the rocket.

Challenge Problems

114. a. $617\ \text{N}\cdot\text{s}, 108°$; b. $F_x = 2.91 \times 10^4\ \text{N}, F_y = 2.6 \times 10^5\ \text{N}$; c. $F_x = 5850\ \text{N}, F_y = 5265\ \text{N}$

116. Conservation of momentum demands $m_1 v_{1,\text{i}} + m_2 v_{2,\text{i}} = m_1 v_{1,\text{f}} + m_2 v_{2,\text{f}}$. We are given that $m_1 = m_2$, $v_{1,\text{i}} = v_{2,\text{f}}$, and $v_{2,\text{i}} = v_{1,\text{f}} = 0$. Combining these equations with the equation given by conservation of momentum gives $v_{1,\text{i}} = v_{1,\text{i}}$, which is true, so conservation of momentum is satisfied. Conservation of energy demands $\frac{1}{2}m_1 v_{1,\text{i}}^2 + \frac{1}{2}m_2 v_{2,\text{i}}^2 = \frac{1}{2}m_1 v_{1,\text{f}}^2 + \frac{1}{2}m_2 v_{2,\text{f}}^2$. Again combining this equation with the conditions given above give $v_{1,\text{i}} = v_{1,\text{i}}$, so conservation of energy is satisfied.

118. Assume origin on centerline and at floor, then $(x_{\text{CM}}, y_{\text{CM}}) = (0{,}86\ \text{cm})$

Chapter 10

Check Your Understanding

10.1 a. $40.0\ \text{rev/s} = 2\pi(40.0)\ \text{rad/s}, \bar{\alpha} = \frac{\Delta\omega}{\Delta t} = \frac{2\pi(40.0) - 0\ \text{rad/s}}{20.0\ \text{s}} = 2\pi(2.0) = 4.0\pi\ \text{rad/s}^2$; b. Since the angular velocity increases linearly, there has to be a constant acceleration throughout the indicated time. Therefore, the instantaneous angular acceleration at any time is the solution to $4.0\pi\ \text{rad/s}^2$.

10.2 a. Using Equation 10.11, we have $7000\ \text{rpm} = \frac{7000.0(2\pi\ \text{rad})}{60.0\ \text{s}} = 733.0\ \text{rad/s}$,

$\alpha = \frac{(\omega - \omega_0)}{t} = \frac{(0 - 733.0\ \text{rad/s})}{10.0\ \text{s}} = -73.3\ \text{rad/s}^2$;

b. Using Equation 10.13, we have

$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta \Rightarrow \Delta\theta = \frac{\omega^2 - \omega_0^2}{2\alpha} = \frac{0 - (733.0\ \text{rad/s})^2}{2(-73.3\ \text{rad/s}^2)} = 3665.2\ \text{rad}$

10.3 The angular acceleration is $\alpha = \frac{(5.0 - 0)\text{rad/s}}{20.0\ \text{s}} = 0.25\ \text{rad/s}^2$. Therefore, the total angle that the boy passes through is

$\Delta\theta = \frac{\omega^2 - \omega_0^2}{2\alpha} = \frac{(5.0)^2 - 0}{2(0.25)} = 50\ \text{rad}.$

Thus, we calculate

$s = r\theta = 5.0\ \text{m}(50.0\ \text{rad}) = 250.0\ \text{m}.$

10.4 The initial rotational kinetic energy of the propeller is

$K_0 = \frac{1}{2}I\omega^2 = \frac{1}{2}(800.0\ \text{kg-m}^2)(4.0 \times 2\pi\ \text{rad/s})^2 = 2.53 \times 10^5\ \text{J}.$

At 5.0 s the new rotational kinetic energy of the propeller is

$K_\text{f} = 2.03 \times 10^5\ \text{J}.$

and the new angular velocity is

$\omega = \sqrt{\frac{2(2.03 \times 10^5\ \text{J})}{800.0\ \text{kg-m}^2}} = 22.53\ \text{rad/s}$

which is 3.58 rev/s.

10.5 $I_{\text{parallel-axis}} = I_{\text{center of mass}} + md^2 = mR^2 + mR^2 = \left(\frac{1}{3}\right)mR^2$

10.6 The angle between the lever arm and the force vector is $80°$; therefore, $r_\perp = 100\text{m}(\sin 80°) = 98.5\ \text{m}$.

The cross product $\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}$ gives a negative or clockwise torque.

The torque is then $\tau = -r_\perp F = -98.5\ \text{m}(5.0 \times 10^5\text{N}) = -4.9 \times 10^7\text{N}\cdot\text{m}$.

10.7 a. The angular acceleration is $\alpha = \frac{20.0(2\pi)\text{rad/s} - 0}{10.0\ \text{s}} = 12.56\ \text{rad/s}^2$. Solving for the torque, we have $\sum_i \tau_i = I\alpha = (30.0\ \text{kg}\cdot\text{m}^2)(12.56\ \text{rad/s}^2) = 376.80\ \text{N}\cdot\text{m}$; b. The angular acceleration is $\alpha = \frac{0 - 20.0(2\pi)\text{rad/s}}{20.0\ \text{s}} = -6.28\ \text{rad/s}^2$. Solving for the torque, we have

$$\sum_i \tau_i = I\alpha = (30.0\,\text{kg-m}^2)(-6.28\,\text{rad/s}^2) = -188.50\,\text{N}\cdot\text{m}$$

10.8 3 MW

Conceptual Questions

1. The second hand rotates clockwise, so by the right-hand rule, the angular velocity vector is into the wall.
3. They have the same angular velocity. Points further out on the bat have greater tangential speeds.
5. straight line, linear in time variable
7. constant
9. The centripetal acceleration vector is perpendicular to the velocity vector.
11. a. both; b. nonzero centripetal acceleration; c. both
13. The hollow sphere, since the mass is distributed further away from the rotation axis.
15. a. It decreases. b. The arms could be approximated with rods and the discus with a disk. The torso is near the axis of rotation so it doesn't contribute much to the moment of inertia.
17. Because the moment of inertia varies as the square of the distance to the axis of rotation. The mass of the rod located at distances greater than $L/2$ would provide the larger contribution to make its moment of inertia greater than the point mass at $L/2$.
19. magnitude of the force, length of the lever arm, and angle of the lever arm and force vector
21. The moment of inertia of the wheels is reduced, so a smaller torque is needed to accelerate them.
23. yes
25. $|\vec{\mathbf{r}}|$ can be equal to the lever arm but never less than the lever arm
27. If the forces are along the axis of rotation, or if they have the same lever arm and are applied at a point on the rod.

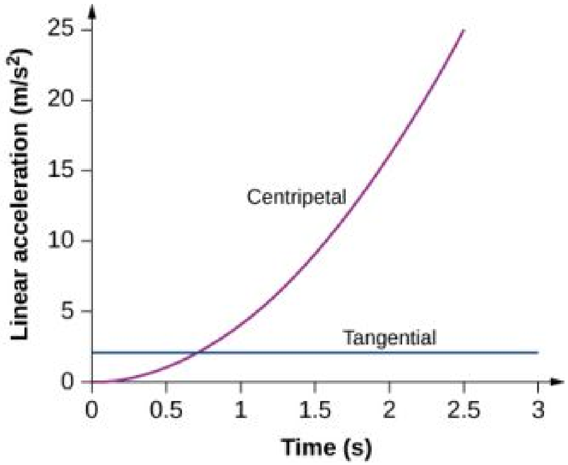
Problems

29. $\omega = \frac{2\pi\,\text{rad}}{45.0\,\text{s}} = 0.14\,\text{rad/s}$
31. a. $\theta = \frac{s}{r} = \frac{3.0\,\text{m}}{1.5\,\text{m}} = 2.0\,\text{rad}$; b. $\omega = \frac{2.0\,\text{rad}}{1.0\,\text{s}} = 2.0\,\text{rad/s}$; c. $\frac{v^2}{r} = \frac{(3.0\,\text{m/s})^2}{1.5\,\text{m}} = 6.0\,\text{m/s}^2$.
33. The propeller takes only $\Delta t = \frac{\Delta\omega}{\alpha} = \frac{0\,\text{rad/s} - 10.0(2\pi)\,\text{rad/s}}{-2.0\,\text{rad/s}^2} = 31.4\,\text{s}$ to come to rest, when the propeller is at 0 rad/s, it would start rotating in the opposite direction. This would be impossible due to the magnitude of forces involved in getting the propeller to stop and start rotating in the opposite direction.
35. a. $\omega = 25.0(2.0\,\text{s}) = 50.0\,\text{rad/s}$; b. $\alpha = \frac{d\omega}{dt} = 25.0\,\text{rad/s}^2$
37. a. $\omega = 54.8\,\text{rad/s}$;
b. $t = 11.0\,\text{s}$
39. a. $0.87\,\text{rad/s}^2$;
b. $\theta = 12{,}600\,\text{rad}$
41. a. $\omega = 42.0\,\text{rad/s}$;

b. $\theta = 220\,\text{rad}$; c.
$$v_t = 42\,\text{m/s}$$
$$a_t = 4.0\,\text{m/s}^2$$
43. a. $\omega = 7.0\,\text{rad/s}$;
b. $\theta = 22.5\,\text{rad}$; c. $a_\text{t} = 0.10\,\text{m/s}^2$
45. $\alpha = 28.6\,\text{rad/s}^2$.
47. $r = 0.78\,\text{m}$
49. a. $\alpha = -0.314\,\text{rad/s}^2$,
b. $a_\text{c} = 197.4\,\text{m/s}^2$; c. $a = \sqrt{a_\text{c}^2 + a_\text{t}^2} = \sqrt{197.4^2 + (-6.28)^2} = 197.5\,\text{m/s}^2$
$\theta = \tan^{-1}\frac{-6.28}{197.4} = -1.8°$ in the counterclockwise direction from the centripetal acceleration vector
51. $ma = 40.0\,\text{kg}(4.75\,\text{m/s}^2) = 190\,\text{N}$
The maximum friction force is $\mu_\text{S} N = 0.6(40.0\,\text{kg})(9.8\,\text{m/s}^2) = 235.2\,\text{N}$ so the child does not fall off yet.

53.
$$\begin{aligned} v_t &= r\omega = 1.0(2.0t)\text{ m/s} \\ a_c &= \frac{v_t^2}{r} = \frac{(2.0t)^2}{1.0\text{ m}} = 4.0t^2\text{m/s}^2 \\ a_t(t) &= r\alpha(t) = r\frac{d\omega}{dt} = 1.0\text{ m}(2.0) = 2.0\text{ m/s}^2. \end{aligned}$$

Plotting both accelerations gives



The tangential acceleration is constant, while the centripetal acceleration is time dependent, and increases with time to values much greater than the tangential acceleration after $t = 1$s. For times less than 0.7 s and approaching zero the centripetal acceleration is much less than the tangential acceleration.

55. a. $K = 2.56 \times 10^{29}$ J;
b. $K = 2.68 \times 10^{33}$ J

57. $K = 434.0\,\text{J}$

59. a. $v_f = 86.5$ m/s;
b. The rotational rate of the propeller stays the same at 20 rev/s.

61. $K = 3.95 \times 10^{42}$ J

63. a. $I = 0.315\,\text{kg}\cdot\text{m}^2$;
b. $K = 621.8\,\text{J}$

65. $I = \frac{7}{36}mL^2$

67. $v = 7.14$ m/s.

69. $\theta = 10.2°$

71. $F = 30\,\text{N}$

73. a. $0.85\,\text{m}\,(55.0\,\text{N}) = 46.75\,\text{N}\cdot\text{m}$; b. It does not matter at what height you push.

75. $m_2 = \frac{4.9\,\text{N}\cdot\text{m}}{9.8(0.3\text{ m})} = 1.67\,\text{kg}$

77. $\tau_{net} = -9.0\,\text{N}\cdot\text{m} + 3.46\,\text{N}\cdot\text{m} + 0 - 3.38\,\text{N}\cdot\text{m} = -8.92\,\text{N}\cdot\text{m}$

79. $\tau = 5.66\,\text{N}\cdot\text{m}$

81. $\sum \tau = 57.82\,\text{N}\cdot\text{m}$

83. $\vec{\mathbf{r}} \times \vec{\mathbf{F}} = 4.0\hat{\mathbf{i}} + 2.0\hat{\mathbf{j}} - 16.0\hat{\mathbf{k}}\text{N}\cdot\text{m}$

85. a. $\tau = (0.280\,\text{m})(180.0\,\text{N}) = 50.4\,\text{N}\cdot\text{m}$; b. $\alpha = 17.14\,\text{rad/s}^2$;
c. $\alpha = 17.04\,\text{rad/s}^2$

87. $\tau = 8.0\,\text{N}\cdot\text{m}$

89. $\tau = -43.6\,\text{N}\cdot\text{m}$

91. a. $\alpha = 1.4 \times 10^{-10}\,\text{rad/s}^2$;
b. $\tau = 1.36 \times 10^{28}$N-m; c. $F = 2.1 \times 10^{21}$ N

93. $a = 3.6\,\text{m/s}^2$

95. a. $a = r\alpha = 14.7\,\text{m/s}^2$; b. $a = \frac{L}{2}\alpha = \frac{3}{4}g$

97. $\tau = \frac{P}{\omega} = \frac{2.0 \times 10^6\text{W}}{2.1\text{ rad/s}} = 9.5 \times 10^5\text{N}\cdot\text{m}$

99. a. $K = 888.50\,\text{J}$;
b. $\Delta\theta = 294.6\,\text{rev}$

101. a. $I = 114.6\,\text{kg} \cdot \text{m}^2$;
b. $P = 104{,}700\,\text{W}$

103. $v = L\omega = \sqrt{3Lg}$

105. a. $a = 5.0\,\text{m/s}^2$; b. $W = 1.25\,\text{N} \cdot \text{m}$

Additional Problems

107. $\Delta t = 10.0\,\text{s}$

109. a. $0.06\,\text{rad/s}^2$; b. $\theta = 16.7$ revolutions

111. $s = 405.26\,\text{m}$

113. a. $I = 0.363\,\text{kg} \cdot \text{m}^2$;
b. $I = 2.34\,\text{kg} \cdot \text{m}^2$

115. $\omega = \sqrt{\frac{6.68\,\text{J}}{4.4\,\text{kgm}^2}} = 1.23\,\text{rad/s}$

117. $F = 23.3\,\text{N}$

119. $\alpha = \frac{190.0\,\text{N-m}}{2.94\,\text{kg-m}^2} = 64.4\,\text{rad/s}^2$

Challenge Problems

121. a. $\omega = 2.0t - 1.5t^2$; b. $\theta = t^2 - 0.5t^3$; c. $\theta = -400.0\,\text{rad}$; d. the vector is at $-0.66(360°) = -237.6°$

123. $I = \frac{2}{5}mR^2$

125. a. $\omega = 8.2\,\text{rad/s}$; b. $\omega = 8.0\,\text{rad/s}$

Chapter 11

Check Your Understanding

11.1 a. $\mu_s \geq \frac{\tan\theta}{1+(mr^2/I_{CM})}$; inserting the angle and noting that for a hollow cylinder $I_{CM} = mr^2$, we have $\mu_s \geq \frac{\tan 60°}{1+(mr^2/mr^2)} = \frac{1}{2}\tan 60° = 0.87$; we are given a value of 0.6 for the coefficient of static friction, which is less than 0.87, so the condition isn't satisfied and the hollow cylinder will slip; b. The solid cylinder obeys the condition $\mu_s \geq \frac{1}{3}\tan\theta = \frac{1}{3}\tan 60° = 0.58$. The value of 0.6 for μ_S satisfies this condition, so the solid cylinder will not slip.

11.2 From the figure, we see that the cross product of the radius vector with the momentum vector gives a vector directed out of the page. Inserting the radius and momentum into the expression for the angular momentum, we have
$\vec{\mathbf{l}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}} = (0.4\,\text{m}\hat{\mathbf{i}}) \times (1.67 \times 10^{-27}\,\text{kg}(4.0 \times 10^6\,\text{m/s})\hat{\mathbf{j}}) = 2.7 \times 10^{-21}\,\text{kg} \cdot \text{m}^2/\text{s}\hat{\mathbf{k}}$

11.3 $I_{\text{sphere}} = \frac{2}{5}mr^2, \quad I_{\text{cylinder}} = \frac{1}{2}mr^2$; Taking the ratio of the angular momenta, we have:
$\frac{L_{\text{cylinder}}}{L_{\text{sphere}}} = \frac{I_{\text{cylinder}}\omega_0}{I_{\text{sphere}}\omega_0} = \frac{\frac{1}{2}mr^2}{\frac{2}{5}mr^2} = \frac{5}{4}$. Thus, the cylinder has 25% more angular momentum. This is because the cylinder has more mass distributed farther from the axis of rotation.

11.4 Using conservation of angular momentum, we have
$I(4.0\,\text{rev/min}) = 1.25I\omega_f, \quad \omega_f = \frac{1.0}{1.25}(4.0\,\text{rev/min}) = 3.2\,\text{rev/min}$

11.5 The Moon's gravity is 1/6 that of Earth's. By examining Equation 11.12, we see that the top's precession frequency is linearly proportional to the acceleration of gravity. All other quantities, mass, moment of inertia, and spin rate are the same on the Moon. Thus, the precession frequency on the Moon is
$\omega_P(\text{Moon}) = \frac{1}{6}\omega_P(\text{Earth}) = \frac{1}{6}(5.0\,\text{rad/s}) = 0.83\,\text{rad/s}.$

Conceptual Questions

1. No, the static friction force is zero.

3. The wheel is more likely to slip on a steep incline since the coefficient of static friction must increase with the angle to keep rolling motion without slipping.
5. The cylinder reaches a greater height. By Equation 11.4, its acceleration in the direction down the incline would be less.
7. All points on the straight line will give zero angular momentum, because a vector crossed into a parallel vector is zero.
9. The particle must be moving on a straight line that passes through the chosen origin.
11. Without the small propeller, the body of the helicopter would rotate in the opposite sense to the large propeller in order to conserve angular momentum. The small propeller exerts a thrust at a distance *R* from the center of mass of the aircraft to prevent this from happening.
13. The angular velocity increases because the moment of inertia is decreasing.
15. More mass is concentrated near the rotational axis, which decreases the moment of inertia causing the star to increase its angular velocity.
17. A torque is needed in the direction perpendicular to the angular momentum vector in order to change its direction. These forces on the space vehicle are external to the container in which the gyroscope is mounted and do not impart torques to the gyroscope's rotating disk.

Problems

19. $v_{\text{CM}} = R\omega \Rightarrow \omega = 66.7 \text{ rad/s}$
21. $\alpha = 3.3 \text{ rad/s}^2$
23. $I_{\text{CM}} = \frac{2}{5}mr^2,\ a_{\text{CM}} = 3.5 \text{ m/s}^2;\ x = 15.75 \text{ m}$
25. positive is down the incline plane;
$a_{\text{CM}} = \frac{mg\sin\theta}{m+(I_{\text{CM}}/r^2)} \Rightarrow I_{\text{CM}} = r^2\left[\frac{mg\sin 30}{a_{\text{CM}}} - m\right]$,
$x - x_0 = v_0 t - \frac{1}{2}a_{\text{CM}}t^2 \Rightarrow a_{\text{CM}} = 2.96 \text{ m/s}^2$,
$I_{\text{CM}} = 0.66\, mr^2$
27. $\alpha = 67.9 \text{ rad/s}^2$,
$(a_{\text{CM}})_x = 1.5 \text{ m/s}^2$
29. $W = -1080.0 \text{ J}$
31. Mechanical energy at the bottom equals mechanical energy at the top;
$\frac{1}{2}mv_0^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{v_0}{r}\right)^2 = mgh \Rightarrow h = \frac{1}{g}\left(\frac{1}{2} + \frac{1}{4}\right)v_0^2$,
$h = 7.7$ m, so the distance up the incline is 22.5 m.
33. Use energy conservation
$\frac{1}{2}mv_0^2 + \frac{1}{2}I_{\text{Cyl}}\omega_0^2 = mgh_{\text{Cyl}}$,
$\frac{1}{2}mv_0^2 + \frac{1}{2}I_{\text{Sph}}\omega_0^2 = mgh_{\text{Sph}}$.
Subtracting the two equations, eliminating the initial translational energy, we have
$h_{\text{Cyl}} = \frac{v_0^2}{g}$,
$h_{\text{Sph}} = \frac{5v_0^2}{6g}$,
The ratio of the height reached by the sphere to the height reached by the cylinder is $\frac{h_{\text{Sph}}}{h_{\text{Cyl}}} = \frac{5}{6}$.
So the sphere reaches a lower height of $\frac{5}{6}\left(1.0 \text{ m}\right) = 0.83 \text{ m}$.
35. The magnitude of the cross product of the radius to the bird and its momentum vector yields $rp\sin\theta$, which gives $r\sin\theta$ as the altitude of the bird *h*. The direction of the angular momentum is perpendicular to the radius and momentum vectors, which we choose arbitrarily as $\hat{\mathbf{k}}$, which is in the plane of the ground:
$\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}} = hmv\hat{\mathbf{k}} = (300.0 \text{ m})(2.0 \text{ kg})(20.0 \text{ m/s})\hat{\mathbf{k}} = 12{,}000.0 \text{ kg}\cdot\text{m}^2/\text{s}\hat{\mathbf{k}}$
37. a. $\vec{\mathbf{l}} = 45.0 \text{ kg}\cdot\text{m}^2/\text{s}\hat{\mathbf{k}}$;
b. $\vec{\boldsymbol{\tau}} = 10.0 \text{ N}\cdot\text{m}\hat{\mathbf{k}}$
39. a. $\vec{\mathbf{l}}_1 = -0.4 \text{ kg}\cdot\text{m}^2/\text{s}\hat{\mathbf{k}},\ \vec{\mathbf{l}}_2 = \vec{\mathbf{l}}_4 = 0$,
$\vec{\mathbf{l}}_3 = 1.35 \text{ kg}\cdot\text{m}^2/\text{s}\hat{\mathbf{k}}$; b. $\vec{\mathbf{L}} = 0.95 \text{ kg}\cdot\text{m}^2/\text{s}\hat{\mathbf{k}}$

41. a. $L = 1.0 \times 10^{11}\ \text{kg} \cdot \text{m}^2/\text{s}$; b. No, the angular momentum stays the same since the cross-product involves only the perpendicular distance from the plane to the ground no matter where it is along its path.

43. a. $\vec{\mathbf{v}} = -gt\hat{\mathbf{j}},\ \vec{\mathbf{r}}_{\perp} = -d\hat{\mathbf{i}},\ \vec{\mathbf{l}} = mdgt\hat{\mathbf{k}}$;
b. $\vec{\mathbf{F}} = -mg\hat{\mathbf{j}},\ \sum \vec{\boldsymbol{\tau}} = dmg\hat{\mathbf{k}}$; c. yes

45. a. $mgh = \frac{1}{2}m(r\omega)^2 + \frac{1}{2}\frac{2}{5}mr^2\omega^2$;
$\omega = 51.2\ \text{rad/s}$;
$L = 16.4\ \text{kg} \cdot \text{m}^2/\text{s}$;
b. $\omega = 72.5\ \text{rad/s}$;
$L = 23.2\ \text{kg} \cdot \text{m}^2/\text{s}$

47. a. $I = 720.0\ \text{kg} \cdot \text{m}^2$; $\alpha = 4.20\ \text{rad/s}^2$;
$\omega(10\ \text{s}) = 42.0\ \text{rad/s}$; $L = 3.02 \times 10^4\ \text{kg} \cdot \text{m}^2/\text{s}$;
$\omega(20\ \text{s}) = 84.0\ \text{rad/s}$;
b. $\tau = 3.03 \times 10^3\ \text{N} \cdot \text{m}$

49. a. $L = 1.131 \times 10^7\ \text{kg} \cdot \text{m}^2/\text{s}$;
b. $\tau = 3.77 \times 10^4\ \text{N} \cdot \text{m}$

51. $\omega = 28.6\ \text{rad/s} \Rightarrow L = 2.6\ \text{kg} \cdot \text{m}^2/\text{s}$

53. $L_f = \frac{2}{5}M_S(3.5 \times 10^3\text{km})^2 \frac{2\pi}{T_f}$,

$$(7.0 \times 10^5\ \text{km})^2 \frac{2\pi}{28\ \text{days}} = (3.5 \times 10^3\text{km})^2 \frac{2\pi}{T_f} \Rightarrow T_f$$

$$= 28\ \text{days}\frac{(3.5 \times 10^3\text{km})^2}{(7.0 \times 10^5\text{km})^2} = 7.0 \times 10^{-4}\text{day} = 60.5\ \text{s}$$

55. $f_\text{f} = 2.1\ \text{rev/s} \Rightarrow f_0 = 0.5\ \text{rev/s}$

57. $r_P m v_P = r_A m v_A \Rightarrow v_P = 10.2\ \text{km/s}$

59. a. $I_\text{disk} = 5.0 \times 10^{-4}\ \text{kg} \cdot \text{m}^2$,
$I_\text{bug} = 2.0 \times 10^{-4}\ \text{kg} \cdot \text{m}^2$,

$$(I_\text{disk} + I_\text{bug})\omega_1 = I_\text{disk}\omega_2,$$

$\omega_2 = 14.0\ \text{rad/s}$
b. $\Delta K = 0.014\ \text{J}$;
c. $\omega_3 = 10.0\ \text{rad/s}$ back to the original value;
d. $\frac{1}{2}(I_\text{disk} + I_\text{bug})\omega_3^2 = 0.035\ \text{J}$ back to the original value;
e. work of the bug crawling on the disk

61. $L_\text{i} = 400.0\ \text{kg} \cdot \text{m}^2/\text{s}$,
$L_\text{f} = 500.0\ \text{kg} \cdot \text{m}^2\omega$,
$\omega = 0.80\ \text{rad/s}$

63. $I_0 = 340.48\ \text{kg} \cdot \text{m}^2$,
$I_\text{f} = 268.8\ \text{kg} \cdot \text{m}^2$,
$\omega_\text{f} = 25.33\ \text{rpm}$

65. a. $L = 280\ \text{kg} \cdot \text{m}^2/\text{s}$,
$I_\text{f} = 89.6\ \text{kg} \cdot \text{m}^2$,
$\omega_\text{f} = 3.125\ \text{rad/s}$; b. $K_\text{i} = 437.5\ \text{J}$,
$K_\text{f} = 437.5\ \text{J}$

67. Moment of inertia in the record spin: $I_0 = 0.5\ \text{kg} \cdot \text{m}^2$,
$I_\text{f} = 1.1\ \text{kg} \cdot \text{m}^2$,

$\omega_f = \frac{I_0}{I_f}\omega_0 \Rightarrow f_f = 155.5\,\text{rev/min}$

69. Her spin rate in the air is: $f_f = 2.0\,\text{rev/s}$;
She can do four flips in the air.

71. Moment of inertia with all children aboard:
$I_0 = 2.4 \times 10^5\,\text{kg} \cdot \text{m}^2$;
$I_f = 1.5 \times 10^5\,\text{kg} \cdot \text{m}^2$;
$f_f = 0.3\text{rev/s}$

73. $I_0 = 1.00 \times 10^{10}\,\text{kg} \cdot \text{m}^2$,
$I_f = 9.94 \times 10^9\,\text{kg} \cdot \text{m}^2$,
$f_f = 3.32\,\text{rev/min}$

75. $I = 2.5 \times 10^{-3}\,\text{kg} \cdot \text{m}^2$,
$\omega_P = 1.17\,\text{rad/s}$

77. a. $L_{\text{Earth}} = 7.06 \times 10^{33}\,\text{kg} \cdot \text{m}^2/\text{s}$,
$\Delta L = 5.63 \times 10^{33}\,\text{kg} \cdot \text{m}^2/\text{s}$;
b. $\tau = 1.4 \times 10^{22}\,\text{N} \cdot \text{m}$;
c. The two forces at the equator would have the same magnitude but different directions, one in the north direction and the other in the south direction on the opposite side of Earth. The angle between the forces and the lever arms to the center of Earth is $90°$, so a given torque would have magnitude
$\tau = FR_E \sin 90° = FR_E$. Both would provide a torque in the same direction:
$\tau = 2FR_E \Rightarrow F = 1.1 \times 10^{15}\,\text{N}$

Additional Problems

79. $a_{\text{CM}} = -\frac{3}{10}g$,
$v^2 = v_0^2 + 2a_{\text{CM}}x \Rightarrow v^2 = (7.0\,\text{m/s})^2 - 2\left(\frac{3}{10}g\right)x,\ v^2 = 0 \Rightarrow x = 8.34\,\text{m}$;
b. $t = \frac{v - v_0}{a_{\text{CM}}},\quad v = v_0 + a_{\text{CM}}t \Rightarrow t = 2.38\,\text{s}$;
The hollow sphere has a larger moment of inertia, and therefore is harder to bring to a rest than the marble, or solid sphere. The distance travelled is larger and the time elapsed is longer.

81. a. $W = -500.0\,\text{J}$;
b. $K + U_{\text{grav}} = \text{constant}$,
$500\,\text{J} + 0 = 0 + (6.0\,\text{kg})(9.8\,\text{m/s}^2)h$,
$h = 8.5\,\text{m},\ d = 17.0\,\text{m}$;
The moment of inertia is less for the hollow sphere, therefore less work is required to stop it. Likewise it rolls up the incline a shorter distance than the hoop.

83. a. $\tau = 34.0\,\text{N} \cdot \text{m}$;
b. $l = mr^2\omega \Rightarrow \omega = 3.6\,\text{rad/s}$

85. a. $d_M = 3.85 \times 10^8\,\text{m}$ average distance to the Moon; orbital period $27.32\text{d} = 2.36 \times 10^6\text{s}$; speed of the Moon $\frac{2\pi 3.85 \times 10^8\,\text{m}}{2.36 \times 10^6\text{s}} = 1.0 \times 10^3\,\text{m/s}$; mass of the Moon $7.35 \times 10^{22}\,\text{kg}$,
$L = 2.90 \times 10^{34}\,\text{kgm}^2/\text{s}$;
b. radius of the Moon $1.74 \times 10^6\,\text{m}$; the orbital period is the same as (a): $\omega = 2.66 \times 10^{-6}\,\text{rad/s}$,
$L = 2.37 \times 10^{29}\,\text{kg} \cdot \text{m}^2/\text{s}$;
The orbital angular momentum is 1.22×10^5 times larger than the rotational angular momentum for the Moon.

87. $I = 0.135\,\text{kg} \cdot \text{m}^2$,
$\alpha = 4.19\,\text{rad/s}^2, \omega = \omega_0 + \alpha t$,
$\omega(5\,\text{s}) = 21.0\,\text{rad/s},\ L = 2.84\,\text{kg} \cdot \text{m}^2/\text{s}$,
$\omega(10\,\text{s}) = 41.9\,\text{rad/s},\ L = 5.66\,\text{kg} \cdot \text{m/s}^2$

89. In the conservation of angular momentum equation, the rotation rate appears on both sides so we keep the (rev/min) notation as the angular velocity can be multiplied by a constant to get (rev/min):

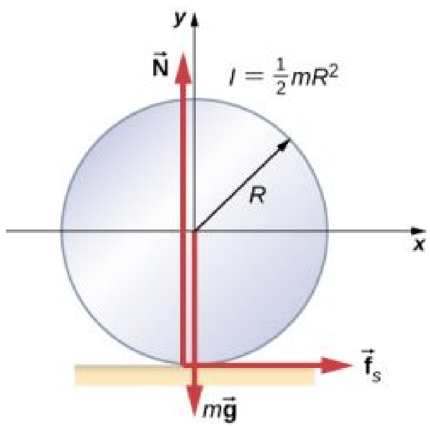
$$\begin{aligned} L_{\mathrm{i}} &= -0.04\,\mathrm{kg}\cdot\mathrm{m}^2(300.0\,\mathrm{rev/min}), \\ L_{\mathrm{f}} &= 0.08\,\mathrm{kg}\cdot\mathrm{m}^2 f_{\mathrm{f}} \Rightarrow f_{\mathrm{f}} = -150.0\,\mathrm{rev/min\ clockwise} \end{aligned}$$

91. $I_0\omega_0 = I_{\mathrm{f}}\omega_{\mathrm{f}}$,
$I_0 = 6120.0\,\mathrm{kg}\cdot\mathrm{m}^2$,
$I_{\mathrm{f}} = 1180.0\,\mathrm{kg}\cdot\mathrm{m}^2$,
$\omega_{\mathrm{f}} = 31.1\,\mathrm{rev/min}$

93. $L_{\mathrm{i}} = 1.00\times10^7\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}$,
$I_{\mathrm{f}} = 2.025\times10^5\,\mathrm{kg}\cdot\mathrm{m}^2$,
$\omega_{\mathrm{f}} = 7.86\,\mathrm{rev/s}$

Challenge Problems

95. Assume the roll accelerates forward with respect to the ground with an acceleration a'. Then it accelerates backwards relative to the truck with an acceleration $(a - a')$.



Also, $R\alpha = a - a'$ $I = \frac{1}{2}mR^2$ $\sum F_x = f_s = ma'$,
$\sum \tau = f_s R = I\alpha = I\frac{a-a'}{R}$ $f_s = \frac{I}{R^2}(a-a') = \frac{1}{2}m(a-a')$,
Solving for a': $f_s = \frac{1}{2}m(a-a')$; $a' = \frac{a}{3}$,
$x - x_0 = v_0 t + \frac{1}{2}at^2$; $d = \frac{1}{3}at^2$; $t = \sqrt{\frac{3d}{a}}$;
therefore, $s = 1.5d$

97. a. The tension in the string provides the centripetal force such that $T\sin\theta = mr_\perp\omega^2$. The component of the tension that is vertical opposes the gravitational force such that $T\cos\theta = mg$. This gives $T = 5.7\,\mathrm{N}$. We solve for $r_\perp = 0.16\,\mathrm{m}$. This gives the length of the string as $r = 0.32\,\mathrm{m}$.
At $\omega = 10.0\,\mathrm{rad/s}$, there is a new angle, tension, and perpendicular radius to the rod. Dividing the two equations involving the tension to eliminate it, we have $\frac{\sin\theta}{\cos\theta} = \frac{(0.32\,\mathrm{m}\sin\theta)\omega^2}{g} \Rightarrow \frac{1}{\cos\theta} = \frac{0.32\,\mathrm{m}\omega^2}{g}$;
$\cos\theta = 0.31 \Rightarrow \theta = 72.2°$; b. $l_{\mathrm{initial}} = 0.08\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}$,
$l_{\mathrm{final}} = 0.46\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}$; c. No, the cosine of the angle is inversely proportional to the square of the angular velocity, therefore in order for $\theta \to 90°$, $\omega \to \infty$. The rod would have to spin infinitely fast.

Chapter 12

Check Your Understanding

12.1 $x = 1.3\,\mathrm{m}$
12.2 (b), (c)
12.3 316.7 g; 5.8 N
12.4 $T = 1963\,\mathrm{N};\ \mathrm{F} = 1732\,\mathrm{N}$
12.5 $\mu_s < 0.5\cot\beta$

12.6 $\vec{\mathbf{F}}_{\text{door on } A} = 100.0\,\text{N}\hat{\mathbf{i}} - 200.0\,\text{N}\hat{\mathbf{j}}$; $\vec{\mathbf{F}}_{\text{door on } B} = -100.0\,\text{N}\hat{\mathbf{i}} - 200.0\,\text{N}\hat{\mathbf{j}}$
12.7 711.0 N; 466.0 N
12.8 1167 N; 980 N directed upward at $18°$ above the horizontal
12.9 $206.8\,\text{kPa};\ 4.6 \times 10^{-5}$
12.10 5.0×10^{-4}
12.11 63 mL
12.12 Fluids have different mechanical properties than those of solids; fluids flow.

Conceptual Questions

1. constant
3. magnitude and direction of the force, and its lever arm
5. True, as the sum of forces cannot be zero in this case unless the force itself is zero.
7. False, provided forces add to zero as vectors then equilibrium can be achieved.
9. It helps a wire-walker to maintain equilibrium.
11. (Proof)
13. In contact with the ground, stress in squirrel's limbs is smaller than stress in human's limbs.
15. tightly
17. compressive; tensile
19. no
23. It acts as "reinforcement," increasing a range of strain values before the structure reaches its breaking point.

Problems

25. $46.8\,\text{N} \cdot \text{m}$
27. 4,472 N, 153.4°
29. 23.3 N
31. 80.0 kg
33. 40 kg
35. right cable, 444.3 N; left cable, 888.5 N; weight of equipment 156.8 N; 16.0 kg
37. 784 N, 132.8 N
39. a. 539 N; b. 461 N; c. do not depend on the angle
41. tension 778 N; at hinge 778 N at $45°$ above the horizontal; no
43. 1500 N; 1620 N at $30°$
45. 1.2 mm
47. 32.9 cm
49. $4.0 \times 10^2\ \text{N/cm}^2$
51. $11.9\,\mu\text{m}$
53. 0.57 mm
55. 8.59 mm
57. $1.35 \times 10^9\,\text{Pa}$
59. 259.0 N
61. 0.01%
63. 1.44 cm
65. 0.63 cm

Additional Problems

69. $\tan^{-1}(1/\mu_s) = 51.3°$
71. a. at corner 66.7 N at $30°$ with the horizontal; at floor 177 N at $109°$ with the horizontal; b. $\mu_s = 0.346$
73. a. $1.10 \times 10^9\,\text{N/m}^2$; b. 5.5×10^{-3}; c. 11.0 mm, 31.4 mm

Challenge Problems

75. $F = Mg\tan\theta;\ f = 0$
77. with the horizontal, $\theta = 42.2°$; $\alpha = 17.8°$ with the steeper side of the wedge

79. $W(l_1/l_2 - 1); Wl_1/l_2 + mg$

81. a. 1.1 mm; b. 6.6 mm to the right; c. 1.11×10^5 N

Chapter 13

Check Your Understanding

13.1 The force of gravity on each object increases with the square of the inverse distance as they fall together, and hence so does the acceleration. For example, if the distance is halved, the force and acceleration are quadrupled. Our average is accurate only for a linearly increasing acceleration, whereas the acceleration actually increases at a greater rate. So our calculated speed is too small. From Newton's third law (action-reaction forces), the force of gravity between any two objects must be the same. But the accelerations will not be if they have different masses.

13.2 The tallest buildings in the world are all less than 1 km. Since *g* is proportional to the distance squared from Earth's center, a simple ratio shows that the change in *g* at 1 km above Earth's surface is less than 0.0001%. There would be no need to consider this in structural design.

13.3 The value of *g* drops by about 10% over this change in height. So $\Delta U = mg(y_2 - y_1)$ will give too large a value. If we use $g = 9.80$ m/s, then we get

$\Delta U = mg(y_2 - y_1) = 3.53 \times 10^{10}$ J

which is about 6% greater than that found with the correct method.

13.4 The probe must overcome both the gravitational pull of Earth and the Sun. In the second calculation of our example, we found the speed necessary to escape the Sun from a distance of Earth's orbit, not from Earth itself. The proper way to find this value is to start with the energy equation, Equation 13.5, in which you would include a potential energy term for both Earth and the Sun.

13.5 You change the direction of your velocity with a force that is perpendicular to the velocity at all points. In effect, you must constantly adjust the thrusters, creating a centripetal force until your momentum changes from tangential to radial. A simple momentum vector diagram shows that the net *change* in momentum is $\sqrt{2}$ times the magnitude of momentum itself. This turns out to be a very inefficient way to reach Mars. We discuss the most efficient way in Kepler's Laws of Planetary Motion.

13.6 In Equation 13.7, the radius appears in the denominator inside the square root. So the radius must increase by a factor of 4, to decrease the orbital velocity by a factor of 2. The circumference of the orbit has also increased by this factor of 4, and so with half the orbital velocity, the period must be 8 times longer. That can also be seen directly from Equation 13.8.

13.7 The assumption is that orbiting object is much less massive than the body it is orbiting. This is not really justified in the case of the Moon and Earth. Both Earth and the Moon orbit about their common center of mass. We tackle this issue in the next example.

13.8 The stars on the "inside" of each galaxy will be closer to the other galaxy and hence will feel a greater gravitational force than those on the outside. Consequently, they will have a greater acceleration. Even without this force difference, the inside stars would be orbiting at a smaller radius, and, hence, there would develop an elongation or stretching of each galaxy. The force difference only increases this effect.

13.9 The semi-major axis for the highly elliptical orbit of Halley's comet is 17.8 AU and is the average of the perihelion and aphelion. This lies between the 9.5 AU and 19 AU orbital radii for Saturn and Uranus, respectively. The radius for a circular orbit is the same as the semi-major axis, and since the period increases with an increase of the semi-major axis, the fact that Halley's period is between the periods of Saturn and Uranus is expected.

13.10 Consider the last equation above. The values of r_1 and r_2 remain nearly the same, but the diameter of the Moon, $(r_2 - r_1)$, is one-fourth that of Earth. So the tidal forces on the Moon are about one-fourth as great as on Earth.

13.11 Given the incredible density required to force an Earth-sized body to become a black hole, we do not expect to see such small black holes. Even a body with the mass of our Sun would have to be compressed by a factor of 80 beyond that of a neutron star. It is believed that stars of this size cannot become black holes. However, for stars with a few solar masses, it is believed that gravitational collapse at the end of a star's life could form a black hole. As we will discuss later, it is now believed that black holes are common at the center of galaxies. These galactic black holes typically contain the mass of many millions of stars.

Conceptual Questions

1. The ultimate truth is experimental verification. Field theory was developed to help explain how force is exerted without objects being in contact for both gravity and electromagnetic forces that act at the speed of light. It has only been since the twentieth century that we have been able to measure that the force is not conveyed immediately.
3. The centripetal acceleration is not directed along the gravitational force and therefore the correct line of the building (i.e., the plumb bob line) is not directed towards the center of Earth. But engineers use either a plumb bob or a transit, both of which respond to both the direction of gravity and acceleration. No special consideration for their location on Earth need be made.
5. As we move to larger orbits, the change in potential energy increases, whereas the orbital velocity decreases. Hence, the ratio is highest near Earth's surface (technically infinite if we orbit at Earth's surface with no elevation change), moving to zero as we reach infinitely far away.
7. The period of the orbit must be 24 hours. But in addition, the satellite must be located in an equatorial orbit and orbiting in the same direction as Earth's rotation. All three criteria must be met for the satellite to remain in one position relative to Earth's surface. At least three satellites are needed, as two on opposite sides of Earth cannot communicate with each other. (This is not technically true, as a wavelength could be chosen that provides sufficient diffraction. But it would be totally impractical.)
9. The speed is greatest where the satellite is closest to the large mass and least where farther away—at the periapsis and apoapsis, respectively. It is conservation of angular momentum that governs this relationship. But it can also be gleaned from conservation of energy, the kinetic energy must be greatest where the gravitational potential energy is the least (most negative). The force, and hence acceleration, is always directed towards *M* in the diagram, and the velocity is always tangent to the path at all points. The acceleration vector has a tangential component along the direction of the velocity at the upper location on the *y*-axis; hence, the satellite is speeding up. Just the opposite is true at the lower position.
11. The laser beam will hit the far wall at a lower elevation than it left, as the floor is accelerating upward. Relative to the lab, the laser beam "falls." So we would expect this to happen in a gravitational field. The mass of light, or even an object with mass, is not relevant.

Problems

13. 7.4×10^{-8} N
15. a. 7.01×10^{-7} N; b. The mass of Jupiter is

$$m_{\mathrm{J}} = 1.90 \times 10^{27}\ \mathrm{kg}$$
$$F_{\mathrm{J}} = 1.35 \times 10^{-6}\ \mathrm{N}$$
$$\frac{F_{\mathrm{f}}}{F_{\mathrm{J}}} = 0.521$$

17. a. 9.25×10^{-6} N; b. Not very, as the ISS is not even symmetrical, much less spherically symmetrical.
19. a. $1.41 \times 10^{-15}\ \mathrm{m/s^2}$; b. $1.69 \times 10^{-4}\ \mathrm{m/s^2}$
21. a. $1.62\ \mathrm{m/s^2}$; b. $3.75\ \mathrm{m/s^2}$
23. a. 147 N; b. 25.5 N; c. 15 kg; d. 0; e. 15 kg
25. $12\ \mathrm{m/s^2}$
27. $(3/2)R_{\mathrm{E}}$
29. 5000 m/s
31. 1440 m/s
33. 11 km/s
35. a. 5.85×10^{10} J; b. -5.85×10^{10} J; No. It assumes the kinetic energy is recoverable. This would not even be reasonable if we had an elevator between Earth and the Moon.
37. a. 0.25; b. 0.125
39. a. 5.08×10^{3} km; b. This less than the radius of Earth.
41. 1.89×10^{27} kg
43. a. 4.01×10^{13} kg; b. The satellite must be outside the radius of the asteroid, so it can't be larger than this. If

it were this size, then its density would be about $1200\ \text{kg/m}^3$. This is just above that of water, so this seems quite reasonable.

45. a. $1.66 \times 10^{-10}\ \text{m/s}^2$; Yes, the centripetal acceleration is so small it supports the contention that a nearly inertial frame of reference can be located at the Sun. b. $2.17 \times 10^5\,\text{m/s}$

47. $1.98 \times 10^{30}\ \text{kg}$; The values are the same within 0.05%.

49. Compare Equation 13.8 and Equation 13.11 to see that they differ only in that the circular radius, *r*, is replaced by the semi-major axis, *a*. Therefore, the mean radius is one-half the sum of the aphelion and perihelion, the same as the semi-major axis.

51. The semi-major axis, 3.78 AU is found from the equation for the period. This is one-half the sum of the aphelion and perihelion, giving an aphelion distance of 4.95 AU.

53. 1.75 years

55. 19,800 N; this is clearly not survivable

57. $1.19 \times 10^7\,\text{km}$

Additional Problems

59. a. $1.85 \times 10^{14}\ \text{N}$; b. Don't do it!

61. $1.49 \times 10^8\,\text{km}$

63. The value of g for this planet is 3.8 m/s^2, which is about one-fourth that of Earth. So they are weak high jumpers.

65. At the North Pole, 983 N; at the equator, 980 N

67. a. The escape velocity is still 43.6 km/s. By launching from Earth in the direction of Earth's tangential velocity, you need $43.4 - 29.8 = 13.8\ \text{km/s}$ relative to Earth. b. The total energy is zero and the trajectory is a parabola.

69. 61.5 km/s

71. a. $1.3 \times 10^7\ \text{m}$; b. $1.56 \times 10^{10}\ \text{J}$; $-3.12 \times 10^{10}\ \text{J}$; $-1.56 \times 10^{10}\ \text{J}$

73. a. $6.24 \times 10^3\ \text{s}$ or about 1.8 hours. This was using the 520 km average diameter. b. Vesta is clearly not very spherical, so you would need to be above the largest dimension, nearly 580 km. More importantly, the nonspherical nature would disturb the orbit very quickly, so this calculation would not be very accurate even for one orbit.

75. a. 323 km/s; b. No, you need only the difference between the solar system's orbital speed and escape speed, so about $323 - 228 = 95\ \text{km/s}$.

77. Setting $e = 1$, we have $\frac{\alpha}{r} = 1 + \cos\theta \to \alpha = r + r\cos\theta = r + x$; hence, $r^2 = x^2 + y^2 = (\alpha - x)^2$. Expand and collect to show $x = \frac{1}{-2\alpha}y^2 + \frac{\alpha}{2}$.

79. Substitute directly into the energy equation using $pv_p = qv_q$ from conservation of angular momentum, and solve for v_p.

Challenge Problems

81. $g = \frac{4}{3}G\rho\pi r \to F = mg = \left[\frac{4}{3}Gm\rho\pi\right] r$, and from $F = m\frac{d^2r}{dt^2}$, we get $\frac{d^2r}{dt^2} = \left[\frac{4}{3}G\rho\pi\right] r$ where the first term is ω^2. Then $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{3}{4G\rho\pi}}$ and if we substitute $\rho = \frac{M}{4/3\pi R^3}$, we get the same expression as for the period of orbit *R*.

83. Using the mass of the Sun and Earth's orbital radius, the equation gives $2.24 \times 10^{15}\,\text{m}^2/\text{s}$. The value of $\pi R_{\text{ES}}^2/(1\ \text{year})$ gives the same value.

85. $\Delta U = U_f - U_i = -\frac{GM_{\text{E}}m}{r_f} + \frac{GM_{\text{E}}m}{r_i} = GM_{\text{E}}m\left(\frac{r_f - r_i}{r_f r_i}\right)$ where $h = r_f - r_i$. If $h << \text{R}_\text{E}$, then $r_f r_i \approx R_{\text{E}}^2$, and upon substitution, we have

$$\Delta U = GM_{\text{E}}m\left(\frac{h}{R_{\text{E}}^2}\right) = m\left(\frac{GM_{\text{E}}}{R_{\text{E}}^2}\right)h$$ where we recognize the expression with the parenthesis as the definition of *g*.

87. a. Find the difference in force,

$F_{\text{tidal}} == \frac{2GMm}{R^3}\Delta r$;

b. For the case given, using the Schwarzschild radius from a previous problem, we have a tidal force of 9.5×10^{-3} N. This won't even be noticed!

Chapter 14

Check Your Understanding

14.1 The pressure found in part (a) of the example is completely independent of the width and length of the lake; it depends only on its average depth at the dam. Thus, the force depends only on the water's average depth and the dimensions of the dam, not on the horizontal extent of the reservoir. In the diagram, note that the thickness of the dam increases with depth to balance the increasing force due to the increasing pressure.

14.2 The density of mercury is 13.6 times greater than the density of water. It takes approximately 76 cm (29.9 in.) of mercury to measure the pressure of the atmosphere, whereas it would take approximately 10 m (34 ft.) of water.

14.3 Yes, it would still work, but since a gas is compressible, it would not operate as efficiently. When the force is applied, the gas would first compress and warm. Hence, the air in the brake lines must be bled out in order for the brakes to work properly.

Conceptual Questions

1. Mercury and water are liquid at room temperature and atmospheric pressure. Air is a gas at room temperature and atmospheric pressure. Glass is an amorphous solid (non-crystalline) material at room temperature and atmospheric pressure. At one time, it was thought that glass flowed, but flowed very slowly. This theory came from the observation that old glass planes were thicker at the bottom. It is now thought unlikely that this theory is accurate.

3. The density of air decreases with altitude. For a column of air of a constant temperature, the density decreases exponentially with altitude. This is a fair approximation, but since the temperature does change with altitude, it is only an approximation.

5. Pressure is force divided by area. If a knife is sharp, the force applied to the cutting surface is divided over a smaller area than the same force applied with a dull knife. This means that the pressure would be greater for the sharper knife, increasing its ability to cut.

7. If the two chunks of ice had the same volume, they would produce the same volume of water. The glacier would cause the greatest rise in the lake, however, because part of the floating chunk of ice is already submerged in the lake, and is thus already contributing to the lake's level.

9. The pressure is acting all around your body, assuming you are not in a vacuum.

11. Because the river level is very high, it has started to leak under the levee. Sandbags are placed around the leak, and the water held by them rises until it is the same level as the river, at which point the water there stops rising. The sandbags will absorb water until the water reaches the height of the water in the levee.

13. Atmospheric pressure does not affect the gas pressure in a rigid tank, but it does affect the pressure inside a balloon. In general, atmospheric pressure affects fluid pressure unless the fluid is enclosed in a rigid container.

15. The pressure of the atmosphere is due to the weight of the air above. The pressure, force per area, on the manometer will be the same at the same depth of the atmosphere.

17. Not at all. Pascal's principle says that the change in the pressure is exerted through the fluid. The reason that the full tub requires more force to pull the plug is because of the weight of the water above the plug.

19. The buoyant force is equal to the weight of the fluid displaced. The greater the density of the fluid, the less fluid that is needed to be displaced to have the weight of the object be supported and to float. Since the density of salt water is higher than that of fresh water, less salt water will be displaced, and the ship will float higher.

21. Consider two different pipes connected to a single pipe of a smaller diameter, with fluid flowing from the two pipes into the smaller pipe. Since the fluid is forced through a smaller cross-sectional area, it must move faster as the flow lines become closer together. Likewise, if a pipe with a large radius feeds into a pipe with a small radius, the stream lines will become closer together and the fluid will move faster.

23. The mass of water that enters a cross-sectional area must equal the amount that leaves. From the continuity equation, we know that the density times the area times the velocity must remain constant. Since the density of the water does not change, the velocity times the cross-sectional area entering a region must equal the cross-sectional area times the velocity leaving the region. Since the velocity of the fountain stream decreases as it rises due to gravity, the area must increase. Since the velocity of the faucet stream speeds up as it falls, the area must decrease.

25. When the tube narrows, the fluid is forced to speed up, thanks to the continuity equation and the work done on the fluid. Where the tube is narrow, the pressure decreases. This means that the entrained fluid will be pushed into the narrow area.

27. The work done by pressure can be used to increase the kinetic energy and to gain potential energy. As the height becomes larger, there is less energy left to give to kinetic energy. Eventually, there will be a maximum height that cannot be overcome.

29. Because of the speed of the air outside the building, the pressure outside the house decreases. The greater pressure inside the building can essentially blow off the roof or cause the building to explode.

31. The air inside the hose has kinetic energy due to its motion. The kinetic energy can be used to do work against the pressure difference.

33. Potential energy due to position, kinetic energy due to velocity, and the work done by a pressure difference.

35. The water has kinetic energy due to its motion. This energy can be converted into work against the difference in pressure.

37. The water in the center of the stream is moving faster than the water near the shore due to resistance between the water and the shore and between the layers of fluid. There is also probably more turbulence near the shore, which will also slow the water down. When paddling up stream, the water pushes against the canoe, so it is better to stay near the shore to minimize the force pushing against the canoe. When moving downstream, the water pushes the canoe, increasing its velocity, so it is better to stay in the middle of the stream to maximize this effect.

39. You would expect the speed to be slower after the obstruction. Resistance is increased due to the reduction in size of the opening, and turbulence will be created because of the obstruction, both of which will clause the fluid to slow down.

Problems

41. 1.610 cm^3

43. The mass is 2.58 g. The volume of your body increases by the volume of air you inhale. The average density of your body decreases when you take a deep breath because the density of air is substantially smaller than the average density of the body.

45. 3.99 cm

47. 2.86 times denser

49. 15.6 g/cm^3

51. $0.760 \text{ m} = 76.0 \text{ cm} = 760 \text{ mm}$

53. proof

55. a. Pressure at $h = 7.06 \times 10^6 \text{N;}$
b. The pressure increases as the depth increases, so the dam must be built thicker toward the bottom to withstand the greater pressure.

57. 4.08 m

59. 251 atm

61. $5.76 \times 10^3 \text{ N extra force}$

63. If the system is not moving, the friction would not play a role. With friction, we know there are losses, so that $W_\text{o} = W_\text{i} - W_\text{f}$; therefore, the work output is less than the work input. In other words, to account for friction, you would need to push harder on the input piston than was calculated.

65. a. 99.5% submerged; b. 96.9% submerged

67. a. 39.5 g; b. 50 cm^3; c. 0.79 g/cm^3; ethyl alcohol

69. a. 960 kg/m^3; b. 6.34%; She floats higher in seawater.

71. a. 0.24; b. 0.72; c. Yes, the cork will float in ethyl alcohol.

73.
$$\begin{aligned} \text{net } F &= F_2 - F_1 = p_2 A - p_1 A = (p_2 - p_1)A = (h_2 \rho_\text{fl} g - h_1 \rho_\text{fl} g)\, A \\ &= (h_2 - h_1)\rho_\text{fl} g A, \text{ where } \rho_\text{fl} = \text{density of fluid.} \\ \text{net } F &= (h_2 - h_1) A \rho_\text{fl} g = V_\text{fl} \rho_\text{fl} g = m_\text{fl} g = w_\text{fl} \end{aligned}$$

75. $2.78\ \text{cm}^3/\text{s}$
77. a. 0.75 m/s; b. 0.13 m/s
79. a. 12.6 m/s; b. $0.0800\ \text{m}^3/\text{s}$; c. No, the flow rate and the velocity are independent of the density of the fluid.
81. If the fluid is incompressible, the flow rate through both sides will be equal:

$$Q = A_1\bar{v}_1 = A_2\bar{v}_2, \text{ or } \pi\frac{d_1^2}{4}\bar{v}_1 = \pi\frac{d_2^2}{4}\bar{v}_2 \Rightarrow \bar{v}_2 = \bar{v}_1(d_1^2/d_2^2) = \bar{v}_1(d_1/d_2)^2$$

83.
$$\begin{aligned} F &= pA \Rightarrow p = \frac{F}{A}, \\ [p] &= \text{N/m}^2 = \text{N}\cdot\text{m/m}^3 = \text{J/m}^3 = \text{energy/volume} \end{aligned}$$
85. −135 mm Hg
87. a. $1.58 \times 10^6\ \text{N/m}^2$; b. 163 m
89. a. $v_2 = 3.28\,\frac{\text{m}}{\text{s}}$;
b. $t = 0.55\ \text{s}$
$x = vt = 1.81\ \text{m}$
91. a. $3.02 \times 10^{-3}\ \text{N}$; b. 1.03×10^{-3}
93. proof
95. 40 m/s
97. $0.537r$; The radius is reduced to 53.7% of its normal value.
99. a. $2.40 \times 10^9\ \text{N}\cdot\text{s/m}^5$; b. $48.3\ (\text{N/m}^2)\cdot\text{s}$; c. $2.67 \times 10^4\ \text{W}$
101. a. Nozzle: $v = 25.5\,\frac{\text{m}}{\text{s}}$
$N_\text{R} = 1.27 \times 10^5 > 2000 \Rightarrow$
Flow is not laminar.
b. Hose: $v = 1.96\,\frac{\text{m}}{\text{s}}$
$N_\text{R} = 35{,}100 > 2000 \Rightarrow$
Flow is not laminar.
103. $3.16 \times 10^{-4}\ \text{m}^3/\text{s}$

Additional Problems

105. 30.6 m

107. a.
$$p_{120} = 1.60 \times 10^4\ \text{N/m}^2\ ;$$
$$p_{80} = 1.07 \times 10^4\ \text{N/m}^2$$
b. Since an infant is only approximately 20 inches tall, while an adult is approximately 70 inches tall, the blood pressure for an infant would be expected to be smaller than that of an adult. The blood only feels a pressure of 20 inches rather than 70 inches, so the pressure should be smaller.
109. a. 41.4 g; b. 41.4 cm^3; c. 1.09 g/cm^3. This is clearly not the density of the bone everywhere. The air pockets will have a density of approximately $1.29 \times 10^{-3}\ \text{g/cm}^3$, while the bone will be substantially denser.
111. 12.3 N
113. a. $3.02 \times 10^{-2}\ \text{cm/s}$. (This small speed allows time for diffusion of materials to and from the blood.) b. 2.37×10^{10} capillaries. (This large number is an overestimate, but it is still reasonable.)
115. a. $2.76 \times 10^5\ \text{N/m}^2$; b. $P_2 = 2.81 \times 10^5\ \text{N/m}^2$
117. $8.7 \times 10^{-2}\ \text{mm}^3/\text{s}$
119. a. 1.52; b. Turbulence would decrease the flow rate of the blood, which would require an even larger increase in the pressure difference, leading to higher blood pressure.

Challenge Problems

121. $p = 0.99 \times 10^5\ \text{Pa}$
123. $800\ \text{kg/m}^3$
125. 11.2 m/s

127. a. 71.8 m/s; b. 257 m/s
129. a. $150\ \text{cm}^3/\text{s}$; b. $33.3\ \text{cm}^3/\text{s}$; c. $25.0\ \text{cm}^3/\text{s}$; d. $0.0100\ \text{cm}^3/\text{s}$; e. $0.0300\ \text{cm}^3/\text{s}$
131. a. $1.20 \times 10^5\ \text{N/m}^2$; b. The flow rate in the main increases by 90%. c. There are approximately 38 more users in the afternoon.

Chapter 15

Check Your Understanding

15.1 The ruler is a stiffer system, which carries greater force for the same amount of displacement. The ruler snaps your hand with greater force, which hurts more.
15.2 You could increase the mass of the object that is oscillating. Other options would be to reduce the amplitude, or use a less stiff spring.
15.3 A ketchup bottle sits on a lazy Susan in the center of the dinner table. You set it rotating in uniform circular motion. A set of lights shine on the bottle, producing a shadow on the wall.
15.4 The movement of the pendulums will not differ at all because the mass of the bob has no effect on the motion of a simple pendulum. The pendulums are only affected by the period (which is related to the pendulum's length) and by the acceleration due to gravity.
15.5 Friction often comes into play whenever an object is moving. Friction causes damping in a harmonic oscillator.
15.6 The performer must be singing a note that corresponds to the natural frequency of the glass. As the sound wave is directed at the glass, the glass responds by resonating at the same frequency as the sound wave. With enough energy introduced into the system, the glass begins to vibrate and eventually shatters.

Conceptual Questions

1. The restoring force must be proportional to the displacement and act opposite to the direction of motion with no drag forces or friction. The frequency of oscillation does not depend on the amplitude.
3. Examples: Mass attached to a spring on a frictionless table, a mass hanging from a string, a simple pendulum with a small amplitude of motion. All of these examples have frequencies of oscillation that are independent of amplitude.
5. Since the frequency is proportional to the square root of the force constant and inversely proportional to the square root of the mass, it is likely that the truck is heavily loaded, since the force constant would be the same whether the truck is empty or heavily loaded.
7. In a car, elastic potential energy is stored when the shock is extended or compressed. In some running shoes elastic potential energy is stored in the compression of the material of the soles of the running shoes. In pole vaulting, elastic potential energy is stored in the bending of the pole.
9. The overall system is stable. There may be times when the stability is interrupted by a storm, but the driving force provided by the sun bring the atmosphere back into a stable pattern.
11. The maximum speed is equal to $v_{\text{max}} = A\omega$ and the angular frequency is independent of the amplitude, so the amplitude would be affected. The radius of the circle represents the amplitude of the circle, so make the amplitude larger.
13. The period of the pendulum is $T = 2\pi\sqrt{L/g}$. In summer, the length increases, and the period increases. If the period should be one second, but period is longer than one second in the summer, it will oscillate fewer than 60 times a minute and clock will run slow. In the winter it will run fast.
15. A car shock absorber.
17. The second law of thermodynamics states that perpetual motion machines are impossible. Eventually the ordered motion of the system decreases and returns to equilibrium.
19. All harmonic motion is damped harmonic motion, but the damping may be negligible. This is due to friction and drag forces. It is easy to come up with five examples of damped motion: (1) A mass oscillating on a hanging on a spring (it eventually comes to rest). (2) Shock absorbers in a car (thankfully they also come to rest). (3) A pendulum is a grandfather clock (weights are added to add energy to the oscillations). (4) A child on a swing (eventually comes to rest unless energy is added by pushing the child). (5) A marble rolling in a bowl (eventually comes to rest). As for the undamped motion, even a mass on a spring in a vacuum will eventually come to rest due to internal forces in the spring. Damping may be negligible, but cannot be eliminated.

Problems

21. Proof
23. 0.400 s/beat
25. 12,500 Hz
27. a. 340 km/hr; b. 11.3×10^3 rev/min
29. $f = \frac{1}{3} f_0$
31. 0.009 kg; 2%
33. a. 1.57×10^5 N/m; b. 77 kg, yes, he is eligible to play
35. a. 6.53×10^3 N/m; b. yes, when the man is at his lowest point in his hopping the spring will be compressed the most
37. a. 1.99 Hz; b. 56.9 cm; c. 77.6 cm
39. a. 0.335 m/s; b. 5.61×10^{-4} J
41. a. $x(t) = 2\,\text{m}\cos\left(0.52\text{s}^{-1} t\right)$; b. $v(t) = (-1.05\text{ m/s})\sin\left(0.52\text{s}^{-1} t\right)$
43. 24.8 cm
45. 4.01 s
47. 1.58 s
49. 9.82002 m/s^2
51. 6%
53. 141 J
55. a. 4.90×10^{-3} m; b. 1.15×10^{-2} m

Additional Problems

57. 94.7 kg
59. a. 314 N/m; b. 1.00 s; c. 1.25 m/s
61. ratio of 2.45
63. The length must increase by 0.0116%.
65. $\theta = (0.31\text{ rad})\sin\left(3.13\text{ s}^{-1} t\right)$
67. a. 0.99 s; b. 0.11 m

Challenge Problems

69. a. 3.95×10^6 N/m; b. 7.90×10^6 J
71. $F \approx -\text{constant } r'$
73. a. 7.54 cm; b. 3.25×10^4 N/m

Chapter 16

Check Your Understanding

16.1 The wavelength of the waves depends on the frequency and the velocity of the wave. The frequency of the sound wave is equal to the frequency of the wave on the string. The wavelengths of the sound waves and the waves on the string are equal only if the velocities of the waves are the same, which is not always the case. If the speed of the sound wave is different from the speed of the wave on the string, the wavelengths are different. This velocity of sound waves will be discussed in Sound.

16.2 In a transverse wave, the wave may move at a constant propagation velocity through the medium, but the medium oscillates perpendicular to the motion of the wave. If the wave moves in the positive *x*-direction, the medium oscillates up and down in the *y*-direction. The velocity of the medium is therefore not constant, but the medium's velocity and acceleration are similar to that of the simple harmonic motion of a mass on a spring.

16.3 Yes, a cosine function is equal to a sine function with a phase shift, and either function can be used in a wave function. Which function is more convenient to use depends on the initial conditions. In Figure 16.11, the wave has an initial height of $y(0.00, 0.00) = 0$ and then the wave height increases to the maximum height at the crest. If the initial height at the initial time was equal to the amplitude of the wave $y(0.00, 0.00) = +A$, then it might be more convenient to model the wave with a cosine function.

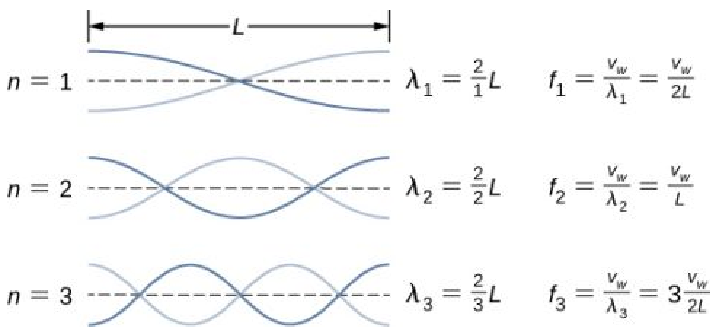
16.4 This wave, with amplitude $A = 0.5\text{ m}$, wavelength $\lambda = 10.00\text{ m}$, period $T = 0.50\text{ s}$, is a solution to the wave equation with a wave velocity $v = 20.00\text{ m/s}$.

16.5 Since the speed of a wave on a taunt string is proportional to the square root of the tension divided by the linear density, the wave speed would increase by $\sqrt{2}$.

16.6 At first glance, the time-averaged power of a sinusoidal wave on a string may look proportional to the linear density of the string because $P = \frac{1}{2}\mu A^2\omega^2 v$; however, the speed of the wave depends on the linear density. Replacing the wave speed with $\sqrt{\frac{F_T}{\mu}}$ shows that the power is proportional to the square root of tension and proportional to the square root of the linear mass density:

$P = \frac{1}{2}\mu A^2\omega^2 v = \frac{1}{2}\mu A^2\omega^2\sqrt{\frac{F_T}{\mu}} = \frac{1}{2}A^2\omega^2\sqrt{\mu F_T}.$

16.7 Yes, the equations would work equally well for symmetric boundary conditions of a medium free to oscillate on each end where there was an antinode on each end. The normal modes of the first three modes are shown below. The dotted line shows the equilibrium position of the medium.



Note that the first mode is two quarters, or one half, of a wavelength. The second mode is one quarter of a wavelength, followed by one half of a wavelength, followed by one quarter of a wavelength, or one full wavelength. The third mode is one and a half wavelengths. These are the same result as the string with a node on each end. The equations for symmetrical boundary conditions work equally well for fixed boundary conditions and free boundary conditions. These results will be revisited in the next chapter when discussing sound wave in an open tube.

Conceptual Questions

1. A wave on a guitar string is an example of a transverse wave. The disturbance of the string moves perpendicular to the propagation of the wave. The sound produced by the string is a longitudinal wave where the disturbance of the air moves parallel to the propagation of the wave.

3. Propagation speed is the speed of the wave propagating through the medium. If the wave speed is constant, the speed can be found by $v = \frac{\lambda}{T} = \lambda f$. The frequency is the number of wave that pass a point per unit time. The wavelength is directly proportional to the wave speed and inversely proportional to the frequency.

5. No, the distance you move your hand up and down will determine the amplitude of the wave. The wavelength will depend on the frequency you move your hand up and down, and the speed of the wave through the spring.

7. Light from the Sun and stars reach Earth through empty space where there is no medium present.

9. The wavelength is equal to the velocity of the wave times the frequency and the wave number is equal to $k = \frac{2\pi}{\lambda}$, so yes, the wave number will depend on the frequency and also depend on the velocity of the wave propagating through the spring.

11. The medium moves in simple harmonic motion as the wave propagates through the medium, continuously changing speed, therefore it accelerates. The acceleration of the medium is due to the restoring force of the medium, which acts in the opposite direction of the displacement.

13. The wave speed is proportional to the square root of the tension, so the speed is doubled.

15. Since the speed of a wave on a string is inversely proportional to the square root of the linear mass density, the speed would be higher in the low linear mass density of the string.

17. The tension in the wire is due to the weight of the electrical power cable.

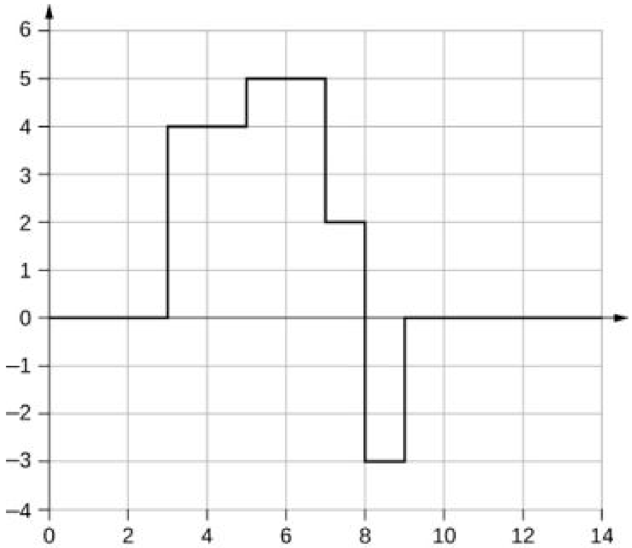
19. The time averaged power is $P = \frac{E_\lambda}{T} = \frac{1}{2}\mu A^2 \omega^2 \frac{\lambda}{T} = \frac{1}{2}\mu A^2 \omega^2 v$. If the frequency or amplitude is halved, the power decreases by a factor of 4.

21. As a portion on the string moves vertically, it exerts a force on the neighboring portion of the string, doing work on the portion and transferring the energy.

23. The intensity of a spherical wave is $I = \frac{P}{4\pi r^2}$, if no energy is dissipated the intensity will decrease by a factor of nine at three meters.

25. At the interface, the incident pulse produces a reflected pulse and a transmitted pulse. The reflected pulse would be out of phase with respect to the incident pulse, and would move at the same propagation speed as the incident pulse, but would move in the opposite direction. The transmitted pulse would travel in the same direction as the incident pulse, but at half the speed. The transmitted pulse would be in phase with the incident pulse. Both the reflected pulse and the transmitted pulse would have amplitudes less than the amplitude of the incident pulse.

27.



29. It may be as easy as changing the length and/or the density a small amount so that the parts do not resonate at the frequency of the motor.

31. Energy is supplied to the glass by the work done by the force of your finger on the glass. When supplied at the right frequency, standing waves form. The glass resonates and the vibrations produce sound.

33. For the equation $y(x, t) = 4.00\,\text{cm}\sin\left(3\,\text{m}^{-1}x\right)\cos\left(4\,\text{s}^{-1}t\right)$, there is a node because when $x = 0.00\,\text{m}$, $\sin\left(3\,\text{m}^{-1}\,(0.00\,\text{m})\right) = 0.00$, so $y(0.00\,\text{m}, t) = 0.00\,\text{m}$ for all time. For the equation $y(x, t) = 4.00\,\text{cm}\sin\left(3\,\text{m}^{-1}x + \frac{\pi}{2}\right)\cos\left(4\,\text{s}^{-1}t\right)$, there is an antinode because when $x = 0.00\,\text{m}$, $\sin\left(3\,\text{m}^{-1}\,(0.00\,\text{m}) + \frac{\pi}{2}\right) = +1.00$, so $y(0.00\,\text{m}, t)$ oscillates between +*A* and −*A* as the cosine term oscillates between +1 and -1.

Problems

35. $2d = vt \Rightarrow d = 11.25\,\text{m}$

37. $$\begin{aligned} v &= f\lambda, \text{ so that } f = 0.125\,\text{Hz, so that} \\ N &= 7.50\,\text{times} \end{aligned}$$

39. $v = f\lambda \Rightarrow \lambda = 0.400\,\text{m}$

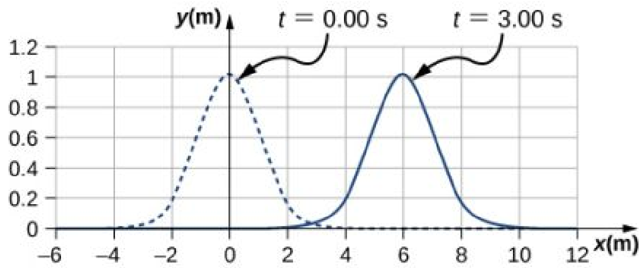
41. $v = f\lambda \Rightarrow f = 2.50 \times 10^9\,\text{Hz}$

43. a. The P-waves outrun the S-waves by a speed of $v = 3.20\,\text{km/s}$; therefore, $\Delta d = 0.320\,\text{km}$. b. Since the uncertainty in the distance is less than a kilometer, our answer to part (a) does not seem to limit the detection of nuclear bomb detonations. However, if the velocities are uncertain, then the uncertainty in the distance would increase and could then make it difficult to identify the source of the seismic waves.

45. $$\begin{aligned} v &= 1900\,\text{m/s} \\ \Delta t &= 1.05\,\mu\,\text{s} \end{aligned}$$

47. $y(x, t) = -0.037 \text{ cm}$

49.



The pulse will move $\Delta x = 6.00 \text{ m}$.

51. a. $A = 0.25 \text{ m}$; b. $k = 0.30 \text{ m}^{-1}$; c. $\omega = 0.90 \text{ s}^{-1}$; d. $v = 3.0 \text{ m/s}$; e. $\phi = \pi/3 \text{ rad}$; f. $\lambda = 20.93 \text{ m}$; g. $T = 6.98 \text{ s}$

53. $A = 0.30 \text{ m},\ \lambda = 4.50 \text{ m},\ v = 18.00 \text{ m/s},\ f = 4.00 \text{ Hz},\ T = 0.25 \text{ s}$

55. $y(x, t) = 0.23 \text{ m} \sin\left(3.49 \text{ m}^{-1} x - 0.63 \text{ s}^{-1} t\right)$

57. They have the same angular frequency, frequency, and period. They are traveling in opposite directions and $y_2(x, t)$ has twice the wavelength as $y_1(x, t)$ and is moving at half the wave speed.

59. Each particle of the medium moves a distance of 4*A* each period. The period can be found by dividing the velocity by the wavelength: $t = 10.42 \text{ s}$

61. a. $\mu = 0.040 \text{ kg/m}$; b. $v = 15.75 \text{ m/s}$

63. $v = 180 \text{ m/s}$

65. $v = 547.723 \text{ m/s},\ \Delta t = 5.48 \text{ ms}$

67. 0.707

69. $v_1 t + v_2 t = 2.00 \text{ m},\ t = 2.21 \text{ ms}$

71. $v = 288.68 \text{ m/s},\ \lambda = 0.73 \text{ m}$

73. a. $A = 0.0125 \text{ cm}$; b. $F_T = 0.96 \text{ N}$

75. $v = 74.54 \text{ m/s},\ P_\lambda = 91.85 \text{ W}$

77. a. $I = 20.0 \text{ W/m}^2$; b.

$$I = \frac{P}{A},\ A = 10.0 \text{ m}^2$$
$$A = 4\pi r^2,\ r = 0.892 \text{ m}$$

79. $I = 650 \text{ W/m}^2$

81.

$$P \propto E \propto I \propto X^2 \Rightarrow \frac{P_2}{P_1} = \left(\frac{X_2}{X_1}\right)^2$$
$$P_2 = 2.50 \text{ kW}$$

83.

$$I \propto X^2 \Rightarrow \frac{I_1}{I_2} = \left(\frac{X_1}{X_2}\right)^2 \Rightarrow$$
$$I_2 = 3.38 \times 10^{-5} \text{ W/m}^2$$

85. $f = 100.00 \text{ Hz},\ A = 1.10 \text{ cm}$

87. a. $I_2 = 0.063 I_1$; b.

$$\begin{aligned} I_1 4\pi r_1^2 &= I_2 4\pi r_2^2 \\ r_2 &= 3.16 \text{ m} \end{aligned}$$

89. $2\pi r_1 A_1^2 = 2\pi r_2 A_2^2,\ A_1 = \left(\frac{r_2}{r_1}\right)^{1/2} A_1 = 0.17 \text{ m}$

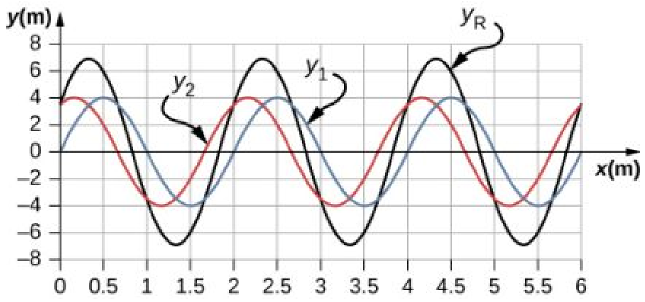
91. $y(x, t) = 0.63 \text{ m}$

93. $A_R = 2A\cos\left(\frac{\phi}{2}\right),\ \phi = 1.17\text{ rad}$

95. $y_R = 1.90\text{ cm}$

97.
$$\begin{aligned}\omega &= 6.28\text{ s}^{-1},\ k = 3.00\text{ m}^{-1},\ \phi = \tfrac{\pi}{8}\text{ rad},\\ A_R &= 2A\cos\left(\tfrac{\phi}{2}\right),\ A = 0.37\text{ m}\end{aligned}$$

99. a.



;
b. $\lambda = 2.0\text{ m}, A = 4\text{ m}$; c. $\lambda_R = 2.0\text{ m}, A_R = 6.93\text{ m}$

101. $y_R(x,t) = 2A\cos\left(\frac{\phi}{2}\right)\cos\left(kx - \omega t + \frac{\phi}{2}\right)$; The result is not surprising because $\cos(\theta) = \sin\left(\theta + \frac{\pi}{2}\right)$.

103.
$$\begin{aligned}\lambda_n &= \tfrac{2.00}{n}L, \quad f_n = \tfrac{v}{\lambda_n}\\ \lambda_1 &= 4.00\text{ m}, \quad f_1 = 12.5\text{ Hz}\\ \lambda_2 &= 2.00\text{ m}, \quad f_2 = 25.00\text{ Hz}\\ \lambda_3 &= 1.33\text{ m}, \quad f_3 = 37.59\text{ Hz}\end{aligned}$$

105.
$$\begin{aligned}v &= 158.11\text{ m/s}, \quad \lambda = 4.44\text{ m}, \quad f = 35.61\text{ Hz}\\ \lambda_s &= 9.63\text{ m}\end{aligned}$$

107. $y(x,t) = \left[0.60\text{ cm}\sin\left(3\text{ m}^{-1}x\right)\right]\cos\left(4\text{ s}^{-1}t\right)$

109.
$$\begin{aligned}\lambda_{100} &= 0.06\text{ m}\\ v &= 56.8\text{ m/s}, \quad f_n = nf_1, \quad n = 1,2,3,4,5...\\ f_{100} &= 947\text{ Hz}\end{aligned}$$

111. $T = 2\Delta t, \quad v = \frac{\lambda}{T}, \quad \lambda = 2.12\text{ m}$

113.
$$\begin{aligned}&\lambda_1 = 6.00\text{ m}, \quad \lambda_2 = 3.00\text{ m}, \quad \lambda_3 = 2.00\text{ m}, \quad \lambda_4 = 1.50\text{ m}\\ &v = 258.20\text{ m/s} = \lambda f\\ &f_1 = 43.03\text{ Hz}, \quad f_2 = 86.07\text{ Hz}, \quad f_3 = 129.10\text{ Hz}, \quad f_4 = 172.13\text{ Hz}\end{aligned}$$

115. $v = 134.16\text{ ms}, \lambda = 1.4\text{ m}, f = 95.83\text{ Hz}, T = 0.0104\text{ s}$

Additional Problems

117. $\lambda = 0.10\text{ m}$

119. a. $f = 4.74 \times 10^{14}\text{ Hz}$; b. $\lambda = 422\text{ nm}$

121. $\lambda = 16.00\text{ m}, \quad f = 0.10\text{ Hz}, \quad T = 10.00\text{ s}, \quad v = 1.6\text{ m/s}$

123. $\lambda = (v_b + v)\,t_b, \quad v = 3.75\text{ m/s}, \quad \lambda = 3.00\text{ m}$

125.
$$\frac{\partial^2(y_1+y_2)}{\partial t^2} = -A\omega^2\sin(kx-\omega t) - A\omega^2\sin(kx-\omega t+\phi)$$
$$\frac{\partial^2(y_1+y_2)}{\partial x^2} = -Ak^2\sin(kx-\omega t) - Ak^2\sin(kx-\omega t+\phi)$$
$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}$$

$$-A\omega^2\sin(kx-\omega t) - A\omega^2\sin(kx-\omega t+\phi) = \left(\frac{1}{v^2}\right)\left(-Ak^2\sin(kx-\omega t) - Ak^2\sin(kx-\omega t+\phi)\right)$$
$$v = \frac{\omega}{k}$$

127. $y(x,t) = 0.40\sin(0.15\,\text{x} + 1.5\,t)$

129. $v = 223.61\text{ m/s},\ k = 1.57\text{ m}^{-1},\ \omega = 351.07\text{ s}^{-1}$

131.
$$P = \frac{1}{2}A^2(2\pi f)^2\sqrt{\mu F_T}$$
$$\mu = 2.00\times10^{-4}\text{ kg/m}$$

133. $P = \frac{1}{2}\mu A^2\omega^2\frac{\lambda}{T},\ \mu = 0.0018\text{ kg/m}$
$F_T = v^2\mu = \left(\frac{\lambda}{T}\right)^2\mu = 110\text{ N}$

135. a. $A_R = 2A\cos\left(\frac{\phi}{2}\right),\ \cos\left(\frac{\phi}{2}\right) = 1,\ \phi = 0, 2\pi, 4\pi, \ldots$; b.
$A_R = 2A\cos\left(\frac{\phi}{2}\right),\ \cos\left(\frac{\phi}{2}\right) = 0,\ \phi = 0, \pi, 3\pi, 5\pi\ldots$

137. $y_R(x,t) = 0.6\text{ m}\sin\left(4\text{ m}^{-1}x\right)\cos\left(3\text{ s}^{-1}t\right)$

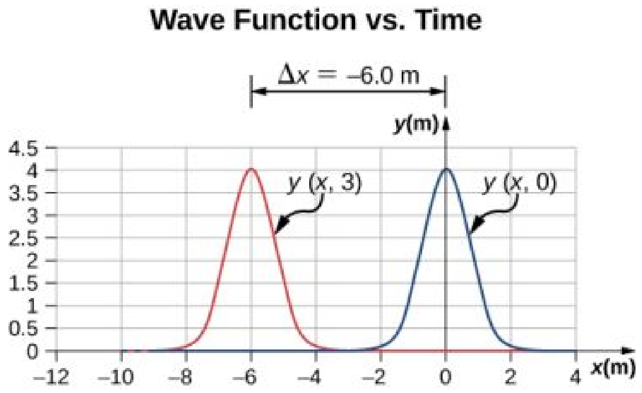
139. a. $(1)\ F_T - 20.00\text{ kg}\left(9.80\text{ m/s}^2\right)\cos 45° = 0$
$(2)\ m\left(9.80\text{ m/s}^2\right) - F_T = 0$
$m = 14.14\text{ kg}$; b.
$$\begin{aligned} F_T &= 138.57\text{ N} \\ v &= 74.45\text{ m/s} \end{aligned}$$

141. $F_T = 2\text{ N}, v = 6.73\text{ m/s}$

143. a. $f_n = \frac{nv}{2L},\ v = \frac{2Lf_{n+1}}{n+1},\ \frac{n+1}{n} = \frac{2Lf_{n+1}}{2Lf_n},\ 1 + \frac{1}{n} = 1.2,\ n = 5$; b. $F_T = 245.76\text{ N}$
$\lambda_n = \frac{2}{n}L,\ \lambda_5 = 1.6\text{ m},\ \lambda_6 = 1.33\text{ m}$

Challenge Problems

145. a. Moves in the negative *x* direction at a propagation speed of $v = 2.00\text{ m/s}$. b. $\Delta x = -6.00\text{ m}$; c.



$$\sin(kx - \omega t) = \sin\left(kx + \frac{\phi}{2}\right)\cos\left(\omega t + \frac{\phi}{2}\right) - \cos\left(kx + \frac{\phi}{2}\right)\sin\left(\omega t + \frac{\phi}{2}\right)$$

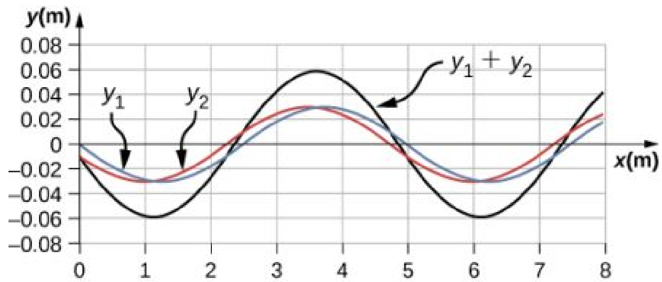
147.
$$\sin(kx - \omega t + \phi) = \sin\left(kx + \frac{\phi}{2}\right)\cos\left(\omega t + \frac{\phi}{2}\right) + \cos\left(kx + \frac{\phi}{2}\right)\sin\left(\omega t + \frac{\phi}{2}\right)$$

$$\sin(kx - \omega t) + \sin(kx + \omega t + \phi) = 2\sin\left(kx + \frac{\phi}{2}\right)\cos\left(\omega t + \frac{\phi}{2}\right)$$

$$y_R = 2A\sin\left(kx + \frac{\phi}{2}\right)\cos\left(\omega t + \frac{\phi}{2}\right)$$

149.
$$\sin\left(kx + \frac{\phi}{2}\right) = 0,\ kx + \frac{\phi}{2} = 0, \pi, 2\pi,\ 1.26\ \text{m}^{-1}x + \frac{\pi}{20} = \pi, 2\pi, 3\pi;$$

$$x = 2.37\ \text{m}, 4.86\ \text{m}, 7.35\ \text{m}$$



Chapter 17

Check Your Understanding

17.1 Sound and light both travel at definite speeds, and the speed of sound is slower than the speed of light. The first shell is probably very close by, so the speed difference is not noticeable. The second shell is farther away, so the light arrives at your eyes noticeably sooner than the sound wave arrives at your ears.

17.2 10 dB: rustle of leaves; 50 dB: average office; 100 dB: noisy factory

17.3 Amplitude is directly proportional to the experience of loudness. As amplitude increases, loudness increases.

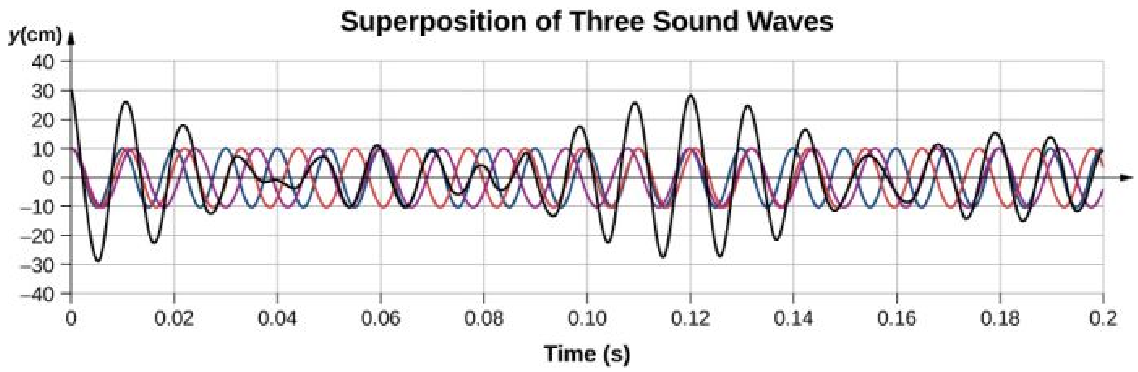
17.4 In the example, the two speakers were producing sound at a single frequency. Music has various frequencies and wavelengths.

17.5 Regular headphones only block sound waves with a physical barrier. Noise-canceling headphones use destructive interference to reduce the loudness of outside sounds.

17.6 When the tube resonates at its natural frequency, the wave's node is located at the closed end of the tube, and the antinode is located at the open end. The length of the tube is equal to one-fourth of the wavelength of this wave. Thus, if we know the wavelength of the wave, we can determine the length of the tube.

17.7 Compare their sizes. High-pitch instruments are generally smaller than low-pitch instruments because they generate a smaller wavelength.

17.8 An easy way to understand this event is to use a graph, as shown below. It appears that beats are produced, but with a more complex pattern of interference.



17.9 If I am driving and I hear Doppler shift in an ambulance siren, I would be able to tell when it was getting closer and also if it has passed by. This would help me to know whether I needed to pull over and let the ambulance through.

Conceptual Questions

1. Sound is a disturbance of matter (a pressure wave) that is transmitted from its source outward. Hearing is the human perception of sound.

3. Consider a sound wave moving through air. The pressure of the air is the equilibrium condition, it is the change in pressure that produces the sound wave.

5. The frequency does not change as the sound wave moves from one medium to another. Since the speed changes and the frequency does not, the wavelength must change. This is similar to the driving force of a harmonic oscillator or a wave on the string.

7. The transducer sends out a sound wave, which reflects off the object in question and measures the time it takes for the sound wave to return. Since the speed of sound is constant, the distance to the object can found by multiplying the velocity of sound by half the time interval measured.

9. The ear plugs reduce the intensity of the sound both in water and on land, but Navy researchers have found that sound under water is heard through vibrations mastoid, which is the bone behind the ear.

11. The fundamental wavelength of a tube open at each end is $2L$, where the wavelength of a tube open at one end and closed at one end is $4L$. The tube open at one end has the lower fundamental frequency, assuming the speed of sound is the same in both tubes.

13. The wavelength in each is twice the length of the tube. The frequency depends on the wavelength and the speed of the sound waves. The frequency in room *B* is higher because the speed of sound is higher where the temperature is higher.

15. When resonating at the fundamental frequency, the wavelength for pipe *C* is $4L$, and for pipes *A* and *B* is $2L$. The frequency is equal to $f = v/\lambda$. Pipe *C* has the lowest frequency and pipes *A* and *B* have equal frequencies, higher than the one in pipe *C*.

17. Since the boundary conditions are both symmetric, the frequencies are $f_n = \frac{nv}{2L}$. Since the speed is the same in each, the frequencies are the same. If the wave speed were doubled in the string, the frequencies in the string would be twice the frequencies in the tube.

19. The frequency of the unknown fork is 255 Hz. No, if only the 250 Hz fork is used, listening to the beat frequency could only limit the possible frequencies to 245 Hz or 255 Hz.

21. The beat frequency is 0.7 Hz.

23. Observer 1 will observe the highest frequency. Observer 2 will observe the lowest frequency. Observer 3 will hear a higher frequency than the source frequency, but lower than the frequency observed by observer 1, as the source approaches and a lower frequency than the source frequency, but higher than the frequency observed by observer 1, as the source moves away from observer 3.

25. Doppler radar can not only detect the distance to a storm, but also the speed and direction at which the storm is traveling.

27. The speed of sound decreases as the temperature decreases. The Mach number is equal to $M = \frac{v_s}{v}$, so the plane should slow down.

Problems

29. $s_{\max} = 4.00\ \text{nm}, \quad \lambda = 1.72\ \text{m}, \quad f = 200\ \text{Hz}, \quad v = 343.17\ \text{m/s}$

31. a. $\lambda = 68.60\ \mu\text{m}$; b. $\lambda = 360.00\ \mu\text{m}$

33. a. $k = 183.09\text{m}^{-1}$;
b. $\Delta P = -1.11\ \text{Pa}$

35.
$$s_1 = 7.00\ \text{nm}, \ s_2 = 3.00\ \text{nm}, \ kx_1 + \phi = 0\ \text{rad}$$
$$kx_2 + \phi = 1.128\ \text{rad}$$
$$k(x_2 - x_1) = 1.128\ \text{rad}, \ k = 5.64\ \text{m}^{-1}$$
$$\lambda = 1.11\ \text{m}, \ f = 306.31\ \text{Hz}$$

37.
$$k = 5.28 \times 10^3\ \text{m}$$
$$s(x, t) = 4.50\ \text{nm} \cos\left(5.28 \times 10^3\ \text{m}^{-1} x - 2\pi(5.00\ \text{MHz})t\right)$$

39. $\lambda = 3.43\ \text{mm}$

$$\lambda = 6.00\ \text{m}$$

41. $s_{\max} = 2.00\ \text{mm}$

$$v = 600\ \text{m/s}$$

$$T = 0.01\ \text{s}$$

43. (a) $f = 100\ \text{Hz}$, (b) $\lambda = 3.43\ \text{m}$

45. $f = 3400\ \text{Hz}$

47. a. $v = 5.96 \times 10^3\ \text{m/s}$; b. steel (from value in Table 17.1)

49. $v = 363\ \frac{\text{m}}{\text{s}}$

51. $\Delta x = 924\ \text{m}$

53.
$$V = 0.05\ \text{m}^3$$
$$m = 392.5\ \text{kg}$$
$$\rho = 7850\ \text{kg/m}^3$$
$$v = 5047.54\ \text{m/s}$$

55.
$$T_C = 35°\text{C}, \ v = 351.58\ \text{m/s}$$
$$\Delta x_1 = 35.16\ \text{m}, \ \Delta x_2 = 52.74\ \text{m}$$
$$\Delta x = 63.39\ \text{m}$$

57. a. $t_{5.00°\text{C}} = 0.0180\ \text{s}, \quad t_{35.0°\text{C}} = 0.0171\ \text{s}$; b. $\%\ \text{uncertainty} = 5.00\%$; c. This uncertainty could definitely cause difficulties for the bat, if it didn't continue to use sound as it closed in on its prey. A 5% uncertainty could be the difference between catching the prey around the neck or around the chest, which means that it could miss grabbing its prey.

59. 1.26×10^{-3} W/m^2
61. 85 dB
63. a. 93 dB; b. 83 dB
65. 1.58×10^{-13} W/m^2
67. A decrease of a factor of 10 in intensity corresponds to a reduction of 10 dB in sound level: $120\,\text{dB} - 10\,\text{dB} = 110\,\text{dB}$.
69. We know that 60 dB corresponds to a factor of 10^6 increase in intensity. Therefore,

$$I \propto X^2 \Rightarrow \frac{I_2}{I_1} = \left(\frac{X_2}{X_1}\right)^2, \text{so that } X_2 = 10^{-6}\text{ atm.}$$

120 dB corresponds to a factor of 10^{12} increase$\Rightarrow$ $10^{-9}\text{ atm}(10^{12})^{1/2} = 10^{-3}\text{ atm.}$
71. 28.2 dB
73. 1×10^6 km
75. $73\,\text{dB} - 70\,\text{dB} = 3\,\text{dB}$; Such a change in sound level is easily noticed.
77. 2.5; The 100-Hz tone must be 2.5 times more intense than the 4000-Hz sound to be audible by this person.
79. 0.974 m
81. 11.0 kHz; The ear is not particularly sensitive to this frequency, so we don't hear overtones due to the ear canal.
83. a. $v = 344.08\text{ m/s}, \ \lambda_1 = 16.00\text{ m}, \ f_1 = 21.51\text{ Hz};$
b. $\lambda_3 = 5.33\text{ m}, \ f_3 = 64.56\text{ Hz}$

85.

$$v_{\text{string}} = 149.07\text{ m/s}, \ \lambda_1 = 4.00\text{ m}, \ f_1 = 37.3\text{ Hz}$$
$$\lambda_1 = \frac{v}{f_1}, \ L = 13.8\text{ m}$$

87. a. $22.0°\text{C}$; b. 1.01 m

89.

$$\text{first overtone} = 180\text{ Hz};$$
$$\text{second overtone} = 270\text{ Hz};$$
$$\text{third overtone} = 360\text{ Hz}$$

91. 1.56 m
93. The pipe has symmetrical boundary conditions;

$$\lambda_n = \frac{2}{n}L, \ f_n = \frac{nv}{2L}, \ n = 1, 2, 3$$
$$\lambda_1 = 6.00\text{ m}, \ \lambda_2 = 3.00\text{ m}, \ \lambda_3 = 2.00\text{ m}$$
$$f_1 = 57.17\text{ Hz}, \ f_2 = 114.33\text{ Hz}, \ f_3 = 171.50\text{ Hz}$$

95.

$$\lambda_6 = 0.5\text{ m}$$
$$v = 1000\text{ m/s}$$
$$F_T = 6500\text{ N}$$

97. $f = 6.40\text{ kHz}$
99. 1.03 or 3%

101.
$$\begin{aligned} f_B &= |f_1 - f_2| \\ |128.3\text{ Hz} - 128.1\text{ Hz}| &= 0.2\text{ Hz}; \\ |128.3\text{ Hz} - 127.8\text{ Hz}| &= 0.5\text{ Hz}; \\ |128.1\text{ Hz} - 127.8\text{ Hz}| &= 0.3\text{ Hz} \end{aligned}$$

103. $v_A = 135.87\text{ m/s},\ v_B = 141.42\text{ m/s},$
$\lambda_A = \lambda_B = 0.40\text{ m}$
$\Delta f = 13.9\text{ Hz}$

105. $v = 155.54\text{ m/s},$
$f_{\text{string}} = 971.17\text{ Hz},\ n = 16.23$
$f_{\text{string}} = 1076.83\text{ Hz},\ n = 18.00$
The frequency is 1076.83 Hz and the wavelength is 0.14 m.

107.
$f_2 = f_1 \pm f_B = 260.00\text{ Hz} \pm 1.50\text{ Hz},$
so that $f_2 = 261.50\text{ Hz}$ or $f_2 = 258.50\text{ Hz}$

109.
$f_{\text{ace}} = \frac{f_1+f_2}{2};\ f_B = f_1 - f_2 (\text{assume} f_1 > f_2)$
$f_{\text{ace}} = \frac{(f_B+f_2)+f_2}{2} \Rightarrow$
$f_2 = 4099.750\text{ Hz}$
$f_1 = 4100.250\text{ Hz}$

111. a. 878 Hz; b. 735 Hz
113. $3.79 \times 10^3\text{ Hz}$
115. a. 12.9 m/s; b. 193 Hz
117. The first eagle hears $4.23 \times 10^3\text{ Hz}$. The second eagle hears $3.56 \times 10^3\text{ Hz}$.

119.
$v_s = 31.29\text{ m/s}$
$f_o = 1.12\text{ kHz}$

121. An audible shift occurs when $\frac{f_{\text{obs}}}{f_s} \geq 1.003;$
$$f_{\text{obs}} = f_s \frac{v}{v - v_s} \Rightarrow \frac{f_{\text{obs}}}{f_s} = \frac{v}{v - v_s} \Rightarrow$$
$$v_s = 0.990\text{ m/s}$$

123. $\theta = 30.02°$
$v_s = 680.00\text{ m/s}$
$\tan\theta = \frac{y}{v_s t},\ \ t = 21.65\text{ s}$

125. $\sin\theta = \frac{1}{M}, \quad \theta = 56.47°$

$y = 9.31 \text{ km}$

127. $s_1 = 6.34 \text{ nm}$

$s_2 = 2.30 \text{ nm}$

$kx_1 + \phi = 0 \text{ rad}$

$kx_2 + \phi = 1.20 \text{ rad}$

$k(x_2 - x_1) = 1.20 \text{ rad}$

$k = 3.00 \text{ m}^{-1}$

$\omega = 1019.62 \text{ s}^{-1}$

$s_1 = s_{\max}\cos(kx_1 - \phi)$

$\phi = 5.66 \text{ rad}$

$s(x,t) = 6.30 \text{ nm}\cos\left(3.00 \text{ m}^{-1} x - 1019.62 \text{ s}^{-1} t + 5.66\right)$

Additional Problems

129. $v_s = 346.40 \text{ m/s}$;

$\lambda_n = \frac{2}{n}L \qquad f_n = \frac{v_s}{\lambda_n}$

$\lambda_1 = 1.60 \text{ m} \quad f_1 = 216.50 \text{ Hz}$

$\lambda_2 = 0.80 \text{ m} \quad f_1 = 433.00 \text{ Hz}$

131. a. $\lambda_6 = 0.40 \text{ m}$; b. $\lambda_s = 2.40 \text{ m}$

$v = 57.15 \frac{\text{m}}{\text{s}}$

$f_6 = 142.89 \text{ Hz}$

133. $v = 344.08 \frac{\text{m}}{\text{s}}$

$v_A = 29.05 \frac{\text{m}}{\text{s}}, \quad v_B = 33.52 \text{ m/s}$

$f_A = 961.18 \text{ Hz},$

$f_B = 958.89 \text{ Hz}$

$f_{A,\text{beat}} = 161.18 \text{ Hz}, \quad f_{B,\text{beat}} = 158.89 \text{ Hz}$

135. $v = 345.24 \frac{\text{m}}{\text{s}}$; a. $I = 31.62 \frac{\mu\text{W}}{\text{m}^2}$; b. $I = 0.16 \frac{\mu\text{W}}{\text{m}^2}$; c. $s_{\max} = 104.39 \ \mu\text{m}$; d. $s_{\max} = 7.43 \ \mu\text{m}$

137. $\frac{f_A}{f_D} = \frac{v+v_s}{v-v_s}, \quad (v - v_s)\frac{f_A}{f_D} = v + v_s, \quad v = 347.39 \frac{\text{m}}{\text{s}}$

$T_C = 27.70°$

Challenge Problems

139.

$$\sqrt{x^2 + d^2} - x = \lambda, \quad x^2 + d^2 = (\lambda + x)^2$$

$$x^2 + d^2 = \lambda^2 + 2x\lambda + x^2, \quad d^2 = \lambda^2 + 2x\lambda$$

$$x = \frac{d^2 - \left(\frac{v}{f}\right)^2}{2\frac{v}{f}}$$

141. a. For maxima

$$\Delta r = d \sin\theta$$

$$d \sin\theta = n\lambda \quad n = 0, \pm 1, \pm 2...., \quad \theta = \sin^{-1}\left(n\frac{\lambda}{d}\right) \quad n = 0, \pm 1, \pm 2....$$

b. For minima,

$$\Delta r = d \sin\theta$$

$$d \sin\theta = \left(n + \frac{1}{2}\right)\lambda \quad n = 0, \pm 1, \pm 2....$$

$$\theta = \sin^{-1}\left(\left(n + \frac{1}{2}\right)\frac{\lambda}{d}\right) \quad n = 0, \pm 1, \pm 2....$$

143. a. $v_{\text{string}} = 160.73\,\frac{\text{m}}{\text{s}}, \quad f_{\text{string}} = 535.77\text{ Hz}$; b. $f_{\text{fork}} = 512\text{ Hz}$; c. $f_{\text{fork}} = \frac{n\sqrt{\frac{F_T}{\mu}}}{2L}, \quad F_T = 141.56\text{ N}$

145. a. $f = 268.62\text{ Hz}$; b. $\Delta f \approx \frac{1}{2}\frac{\Delta F_T}{F_T} f = 1.34\text{ Hz}$

147. a. $v = 466.07\,\frac{\text{m}}{\text{s}}$; b. $\lambda_9 = 51.11\text{ mm}$; c. $f_9 = 9.12\text{ kHz}$; d. $f_{\text{sound}} = 9.12\text{ kHz}$; e. $\lambda_{\text{air}} = 37.86\text{ mm}$

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